

Table 37.1 The Nine Phyla of Extant Vascular Plants

Phylum	Examples		Key Characteristics	Approximate Number of Living Species
Anthophyta	Flowering plants (angiosperms)		Heterosporous seed plants. Sperm not motile; conducted to egg by a pollen tube. Seeds enclosed within a fruit. Leaves greatly varied in size and form. Herbs, vines, shrubs, trees. About 14,000 genera.	250,000
Pterophyta	Ferns		Primarily homosporous (a few heterosporous) vascular plants. Sperm motile. External water necessary for fertilization. Leaves are megaphylls that uncoil as they mature. Sporophytes and virtually all gametophytes photosynthetic. About 365 genera.	11,000
Lycophyta	Club mosses		Homosporous or heterosporous vascular plants. Sperm motile. External water necessary for fertilization. Leaves are microphylls. About 12–13 genera.	1,150
Coniferophyta	Conifers (including pines, spruces, firs, yews, redwoods, and others)		Heterosporous seed plants. Sperm not motile; conducted to egg by a pollen tube. Leaves mostly needlelike or scalelike. Trees, shrubs. About 50 genera.	601
Cycadophyta	Cycads		Heterosporous vascular seed plants. Sperm flagellated and motile but confined within a pollen tube that grows to the vicinity of the egg. Palmlike plants with pinnate leaves. Secondary growth slow compared with that of the conifers. Ten genera.	206
Gnetophyta	Gnetophytes		Heterosporous vascular seed plants. Sperm not motile; conducted to egg by a pollen tube. The only gymnosperms with vessels. Trees, shrubs, vines. Three very diverse genera (<i>Ephedra</i> , <i>Gnetum</i> , <i>Welwitschia</i>).	65
Arthropphyta	Horsetails		Homosporous vascular plants. Sperm motile. External water necessary for fertilization. Stems ribbed, jointed, either photosynthetic or nonphotosynthetic. Leaves scalelike, in whorls, nonphotosynthetic at maturity. One genus.	15
Psilophyta	Whisk ferns		Homosporous vascular plants. Sperm motile. External water necessary for fertilization. No differentiation between root and shoot. No leaves; one of the two genera has scalelike enations and the other leaflike appendages.	6
Ginkgophyta	<i>Ginkgo</i>		Heterosporous vascular seed plants. Sperm flagellated and motile but conducted to the vicinity of the egg by a pollen tube. Deciduous tree with fan-shaped leaves that have evenly forking veins. Seeds resemble a small plum with fleshy, ill-scented outer covering.	1

Seedless Vascular Plants

The earliest vascular plants lacked seeds. Members of four phyla of living vascular plants lack seeds, as do at least three other phyla known only from fossils. As we explore the adaptations of the vascular plants, we focus on both reproductive strategies and the advantages of increasingly complex transport systems. We will begin with the most familiar phylum of seedless vascular plants, the ferns.

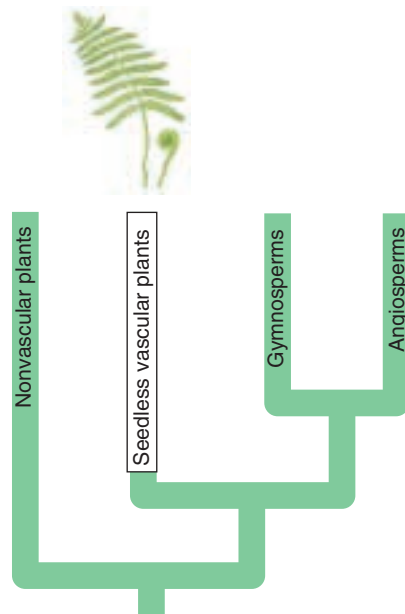
Ferns (Pterophyta)

Ferns are the most abundant group of seedless vascular plants, with about 12,000 living species. The fossil record indicates that they originated during the Devonian period about 350 million years ago and became abundant and varied in form during the next 50 million years. Their apparent ancestors had no broad leaves and were established on land as much as 375 million years ago.

Today, ferns flourish in a wide range of habitats throughout the world; about 75% of the species, however, occur in the tropics. The conspicuous sporophytes may be less than a centimeter in diameter—as seen in small aquatic ferns such as *Azolla*—or more than 24 meters tall and with leaves up to 5 meters or more long in the tree ferns (figure 37.9). The sporophytes and the smaller gametophytes, which rarely reach 6 millimeters in diameter, are both photosynthetic. The fern life cycle differs from that of a moss primarily in the much greater development, independence, and dominance of the fern's sporophyte. The fern's sporophyte is much more complex than that of the moss's; the fern sporophyte has vascular tissue and well-differentiated roots, stems, and leaves. The gametophyte, however, lacks vascular tissue.

Fern sporophytes typically have a horizontal underground stem called a **rhizome**, with roots emerging from the sides. The leaves, referred to as **fronds**, usually develop at the tip of the rhizome as tightly rolled-up coils (“fiddleheads”) that unroll and expand. Many fronds are highly dissected and feathery, making the ferns that produce them prized as ornamentals. Some ferns, such as *Marsilea*, have fronds that resemble a four-leaf clover, but *Marsilea* fronds still begin as coiled fiddleheads. Other ferns produce a mixture of photosynthetic fronds and nonphotosynthetic reproductive fronds that tend to be brownish in color.

Most ferns are homosporous, producing distinctive, **sporangia**, usually in clusters called **sori**, typically on the backs of the fronds. Sori are often protected during their development by a transparent, umbrella-like covering. At first



glance, one might mistake the sori for an infection on the plant. Diploid **spore mother cells** in each sporangium undergo meiosis, producing haploid spores. At maturity, the spores are catapulted from the sporangium by a snapping action, and those that land in suitable damp locations may germinate, producing gametophytes which are often heart-shaped, are only one cell thick (except in the center) and have rhizoids that anchor them to their substrate. These rhizoids are not true roots as they lack vascular tissue, but as with many of the nonvascular plants they do aid in transporting water and nutrients from the soil. Flask-shaped **archegonia** and globular **antheridia** are produced on either the same or different gametophyte.



FIGURE 37.9
A tree fern (phylum Pterophyta) in the forests of Malaysia. The ferns are by far the largest group of seedless vascular plants. Trees in other phyla are now extinct.

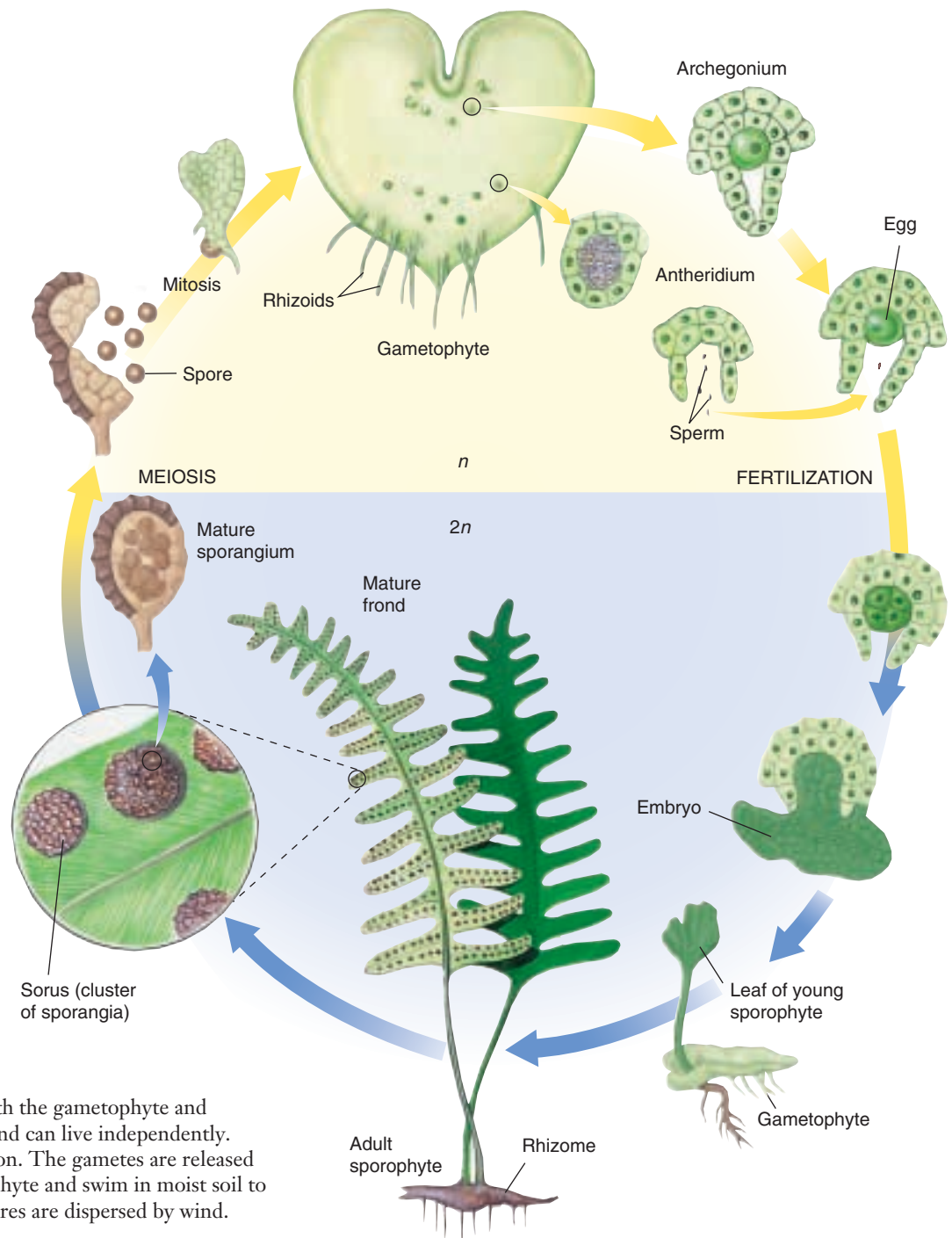


FIGURE 37.10

Life cycle of a typical fern. Both the gametophyte and sporophyte are photosynthetic and can live independently. Water is necessary for fertilization. The gametes are released on the underside of the gametophyte and swim in moist soil to neighboring gametophytes. Spores are dispersed by wind.

The sperm formed in the antheridia have flagella, with which they swim toward the archegonia when water is present, often in response to a chemical signal secreted by the archegonia. One sperm unites with the single egg toward the base of an archegonium, forming a *zygote*. The zygote then develops into a new sporophyte, completing the life

cycle (figure 37.10). There are still multicellular gametangia. As discussed earlier, the shift to a dominant sporophyte generation allows ferns to achieve significant height without interfering with sperm swimming efficiently to the egg. The multicellular archegonia provide some protection for the developing embryo.

Whisk Ferns (Psilophyta)

The three other phyla of seedless vascular plants, the Psilophyta, (whisk ferns), Lycophyta (club mosses), and Arthrophyta (horsetails), have many features in common with ferns. For example, they all form antheridia and archegonia. Free water is required for the process of fertilization, during which the sperm, which have flagella, swim to and unite with the eggs. In contrast, most seed plants have non-flagellated sperm; none form antheridia, although a few form archegonia.

The origins of the two genera of whisk ferns, which occur in the tropics and subtropics, are not clear, but they are considered to be living remnants of the very earliest vascular plants. Certainly they are the simplest of all extant vascular plants, consisting merely of evenly forking green stems without roots or leaves. The two or three species of the genus *Psilotum* do, however, have tiny, green, spirally arranged, flaps of tissue lacking veins and stomata. Another genus, *Tmespiteris*, has leaflike appendages.

The gametophytes of whisk ferns were unknown for many years until their discovery in the soil beneath the sporophytes. They are essentially colorless and are less than 2 millimeters in diameter, but they can be up to 18 millimeters long. They form saprobic or parasitic associations with fungi, which furnish their nutrients. Some develop elements of vascular tissue and have the distinction of being the only gametophytes known to do so.

Club Mosses (Lycophyta)

The club mosses are worldwide in distribution but are most abundant in the tropics and moist temperate regions. Several genera of club mosses, some of them tree-like, became extinct about 270 million years ago. Members of the four genera and nearly 1000 living species of club mosses superficially resemble true mosses, but once their internal structure and reproductive processes became known it was clear that these vascular plants are quite unrelated to mosses. Modern club mosses are either homosporous or heterosporous. The sporophytes have leafy stems that are seldom more than 30 centimeters long.

Horsetails (Arthrophyta)

The 15 living species of horsetails, also called scouring rushes, are all heterosporous and herbaceous. They constitute a single genus, *Equisetum*. Fossil forms of *Equisetum* extend back 300 million years to an era when some of their relatives were treelike. Today, they are widely scattered around the world, mostly in damp places. Some that grow among the coastal redwoods of California may reach a height of 3 meters, but most are less than a meter tall (figure 37.11).



FIGURE 37.11

A horsetail, *Equisetum telmateia*, a representative of the only living genus of the phylum Arthrophyta. This species forms two kinds of erect stems; one is green and photosynthetic, and the other, which terminates in a spore-producing “cone,” is mostly light brown.

Horsetail sporophytes consist of ribbed, jointed, photosynthetic stems that arise from branching underground rhizomes with roots at their nodes. A whorl of nonphotosynthetic, scalelike leaves emerges at each node. The stems, which are hollow in the center, have silica deposits in the epidermal cells of the ribs, and the interior parts of the stems have two sets of vertical, somewhat tubular canals. The larger outer canals, which alternate with the ribs, contain air, while the smaller inner canals opposite the ribs contain water.

Ferns and other seedless vascular plants have a much larger and more conspicuous sporophyte, with vascular tissue. Many have well-differentiated roots, stem, and leaves. The shift to a dominant sporophyte led to the evolution of trees.

37.4 Seeds protect and aid in the dispersal of plant embryos.

Seed Plants

Seed plants first appeared about 425 million years ago. Their ancestors appear to have been spore-bearing plants known as progymnosperms. Progymnosperms shared several features with modern gymnosperms, including secondary xylem and phloem (which allows for an increase in girth later in development). Some progymnosperms had leaves. Their reproduction was very simple, and it is not certain which particular group of progymnosperms gave rise to seed plants.

From an evolutionary and ecological perspective, the seed represents an important advance. The embryo is protected by an extra layer of sporophyte tissue creating the ovule. During development this tissue hardens to produce the seed coat. In addition to protection from drought, dispersal is enhanced. Perhaps even more significantly, a dormant phase is introduced into the life cycle that allows the embryo to survive until environmental conditions are favorable for further growth.

Seed plants produce two kinds of gametophytes—male and female, each of which consists of just a few cells. Pollen grains, multicellular male gametophytes, are conveyed to the egg in the female gametophyte by wind or a pollinator. The sperm move toward the egg through a growing pollen tube. This eliminates the need for water. In contrast to the seedless plants, the whole male gametophyte rather than just the sperm moves to the female gametophyte. A female gametophyte develops within an ovule. In flowering plants (angiosperms), the ovules are completely enclosed within diploid sporophyte tissue (ovaries which develop into the fruit). In gymnosperms (mostly cone-bearing seed plants), the ovules are not completely enclosed by sporophyte tissue at the time of pollination.

A common ancestor with seeds gave rise to the gymnosperms and the angiosperms. Seeds can allow for a pause in the life cycle until environmental conditions are more optimal.

A Vocabulary of Plant Terms

androecium The stamens of a flower.

anther The pollen-producing portion of a stamen. This is a sporophyte structure where male gametophytes are produced by meiosis.

antheridium The male sperm-producing structure found in the gametophytes of seedless plants and certain fungi.

archegonium The multicellular egg-producing structure in the gametophytes of seedless plants and gymnosperms.

carpel A leaflike organ in angiosperms that encloses one or more ovules; a unit of a gynoecium.

double fertilization The process, unique to angiosperms, in which one sperm fuses with the egg, forming a zygote, and the other sperm fuses with the two polar nuclei, forming the primary endosperm nucleus.

endosperm The usually triploid (although it can have a much higher ploidy level) food supply of some angiosperm seeds.

filament The stalklike structure that supports the anther of a stamen.

gametophyte The multicellular, haploid phase of a plant life cycle in which gametes are produced by mitosis.

gynoecium The carpel(s) of a flower.

heterosporous Refers to a plant that produces two types of spores: microspores and megaspores.

homosporous Refers to a plant that produces only one type of spore.

integument The outer layer(s) of an ovule; integuments become the seed coat of a seed.

micropyle The opening in the ovule integument through which the pollen tube grows.

nucellus The tissue of an ovule in which an embryo sac develops.

ovary The basal, swollen portion of a carpel (gynoecium); it contains the ovules and develops into the fruit.

ovule A seed plant structure within an ovary; it contains a female gametophyte surrounded by the nucellus and one or two integuments. At maturity, an ovule becomes a seed.

pollen grain A binucleate or trinucleate seed plant structure produced from a mi-

crospore in a microsporangium.

pollination The transfer of a pollen grain from an anther to a stigma in angiosperms, or to the vicinity of the ovule in gymnosperms.

primary endosperm nucleus The triploid nucleus resulting from the fusion of a single sperm with the two polar nuclei.

seed A reproductive structure that develops from an ovule in seed plants. It consists of an embryo and a food supply surrounded by a seed coat.

seed coat The protective layer of a seed; it develops from the integument or integuments.

spore A haploid reproductive cell, produced when a diploid spore mother cell undergoes meiosis; it gives rise by mitosis to a gametophyte.

sporophyte The multicellular, diploid phase of a plant life cycle; it is the generation that ultimately produces spores.

stamen A unit of an androecium; it consists of a pollen-bearing anther and usually a stalklike filament.

stigma The uppermost pollen-receptive portion of a gynoecium.

Gymnosperms

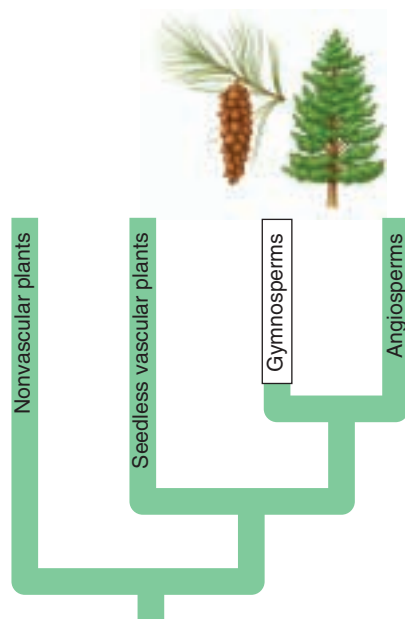
There are several groups of living gymnosperms (conifers, cycads, gnetophytes, and *Ginkgo*), none of which are directly related to one another, but all of which lack the flowers and fruits of angiosperms. In all of them the **ovule**, which becomes a seed, rests exposed on a scale (modified leaf) and is not completely enclosed by sporophyte tissues at the time of pollination. The name gymnosperm combines the Greek root *gymnos*, or “naked,” with *sperma*, or “seed.” In other words, gymnosperms are naked-seeded plants. However, although the ovules are naked at the time of pollination, the seeds of gymnosperms are sometimes enclosed by other sporophyte tissues by the time they are mature.

Details of reproduction vary somewhat in gymnosperms, and their forms vary greatly. For example, cycads and *Ginkgo* have motile sperm, even though the sperm are borne within a pollen tube, while many others have sperm with no flagella. The female cones range from tiny woody structures weighing less than 25 grams with a diameter of a few millimeters, to massive structures weighing more than 45 kilograms growing to lengths more than a meter.

Conifers (Coniferophyta)

The most familiar gymnosperms are conifers (phylum Coniferophyta), which include pines (figure 37.12), spruces, firs, cedars, hemlocks, yews, larches, cypresses, and others. The coastal redwood (*Sequoia sempervirens*), a conifer native to northwestern California and southwestern Oregon, is the tallest living vascular plant; it may attain nearly 100 meters (300 feet) in height. Another conifer, the bristlecone pine (*Pinus longaeva*) of the White Mountains of California is the oldest living tree; one is 4900 years of age. Conifers are found in the colder temperate and sometimes drier regions of the world, especially in the northern hemisphere. They are sources of timber, paper, resin, turpentine, taxol (used to treat cancer) and other economically important products.

Pines. More than 100 species of pines exist today, all native to the northern hemisphere, although the range of one species does extend a little south of the equator. Pines and spruces are members of the vast coniferous forests that lie between the arctic tundra and the temperate deciduous forests and prairies to their south. During the past century, pines have been extensively planted in the southern hemisphere.



Pines have tough, needlelike leaves produced mostly in clusters of two to five. The leaves, which have a thick cuticle and recessed stomata, represent an evolutionary adaptation for retarding water loss. This is important because many of the trees grow in areas where the topsoil is frozen for part of the year, making it difficult for the roots to obtain water. The leaves and other parts of the sporophyte have canals into which surrounding cells secrete resin. The resin, apparently secreted in response to wounding, deters insect and fungal attacks. The resin of certain pines is harvested commercially for its volatile liquid portion, called *turpentine*, and for the solid *rosin*, which is used on stringed instruments. The wood of pines consists primarily of xylem tissue that lacks some of the more rigid cell types found in

other trees. Thus it is considered a “soft” rather than a “hard” wood. The thick bark of pines represents another adaptation for surviving fires and subzero temperatures. Some cones actually depend on fire to open, releasing seed to reforest burnt areas.



FIGURE 37.12

Conifers. Slash pines, *Pinus palustris*, in Florida, are representative of the Coniferophyta, the largest phylum of gymnosperms.

As mentioned earlier, all seed plants are heterosporous, so the spores give rise to two types of gametophytes (figure 37.13). The male gametophytes of pines develop from pollen grains, which are produced in male cones that develop in clusters of 30 to 70, typically at the tips of the lower branches; there may be hundreds of such clusters on any single tree. The male cones generally are 1 to 4 centimeters long and consist of small, papery scales arranged in a spiral or in whorls. A pair of microsporangia form as sacs within each scale. Numerous microspore mother cells in the microsporangia undergo meiosis, each becoming four microspores. The microspores develop into four-celled pollen grains with a pair of air sacs that give them added buoyancy when released into the air. A single cluster of male pine cones may produce more than 1 million pollen grains.

Female cones typically are produced on the upper branches of the same tree that produces male cones. Female cones are larger than male cones, and their scales become woody. Two ovules develop toward the base of each scale. Each ovule contains a megasporangium called the **nucellus**. The nucellus itself is completely surrounded by a thick layer of cells called the integument that has a small opening (the **micropyle**) toward one end. One of the layers of the integument later becomes the seed coat. A single megaspore mother cell within each megasporangium undergoes meiosis, becoming a row of four megaspores. Three of the megaspores break down, but the remaining one, over the better part of a year, slowly develops into a female gametophyte. The female gametophyte at maturity may consist of thousands of cells, with two to six archegonia formed at the micropylar end. Each archegonium contains an egg so large it can be seen without a microscope.

Female cones usually take two or more seasons to mature. At first they may be reddish or purplish in color, but they soon turn green, and during the first spring, the scales spread apart. While the scales are open, pollen grains carried by the wind drift down between them, some catching in sticky fluid oozing out of the micropyle. As the sticky fluid evaporates, the pollen grains are slowly drawn down through the micropyle to the top of the nucellus, and the scales close shortly thereafter. The archegonia and the rest of the female gametophyte

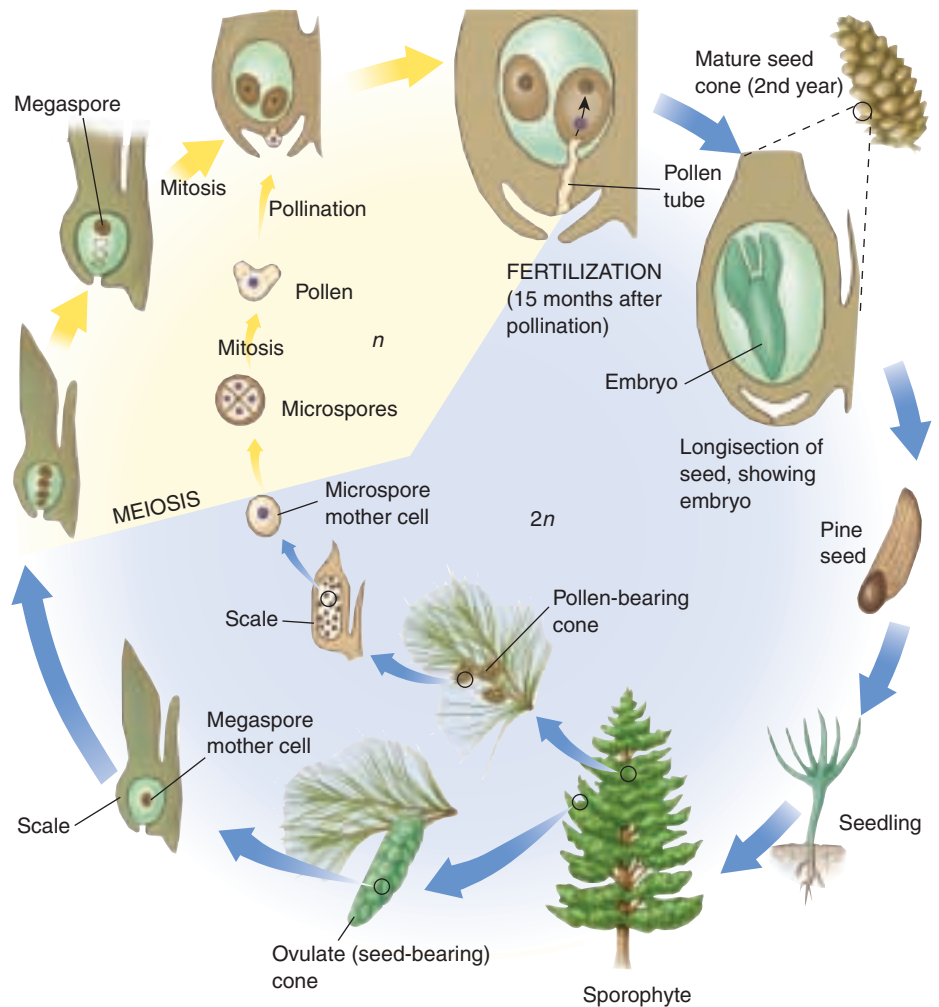


FIGURE 37.13

Life cycle of a typical pine. The male and female gametophytes have been dramatically reduced in size. Wind generally disperses sperm that is within the male gametophyte (pollen). Pollen tube growth delivers the sperm to the egg on the female cone. Additional protection for the embryo is provided by the ovule which develops into the seed coat.

are not mature until about a year later. While the female gametophyte is developing, a pollen tube emerges from a pollen grain at the bottom of the micropyle and slowly digests its way through the nucellus to the archegonia. While the pollen tube is growing, one of the pollen grain's four cells, the generative cell, divides by mitosis, with one of the resulting two cells dividing once more. These last two cells function as sperm. The germinated pollen grain with its two sperm is the mature male gametophyte.

About 15 months after pollination, the pollen tube reaches an archegonium, and discharges its contents into it. One sperm unites with the egg, forming a zygote. The other sperm and cells of the pollen grain degenerate. The zygote develops into an embryo within a seed. After dispersal and germination of the seed, the young sporophyte of the next generation grows into a tree.



(a)



(b)



(c)

FIGURE 37.14

Three phyla of gymnosperms.

(a) An African cycad, *Encephalartos transvenosus*.

(b) *Welwitschia mirabilis*, one of the three genera of gnetophytes.

(c) Maidenhair tree, *Ginkgo biloba*, the only living representative of the phylum Ginkgophyta.

Cycads (Cycadophyta)

Cycads are slow-growing gymnosperms of tropical and subtropical regions. The sporophytes of most of the 100 known species resemble palm trees (figure 37.14a) with trunks that can attain heights of 15 meters or more. Unlike palm trees—which are flowering plants—cycads produce cones and have a life cycle similar to that of pines. The female cones, which develop upright among the leaf bases, are huge in some species and can weigh up to 45 kilograms. The sperm of cycads, although conveyed to an archegonium by a pollen tube, have thousands of spirally arranged flagella. Several species are facing extinction in the wild and soon may exist only in botanical gardens.

Gnetophytes (Gnetophyta)

There are three genera and about 70 living species of Gnetophyta. Gnetophytes are the closest living relatives of angiosperms and probably share a common ancestor with that group. They are the only gymnosperms with vessels (a particularly efficient conducting cell type) in their xylem—a common feature in angiosperms. The members of the three genera differ greatly from one another in form. One of the most bizarre of all plants is *Welwitschia*, which occurs in the Namib and Mossamedes deserts of southwestern Africa (figure 37.14b). The stem is shaped like a large, shallow cup that tapers into a taproot below the surface. It has two strap-shaped, leathery leaves that grow continuously from their base, splitting as they flap in the wind. The reproductive structures of *Welwitschia* are conelike, appear toward the bases of the leaves around the rims of the stems, and are produced on separate male and female plants.

More than half of the gnetophyte species are in the genus *Ephedra*, which is common in arid regions of the western United States and Mexico. The plants are shrubby, with stems that superficially resemble those of horsetails as they are jointed and have tiny, scalelike leaves at each node. Male and female reproductive structures may be produced

on the same or different plants. The drug ephedrine, widely used in the treatment of respiratory problems, was in the past extracted from Chinese species of *Ephedra*, but it has now been largely replaced with synthetic preparations. Mormon tea is brewed from *Ephedra* stems in the southwestern United States.

The best known species of *Gnetum* is a tropical tree, but most species are vinelike. All species have broad leaves similar to those of angiosperms. One *Gnetum* species is cultivated in Java for its tender shoots, which are cooked as a vegetable.

Ginkgo (Ginkgophyta)

The fossil record indicates that members of the Ginkgo family were once widely distributed, particularly in the northern hemisphere; today only one living species, the maidenhair tree (*Ginkgo biloba*), remains. The tree, which sheds its leaves in the fall, was first encountered by Europeans in cultivation in Japan and China; it apparently no longer exists in the wild (figure 37.14c). The common name comes from the resemblance of its fan-shaped leaves to the leaflets of maidenhair ferns. Like the sperm of cycads, those of *Ginkgo* have flagella. The *Ginkgo* is dioecious, that is, the male and female reproductive structures of *Ginkgo* are produced on separate trees. The fleshy outer coverings of the seeds of female *Ginkgo* plants exude the foul smell of rancid butter caused by butyric and isobutyric acids. In the Orient, however, the seeds are considered a delicacy. In Western countries, because of the seed odor, male plants vegetatively propagated from shoots are preferred for cultivation. Because it is resistant to air pollution, *Ginkgo* is commonly planted along city streets.

Gymnosperms are mostly cone-bearing seed plants. In gymnosperms, the ovules are not completely enclosed by sporophyte tissue at pollination.

Angiosperms

The 250,000 known species of flowering plants are called angiosperms because their ovules, unlike those of gymnosperms, are enclosed within diploid tissues at the time of pollination. The name *angiosperm* derives from the Greek words *angeion*, “vessel,” and *sperma*, “seed.” The “vessel” in this instance refers to the carpel, which is a modified leaf that encapsulates seeds. The carpel develops into the fruit, a unique angiosperm feature. While some gymnosperms, including yew, have fleshlike tissue around their seeds, it is of a different origin and not a true fruit.

The origins of the angiosperms puzzled even Darwin (his “abominable mystery”). Recently, consensus has been reached on the most basal, living angiosperm—*Amborella trichopoda* (figure 37.15). This has ended the debate between the supporters of magnolias and those of water lilies as the closest relatives of the original angiosperm. *Amborella*, with small, cream-colored flowers, is even more primitive than either the magnolias or water lilies. This small shrub found only on the island of New Caledonia in the South Pacific is the last remaining species of the earliest extant lineage of the angiosperms. About 135 million years ago a close relative of *Amborella* developed floral parts and branched off from the gymnosperms. While *Amborella* is not the original angiosperm, it is sufficiently close that much will be learned from studying its reproductive biology that will help us understand the early radiation of the angiosperms.

Flowering plants (phylum Anthophyta) exhibit an almost infinite variety of shapes, sizes, and textures. They vary, for example, from the huge Tasmanian *Eucalyptus* trees, which have nearly as much mass as the giant redwoods, to the tiniest duckweeds, which are less than 1 millimeter long. In addition to the typical flattened green leaves with which everyone is familiar, flowering plant leaves may be succulent, floating, submerged, cup-shaped, spinelike, scalelike, feathery, papery, hairy, or insect-trapping, and of almost any color. Some are so tiny one needs a microscope to examine them, while others, such as those of the Seychelles Island palm, can be up to 6 meters long. Their flowers vary from the simple blossoms of buttercups to the extraordinarily complex flowers of some orchids, which may lure their pollinators with drugs, forcibly attach bags of pollen to their bodies, or dunk them in fluid they secrete. The flowers may weigh less than 1 gram and remain functional for only a few minutes, or they can weigh up to 9 kilograms and be functional for months. Plants of several families are parasitic or partially parasitic (for example, dodder, or

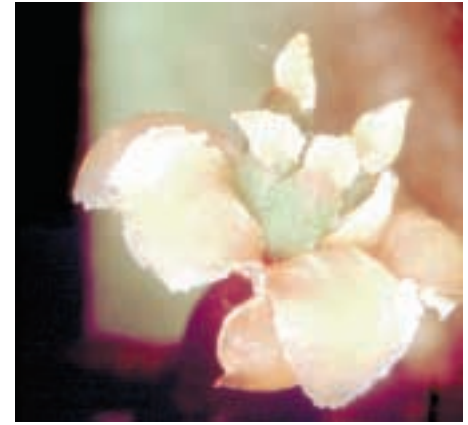
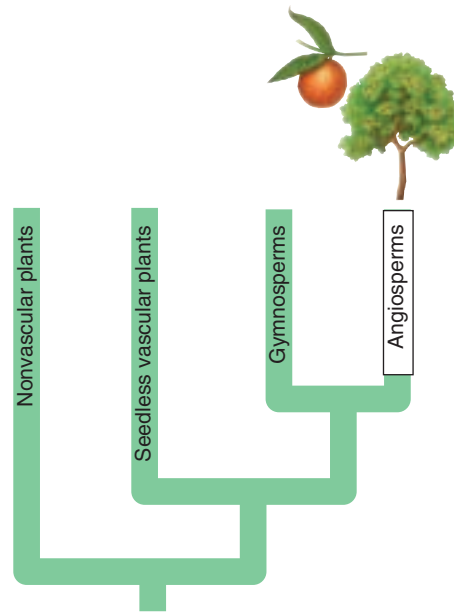


FIGURE 37.15

A flowering plant. *Amborella trichopoda*. This plant is believed to be the closest living relative to the original angiosperm.

mistletoe) on other plants, or *mycotrophic* (deriving their nutrients from fungi that form a mutualism with plant roots). Others, such as many orchids, are *epiphytic* (attached to other plants, with no roots in the ground, and not in any way parasitic).

The Structure of Flowers

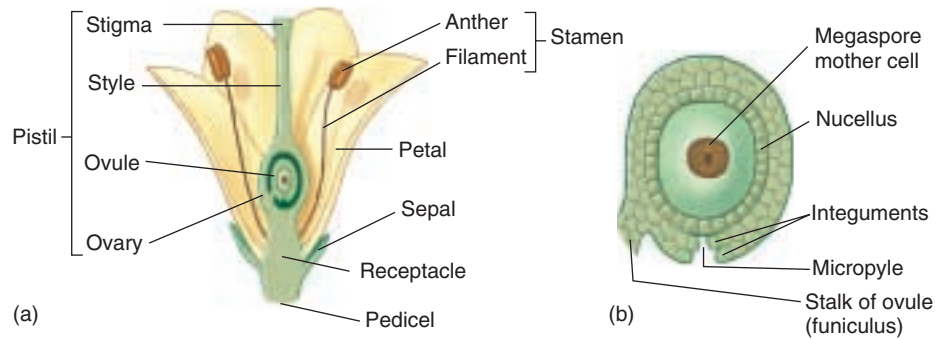
Flowers are considered to be modified stems bearing modified leaves. Regardless of their size and shape, they all share certain features (see figure 37.16). Each flower originates as a **primordium** that develops into a bud at the end of a stalk called a **pedicel**. The pedicel expands slightly at the tip into a base, the **receptacle**, to which the remaining flower parts are attached. The other flower parts typically are attached in circles called *whorls*. The outermost whorl is composed of **sepals**. In most flowers there are three to five sepals, which are green and somewhat leaflike; they often function in protecting the immature flower and in some species may drop off as the flower opens. The next whorl consists of **petals** that are often colored and attract pollinators such as insects and birds. The petals, which commonly number three to five, may be separate, fused together, or missing altogether in *wind-pollinated flowers*.

The third whorl consists of **stamens**, collectively called the **androecium**, a term derived from the Greek words *andros*, “male,” and *oikos*, “house.” Each stamen consists of a pollen-bearing **anther** and a stalk called a **filament**, which may be missing in some flowers. The **gynoecium**, consisting of one or more **carpels**, is at the center of the flower. The term *gynoecium* derives from the Greek words *gynos*, which means “female,” and *oikos*, or “house.” The first carpel is believed to have been formed from a leaflike structure with ovules along its margins. The edges of the blade then rolled inward and fused together, forming a carpel.

FIGURE 37.16

Diagram of an angiosperm flower.

(a) The main structures of the flower are labeled. (b) Details of an ovule. The ovary as it matures will become a fruit; as the ovule's outer layers (integuments) mature, they will become a seed coat.



Primitive flowers can have several to many separate carpels, but in most flowers, two to several carpels are fused together. Such fusion can be seen in an orange sliced in half; each segment represents one carpel. A carpel has three major regions (figure 37.16). The **ovary** is the swollen base, which contains from one to hundreds of **ovules**; the ovary later develops into a **fruit**. The tip of the pistil is called a **stigma**. Most stigmas are sticky or feathery, causing pollen grains that land on them to adhere. Typically there is a neck or stalk called a **style** connecting the stigma and the ovary; in some flowers, the style may be very short or even missing. Many flowers have nectar-secreting glands called *nectaries*, often located toward the base of the ovary. Nectar is a fluid containing sugars, amino acids, and other molecules used to attract insects, birds, and other animals to flowers.

The Angiosperm Life Cycle

While a flower bud is developing, a single megaspore mother cell in the ovule undergoes meiosis, producing four megaspores (figure 37.17). In most flowering plants, three of the megaspores soon disappear while the nucleus of the remaining one divides mitotically, and the cell slowly expands until it becomes many times its original size. While this expansion is occurring, each of the daughter nuclei divide twice, resulting in eight haploid nuclei arranged in two groups of four. At the same time, two layers of the ovule, the **integuments**, differentiate and become the *seed coat* of a seed. The integuments, as they develop, leave a small gap or pore at one end—the *micropyle* (see figure 37.16). One nucleus from each group of four migrates toward the center, where they function as **polar nuclei**. Polar nuclei may fuse together, forming a single diploid nucleus, or they may form a single cell with two haploid nuclei. Cell walls also form around the remaining nuclei. In the group closest to the micropyle, one cell functions as the **egg**; the other two nuclei are called **synergids**. At the other end, the three cells are now called **antipodals**; they have no apparent function and eventually break down and disappear. The large sac with eight nuclei in seven cells is called an **embryo sac**; it constitutes the female gametophyte. Although it is completely dependent on the sporophyte for nutrition, it is a multicellular, haploid individual.

While the female gametophyte is developing, a similar but less complex process takes place in the anthers. Most anthers have patches of tissue (usually four) that eventually become chambers lined with nutritive cells. The tissue in each patch is composed of many diploid microspore mother cells that undergo meiosis more or less simultaneously, each producing four microspores. The four microspores at first remain together as a quartet or tetrad, and the nucleus of each microspore divides once; in most species the microspores of each quartet then separate. At the same time, a two-layered wall develops around each microspore. As the anther matures, the wall between adjacent pairs of chambers breaks down, leaving two larger sacs. At this point, the binucleate microspores have become **pollen grains**. The outer pollen grain wall layer often becomes beautifully sculptured, and it contains chemicals that may react with others in a stigma to signal whether or not development of the male gametophyte should proceed to completion. The pollen grain has areas called *apertures*, through which a pollen tube may later emerge.

Pollination is simply the mechanical transfer of pollen from its source (an anther) to a receptive area (the stigma of a flowering plant). Most pollination takes place between flowers of different plants and is brought about by insects, wind, water, gravity, bats, and other animals. In as many as a quarter of all angiosperms, however, a pollen grain may be deposited directly on the stigma of its own flower, and self-pollination occurs. Pollination may or may not be followed by *fertilization*, depending on the genetic compatibility of the pollen grain and the flower on whose stigma it has landed. (In some species, complex, genetically controlled mechanisms prevent self-fertilization to enhance genetic diversity in the progeny.) If the stigma is receptive, the pollen grain's dense cytoplasm absorbs substances from the stigma and bulges through an aperture. The bulge develops into a *pollen tube* that responds to chemicals released by the embryo sac. It follows a diffusion gradient of the chemicals and grows down through the style and into the micropyle. The pollen tube usually takes several hours to two days to reach the micropyle, but in a few instances, it may take up to a year. One of the pollen grain's two nuclei, the *generative nucleus*, lags behind. This nucleus divides, ei-

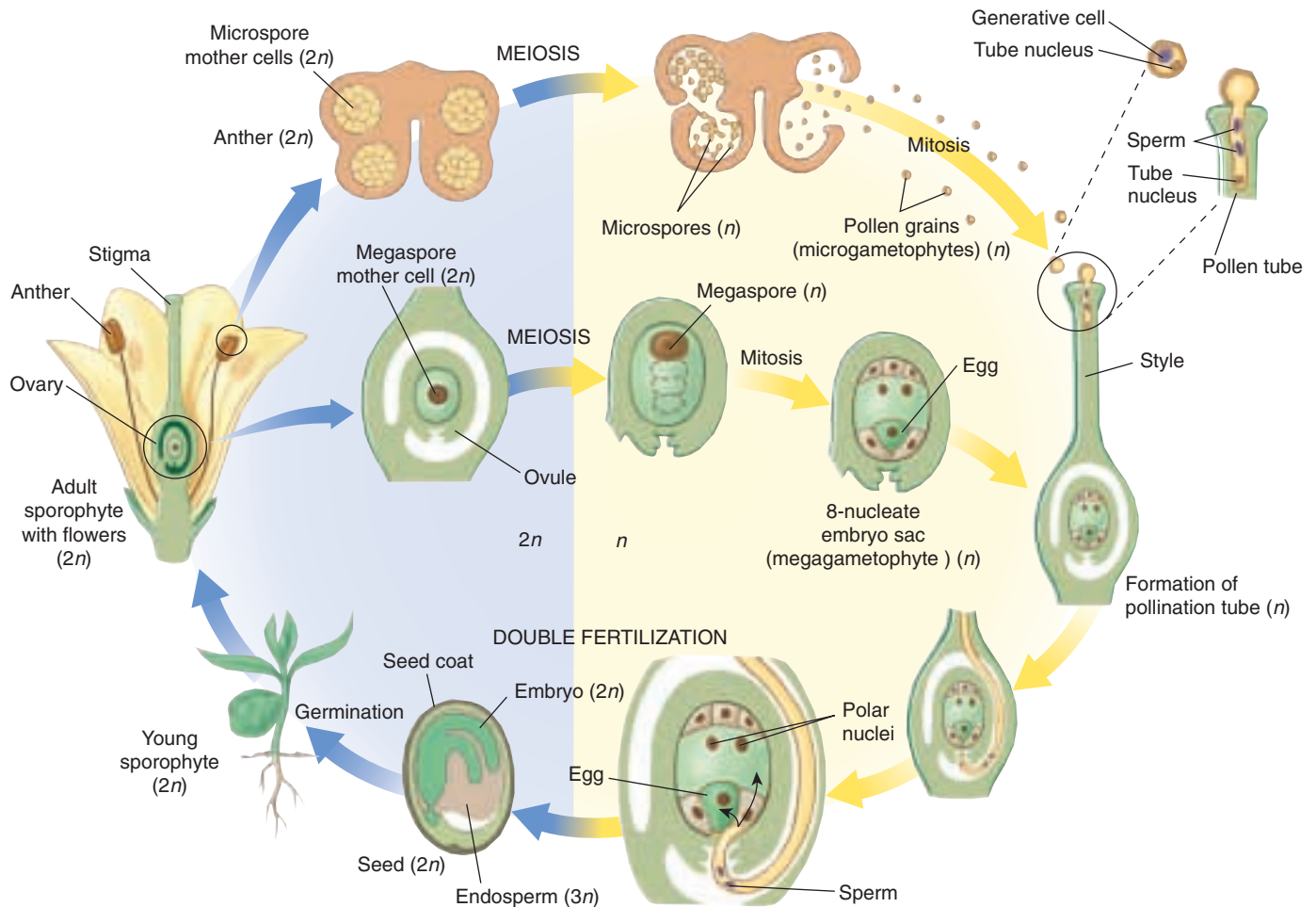


FIGURE 37.17

Life cycle of a typical angiosperm. As in pines, water is no longer required for fertilization. In most species of angiosperms, animals carry pollen to the carpel. The outer wall of the carpel forms the fruit that entices animals to disperse seed.

ther in the pollen grain or in the pollen tube, producing two sperm nuclei. Unlike sperm in mosses, ferns, and some gymnosperms, the sperm of flowering plants have no flagella. At this point, the pollen grain with its tube and sperm has become a mature male gametophyte.

As the pollen tube enters the embryo sac, it destroys a synergid in the process and then discharges its contents. Both sperm are functional, and an event called **double fertilization**, unique to angiosperms, follows. One sperm unites with the egg and forms a zygote, which develops into an embryo sporophyte plant. The other sperm and the two polar nuclei unite, forming a triploid primary endosperm nucleus. The primary endosperm nucleus begins dividing rapidly and repeatedly, becoming triploid **endosperm** tissue that may soon consist of thousands of cells. Endosperm tissue can become an extensive part of the seed in grasses such as corn (see figure 41.7). But in most flowering plants, it provides nutrients for the embryo that develops from the zygote; in many species, such as

peas and beans, it disappears completely by the time the seed is mature. Following double fertilization, the integuments harden and become the seed coat of a seed. The haploid cells remaining in the embryo sac (antipodals, synergid, tube nucleus) degenerate. There is some evidence for a type of double fertilization in gymnosperms believed to be closely related to the angiosperms. Further studies of this and of fertilization in *Amborella*, the most basal, extant angiosperm, may provide clues to the evolution of this double fertilization event.

Angiosperms are characterized by ovules that at pollination are enclosed within an ovary at the base of a carpel—a structure unique to the phylum; a fruit develops from the ovary. Evolutionary innovations including flowers to attract pollinators, fruits to protect and aid in embryo dispersal, and double fertilization providing additional nutrients for the embryo all have contributed to the widespread success of this phylum.



Summary

Questions

Media Resources

37.1 Plants have multicellular haploid and diploid stages in their life cycles.

- Plants evolved from a multicellular, freshwater green algae 450 million years ago. The evolution of their conducting tissues, cuticle, stomata, and seeds has made them progressively less dependent on external water for reproduction.
- All plants have a haplodiplontic life cycle in which haploid gametophytes alternate with diploid sporophytes.

- Where did the most recent ancestors of land plants live? What were they like? What adaptations were necessary for the “move” onto land?
- What does it mean for a plant to alternate generations? Distinguish between sporophyte and gametophyte.



- Life Cycles of Plants



- Student Research: Plant Biodiversity in New Hampshire
- Book Review: *A Rum Affair* by Sabbagh
- Book Review: The Orchid Thief by Orlean

37.2 Nonvascular plants are relatively unspecialized, but successful in many terrestrial environments.

- Three phyla of plants lack well-developed vascular tissue, are the simplest in structure, and have been grouped as bryophytes. This grouping does not reflect a common ancestry or close relationship.
- Sporophytes of mosses, liverworts, and hornworts are usually nutritionally dependent on the gametophytes, which are more conspicuous and photosynthetic.

- Distinguish between male gametophytes and female gametophytes. Which specific haploid spores give rise to each of these?
- What reproductive limitations would a moss tree (if one existed) face?



- Non-Vascular Plants

37.3 Seedless vascular plants have well-developed conducting tissues in their sporophytes.

- Nine of the 12 plant phyla contain vascular plants, which have two kinds of well-defined conducting tissues: xylem, which is specialized to conduct water and dissolved minerals; and phloem, which is specialized to conduct the sugars produced by photosynthesis and plant growth regulators.
- In ferns and other seedless vascular plants, the sporophyte generation is dominant. The fern sporophyte has vascular tissue and well-differentiated roots, stems, and leaves.

- In what ways are the gametophytes of seedless plants different from the gametophytes of seed plants?
- Which generation(s) of the fern are nutritionally independent?



- Seedless Vascular Plants

37.4 Seeds protect and aid in the dispersal of plant embryos.

- Seeds were an important evolutionary advance providing for a dormant stage in development.
- In gymnosperms, ovules are exposed directly to pollen at the time of pollination; in angiosperms, ovules are enclosed within an ovary, and a pollen tube grows from the stigma to the ovule.
- The pollen of gymnosperms is usually disseminated by wind. In most angiosperms the pollen is transported by insects and other animals. Both flowers and fruits are found only in angiosperms and may account for the extensive colonization of terrestrial environments by the flowering plants.

- What is a seed? Why is the seed a crucial adaptation to terrestrial life?
- What is the principal difference between gymnosperms and angiosperms?
- If all the offspring of a plant were to develop in a small area, they would suffer from limited resources. Compare dispersal strategies in moss, pine, and angiosperms.



- Gymnosperms
- Angiosperms

38

The Plant Body

Concept Outline

38.1 Meristems elaborate the plant body plan after germination.

Meristems. Growth occurs in the continually dividing cells that function like stem cells in animals.

Organization of the Plant Body. The plant body is a series of iterative units stacked above and below the ground.

Primary and Secondary Growth. Different meristems allow plants to grow in both height and circumference.

38.2 Plants have three basic tissues, each composed of several cell types.

Dermal Tissue. This tissue forms the “skin” of the plant body, protecting it and preventing water loss.

Ground Tissue. Much of a young plant is ground tissue, which supports the plant body and stores food and water.

Vascular Tissue. Special piping tissues conduct water and sugars through the plant body.

38.3 Root cells differentiate as they become distanced from the dividing root apical meristem.

Root Structure. Roots have a durable cap, behind which primary growth occurs.

Modified Roots. Roots can have specialized functions.

38.4 Stems are the backbone of the shoot, transporting nutrients and supporting the aerial plant organs.

Stem Structure. The stem supports the leaves and is anchored by the roots. Vascular tissues are organized within the stem in different ways.

Modified Stems. Specialized stems are adapted for storage and vegetative (asexual) propagation.

38.5 Leaves are adapted to support basic plant functions.

Leaf External Structure. Leaves have flattened blades and slender stalks.

Leaf Internal Structure. Leaves contain cells that carry out photosynthesis, gas exchange, and evaporation.

Modified Leaves. In some plants, leaf development has been modified to provide for a unique need.

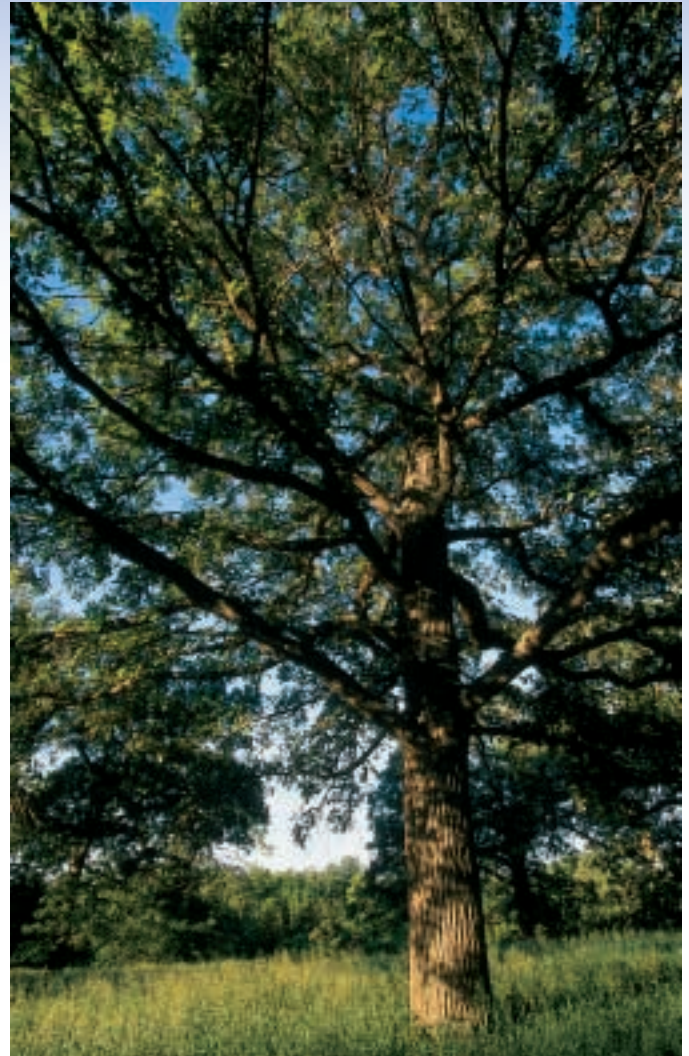


FIGURE 38.1

All vascular plants share certain characteristics. Vascular plants such as this tree require an elaborate system of support and fluid transport to grow this large. Smaller plants have similar (though simpler) structures. Much of this support system is actually underground in the form of extensive branching root systems.

Although the similarities between a cactus, an orchid plant, and a tree might not be obvious at first sight, most plants have a basic unity of structure (figure 38.1). This unity is reflected in how the plants are constructed; in the way they grow, manufacture, and transport their food; and in how their development is regulated. This chapter addresses the question of how a vascular plant is “built.” We will focus on the diversity of cell, tissue, and organ types that compose the adult body. The roots and shoots which give the adult plant its distinct above and below ground architecture are the final product of a basic body plan first established during embryogenesis, a process we will explore in detail in chapter 40.

38.1 Meristems elaborate the plant body plan after germination.

Meristems

The plant body that develops after germination depends on the activities of meristematic tissues. Meristematic tissues are lumps of small cells with dense cytoplasm and proportionately large nuclei that act like stem cells in animals. That is, one cell divides to give rise to two cells. One remains meristematic, while the other is free to differentiate and contribute to the plant body. In this way, the population of meristem cells is continually renewed. Molecular genetic evidence supports the hypothesis that stem cells and meristem cells may also share some common molecular mechanisms.

Elongation of both root and shoot takes place as a result of repeated cell divisions and subsequent elongation of the cells produced by the **apical meristems**. In some vascular plants, including shrubs and most trees, **lateral meristems** produce an increase in girth.

Apical Meristems

Apical meristems are located at the tips of stems (figure 38.2) and at the tips of roots (figure 38.3), just behind the root cap. The plant tissues that result from primary growth are called **primary tissues**. During periods of growth, the cells of apical meristems divide and continually add more cells to the tips of a seedling's body. Thus, the seedling lengthens. Primary growth in plants is brought about by the apical meristems. The elongation of the root and stem forms what is known as the **primary plant body**, which is made up of primary tissues. The primary plant body comprises the young, soft shoots and roots of a tree or shrub, or the entire plant body in some herbaceous plants.

Both root and shoot apical meristems are composed of delicate cells that need protection. The root apical meristem is protected from the time it emerges by the root cap. Root cap cells are produced by the root meristem and are sloughed off and replaced as the root moves through the soil. A variety of adaptive mechanisms protect shoot apical meristem during germination (figure 38.4). The epicotyl or hypocotyl ("stemlike" tissue above or below the cotyledons) may bend as the seedling emerges to minimize the force on the shoot tip. In the monocots (a late evolving group of angiosperms) there is often a coleoptile (sheath of tissue) that forms a protective tube around the emerging shoot. Later in development, the leaf primordia cover the shoot apical meristem which is particularly susceptible to desiccation.

The apical meristem gives rise to three types of embryonic tissue systems called **primary meristems**. Cell division continues in these partly differentiated tissues as they develop into the primary tissues of the plant body. The

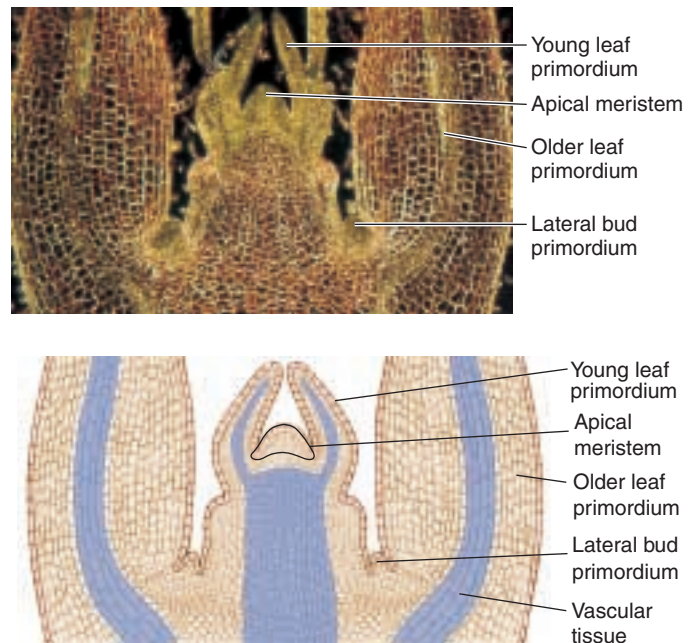


FIGURE 38.2

An apical shoot meristem. This longitudinal section through a shoot apex in *Coleus* shows the tip of a stem. Between the young leaf primordia is the apical meristem.

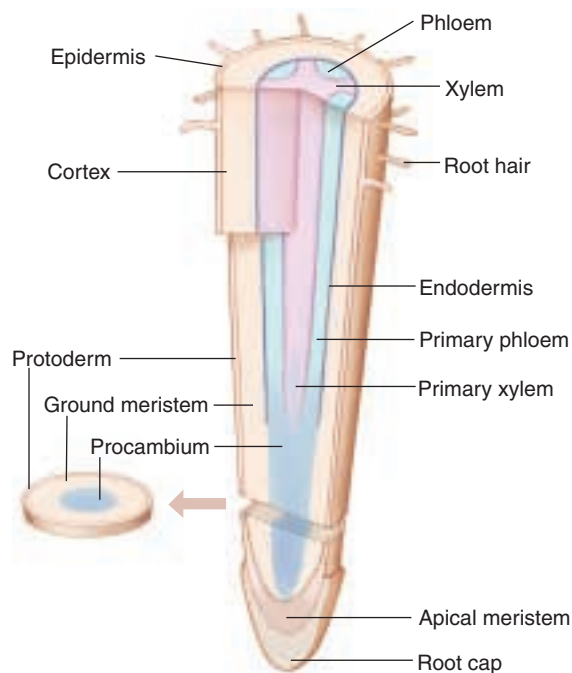


FIGURE 38.3

An apical root meristem. This diagram of meristems in the root shows their relation to the root tip.

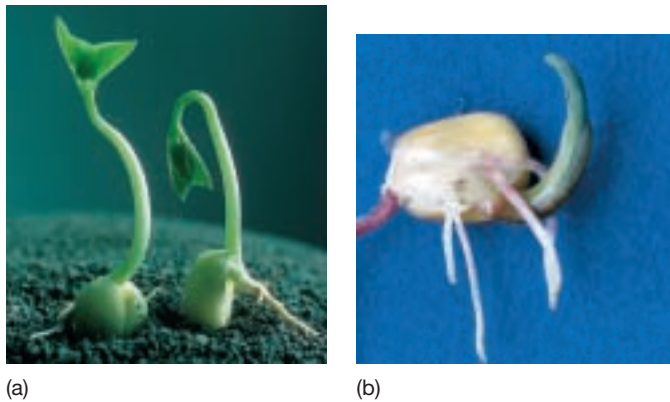


FIGURE 38.4

Developing seedling. Apical meristems are protected early in development. (a) In this soybean, a dicot, a bent epicotyl (stem above the cotyledons), rather than the shoot tip, pushes through the soil before straightening. (b) In corn, a monocot, a sleeve of tissue called the coleoptile sheaths the shoot tip until it has made it to daylight.

three primary meristems are the **protoderm**, which forms the epidermis; the **procambium**, which produces primary vascular tissues (primary xylem and primary phloem); and the **ground meristem**, which differentiates further into ground tissue, which is composed of parenchyma cells. In some plants, such as horsetails and corn, **intercalary meristems** arise in stem internodes, adding to the internode lengths. If you walk through a corn field (when the corn is about knee high) on a quiet summer night, you may hear a soft popping sound. This is caused by the rapid growth of intercalary meristems. The amount of stem elongation that occurs in a very short time is quite surprising.

Lateral Meristems

Many herbaceous plants exhibit only primary growth, but others also exhibit **secondary growth**. Most trees, shrubs, and some herbs have active **lateral meristems**, which are cylinders of meristematic tissue within the stems and roots (figure 38.5). Although secondary growth increases girth in many nonwoody plants, its effects are most dramatic in woody plants which have two lateral meristems. Within the bark of a woody stem is the **cork cambium**, a lateral meristem that produces the cork cells of the outer bark. Just beneath the bark is the **vascular cambium**, a lateral meristem that produces secondary vascular tissue. The vascular cambium forms between the xylem and phloem in vascular bundles, adding secondary vascular tissue on opposite sides of the vascular cambium. *Secondary xylem* is the main component of wood. *Secondary phloem* is very close to the outer surface of a woody stem. Removing the bark of a tree damages the phloem and may eventually kill the tree. Tissues formed from lateral meristems, which comprise most of the

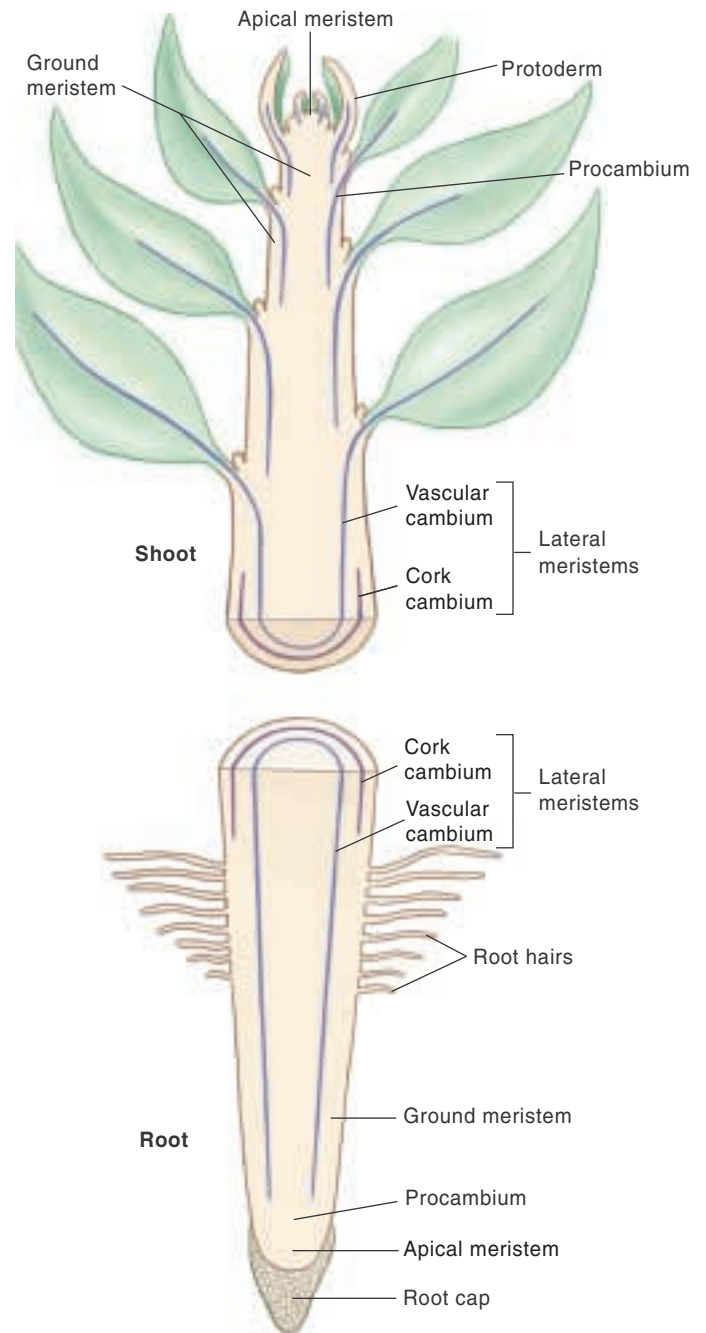


FIGURE 38.5

Apical and lateral meristems. Apical meristems produce primary growth, the elongation of the root and stem. In some plants, the lateral meristems produce an increase in the girth of a plant. This type of growth is secondary because the meristems were not directly produced by apical meristems.

trunk, branches, and older roots of trees and shrubs, are known as **secondary tissues** and are collectively called the secondary plant body.

Meristems are actively dividing, embryonic tissues responsible for both primary and secondary growth.

Organization of the Plant Body

Coordination of primary and secondary meristematic growth produces the body of the adult sporophyte plant. Plant bodies do not have a fixed size. Parts such as leaves, roots, branches, and flowers all vary in size and number from plant to plant—even within a species. The development of the form and structure of plant parts may be relatively rigidly controlled, but some aspects of leaf, stem, and root development are quite flexible. As a plant grows, the number, location, size, and even structure of leaves and roots are often influenced by the environment.

A vascular plant consists of a **root system** and a **shoot system** (figure 38.6). The root system anchors the plant and penetrates the soil, from which it absorbs water and ions crucial to the plant's nutrition. The shoot system consists of the **stems** and their **leaves**. The stem serves as a framework for positioning the leaves, the principal sites of photosynthesis. The arrangement, size, and other features of the leaves are of critical importance in the plant's production of food. Flowers, other reproductive organs, and, ultimately, fruits and seeds are also formed on the shoot (see chapters 40 and 42). The reiterative unit of the vegetative shoot consists of the internode, node, leaf, and axillary buds. Axillary buds are apical meristems derived from the primary apical meristem that allow the plant to branch or replace the main shoot if it is munched by an herbivore. A vegetative axillary bud has the capacity to reiterate the development of the primary shoot. When the plant has transited to the reproductive phase of development (see chapter 41), these axillaries may produce flowers or floral shoots.

Three basic types of tissues exist in plants: *ground tissue*, *dermal*, and *vascular tissue*. Each of the three basic tissues has its own distinctive, functionally related cell types. Some of these cell types will be discussed later in this chapter. In plants limited to primary growth, the dermal system is composed of the **epidermis**. This tissue is one cell thick in most plants, and forms the outer protective covering of the plant. In young exposed parts of the plant, the epidermis is covered with a fatty **cutin** layer constituting the **cuticle**; in plants such as the desert succulents, a layer of wax may be added outside the cuticle. In plants with secondary growth, the bark forms the outer protective layer and is considered a part of the dermal tissue system.

Ground tissue consists primarily of thin-walled **parenchyma** cells that are initially (but briefly) more or less spherical. However, the cells, which have living protoplasts, push up against each other shortly after they are produced and assume other shapes, often ending up with 11 to 17 sides. Parenchyma cells may live for many years; they function in storage, photosynthesis, and secretion.

Vascular tissue includes two kinds of conducting tissues: (1) **xylem**, which conducts water and dissolved miner-

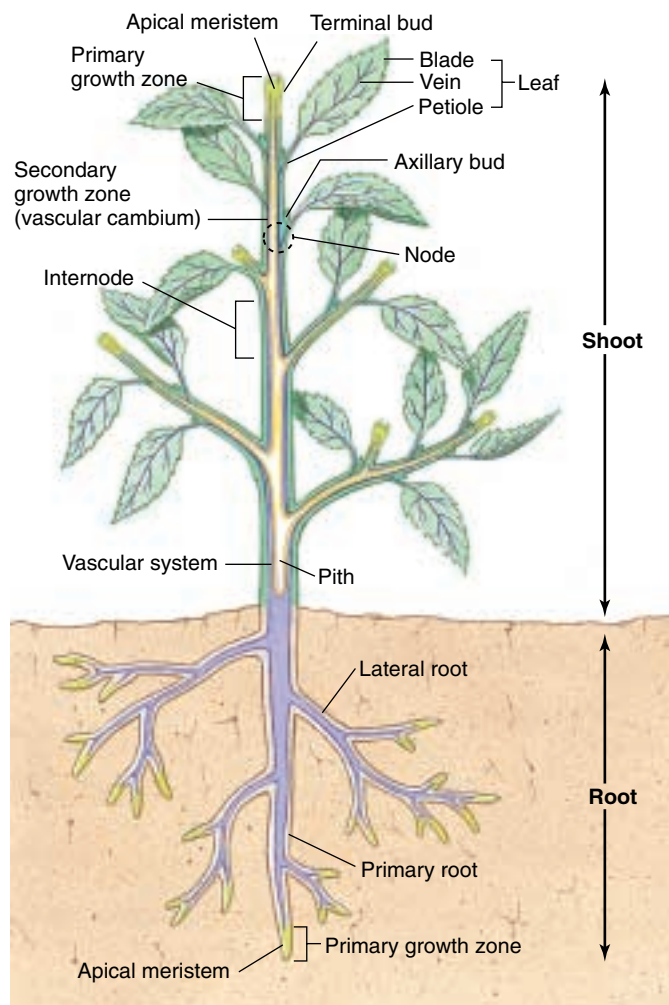


FIGURE 38.6

Diagram of a plant body. Branching in both the root and shoot system increases the number of apical meristems. A significant increase in stem/root circumference and the formation of bark can only occur if there is secondary growth initiated by vascular and cork cambium (secondary meristems). The lime green areas are zones of active elongation; secondary growth occurs in the lavender areas.

als; and (2) **phloem**, which conducts carbohydrates—mainly sucrose—used by plants for food. The phloem also transports hormones, amino acids, and other substances that are necessary for plant growth. Xylem and phloem differ in structure as well as in function.

Root and shoot meristems give rise to a plant body with an extensive underground, branching root system and aboveground shoot system with reiterative units of advantageously placed leaves joined at the node of the plant, internode, and axillary buds.

Primary and Secondary Growth

Primary and secondary growth play important roles in establishing the basic body plan of the organism. Here we will look at how these meristems give rise to highly differentiated tissues that support the growing plant body. In the earliest vascular plants, the vascular tissues produced by primary meristems played the same conducting roles as they do in contemporary vascular plants. There was no differentiation of the plant body into stems, leaves, and roots. The presence of these three kinds of organs is a property of most modern plants. It reflects increasing specialization in relation to the demands of a terrestrial existence.

With the evolution of secondary growth, vascular plants could develop thick trunks and become treelike (figure 38.7). This evolutionary advance in the sporophyte generation made possible the development of forests and the domination of the land by plants. As discussed in chapter 37, reproductive constraints would have made secondary growth and increased height nonadaptive if it had occurred in the gametophyte generation. Judging from the fossil record, secondary growth evolved independently in several groups of vascular plants by the middle of the Devonian period 380 million years ago.

There were two types of conducting systems in the earliest plants—systems that have become characteristic of vascular plants as a group. *Sieve-tube members* conduct carbohydrates away from areas where they are manufactured or stored. *Vessel members* and *tracheids* are thick-walled cells that transport water and dissolved minerals up from the roots. Both kinds of cells are elongated and occur in linked strands making tubes. Sieve-tube members are characteristic of phloem tissue; vessel members and tracheids are characteristic of xylem tissue. In primary tissues, which result from primary growth, these two types of tissue are typically associated with each other in the same vascular strands. In secondary growth, the phloem is found on the periphery, while a very thick xylem core develops more centrally. You will see that roots and shoots of many vascular plants have different patterns of vascular tissue and secondary growth. Keep in mind that water and nutrients travel between the most distant tip of a redwood root and the tip of the shoot. For the system to work, these tissues connect, which they do in the transition zone between the root and the shoot. In the next section, we will consider the three tissue systems that are present in all plant organs, whether the plant has secondary growth or not.

Plants grow from the division of meristematic tissue. Primary growth results from cell division at the apical meristem at the tip of the plant, making the shoot longer. Secondary growth results from cell division at the lateral meristem in a cylinder encasing the shoot, and increases the shoot's girth.

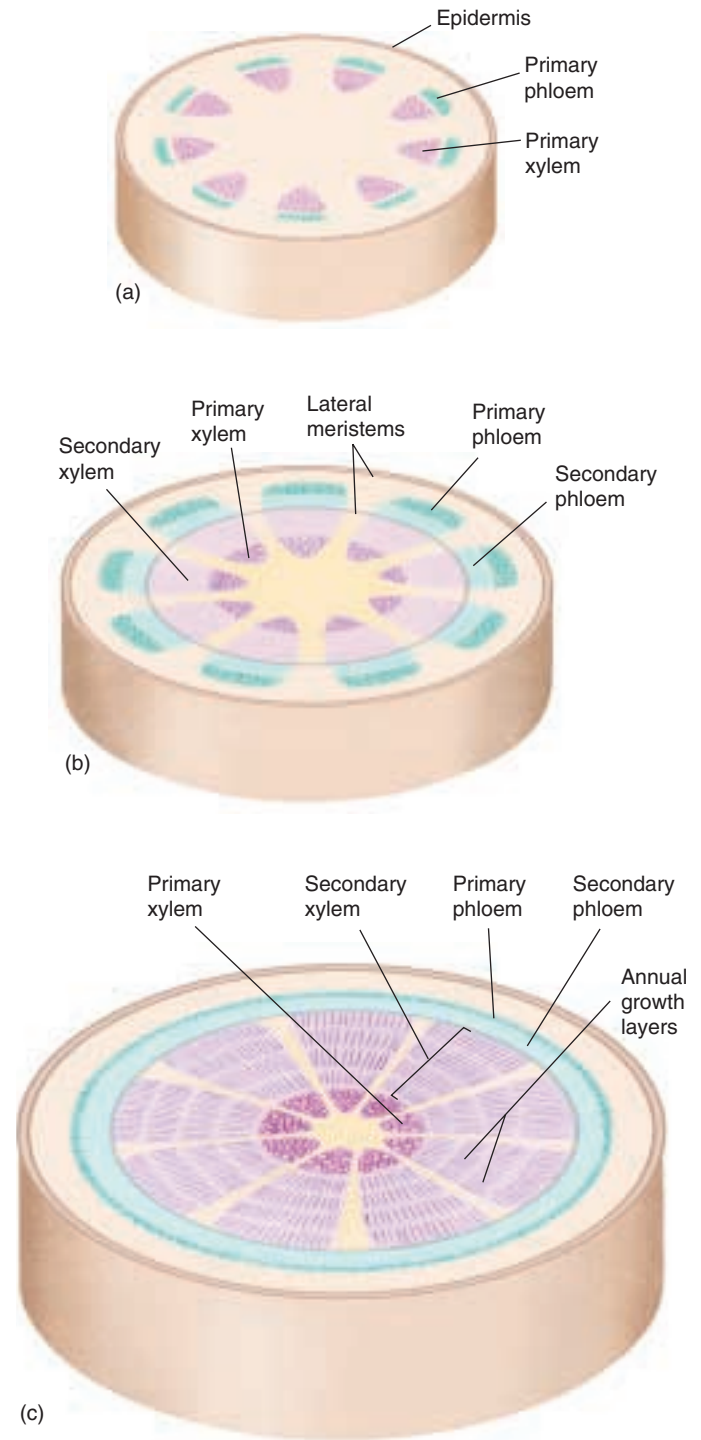


FIGURE 38.7
Secondary growth. (a) Before secondary growth begins, primary tissues continue to elongate as the apical meristems produce primary growth. (b) As secondary growth begins, the lateral meristems produce secondary tissues, and the stem's girth increases. (c) In this three-year-old stem, the secondary tissues continue to widen, and the trunk has become thick and woody. Note that the lateral meristems form cylinders that run axially in roots and shoots that have them.

38.2 Plants have three basic tissues, each composed of several cell types.

Dermal Tissue

Epidermal cells, which originate from the protoderm, cover all parts of the primary plant body. This is probably the earliest tissue system to appear in embryogenesis. The exposed outer walls have a cuticle that varies in thickness, depending on the species and environmental conditions. A number of types of specialized cells occur in the epidermis, including **guard cells**, **trichomes**, and **root hairs**.

Guard cells are paired sausage- or dumbbell-shaped cells flanking a **stoma** (plural, **stomata**), a mouth-shaped epidermal opening. Guard cells, unlike other epidermal cells, contain chloroplasts. Stomata occur in the epidermis of leaves (figure 38.8), and sometimes on other parts of the plant, such as stems or fruits. The passage of oxygen and carbon dioxide, as well as diffusion of water in vapor form, takes place almost exclusively through the stomata. There are between 1000 to more than 1 million stomata per square centimeter of leaf surface. In many plants, stomata are more numerous on the lower epidermis than on the upper epidermis of the leaf. Some plants have stomata only on the lower epidermis, and a few, such as water lilies, have them only on the upper epidermis.

Guard cell formation is the result of an asymmetrical cell division just like we saw in the first cell division in an algal and angiosperm zygote. The patterning of these asymmetrical divisions resulting in stomatal distribution has intrigued developmental biologists. Research on mutants that get “confused” about where to position stomata are providing information on the timing of stomatal initiation and the kind of intercellular communication that triggers guard cell formation. For example, the *too many mouths* mutation may be caused by a failure of developing stomata to suppress stomatal formation in neighboring cells (figure 38.9).

The stomata open and shut in response to external factors such as light, temperature, and availability of water. During periods of active photosynthesis, the stomata are open, allowing the free passage of carbon dioxide into and oxygen out of the leaf. We will consider the mechanism that governs such movements in chapter 39.

Trichomes are hairlike outgrowths of the epidermis (figure 38.10). They occur frequently on stems, leaves, and reproductive organs. A “fuzzy” or “woolly” leaf is covered with trichomes that can be seen clearly with a microscope

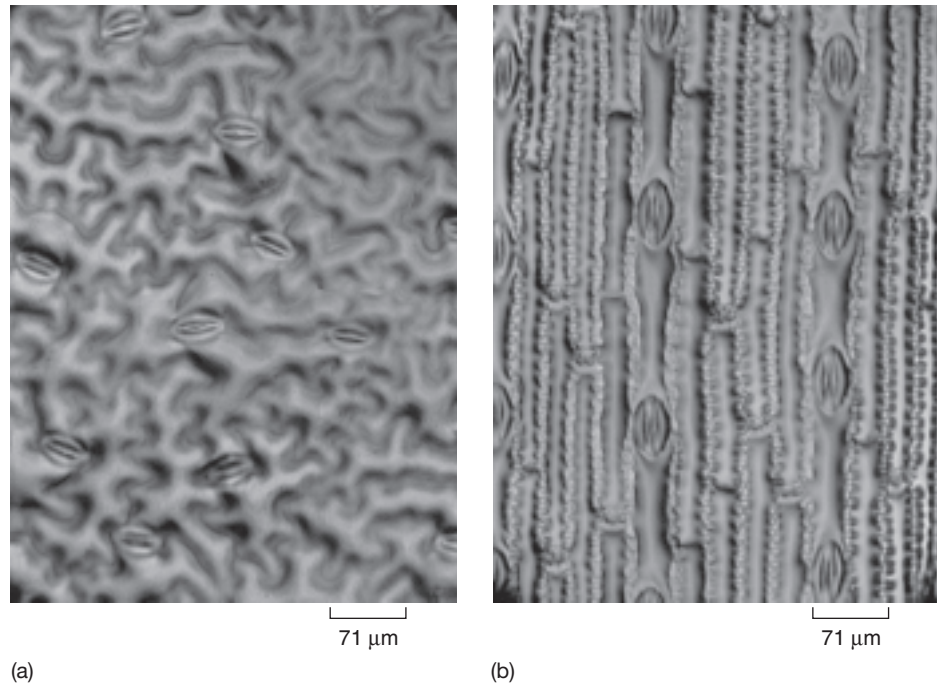


FIGURE 38.8

Epidermis of a dicot and monocot leaf (250 $\times$). Stomata are evenly distributed over the epidermis of monocots and dicots, but the patterning is quite different. (a) A pea (dicot) leaf with a random arrangement of stomata. (b) A corn leaf with stomata evenly spaced in rows. These photos also show the variety of cell shapes in plants. Some plant cells are boxlike, as seen in corn (b), while others are irregularly shaped, as seen in peas (a).



FIGURE 38.9

The *too many mouths* stomatal mutant. This *Arabidopsis* plant lacks an essential signal for spacing guard cells.

under low magnification. Trichomes play an important role in keeping the leaf surface cool and in reducing the rate of evaporation. Trichomes vary greatly in form in different kinds of plants; some consist of a single cell, while others may consist of several cells. Some are glandular, often secreting sticky or toxic substances to deter herbivory.

Trichome development has been investigated extensively in *Arabidopsis*. Four genes are needed to specify the site of trichome formation and initiate it (figure 38.11). Next, eight genes are necessary for extension growth. Loss of function of any one of these genes results in a trichome with a distorted root hair. This is an example of taking a very simple system and trying to genetically dissect all the component parts. Understanding the formation of more complex plant parts is a major challenge.

Root hairs, which are tubular extensions of individual epidermal cells, occur in a zone just behind the tips of young, growing roots (see figure 38.3). Because a root hair is simply an extension and not a separate cell, there is no crosswall isolating it from the epidermal cell. Root hairs keep the root in intimate contact with the surrounding soil particles and greatly increase the root's surface area and the efficiency of absorption. As the root grows, the extent of the root hair zone remains roughly constant as root hairs at the older end slough off while new ones are produced at the other end. Most of the absorption of water and minerals occurs through root hairs, especially in herbaceous plants. Root hairs should not be confused with lateral roots which are multicellular and have their origins deep within the root.

In the case of secondary growth, the cork cambium (discussed in the section on stems in this chapter) produces the bark of a tree trunk or root. This replaces the epidermis which gets stretched and broken with the radial expansion of the axis. Epidermal cells generally lack the plasticity of other cells, but in some cases, they can fuse to the epidermal cells of another organ or organelle and dedifferentiate.

Some epidermal cells are specialized for protection, others for absorption. Spacing of these specialized cells within the epidermis maximizes their function and is an intriguing developmental puzzle.

FIGURE 38.11

Trichome mutations. Mutants have revealed genes involved in a signal transduction pathway that regulates the spacing and development of trichomes. These include (a) *DISTORTED1* (*DIS1*) and (b) *DIS2* mutants in which trichomes are swollen and twisted.

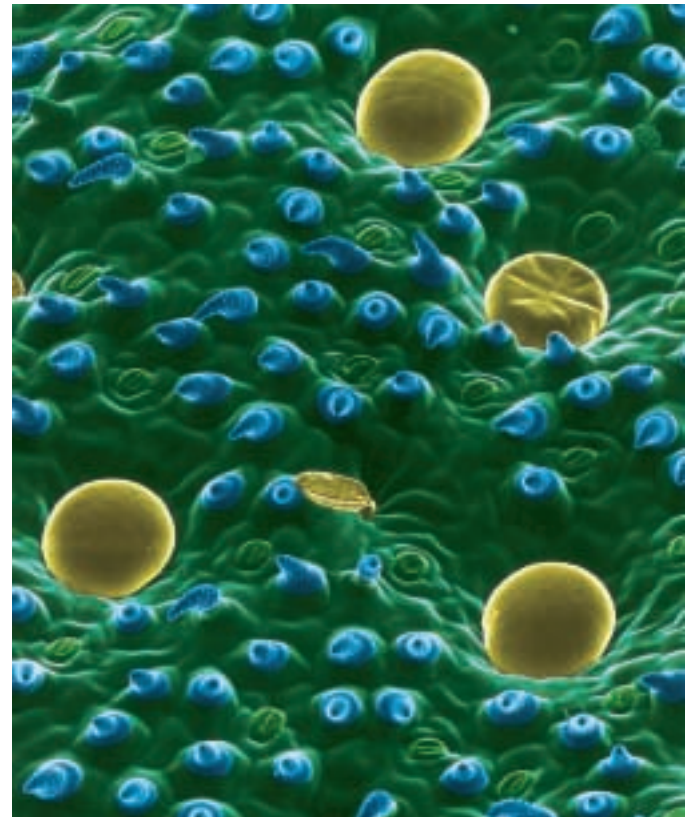


FIGURE 38.10

Trichomes. A covering of trichomes, teardrop-shaped blue structures above, creates a layer of more humid air near the leaf surface, enabling the plant to conserve available water supplies.



(a)

32 μm



(b)

57 μm

Ground Tissue

Parenchyma

Parenchyma cells, which have large vacuoles, thin walls, and an average of 14 sides at maturity, are the most common type of plant cell. They are the most abundant cells of primary tissues and may also occur, to a much lesser extent, in secondary tissues (figure 38.12*a*). Most parenchyma cells have only primary walls, which are walls laid down while the cells are still maturing. Parenchyma are less specialized than other plant cells, although there are many variations that do have special functions such as nectar and resin secretion, or storage of latex, proteins, and metabolic wastes.

Parenchyma cells, which have functional nuclei and are capable of dividing, commonly also store food and water, and usually remain alive after they mature; in some plants (for example, cacti), they may live to be over 100 years old. The majority of cells in fruits such as apples are parenchyma. Some parenchyma contain chloroplasts, especially in leaves and in the outer parts of herbaceous stems. Such photosynthetic parenchyma tissue is called *chlorenchyma*.

Collenchyma

Collenchyma cells, like parenchyma cells, have living protoplasts and may live for many years. The cells, which are usually a little longer than wide, have walls that vary in thickness (figure 38.12*b*). Collenchyma cells, which are relatively flexible, provide support for plant organs, allowing them to bend without breaking. They often form strands or continuous cylinders beneath the epidermis of stems or leaf petioles (stalks) and along the veins in leaves. Strands of

collenchyma provide much of the support for stems in which secondary growth has not taken place. The parts of celery that we eat (petioles, or leaf stalks), have “strings” that consist mainly of collenchyma and vascular bundles (conducting tissues).

Sclerenchyma

Sclerenchyma cells have tough, thick walls; they usually lack living protoplasts when they are mature. Their secondary cell walls are often impregnated with **lignin**, a highly branched polymer that makes cell walls more rigid. Cell walls containing lignin are said to be **lignified**. Lignin is common in the walls of plant cells that have a supporting or mechanical function. Some kinds of cells have lignin deposited in primary as well as secondary cell walls.

There are two types of sclerenchyma: fibers and sclereids. **Fibers** are long, slender cells that are usually grouped together in strands. Linen, for example, is woven from strands of sclerenchyma fibers that occur in the phloem of flax. **Sclereids** are variable in shape but often branched. They may occur singly or in groups; they are not elongated, but may have various forms, including that of a star. The gritty texture of a pear is caused by groups of sclereids that occur throughout the soft flesh of the fruit (figure 38.12*c*). Both of these tough, thick-walled cell types serve to strengthen the tissues in which they occur.

Parenchyma cells are the most common type of plant cells and have various functions. Collenchyma cells provide much of the support in young stems and leaves. Sclerenchyma cells strengthen plant tissues and may be nonliving at maturity.

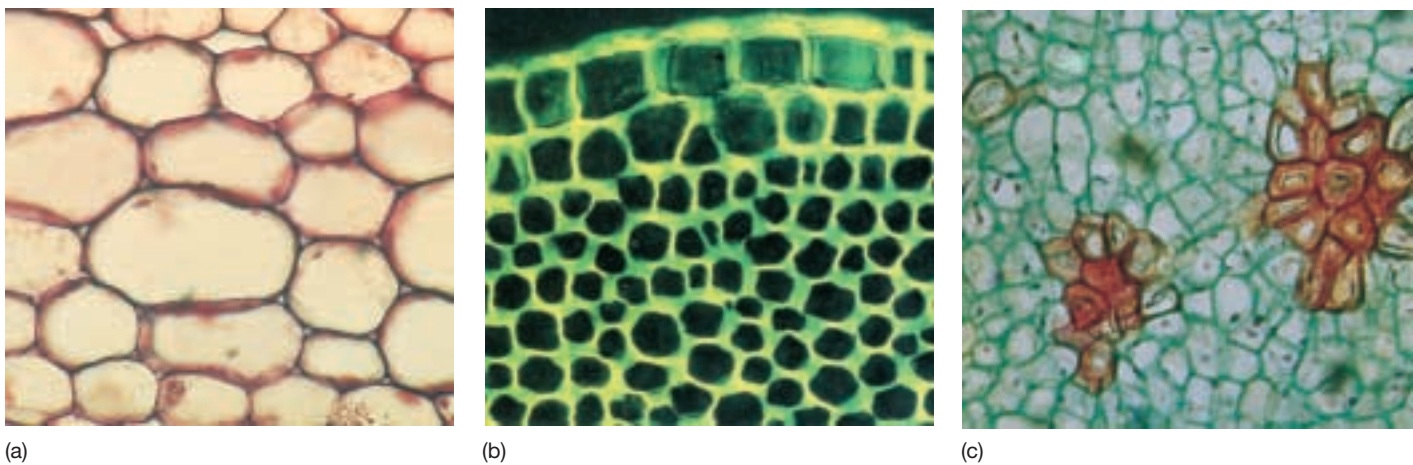


FIGURE 38.12

The three types of ground tissue. (*a*) Parenchyma cells. Only primary cell walls are seen in this cross-section of parenchyma cells from grass. (*b*) Collenchyma cells. Thickened side walls are seen in this cross-section of collenchyma cells from a young branch of elderberry (*Sambucus*). In other kinds of collenchyma cells, the thickened areas may occur at the corners of the cells or in other kinds of strips. (*c*) Sclereids. Clusters of sclereids (“stone cells”), stained red in this preparation, in the pulp of a pear. The surrounding thin-walled cells, stained light blue, are *parenchyma*. These sclereid clusters give pears their gritty texture.

Vascular Tissue

Xylem

Xylem, the principal water-conducting tissues of plants, usually contains a combination of **vessels**, which are continuous tubes formed from dead, hollow, cylindrical cells (**vessel members**) arranged end to end, and **tracheids**, which are dead cells that taper at the ends and overlap one another (figure 38.13). In some plants, such as gymnosperms, tracheids are the only water-conducting cells present; water passes in an unbroken stream through the xylem from the roots up through the shoot and into the leaves. When the water reaches the leaves, much of it passes into a film of water on the outside of the parenchyma cells, and then it diffuses in the form of water vapor into the intercellular spaces and out of the leaves into the surrounding air, mainly through the stomata. This diffusion of water vapor from a plant is known as **transpiration**. In addition to conducting water, dissolved minerals, and inorganic ions such as nitrates and phosphates throughout the plant, xylem supplies support for the plant body.

Primary xylem is derived from the procambium, which comes from the apical meristem. **Secondary xylem** is formed by the vascular cambium, a lateral meristem that develops later. Wood consists of accumulated secondary xylem.

Vessel members are found almost exclusively in angiosperms. In primitive angiosperms, vessel members tend to resemble fibers and are relatively long. In more advanced angiosperms, vessel members tend to be shorter and wider, resembling microscopic, squat coffee cans with both ends removed. Both vessel members and tracheids have thick, lignified secondary walls and no living protoplasts at maturity. Lignin is produced by the cell and secreted to strengthen the cellulose cell walls before the protoplast

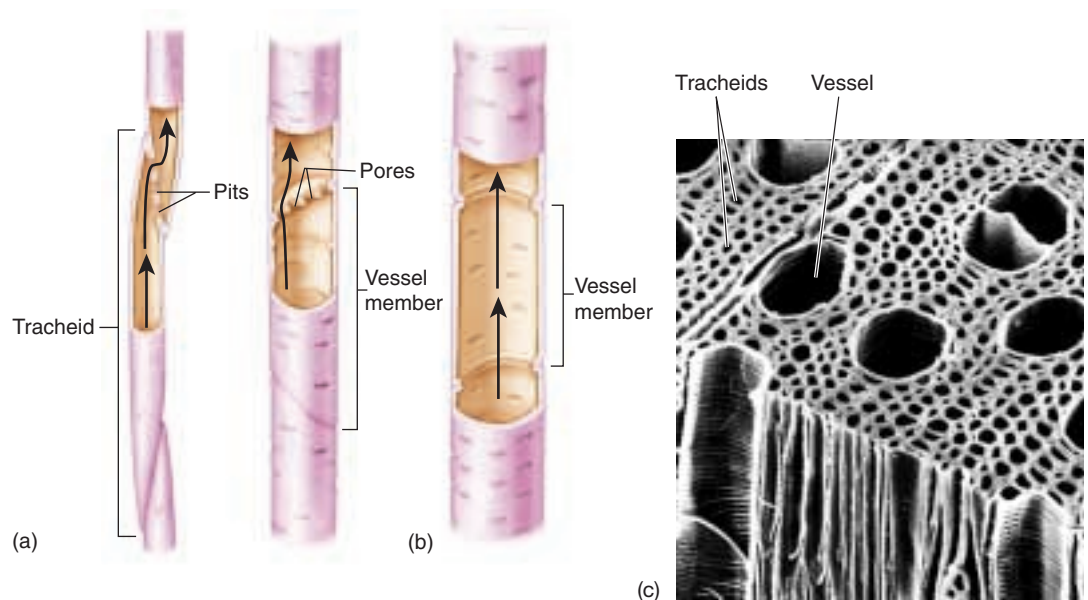
dies, leaving only the cell wall. When the continuous stream of water in a plant flows through tracheids, it passes through **pits**, which are small, mostly rounded-to-elliptical areas where no secondary wall material has been deposited. The pits of adjacent cells occur opposite one another. In contrast, vessel members, which are joined end to end, may be almost completely open or may have bars or strips of wall material across the open ends.

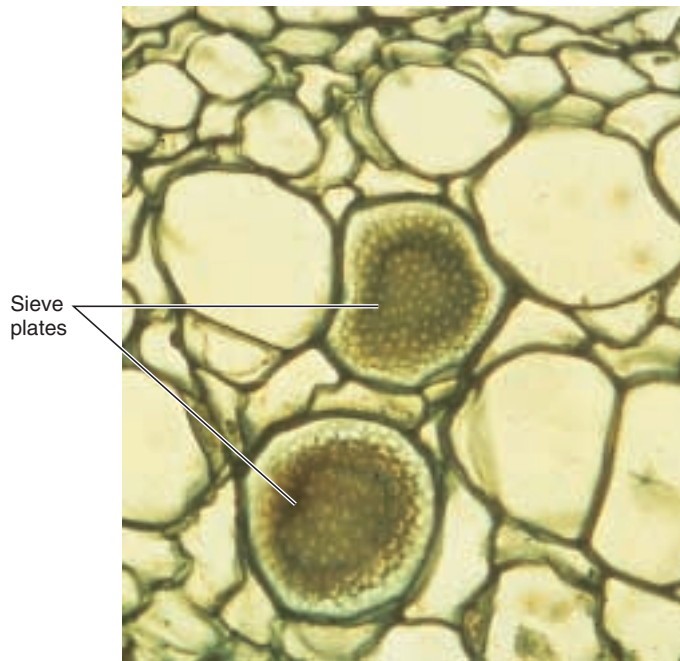
Vessels appear to conduct water more efficiently than do the overlapping strands of tracheids. We know this partly because vessel members have evolved from tracheids independently in several groups of plants, suggesting that they are favored by natural selection. It is also probable that some types of fibers have evolved from tracheids, becoming specialized for strengthening rather than conducting. Some ancient flowering plants have only tracheids, but virtually all modern angiosperms have vessels. Plants, with a mutation that prevents the differentiation of xylem, but does not affect tracheids, wilt soon after germination and are unable to transport water efficiently.

In addition to conducting cells, xylem typically includes fibers and parenchyma cells (ground tissue cells). The parenchyma cells, which are usually produced in horizontal rows called **rays** by special *ray initials* of the vascular cambium, function in lateral conduction and food storage. An initial is another term for a meristematic cell. It divides to produce another initial and a cell that differentiates into a ray cell. In cross-sections of woody stems and roots, the rays can be seen radiating out from the center of the xylem like the spokes of a wheel. Fibers are abundant in some kinds of wood, such as oak (*Quercus*), and the wood is correspondingly dense and heavy. The arrangements of these and other kinds of cells in the xylem make it possible to identify most plant genera and many species from their wood alone. These fibers are a major component in modern paper. Earlier paper was made from fibers in phloem.

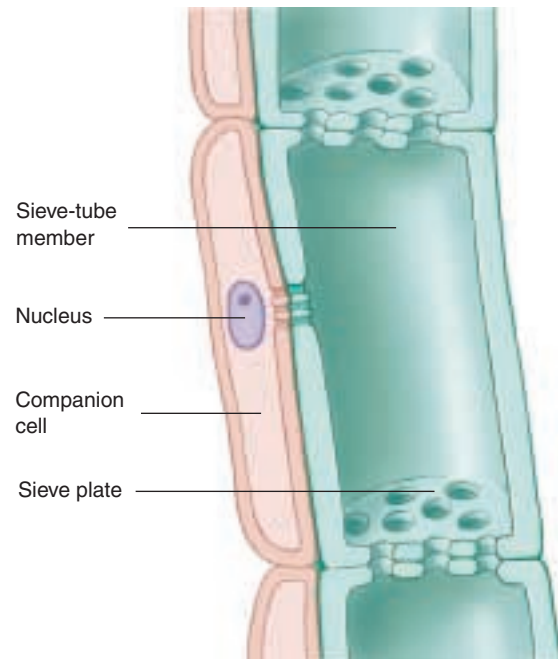
FIGURE 38.13

Comparison between vessel members and tracheids. (a) In tracheids, the water passes from cell to cell by means of pits, (b) while in vessel members, it moves by way of perforation plates or between bars of wall material. In gymnosperm wood, tracheids both conduct water and provide support; in most kinds of angiosperms, vessels are present in addition to tracheids, or present exclusively. These two types of cells conduct the water, and fibers provide additional support. (c) Scanning micrograph of the wood of red maple, *Acer rubrum* (350 $\times$).





(a)



(b)

FIGURE 38.14

A sieve-tube member. (a) Looking down into sieve plates in squash phloem reveals the perforations sucrose and hormones move through. (b) Sieve-tube member cells are stacked with the sieve plates forming the connection. The narrow cell with the nucleus at the left of the sieve-tube member is a companion cell. This cell nourishes the sieve-tube members, which have plasma membranes, but no nuclei.

Phloem

Phloem, which is located toward the outer part of roots and stems, is the principal food-conducting tissue in vascular plants. If a plant is *girdled* (by removing a substantial strip of bark down to the vascular cambium), the plant eventually dies from starvation of the roots.

Food conduction in phloem is carried out through two kinds of elongated cells: **sieve cells** and **sieve-tube members** (figure 38.14). Seedless vascular plants and gymnosperms have only sieve cells; most angiosperms have sieve-tube members. Both types of cells have clusters of pores known as **sieve areas**. Sieve areas are more abundant on the overlapping ends of the cells and connect the protoplasts of adjoining sieve cells and sieve-tube members. Both of these types of cells are living, but most sieve cells and all sieve-tube members lack a nucleus at maturity. This type of cell differentiation has parallels to the differentiation of human red blood cells which also lack a nucleus at maturity.

In sieve-tube members, some sieve areas have larger pores and are called **sieve plates**. Sieve-tube members occur end to end, forming longitudinal series called **sieve tubes**. Sieve cells are less specialized than sieve-tube mem-

bers, and the pores in all of their sieve areas are roughly of the same diameter. In an evolutionary sense, sieve-tube members are more advanced, more specialized, and, presumably, more efficient.

Each sieve-tube member is associated with an adjacent specialized parenchyma cell known as a **companion cell**. Companion cells apparently carry out some of the metabolic functions that are needed to maintain the associated sieve-tube member. In angiosperms, a common initial cell divides asymmetrically to produce a sieve-tube member cell and its companion cell. Companion cells have all of the components of normal parenchyma cells, including nuclei, and their numerous **plasmodesmata** (cytoplasmic connections between adjacent cells) connect their cytoplasm with that of the associated sieve-tube members. Fibers and parenchyma cells are often abundant in phloem.

Xylem conducts water and dissolved minerals from the roots to the shoots and the leaves. Phloem carries organic materials from one part of the plant to another.

38.3 Root cells differentiate as they become distanced from the dividing root apical meristem.

Root Structure

The three tissue systems are found in the three kinds of vegetative organs in plants: **roots, stems, and leaves**. Roots have a simpler pattern of organization and development than stems, and we will consider them first. Four zones or regions are commonly recognized in developing roots. The zones are called the **root cap**, the **zone of cell division**, the **zone of elongation**, and the **zone of maturation** (figure 38.15). In three of the zones, the boundaries are not clearly defined. When apical initials divide, daughter cells that end up on the tip end of the root become root cap cells. Cells that divide in the opposite direction pass through the three other zones before they finish differentiating. As you consider the different zones, visualize the tip of the root moving away from the soil surface by growth. This will counter the static image of a root that diagrams and photos convey.

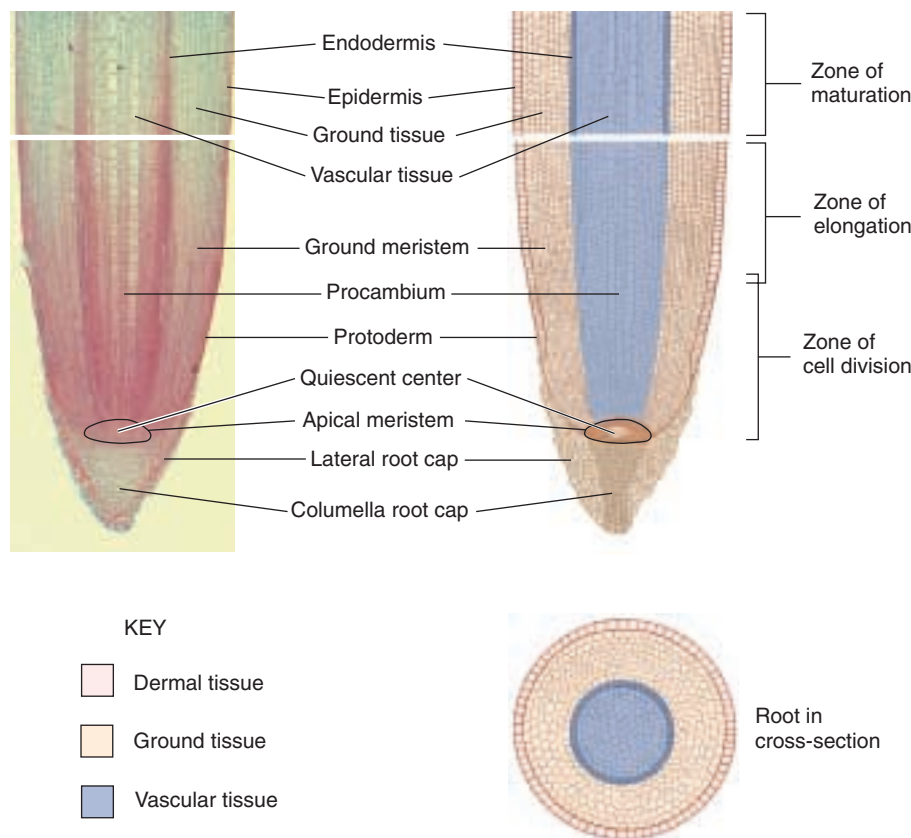


FIGURE 38.15
Root structure. A root tip in corn, *Zea mays*. This longitudinal section of a root shows the root cap, apical meristem, procambium, protoderm, epidermis, and ground meristem.

The Root Cap

The root cap has no equivalent in stems. It is composed of two types of cells, the inner columella (they look like columns) cells and the outer, lateral root cap cells that are continuously replenished by the root apical meristem. In some plants with larger roots it is quite obvious. Its most obvious function is to protect the delicate tissues behind it as growth extends the root through mostly abrasive soil particles. Golgi bodies in the outer root cap cells secrete and release a slimy substance that passes through the cell walls to the outside. The cells, which have an average life of less than a week, are constantly being replaced from the inside, forming a mucilaginous lubricant that eases the root through the soil. The slimy mass also provides a medium for the growth of beneficial nitrogen-fixing bacteria in the roots of some plants such as legumes.

A new root cap is produced when an existing one is artificially or accidentally removed from a root. The root cap also functions in the *perception of gravity*. The columella cells are highly specialized with the endoplasmic reticulum in the periphery and the nucleus located at either the middle or the top of the cell. There are no large vacuoles. Columella cells contain amyloplasts (plastids with starch grains) that collect on the sides of cells facing the pull of gravity. When a potted plant is placed on its side, the amyloplasts drift or tumble down to the side nearest the source of gravity, and the root bends in that direction. Lasers have been used to ablate (kill) individual columella cells in *Arabidopsis*. It turns out that only two of the columella cells are essential for gravity sensing! The precise nature of the gravitational response is not known, but some evidence indicates that calcium ions in the amyloplasts influence the distribution of growth hormones (auxin in this case) in the cells. There may be multiple signaling mechanisms because bending has been observed in the absence of auxin. A current hypothesis is that an electrical signal moves from the columella cell to cells in the distal region of the elongation zone (the region closest to the zone of cell division).

The Zone of Cell Division

The apical meristem is shaped like an inverted, concave dome of cells and is located in the center of the root tip in the area protected by the root cap. Most of the activity in this *zone of cell division* takes place toward the edges of the dome, where the cells divide every 12 to 36 hours, often rhythmically, reaching a peak of division once or twice a day. Most of the cells are essentially cuboidal with small vacuoles and proportionately large, centrally located nuclei. These rapidly dividing cells are daughter cells of the apical meristem. The *quiescent center* is a group of cells in the center of the root apical meristem. They divide very infrequently. This makes sense if you think about a solid ball expanding. The outer surface would have to increase far more rapidly than the very center.

The apical meristem daughter cells soon subdivide into the three primary tissue systems previously discussed: *protoderm*, *procambium*, and *ground meristem*. Genes have been identified in the relatively simple root of *Arabidopsis* that regulate the patterning of these tissue systems. The patterning of these cells begins in this zone, but it is not until the cells reach the zone of maturation that the anatomical and morphological expression of this patterning is fully revealed. *WEREWOLF*, for example, is required for the pat-

tern of the two root epidermal cell types, those with and without root hairs (figure 38.16a). The *SCARECROW* gene is important in ground cell differentiation (figure 38.16b). It is necessary for an asymmetric cell division that gives rise to two cylinders of cells from one. The outer cell layer becomes ground tissue and serves a storage function. The inner cell layer forms the endodermis which regulates the intercellular flow of water and solutes into the vascular core of the root. Cells in this region develop according to their position. If that position changes because of a mistake in cell division or the ablation of another cell, the cell develops according to its new position.

The Zone of Elongation

In the *zone of elongation*, the cells produced by the primary meristems become several times longer than wide, and their width also increases slightly. The small vacuoles present merge and grow until they occupy 90% or more of the volume of each cell. No further increase in cell size occurs above the zone of elongation, and the mature parts of the root, except for an increase in girth, remain stationary for the life of the plant.

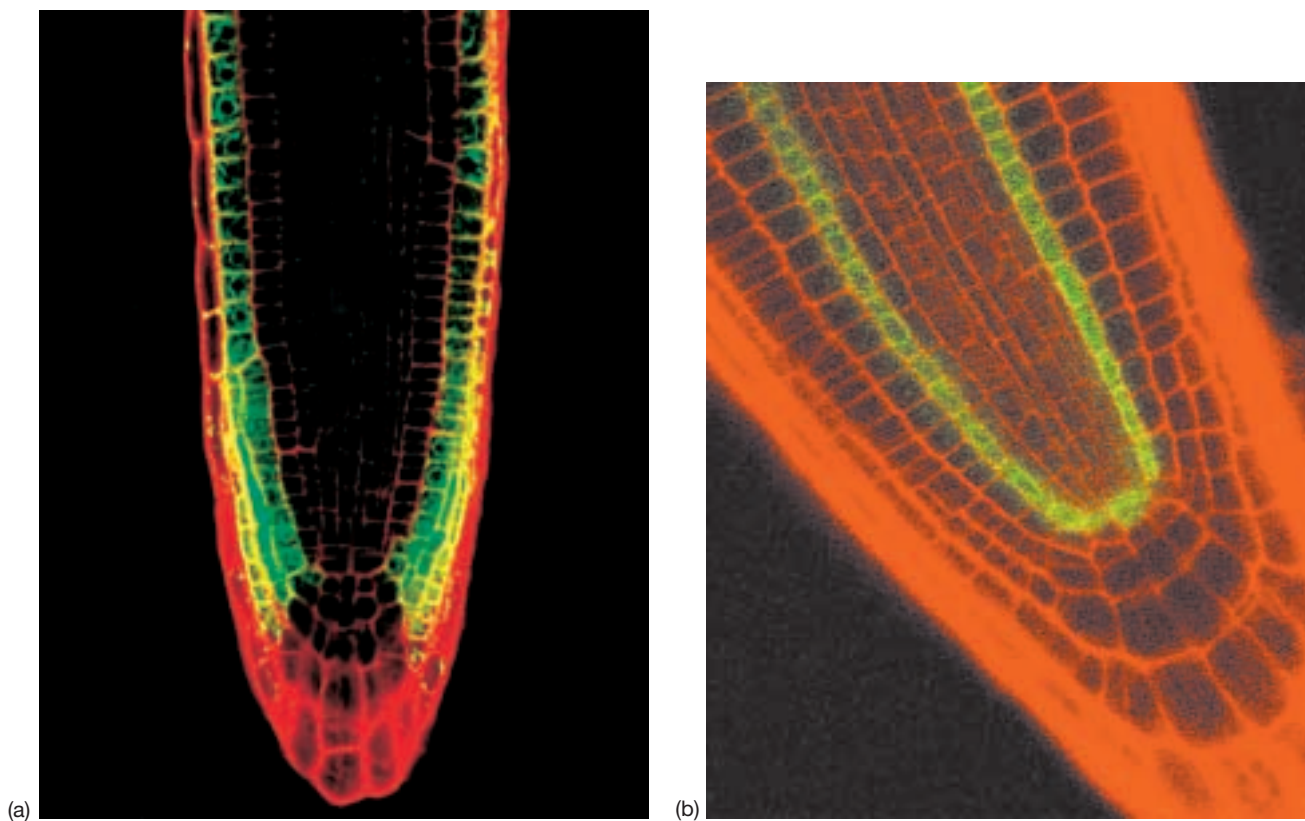


FIGURE 38.16

Tissue-specific gene expression. (a) Epidermal-specific gene expression. The promoter of the *WEREWOLF* gene of *Arabidopsis* was attached to a green fluorescent protein and used to make a transgenic plant. The green fluorescence shows the epidermal cells where the gene is expressed. The red was used to visually indicate cell boundaries. (b) Ground tissue-specific gene expression. The *SCARECROW* gene is needed for an asymmetric cell division allowing for the formation of side-by-side ground tissue and endodermal cells. These two layers are blue in wild-type, but in the mutant, only one cell layer is blue because the asymmetric cell division does not occur.

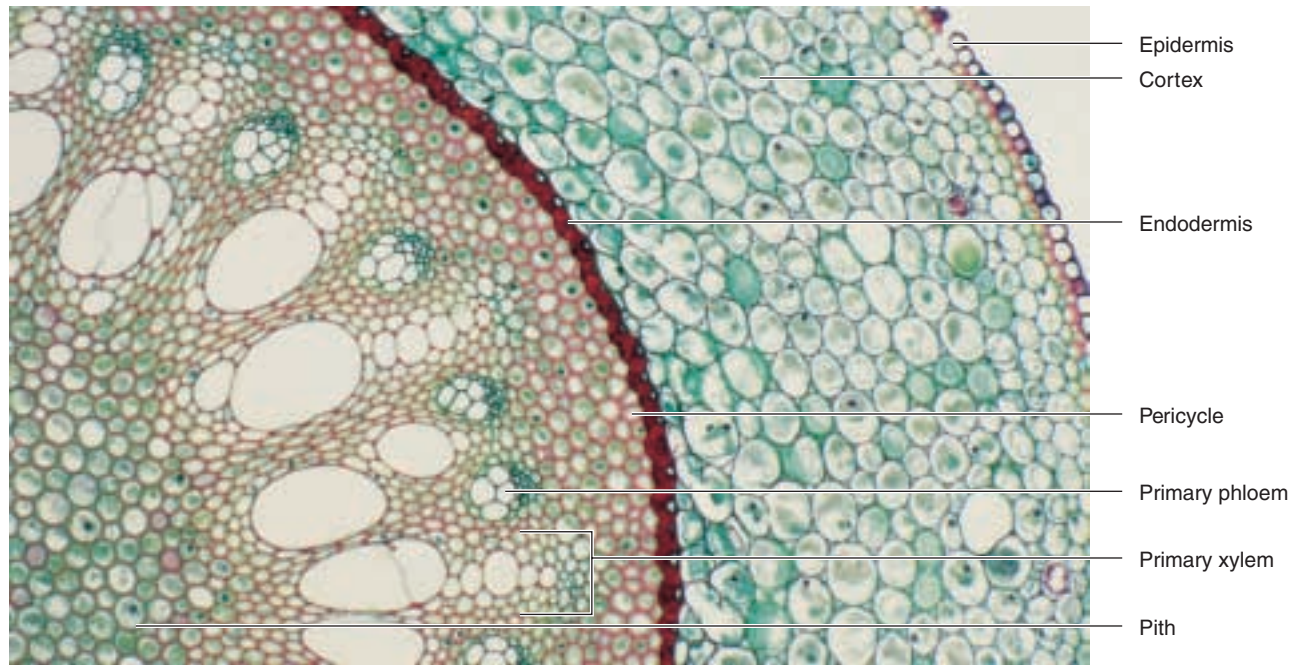


FIGURE 38.17
Cross-section of the zone of maturation of a young monocot root. Greenbrier (*Smilax*), a monocot (100 $\times$).

The Zone of Maturation

The cells that have elongated in the zone of elongation become differentiated into specific cell types in the *zone of maturation*. The cells of the root surface cylinder mature into *epidermal cells*, which have a very thin cuticle. Many of the epidermal cells each develop a **root hair**; the protuberance is not separated by a crosswall from the main part of the cell and the nucleus may move into it. Root hairs, which can number over 35,000 per square centimeter of root surface and many billions per plant, greatly increase the surface area and therefore the absorptive capacity of the root. The root hairs usually are alive and functional for only a few days before they are sloughed off at the older part of the zone of maturation, while new ones are being produced toward the zone of elongation. Symbiotic bacteria that fix atmospheric nitrogen into a form usable by legumes enter the plant via root hairs and “instruct” the plant to create a nodule around it.

Parenchyma cells are produced by the ground meristem immediately to the interior of the epidermis. This tissue, called the **cortex**, may be many cells wide and functions in food storage. The inner boundary of the cortex differentiates into a single-layered cylinder of **endodermis** (figure 38.17), whose primary walls are impregnated with **suberin**, a fatty substance that is impervious to water. The suberin is produced in bands, called **Casparian strips** that surround each adjacent endodermal cell wall perpendicular to the root’s surface (figure 38.18). This blocks transport between cells. The two surfaces that

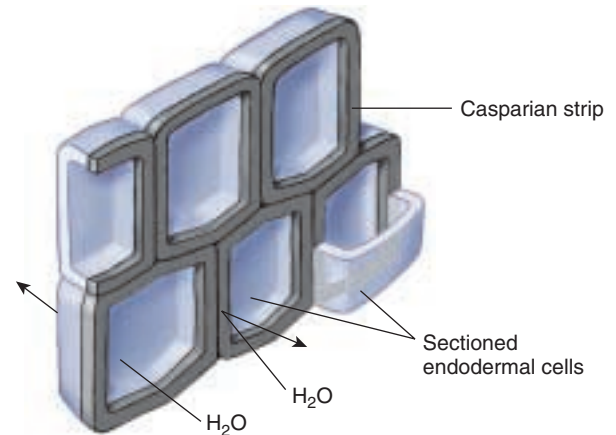


FIGURE 38.18
Casparian strip. The Casparian strip is a water-proofing band that protects cells inside the endodermis from flooding.

are parallel to the root surface are the only way into the core of the root and the cell membranes control what passes through.

All the tissues interior to the endodermis are collectively referred to as the **stele**. Immediately adjacent and interior to the endodermis is a cylinder of parenchyma cells known as the **pericycle**. Pericycle cells can divide, even after they mature. They can give rise to *lateral* (branch) *roots* or, in dicots, to part of the *vascular cambium*.

Modified Roots

Most plants produce either a **taproot system** in which there is a single large root with smaller branch roots, or a **fibrous root system** in which there are many smaller roots of similar diameter. Some plants, however, have intriguing root modifications with specific functions in addition to those of anchorage and absorption.

Aerial roots. Some plants, such as epiphytic orchids (orchids that are attached to tree branches and grow unconnected to the ground without being parasitic in any way) have roots that extend out into the air. Some aerial roots have an epidermis that is several cells thick, an adaptation to reduce water loss. These aerial roots may also be green and photosynthetic, as in the vanilla orchid. Some monocots, such as corn, produce thick roots from the lower parts of the stem; these *prop roots* grow down to the ground and brace the plants against wind. Climbing plants such as ivy also produce roots from their stems; these anchor the stems to tree trunks or a brick wall. Any root that arises along a stem or in some place other than the root of the plant is called an *adventitious root*. Adventitious root formation in ivy depends on the developmental stage of the shoot. When the shoot transitions to the adult phase of development, it is no longer capable of initiating these roots.

Pneumatophores. Some plants that grow in swamps and other wet places may produce spongy outgrowths from their underwater roots called *pneumatophores* from their underwater roots (figure 38.21a). The pneumatophores commonly extend several centimeters above water, facilitating the oxygen supply to the roots beneath.

Contractile roots. The roots from the bulbs of lilies and of several other plants such as dandelions contract by spiraling to pull the plant a little deeper into the soil each year until they reach an area of relatively stable temperatures. The roots may contract to a third of their original length as they spiral like a corkscrew due to cellular thickening and constricting.

Parasitic roots. The stems of certain plants that lack chlorophyll, such as dodder (*Cuscuta*), produce peglike

roots called *haustoria* that penetrate the host plants around which they are twined. The haustoria establish contact with the conducting tissues of the host and effectively parasitize their host.

Food storage roots. The xylem of branch roots of sweet potatoes and similar plants produce at intervals many extra parenchyma cells that store large quantities of carbohydrates. Carrots, beets, parsnips, radishes, and turnips have combinations of stem and root that also function in food storage. Cross sections of these roots reveal multiple rings of secondary growth.

Water storage roots. Some members of the pumpkin family (Cucurbitaceae), especially those that grow in arid regions, may produce water-storage roots weighing 50 or more kilograms (figure 38.21b).

Buttress roots. Certain species of fig and other tropical trees produce huge buttress roots toward the base of the trunk, which provide considerable stability (figure 38.21c).

Some plants have modified roots that carry out photosynthesis, gather oxygen, parasitize other plants, store food or water, or support the stem.



(a)



(b)



(c)

FIGURE 38.21
Three types of modified roots. (a) Pneumatophores (foreground) are spongy outgrowths from the roots below. (b) A water storage root weighing over 25 kilograms (60 pounds). (c) Buttress roots of a tropical fig tree.

38.4 Stems are the backbone of the shoot, transporting nutrients and supporting the aerial plant organs.

Stem Structure

External Form

The shoot apical meristem initiates stem tissue and intermittently produces bulges (**primordia**) that will develop into leaves, other shoots, or even flowers (figure 38.22). The stem is an axis from which organs grow. Leaves may be arranged in a spiral around the stem, or they may be in pairs opposite one another; they also may occur in *whorls* (circles) of three or more. Spirals are the most common and, for reasons still not understood, sequential leaves tend to be placed 137.5° apart. This angle relates to the golden mean, a mathematical ratio, that is found in nature (the angle of coiling in some shells, for example), classical architecture (the Parthenon wall dimensions), and even modern art (Mondrian). The pattern of leaf arrangement is called **phyllotaxy** and may optimize exposure of leaves to the sun.

The *region* or *area* (no structure is involved) of leaf attachment is called a **node**; the area of stem between two nodes is called an **internode**. A leaf usually has a flattened **blade** and sometimes a **petiole** (stalk). When the petiole is missing, the leaf is then said to be **sessile**. Note that the word *sessile* as applied to plants has a different meaning than it does when applied to animals (probably obvious, as plants don't get up and move around!); in plants, it means *immobile* or *attached*. The space between a petiole (or blade) and the stem is called an **axil**. An **axillary bud** is produced in each axil. This bud is a product of the primary shoot apical meristem, which, with its associated leaf primordia, is called a **terminal bud**. Axillary buds frequently develop into branches or may form meristems that will develop into flowers. (Refer back to figure 38.6 to review these terms.)

Herbaceous stems do not produce a cork cambium. The stems are usually green and photosynthetic, with at least the outer cells of the cortex containing chloroplasts. Herbaceous stems commonly have stomata, and may have various types of trichomes (hairs).

Woody stems can persist over a number of years and develop distinctive markings in addition to the original organs that form. Terminal buds usually extend the length of the shoot during the growing season. Some buds, such as those of geraniums, are unprotected, but



FIGURE 38.22
A shoot apex
(200×). Scanning
electron
micrograph of the
apical meristem of
wheat (*Triticum*).

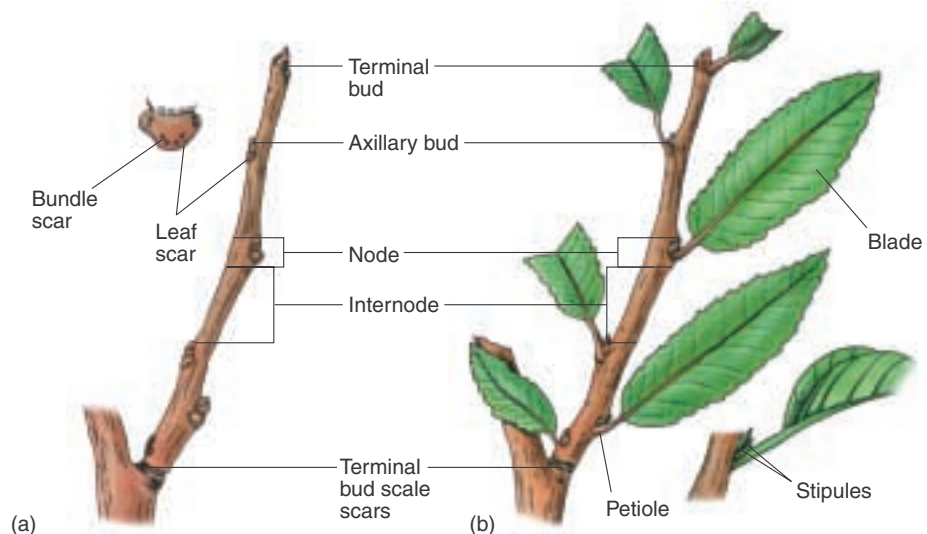


FIGURE 38.23
A woody twig. (a) In winter. (b) In summer.

most buds of woody plants have protective winter *bud scales* that drop off, leaving tiny *bud scars* as the buds expand. Some twigs have tiny scars of a different origin. A pair of butterfly-like appendages called **stipules** (part of the leaf) develop at the base of some leaves. The stipules can fall off and leave *stipule scars*. When leaves of deciduous trees drop in the fall, they leave *leaf scars* with tiny *bundle scars*, marking where vascular connections were. The shapes, sizes, and other features of leaf scars can be distinctive enough to identify the plants in winter (figure 38.23).

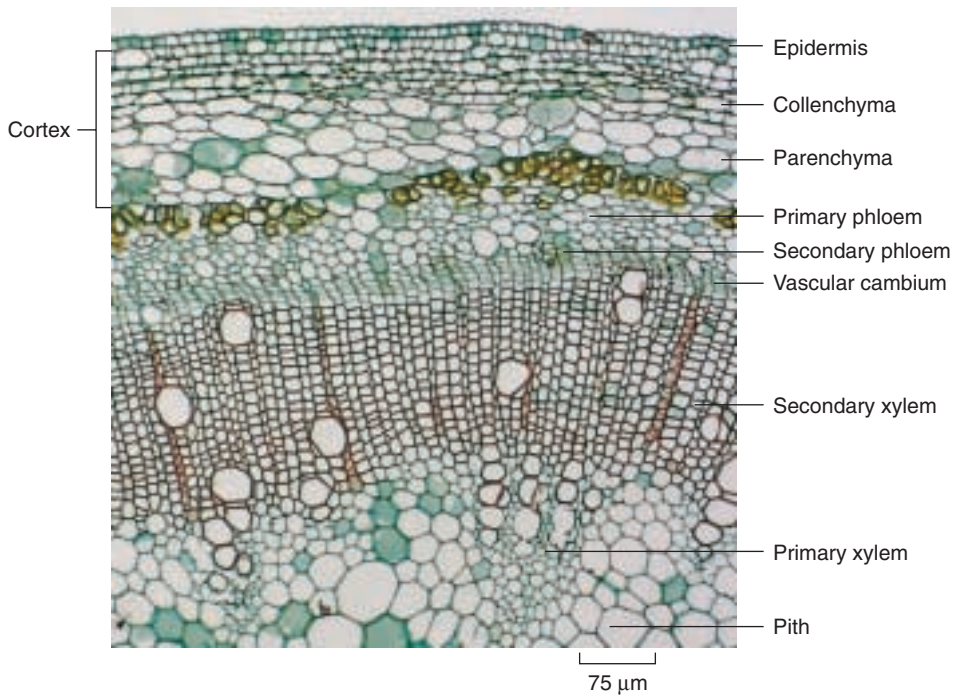


FIGURE 38.24
Early stage in differentiation of vascular cambium in the castor bean, *Ricinus* (25x). The outer part of the cortex consists of collenchyma, and the inner part of parenchyma.

Internal Form

As in roots, there is an *apical meristem* at the tip of each stem, which produces *primary tissues* that contribute to the stem's increases in length. Three primary meristems develop from the apical meristem. The *protoderm* gives rise to the *epidermis*. The *ground meristem* produces parenchyma cells. Parenchyma cells in the center of the stem constitute the *pith*; parenchyma cells away from the center constitute the *cortex*. The *procambium* produces cylinders of *primary xylem* and *primary phloem*, which are surrounded by ground tissue.

A strand of xylem and phloem, called a *trace*, branches off from the main cylinder of xylem and phloem and enters the developing leaf, flower, or shoot. These spaces in the main cylinder of conducting tissues are called **gaps**. In dicots, a **vascular cambium** develops between the primary xylem and primary phloem (figure 38.24). In many ways, this is a connect-the-dots game where the vascular cambium connects the ring of primary vascular bundles. In monocots, these bundles are scattered throughout the ground tissue (figure 38.25) and there is no logical way to connect them that would allow a uniform increase in girth. This is why monocots do not have secondary growth.

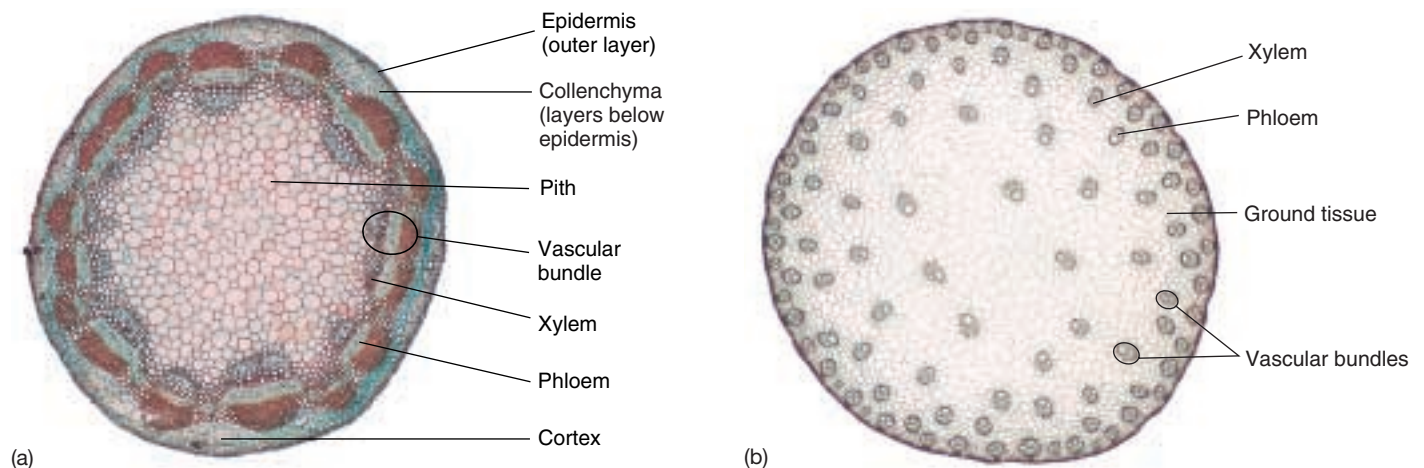


FIGURE 38.25
Stems. Transverse sections of a young stem in (a) a dicot, the common sunflower, *Helianthus annuus*, in which the vascular bundles are arranged around the outside of the stem (10x); and (b) a monocot, corn, *Zea mays*, with the scattered vascular bundles characteristic of the class (5x).

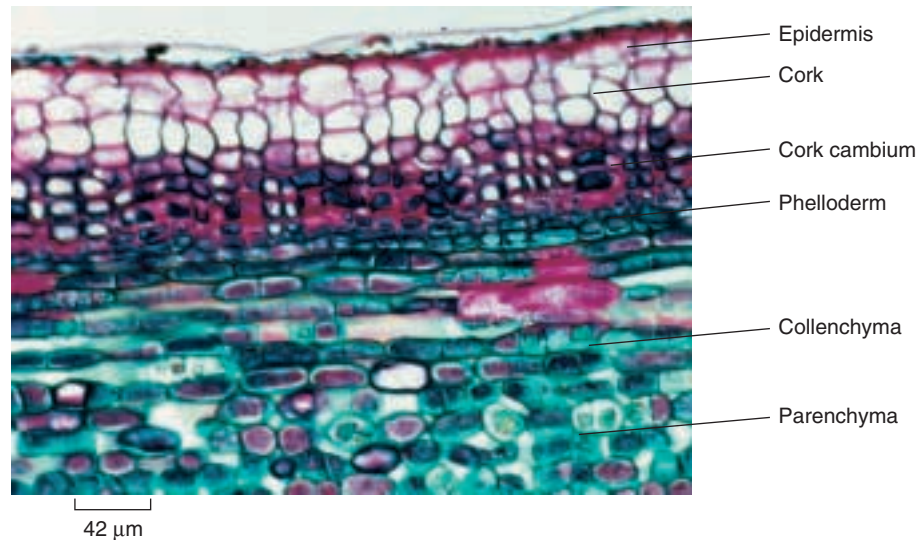


FIGURE 38.26
Section of periderm (50×). An early stage in the development of periderm in cottonwood (*Populus* sp.).

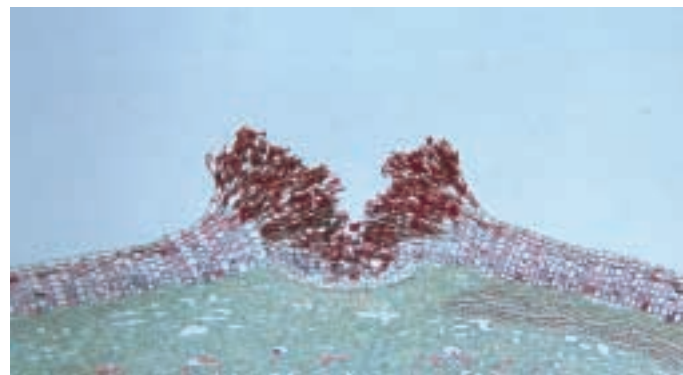
The cells of the vascular cambium divide indefinitely, producing **secondary tissues** (mainly *secondary xylem* and *secondary phloem*). The production of xylem is extensive in trees and is called wood. Rings in the stump of a tree reveal annual patterns of growth; cell size varies depending on growth conditions. In woody dicots, a second cambium, the *cork cambium*, arises in the outer cortex (occasionally in the epidermis or phloem) and produces box-like *cork cells* to the outside and also may produce parenchyma-like *phelloderm* cells to the inside; the cork cambium, cork, and phelloderm are collectively referred to as the *periderm* (figure 38.26). Cork tissues, whose cells become impregnated with *suberin* shortly after they are formed and then die, constitute the **outer bark**. The cork tissue, whose suberin is impervious to moisture, cuts off water and food to the epidermis, which dies and sloughs off. In young stems, gas exchange between stem tissues and the air takes place through stomata, but as the cork cambium produces cork, it also produces patches of unsubserved cells beneath the stomata. These unsubserved cells, which permit gas exchange to continue, are called **lenticels** (figure 38.27).

The stem results from the dynamic growth of the shoot apical meristem which initiates stem tissue and organs including leaves. Shoot apical meristems initiate new apical meristems at the junction of leaf and stem. These meristems can form buds which reiterate the growth pattern of the terminal bud or they can make flowers directly.



(a)

FIGURE 38.27
Lenticels. (a) Lenticels, the numerous, small, pale, raised areas shown here on cherry tree bark (*Prunus cerasifera*), allow gas exchange between the external atmosphere and the living tissues immediately beneath the bark of woody plants. Highly variable in form in different species, lenticels are an aid to the identification of deciduous trees and shrubs in winter. (b) Transverse section through a lenticel (extruding area) in a stem of elderberry, *Sambucus canadensis* (30×).



(b)

Modified Stems

Although most stems grow erect, there are some modifications that serve special purposes, including that of natural *vegetative propagation*. In fact, the widespread artificial vegetative propagation of plants, both commercial and private, frequently involves the cutting of modified stems into segments, which are then planted and produce new plants. As you become acquainted with the following modified stems, keep in mind that stems have *leaves at nodes*, with *internodes* between the nodes, and *buds* in the *axils* of the leaves, while roots have no leaves, nodes, or axillary buds.

Bulbs. Onions, lilies, and tulips have swollen underground stems that are really large buds with adventitious roots at the base (figure 38.28a). Most of a *bulb* consists of fleshy leaves attached to a small, knoblike stem. In onions, the fleshy leaves are surrounded by papery, scalelike leaf bases of the long, green aboveground leaves.

Corms. Crocuses, gladioluses, and other popular garden plants produce *corms* that superficially resemble bulbs. Cutting a corm in half, however, reveals no fleshy leaves. Instead, almost all of a corm consists of stem, with a few papery, brown nonfunctional leaves on the outside, and adventitious roots below.

Rhizomes. Perennial grasses, ferns, irises, and many other plants produce *rhizomes*, which typically are horizontal stems that grow underground, often close to the surface (figure 38.28b). Each node has an inconspicuous scalelike leaf with an axillary bud; much larger photosynthetic leaves may be produced at the rhizome tip. Adventitious roots are produced throughout the length of the rhizome, mainly on the lower surface.

Runners and stolons. Strawberry plants produce horizontal stems with long internodes, which, unlike rhizomes, usually grow along the surface of the ground. Several *runners* may radiate out from a single plant (figure 38.28c). Some botanists use the term *stolon* synonymously with runner; others reserve the term stolon for a stem with long internodes that grows underground, as seen in Irish (white) potato plants. An Irish potato itself, however, is another type of modified stem—a *tuber*.

Tubers. In Irish potato plants, carbohydrates may accumulate at the tips of stolons, which swell, becoming *tubers*; the stolons die after the tubers mature (figure 38.28d). The “eyes” of a potato are axillary buds formed in the axils of scalelike leaves. The scalelike leaves, which are present when the potato is starting to form, soon drop off; the tiny ridge adjacent to each “eye” of a mature potato is a leaf scar.

Tendrils. Many climbing plants, such as grapes and Boston ivy, produce modified stems known as *tendrils*, which twine around supports and aid in climbing (figure 38.28e). Some tendrils, such as those of peas and pumpkins, are actually modified leaves or leaflets.

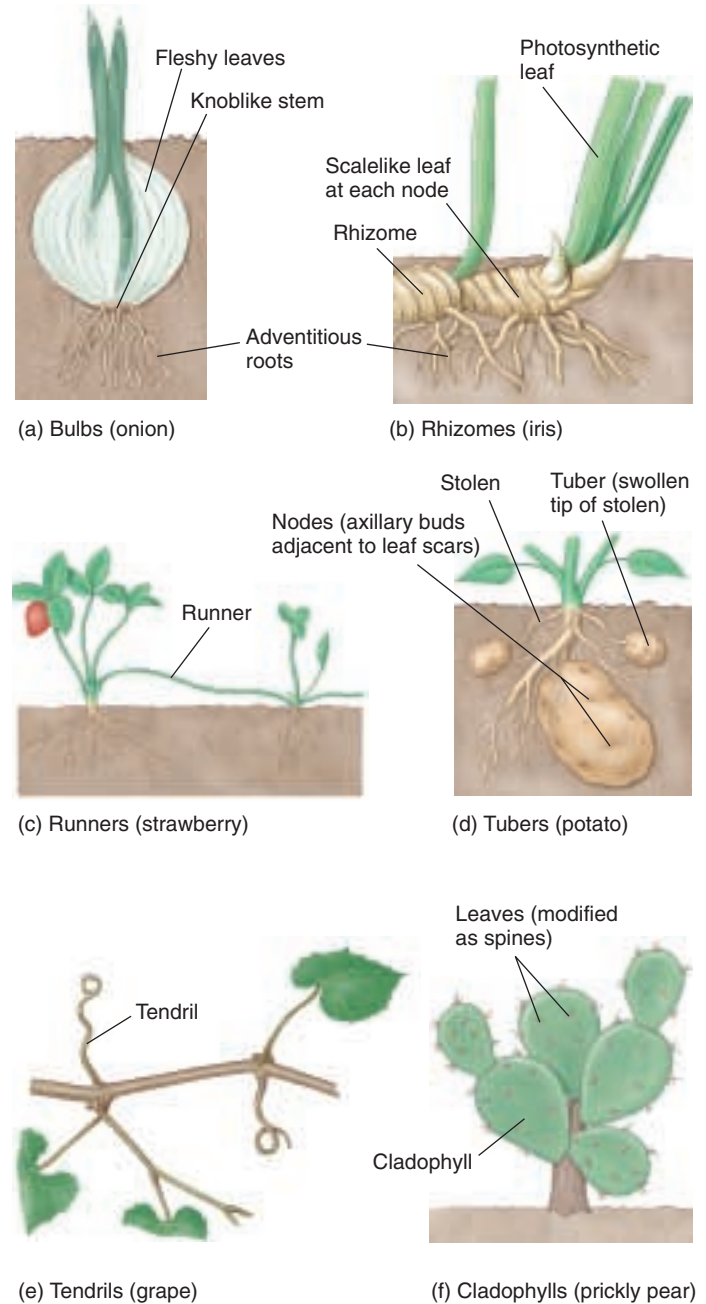


FIGURE 38.28
Types of modified stems.

Cladophylls. Cacti and several other plants produce flattened, photosynthetic stems called *cladophylls* that resemble leaves (figure 38.28f). In cacti, the real leaves are modified as spines.

Some plants possess modified stems that serve special purposes including food storage, support, or vegetative propagation.

38.5 Leaves are adapted to support basic plant functions.

Leaf External Structure

Leaves, which are initiated as *primordia* by the apical meristems (see figure 38.2), are vital to life as we know it. They are the principal sites of photosynthesis on land. Leaves expand primarily by cell enlargement and some cell division. Like our arms and legs, they are determinate structures which means growth stops at maturity. Because leaves are crucial to a plant, features such as their arrangement, form, size, and internal structure are highly significant and can differ greatly. Different patterns have adaptive value in different environments.

Leaves are really an extension of the shoot apical meristem and stem development. Leaves first emerge as primordia as discussed in the section on stems. At that point, they are not committed to be leaves. Experiments where very young leaf primordia in fern and in coleus are isolated and grown in culture demonstrate this. If the primordia are young enough, they will form an entire shoot rather than a leaf. So, positioning the primordia and beginning the initial cell divisions occurs before those cells are committed to the leaf developmental pathway.

Leaves fall into two different morphological groups which may reflect differences in evolutionary origin. A *microphyll* is a leaf with one vein that does not leave a gap when it branches from the vascular cylinder of the stem; microphylls are mostly small and are associated primarily with the phylum Lycophyta (see chapter 37). Most plants have leaves called *megaphylls*, which have several to many veins; a megaphyll's conducting tissue leaves a gap in the stem's vascular cylinder as it branches from it.

Most dicot leaves have a flattened *blade*, and a slender stalk, the *petiole*. The flattening of the leaf blade reflects a shift from radial symmetry to dorsal-ventral (top-bottom) symmetry. We're just beginning to understand how this shift occurs by analyzing mutants like *phantastica* which prevents this transition (figure 38.29). In addition, a pair of *stipules* may be present at the base of the petiole. The stipules, which may be leaflike or modified as *spines* (as in the black locust—*Robinia pseudo-acacia*) or *glands* (as in cherry trees—*Prunus cerasifera*), vary considerably in size from microscopic to almost half the size of the leaf blade. Development of stipules appears to be independent of development of the rest of the leaf.

Grasses and other monocot leaves usually lack a petiole and tend to sheathe the stem toward the base. **Veins** (a term used for the vascular bundles in leaves), consisting of both xylem and phloem, are distributed throughout the leaf blades. The main veins are parallel in most monocot leaves; the veins of dicots, on the other hand, form an often intricate network (figure 38.30).



FIGURE 38.29

The *phantastica* mutant in snapdragon. Snapdragon leaves are usually flattened with a top and bottom side (plant on *left*). In the *phantastica* mutant (plant on *right*), the leaf never flattens but persists as a radially symmetrical bulge.

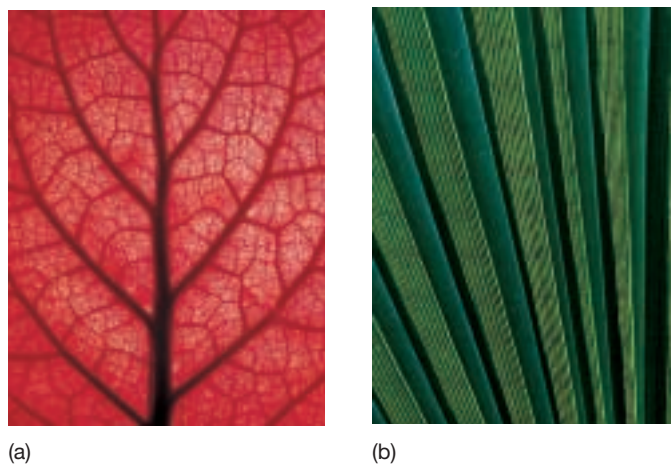


FIGURE 38.30

Dicot and monocot leaves. The leaves of dicots, such as this (a) African violet relative from Sri Lanka, have netted, or reticulate, veins; (b) those of monocots, like this cabbage palmetto, have parallel veins. The dicot leaf has been cleared with chemicals and stained with a red dye to make the veins show up more clearly.



(a)



(b)



(c)

FIGURE 38.31

Simple versus compound leaves. (a) A simple leaf, its margin deeply lobed, from the tulip tree (*Liriodendron tulipifera*). (b) A pinnately compound leaf, from a mountain ash (*Sorbus* sp.). A compound leaf is associated with a single lateral bud, located where the petiole is attached to the stem. (c) Palmately compound leaves of a Virginia creeper (*Parthenocissus quinquefolia*).

Leaf blades come in a variety of forms from oval to deeply lobed to having separate leaflets. In **simple leaves** (figure 38.31a), such as those of lilacs or birch trees, the blades are undivided, but simple leaves may have teeth, indentations, or lobes of various sizes, as in the leaves of maples and oaks. In **compound leaves**, such as those of ashes, box elders, and walnuts, the blade is divided into **leaflets**. The relationship between the development of compound and simple leaves is an open question. Two explanations are being debated: (1) a compound leaf is a highly lobed simple leaf, or (2) a compound leaf utilizes a shoot development program. There are single mutations that convert compound leaves to simple leaves which are being used to address this debate. If the leaflets are arranged in pairs along a common axis (the axis is called a *rachis*—the equivalent of the main central vein, or *midrib*, in simple leaves), the leaf is **pinnately compound** (figure 38.31b). If, however, the leaflets radiate out from a common point at the blade end of the petiole, the leaf is **palmately compound** (figure 38.31c). Palmately compound leaves occur in buckeyes (*Aesculus* spp.) and Virginia creeper (*Parthenocissus quinquefolia*). The leaf blades themselves may have similar arrangements of their veins, and are said to be **pinnately** or **palmately** veined.

Leaves, regardless of whether they are simple or compound, may be **alternately** arranged (alternate leaves usually spiral around a shoot) or they may be in **opposite** pairs. Less often, three or more leaves may be in a **whorl**, a circle of leaves at the same level at a node (figure 38.32).

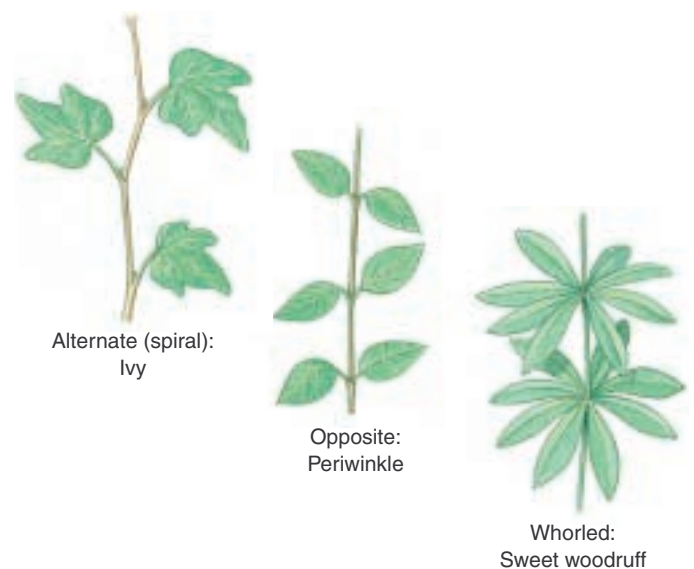


FIGURE 38.32

Types of leaf arrangements. The three common types of leaf arrangements are alternate, opposite, and whorled.

Leaves are the principal sites of photosynthesis. Their blades may be arranged in a variety of ways. In simple leaves the blades are undivided, while in compound leaves the leaf is composed of two or more leaflets.

Leaf Internal Structure

The entire surface of a leaf is covered by a transparent epidermis, most of whose cells have no chloroplasts. The epidermis itself has a waxy *cuticle* of variable thickness, and may have different types of glands and trichomes (hairs) present. The lower epidermis (and occasionally the upper epidermis) of most leaves contains numerous slit-like or mouth-shaped *stomata* (figure 38.33). Stomata, as discussed earlier, are flanked by *guard cells* and function in gas exchange and regulation of water movement through the plant.

The tissue between the upper and lower epidermis is called **mesophyll**. Mesophyll is interspersed with veins (vascular bundles) of various sizes. In most dicot leaves, there are two distinct types of mesophyll. Closest to the upper epidermis are one to several (usually two) rows of tightly packed, barrel-shaped to cylindrical *chlorenchyma* cells (parenchyma with chloroplasts) that constitute the **palisade mesophyll** (figure 38.34). Some plants, including species of *Eucalyptus*, have leaves that hang down, rather than extending horizontally. They have palisade parenchyma on both sides of the leaf, and there is, in effect, no upper side. In nearly all leaves there are loosely arranged **spongy mesophyll** cells between the palisade mesophyll and the lower epidermis, with many air spaces throughout the tissue. The interconnected intercellular spaces, along with the stomata, function in gas exchange and the passage of water vapor from

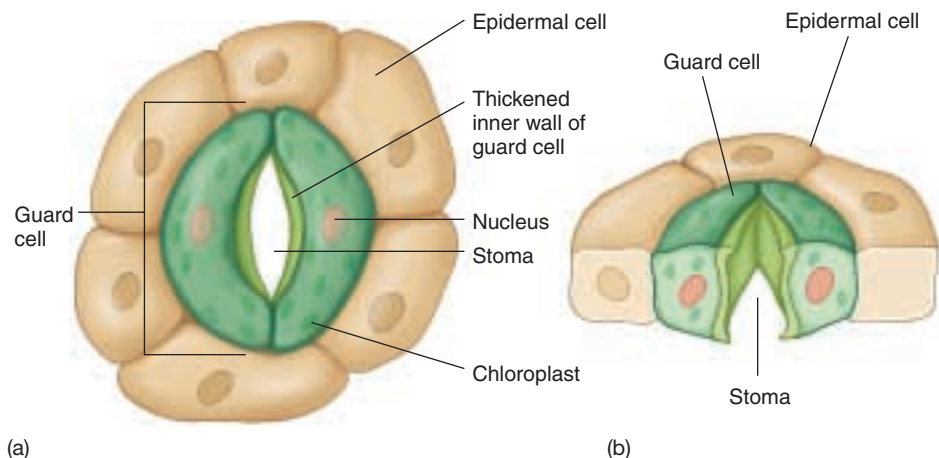


FIGURE 38.33
A stoma. (a) Surface view. (b) View in cross-section.

the leaves. The mesophyll of monocot leaves is not differentiated into palisade and spongy layers and there is often little distinction between the upper and lower epidermis. This anatomical difference often correlates with a modified photosynthetic pathway that maximizes the amount of CO_2 relative to O_2 to reduce energy loss through photorespiration (refer to chapter 10). Leaf anatomy directly relates to its juggling act to balance water loss, gas exchange, and transport of photosynthetic products to the rest of the plant.

Leaves are basically flattened bags of epidermis containing vascular tissue and tightly packed palisade mesophyll rich in chloroplasts and loosely packed spongy mesophyll with many interconnected air spaces that function in gas and water vapor exchange.

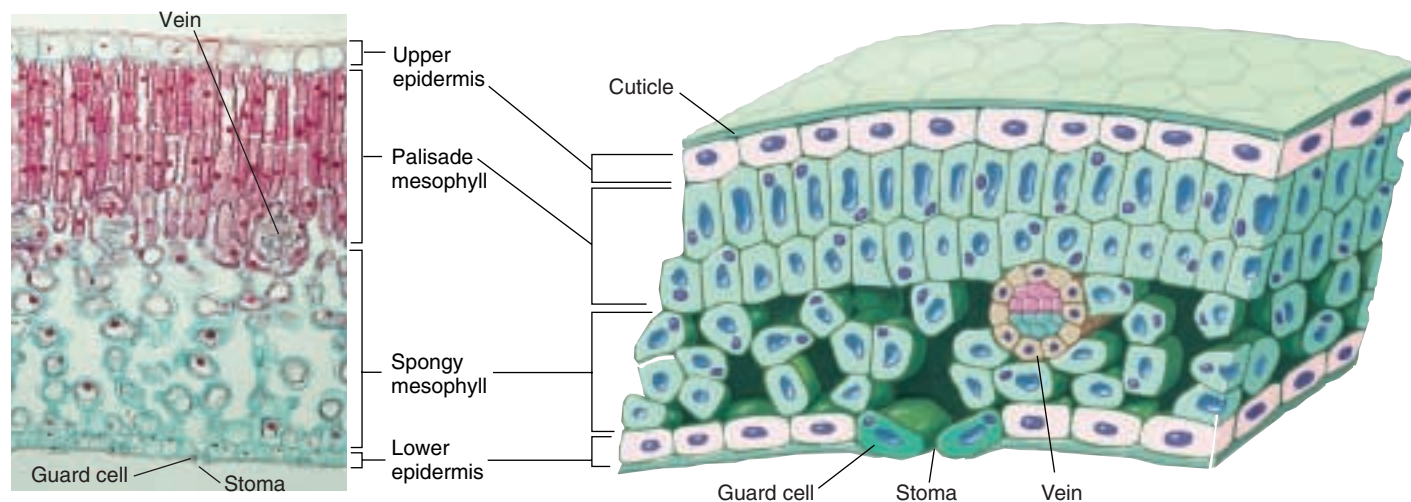


FIGURE 38.34
A leaf in cross-section. Transection of a leaf showing the arrangement of palisade and spongy mesophyll, a vascular bundle or vein, and the epidermis with paired guard cells flanking the stoma.

Modified Leaves

As plants colonized a wide variety of environments, from deserts to lakes to tropical rain forests, modifications of plant organs that would adapt the plants to their specific habitats arose. Leaves, in particular, have evolved some remarkable adaptations. A brief discussion of a few of these modifications follows.

Floral leaves (bracts). Poinsettias and dogwoods have relatively inconspicuous, small, greenish-yellow flowers. However, both plants produce large modified leaves, called **bracts** (mostly colored red in poinsettias and white or pink in dogwoods). These bracts surround the true flowers and perform the same function as showy petals (figure 38.35). It should be noted, however, that bracts can also be quite small and not as conspicuous as those of the examples mentioned.

Spines. The leaves of many cacti, barberries, and other plants are modified as **spines** (see figure 38.28f). In the case of cacti, the reduction of leaf surface reduces water loss and also may deter predators. Spines should not be confused with *thorns*, such as those on honey locust (*Gleditsia triacanthos*), which are modified stems, or with the *prickles* on raspberries and rose bushes, which are simply outgrowths from the epidermis or the cortex just beneath it.

Reproductive leaves. Several plants, notably *Kalanchoë*, produce tiny but complete plantlets along their margins. Each plantlet, when separated from the leaf, is capable of growing independently into a full-sized plant. The walking fern (*Asplenium rhizophyllum*) produces new plantlets at the tips of its fronds. While leaf tissue isolated from many species will regenerate a whole plant, this in vivo regeneration is unique among just a few species.

Window leaves. Several genera of plants growing in arid regions produce succulent, cone-shaped leaves with transparent tips. The leaves often become mostly buried in sand blown by the wind, but the transparent tips, which have a thick epidermis and cuticle, admit light to the hollow interiors. This allows photosynthesis to take place beneath the surface of the ground.

Shade leaves. Leaves produced where they receive significant amounts of shade tend to be larger in surface area, but thinner and with less mesophyll than leaves on the same tree receiving more direct light. This plasticity in development is remarkable, as both types of leaves on the plant have exactly the same genes. Environmental signals can have a major effect on development.

Insectivorous leaves. Almost 200 species of flowering plants are known to have leaves that trap insects, with some digesting their soft parts. Plants with insectivorous leaves often grow in acid swamps deficient in needed elements, or containing elements in forms not readily available to the plants; this inhibits the plants' capacities



FIGURE 38.35

Modified leaves. In this dogwood “flower,” the white-colored bracts (modified leaves) surround the several true flowers without petals in the center.

to maintain metabolic processes sufficient to meet their growth requirements. Their needs are, however, met by the supplementary absorption of nutrients from the animal kingdom.

Pitcher plants (for example, *Sarracenia*, *Darlingtonia*, *Nepenthes*) have cone-shaped leaves in which rainwater can accumulate. The insides of the leaves are very smooth, but there are stiff, downward-pointing hairs at the rim. An insect falling into such a leaf finds it very difficult to escape and eventually drowns. The nutrients released when bacteria, and in most species digestive enzymes, decompose the insect bodies are absorbed into the leaf. Other plants, such as sundews (*Drosera*), have glands that secrete sticky mucilage that trap insects, which are then digested by enzymes. The Venus flytrap (*Dionaea muscipula*) produces leaves that look hinged at the midrib. When tiny trigger hairs on the leaf blade are stimulated by a moving insect, the two halves of the leaf snap shut, and digestive enzymes break down the soft parts of the trapped insect into nutrients that can be absorbed through the leaf surface. Nitrogen is the most common nutrient needed. Curiously, the Venus flytrap will not survive in a nitrogen-rich environment, perhaps a trade-off made in the intricate evolutionary process that resulted in its ability to capture and digest insects.

The leaves of plants exhibit a variety of adaptations, including spines, vegetative reproduction, and even leaves that are carnivorous.



Summary

Questions

Media Resources

38.1 Meristems elaborate the plant body plan after germination.

- A plant body is basically an axis that includes two parts: root and shoot—with associated leaves. There are four basic types of tissues in plants: meristems, ground tissue, epidermis, and vascular tissue.

1. What are the three major tissue systems in plants? What are their functions?



- Art Activity: Plant Body Organization
- Art Activity: Stem Tip Structure
- Art Activity: Primary Meristem Structure

38.2 Plants have three basic tissues, each composed of several cell types.

- Ground tissue supports the plant and stores food and water.
- Epidermis forms an outer protective covering for the plant.
- Vascular tissue conducts water, carbohydrates, and dissolved minerals to different parts of the plant. Xylem conducts water and minerals from the roots to shoots and leaves, and phloem conducts food molecules from sources to all parts of the plant.

2. What is the function of xylem? How do primary and secondary xylem differ in origin? What are the two types of conducting cells within xylem?



- Characteristics of Plants
- Meristems
- Cambia

3. What is the function of phloem? How do the two types of conducting cells in phloem differ?



- Effect of Water on Leaves
- Girth Increase in Woody Dicots
- Vascular System of Plants



- Activity: Vascular Tissue
- Ground Tissue
- Dermal Tissue
- Vascular Tissue

38.3 Root cells differentiate as they become distanced from the dividing root apical meristem.

- Roots have four growth zones: the root cap, zone of cell division, zone of elongation, and zone of maturation.
- Some plants have modified roots, adapted for photosynthesis, food or water storage, structural support, or parasitism.

4. Compare monocot and dicot roots. How does the arrangement of the tissues differ?



- Student Research: Leaf Structure Wetness

5. How are lateral branches of roots formed?



- Art Activity: Dicot Root Structure



- Roots

38.4 Stems are the backbone of the shoot, transporting nutrients and supporting the aerial plant organs.

- Plants branch by means of buds derived from the primary apical meristem. They are found in the junction between the leaf and the stem.
- The vascular cambium is a cylinder of dividing cells found in both roots and shoots. As a result of their activity, the girth of a plant increases.

6. What types of cells are produced when the vascular cambium divides outwardly, inwardly, or laterally?



- Art Activity: Dicot Stem Structure
- Art Activity: Secondary Growth
- Art Activity: Herbaceous Dicot Stem Anatomy

7. Why don't monocots have secondary growth?



- Activity: Cambium
- Stems

38.5 Leaves are adapted to support basic plant functions.

- Leaves emerge as bulges on the meristem in a variety of patterns, but most form a spiral around the stem. The bulge lengthens and loses its radial symmetry as it flattens.
- Photosynthesis occurs in the ground tissue system which is called mesophyll in the leaf. Vascular tissue forms the venation patterns in the leaves, serving as the endpoint for water conduction and often the starting point for the transport of photosynthetically produced sugars.

8. How do simple and compound leaves differ from each other? Name and describe the three common types of leaf growth patterns.



- Art Activity: Plant Anatomy
- Art Activity: Leaf Structure



- Leaves

39

Nutrition and Transport in Plants

Concept Outline

39.1 Plants require a variety of nutrients in addition to the direct products of photosynthesis.

Plant Nutrients. Plants require a few macronutrients in large amounts and several micronutrients in trace amounts.

Soil. Plant growth is significantly influenced by the nature of the soil.

39.2 Some plants have novel strategies for obtaining nutrients.

Nutritional Adaptations. Venus flytraps and other carnivorous plants lure and capture insects and then digest them to obtain energy and nutrients. Some plants entice bacteria to produce organic nitrogen for them. These bacteria may be free-living or form a symbiotic relationship with a host plant. About 90% of all vascular plants rely on fungal associations to gather essential nutrients, especially phosphorus.

39.3 Water and minerals move upward through the xylem.

Overview of Water and Mineral Movement through Plants. The bulk movement of water and dissolved minerals is the result of movement between cells, across cell membranes, and through tubes of xylem.

Water and Mineral Absorption. Water and minerals enter the plant through the roots.

Water and Mineral Movement. A combination of the properties of water, structure of xylem, and transpiration of water through the leaves results in the passive movement of water to incredible heights. Water leaves the plant through openings in the leaves called stomata. Too much water is harmful to a plant, although many plants have adaptations that make them tolerant of flooding.

39.4 Dissolved sugars and hormones are transported in the phloem.

Phloem Transport Is Bidirectional. Sucrose and hormones can move from shoot to root or root to shoot in the phloem. Phloem transport requires energy to load and unload sieve tubes.



FIGURE 39.1

A carnivorous plant. Most plants absorb water and essential nutrients from the soil, but carnivorous plants are able to obtain some nutrients directly from small animals.

Vast energy inputs are required for the ongoing construction of a plant such as described in chapter 38. In this chapter, we address two major questions: (1) what inputs, besides energy from the sun, does a plant need to survive? and (2) how do all parts of the complex plant body share the essentials of life? Plants, like animals, need various nutrients to remain alive and healthy. Lack of an important nutrient may slow a plant's growth or make the plant more susceptible to disease or even death. Plants acquire these nutrients through photosynthesis and from the soil, although some take a more direct approach (figure 39.1). Carbohydrates produced in leaves must be carried throughout the plant, and minerals and water absorbed from the ground must be transported up to the leaves and other parts of the plant. As discussed in chapter 38, these two types of transport take place in specialized tissues, xylem and phloem.

39.1 Plants require a variety of nutrients in addition to the direct products of photosynthesis.

Plant Nutrients

The major source of plant nutrition is the fixation of atmospheric CO₂ into simple sugar using the energy of the sun. CO₂ enters through the stomata. O₂ is a product of photosynthesis and atmospheric component that also moves through the stomata. It is used in cellular respiration to release energy from the chemical bonds in the sugar to support growth and maintenance in the plant. However, CO₂ and light energy are not sufficient for the synthesis of all the molecules a plant needs. Plants require a number of inorganic nutrients (table 39.1). Some of these are macronutrients, which the plants need in relatively large amounts, and others are micronutrients, which are required in trace amounts. There are nine macronutrients: carbon, hydrogen, and oxygen—the three elements found in all organic compounds—as well as nitrogen

(essential for amino acids), potassium, calcium, phosphorus, magnesium (the center of the chlorophyll molecule), and sulfur. Each of these nutrients approaches or, as in the case with carbon, may greatly exceed 1% of the dry weight of a healthy plant. The seven micronutrient elements—iron, chlorine, copper, manganese, zinc, molybdenum, and boron—constitute from less than one to several hundred parts per million in most plants (figure 39.2). The macronutrients were generally discovered in the last century, but the micronutrients have been detected much more recently as technology developed to identify and work with such small quantities.

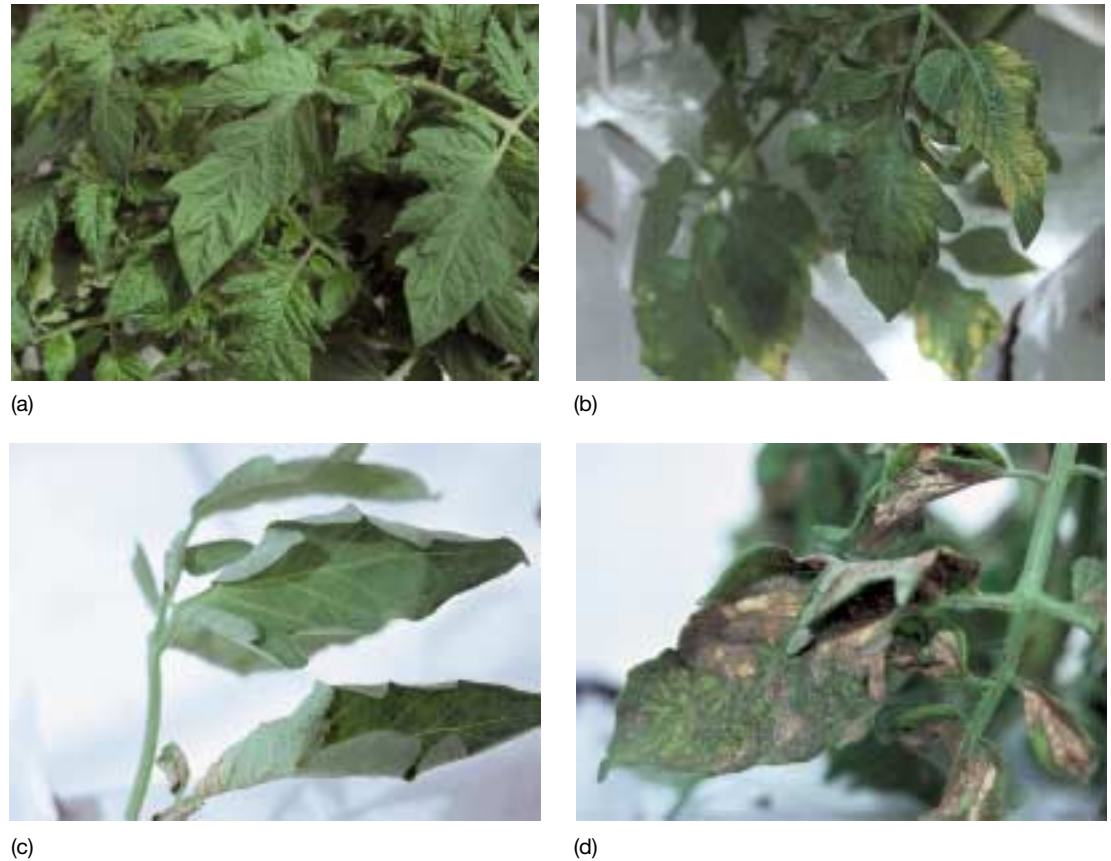
Nutritional requirements are assessed in hydroponic cultures; the plants roots are suspended in aerated water containing nutrients. The solutions contain all the necessary nutrients in the right proportions but with certain known or suspected nutrients left out. The plants are then

Table 39.1 Essential Nutrients in Plants

Elements	Principal Form in which Element Is Absorbed	Approximate Percent of Dry Weight	Examples of Important Functions
MACRONUTRIENTS			
Carbon	(CO ₂)	44	Major component of organic molecules
Oxygen	(O ₂ , H ₂ O)	44	Major component of organic molecules
Hydrogen	(H ₂ O)	6	Major component of organic molecules
Nitrogen	(NO ₃ ⁻ , NH ₄ ⁺)	1–4	Component of amino acids, proteins, nucleotides, nucleic acids, chlorophyll, coenzymes, enzymes
Potassium	(K ⁺)	0.5–6	Protein synthesis, operation of stomata
Calcium	(Ca ⁺⁺)	0.2–3.5	Component of cell walls, maintenance of membrane structure and permeability, activates some enzymes
Magnesium	(Mg ⁺⁺)	0.1–0.8	Component of chlorophyll molecule, activates many enzymes
Phosphorus	(H ₂ PO ₄ ⁻ , HPO ₄ ⁼)	0.1–0.8	Component of ADP and ATP, nucleic acids, phospholipids, several coenzymes
Sulfur	(SO ₄ ⁼)	0.05–1	Components of some amino acids and proteins, coenzyme A
MICRONUTRIENTS (CONCENTRATIONS IN PPM)			
Chlorine	(Cl ⁻)	100–10,000	Osmosis and ionic balance
Iron	(Fe ⁺⁺ , Fe ⁺⁺⁺)	25–300	Chlorophyll synthesis, cytochromes, nitrogenase
Manganese	(Mn ⁺⁺)	15–800	Activator of certain enzymes
Zinc	(Zn ⁺⁺)	15–100	Activator of many enzymes, active in formation of chlorophyll
Boron	(BO ₃ ⁻ or B ₄ O ₇ ⁼)	5–75	Possibly involved in carbohydrate transport, nucleic acid synthesis
Copper	(Cu ⁺⁺)	4–30	Activator or component of certain enzymes
Molybdenum	(MoO ₄ ⁼)	0.1–5	Nitrogen fixation, nitrate reduction

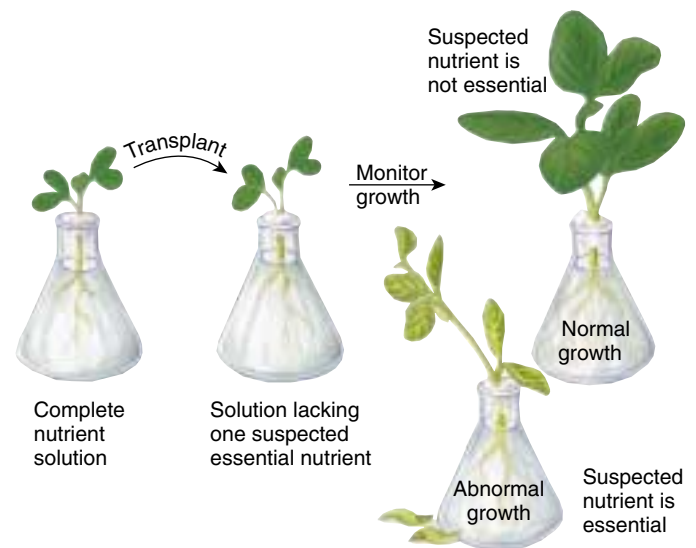
FIGURE 39.2

Mineral deficiencies in plants. (a) Leaves of a healthy Marglobe tomato (*Lycopersicon esculentum*) plant. (b) Chlorine-deficient plant with necrotic leaves (leaves with patches of dead tissue). (c) Copper-deficient plant with blue-green, curled leaves. (d) Zinc-deficient plant with small, necrotic leaves. (f) Manganese-deficient plant with chlorosis (yellowing) between the veins. The agricultural implications of deficiencies such as these are obvious; a trained observer can determine the nutrient deficiencies that are affecting a plant simply by inspecting it.



allowed to grow and are studied for the presence of abnormal symptoms that might indicate a need for the missing element (figure 39.3). However, the water or vessels used often contain enough micronutrients to allow the plants to grow normally, even though these substances were not added deliberately to the solutions. To give an idea of how small the quantities of micronutrients may be, the standard dose of molybdenum added to seriously deficient soils in Australia amounts to about 34 grams (about one handful) per hectare, once every 10 years! Most plants grow satisfactorily in hydroponic culture, and the method, although expensive, is occasionally practical for commercial purposes. Analytical chemistry has made it much easier to take plant material and test for levels of different molecules. One application has been the investigation of elevated levels of CO₂ (a result of global warming) on plant growth. With increasing levels of CO₂, the leaves of some plants increase in size, but the amount of nitrogen decreases relative to carbon. This decreases the nutritional value of the leaves to herbivores.

The plant macronutrients carbon, oxygen, and hydrogen constitute about 94% of a plant's dry weight; the other macronutrients—nitrogen, potassium, calcium, phosphorus, magnesium, and sulfur—each approach or exceed 1% of a plant's dry weight.

**FIGURE 39.3**

Identifying nutritional requirements of plants. A seedling is first grown in a complete nutrient solution. The seedling is then transplanted to solution that lacks one suspected essential nutrient. The growth of the seedling is then studied for the presence of abnormal symptoms, such as discolored leaves and stunted growth. If the seedling's growth is normal, the nutrient that was left out may not be essential; if the seedling's growth is abnormal, the lacking nutrient is essential for growth.

Soil

Plant growth is affected by soil composition. Soil is the highly weathered outer layer of the earth's crust. It is composed of a mixture of ingredients, which may include sand, rocks of various sizes, clay, silt, humus, and various other forms of mineral and organic matter; pore spaces containing water and air occur between the particles. The mineral fraction of soils varies according to the composition of the rocks. The crust includes about 92 naturally occurring elements; table 2.1 in chapter 2 lists the most common of these elements and their percentage of the earth's crust by weight. Most elements are combined as inorganic compounds called **minerals**; most rocks consist of several different minerals. The soil is also full of microorganisms that break down and recycle organic debris. About 5 metric tons of carbon is tied up in the organisms that are present in the soil under a hectare (0.06 mile²) of wheat land in England, an amount that approximately equals the weight of 100 sheep!

Most roots are found in **topsoil** (figure 39.4), which is a mixture of mineral particles of varying size (most less than 2 mm thick), living organisms, and **humus**. Humus consists of partly decayed organic material. When topsoil is lost because of erosion or poor landscaping, both the water-holding capacity and the nutrient relationships of the soil are adversely affected.

About half of the total soil volume is occupied by spaces or pores, which may be filled with air or water, depending on moisture conditions. Some of the soil water, because of its properties described below, is unavailable to plants. Due to gravity, some of the water that reaches a given soil will drain through it immediately. Another fraction of the water is held in small soil pores, which are generally less than about 50 micrometers in diameter. This water is readily available to plants. When it is depleted through evaporation or root uptake, the plant will wilt and eventually die unless more water is added to the soil.

Cultivation

In natural communities, nutrients are recycled and made available to organisms on a continuous basis. When these communities are replaced by cultivated crops, the situation changes drastically: the soil is much more exposed to erosion and the loss of nutrients. For this reason, cultivated crops and garden plants usually must be supplied with additional mineral nutrients.

One solution to this is **crop rotation**. For example, a farmer might grow corn in a field one year and soybeans the next year. Both crops remove nutrients from the soil, but the plants have different nutritional requirements, and therefore the soil does not lose the same nutrients two years in a row. Soybean plants even add nitrogen compounds to the soil, released by nitrogen-fixing bacteria growing in nodules on their roots. Sometimes farmers allow a field to lie fallow—that is,

FIGURE 39.4

Most roots occur in topsoil. The uppermost layer in soil is called topsoil, and it contains organic matter, such as roots, small animals, and humus, and mineral particles of various sizes. Subsoil lies underneath the topsoil and contains larger mineral particles and relatively little organic matter. Beneath the subsoil are layers of bedrock, the raw material from which soil is formed over time and weathering.



they do not grow a crop in the field for a year or two. This allows natural processes to rebuild the field's store of nutrients.

Other farming practices that help maintain soil fertility involve plowing under plant material left in fields. You can do the same thing in a lawn or garden by leaving grass clippings and dead leaves. Decomposers in the soil do the rest, turning the plant material into humus.

Fertilizers are also used to replace nutrients lost in cultivated fields. The most important mineral nutrients that need to be added to soils are nitrogen (N), phosphorus (P), and potassium (K). All of these elements are needed in large quantities (see table 39.1) and are the most likely to become deficient in the soil. Both chemical and organic fertilizers are often added in large quantities and can be significant sources of pollution in certain situations (see chapter 30). Organic fertilizers were widely used long before chemical fertilizers were available. Substances such as manure or the remains of dead animals have traditionally been applied to crops, and plants are often plowed under to increase the soil's fertility. There is no basis for believing that organic fertilizers supply any element to plants that inorganic fertilizers cannot provide and they can. However, organic fertilizers build up the humus content of the soil, which often enhances its water- and nutrient-retaining properties. For this reason, nutrient availability to plants at different times of the year may be improved, under certain circumstances, with organic fertilizers.

Soils contain organic matter and various minerals and nutrients. Farming practices like crop rotation, plowing crops under, and fertilization are often necessary to maintain soil fertility.

39.2 Some plants have novel strategies for obtaining nutrients.

Nutritional Adaptations

Carnivorous Plants

Some plants are able to obtain nitrogen directly from other organisms, just as animals do. These carnivorous plants often grow in acidic soils, such as bogs that lack organic nitrogen. By capturing and digesting small animals directly, such plants obtain adequate nitrogen supplies and thus are able to grow in these seemingly unfavorable environments. Carnivorous plants have modified leaves adapted to lure and trap insects and other small animals (figure 39.5). The plants digest their prey with enzymes secreted from various types of glands.

The Venus flytrap (*Dionaea muscipula*), which grows in the bogs of coastal North and South Carolina, has three sensitive hairs on each side of each leaf, which, when touched, trigger the two halves of the leaf to snap together (see figure 39.1). Once the Venus flytrap enfolds a prey item within a leaf, enzymes secreted from the leaf surfaces digest the prey. These flytraps actually shut and open by a growth mechanism. They have a limited number of times they can open and close as a result. In the sundews, the glandular trichomes secrete both sticky mucilage, which traps small animals, and digestive enzymes. Unlike Venus flytraps they do not close rapidly and it is possible that the two share a common ancestor.

Pitcher plants attract insects by the bright, flowerlike colors within their pitcher-shaped leaves and perhaps also by sugar-rich secretions. Once inside the pitchers, insects slide down into the cavity of the leaf, which is filled with water and digestive enzymes.

Bladderworts, *Utricularia*, are aquatic. They sweep small animals into their bladderlike leaves by the rapid action of a springlike trapdoor, and then they digest these animals.



FIGURE 39.5

A carnivorous plant. A tropical Asian pitcher plant, *Nepenthes*. Insects enter the pitchers and are trapped and digested. Complex communities of invertebrate animals and protists inhabit the pitchers.

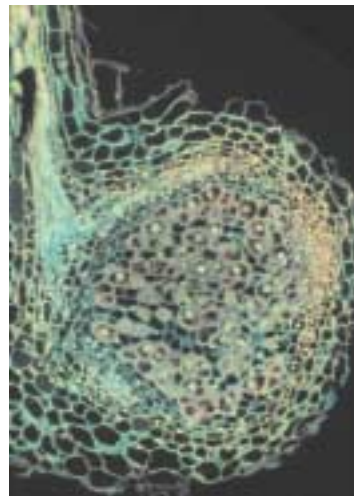


FIGURE 39.6

Nitrogen-fixing nodule. A root hair of alfalfa is invaded by *Rhizobium*, a bacterium (yellow structures) that fixes nitrogen. Through a series of exchanges of chemical signals, the plant cells divide to create a nodule for the bacteria which differentiate and begin producing ammonia.

Nitrogen-Fixing Bacteria

Plants need ammonia (NH_3) to build amino acids, but most of the nitrogen is in the atmosphere in the form of N_2 . Plants lack the biochemical pathways (including the enzyme nitrogenase) necessary to convert gaseous nitrogen to ammonia, but some bacteria have this capacity. Some of these bacteria live in close association with the roots of plants. Others go through an intricate dance and end up being housed in plant tissues created especially for them called nodules (figure 39.6). Only legumes are capable of forming root nodules and there is a very specific recognition required by a bacteria species and its host. Hosting these bacteria costs the plant in terms of energy, but is well worth it when there is little ammonia in the soil. An energy conservation mechanism has evolved in the legumes so that the root hairs will not respond to bacterial signals when nitrogen levels are high.

Mycorrhizae

While symbiotic relationships with nitrogen-fixing bacteria are rare, symbiotic associations with mycorrhizal fungi are found in about 90% of the vascular plants. These fungi have been described in detail in chapter 36. In terms of plant nutrition, it is important to recognize the significant role these organisms play in enhancing phosphorus transfer to the plant. The uptake of some of the micronutrients is also enhanced. Functionally, the mycorrhizae extend the surface area of nutrient uptake substantially

Carnivorous plants obtain nutrients, especially nitrogen, directly by capturing and digesting insects and other organisms. Nitrogen can also be obtained from bacteria living in close association with the roots. Fungi help plants obtain phosphorus and other nutrients from the soil.

39.3 Water and minerals move upward through the xylem.

Overview of Water and Mineral Movement through Plants

Local Changes Result in the Long-Distance, Upward Movement of Water

Most of the nutrients and water discussed above enter the plant through the roots and move upward in the xylem. It is not unusual for a large tree to have leaves more than 10 stories off the ground (figure 39.7). Did you ever wonder how water gets from the roots to the top of a tree that high? Water moves through the spaces between the protoplasts of cells, through plasmodesmata (membrane connections between cells), through cell membranes and through the continuous tubing system in the xylem. We know that there are interconnected, water-conducting xylem elements extending throughout a plant. We also know that water first enters the roots and then moves to the xylem. After that, however, water rises through the xylem because of a combination of factors and some exits through the stomata in the leaves.

While most of our focus will be on the mechanics of water transport through xylem, the movement of water at the cellular level plays a significant role in bulk water transport in the plant as well, although over much shorter distances. You know that the Casparian strip in the root forces water to move through cells. In the case of parenchyma cells it turns out that most water also moves across membranes rather than in the intercellular spaces. For a long time, it was believed that water moved across cell membranes only by osmosis through the lipid bilayer. We now know that osmosis is enhanced by water channels called **aquaporins**. These transport channels are found in both plants and animals. In plants they exist in vacuole and plasma membranes. There are at least 30 different genes coding for aquaporin-like proteins in *Arabidopsis*. Some aquaporins only appear or open during drought stress. Aquaporins allow for faster water movement between cells than osmosis. They are important not only in maintaining water balance within a cell, but in getting water between many plant cells and the xylem. The greatest distances traveled by water molecules and dissolved minerals are in the xylem.

Once water enters the xylem, it can move upward 100 m in the redwoods. Some “pushing” from the pressure of water entering the roots is involved. However, most of the force is “pulling” caused by water evaporating (**transpiration**) through the stomata on the leaves and other plant surfaces. This works because water molecules stick to themselves with hydrogen bonds (cohesion) and to the walls of the tracheid or xylem vessel (adhesion). The result is an unusually stable column of liquid reaching great heights.



FIGURE 39.7

How does water get to the top of this tree? We would expect gravity to make such a tall column of water too heavy to be maintained by capillary action. What pulls the water up?

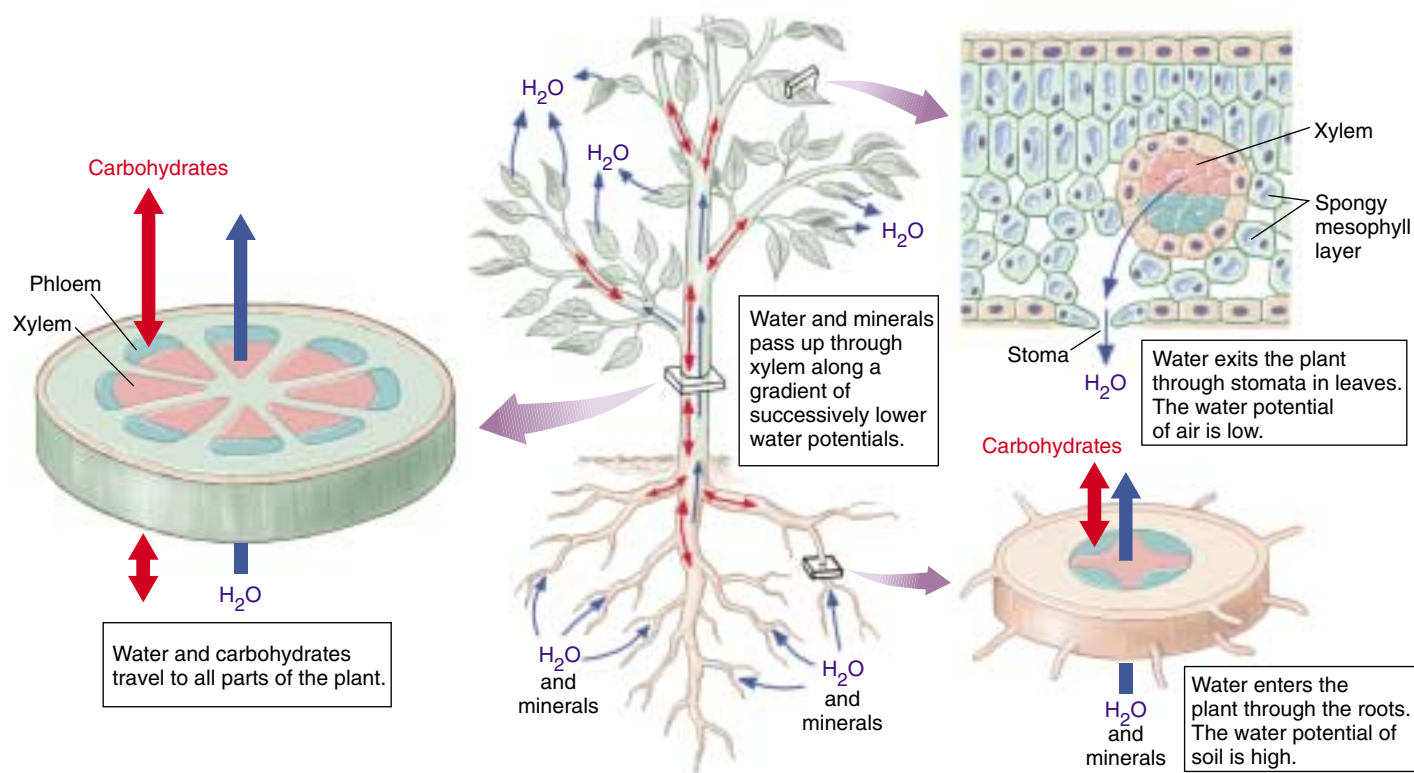


FIGURE 39.8

Water movement through a plant. This diagram illustrates the path of water and inorganic materials as they move into, through, and out of the plant body.

Water Potential

Plant biologists often discuss the forces that act on water within a plant in terms of **potentials**. The *turgor pressure*, which is a physical pressure that results as water enters the cell vacuoles, is referred to as **pressure potential**. Water coming through a garden hose is an example of physical pressure. There is also a potential caused by an uneven distribution of a solute on either side of a membrane, which will result in osmosis (movement of water to the side with the greater concentration of solute). By applying pressure (on the side that has the greater concentration of solute), it is possible to prevent osmosis from taking place. The smallest amount of pressure needed to stop osmosis is referred to as the **solute** (or **osmotic**) **potential** of the solution. Water will enter a cell osmotically until it is stopped by the pressure potential caused by the cell wall. The **water potential** of a plant cell is, in essence, the combination of its pressure potential and solute potential; it represents the total potential energy of the water in a plant. If two adjacent cells have different water potentials, water will move from the cell with the higher water potential to the cell with the lower water potential. Water in a plant moves along a

gradient between the relatively high water potential in the soil to successively lower water potentials in the roots, stems, leaves, and atmosphere.

Water potential in a plant regulates movement of water. At the roots there is a positive water potential (except in the case of severe drought). On the surface of leaves and other organs, water loss called **transpiration** creates a negative pressure. It depends on its osmotic absorption by the roots and the negative pressures created by water loss from the leaves and other plant surfaces (figure 39.8). The negative pressure generated by transpiration is largely responsible for the upward movement of water in xylem.

Aquaporins enhance water transport at the cellular level, which ultimately affects bulk water transport. The loss of water from the leaf surface, called transpiration, literally pulls water up the stem from the roots which have the greater water potential. This works because of the strong cohesive forces between molecules of water that allow them to stay “stuck” together in a liquid column and adhesion to walls of tracheids and vessels.

Water and Mineral Absorption

Most of the water absorbed by the plant comes in through root hairs, which collectively have an enormous surface area (figure 39.8). Root hairs are almost always turgid because their solute potential is greater than that of the surrounding soil due to mineral ions being actively pumped into the cells. Because the mineral ion concentration in the soil water is usually much lower than it is in the plant, an expenditure of energy (supplied by ATP) is required for the accumulation of such ions in root cells. The plasma membranes of root hair cells contain a variety of protein transport channels, through which *proton pumps* (see page 120) transport specific ions against even large concentration gradients. Once in the roots, the ions, which are plant nutrients, are transported via the xylem throughout the plant.

The ions may follow the cell walls and the spaces between them or more often go directly through the plasma membranes and the protoplasm of adjacent cells (figure 39.9). When mineral ions pass between the cell walls, they do so nonselectively. Eventually, on their journey inward, they reach the endodermis and any further passage through the cell walls is blocked by the Casparian

strips. Water and ions must pass through the plasma membranes and protoplasts of the endodermal cells to reach the xylem. However, transport through the cells of the endodermis is selective. The endodermis, with its unique structure, along with the cortex and epidermis, controls which ions reach the xylem.

Transpiration from the leaves (figure 39.10), which creates a pull on the water columns, indirectly plays a role in helping water, with its dissolved ions, enter the root cells. However, at night, when the relative humidity may approach 100%, there may be no transpiration. Under these circumstances, the negative pressure component of water potential becomes small or nonexistent.

Active transport of ions into the roots still continues to take place under these circumstances. This results in an increasingly high ion concentration with the cells, which causes more water to enter the root hair cells by osmosis. In terms of water potential, we say that active transport increases the solute potential of the roots. The result is movement of water into the plant and up the xylem columns despite the absence of transpiration. This phenomenon is called **root pressure**, which in reality is an osmotic phenomenon.

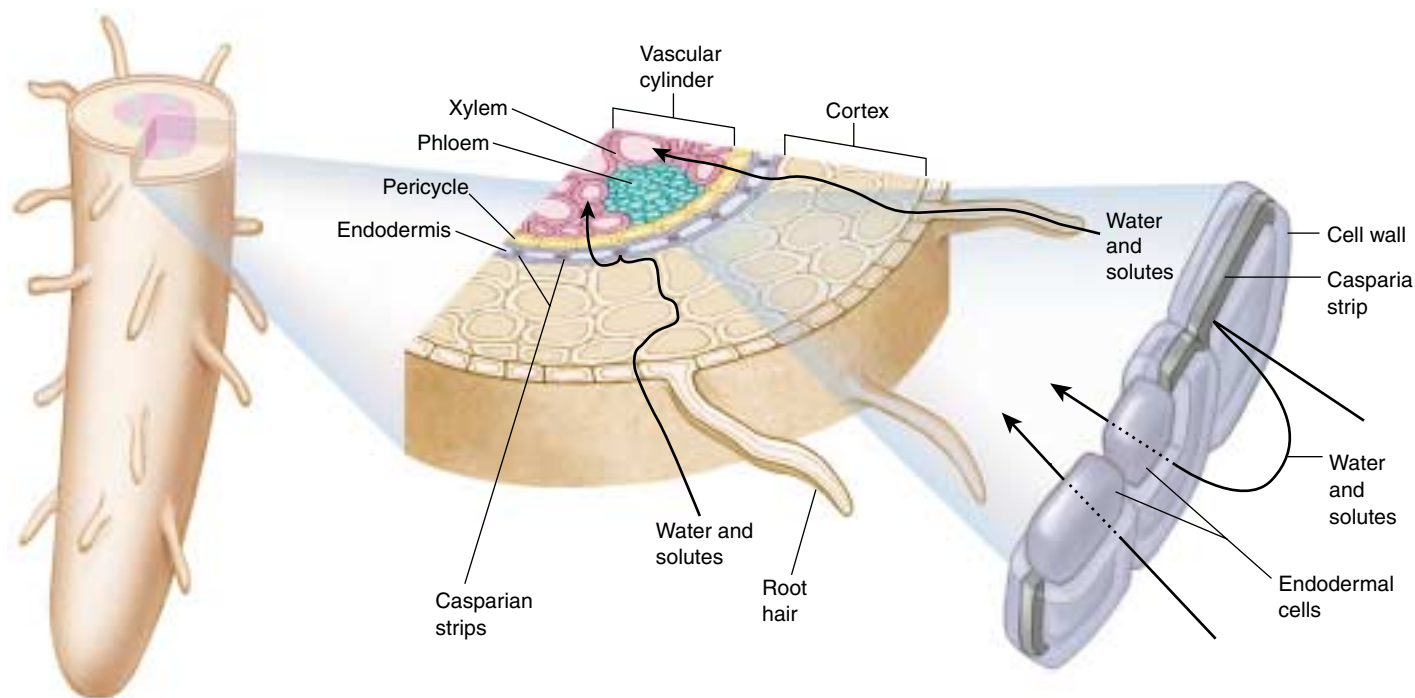


FIGURE 39.9

The pathways of mineral transport in roots. Minerals are absorbed at the surface of the root, mainly by the root hairs. In passing through the cortex, they must either follow the cell walls and the spaces between them or go directly through the plasma membranes and the protoplasts of the cells, passing from one cell to the next by way of the plasmodesmata. When they reach the endodermis, however, their further passage through the cell walls is blocked by the Casparian strips, and they must pass through the membrane and protoplast of an endodermal cell before they can reach the xylem.

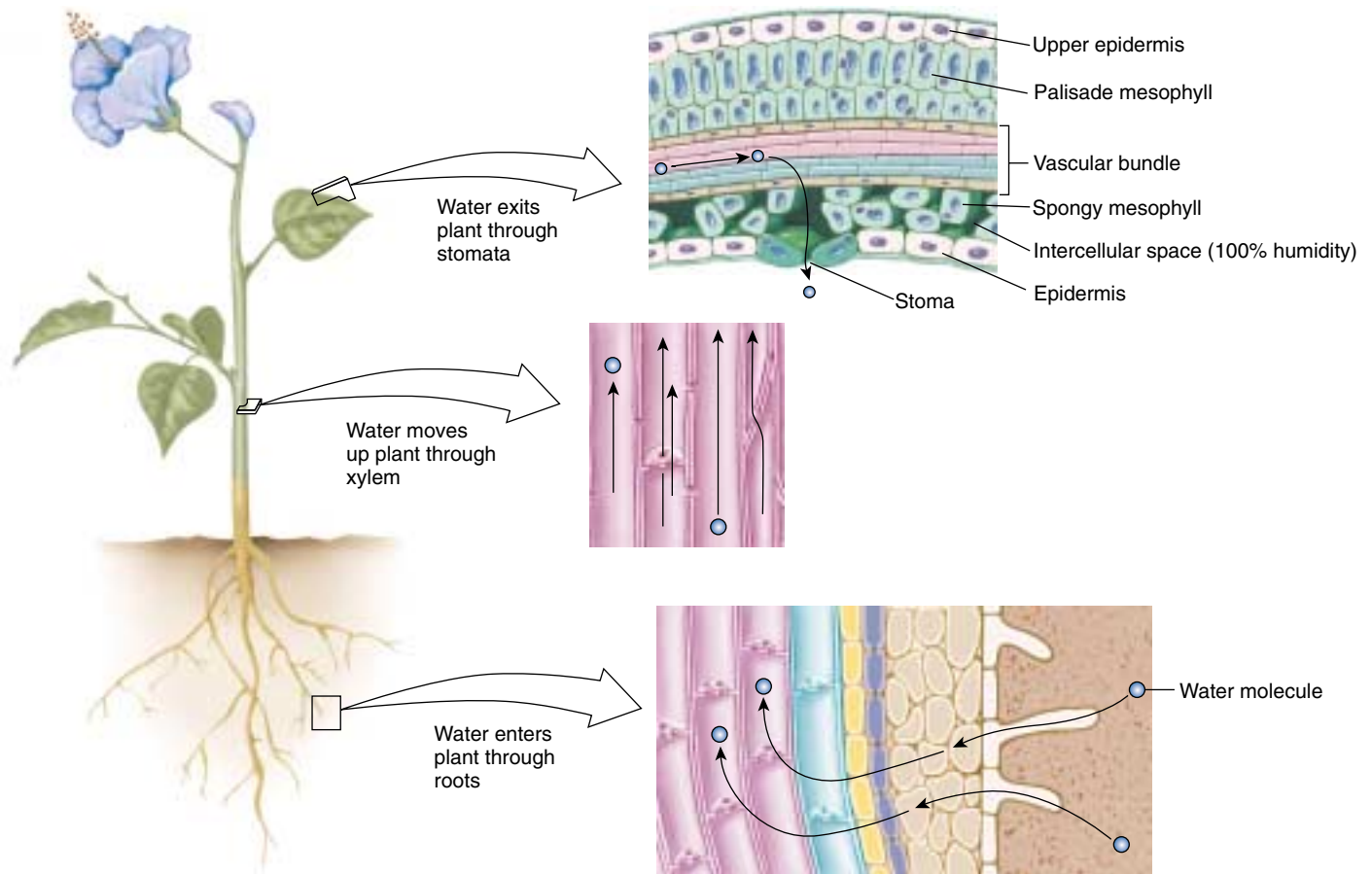


FIGURE 39.10

Transpiration. Water evaporating from the leaves through the stomata causes the movement of water upward in the xylem and the entrance of water through the roots.

Under certain circumstances, root pressure is so strong that water will ooze out of a cut plant stem for hours or even days. When root pressure is very high, it can force water up to the leaves, where it may be lost in a liquid form through a process known as **guttation** (figure 39.11). Guttation does not take place through the stomata, but instead occurs through special groups of cells located near the ends of small veins that function only in this process. Root pressure is never sufficient to push water up great distances.

Water enters the plant by osmosis. Transport of minerals (ions) across the endodermis is selective. Root pressure, which often occurs at night, is caused by the continued, active accumulation of ions in the roots at times when transpiration from the leaves is very low or absent.



FIGURE 39.11

Guttation. In herbaceous plants, water passes through specialized groups of cells at the edges of the leaves; it is visible here as small droplets around the edge of the leaf in this strawberry plant (*Fragaria ananassa*).

Water and Mineral Movement

Water and Mineral Movement through the Xylem

It is clear that root pressure is insufficient to push water to the top of a tall tree, although it can help. So, what does work? Otto Renner proposed the solution in Germany in 1911. Passage of air across leaf surfaces results in loss of water by evaporation, creating a pull at the open upper end of the “tube.” Evaporation from the leaves produces a tension on the entire water column that extends all the way down to the roots. Water has an inherent tensile strength that arises from the cohesion of its molecules, their tendency to form hydrogen bonds with one another. The tensile strength of a column of water varies inversely with the diameter of the column; that is, the smaller the diameter of the column, the greater the tensile strength. Because plants have transporting vessels of very narrow diameter, the cohesive forces in them are strong. The water molecules also adhere to the sides of the tracheid or xylem vessels, further stabilizing the long column of water.

The water column would fail if air bubbles were inserted (visualize a tower of blocks and then pull one out in the middle). Anatomical adaptations decrease the probability of this. Individual tracheids and vessel members are connected by one of more *pits* (cavities) in their walls. Air bubbles are generally larger than the openings, so they cannot pass through them. Furthermore, the cohesive force of water is so great that the bubbles are forced into rigid spheres that have no plasticity and therefore cannot squeeze through the openings. Deformed cells or freezing can cause small bubbles of air to form within xylem cells. Any bubbles that do form are limited to the xylem elements where they originate, and water may continue to rise in parallel columns. This is more likely to occur with seasonal temperature changes. As a result, most of the active xylem in woody plants occurs peripherally, toward the vascular cambium.

Most minerals the plant needs enter the root through active transport. Ultimately, they are removed from the roots and relocated through the xylem to other metabolically active parts of the plant. Phosphorus, potassium, nitrogen, and sometimes iron may be abundant in the xylem during certain seasons. In many plants, such a pattern of ionic concentration helps to conserve these essential nutrients, which may move from mature deciduous parts such as leaves and twigs to areas of active growth. Keep in mind that minerals that are relocated via the xylem must move with the generally upward flow through the xylem. Not all minerals can re-enter the xylem conduit. Calcium, an essential nutrient, cannot be transported elsewhere once it has been deposited in plant parts.

Transpiration of Water from Leaves

More than 90% of the water taken in by the roots of a plant is ultimately lost to the atmosphere through transpiration from the leaves. Water moves into the pockets of air in the leaf from the moist surfaces of the walls of the mesophyll cells. As you saw in chapter 38, these intercellular spaces are in contact with the air outside of the leaf by way of the stomata. Water that evaporates from the surfaces of the mesophyll cells leaves the stomata as vapor. This water is continuously replenished from the tips of the veinlets in the leaves.

Water is essential for plant metabolism, but is continuously being lost to the atmosphere through the stomata. Photosynthesis requires a supply of CO₂ entering the stomata from the atmosphere. This results in two somewhat conflicting requirements: the need to minimize the loss of water to the atmosphere and the need to admit carbon dioxide. Structural features such as stomata and the cuticle have evolved in response to one or both of these requirements.

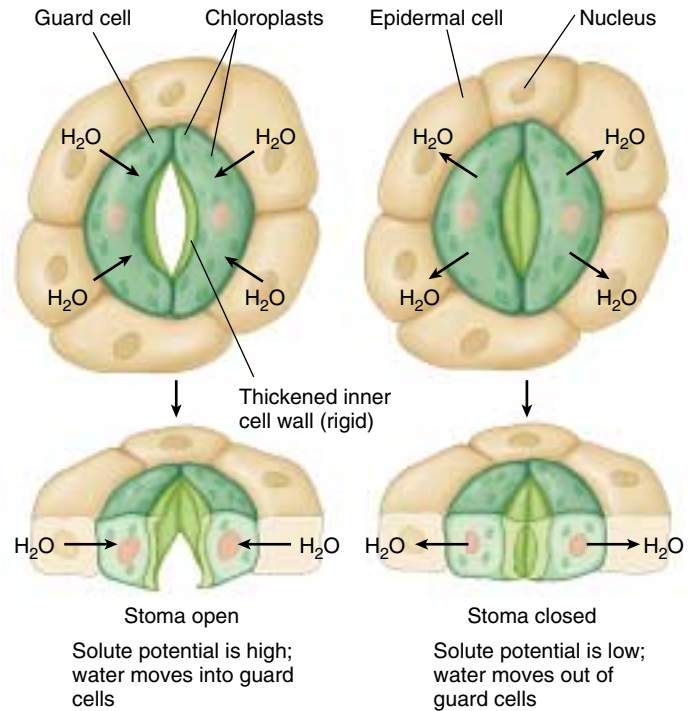
The rate of transpiration depends on weather conditions like humidity and the time of day. After the sun sets, transpiration from the leaves decreases. The sun is the ultimate source of potential energy for water movement. The water potential that is responsible for water movement is largely the product of negative pressure generated by transpiration, which is driven by the warming effects of sunlight.

The Regulation of Transpiration Rate. On a short-term basis, closing the stomata can control water loss. This occurs in many plants when they are subjected to water stress. However, the stomata must be open at least part of the time so that CO₂ can enter. As CO₂ enters the intercellular spaces, it dissolves in water before entering the plant's cells. The gas dissolves mainly in water on the walls of the intercellular spaces below the stomata. The continuous stream of water that reaches the leaves from the roots keeps these walls moist. A plant must respond both to the need to conserve water and to the need to admit CO₂.

Stomata open and close because of changes in the turgor pressure of their guard cells. The sausage- or dumbbell-shaped guard cells stand out from other epidermal cells not only because of their shape, but also because they are the only epidermal cells containing chloroplasts. Their distinctive wall construction, which is thicker on the inside and thinner elsewhere, results in a bulging out and bowing when they become turgid. You can make a model of this for yourself by taking two elongated balloons, tying the closed ends together, and inflating both balloons slightly. When you hold the two open ends together, there should be very little space between the two balloons. Now place



(a)



(b)

FIGURE 39.12

How a stoma opens and closes. (a) When potassium ions from surrounding cells are pumped into guard cells, the guard cell turgor pressure increases as water enters by osmosis. The increased turgor pressure causes the guard cells to bulge, with the thick walls on the inner side of each guard cell bowing outward, thereby opening the stoma. (b) When the potassium ions leave the guard cells and their solute potential becomes low, they lose water and turgor, and the stoma closes.

duct tape on the inside edge of both balloons and inflate each one a bit more. Hold the open ends together. You should now be holding a doughnut-shaped pair of “guard cells” with a “stoma” in the middle. Real guard cells rely on the influx and efflux of water, rather than air, to open and shut.

Loss of turgor in guard cells causes the uptake of potassium (K^+) ions through ATP-powered ion transport channels in their plasma membranes. This creates a solute potential within the guard cells that causes water to enter osmotically. As a result, these cells accumulate water and become turgid, opening the stomata (figure 39.12a). Keeping the stomata open requires a constant expenditure of ATP, and the guard cells remain turgid only as long as ions are pumped into the cells. When stomata close, sucrose, rather than K^+ , leaves the cell through sucrose transporters. Water then leaves the guard cells, which lose turgor, and the stomata close (figure 39.12b). Why closing depends on sucrose transport out of the cell and opening on K^+ uptake is an open question. Experimental evidence is consistent with several pathways regulating stomatal opening and closing.

Photosynthesis in the guard cells apparently provides an immediate source of ATP, which drives the active transport

of K^+ by way of a specific K^+ channel; this K^+ channel has now been isolated and studied. In some species, Cl^- accompanies the K^+ in and out of the guard cells, thus maintaining electrical neutrality. In most species, both Cl^- and malate $^{2-}$ move in the opposite direction of K^+ .

When a whole plant wilts because there is insufficient water available, the guard cells may also lose turgor, and as a result, the stomata may close. The guard cells of many plant species regularly become turgid in the morning, when photosynthesis occurs, and lose turgor in the evening, regardless of the availability of water. When they are turgid, the stomata open, and CO_2 enters freely; when they are flaccid, CO_2 is largely excluded, but water loss is also retarded.

Absciscic acid, a plant hormone discussed in chapter 41, plays a primary role in allowing K^+ to pass rapidly out of guard cells, causing the stomata to close in response to drought. This hormone is released from chloroplasts and produced in leaves. It binds to specific receptor sites in the plasma membranes of guard cells. Plants likely control the duration of stomatal opening through the integration of several stimuli, including blue light. In the next chapter, we will explore the interactions between the environment and the plant in more detail.

Other Factors Regulating Transpiration. Factors such as CO₂ concentration, light, and temperature can also affect stomatal opening. When CO₂ concentrations are high, guard cells of many plant species lose turgor, and their stomata close. Additional CO₂ is not needed at such times, and water is conserved when the guard cells are closed. The stomata also close when the temperature exceeds 30° to 34°C when transpiration would increase substantially. In the dark, stomata will open at low concentrations of CO₂. In chapter 10, we mentioned CAM photosynthesis, which occurs in some succulent like cacti. In this process, CO₂ is taken in at night and fixed during the day. CAM photosynthesis conserves water in dry environments where succulent plants grow.

Many mechanisms to regulate the rate of water loss have evolved in plants. One involves dormancy during dry times of the year; another involves loss of leaves. Deciduous plants are common in areas that periodically experience a severe drought. Plants are often deciduous in regions with severe winters, when water is locked up in ice and snow and thus unavailable to them. In a general sense, annual plants conserve water when conditions are unfavorable, simply by going into dormancy as seeds.

Thick, hard leaves, often with relatively few stomata—and frequently with stomata only on the lower side of the leaf—lose water far more slowly than large, pliable leaves with abundant stomata. Temperatures are significantly reduced in leaves covered with masses of woolly-looking trichomes. These trichomes also increase humidity at the leaf surface. Plants in arid or semiarid habitats often have their stomata in crypts or pits in the leaf surface. Within these depressions the water vapor content of the air may be high, reducing the rate of water loss.

Plant Responses to Flooding

Plants can also receive too much water, and ultimately “drown.” Flooding rapidly depletes available oxygen in the soil and interferes with the transport of minerals and carbohydrates in the roots. Abnormal growth often results. Hormone levels change in flooded plants—ethylene (the only hormone that is a gas) increases, while gibberellins and cytokinins usually decrease. Hormonal changes contribute to the abnormal growth patterns.

Oxygen-deprivation is among the most significant problems. Standing water has much less oxygen than moving water. Generally, standing water flooding is more harmful to a plant (riptides excluded). Flooding that occurs when a plant is dormant is much less harmful than flooding when it is growing actively.

Physical changes that occur in the roots as a result of oxygen deprivation may halt the flow of water through the plant. Paradoxically, even though the roots of a plant may be standing in water, its leaves may be drying out. One adaptive solution is that stomata of flooded plants often close to maintain leaf turgor.



FIGURE 39.13

Adaptation to flooded conditions. The “knees” of the bald cypress (*Taxodium*) form whenever it grows in wet conditions, increasing its ability to take in oxygen.

Adapting to Life in Fresh Water. Algal ancestors of plants adapted to a freshwater environment from a salt-water environment before the “move” onto land. This involved a major change in controlling salt balance. Since that time, many have “moved” back into fresh water and grow in places that are often or always flooded naturally; they have adapted to these conditions during the course of their evolution (figure 39.13). One of the most frequent adaptations among such plants is the formation of **aerenchyma**, loose parenchymal tissue with large air spaces in it (figure 39.14). Aerenchyma is very prominent in water lilies and many other aquatic plants. Oxygen may be transported from the parts of the plant above water to those below by way of passages in the aerenchyma. This supply of oxygen allows oxidative respiration to take place even in the submerged portions of the plant.

Some plants normally form aerenchyma, whereas others, subject to periodic flooding, can form it when necessary. In corn, ethylene, which becomes abundant under the anaerobic conditions of flooding, induces aerenchyma formation. Plants also respond to flooded conditions by forming larger lenticels (which facilitate gas exchange) and additional adventitious roots.

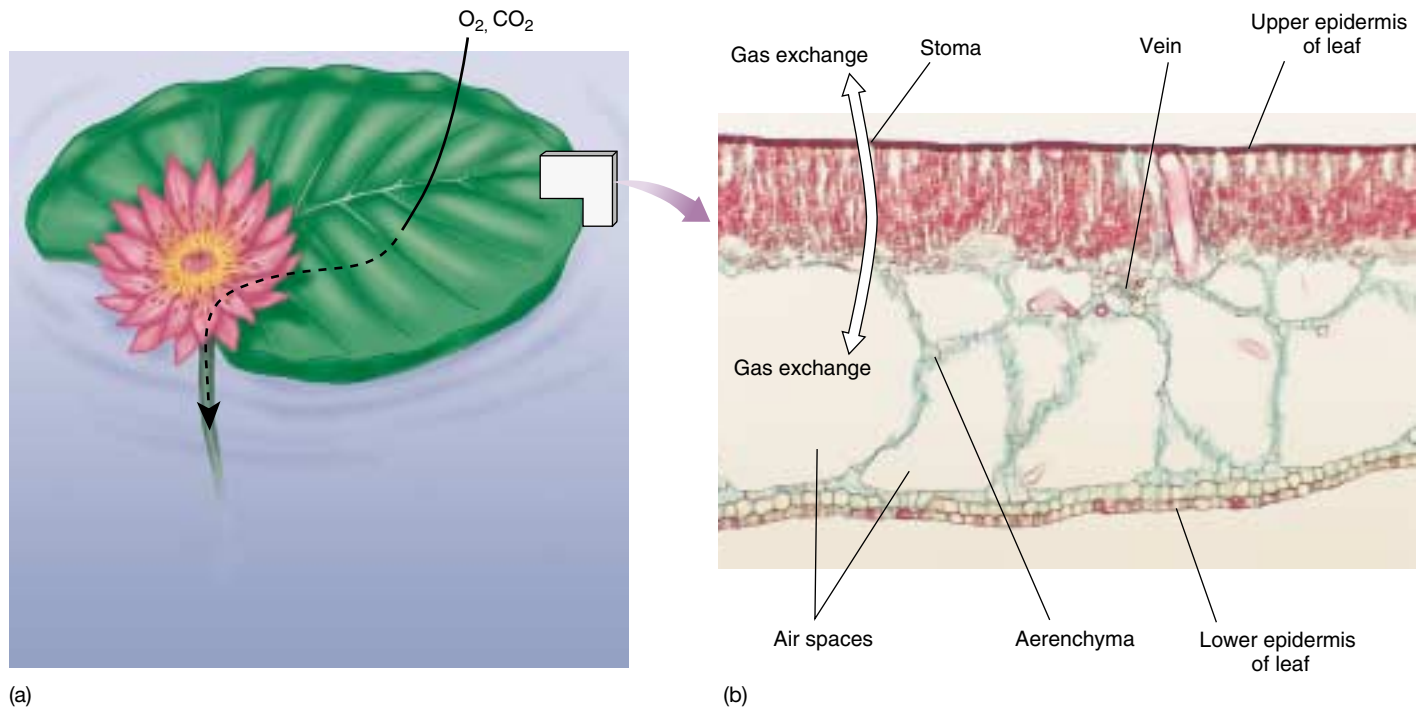


FIGURE 39.14

Aerenchyma tissue. Gas exchange in aquatic plants. (a) Water lilies float on the surface of ponds where oxygen is collected and transported to submerged portions of the plant. (b) Large air spaces in the leaves add buoyancy. The specialized parenchyma tissue that forms these open spaces is called aerenchyma. Gas exchange occurs through stomata found only on the upper surface of the leaf.

Adapting to Life in Salt Water. Plants such as mangroves that are normally flooded with salt water must not only provide a supply of oxygen for their submerged parts, but also control their salt balance. The salt must be excluded, actively secreted, or diluted as it enters. The arching silt roots of mangroves are connected to long, spongy, air-filled roots that emerge above the mud. These roots, called pneumatophores (see chapter 38), have large lenticels on their above-water portions through which oxygen enters; it is then transported to the submerged roots (figure 39.15). In addition, the succulent leaves of mangroves contain large quantities of water, which dilute the salt that reaches them. Many plants that grow in such conditions also secrete large quantities of salt.

Transpiration from leaves pulls water and minerals up the xylem. This works because of the physical properties of water and the narrow diameters of the conducting tubes. Stomata open when their guard cells become turgid. Opening and closing of stomata is osmotically regulated. Biochemical, anatomical, and morphological adaptations have evolved to reduce water loss through transpiration. Plants are harmed by excess water. However, plants can survive flooded conditions, and even thrive in them, if they can deliver oxygen to their submerged parts.

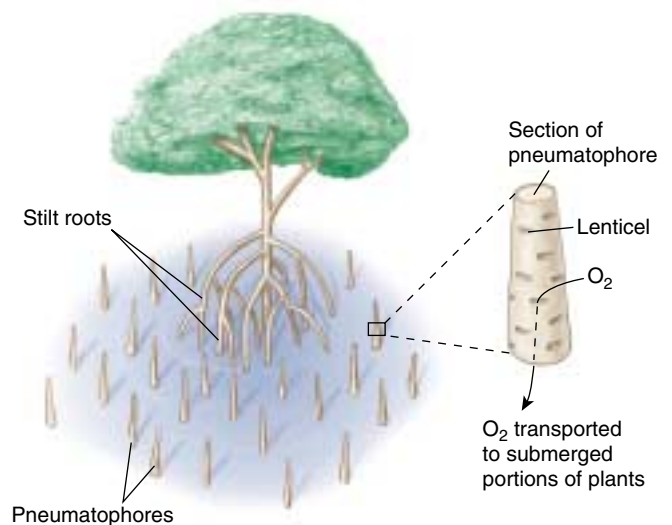


FIGURE 39.15

How mangroves get oxygen to their submerged part.

Mangrove plants grow in areas that are commonly flooded, and much of each plant is usually submerged. However, modified roots called pneumatophores supply the submerged portions of the plant with oxygen because these roots emerge above the water and have large lenticels. Oxygen can enter the roots through the lenticels, pass into the abundant aerenchyma, and move to the rest of the plant.

39.4 Dissolved sugars and hormones are transported in the phloem.

Phloem Transport Is Bidirectional

Most carbohydrates manufactured in leaves and other green parts are distributed through the phloem to the rest of the plant. This process, known as **translocation**, is responsible for the availability of suitable carbohydrate building blocks in roots and other actively growing regions of the plant. Carbohydrates concentrated in storage organs such as tubers, often in the form of starch, are also converted into transportable molecules, such as sucrose, and moved through the phloem. The pathway that sugars and other substances travel within the plant has been demonstrated precisely by using radioactive tracers, despite the fact that living phloem is delicate and the process of transport within it is easily disturbed. Radioactive carbon dioxide ($^{14}\text{CO}_2$) gets incorporated into glucose as a result of photosynthesis. The glucose is used to make sucrose, which is transported in the phloem. Such studies have shown that sucrose moves both up and down in the phloem.

Aphids, a group of insects that extract plant sap for food, have been valuable tools in understanding translocation. Aphids thrust their *stylets* (piercing mouthparts) into phloem cells of leaves and stems to obtain abundant sugars there. When a feeding aphid is removed by cutting its stylet, the liquid from the phloem continues to flow through the detached mouthpart and is thus available in pure form for analysis (figure 39.16). The liquid in the phloem contains 10 to 25% dry matter, almost all of which is sucrose. Using aphids to obtain the critical samples and radioactive tracers to mark them, it has been demonstrated that movement of substances in phloem can be remarkably fast; rates of 50 to 100 centimeters per hour have been measured.

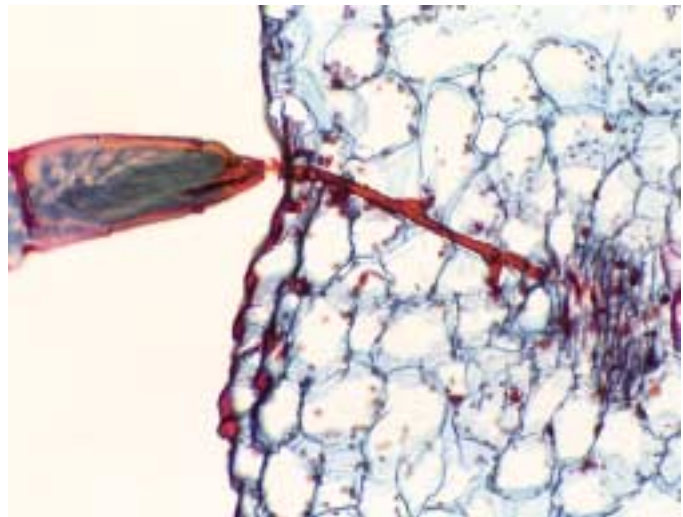
While the primary focus of this chapter is on nutrient and water transport, it is important to note that phloem also transports plant hormones. As we will explore in the next chapter, environmental signals can result in the rapid translocation of hormones in the plant.

Energy Requirements for Phloem Transport

The most widely accepted model of how carbohydrates in solution move through the phloem has been called the **mass-flow hypothesis**, **pressure flow hypothesis**, or **bulk flow hypothesis**. Experimental evidence supports much of this model. Dissolved carbohydrates flow from a **source** and are released at a **sink** where they are utilized. Carbohydrate sources include photosynthetic tissues, such as the mesophyll of leaves, and food-storage tissues, such as the cortex of roots. Sinks occur primarily at the growing tips of roots and stems and in developing fruits.



(a)



(a)

FIGURE 39.16

Feeding on phloem. (a) Aphids, like this individual of *Macrosiphon rosae* shown here on the edge of a rose leaf, feed on the food-rich contents of the phloem, which they extract through their piercing mouthparts (b), called stylets. When an aphid is separated from its stylet and the cut stylet is left in the plant, the phloem fluid oozes out of it and can then be collected and analyzed.

In a process known as *phloem loading*, carbohydrates (mostly sucrose) enter the sieve tubes in the smallest veinlets at the source. This is an energy-requiring step, as active transport is needed. Companion cells and parenchyma cells adjacent to the sieve tubes provide the ATP energy to drive this transport. Then, because of the

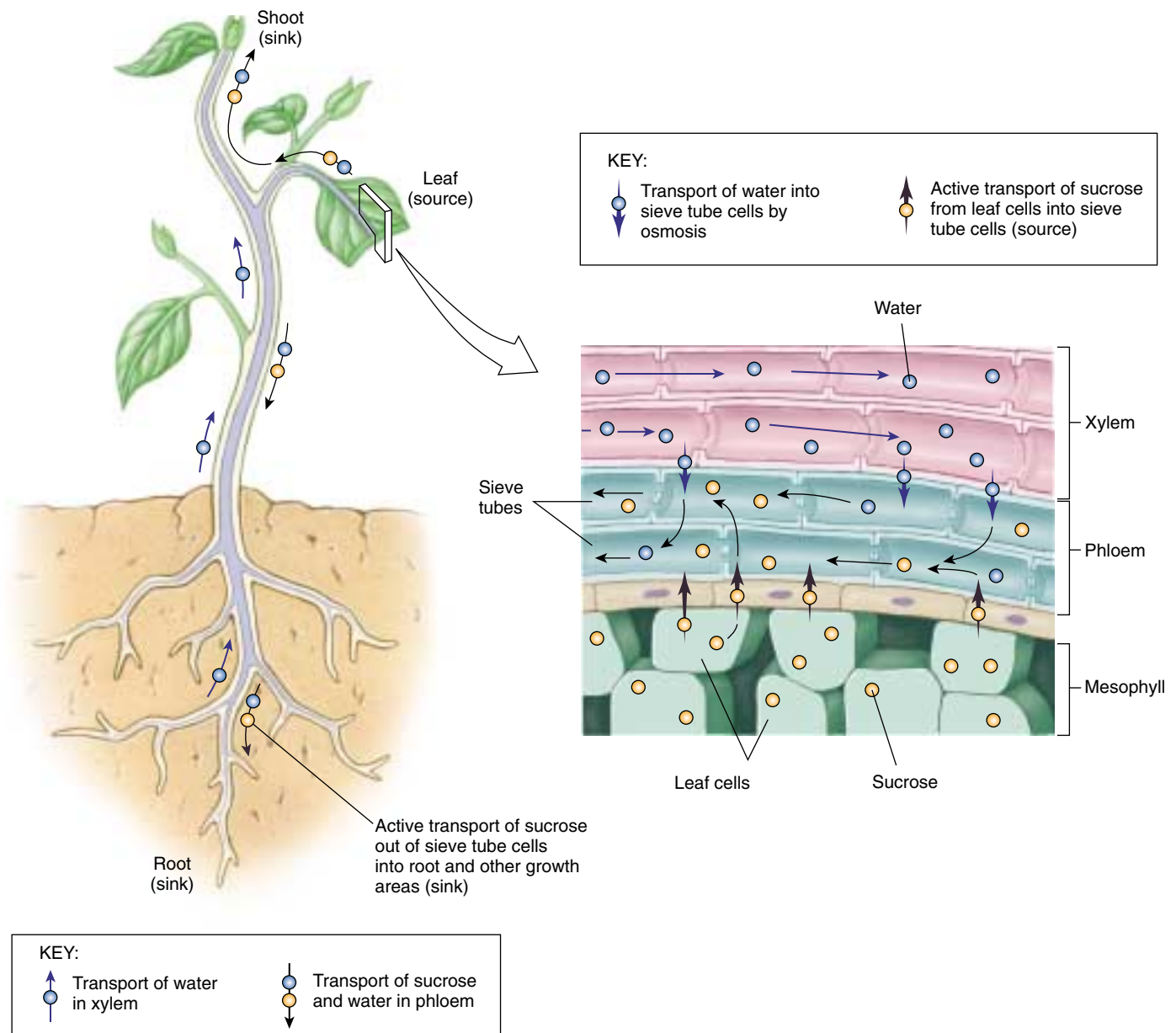


FIGURE 39.17

Diagram of mass flow. In this diagram, red dots represent sucrose molecules, and blue dots symbolize water molecules. Moving from the mesophyll cells of a leaf or another part of the plant into the conducting cells of the phloem, the sucrose molecules are then transported to other parts of the plant by mass flow and unloaded where they are required.

difference between the water potentials in the sieve tubes and in the nearby xylem cells, water flows into the sieve tubes by osmosis. Turgor pressure in the sieve tubes increases. The increased turgor pressure drives the fluid throughout the plant's system of sieve tubes. At the sink, carbohydrates are actively removed. Water moves from the sieve tubes by osmosis and the turgor pressure there drops, causing a mass flow from the higher pressure at the source to the lower pressure sink (figure 39.17). Most

of the water at the sink diffuses then back into the xylem, where it may either be recirculated or lost through transpiration.

Transport of sucrose and other carbohydrates through sieve tubes does not require energy. The loading and unloading of these substances from the sieve tubes does.



Summary

Questions

Media Resources

39.1 Plants require a variety of nutrients in addition to the direct products of photosynthesis.

- Plants require a few macronutrients in large amounts and several micronutrients in trace amounts. Most of these are obtained from the soil through the roots.
- Plant growth is significantly influenced by the nature of the soil. Soils vary in terms of nutrient composition and water-holding capacity.

1. What is the difference between a macronutrient and a micronutrient? Explain how a plant would utilize each of the macronutrients.



- Nutrients
- Soil

39.2 Some plants have novel strategies for obtaining nutrients.

- Some plants entice bacteria to produce organic nitrogen for them. These bacteria may be free-living or form a symbiotic relationship with a host plant.
- About 90% of all vascular plants rely on fungal associations to gather essential nutrients.

2. The atmosphere is full of nitrogen yet it is inaccessible to most plants. Why is that? What solution has evolved in legumes?

39.3 Water and minerals move upward through the xylem.

- Water and minerals enter the plant through the roots. Energy is required for active transport.
- The bulk movement of water and minerals is the result of movement between cells, across cell membranes, and through tubes of xylem. Aquaporins are water channels that enhance osmosis.
- A combination of the properties of water, structure of xylem, and transpiration of water through the leaves results in the passive movement of water to incredible heights. The ultimate energy source for pulling water through xylem vessels and tracheids is the sun.
- Water leaves the plant through openings in the leaves called stomata. Stomata open when their guard cells are turgid and bulge, causing the thickened inner walls of these cells to bow away from the opening.
- Plants can tolerate long submersion in water, if they can deliver oxygen to their submerged tissues.

3. What is pressure potential? How does it differ from solute potential? How do these pressures cause water to rise in a plant?

4. What proportion of water that enters a plant leaves it via transpiration?

5. Why are root hairs usually turgid? Does the accumulation of minerals within a plant root require the expenditure of energy? Why or why not?

6. Under what environmental condition is water transport through the xylem reduced to near zero? How much transpiration occurs under these circumstances?

7. Does stomatal control require energy? Explain.



- Activity: Water Movement
- Uptake by Roots
- Water Movement



- Student Research: Heavy Metal Uptake

39.4 Dissolved sugars and hormones are transported in the phloem.

- Sucrose and hormones can move up and down in the phloem between sources and sinks.
- The movement of water containing dissolved sucrose and other substances in the phloem requires energy. Sucrose is loaded into the phloem near sites of synthesis, or sources, using energy supplied by the companion cells or other nearby parenchyma cells.

8. What is translocation? What is the driving force behind translocation?

9. Describe the movement of carbohydrates through a plant, beginning with the source and ending with the sink. Is this process active or passive?

Plant Growth and Reproduction

The Control of Patterning in Plant Root Development

Did you ever think of how a root grows? Down in the dark, with gravity its only cue, the very tip of the root elongates, periodically forming a node from which root branches will extend. How does the root determine the position of its branches, and the spacing between them? The serial organization of the root's branches is controlled by events that happen on a microscopic scale out at the very tip of the root, the so-called root apex. There, within a space of a millimeter or less, molecular events occur that orchestrate how the root will grow and what it will be like.

The problem of understanding how a plant's root apex controls the way a root develops is one example of a much larger issue, perhaps the most challenging research problem in modern botany: What mechanism mediates central pattern formation in the plant kingdom? Almost nothing was known of these mechanisms a decade ago, but intensive research is now rapidly painting in the blank canvas.

Much of the most exciting research on plant pattern formation is being performed on a small weedy relative of the mustard plant, the wall cress *Arabidopsis thaliana* (see photo above). With individual plants no taller than your thumb that grow quickly in laboratory test-tubes, *Arabidopsis* is an ideal model for studying plant development. Its genome, about the size of the fruit fly *Drosophila*, has been completely sequenced, greatly aiding research into the molecular events underlying pattern formation.

To gain some insight into the sort of research being done, we will focus on work being done by John Schiefelbein and colleagues at the University of Michigan. Schiefelbein has focused on one sharply defined aspect of plant root pattern formation in *Arabidopsis*, the formation of root hairs on the epidermis, the root's outer layer of cells. These root hairs constitute the principal absorbing surface of the root, and their position is under tight central control.

In a nutshell, the problem of properly positioning root hairs is one of balancing cell production and cell differentiation. Cells in the growth zone beneath the surface of the root—a sheath called a meristem—are constantly dividing. The cells that are produced by the meristem go on to differ-



Arabidopsis thaliana. An important plant for studying root development because it offers a simple pattern of cellular organization in the root.

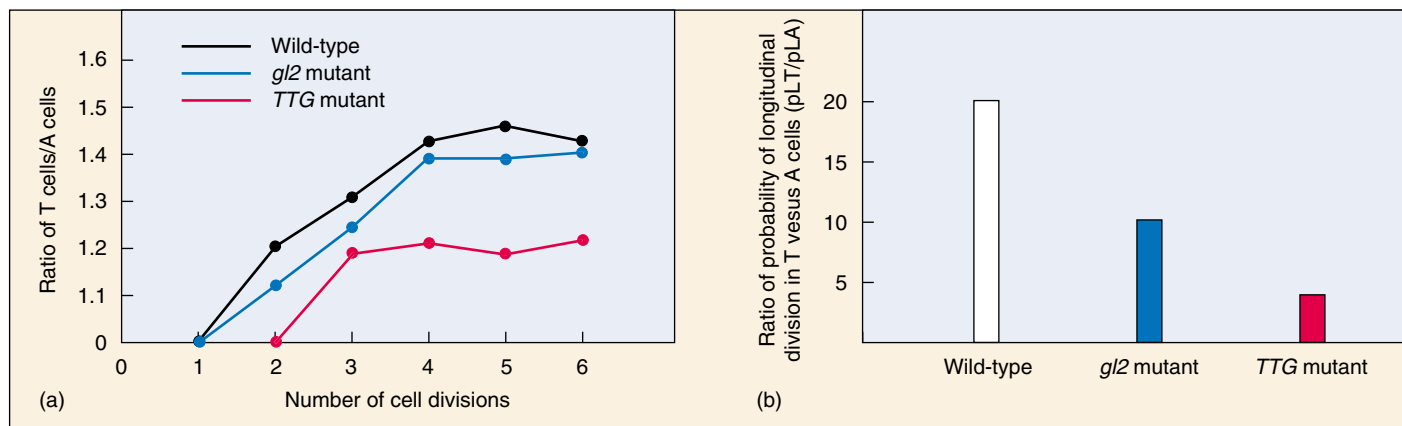
entiate into two kinds of cells: trichoblasts which form hair-bearing epidermal cells, and atrichoblasts which form hairless epidermal cells. The positioning of trichoblasts among atrichoblasts determines the pattern of root hairs on the developing root.

When researchers looked very carefully at the dividing root meristem, they found that the initial cells determined to be trichoblasts and atrichoblasts alternate with one another in a ring of 16 cells around the circumference of the root. As the cells divide, more and more cells are added, forming columns of cells extending out in 16 files. As the files extend farther and farther out, occasional side-ways divisions fill in the gaps that develop, forming new files.

Maintaining this simple architecture requires that the root maintain a tight control of the plane and rate of cell division. Because this rate is different for the two cell types, the root must also control the rate at which the cell types differentiate. Schiefelbein set out to learn how the root apex coordinates these two processes.

To get a handle on the process, Schiefelbein seized on a recently characterized root pattern mutant called *transparent testa glabra* (*TTG*). This mutant changes the pattern of root hairs in *Arabidopsis*, and it has been proposed that it controls whether a cell becomes a trichoblast or an atrichoblast. But does it control the rate and orientation in the root meristem epidermis?

To answer this question, Schiefelbein's team used clonal analysis to microscopically identify individual cell types in the root epidermis, and set out to see if they indeed divide at different rates, and if the *TTG* mutation affects these rates differently. If so, there must be a link between cell differentiation and the control of cell division in plants.



Comparing the differentiation and cell division of trichoblast (T) cells versus atrichoblast (A) cells in root epidermis. (a) As cell divisions proceeded, T cells and A cells were identified in the root epidermis of wild-type plants and two mutants, *gl2* and *TTG*. Comparing the ratio of T cells to A cells, there is an increase in the number of A cells compared to T cells in the *TTG* mutant. (b) The rate of cell division was also examined by comparing the ratio of probabilities of longitudinal anticlinal cell division in T cells and A cells among the wild-type and mutant plants. This ratio was lowest in *TTG* mutants, indicating that this mutation affects cell division.

The Experiment

Two developmental mutants of *A. thaliana* were compared to investigate whether the control of cell differentiation and the rate of cell division were linked. One, *TTG*, alters early events in root epidermal cell differentiation, while the other, *glabra2* (*gl2*) acts later.

The investigators first set out to map the surface of the roots of each mutant type, as well as those of nonmutant wild type. To avoid confusion in studying files of cells, it is necessary to clearly identify the starting point of each file of cells. To do this, roots were selected that contained clones of trichoblast and atrichoblast produced by longitudinal cell divisions perpendicular to the surface of the root. Called longitudinal anticlinal cell divisions, these clones are rare but easily recognized when stained with propidium iodide. Careful mapping of individual cells with a confocal microscope allowed investigators to determine the number and location of trichoblast and atrichoblast cells present in the epidermal tissue of each clone.

The Results

The researchers made two important observations based on their visual identification of individual trichoblast and atrichoblast cells in the various plant types examined.

1. *The two cell types are produced at different rates.* Among plants that had been cultured for up to six cell divisions, they observed a significant difference in the ratio of trichoblast (T) versus atrichoblast (A) cells following two or more cell divisions. In their study you can readily see that the *TTG* mutant produces a significantly lower ratio of T cells to A cells compared to the wild-type plants or *gl2* mutants (see graph *a* above). This strongly suggests that *TTG* is involved in controlling the rate of cell division in the T cell file.

2. *TTG controls the rate of longitudinal cell division.* The re-

search team went on to examine longitudinal cell divisions that fill in the gaps as cell division causes files of cells to extend outward from the meristem. The researchers set out to determine the probability of such longitudinal anticlinal cell division occurring in the three types of plants shown in graph *a*. The more rapidly cell files are produced, the more often longitudinal divisions would be required to fill in gaps between files. For proper root hair position to be maintained, the rate of this longitudinal division would have to be tightly coordinated with the rate of vertical division within the file.

The investigators found that longitudinal cell division, always rare, was usually seen, when it did occur, in T cell files. Did the *TTG* mutation affect this process as well as file-extending cell divisions? This was determined by examining the ratio of the probability of longitudinal anticlinal divisions in T cells versus A cells (pLT/pLA).

Researchers compared the ratio in wild-type plants with that in the two mutants, *TTG* and *gl2*. Did the *TTG* mutation alter longitudinal division? Yes! Their results indicate at least a 60% reduction in the pLT/pLA ratio of the *TTG* mutant compared to wild type and *gl2* plants (see graph *b* above). The percent of clones in the A file of the *TTG* mutants exhibiting this type of cell division was twice that seen in the wild-type or *gl2* mutants.

This observation directly supports the hypothesis that the *TTG* gene is not only required for cell division in the T cell file, but also controls longitudinal cell divisions which are characteristically more frequent in trichoblasts.

The research team concluded from these studies that *TTG* is probably the earliest point of control of root epidermis cell fate specification, and that this control most likely acts by negatively controlling trichoblast cell fate.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab11.mhtml

40

Early Plant Development

Concept Outline

40.1 Plant embryo development establishes a basic body plan.

Establishing the Root-Shoot Axis. Asymmetric cell division starts patterning the embryo. Early in embryogenesis the root-shoot axis is established.

Establishing Three Tissue Systems. Three tissue systems are established without any cell movement. While the embryo is still a round ball, the root-shoot axis is established. The shape of the plant is determined by planes of cell division and direction of cell elongation. Nutrients are used during embryogenesis, but proteins, lipids, and carbohydrates are also set aside to support the plant during germination before it becomes photosynthetic.

40.2 Seed formation protects the dormant embryo from water loss.

How Seeds Form. Seeds allow plants to survive unfavorable conditions and invade new habitats.

40.3 Fruit formation enhances the dispersal of seeds.

How Fruits Form. Seed-containing fruits are carried far by animals, wind and water, allowing angiosperms to colonize large areas.

40.4 Germination initiates post-seed development.

Mechanisms of Germination. External signals including water, light, abrasion, and temperature can trigger germination. Rupturing the seed coat and adequate oxygen are essential. Stored reserves in the endosperm or cotyledon are made available to the embryo during germination.



FIGURE 40.1

This plant has recently emerged from its seed. It is extending its shoot and leaves up into the air, toward light.

In chapter 37 we emphasized evolutionary changes in reproduction and physiology that gave rise to the highly successful flowering plants (angiosperms). Chapters 38 and 39 explored the morphological and anatomical development of the angiosperm sporophyte, where most of these innovations occurred. In the next few chapters, we continue our focus on the sporophyte generation of the angiosperms. In many cases, we will use the model plant *Arabidopsis*, a weedy member of the mustard family. Its very small genome has allowed plant biologists to study how genes regulate plant growth and development. In this chapter, we will follow the development of the embryo through seed germination (figure 40.1). The next few chapters will also continue to emphasize the roles of gene expression, hormones, and environmental signals in regulating plant development and function.

40.1 Plant embryo development establishes a basic body plan.

Establishing the Root-Shoot Axis

In plants, three-dimensional shape and form arises by regulating the amount and pattern of cell division. Even the very first cell division is asymmetric resulting in two different cell types. Early in embryo development most cells can give rise to a wide range of cell and organ types, including leaves. As development proceeds, the cells with multiple potentials are restricted to regions called meristems. Many meristems are established by the time embryogenesis ends and the seed becomes dormant. Apical meristems will continue adding cells to the growing root and shoot tips after germination. These generate the large numbers of cells needed to form leaves, flowers, and all other components of the mature plant. Apical meristem cells of corn, for example, divide every 12 hours, producing half a million cells a day in an actively growing corn plant. Lateral meristems can cause an increase in the girth of some plants, while intercalary meristems within the stems allow for elongation.

In addition to developing the root-shoot axis in embryogenesis, cell differentiation occurs and three basic tissue systems are established. These are the dermal, ground, and vascular tissue systems and they are radially patterned. These tissue systems contain various cell types that can be highly differentiated for specific functions.

These tissue systems are organized radially around the root-shoot axis.

While the embryo is developing, two other critical events are occurring. A food supply is established that will support the embryo during germination while it gains photosynthetic capacity. This starts with the second fertilization event that produces endosperm in angiosperms. Secondly, ovule tissue (from the parental sporophyte) differentiates to form a hard, protective covering around the embryo. The seed (ovule containing the embryo) then enters a dormant phase, signaling the end of embryogenesis. Environmental signals (for example, water, temperature, and light) can break dormancy and trigger a cascade of internal events resulting in germination.

Early Cell Division and Patterning

The first division of the fertilized egg in a flowering plant is asymmetric and generates cells with two different fates (figure 40.2). One daughter cell is small, with dense cytoplasm. That cell, which will become the embryo, begins to divide repeatedly in different planes, forming a ball of cells. The other daughter cell divides repeatedly, forming an elongated structure called a *suspensor*, which links the embryo to the nutrient tissue of the seed. The suspensor also

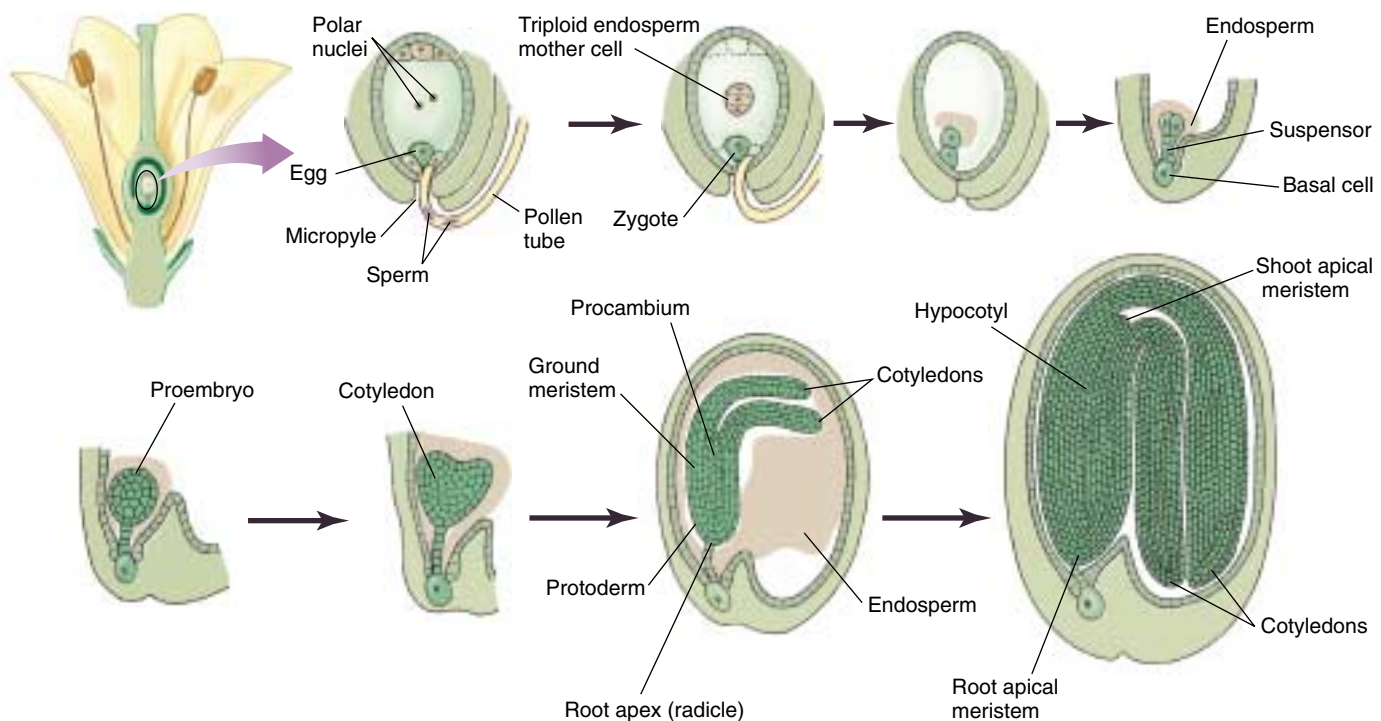
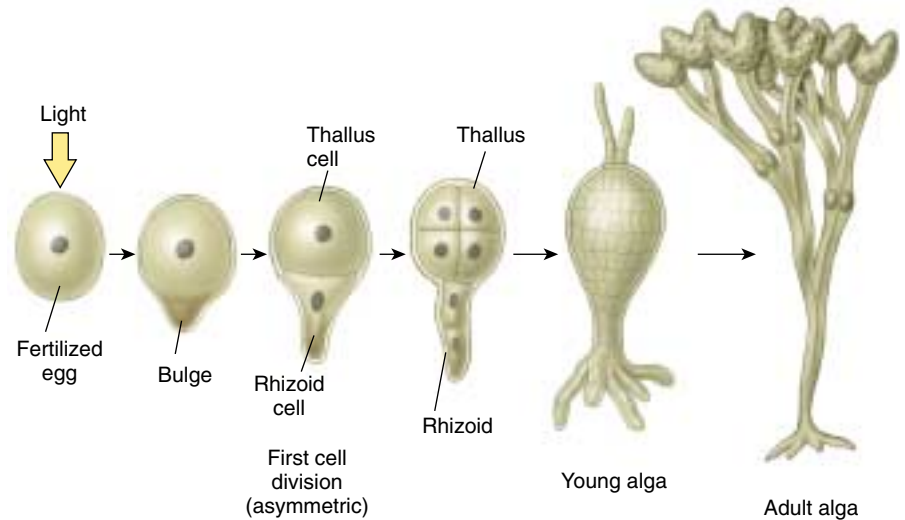


FIGURE 40.2

Stages of development in an angiosperm embryo. The very first cell division is asymmetric. Differentiation begins almost immediately after fertilization.

FIGURE 40.3**Asymmetric cell division in a *Fucus* zygote.**

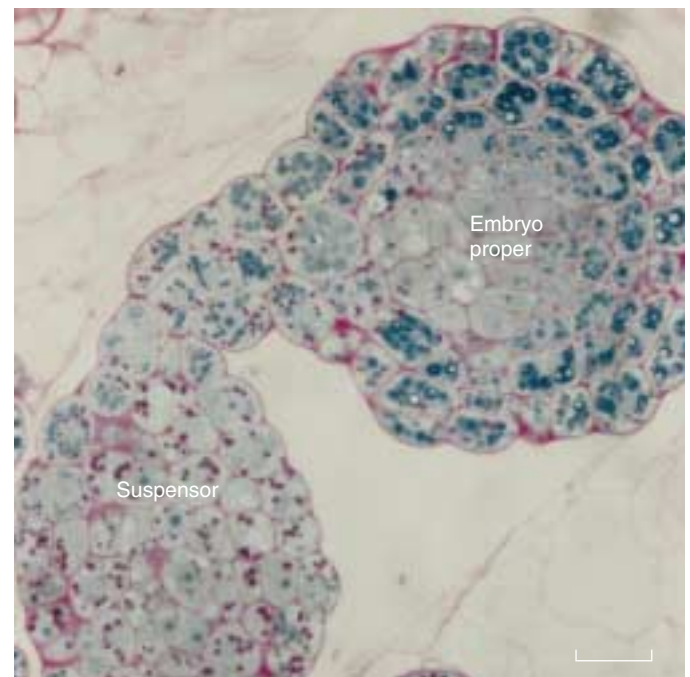
An unequal distribution of material in the fertilized egg leads to a bulge where the first cell division will occur. This division results in a smaller cell that will go on to divide and produce the rhizoid that anchors the plant; the larger cell divides to form the thallus or main plant body. The point of sperm entry determines where the smaller rhizoid cell will form, but light and gravity can modify this to ensure that the rhizoid will point downward where it can anchor this brown alga. Calcium-mediated currents set up an internal gradient of charged molecules which lead to a weakening of the cell wall where the rhizoid will form. The fate of the two resulting cells is held in memory by cell wall components.



provides a route for nutrients to reach the developing embryo. The root-shoot axis also forms at this time. Cells near the suspensor are destined to form a root, while those at the other end of the axis ultimately become a shoot.

Investigating the asymmetry of the first cell division is difficult because the fertilized egg is embedded within the gametophyte, which is surrounded by sporophyte tissue (ovule and carpel tissue). One approach has been to use the brown algae *Fucus* as a model system. Although there is a huge evolutionary difference between brown algae and the angiosperms, there are similarities in early embryogenesis which may have ancient origins. The egg is released prior to fertilization so there are no extra tissues surrounding the zygote (fertilized egg). One side of the zygote begins to bulge establishing the vertical axis. Cell division occurs and the original bulge becomes the smaller of the two cells. It develops into a rhizoid that anchors the alga and the other cell develops into the main body, or thallus, of the sporophyte. This axis is first established by the point of sperm entry, but can be changed by environmental signals, especially light and gravity which ensure that the rhizoid is down and the thallus is up. Internal gradients are established that specify where the rhizoid will form in response to environmental signals (figure 40.3). The ability to “remember” where the rhizoid will form depends on the cell wall. Enzymatic removal of the cell wall in *Fucus* cells specified to form either rhizoids or plant body, resulted in cells that could give rise to both. Cell walls contain a wide variety of carbohydrates and proteins attached to the wall’s structural fibers. Attempting to pin down the identities of these suspected developmental signals is an area of active research.

Another approach to investigating the initial asymmetry in embryos has been to study mutants with abnormal suspensors. By understanding what is going wrong, it is often possible to infer normal developmental mechanisms. For example, the *suspensor* mutant in *Arabidopsis* has aberrant

**FIGURE 40.4**

The embryo suppresses development of the suspensor as a second embryo. This *suspensor* mutant of *Arabidopsis* has a defect appear in embryo development followed by embryo-like development of the suspensor.

development in the embryo followed by embryo-like development of the suspensor (figure 40.4). From this, one can conclude that the embryo normally prevents the suspensor from developing into a second embryo.

Early in embryogenesis the root-shoot axis is established.

Establishing Three Tissue Systems

Three basic tissues differentiate while the plant embryo is still a ball of cells, the globular stage (figure 40.5), but no cell movements are involved. The *protoderm* consists of the outermost cells in a plant embryo and will become *dermal tissue*. These cells almost always divide with their cell plate perpendicular to the surface. This perpetuates a single outer layer of cells. Dermal tissue produces cells that protect the plant from desiccation, including the stomata that open and close to facilitate gas exchange and minimize water loss. The bulk of the embryonic interior consists of *ground tissue* cells that eventually function in food and water storage. Lastly, *procambium* at the core of the embryo is destined to form the future *vascular tissue* responsible for water and nutrient transport.

Root and Shoot Formation

The root-shoot axis is established during the globular stage of development. The shoot apical meristem will later give rise to leaves and eventually reproductive structures. While both the shoot and root meristems are apical meristems, their formation is controlled independently. This conclusion is supported by mutant analysis in *Arabidopsis* where the *shootmeristemless* (*stm*) mutant fails to produce a viable shoot, but does produce a root (figure 40.6). Similarly, root meristem-specific genes have been identified. For example, monopterous mutants of *Arabidopsis* lack roots. The hormone auxin may play a role in root-shoot axis formation. Auxin is one of six classes of hormones that



FIGURE 40.6
Shoot-specific genes specify formation of the shoot apical meristem. The *shootmeristemless* mutant of *Arabidopsis* has a normal root meristem, but fails to produce a shoot meristem.

regulate plant development and function that we will explore in more detail later in this unit.

As you study the development of roots and shoots after germination, you will notice that many of the same patterns of tissue differentiation seen in the embryo are reiterated in the apical meristems. Remember that there are also many events discussed earlier in this chapter that are unique to embryogenesis. For example, the *LEAFY COTYLEDON*

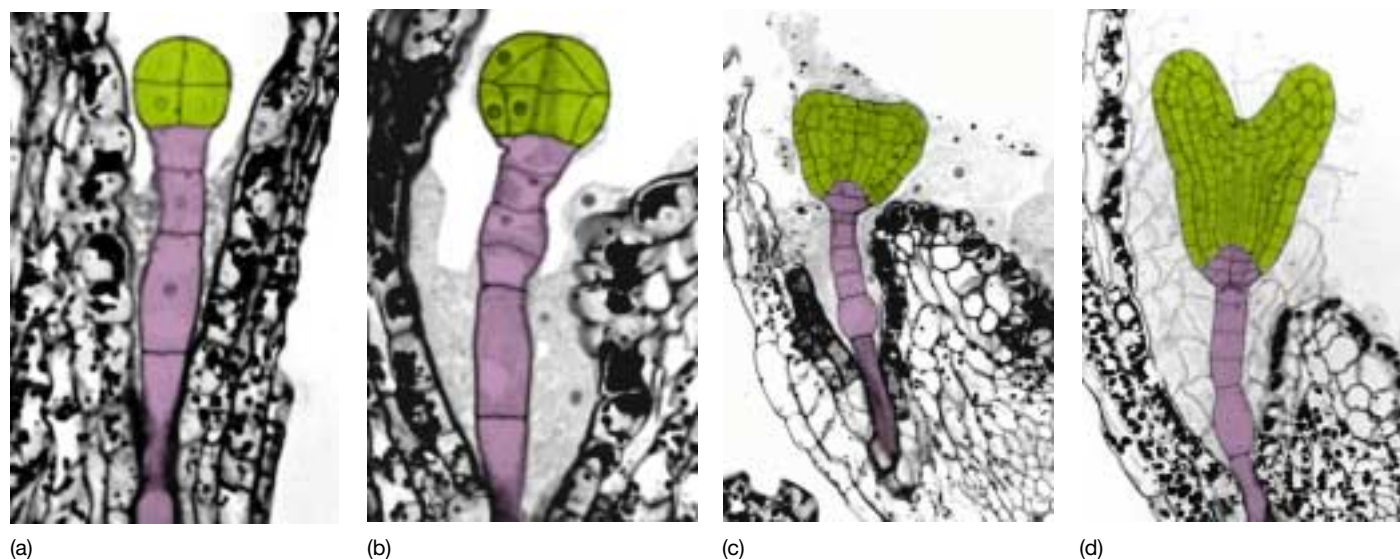


FIGURE 40.5
Early developmental stages of *Arabidopsis thaliana*. (a) Early cell division has produced the embryo and suspensor. (b) Globular stage. (c,d) Heart-shaped stage.

gene in *Arabidopsis* is active in early and late embryo development and may be responsible for maintaining an embryonic environment. It is possible to turn this gene on later in development using recombinant DNA techniques (see chapter 43). In that case, embryos can form on leaves!

Morphogenesis

The globular stage gives rise to a heart-shaped embryo in one group of angiosperms (the dicots, see figure 40.5) and a ball with a bulge on a single side in another group (the monocots). The bulges are cotyledons (“first leaves”) and are produced by the embryonic cells, not the shoot apical meristem that begins forming during the globular stage. This process, called *morphogenesis* (generation of form), results from changes in planes and rates of cell division. Because plant cells cannot move, the form of a plant body is largely determined by the plane in which cells divide and by controlled changes in cell shape as they expand osmotically after they form. Both microtubules and actin play a role in establishing the position of the cell plate which determines the direction of division. Plant growth-regulators and other factors influence the orientation of bundles of microtubules on the interior of the plasma membrane. These microtubules also guide cellulose deposition as the cell wall forms around the outside of a new cell, determining its final shape. For example, if you start with a box and reinforce four of the six sides more heavily with cellulose, the cell will expand and grow in the direction of the two sides with less reinforcement. Much is being learned at the cell biological level about morphogenesis from mutants that divide, but cannot control their plane of cell division or the direction of cell expansion.

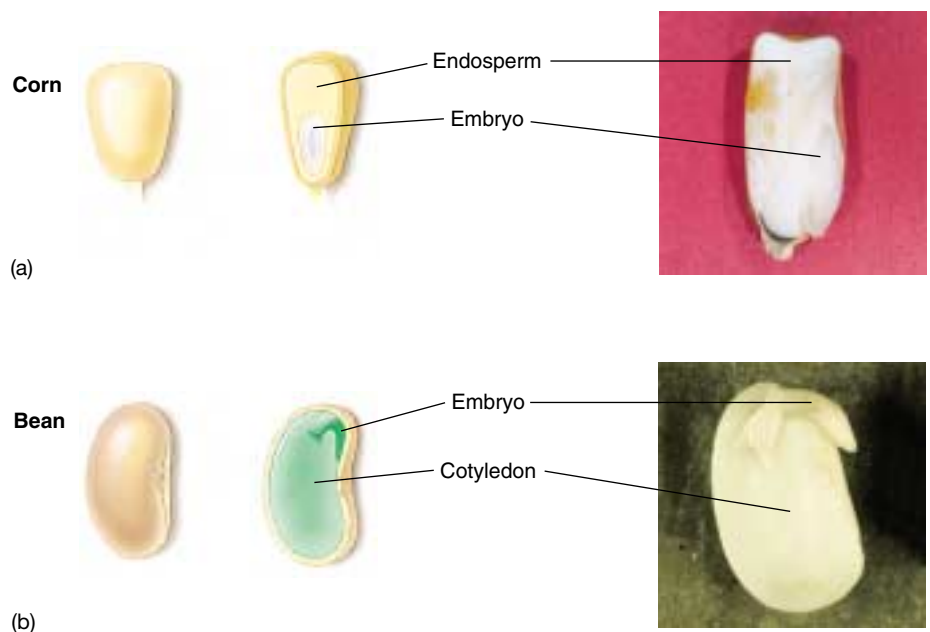
Food Storage

Throughout embryogenesis there is the production of starch, lipids, and proteins. The seed storage proteins are so abundant that the genes coding for them were the first cloning targets for plant molecular biologists. As noted in chapter 37, the evolutionary trend in the plants has been toward increased protection of the embryo. One way this is accomplished is through parental sporophyte input transferred by the suspensor in angiosperms (in gymnosperms the suspensor serves only to push the embryo closer to the gametophytic nutrient source produced by multiple nuclear divisions without cell division). This happens concurrently with the development of the endosperm (present only in angiosperms, although double fertilization has been observed in the gymnosperm *Ephedra*) which may be extensive or minimal. Endosperm in coconut is the “milk” and is in liquid form. In corn the endosperm is solid and in popping corn expands with heat to form the edible part of popcorn. In peas and beans, the endosperm is used up during embryo development and nutrients are stored in thick, fleshy cotyledons (figure 40.7). The photosynthetic machinery is built in response to light. So, it is critical that seeds have stored nutrients to aid in germination until the growing sporophyte can photosynthesize. A seed buried too deeply will use up all its reserves before reaching the surface and sunlight.

After the root-shoot axis is established, a radial, three-tissue system, and a stored food supply, are formed through controlled cell division and expansion.

FIGURE 40.7

Endosperm in corn and bean. The corn kernel has endosperm that is still present at maturity, while the endosperm in the bean has disappeared; the bean embryo's cotyledons take over food storage functions.



40.2 Seed formation protects the dormant embryo from water loss.

How Seeds Form

A protective seed coat forms from the outer layers of ovule cells, and the embryo within is now either surrounded by nutritive tissue or has amassed stored food in its cotyledons. The resulting package, known as a *seed*, is resistant to drought and other unfavorable conditions; in its dormant state, it is a vehicle for dispersing the embryo to distant sites and allows a plant embryo to survive in environments that might kill a mature plant. In some embryos, the cotyledons are bent over to fit within the constraints of the hardening ovule wall with the inward cotyledon being slightly smaller for efficient packing (figure 40.8). Remember that the ovule wall is actually tissue from the previous sporophyte generation.

Adaptive Importance of Seeds

Early in the development of an angiosperm embryo, a profoundly significant event occurs: the embryo stops developing. In many plants, development of the embryo is arrested soon after the meristems and cotyledons differentiate. The *integuments*—the outer cell layers of the ovule—develop into a relatively impermeable seed coat, which encloses the seed with its dormant embryo and stored food. Seeds are important adaptively in at least four ways:

1. They maintain dormancy under unfavorable conditions and postpone development until better conditions arise. If conditions are marginal, a plant can “afford” to have some seeds germinate, because others will remain dormant.
2. The seed affords maximum protection to the young plant at its most vulnerable stage of development.
3. The seed contains stored food that permits development of a young plant prior to the availability of an adequate food supply from photosynthetic activity.
4. Perhaps most important, the dispersal of seeds facilitates the migration and dispersal of plant genotypes into new habitats.

Once a seed coat forms, most of the embryo’s metabolic activities cease. A mature seed contains only about 5% to 20% water. Under these conditions, the seed and the young plant within it are very stable; it is primarily the progressive and severe desiccation of the embryo and the associated reduction in metabolic activity that are responsible for its arrested growth. Germination cannot take place until water and oxygen reach the embryo, a process that sometimes involves cracking the seed coat through abrasion or alternate freezing and thawing. Seeds of some plants have been known to remain viable for hundreds and, in rare instances, thousands of years (figure 40.9).

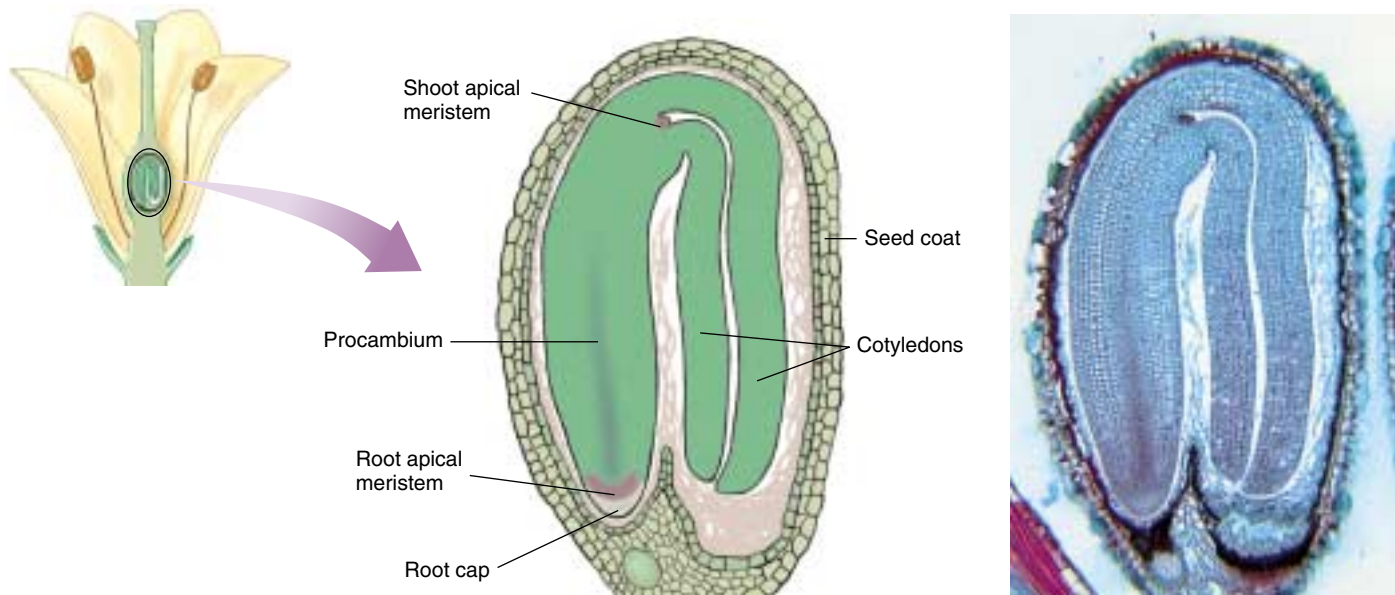


FIGURE 40.8

A mature angiosperm embryo. Note that two cotyledons have grown in a bent shape to accommodate the tight confines of the seed. In some embryos, the shoot apical meristem will have already initiated a few leaf primordia as well.

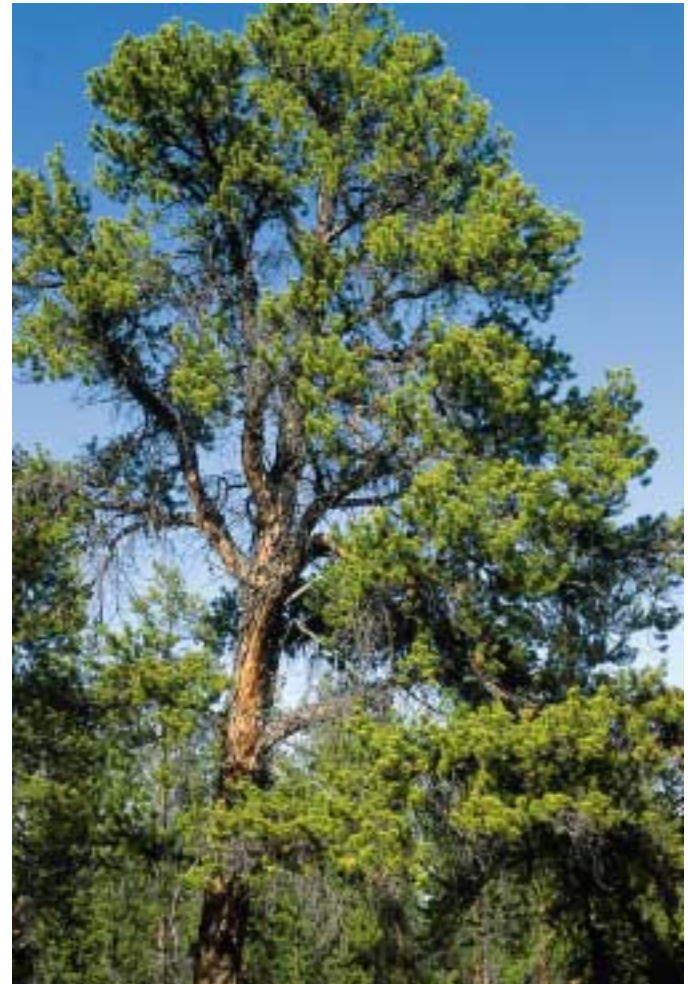


FIGURE 40.9

Seeds can remain dormant for long periods. This seedling was grown from a lotus seed recovered from the mud of a dry lake bed in Manchuria, northern China. The radiocarbon age of this seed indicates that it was formed around A.D. 1515. The coin is included in the photo to give some idea of the size.

Specific adaptations often help ensure that the plant will germinate only under appropriate conditions. Sometimes, seeds lie within tough cones that do not open until they are exposed to the heat of a fire (figure 40.10). This causes the plant to germinate in an open, fire-cleared habitat; nutrients will be relatively abundant, having been released from plants burned in the fire. Seeds of other plants will germinate only when inhibitory chemicals have been leached from their seed coats, thus guaranteeing their germination when sufficient water is available. Still other plants will germinate only after they pass through the intestines of birds or mammals or are regurgitated by them, which both weakens the seed coats and ensures the dispersal of the plants involved. Sometimes seeds of plants thought to be extinct in a particular area may germinate under unique or improved environmental circumstances, and the plants may then reappear.

Seed dormancy is an important evolutionary factor in plants, ensuring their survival in unfavorable conditions and allowing them to germinate when the chances of survival for the young plants are the greatest.



(a)



(b)



(c)

FIGURE 40.10

Fire induces seed germination in some pines. (a) Fire will destroy these adult jack pines, but stimulate growth of the next generation. (b) Cones of a jack pine are tightly sealed and cannot release the seeds protected by the scales. (c) High temperatures lead to the release of the seeds.

40.3 Fruit formation enhances the dispersal of seeds.

How Fruits Form

Paralleling the evolution of angiosperm flowers, and nearly as spectacular, has been the evolution of their **fruits**, which are defined simply as mature ovaries (carpels). During seed formation, the flower ovary begins to develop into fruit. Fruits form in many ways and exhibit a wide array of spe-

cializations in relation to their dispersal. The differences among some of the fruit types seen today are shown in figure 40.11. Three layers of ovary wall can have distinct fates which accounts for the diversity of fruit types from fleshy to dry and hard. An array of mechanisms allow for the release of the seed(s) within the fruits. Developmentally, fruits are fascinating organs that contain three generations



Follicles

Split along one carpel edge only; milkweed, larkspur.



Legumes

Split along two carpel edges with seeds attached to carpel edges; peas, beans.



Samaras

Not split and with a wing formed from the outer tissues; maples, elms, ashes.



Drupes

Single seed enclosed in a hard pit; peaches, plums, cherries.



True berries

More than one seed and a thin skin; blueberries, tomatoes, grapes, peppers.



Hesperidiums

More than one seed and a leathery skin; oranges, lemons, limes.



Aggregate fruits

Derived from many ovaries of a single flower; strawberries, blackberries.



FIGURE 40.11

Examples of some kinds of fruits. Distinguishing features of each of these fruit types are listed below each photo. Follicles, legumes, and samaras are examples of dry fruits. Drupes, true berries, and hesperidiums are simple fleshy fruits; they develop from a flower with a single pistil. Aggregate and multiple fruits are compound fleshy fruits; they develop from flowers with more than one ovary or from more than one flower.

Multiple fruits

Develop from a cluster of flowers; mulberries, pineapples.

in one package. The fruit and seed coat are from the prior sporophyte generation. Within the seed are remnants of the gametophyte generation that produced the egg that was fertilized to give rise to the next sporophyte generation, the embryo.

The Dispersal of Fruits

Aside from the many ways fruits can form, they also exhibit a wide array of specialized dispersal methods. Fruits with fleshy coverings, often shiny black or bright blue or red, normally are dispersed by birds or other vertebrates (figure 40.12*a*). Like red flowers, red fruits signal an abundant food supply. By feeding on these fruits, birds and other animals may carry seeds from place to place and thus transfer plants from one suitable habitat to another.

Fruits with hooked spines, like those of burgrass (figure 40.12*b*), are typical of several genera of plants that occur in the northern deciduous woods. Such fruits are often disseminated by mammals, including humans. Squirrels and similar mammals disperse and bury fruits such as acorns and other nuts. Other fruits, such as those of maples, elms, and ashes, have wings which aid in their distribution by the wind. The dandelion provides another familiar example of a fruit type that is dispersed by wind (figure 40.13), and the dispersal of seeds from plants such as milkweeds, willows, and cottonwoods is similar. Orchids have minute, dustlike seeds, which are likewise blown away by the wind.

Coconuts and other plants that characteristically occur on or near beaches are regularly spread throughout a region by water (figure 40.14). This sort of dispersal is especially important in the colonization of distant island groups, such as the Hawaiian Islands. It has been calculated that seeds of about 175 original angiosperms, nearly a third from North America, must have reached Hawaii to have evolved into the roughly 970 species found there today. Some of these seeds blew through the air, others were transported on the feathers or in the guts of birds, and still others drifted across the Pacific. Although the distances are rarely as great as the distance between Hawaii and the mainland, dispersal is just as important for mainland plant species that have discontinuous habitats, such as mountaintops, marshes, or north-facing cliffs.

Fruits, which are characteristic of angiosperms, are extremely diverse. The evolution of specialized structures allows fruits to be dispersed by animals, wind, and water.



(a)



(b)

FIGURE 40.12

Animal-dispersed fruits. (a) The bright red berries of this honeysuckle, *Lonicera hispidula*, are highly attractive to birds, just as are red flowers. After eating the fruits, birds may carry the seeds they contain for great distances either internally or, because of their sticky pulp, stuck to their feet or other body parts. (b) The spiny fruits of this burgrass, *Cenchrus incertus*, adhere readily to any passing animal, as you will know if you have ever stepped on them.



FIGURE 40.13

Wind-dispersed fruits. False dandelion, *Pyrhopappus carolinianus*. The “parachutes” disperse the fruits of both false and true dandelions widely in the wind, much to the gardener’s despair.



FIGURE 40.14

A water-dispersed fruit. This fruit of the coconut, *Cocos nucifera*, is sprouting on a sandy beach. Coconuts, one of the most useful plants for humans in the tropics, have become established on even the most distant islands by drifting on the waves.

40.4 Germination initiates post-seed development.

When conditions are satisfactory, the embryo emerges from its desiccated state, utilizes food reserves, and resumes growth. As the sporophyte pushes through the seed coat it orients with the environment so the root grows down and the shoot grows up. New growth comes from delicate meristems that are protected from environmental rigors. The shoot becomes photosynthetic and the post-embryonic phase of growth and development is underway.

Mechanisms of Germination

Germination is the first step in the development of the plant outside of its seed coat. Germination occurs when a seed absorbs water and its metabolism resumes. The amount of water a seed can absorb is phenomenal and creates a force strong enough to break the seed coat. At this point, it is important that oxygen be available to the developing embryo because plants, like animals, require oxygen for cellular respiration. Few plants produce seeds that germinate successfully under water, although some, such as rice, have evolved a tolerance to anaerobic conditions.

A dormant seed, although it may have imbibed a full supply of water and may be respiring, synthesizing proteins and RNA, and apparently carrying on normal metabolism, may nonetheless fail to germinate without an additional signal from the environment. This signal may be light of the correct wavelengths and intensity, a series of cold days, or simply the passage of time at temperatures appropriate for germination. Seeds of many plants will not germinate unless they have been **stratified**—held for periods of time at low temperatures. This phenomenon prevents seeds of plants that grow in cold areas from germinating until they have passed the winter, thus protecting their seedlings from cold conditions.

Germination can occur over a wide temperature range (5° to 30°C), although certain species and specific habitats may have relatively narrow optimum ranges. Some seeds will not germinate even under the best conditions. In some species, a significant fraction of a season's seeds remain dormant, providing a gene pool of great evolutionary significance to the future plant population.

The Utilization of Reserves

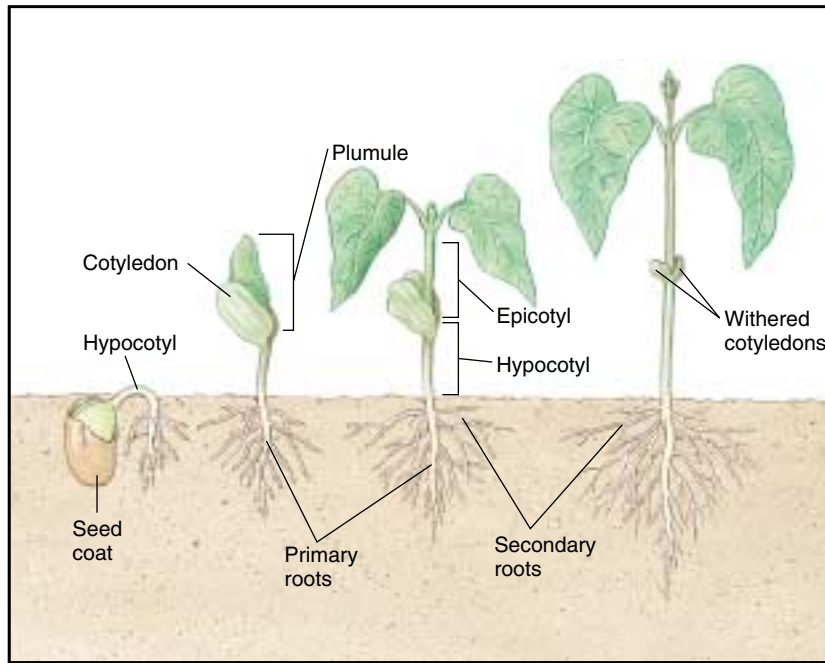
Germination occurs when all internal and external requirements are met. Germination and early seedling

growth require the utilization of metabolic reserves; these reserves are stored in the starch grains of **amyloplasts** (colorless plastids that store starch) and protein bodies. Fats and oils also are important food reserves in some kinds of seeds. They can readily be digested during germination, producing glycerol and fatty acids, which yield energy through cellular respiration; they can also be converted to glucose. Depending on the kind of plant, any of these reserves may be stored in the embryo itself or in the endosperm.

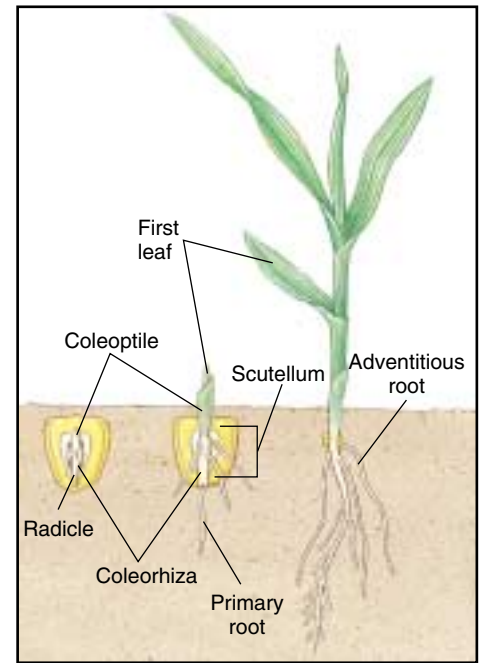
In the kernels of cereal grains, the single cotyledon is modified into a relatively massive structure called the **scutellum** (figure 40.15), from the Latin word meaning “shield.” The abundant food stored in the scutellum is used up first because these plants do not need to use the endosperm during germination. Later, while the seedling is becoming established, the scutellum serves as a nutrient conduit from the endosperm to the embryo. This is one of the best examples of how plant growth regulators modulate development in plants (40.16). The embryo produces gibberellic acid which signals the outer layer of the endosperm called the **aleurone** to produce α -amylase. This enzyme is responsible for breaking the starch in the endosperm down into sugars that are passed by the scutellum to the embryo. Absciscic acid, another plant growth regulator, which is important in establishing dormancy, can inhibit this process. Absciscic acid levels may be reduced further when a seed absorbs water.

The emergence of the embryonic root and shoot from the seed during germination varies widely from species to species. In most plants, the root emerges before the shoot appears and anchors the young seedling in the soil (see figure 40.15). In plants such as peas and corn, the cotyledons may be held below ground; in other plants, such as beans, radishes, and sunflowers, the cotyledons are held above ground. The cotyledons may or may not become green and contribute to the nutrition of the seedling as it becomes established. The period from the germination of the seed to the establishment of the young plant is a very critical one for the plant's survival; the seedling is unusually susceptible to disease and drought during this period.

During germination and early seedling establishment, the utilization of food reserves stored in the embryo or the endosperm is mediated by hormones, which, in some cases, are gibberellins.



(a)



(b)

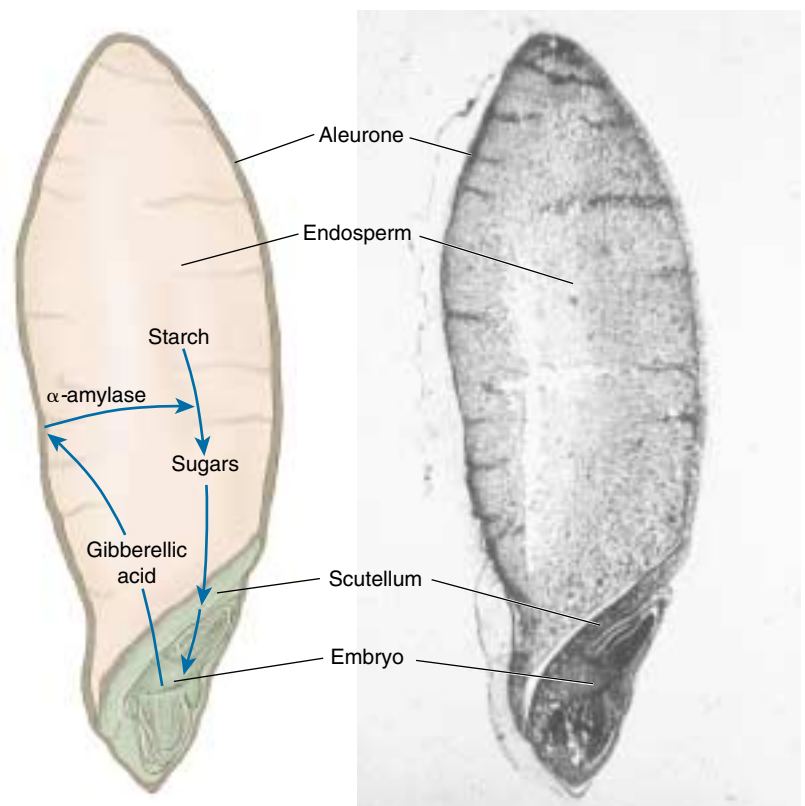
FIGURE 40.15

Shoot development. The stages shown are for a dicot, the common bean, (a) *Phaseolus vulgaris*, and a monocot, corn, (b) *Zea mays*.

FIGURE 40.16

Hormonal regulation of seedling growth.

The germinating barley embryos utilize the starch stored in the endosperm by releasing the hormone gibberellic acid (GA) that triggers the outer layers of the endosperm (aleurone layers) to produce the starch-digesting enzyme α -amylase. The α -amylase breaks down starch into sugar which moves through the scutellum (cotyledon) into the embryo where it provides energy for growth. A second hormone, abscisic acid (ABA), is important in establishing dormancy and becomes diluted as seeds imbibe water. When there is excess ABA, the GA-triggered production of α -amylase is inhibited.





Summary

Questions

Media Resources

40.1 Plant embryo development establishes a basic body plan.

- Plant shape is determined by the direction of cell division and expansion.
- Three tissue systems form radially through regulated cell division and differentiation.
- Shoot and root apical meristems are established to continuously produce new tissues, which then differentiate into body parts.
- Carbohydrates, lipids, and proteins are stored for germination in the endosperm or cotyledons.

1. The pattern of cell division regulates the shape of an embryo. Describe the cell division pattern that results in the single, outer layer of protoderm in the globular stage embryo.
2. What evidence supports the claim that the shoot meristem is genetically specified separately from the root?



- Art Activity: Corn Grain Structure
- Art Activity: Garden Bean Seed Structure

40.2 Seed formation protects the dormant embryo from water loss.

- The ovule wall (integuments) around the embryo hardens to protect the embryo as embryogenesis ends.
- Seed formation allows the embryo to enter a dormant state and continue growth under more optimal conditions.

3. Why are seeds adaptively important? Why may a seed showing proper respiration and synthesis of proteins and nucleic acids and all other normal metabolic activities still fail to germinate?



- Embryos and Seeds

40.3 Fruit formation enhances seed dispersal.

- Fruits are an angiosperm innovation that develop from the ovary wall (a modified leaf) that surrounds the ovule(s).
- Fruits are highly diverse in terms of their dispersal mechanisms, often displaying wings, barbs, or other structures that aid in their transport from place to place. Fruit dispersal methods are especially important in the colonization of islands or other distant patches of suitable habitat.

4. Why is it advantageous for a plant to produce fruit? How does the genotype of the fruit compare with the genotype of the embryo? How does the genotype of the seed wall compare with the fruit wall?



- Activity: Fruits
- Fruits

40.4 Germination initiates post-seed development.

- In a seed, the embryo with its food supply is encased within a sometimes rigid, relatively impermeable seed coat that may need to be abraded before germination can occur. Weather or passage through an animal's digestive tract may be necessary for germination to begin.
- When temperature, light, and water conditions are appropriate, germination can begin. In some cases, a period of chilling is required prior to germination. This adaptation protects seeds from germinating during the cold season.
- At germination, the mobilization of the food reserves is critical. Hormones control this process.

5. Explain how the embryo signals the endosperm to obtain sugars for growth during germination.
6. Why does the root (actually the radicle) of the embryo emerge first?



- Germination

41

How Plants Grow in Response to Their Environment

Concept Outline

41.1 Plant growth is often guided by environmental cues.

Tropisms. Plant growth is often influenced by light, gravity, and contact with other plants and animals.

Dormancy. The ability to cease growth allows plants to wait out the bad times.

41.2 The hormones that guide growth are keyed to the environment.

Plant Hormones. Hormones are grouped into seven classes.

Auxin. Auxin is involved in the elongation of stems.

Cytokinins. Cytokinins stimulate cell division.

Gibberellins. Gibberellins control stem elongation.

Brassinosteroids and Oligosaccharins. There are several recent additions to the plant hormone family.

Ethylene. Ethylene controls leaves and flower abscission.

Abscisic Acid. Abscisic acid suppresses growth of buds and promotes leaf senescence.

41.3 The environment influences flowering.

Plants Undergo Metamorphosis. The transition of a shoot meristem from vegetative to adult is called phase change.

Pathways Leading to Flower Production. Photoperiod is regulated in complex ways.

Identity Genes and the Formation of Floral Meristems and Floral Organs. Floral meristem identity genes activate floral organ identity genes.

41.4 Many short-term responses to the environment do not require growth.

Turgor Movement. Changes in the water pressure within plant cells result in quick and reversible plant movements.

Plant Defense Responses. In addition to generalized defense mechanisms, some plants have highly evolved recognition mechanisms for specific pathogens.



FIGURE 41.1

Plant growth is affected by environmental cues. The branches of this fallen tree are growing straight up in response to gravity and light.

All organisms sense and interact with their environment. This is particularly true of plants. Plant survival and growth is critically influenced by abiotic factors including water, wind, and light. In this chapter, we will explore how a plant senses such factors, and transduces these signals to elicit an optimal physiological, growth, or developmental response. Hormones play an important role in the internal signaling that brings about environmental responses, and is keyed in many ways to the environment. The effect of the local environment on plant growth also accounts for much of the variation in adult form within a species (figure 41.1). Precisely regulated responses to the environment not only allow a plant to survive from day to day but also determine when a flowering plant will produce a flower. The entire process of constructing a flower in turn sets the stage for intricate reproductive strategies that will be discussed in the next chapter.

41.1 Plant growth is often guided by environmental cues.

Tropisms

Growth patterns in plants are often guided by environmental signals. **Tropisms** (from *trope*, the Greek word for “turn”) are positive or negative growth responses of plants to external stimuli that usually come from one direction. Some responses occur independently of the direction of the stimuli and are referred to as nastic movements. For example, a tendril of a pea plant will always coil in one direction when touched. Tropisms, on the other hand, are directional and offer significant compensation for the plant’s inability to get up and walk away from unfavorable environmental conditions. Tropisms contribute the variety of branching patterns we see within a species. Here we will consider three major classes of plant tropisms: phototropism, gravitropism, and thigmotropism. Tropisms are particularly intriguing because they challenge us to connect environmental signals with cellular perception of the signal, transduction into biochemical pathways, and ultimately an altered growth response.

Phototropism

Phototropic responses involve the bending of growing stems and other plant parts toward sources of light (figure 41.2). In general, stems are positively phototropic, growing toward a light source, while most roots do not respond to light or, in exceptional cases, exhibit only a weak negative phototropic response. The phototropic reactions of stems are clearly of adaptive value, giving plants greater exposure to available light. They are also important in determining the development of plant organs and, therefore, the appearance of the plant. Individual leaves may display phototropic responses. The position of leaves is important to the photosynthetic efficiency of the plant. A plant hormone called auxin (discussed later in this chapter) is probably involved in most, if not all, of the phototropic growth responses of plants.

The first step in a phototropic response is perceiving the light. Photoreceptors perceive different wavelengths of light with blue and red being the most common. Blue light receptors are being characterized and we are beginning to understand how plants “see blue.” Much more is known about “seeing red” and translating that perception into a signal transduction pathway leading to an altered growth response. Plants contain a pigment, **phytochrome**, which exists in two interconvertible forms, P_r and P_{fr} . In the first form, phytochrome absorbs red light; in the second, it absorbs far-red light. When a molecule of P_r absorbs a photon of red light (660 nm), it is instantly converted into a molecule of P_{fr} , and when a molecule of P_{fr} absorbs a photon of far-red light (730 nm), it is instantly converted to P_r . P_{fr} is biologically active and P_r is biologically inactive. In other words, when P_{fr} is present, a



FIGURE 41.2
Phototropism. *Impatiens* plant growing toward light.

given biological reaction that is affected by phytochrome will occur. When most of the P_{fr} has been replaced by P_r , the reaction will not occur (figure 41.3). While we refer to phytochrome as a single molecule here, it is important to note that several different phytochromes have now been identified that appear to have specific biological functions.

Phytochrome is a light receptor, but it does not act directly to bring about reactions to light. The existence of phytochrome was conclusively demonstrated in 1959 by Harry A. Borthwick and his collaborators at the U.S. Department of Agriculture Research Center at Beltsville, Maryland. It has since been shown that the molecule consists of two parts: a smaller one that is sensitive to light and a larger portion that is a protein. The protein component initiates a signal transduction leading to a particular tropism. The phytochrome pigment is blue, and its light-sensitive portion is similar in structure to the phycobilins that occur in cyanobacteria and red algae. Phytochrome is present in all groups of plants and in a few genera of green algae, but not in bacteria, fungi, or protists (other than the few green algae). It is likely that phytochrome systems for measuring light evolved among the green algae and were present in the common ancestor of the plants.

Phytochrome is involved in many plant growth responses. For example, seed germination is inhibited by far-red light and stimulated by red light in many plants. Because chlorophyll absorbs red light strongly but does not absorb far-red light, light passing through green leaves inhibits seed germination. Consequently, seeds on the

ground under deciduous plants that lose their leaves in winter are more apt to germinate in the spring after the leaves have decomposed and the seedlings are exposed to direct sunlight. This greatly improves the chances the seedlings will become established.

A second example of these relationships is the elongation of the shoot in an *etiolated* seedling (one that is pale and slender from having been kept in the dark). Such plants become normal when exposed to light, especially red light, but the effects of such exposure are canceled by far-red light. This indicates a relationship similar to that observed in seed germination. There appears to be a link between phytochrome light perception and brassinosteroids in the etiolation response. Etiolation is an energy conservation strategy to help plants growing in the dark reach the light before they die. They don't green-up until there is light, and they divert energy to growing as tall as possible through internode elongation. The de-etiolated (*det2*) *Arabidopsis* mutant has a poor etiolation response. It does not have elongated internodes and greens up a bit in the dark. It turns out that *det2* mutants are defective in an enzyme necessary for brassinosteroid biosynthesis. Researchers suspect that brassinosteroids play a role in how plants respond to light through phytochrome. Thus, because *det2* mutants lack brassinosteroids, they do not respond to light, or lack of light, as normal plants do, and the *det2* mutants grow normally in the dark.

Red and far-red light also are used as signals for plant spacing. The closer plants are together, the more likely they are to grow tall and try to outcompete others for the sunshine. Plants somehow measure the amount of far-red light being bounced back to them from neighboring trees. If their perception is messed up by putting a collar around the stem with a solution that blocks light absorption, the elongation response is no longer seen.

Gravitropism

When a potted plant is tipped over, the shoot bends and grows upward (figure 41.4). The same thing happens when a storm pushes over plants in a field. These are examples of **gravitropism**, the response of a plant to the gravitational field of the earth. We saw in chapter 40 that brown algae orient their first cell division so the rhizoid grows downward. Rhizoids also develop away from a unilateral light source. Separating out phototropic effects is important in the study of gravitropisms.

Gravitropic responses are present at germination when the root grows down and the shoot grows up. Why does a shoot have a negative gravitropic response (growth away from gravity), while a root has a positive gravitropic response? The opportunity to experiment on the space shuttle in a gravity-free environment has accelerated research in this area. Auxins play a primary role in gravitropic responses, but they may not be the only way gravitational information is sent through the plant. When John

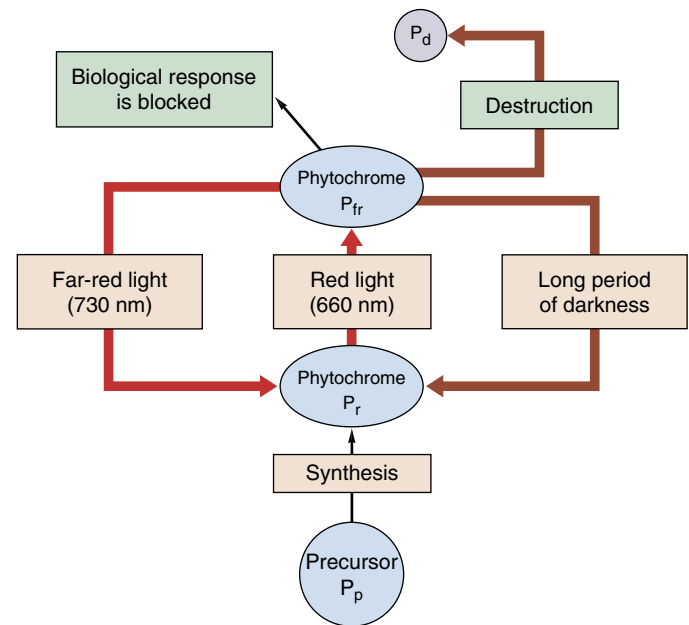


FIGURE 41.3

How phytochrome works. Phytochrome is synthesized in the P_r form from amino acids, designated P_p for phytochrome precursor. When exposed to red light, P_r changes to P_{fr} , which is the active form that elicits a response in plants. P_{fr} is converted to P_r when exposed to far-red light, and it also converts to P_r or is destroyed in darkness. The destruction product is designated P_d .

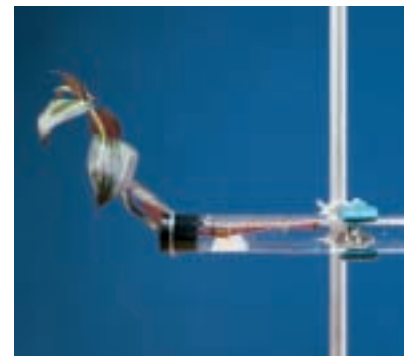


FIGURE 41.4

Plant response to gravity. This plant (*Zebrina pendula*) was placed horizontally and allowed to grow for 7 days. Note the negative gravitational response of the shoot.

Glenn made his second trip into space, he was accompanied by an experiment designed to test the role of gravity and electrical signaling in root bending. Analysis of gravitropic mutants is also adding to our understanding of gravitropism. There are four steps that lead to a gravitropic response:

1. Gravity is perceived by the cell
2. Signals form in the cell that perceives gravity
3. The signal is transduced intra- and intercellularly
4. Differential cell elongation occurs between cells in the "up" and "down" sides of the root or shoot.

One of the first steps in perceiving gravity is that amyloplasts, plastids that contain starch, sink toward the gravitational field. These may interact with the cytoskeleton, but the net effect is that auxin becomes more concentrated on the lower side of the stem axis than on the upper side. The increased auxin concentration on the lower side in stems causes the cells in that area to grow more than the cells on the upper side. The result is a bending upward of the stem against the force of gravity—in other words there is a *negative gravitropic response*. Such differences in hormone concentration have not been as well documented in roots. Nevertheless, the upper sides of roots oriented horizontally grow more rapidly than the lower sides, causing the root ultimately to grow downward; this phenomenon is known as *positive gravitropism*. In shoots, the gravity-sensing cells are in the endoderm. Mutants like *scarecrow* and *short root* in *Arabidopsis* that lack normal endodermal development fail to have a normal gravitropic response. These endodermal cells are the sites of the amyloplasts in the stems.

In roots, the gravity-sensing cells are located in the root cap and the cells that actually do the asymmetric growth are in the distal elongation zone which is closest to the root cap. How the information gets transferred over this distance is an intriguing problem. Auxin may be involved, but when auxin transport is suppressed, there is still a gravitropic response in the distal elongation zone. Some type of electrical signaling involving membrane polarization has been hypothesized and this was tested aboard the space shuttle. So far the verdict is still not in on the exact mechanism.

It may surprise you to learn that in tropical rain forests, roots of some plants may grow up the stems of neighboring plants, instead of exhibiting the normal positive gravitropic responses typical of other roots. The rainwater dissolves nutrients, both while passing through the lush upper canopy of the forest, and also subsequently as it trickles down tree trunks. Such water functions as a more reliable source of nutrients for the roots than the nutrient-poor soils in which the plants are anchored. Explaining this in terms of current hypotheses is a challenge. It has been proposed that roots are more sensitive to auxin than shoots and that auxin may actually inhibit growth on the lower side of a root, resulting in a positive gravitropic response. Perhaps in these tropical plants, the sensitivity to auxin in roots is reduced.

Thigmotropism

Thigmotropism is a name derived from the Greek root *thigma*, meaning “touch.” A thigmotropism is a response of a plant or plant part to contact with the touch of an object, animal, plant, or even the wind. (figure 41.5). When a tendril makes contact with an object, specialized epidermal cells, whose action is not clearly understood, perceive the contact and promote uneven growth, causing the tendril to

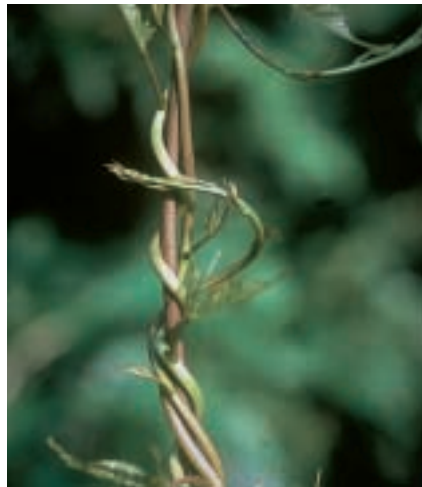


FIGURE 41.5
Thigmotropism. The thigmotropic response of these twining stems causes them to coil around the object with which they have come in contact.

curl around the object, sometimes within as little as 3 to 10 minutes. Both auxin and ethylene appear to be involved in tendril movements, and they can induce coiling in the absence of any contact stimulus. In other plants, such as clematis, bindweed, and dodder, leaf petioles or unmodified stems twine around other stems or solid objects.

Again, *Arabidopsis* is proving valuable as a model system. A gene has been identified that is expressed in 100-fold higher levels 10 to 30 minutes after touch. Given the value of a molecular genetics approach in dissecting the pathways leading from an environmental signal to a growth response, this gene provides a promising first step in understanding how plants respond to touch.

Other Tropisms

The tropisms just discussed are among the best known, but others have been recognized. They include *electrotropism* (responses to electricity); *chemotropism* (response to chemicals); *traumotropism* (response to wounding which we discuss on page 834); *thermotropism* (response to temperature); *aerotropism* (response to oxygen); *skototropism* (response to dark); and *geomagnetotropism* (response to magnetic fields). Roots will often follow a diffusion gradient of water coming from a cracked pipe and enter the crack. Some call such growth movement *hydrotropism*, but there is disagreement whether responses to water and several other “stimuli” are true tropisms.

While plants can't move away or toward optimal conditions, they can grow. Phototropisms are growth responses of plants to a unidirectional source of light. Gravitropism, the response of a plant to gravity, generally causes shoots to grow up (negative gravitropism) and roots to grow down (positive gravitropism). Thigmotropisms are growth responses of plants to contact.

Dormancy

Sometimes modifying the direction of growth is not enough to protect a plant from harsh conditions. The ability to cease growth and go into a dormant stage provides a survival advantage. The extreme example is seed dormancy, but there are intermediate approaches to waiting out the bad times as well. Environmental signals both initiate and end dormant phases in the life of a plant.

In temperate regions, we generally associate dormancy with winter, when freezing temperatures and the accompanying unavailability of water make it impossible for plants to grow. During this season, buds of deciduous trees and shrubs remain dormant, and apical meristems remain well protected inside enfolding scales. Perennial herbs spend the winter underground as stout stems or roots packed with stored food. Many other kinds of plants, including most annuals, pass the winter as seeds.

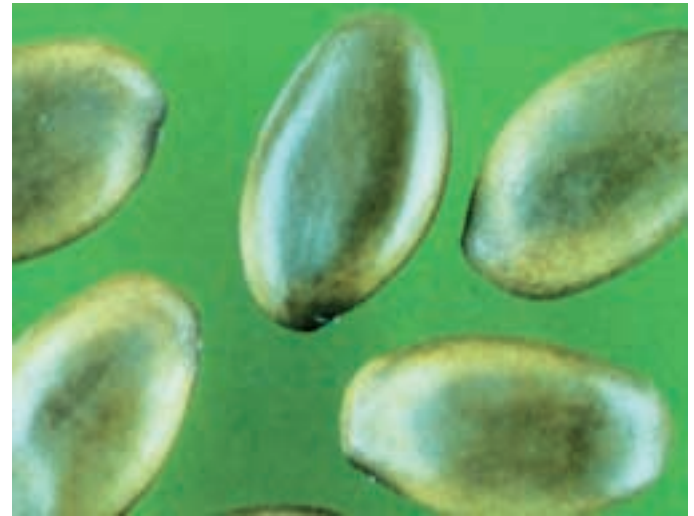
In some seasonally dry climates, seed dormancy occurs primarily during the dry season, often the summer. Rainfalls trigger germination when conditions for survival are more favorable. Annual plants occur frequently in areas of seasonal drought. Seeds are ideal for allowing annual plants to bypass the dry season, when there is insufficient water for growth. When it rains, they can germinate and the plants can grow rapidly, having adapted to the relatively short periods when water is available. Chapter 40 covered some of the mechanisms involved in breaking seed dormancy and allowing germination under favorable circumstances. These include the leaching from the seed coats of chemicals that inhibit germination, or mechanically cracking the seed coats, a procedure that is particularly suitable for promoting growth in seasonally dry areas. Whenever rains occur, they will leach out the chemicals from the seed coats, and the hard coats of other seeds may be cracked when they are being washed down along temporarily flooded arroyos (figure 41.6).

Seeds may remain dormant for a surprisingly long time. Many legumes (plants of the pea and bean family, Fabaceae) have tough seeds that are virtually impermeable to water and oxygen. These seeds often last decades and even longer without special care; they will eventually germinate when their seed coats have been cracked and water is available. Seeds that are thousands of years old have been successfully germinated!

A period of cold is necessary before some kinds of seeds will germinate, as we mentioned in chapter 40. The seeds of other plants will germinate only when adequate water is available and the temperatures are relatively high. For this reason, certain weeds germinate and grow in the cooler part of the year and others in the warmer part of the year. Similarly, a period of cold is needed before the buds of some trees and shrubs will break dormancy and develop normally. For this reason, many plants that normally grow in temperate regions do not thrive in warmer regions near the equator, because even at high elevations



(a)



(b)

FIGURE 41.6

Palo verde (*Cercidium floridum*). This desert tree (a) has tough seeds (b) that germinate only after they are cracked.

in the tropics it still does not get cold enough, and the day-length relationships are different from those of temperate regions.

Mature plants may become dormant in dry or cold seasons that are unfavorable for growth. Dormant plants usually lose their leaves and drought-resistant winter buds are produced. Long unfavorable periods may be bypassed through the production of seeds, which themselves can remain dormant for long periods.

41.2 The hormones that guide growth are keyed to the environment.

Plant Hormones

While initial responses of plants to environmental signals may rely primarily on electrical signaling, longer-term responses that alter morphology rely on complex physiological networks. Many internal signaling pathways involve plant hormones, which are the focus of this section. Hormones are involved in responses to the environment, as well as internally regulated development (examples of which you saw in chapter 40).

Hormones are chemical substances produced in small, often minute, quantities in one part of an organism and then transported to another part, where they bring about physiological or developmental responses. The activity of hormones results from their capacity to stimulate certain physiological processes and to inhibit others (figure 41.7). How they act in a particular instance is influenced both by the hormone and the tissue that receives the message.

In animals, hormones are usually produced at definite sites, usually organs. In plants, hormones are not produced in specialized tissues but, instead, in tissues that also carry

out other, usually more obvious, functions. There are seven major kinds of plant hormones: auxin, cytokinins, gibberellins, brassinosteroids, oligosaccharins, ethylene, and abscisic acid (table 41.1). Current research is focused on the biosynthesis of hormones and on characterizing the hormone receptors that trigger signal transduction pathways. Much of the molecular basis of hormone function remains enigmatic.

Because hormones are involved in so many aspects of plant function and development, we have chosen to integrate examples of hormone activity with specific aspects of plant biology throughout the text. In this section, our goal is to give you a brief overview of these hormones. Use this section as a reference when specific hormones are discussed in the next few chapters.

There are seven major kinds of plant hormones: auxin, cytokinins, gibberellins, brassinosteroids, oligosaccharins, ethylene, and abscisic acid.



(a)

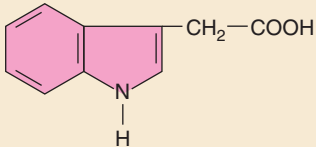
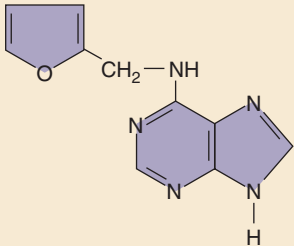
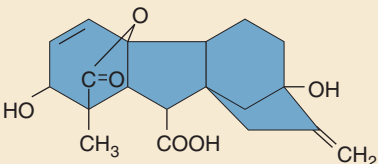
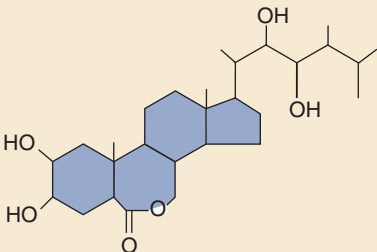
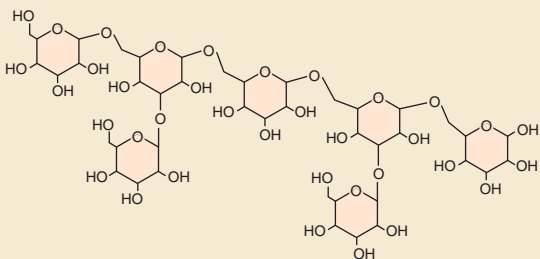
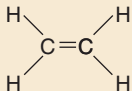
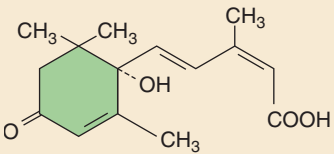


(b)

FIGURE 41.7

Effects of plant hormones. Plant hormones, often acting together, influence many aspects of plant growth and development, including (a) leaf abscission and (b) the formation of mature fruit.

Table 41.1 Functions of the Major Plant Hormones

Hormone		Major Functions	Where Produced or Found in Plant
Auxin (IAA)		Promotion of stem elongation and growth; formation of adventitious roots; inhibition of leaf abscission; promotion of cell division (with cytokinins); inducement of ethylene production; promotion of lateral bud dormancy	Apical meristems; other immature parts of plants
Cytokinins		Stimulation of cell division, but only in the presence of auxin; promotion of chloroplast development; delay of leaf aging; promotion of bud formation	Root apical meristems; immature fruits
Gibberellins		Promotion of stem elongation; stimulation of enzyme production in germinating seeds	Roots and shoot tips; young leaves; seeds
Brassinosteroids		Overlapping functions with auxins and gibberellins	Pollen, immature seeds, shoots, leaves
Oligosaccharins		Pathogen defense, possibly reproductive development	Cell walls
Ethylene		Control of leaf, flower, and fruit abscission; promotion of fruit ripening	Roots, shoot apical meristems; leaf nodes; aging flowers; ripening fruits
Absciscic acid		Inhibition of bud growth; control of stomatal closure; some control of seed dormancy; inhibition of effects of other hormones	Leaves, fruits, root caps, seeds

Auxin

More than a century ago, an organic substance known as **auxin** became the first plant hormone to be discovered. Auxin increases the plasticity of plant cell walls and is involved in elongation of stems. Cells can enlarge in response to changes in turgor pressure when their cell walls have enhanced plasticity from auxin action. The discovery of auxin and its role in plant growth is an elegant example of thoughtful experimental design. The historical story is recounted here for that reason. Recent efforts have uncovered an auxin receptor. Transport mechanisms are also being unraveled. As with all the classes of hormones, we are just beginning to understand, at a cellular and molecular level, how hormones regulate growth and development.

Discovery of Auxin

In his later years, the great evolutionist, Charles Darwin, became increasingly devoted to the study of plants. In 1881, he and his son Francis published a book called *The Power of Movement of Plants*. In this book, the Darwins reported their systematic experiments on the response of growing plants to light—responses that came to be known as **phototropisms**. They used germinating oat (*Avena sativa*) and canary grass (*Phalaris canariensis*) seedlings in their experiments and made many observations in this field.

Charles and Francis Darwin knew that if light came primarily from one direction, the seedlings would bend strongly toward it. If they covered the tip of the shoot with a thin glass tube, the shoot would bend as if it were not covered. However, if they used a metal foil cap to exclude light from the plant tip, the shoot would not bend (figure 41.8). They also found that using an opaque collar to exclude light from the stem below the tip did not keep the area above the collar from bending.

In explaining these unexpected findings, the Darwins hypothesized that when the shoots were illuminated from one side, they bent toward the light in response to an “influence” that was transmitted downward from its source at the tip of the shoot. For some 30 years, the Darwins’ perceptive experiments remained the sole source of information about this interesting phenomenon. Then Danish plant physiologist Peter Boysen-Jensen and the Hungarian plant physiologist Arpad Paal independently demonstrated that the substance that caused the shoots to bend was a chemical. They showed that if the tip of a germinating grass seedling was cut off and then replaced with a small block of agar separating it from the rest of the seedling, the seedling would grow as if there had been no change. Something evidently was passing from the tip of the seedling through the agar into the region where the bending occurred. On the basis of these observations under conditions of uniform illumination or of darkness, Paal suggested that

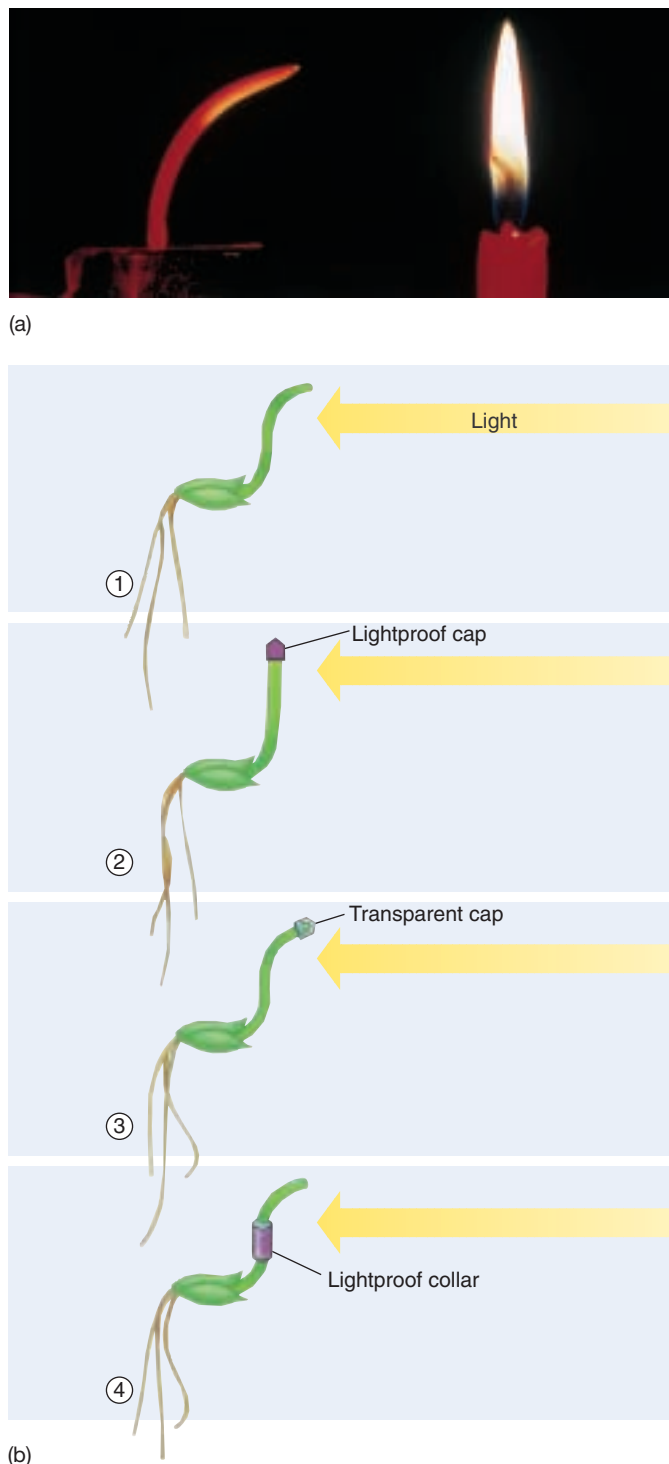
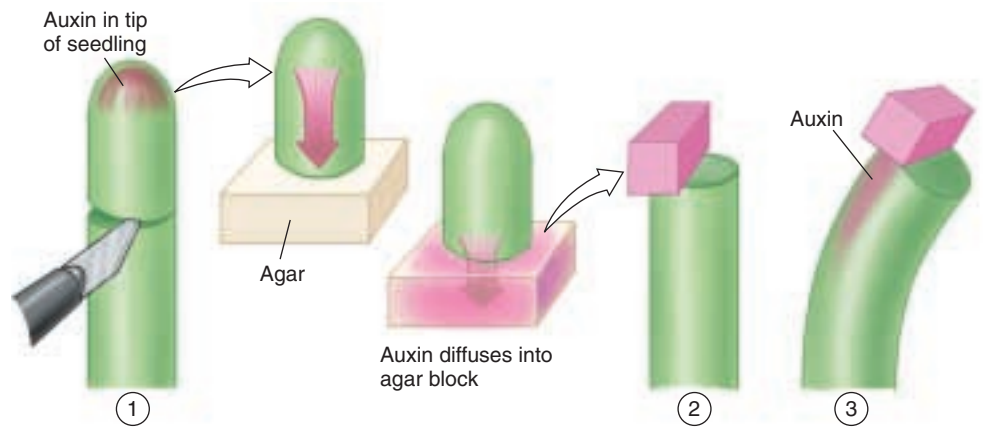


FIGURE 41.8

The Darwins’ experiment. (a) Young grass seedlings normally bend toward the light. (b) The bending (1) did not occur when the tip of a seedling was covered with a lightproof cap (2), but did occur when it was covered with a transparent one (3). When a collar was placed below the tip (4), the characteristic light response took place. From these experiments, the Darwins concluded that, in response to light, an “influence” that caused bending was transmitted from the tip of the seedling to the area below, where bending normally occurs.

FIGURE 41.9

Frits Went's experiment. (1) Went removed the tips of oat seedlings and put them in agar, an inert, gelatinous substance. (2) Blocks of agar were then placed off-center on the ends of other oat seedlings from which the tips had been removed. (3) The seedlings bent away from the side on which the agar block was placed. Went concluded that the substance that he named *auxin* promoted the elongation of the cells and that it accumulated on the side of an oat seedling away from the light.



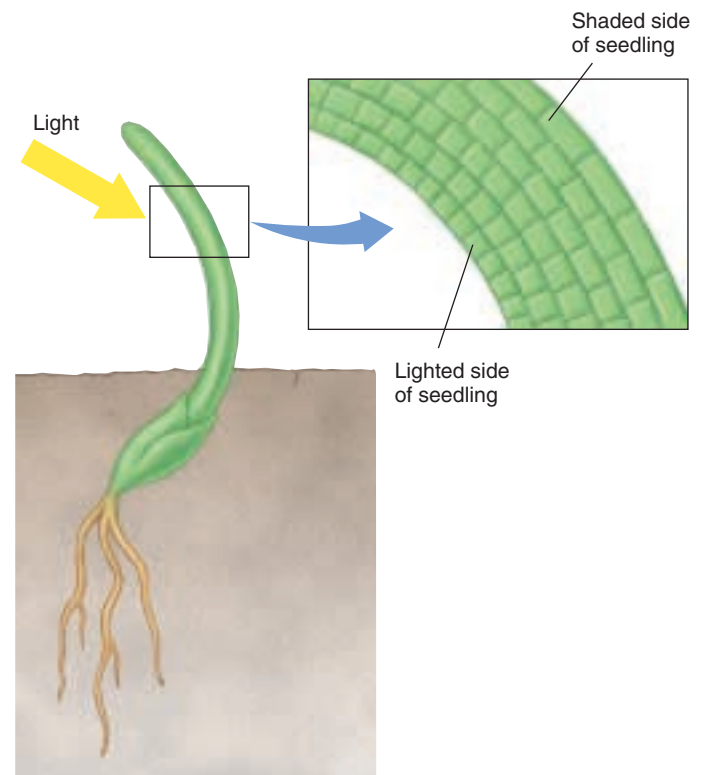
an unknown substance continually moves down from the tips of grass seedlings and promotes growth on all sides. Such a light pattern would not, of course, cause the shoot to bend.

Then, in 1926, Dutch plant physiologist Frits Went carried Paal's experiments an important step further. Went cut off the tips of oat seedlings that had been illuminated normally and set these tips on agar. He then took oat seedlings that had been grown in the dark and cut off their tips in a similar way. Finally, Went cut tiny blocks from the agar on which the tips of the light-grown seedlings had been placed and placed them off-center on the tops of the decapitated dark-grown seedlings (figure 41.9). Even though these seedlings had not been exposed to the light themselves, they bent away from the side on which the agar blocks were placed.

Went then put blocks of pure agar on the decapitated stem tips and noted either no effect or a slight bending toward the side where the agar blocks were placed. Finally, Went cut sections out of the lower portions of the light-grown seedlings to see whether the active principle was present in them. He placed these sections on the tips of decapitated, dark-green oat seedlings and again observed no effect.

As a result of his experiments, Went was able to show that the substance that had diffused into the agar from the tips of light-grown oat seedlings could make seedlings curve when they otherwise would have remained straight. He also showed that this chemical messenger caused the cells on the side of the seedling into which it flowed to grow more than those on the opposite side (figure 41.10). In other words, it enhanced rather than retarded cell elongation. He named the substance that he had discovered **auxin**, from the Greek word *auxein*, which means "to increase."

Went's experiments provided a basis for understanding the responses that the Darwins had obtained some 45 years earlier. The oat seedlings bent toward the light because of

**FIGURE 41.10**

Auxin causes cells on the dark side to elongate. Went determined that a substance called auxin enhanced cell elongation. Plant cells that are in the shade have more auxin and grow faster than cells on the lighted side, causing the plant to bend toward light. Further experiments showed exactly why there is more auxin on the shaded side of a plant.

differences in the auxin concentrations on the two sides of the shoot. The side of the shoot that was in the shade had more auxin, and its cells therefore elongated more than those on the lighted side, bending the plant toward the light.

The Effects of Auxins

Auxin acts to adapt the plant to its environment in a highly advantageous way. It promotes growth and elongation and facilitates the plant's response to its environment. Environmental signals directly influence the distribution of auxin in the plant. How does the environment—specifically, light—exert this influence? Theoretically, it might destroy the auxin, decrease the cells' sensitivity to auxin, or cause the auxin molecules to migrate away from the light into the shaded portion of the shoot. This last possibility has proved to be the case.

In a simple but effective experiment, Winslow Briggs inserted a thin sheet of mica vertically between the half of the shoot oriented toward the light and the half of the shoot oriented away from it (figure 41.11). He found that light from one side does not cause a shoot with such a barrier to bend. When Briggs examined the illuminated plant, he found equal auxin levels on both the light and dark sides of the barrier. He concluded that a normal plant's response to light from one direction involves auxin migrating from the light side to the dark side, and that the mica barrier prevented a response by blocking the migration of auxin.

The effects of auxin are numerous and varied. Auxin promotes the activity of the vascular cambium and the vascular tissues. Also, auxins are present in pollen in large

quantities and play a key role in the development of fruits. Synthetic auxins are used commercially for the same purpose. Fruits will normally not develop if fertilization has not occurred and seeds are not present, but frequently they will if auxins are applied. Pollination may trigger auxin release in some species leading to fruit development occurring even before fertilization.

How Auxin Works

In spite of this long history of research on auxin, its molecular basis of action has been an enigma. The chemical structure of IAA resembles that of the amino acid tryptophan, from which it is probably synthesized by plants (figure 41.12). Unlike animal hormones, a specific signal is not sent to specific cells, eliciting a predictable response. There are most likely multiple auxin perception sites. Auxin is also unique among the plant hormones in that it is transported toward the base of the plant. Two families of genes have been identified in *Arabidopsis* that are involved in auxin transport. For example, one protein is involved in the top to bottom transport of auxin; while two other proteins function in the root tip to regulate the growth response to gravity. We are still a ways from linking the measurable and observable effects of auxin to events that transpire after it travels to a site and binds to a receptor.

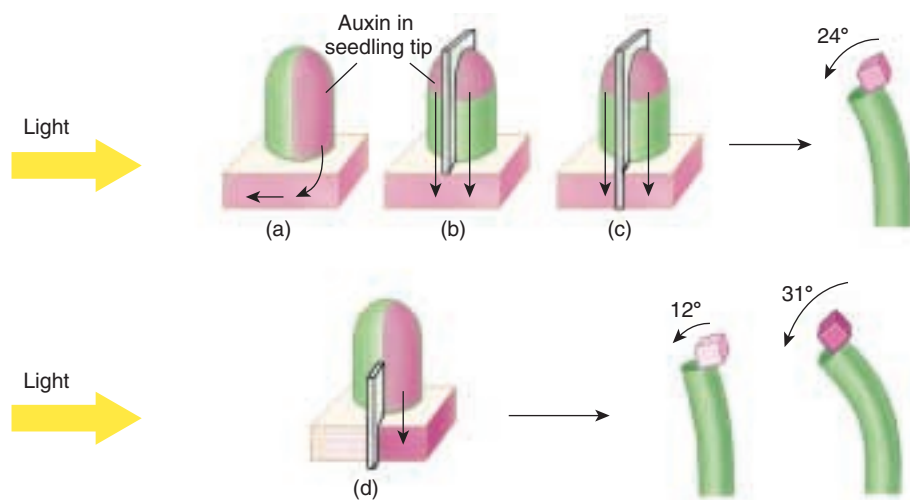


FIGURE 41.11

Phototropism and auxin: the Winslow Briggs experiments. The basic design of these experiments was to place the tip of an oat seedling on an agar block, apply light from one side, and observe the degree of curvature produced when the agar blocks were later placed on the decapitated seedlings. However, Briggs inserted a barrier in various places and noted how this affected the location of auxin. A comparison of (a) and (b) with similar experiments performed in the dark showed that auxin production does not depend on light; all produced approximately 24° of curvature. If a barrier was inserted in the agar block (d), light caused the displacement of the auxin away from the light. Finally, experiment (c) showed that it was displacement that had occurred, and not different rates of auxin production on the dark and light sides, because when displacement was prevented with a barrier, both sides of the agar block produced about 24° of curvature.

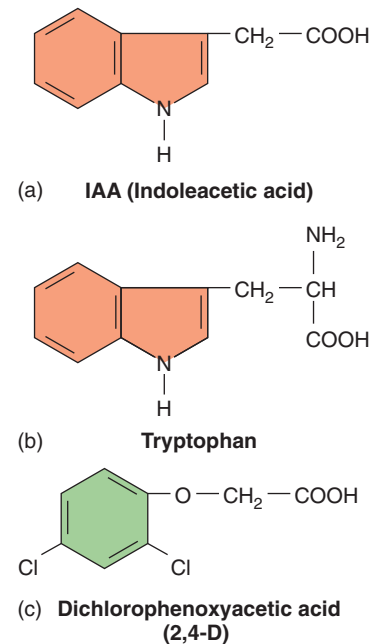
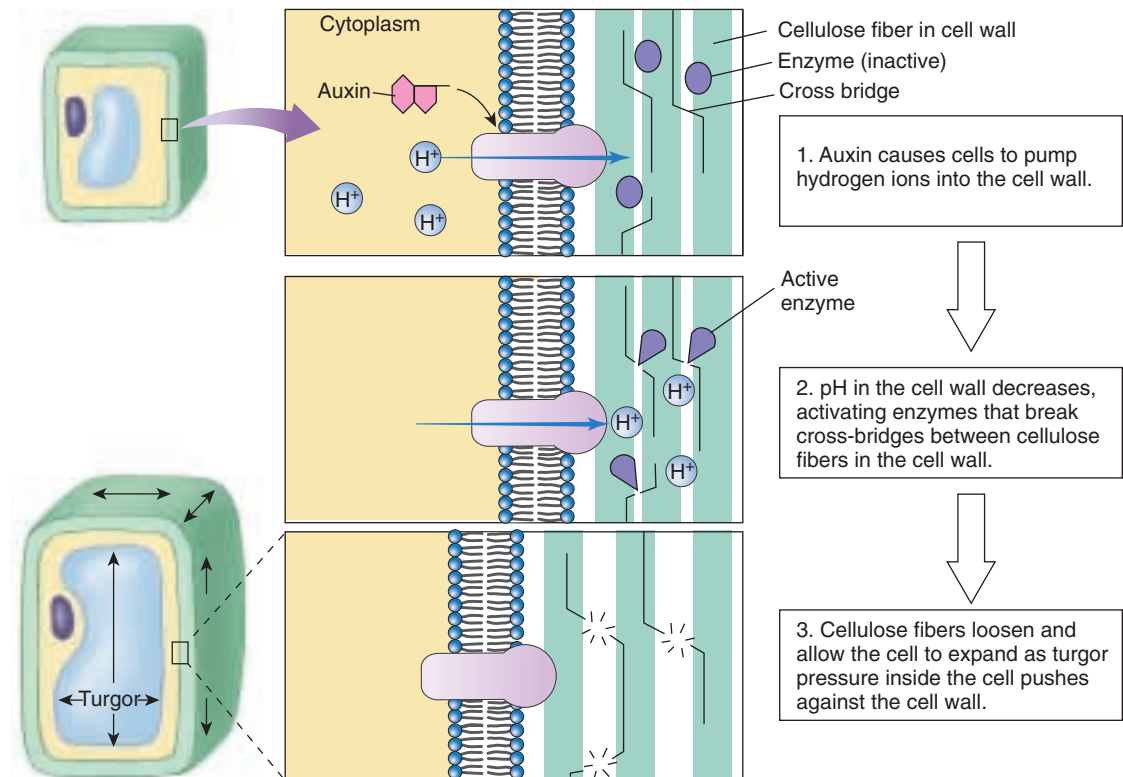


FIGURE 41.12

Auxins. (a) Indoleacetic acid (IAA), the principal naturally occurring auxin. (b) Tryptophan, the amino acid from which plants probably synthesize IAA. (c) Dichlorophenoxyacetic acid (2,4-D), a synthetic auxin, is a widely used herbicide.

FIGURE 41.13**Acid growth hypothesis.**

Auxin stimulates the release of hydrogen ions from the target cells which alters the pH of the cell wall. This optimizes the activity of enzymes which break bonds in the cell wall, allowing them to expand.



One of the downstream effects of auxin is an increase in plasticity of the plant cell wall. This will only work on young cell walls without extensive secondary cell wall formation. A more plastic wall will stretch more while its protoplast is swelling during active cell growth. The **acid growth hypothesis** provides a model linking auxin to cell wall expansion (figure 41.13). Auxin causes responsive cells to release hydrogen ions into the cell wall. This decreases the pH which activates enzymes that can break bonds between cell wall fibers. Remember that different enzymes operate optimally at different pHs. This hypothesis has been experimentally supported in several ways. Buffers that prevent cell wall acidification block cell expansion. Other compounds that release hydrogen ions from the cell can also cause cell expansion. The movement of hydrogen ions has been observed in response to auxin treatment. This hypothesis explains the rapid growth response. There are also delayed responses which most likely involve auxin-stimulated gene expression.

Synthetic Auxins. Synthetic auxins such as NAA (naphthalene acetic acid) and IBA (indolebutyric acid) have many uses in agriculture and horticulture. One of their most important uses is based on their prevention of abscission, the process that causes a leaf or other organ to fall from a plant. Synthetic auxins are used to prevent fruit drop in apples before they are ripe and to hold berries on holly that is being prepared for shipping. Synthetic auxins are also used to promote flowering and fruiting in pineapples and to induce the formation of roots in cuttings.

Synthetic auxins are routinely used to control weeds. When used as herbicides, they are applied in higher concentrations than IAA would normally occur in plants. One of the most important synthetic auxin herbicides is 2,4-dichlorophenoxyacetic acid, usually known as 2,4-D (see figure 41.12c). It kills weeds in grass lawns by selectively eliminating broad-leaved dicots. The stems of the dicot weeds cease all axial growth.

The herbicide 2,4,5-trichlorophenoxyacetic acid, better known as 2,4,5-T, is closely related to 2,4-D. 2,4,5-T was widely used as a broad-spectrum herbicide to kill weeds and seedlings of woody plants. It became notorious during the Vietnam War as a component of a jungle defoliant known as Agent Orange and was banned in 1979 for most uses in the United States. When 2,4,5-T is manufactured, it is unavoidably contaminated with minute amounts of dioxin. Dioxin, in doses as low as a few parts per billion, has produced liver and lung diseases, leukemia, miscarriages, birth defects, and even death in laboratory animals. Vietnam veterans and children of Vietnam veterans exposed to Agent Orange have been among the victims.

Auxin is synthesized in apical meristems of shoots. It causes young stems to bend toward light when it migrates toward the darker side, where it makes young cell walls more plastic and thereby promotes cell elongation. By interacting with other hormones, auxin can promote an increase in girth and is involved in growth responses to gravity and fruit ripening.

Cytokinins

Cytokinins comprise another group of naturally occurring growth hormones in plants. Studies by Gottlieb Haberlandt of Austria around 1913 demonstrated the existence of an unknown chemical in various tissues of vascular plants that, in cut potato tubers, would cause parenchyma cells to become meristematic, and would induce the differentiation of a cork cambium. The role of cytokinins, active components of coconut milk, in promoting the differentiation of organs in masses of plant tissue growing in culture later led to their discovery. Subsequent studies have focused on the role cytokinins play in the differentiation of tissues from callus.

A cytokinin is a plant hormone that, in combination with auxin, stimulates cell division and differentiation in plants. Most cytokinins are produced in the root apical meristems and transported throughout the plant. Developing fruits are also important sites of cytokinin synthesis. In mosses, cytokinins cause the formation of vegetative buds on the gametophyte. In all plants, cytokinins, working with other hormones, seem to regulate growth patterns.

All naturally occurring cytokinins are purines that appear to be derivatives of, or have molecule side chains similar to, those of adenine (figure 41.14). Other chemically diverse molecules, not known to occur naturally, have effects similar to those of cytokinins. Cytokinins promote growth of lateral buds into branches (figure 41.15); though, along with auxin and ethylene, they also play a role in apical dominance (the suppression of lateral bud growth). Con-

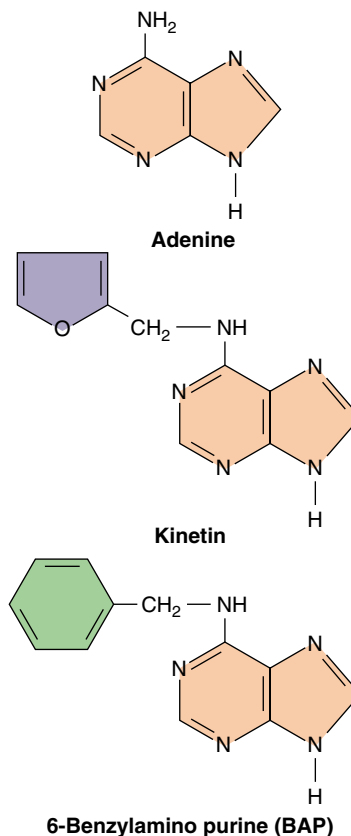


FIGURE 41.14
Some cytokinins. Two commonly used synthetic cytokinins: kinetin and 6-benzylamino purine. Note their resemblance to the purine base adenine.

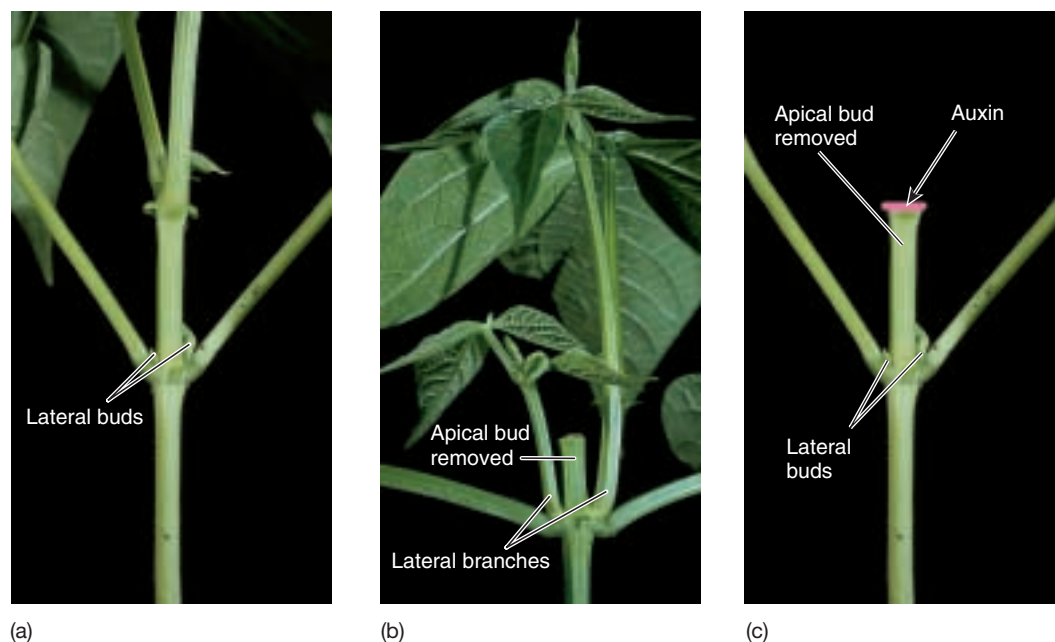


FIGURE 41.15
Cytokinins stimulate lateral bud growth. (a) When the apical meristem of a plant is intact, auxin from the apical bud will inhibit the growth of lateral buds. (b) When the apical bud is removed, cytokinins are able to produce the growth of lateral buds into branches. (c) When the apical bud is removed and auxin is added to the cut surface, axillary bud outgrowth is suppressed.

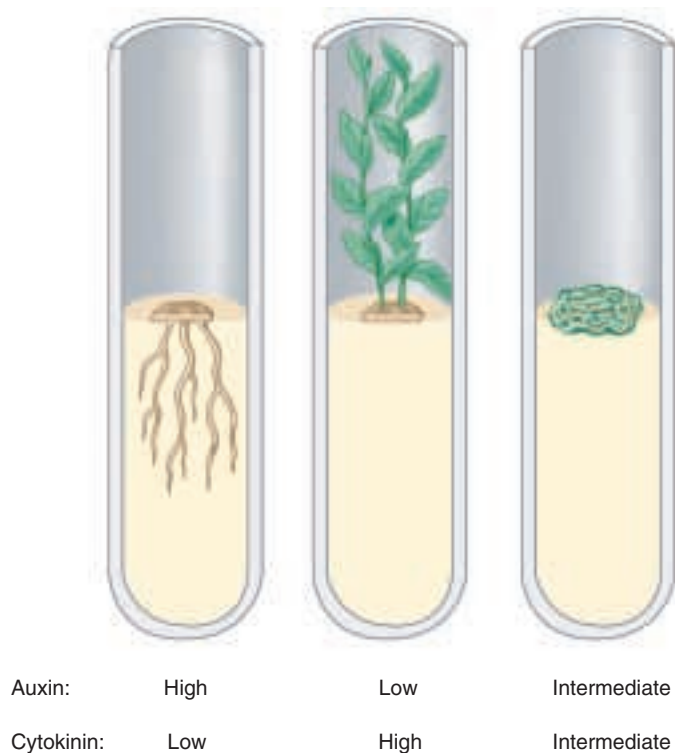


FIGURE 41.16

Relative amounts of cytokinins and auxin affect organ regeneration in culture. In the case of tobacco, (a) high auxin to cytokinin ratios favor root development; (b) high cytokinin to auxin ratios favor shoot development; and (c) intermediate concentrations result in the formation of undifferentiated cells. These developmental responses to cytokinin/auxin ratios in culture are species specific.

versely, cytokinins inhibit formation of lateral roots, while auxins promote their formation. As a consequence of these relationships, the balance between cytokinins and auxin, along with other factors, determines the appearance of a mature plant. In addition, the application of cytokinins to leaves detached from a plant retards their yellowing. They function as anti-aging hormones.

The action of cytokinins, like that of other hormones, has been studied in terms of its effects on growth and differentiation of masses of tissue growing in defined media. Plant tissue can form shoots, roots, or an undifferentiated mass of tissues, depending on the relative amounts of auxin and cytokinin (figure 41.16). In the early cell-growth experiments coconut “milk” was an essential factor. Eventually, it was discovered that coconut “milk” is not only rich in amino acids and other reduced nitrogen compounds required for growth, but it also contains cytokinins. Cytokinins seem to be essential for mitosis and cell division. They apparently promote the synthesis or activation of proteins that are specifically required for mitosis.

Cytokinins have also been used against plants by pathogens. The bacteria *Agrobacterium*, for example, intro-



FIGURE 41.17

Crown gall tumor. Sometimes cytokinins can be used against the plant by a pathogen. In this case *Agrobacterium tumefaciens* (a bacteria) has incorporated a piece of its DNA into the plant genome. This DNA contains genes coding for enzymes necessary for cytokinin and auxin biosynthesis. The increased levels of these hormones in the plant cause massive cell division and the formation of a tumor.

duces genes into the plant genome that increase the rate of cytokinin, as well as auxin, production. This causes massive cell division and the formation of a tumor called crown gall (figure 41.17). How these hormone biosynthesis genes ended up in a bacterium is an intriguing evolutionary question. Coevolution does not always work to the plant’s advantage.

Cytokinins are plant hormones that, in combination with auxin, stimulate cell division and, along with a number of other factors, determine the course of differentiation. In contrast to auxins, cytokinins are purines that are related to or derived from adenine.

Gibberellins

Gibberellins are named after the fungus *Gibberella fujikuroi*, which causes rice plants, on which it is parasitic, to grow abnormally tall. Japanese plant pathologist Eiichi Kurosawa investigated Bakane (“foolish seedling”) disease in the 1920s. He grew *Gibberella* in culture and obtained a substance that, when applied to rice plants, produced bakane. This substance was isolated and the structural formula identified by Japanese chemists in 1939. British chemists reconfirmed the formula in 1954. Although such chemicals were first thought to be only a curiosity, they have since turned out to belong to a large class of more than 100 naturally occurring plant hormones called gibberellins. All are acidic and are usually abbreviated to **GA** (for gibberellic acid), with a different subscript (GA₁, GA₂, and so forth) to distinguish each one. While gibberellins function endogenously as hormones, they also function as pheromones in ferns. In ferns gibberellin-like compounds released from one gametophyte can trigger the development of male reproductive structures on a neighboring gametophyte.

Gibberellins, which are synthesized in the apical portions of stems and roots, have important effects on stem elongation. The elongation effect is enhanced if auxin is also present. The application of gibberellins to dwarf mutants is known to restore the normal growth and development in many plants (figure 41.18). Some dwarf mutants produce insufficient amounts of gibberellin; while others lack the ability to perceive gibberellin. The large number of gibberellins are all part of a complex biosynthetic pathway that has been unraveled using gibberellin-deficient mutants in maize (corn). While many of these gibberellins are intermediate forms in the production of GA₁, recent work shows that different forms may have specific biological roles.

In chapter 41, we noted the role gibberellins stimulate the production of α -amylase and other hydrolytic enzymes needed for utilization of food resources during germination and establishment of cereal seedlings. How are the genes encoding these enzymes transcribed? Experimental studies in the aleurone layer surrounding the endosperms of cereal grains have shown that transcription occurs when the gibberellins initiate a burst of messenger RNA (mRNA) and protein synthesis. GA somehow enhances DNA binding proteins, which in turn allow DNA transcription of a gene. Synthesis of DNA does not seem to occur during the early stages of seed germination but becomes important when the radicle has grown through the seed coats.

Gibberellins also affect a number of other aspects of plant growth and development. These hormones also hasten seed germination, apparently because they can substitute for the effects of cold or light requirements in this process. Gibberellins are used commercially to space grape



FIGURE 41.18

Effects of gibberellins. This rapid cycling member of the mustard family plant (*Brassica rapa*) will “bolt” and flower because of increased gibberellin levels. Mutants such as the rosette mutant shown here (left) are defective in producing gibberellins. They can be rescued by applying gibberellins. Other mutants have been identified that are defective in perceiving gibberellins and they will not respond to gibberellin applications.



FIGURE 41.19

Applications of gibberellins increase the space between grapes. Larger grapes develop because there is more room between individual grapes.

flowers by extending internode length so the fruits have more room to grow (figure 41.19).

Gibberellins are an important class of plant hormones that are produced in the apical regions of shoots and roots. They play the major role in controlling stem elongation for most plants, acting in concert with auxin and other hormones.

Brassinosteroids and Oligosaccharins

Brassinosteroids

Although we've known about brassinosteroids for 30 years, it is only recently that they have claimed their place as a class of plant hormones. They were first discovered in *Brassica* pollen, hence the name. Their historical absence in discussions of hormones may be partially due to their functional overlap with other plant hormones, especially auxins and gibberellins. Additive effects among these three classes have been reported. The application of molecular genetics to the study of brassinosteroids has led to tremendous advances in our understanding of how they are made and, to some extent, how they function in signal transduction pathways. What is particularly intriguing about brassinosteroids are similarities to animal steroid hormones (figure 41.20). One of the genes coding for an enzyme in the brassinosteroid biosynthetic pathway has significant similarity to an enzyme used in the synthesis of testosterone and related steroids. Brassinosteroids have been identified in algae and appear to be quite ubiquitous among the plants. It is plausible that their evolutionary origin predated the plant-animal split.

Brassinosteroids have a broad spectrum of physiological effects—elongation, cell division, bending of stems, vascular tissue development, delayed senescence, membrane polarization, and reproductive development. Environmental signals can trigger brassinosteroid actions. Mutants have been identified that alter the response to brassinosteroid, but signal transduction pathways remain to be uncovered. From an evolutionary perspective, it will be quite interesting to see how these pathways compare with animal steroid signal transduction pathways.

Oligosaccharins

In addition to cellulose, plant cell walls are composed of numerous complex carbohydrates called oligosaccharides. There is some evidence that these cell wall components function as signaling molecules as well as structural wall components. Oligosaccharides that are proposed to have a hormonelike function are called oligosaccharins. Oligosaccharins can be released from the cell wall by enzymes secreted by pathogens. These carbohydrates are believed to signal defense responses, such as the hypersensitive response discussed later in this chapter. Another oligosaccharin has been shown to inhibit auxin-stimulated elongation of pea stems. These molecules are active at concentrations one to two orders of magnitude less than the traditional plant hormones. You have seen how auxin and cytokinin ratios can affect organogenesis in culture. Oligosaccharins also affect the phenotype of regenerated tobacco tissue, inhibiting root formation and stimulating flower production in tissues that are competent to regenerate flowers. How

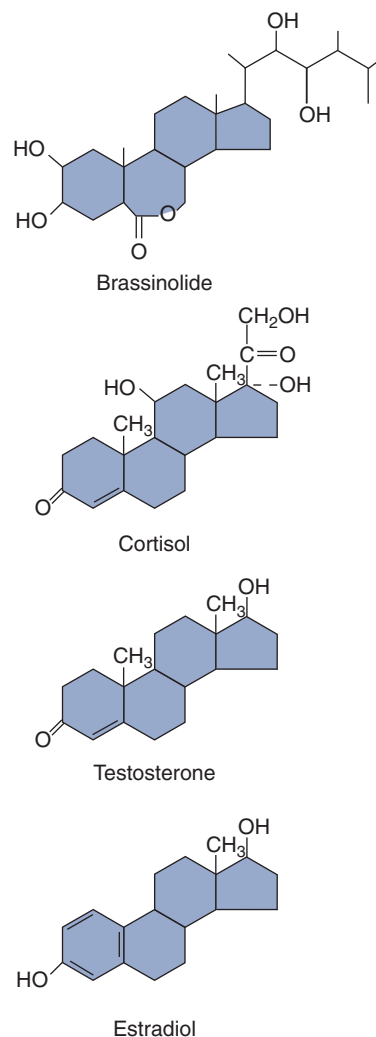


FIGURE 41.20
Brassinosteroids, such as brassinolide, have structural similarities to animal steroid hormones.

the culture results translate to in vivo systems is an open question. The structural biochemistry of oligosaccharins makes them particularly challenging molecules to study. How they interface with cells and initiate signal transduction pathways is an open question.

Brassinosteroids are structurally similar to animal steroid hormones. They have many effects on plant growth and development that parallel those of auxins and gibberellins. Oligosaccharins are complex carbohydrates that are released from cell walls and appear to regulate both pathogen responses and growth and development in some plants.

Ethylene

Long before its role as a plant hormone was appreciated, the simple, gaseous hydrocarbon ethylene ($\text{H}_2\text{C}=\text{CH}_2$) was known to defoliate plants when it leaked from gaslights in streetlamps. Ethylene is, however, a natural product of plant metabolism that, in minute amounts, interacts with other plant hormones. When auxin is transported down from the apical meristem of the stem, it stimulates the production of ethylene in the tissues around the lateral buds and thus retards their growth. Ethylene also suppresses stem and root elongation, probably in a similar way. An ethylene receptor has been identified and characterized. It appears to have evolved early in the evolution of photosynthetic organisms, sharing features with environmental-sensing proteins identified in bacteria.

Ethylene plays a major role in fruit ripening. At first, auxin, which is produced in significant amounts in pollinated flowers and developing fruits, stimulates ethylene production; this, in turn, hastens fruit ripening. Complex carbohydrates are broken down into simple sugars, chlorophylls are broken down, cell walls become soft, and the volatile compounds associated with flavor and scent in ripe fruits are produced.

One of the first observations that led to the recognition of ethylene as a plant hormone was the premature ripening in bananas produced by gases coming from oranges. Such relationships have led to major commercial uses of ethylene. For example, tomatoes are often picked green and artificially ripened later by the application of ethylene. Ethylene is widely used to speed the ripening of lemons and oranges as well. Carbon dioxide has the opposite effect of arresting ripening. Fruits are often shipped in an atmosphere of carbon dioxide. A biotechnology solution has also been developed (figure 41.21). One of the genes necessary for ethylene biosynthesis has been cloned, and its antisense copy has been inserted into the tomato genome. The antisense copy of the gene is a nucleotide sequence that is complementary to the sense copy of the gene. In this transgenic plant, both the sense and antisense sequences for the ethylene biosynthesis gene are transcribed. The sense and antisense mRNA sequences then pair with each other. This blocks translation, which requires single-stranded RNA; ethylene is not synthesized, and the transgenic tomatoes do not ripen. Sturdy green tomatoes can be shipped without ripening and

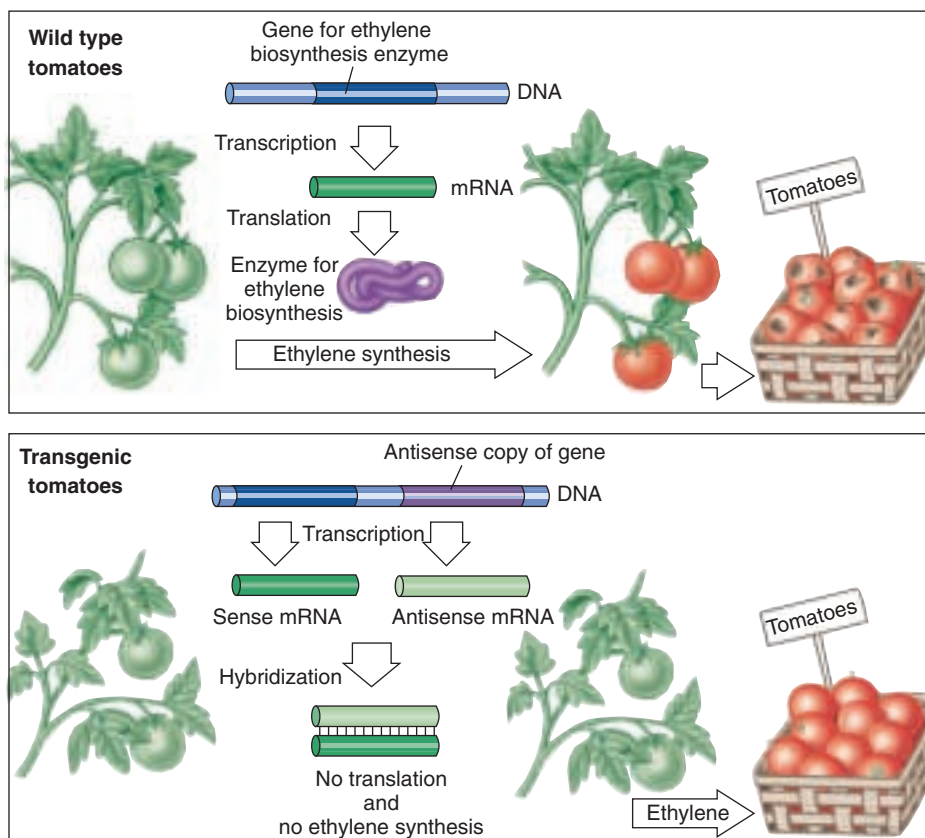


FIGURE 41.21

Genetic regulation of fruit ripening. An antisense copy of the gene for ethylene biosynthesis prevents the formation of ethylene and subsequent ripening of transgenic fruit. The antisense strand is complementary to the sequence for the ethylene biosynthesis gene. After transcription, the antisense mRNA pairs with the sense mRNA, and the double-stranded mRNA cannot be translated into a functional protein. Ethylene is not produced, and the fruit does not ripen. The fruit is sturdier for shipping in its unripened form and can be ripened later with exposure to ethylene. Thus, while wild-type tomatoes may already be rotten and damaged by the time they reach stores, transgenic tomatoes stay fresh longer.

rotting. Exposing these tomatoes to ethylene later will allow them to ripen.

Studies have shown that ethylene plays an important ecological role. Ethylene production increases rapidly when a plant is exposed to ozone and other toxic chemicals, temperature extremes, drought, attack by pathogens or herbivores, and other stresses. The increased production of ethylene that occurs can accelerate the loss of leaves or fruits that have been damaged by these stresses. Some of the damage associated with exposure to ozone is due to the ethylene produced by the plants. The production of ethylene by plants attacked by herbivores or infected with diseases may be a signal to activate the defense mechanisms of the plants. This may include the production of molecules toxic to the pests.

Ethylene, a simple gaseous hydrocarbon, is a naturally occurring plant hormone. Among its numerous effects is the stimulation of ripening in fruit. Ethylene production is also elevated in response to environmental stress.

Abscisic Acid

Abscisic acid, a naturally occurring plant hormone, appears to be synthesized mainly in mature green leaves, fruits, and root caps. The hormone earned its name because applications of it appear to stimulate leaf senescence (aging) and abscission, but there is little evidence that it plays an important role in this process. In fact, it is believed that abscisic acid may cause ethylene synthesis, and that it is actually the ethylene that promotes senescence and abscission. When abscisic acid is applied to a green leaf, the areas of contact turn yellow. Thus, abscisic acid has the exact opposite effect on a leaf from that of the cytokinins; a yellowing leaf will remain green in an area where cytokinins are applied.

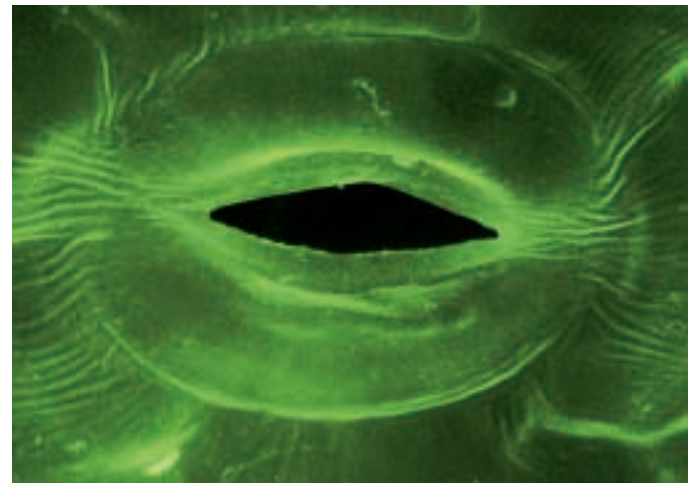
Abscisic acid probably induces the formation of winter buds—dormant buds that remain through the winter—by suppressing growth. The conversion of leaf primordia into bud scales follows (figure 41.22*a*). Like ethylene, it may also suppress growth of dormant lateral buds. It appears that abscisic acid, by suppressing growth and elongation of buds, can counteract some of the effects of gibberellins (which stimulate growth and elongation of buds); it also promotes senescence by counteracting auxin (which tends to retard senescence). Abscisic acid plays a role in seed dormancy and is antagonistic to gibberellins during germination. It is also important in controlling the opening and closing of stomata (figure 41.22*b*).

Abscisic acid occurs in all groups of plants and apparently has been functioning as a growth-regulating substance since early in the evolution of the plant kingdom. Relatively little is known about the exact nature of its physiological and biochemical effects. These effects are very rapid—often taking place within a minute or two—and therefore they must be at least partly independent of gene expression. Some longer-term effects of abscisic acid involve the regulation of gene expression, but the way this occurs is poorly understood. Abscisic acid levels become greatly elevated when the plant is subject to stress, especially drought. Like other plant hormones, abscisic acid probably will prove to have valuable commercial applications when its mode of action is better understood. It is a particularly strong candidate for understanding desiccation tolerance.

Abscisic acid, produced chiefly in mature green leaves and in fruits, suppresses growth of buds and promotes leaf senescence. It also plays an important role in controlling the opening and closing of stomata. Abscisic acid may be critical in ensuring survival under environmental stress, especially water stresses.



(a)



(b)

FIGURE 41.22

Effects of abscisic acid. (a) Abscisic acid plays a role in the formation of these winter buds of an American basswood. These buds will remain dormant for the winter, and bud scales—modified leaves—will protect the buds from desiccation. (b) Abscisic acid also affects the closing of stomata by influencing the movement of potassium ions out of guard cells.

41.3 The environment influences flowering.

Plants Undergo Metamorphosis

Overview of Initiating Flowering

Carefully regulated processes determine when and where flowers will form. Plants must often gain competence to respond to internal or external signals regulating flowering. Once plants are competent to reproduce, a combination of factors including light, temperature, and both promotive and inhibitory internal signals determine when a flower is produced (figure 41.23). These signals turn on genes that specify where the floral organs, sepals, petals, stamens, and carpels will form. Once cells have instructions to become a specific floral organ, yet another developmental cascade leads to the three-dimensional construction of flower parts.

Phase Change

Plants go through developmental changes leading to reproductive maturity just like many animals. This shift from juvenile to adult development is seen in the metamorphosis of a tadpole to an adult frog or caterpillar to a butterfly that can then reproduce. Plants undergo a similar metamorphosis that leads to the production of a flower. Unlike the frog that loses its tail, plants just keep adding on structures to existing structures with their meristems. At germination, most plants are incapable of

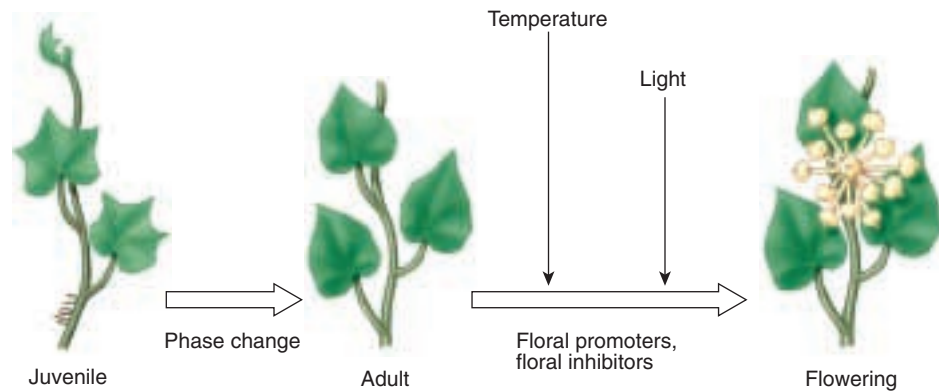


FIGURE 41.23

Factors involved in initiating flowering. This is a model of environmentally cued and internally processed events that result in a shoot meristem initiating flowers.

producing a flower, even if all the environmental cues are optimal. Internal developmental changes allow plants to obtain **competence** to respond to external and/or internal signals that trigger flower formation. This transition is referred to as **phase change**. Phase change can be morphologically obvious or very subtle. Take a look at an oak tree in the winter. The lower leaves will still be clinging to the branches, while the upper ones will be gone (figure 41.24a). Those lower branches were initiated by a juvenile meristem. The fact that they did not respond to environmental cues and drop their leaves indicates that they are young branches and have not made a phase change. Ivy also has distinctive juvenile and adult phases of growth (figure 41.24b). Stem tissue produced by a juvenile meristem initiates adventitious roots that can cling to walls. If



FIGURE 41.24

Phase change. (a) The lower branches of this oak tree represent the juvenile phase of development and cling to their leaves in the winter. The lower leaves are not able to form an abscission layer and break off the tree in the fall. Such visible changes are marks of phase change, but the real test is whether or not the plant is able to flower. (b) Juvenile ivy (*left*) makes adventitious roots and has an alternating leaf arrangement. Adult ivy (*right*) cannot make adventitious roots and has leaves with a different morphology that are arranged on an upright stem in a spiral.

you look at very old brick buildings that are covered with ivy, you will notice the uppermost branches are falling off because they have transitioned to the adult phase of growth and have lost the ability to produce adventitious roots. It is important to remember that even though a plant has reached the adult stage of development, it may or may not produce reproductive structures. Other factors may be necessary to trigger flowering.

Generally it is easier to get a plant to revert from an adult to vegetative state than to induce phase change experimentally. Applications of gibberellins and severe pruning can cause reversion. There is evidence in peas and *Arabidopsis* for genetically controlled repression of flowering. The *embryonic flower* mutant of *Arabidopsis* flowers almost immediately (figure 41.25), which is consistent with the hypothesis that the wild-type allele suppresses flowering. It is possible that flowering is the default state and that mechanisms have evolved to delay flowering. This delay allows the plant to store more energy to be allocated for reproduction.

The best example of inducing the juvenile to adult transition comes from the construction of transgenic plants that overexpress a gene necessary for flowering, that is found in many species. This gene, *LEAFY*, was cloned in *Arabidopsis* and its promoter was replaced with a viral promoter that results in constant, high levels of *LEAFY* transcription. This gene construct was then introduced into cultured aspen cells which were used to regenerate plants. When *LEAFY* is overexpressed in aspen, flowering occurs in weeks instead of years (figure 41.26). Phase change requires both sufficient signal and the ability to perceive the signal. Some plants acquire competence in the shoot to perceive a signal of a certain intensity. Others acquire competence to produce sufficient promotive signal(s) and/or decrease inhibitory signal(s).

Plants become reproductively competent through changes in signaling and perception. The transition to the adult stage of development where reproduction is possible is called phase change. Plants in the adult phase of development may or may not produce reproductive structures (flowers), depending on environmental cues.

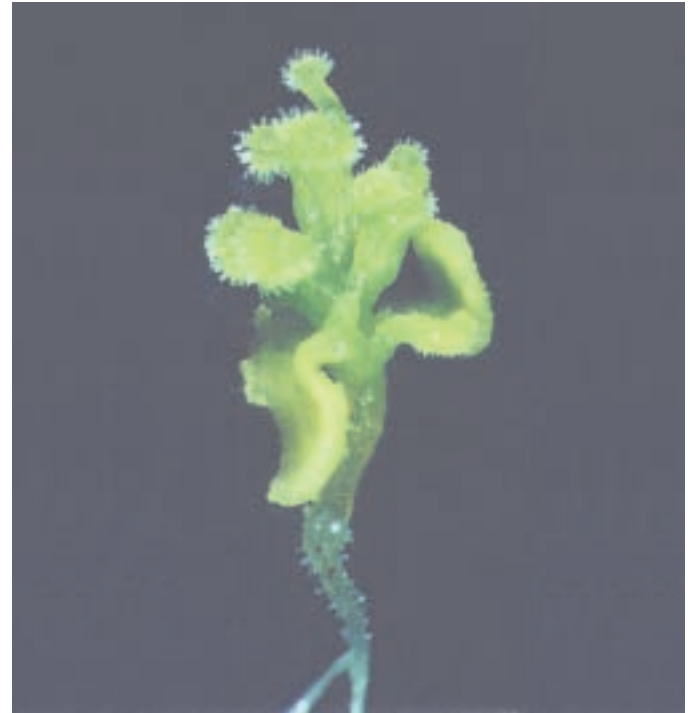


FIGURE 41.25
In *Arabidopsis*, the *embryonic flower* gene may repress flowering. The *embryonic flower* mutant flowers upon germination.



(a)



(b)

FIGURE 41.26
Overexpression of a flowering gene can accelerate phase change. (a) An aspen tree normally grows for several years before producing flowers. (b) Overexpression of the *Arabidopsis* flowering gene, *LEAFY*, causes rapid flowering in a transgenic aspen.

Pathways Leading to Flower Production

The environment can promote or repress flowering. In some cases, it can be relatively neutral. Light can be a signal that long, summer days have arrived in a temperate climate and conditions are favorable for reproduction. In other cases, plants depend on light to accumulate sufficient amounts of sucrose to fuel reproduction, but flower independently of the length of day. Temperature can also be used as a clue. Gibberellins are important and have been linked to the vernalization pathway. Clearly, reproductive success would be unlikely in the middle of a blizzard. Assuming regulation of reproduction first arose in more constant tropical environments, many of the day length and temperature controls would have evolved as plants colonized more temperate climates. Plants can rely primarily on one pathway, but all three pathways can be present. The complexity of the flowering pathways has been dissected physiologically. Now analysis of flowering mutants is providing insight into the molecular mechanisms of the floral pathways. The redundancy of pathways to flowering ensures that there will be another generation.

Light-Dependent Pathway

Flowering requires much energy accumulated via photosynthesis. Thus, all plants require light for flowering, but this is distinct from the **photoperiodic**, or light-dependent, flowering pathway. Aspects of growth and development in most plants are keyed to changes in the proportion of light to dark in the daily 24-hour cycle (day length). This provides a mechanism for organisms to respond to seasonal changes in the relative length of day and night. Day length changes with the seasons; the farther from the equator, the greater the variation. Flowering responses of plants to day length fall into several basic categories. When the daylight becomes shorter than a critical length, flowering is initiated in **short-day plants** (figure 41.27). When the daylight becomes longer than a critical length, flowering is initiated in **long-day plants**. Other plants, such as snapdragons, roses, and many native to the tropics (for example, tomatoes), will flower when mature regardless of day length, as long as they have received enough light for normal growth. These are referred to as **day-neutral plants**. Several grasses (for example, Indian grass, *Sorghastrum nutans*), as well as ivy, have two critical

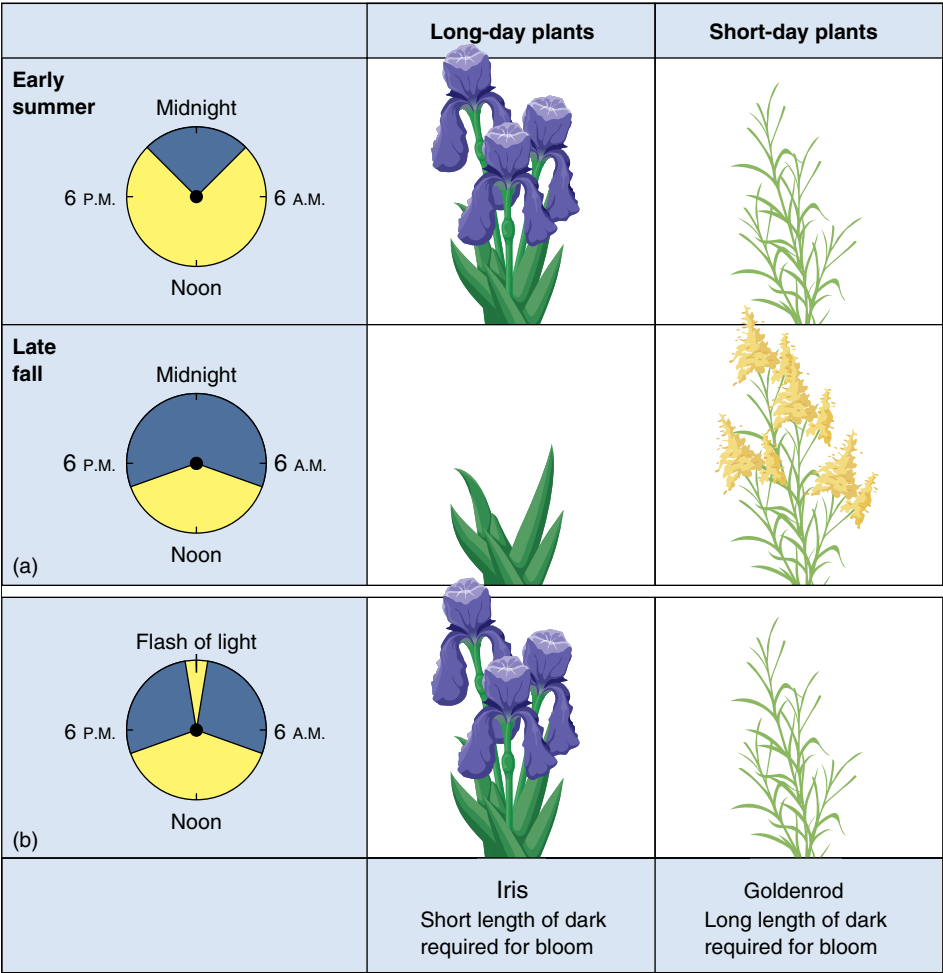


FIGURE 41.27
How flowering responds to day length.
 (a) This iris is a long-day plant that is stimulated by short nights to flower in the spring. The goldenrod is a short-day plant that, throughout its natural distribution in the northern hemisphere, is stimulated by long nights to flower in the fall. (b) If the long night of winter is artificially interrupted by a flash of light, the goldenrod will not flower, and the iris will. In each case, it is the duration of uninterrupted darkness that determines when flowering will occur.

photoperiods; they will not flower if the days are too long, and they also will not flower if the days are too short. In some species, there is a sharp distinction between long and short days. In others, flowering occurs more rapidly or slowly depending on the length of day. These plants rely on other flowering pathways as well and are called facultative long- or short-day plants. The garden pea is an example of a facultative long-day plant. In all of these plants, it is actually the length of darkness (night), not the length of day, that is physiologically significant. Using light as a cue permits plants to flower when abiotic environmental conditions are optimal, pollinators are available, and competition for resources with other plants may be less. For example, the spring ephemerals flower in the woods before the canopy leaves out, blocking sunlight necessary for photosynthesis.

At middle latitudes, most long-day plants flower in the spring and early summer; examples of such plants include clover, irises, lettuce, spinach, and hollyhocks. Short-day plants usually flower in late summer and fall, and include chrysanthemums, goldenrods, poinsettias, soybeans, and many weeds. Commercial plant growers use these responses to day length to bring plants into flower at specific times. For example, photoperiod is manipulated in greenhouses so poinsettias flower just in time for the winter holidays (figure 41.28). The geographic distribution of certain plants may be determined by flowering responses to day length.

Photoperiod is perceived by several different forms of phytochrome and also a blue-light-sensitive molecule (cryptochrome). The conformational change in a light receptor molecule triggers a cascade of events that leads to the production of a flower. There is a link between light and the circadian rhythm regulated by an internal clock that facilitates or inhibits flowering. At a molecular level the gaps between light signaling and production of flowers are rapidly filling in and the control mechanisms are quite complex. Here is one example of how day length affects a specific flowering gene in *Arabidopsis*, a facultative long-day plant that flowers in response to both far-red and blue light. Red light inhibits flowering. The gene *CONSTANS* (*CO*) is expressed under long days but not short days. The loss of *CO* product does not alter when a plant flowers under short days, but delays flowering under long days. What happens is that the gene is positively regulated by cryptochrome that perceives blue light under long days. Cryptochrome appears to inhibit the inhibition of flowering by phytochrome B exposed to red light. Simply put, flowering is promoted by repressing a gene that represses flowering! *CO* is a transcription factor that turns on other genes which results in the expression of *LEAFY*. As discussed in the section on phase change, *LEAFY* is one of the key genes that “tells” a meristem to switch over to flowering. We will see that other pathways also converge on this important gene.



FIGURE 41.28
Manipulation of photoperiod in greenhouses ensures that short-day poinsettias flower in time for the winter holidays. Note that the colorful “petals” are actually sepals. Even after flowering is induced, there are many developmental events leading to the production of species-specific flowers.

The Flowering Hormone: Does It Exist? The Holy Grail in plant biology has been a flowering hormone, quested unsuccessfully for more than 50 years. A considerable amount of evidence demonstrates the existence of substances that promote flowering and substances that inhibit it. Grafting experiments have shown that these substances can move from leaves to shoots. The complexity of their interactions, as well as the fact that multiple chemical messengers are evidently involved, has made this scientifically and commercially interesting search very difficult, and to this day, the existence of a flowering hormone remains strictly hypothetical. We do know that *LEAFY* can be expressed in the vegetative as well as the reproductive portions of plants. Clearly, information about day length gathered by leaves is transmitted to shoot apices. Given that there are multiple pathways to flowering, several signals may be facilitating communication between leaves and shoots. We also know that roots can be a source of floral inhibitors affecting shoot development.

Temperature-Dependent Pathway

Lysenko solved the problem of winter wheat seed rotting in the fields in Russia by chilling the seeds and planting them in the spring. Winter wheat would not flower without a period of chilling, called **vernalization**. Unfortunately a great many problems, including mistreatment of Russian geneticists, resulted from this scientifically significant discovery. Lysenko erroneously concluded that he had converted one species, winter wheat, to another, spring wheat, by simply altering the environment. There was a shift from science to politics. Genetics and Darwinian evolution were suspect for half a century. Social history aside, the valuable lesson here is that cold temperatures can accelerate or permit flowering in many species. As with light, this ensures that plants flower at more optimal times.

Vernalization may be necessary for seeds or plants in later stages of development. Analysis of mutants in *Arabidopsis* and pea indicate that vernalization is a separate flowering pathway that may be linked to the hormone gibberellin. In this pathway, repression may also lead to flowering. High levels of one of the genes in the pathway may block the promotion of flowering by gibberellins. When plants are chilled, there is less of this gene product and gibberellin activity may increase. It is known that gibberellins enhance the expression of *LEAFY*. One proposal is that both the vernalization and autonomous pathways share a common intersection affecting gibberellin promotion of flowering. Weigel has shown that gibberellin actually binds the promoter of the *LEAFY* gene, so its effect on flowering is direct. The connection between gibberellin levels and temperature also needs to be understood.

Autonomous Pathway

The autonomous pathway to flowering is independent of external cues except for basic nutrition. Presumably this was the first pathway to evolve. Day-neutral plants often depend primarily on the autonomous pathway which allows plants to “count” and “remember.” A field of day-neutral tobacco will produce a uniform number of nodes before flowering. If the shoots of these plants are removed at different positions, axillary buds will grow out and produce the same number of nodes as the removed portion of the shoot (figure 41.29). At a certain point in development shoots become committed or **determined** to flower (figure 41.30). The upper axillary buds of flowering tobacco will remember their position when rooted or grafted. The terminal shoot tip becomes florally determined about four nodes before it initiates a flower. In some other species, this commitment is less stable and/or occurs later.

How do shoots know where they are and at some point “remember” that information? It is clear that there are inhibitory signals from the roots. If bottomless pots are continuously placed over a growing tobacco plant and filled

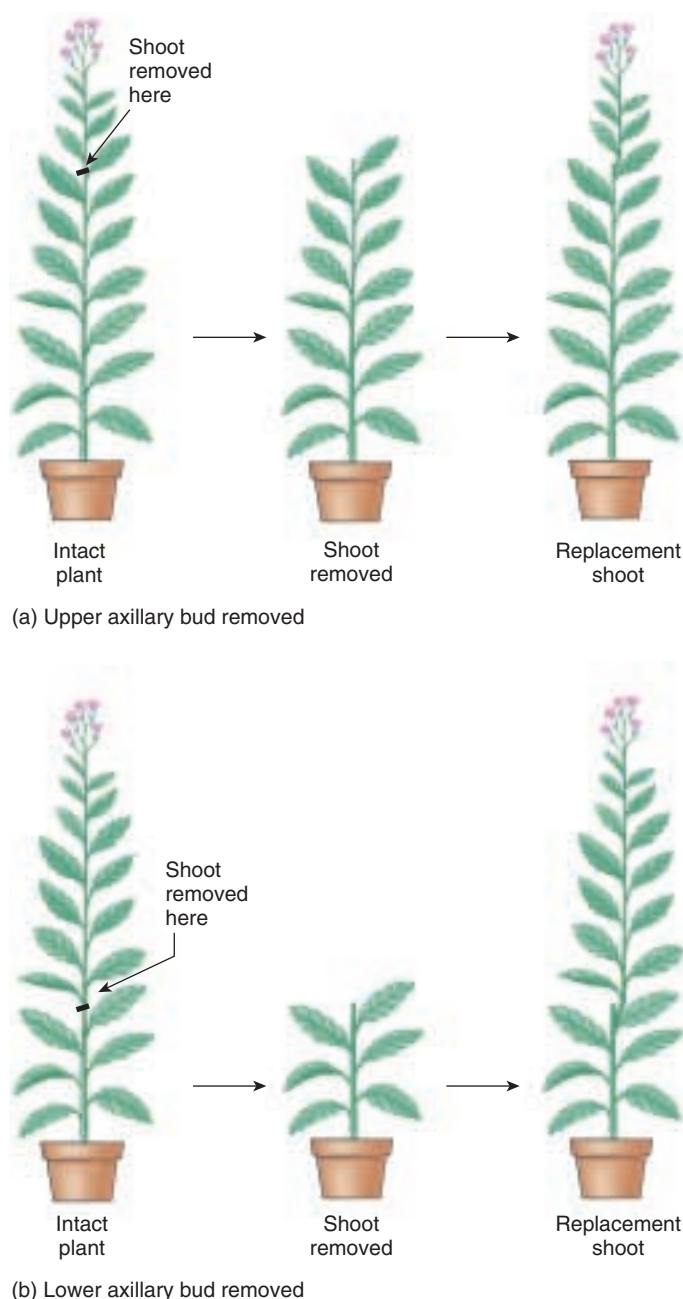


FIGURE 41.29

Plants can count. When axillary buds of flowering tobacco plants are released from apical dominance by removing the main shoot, they replace the number of nodes that were initiated by the main shoot. (After McDaniel 1996.)

with soil, flowering is delayed by the formation of adventitious roots (figure 41.31). Control experiments with leaf removal show that it is the addition of roots and not the loss of leaves that delays flowering. A balance between floral promoting and inhibiting signals may regulate when flowering occurs in the autonomous pathway and the other pathways as well.

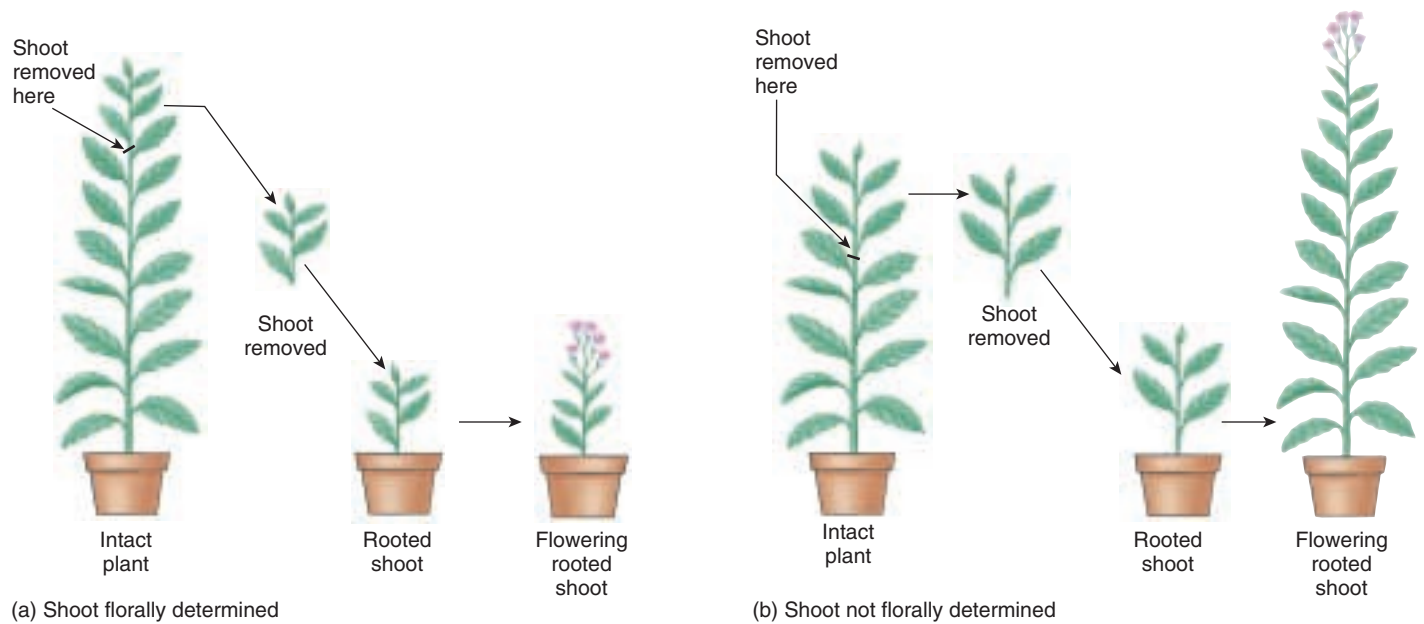


FIGURE 41.30

Plants can remember. At a certain point in the flowering process, shoots become committed to making a flower. This is called floral determination. (a) Florally determined axillary buds “remember” their position when rooted in a pot. That is, they produce the same number of nodes that they would have if they had grown out on the plant, and then they flower. (b) Those that are not yet florally determined cannot remember how many nodes they have left, so they start counting again. That is, they develop like a seedling and then flower. (After McDaniel 1996.)

Determination for flowering is tested at the organ or whole plant level by changing the environment and ascertaining whether or not the fate has changed. How does floral determination correlate with molecular level changes? In *Arabidopsis*, floral determination correlates with the increase of *LEAFY* gene expression and has occurred by the time a second flowering gene, *APETALA1*, is expressed. Because all three flowering pathways appear to converge with increased levels of *LEAFY*, this determination event should occur in species with a variety of balances among the pathways.

Plants use light receptor molecules to measure the length of night. This information is then used to signal pathways that promote or inhibit flowering. Light receptors in the leaves trigger events that result in changes in the shoot meristem. Vernalization is the requirement for a period of chilling before a plant can flower. The autonomous pathway leads to flowering independent of environmental cues. Plants integrate information about position in regulating flowering and both promoters and inhibitors of flowering are important.

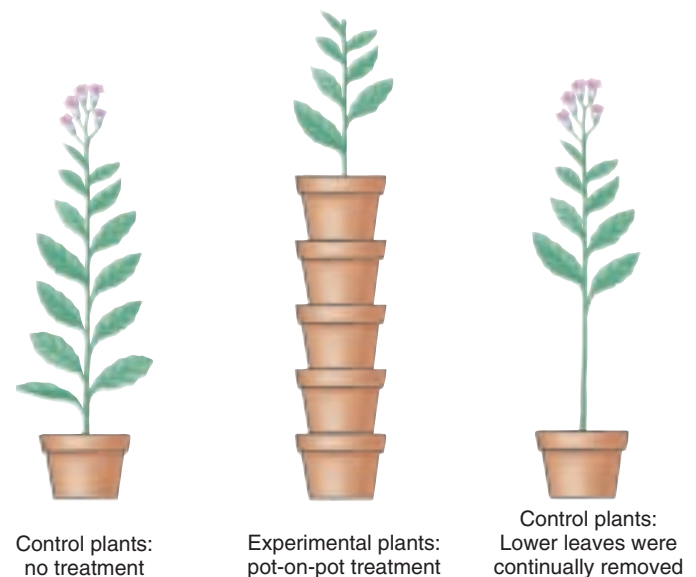


FIGURE 41.31

Roots can inhibit flowering. Adventitious roots formed as bottomless pots were continuously placed over growing tobacco plants, delaying flowering. The delay in flowering is caused by the roots, not the loss of the leaves. This was shown by removing leaves on control plants at the same time and in the same position as leaves on experimental plants that became buried as pots were added.

Identity Genes and the Formation of Floral Meristem and Floral Organs

Arabidopsis and snapdragon are valuable model systems for identifying flowering genes and understanding their interactions. The three pathways, discussed in the previous section, lead to an adult meristem becoming a floral meristem by either activating or repressing the inhibition of floral meristem identity genes (figure 41.32). Two of the key floral meristem identity genes are *LEAFY* and *APETALA1*. These genes establish the meristem as a flower meristem. They then turn on floral organ identity genes. The floral organ identity genes define four concentric whorls moving inward in the floral meristem as sepal, petal, stamen, and carpel. Meyerowitz and Coen proposed a model, called the ABC model, to explain how three classes of floral organ identity genes could specify four distinct organ types (figure 41.33). The ABC model proposes that three classes of organ identity genes (*A*, *B*, and *C*) specify the floral organs in the four floral whorls. By studying mutants the researchers have determined the following:

1. Class *A* genes alone specify the sepals.
2. Class *A* and class *B* genes together specify the petals.
3. Class *B* and class *C* genes together specify the stamens.
4. Class *C* genes alone specify the carpels.

The beauty of their ABC model is that it is entirely testable by making different combinations of floral organ identity

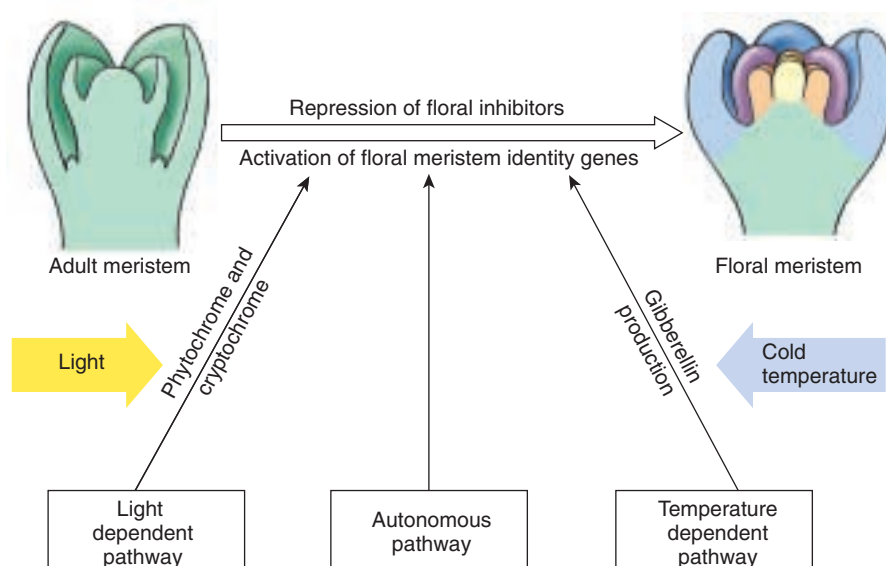
mutants. Each class of genes is expressed in two whorls, yielding four different combinations of the gene products. When any one class is missing, there are aberrant floral organs in predictable positions.

It is important to recognize that this is actually only the beginning of the making of a flower. These organ identity genes are transcription factors that turn on many more genes that will actually give rise to the three-dimensional flower. There are also genes that “paint” the petals. Complex biochemical pathways lead to the accumulation of anthocyanin pigments in vacuoles. These pigments can be orange, red, or purple and the actual color is influenced by pH and by the shape of the petal.

The Formation of Gametes

The ovule within the carpel has origins more ancient than the angiosperms. Floral parts are modified leaves, and within the ovule is the female gametophyte. This next generation develops from placental tissue in the ovary. A megaspore mother cell develops and meiotically gives rise to the embryo sac. Usually two layers of integument tissue form around this embryo sac and will become the seed coat. Genes responsible for the initiation of integuments and also those responsible for the formation of the integument have been identified. Some also affect leaf structure. This chapter has focused on the complex and elegant process that gives rise to the reproductive structure called the flower. It is indeed a metamorphosis, but the subtle shift from mitosis to meiosis in the megaspore mother cell

leading to the development of a haploid, gamete-producing gametophyte is perhaps even more critical. The same can be said for pollen formation in the anther of the stamen. As we will see in the next chapter, the flower houses the haploid generations that will produce gametes. The flower also functions to increase the probability that male and female gametes from different (or sometimes the same plant) will unite.



Floral structures form as a result of floral meristem identity genes turning on floral organ identity genes which specify where sepals, petals, stamens, and carpels will form. This is followed by organ development which involves many complex pathways that account for floral diversity among species.

FIGURE 41.32

Model for flowering. The light-dependent, temperature-dependent, and autonomous flowering pathways promote the formation of floral meristems from adult meristems by repressing floral inhibitors and activating floral meristem identity genes.

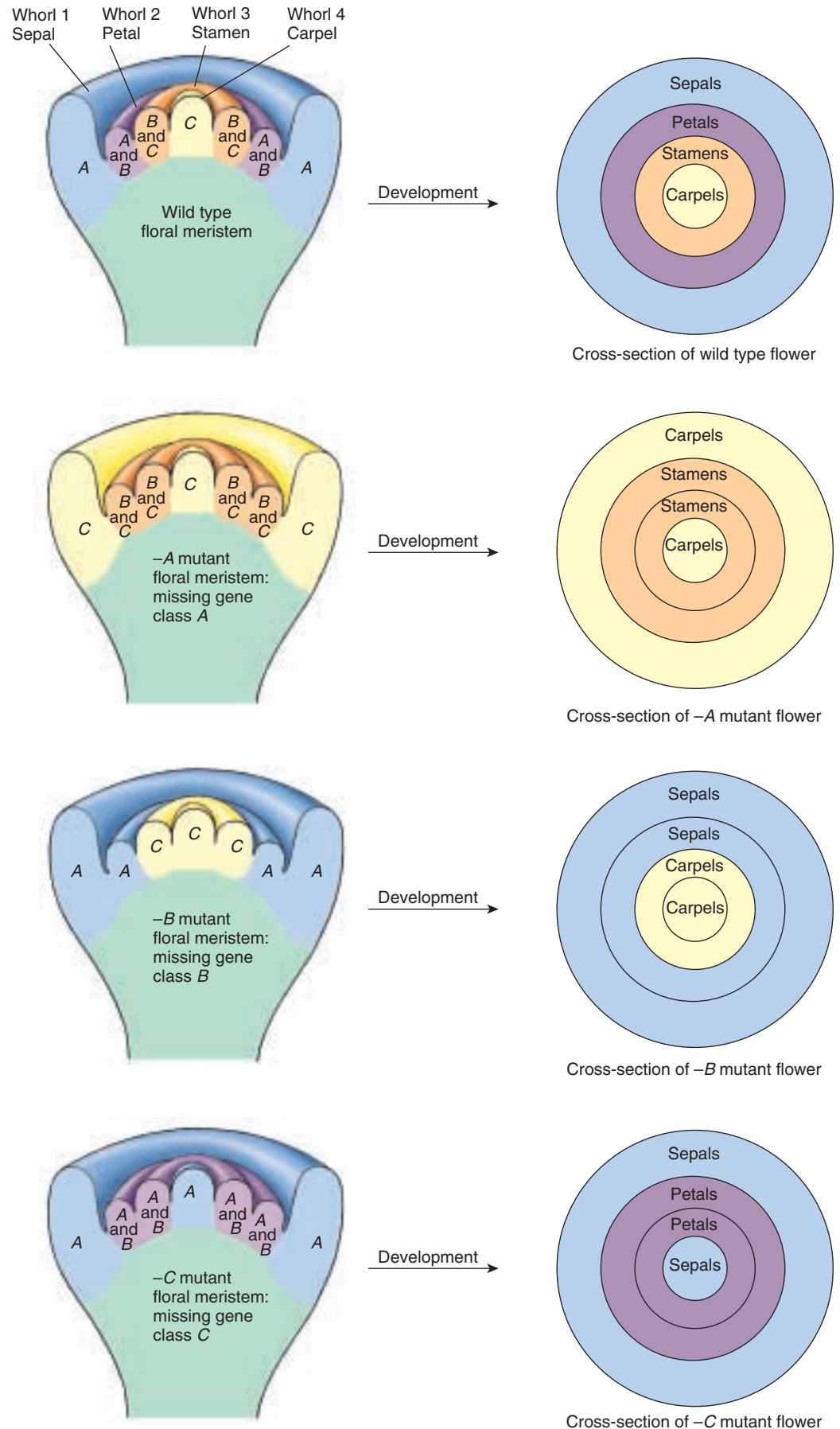


FIGURE 41.33

ABC model for floral organ specification.

Letters labeling whorls indicate which gene classes are active. When *A* function is lost (*-A*), *C* expands to the first and second whorls. When *B* function is lost (*-B*), both outer two whorls have just *A* function, and both inner two whorls have just *C* function; none of the whorls have dual gene function. When *C* function is lost (*-C*), *A* expands into the inner two whorls. These new combinations of gene expression patterns alter which floral structures form in each whorl. (Model proposed by Coen and Meyerowitz, 1991.)

41.4 Many short-term responses to the environment do not require growth.

Larger predators, microbes, water, and wind often present a plant with rapid immediate stress. Response, to be effective, must also be immediate. There is little time for growth, and plants instead invoke a variety of other kinds of responses. Many environmental cues trigger rapid and reversible localized plant movements, for example. The rapid folding of leaves can startle a potential predator. Leaf folding can also prevent water loss by reducing the surface area available for transpiration. Some localized plant movements are triggered by unpredictable environmental signals. Other movements are tied into daily internal rhythms established by cyclic environmental signals like light and temperature. Plants lack a nervous system in the conventional sense. Some of the rapid signaling, however, is the result of electric charge moving through an organ as a wave of membrane ion exchange, not unlike that seen in animals. This is translated into movement by changing the turgor pressure of cells.

Turgor Movement

Turgor is pressure within a living cell resulting from diffusion of water into it. If water leaves turgid cells (ones with

turgor pressure), the cells may collapse, causing plant movement; conversely, water entering a limp cell may also cause movement as the cell once more becomes turgid.

Many other plants, including those of the legume family (Fabaceae), exhibit leaf movements in response to touch or other stimuli. After exposure to a stimulus, the changes in leaf orientation are mostly associated with rapid turgor pressure changes in **pulvini** (singular: pulvinus), which are multicellular swellings located at the base of each leaf or leaflet. When leaves with pulvini, such as those of the sensitive plant (*Mimosa pudica*), are stimulated by wind, heat, touch, or, in some instances, intense light, an electrical signal is generated. The electrical signal is translated into a chemical signal, with potassium ions, followed by water, migrating from the cells in one half of a pulvinus to the intercellular spaces in the other half. The loss of turgor in half of the pulvinus causes the leaf to “fold.” The movements of the leaves and leaflets of the sensitive plant are especially rapid; the folding occurs within a second or two after the leaves are touched (figure 41.34). Over a span of about 15 to 30 minutes after the leaves and leaflets have folded, water usually diffuses back into the same cells from which it left, and the leaf returns to its original position.

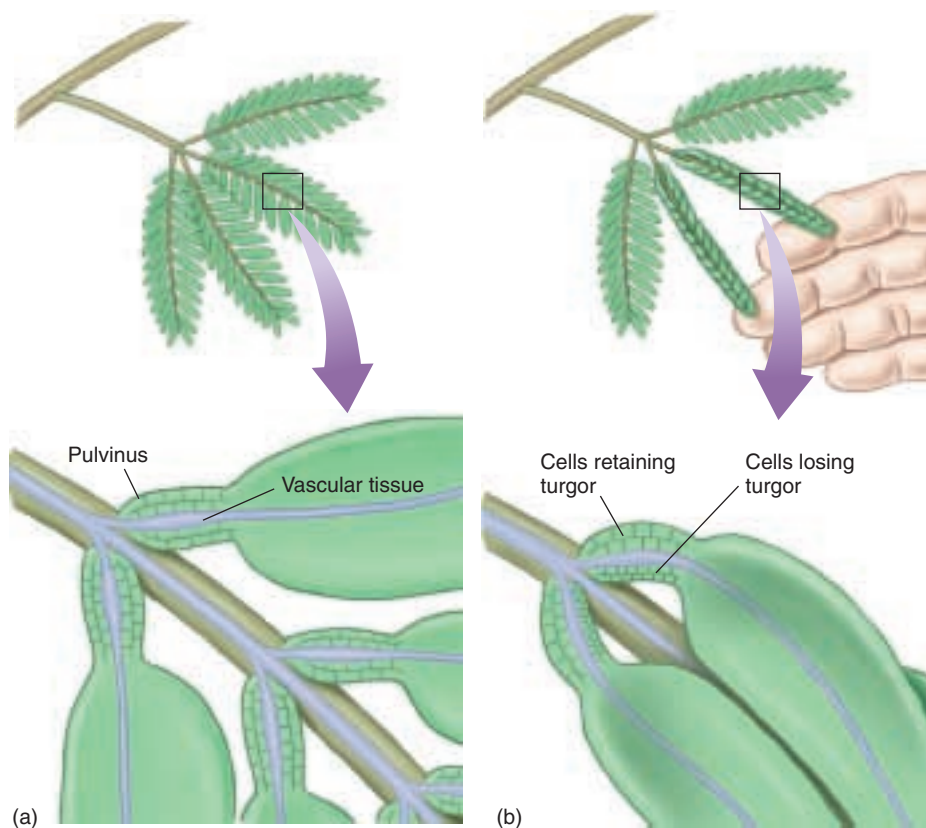


FIGURE 41.34
Sensitive plant (*Mimosa pudica*). (a) The blades of *Mimosa* leaves are divided into numerous leaflets; at the base of each leaflet is a swollen structure called a pulvinus. (b) Changes in turgor cause leaflets to fold in response to a stimulus. (c) When leaves are touched (center two leaves above), they fold due to loss of turgor.

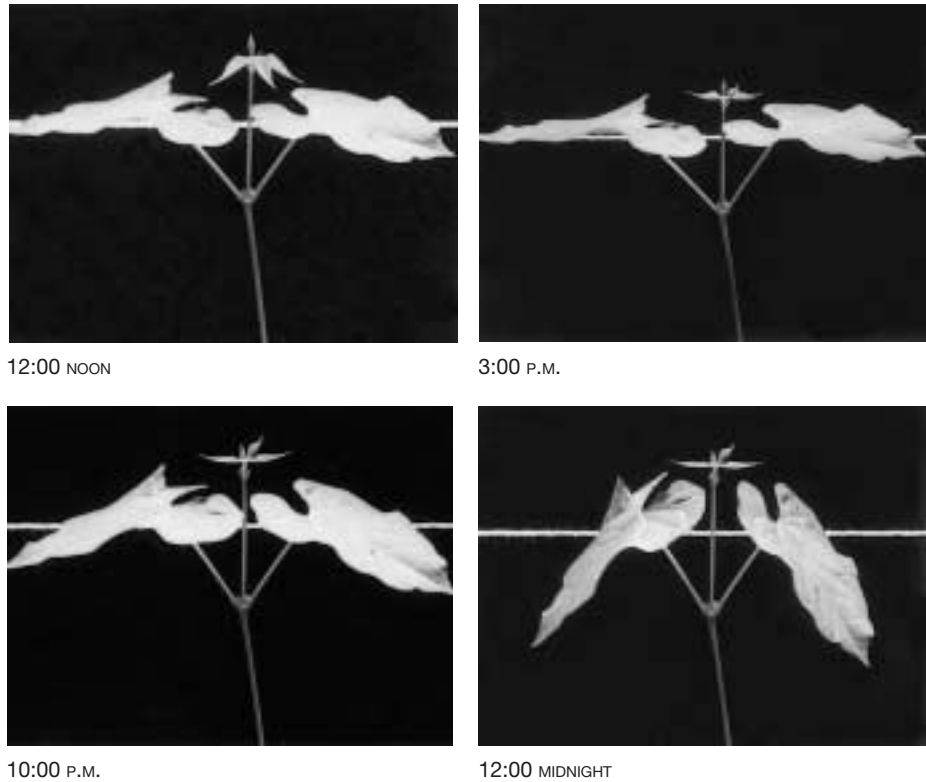


FIGURE 41.35
Sleep movements in bean leaves. In the bean plant, leaf blades are oriented horizontally during the day and vertically at night.

The leaves of some plants with similar mechanisms may track the sun, with their blades oriented at right angles to it; how their orientation is directed is, however, poorly understood. Such leaves can move quite rapidly (as much as 15° an hour). This movement maximizes photosynthesis and is analogous to solar panels that are designed to track the sun.

Some of the most familiar of these reversible changes are seen in leaves and flowers that “open” during the day and “close” at night. For example, the flowers of four o’clocks open at 4 P.M. and evening primrose petals open at night. The blades of plant leaves that exhibit such a daily shift in position may not actually fold; instead, their orientation may be changed as a result of **turgor movements**. Bean leaves are horizontal during the day when their pulvini are turgid, but become more or less vertical at night as the pulvini lose turgor (figure 41.35). These sleep movements reduce water loss from transpiration during the night, but maximize photosynthetic surface area during the day. In these cases, the movement is closely tied to an internal rhythm.

Circadian Clocks

How do leaves know when to “sleep”? They have endogenous circadian clocks that set a rhythm with a period of about 24 hours (actually it is closer to 22 or 23 hours). While there are shorter and much longer, naturally occur-

ring rhythms, circadian rhythms are particularly common and widespread because of the day-night cycle on earth. Jean de Mairan, a French astronomer, first identified circadian rhythms in 1729. He studied the sensitive plant which, in addition to having a touch response, closes its leaflets and leaves at night like the bean plant described above. When de Mairan put the plants in total darkness, they continued “sleeping” and “waking” just as they did when exposed to night and day. This is one of four characteristics of a circadian rhythm: it must continue to run in the absence of external inputs. It must be about 24 hours in duration and can be reset or entrained. (Perhaps you’ve experienced entrainment when traveling to a different time zone in the form of jet-lag recovery.) The fourth characteristic is that the clock can compensate for differences in temperature. This is quite unique when you consider what you know about biochemical reactions; most rates of reactions vary significantly based on temperature. Circadian clocks exist in many organisms and appear to have evolved independently multiple times. The mechanism behind the clock is not fully understood, but is being actively investigated at the molecular level.

Turgor movements of plants are reversible and involve changes in the turgor pressure of specific cells.
Circadian clocks are endogenous timekeepers that keep plant movements and other responses synchronized with the environment.

Plant Defense Responses

Interactions between plants and other organisms can be symbiotic (for example, nitrogen-fixing bacteria and mycorrhizae) or pathogenic. In evolutionary terms, these two types of interactions may simply be opposite sides of the same coin. The interactions have many common aspects and are the result of coevolution between two species that signal and respond to each other. In the case of pathogens, the microbe or pest is “winning,” at least for that second in evolutionary time. In chapter 38, we discussed surface barriers the plant constructs to block invasion. In this section, we will focus on cellular level responses to attacks by microbes and animals.

Recognizing the Invader

Half a century ago, Flor proposed that there is a plant resistance gene (*R*) whose product interacts with that of a pathogen avirulence gene (*avr*). This is called the gene-for-gene model and several pairs of *avr* and *R* genes have been cloned in different species pathogenized by microbes, fungi, and insects, in one case. This has been motivated partially by the agronomic benefit of identifying genes that can be added to protect other plants from invaders. Much is now known about the signal transduction pathways that follow the recognition of the pathogen by the *R* gene. These pathways lead to the triggering of the hypersensitive response (HR) which leads to rapid cell death around the source of the invasion and also a longer-term resistance (figure 41.36). There is not always a gene-for-gene response, but plants still have defense responses to pathogens and also mechanical wounding. Some of the response pathways may be similar. Also, oligosaccharins in the cell walls may serve as recognition and signaling molecules.

While our focus is on invaders outside the plant kingdom, more is being learned about how parasitic plants invade other plants. There are specific molecules released from the root hairs of the host that the parasitic plant recognizes and responds to with invasive action. Less is known about the host response and so far the different defense genes that are activated appear to be ineffective.

Responding to the Invader

When a plant is attacked and there is gene-for-gene recognition, the HR response leads to very rapid cell death around the site of attack. This seals off the wounded tissue

to prevent the pathogen or pest from moving into the rest of the plant. Hydrogen peroxide and nitric oxide are produced and may signal a cascade of biochemical events resulting in the localized death of host cells. They may also have negative effects on the pathogen, although antioxidant abilities have coevolved in pathogens. Other antimicrobial agents produced include the phytoalexins which are chemical defense agents. A variety of pathogenesis-related genes (*PR* genes) are expressed and their proteins can either function as antimicrobial agents or signals for other events that protect the plant.

In the case of virulent invaders (no *R* gene recognition), there are changes in local cell walls that at least partially block the movement of the pathogen or pest farther into the plant. In this case there is not an HR response and the local plant cells are not suicidal.

When an insect takes a mouthful of a leaf, defense responses are also triggered. Mechanical damage causes responses that have some similar components, but the reaction may be slower. Biochemically, it is distinct from some of the events triggered by signals in the insect's mouth. Such responses are collectively called wound responses. Wound responses are a challenge in designing other types of experiments with plants that involve cutting or otherwise mechanically damaging the tissue. It is important to run control experiments to be sure you are answering your question and not observing a wound response.

Preparing for Future Attacks

In addition to the HR or other local responses, plants are capable of a systemic response to a pathogen or pest attack. This is called a systemic acquired response (SAR). Several pathways lead to broad-ranging resistance that lasts for a period of days. The signals that induce SAR include salicylic acid and jasmonic acid. Salicylic acid is the active ingredient in aspirin too! SAR allows the plant to respond more quickly if it is attacked again. However, this is not the same as the human immune response where antibodies (proteins) that recognize specific antigens (foreign proteins) persist in the body. SAR is neither as specific or long lasting.

Plants defend themselves from invasion in ways reminiscent of the animal immune system. When an invader is recognized, localized cell death seals off the infected area.

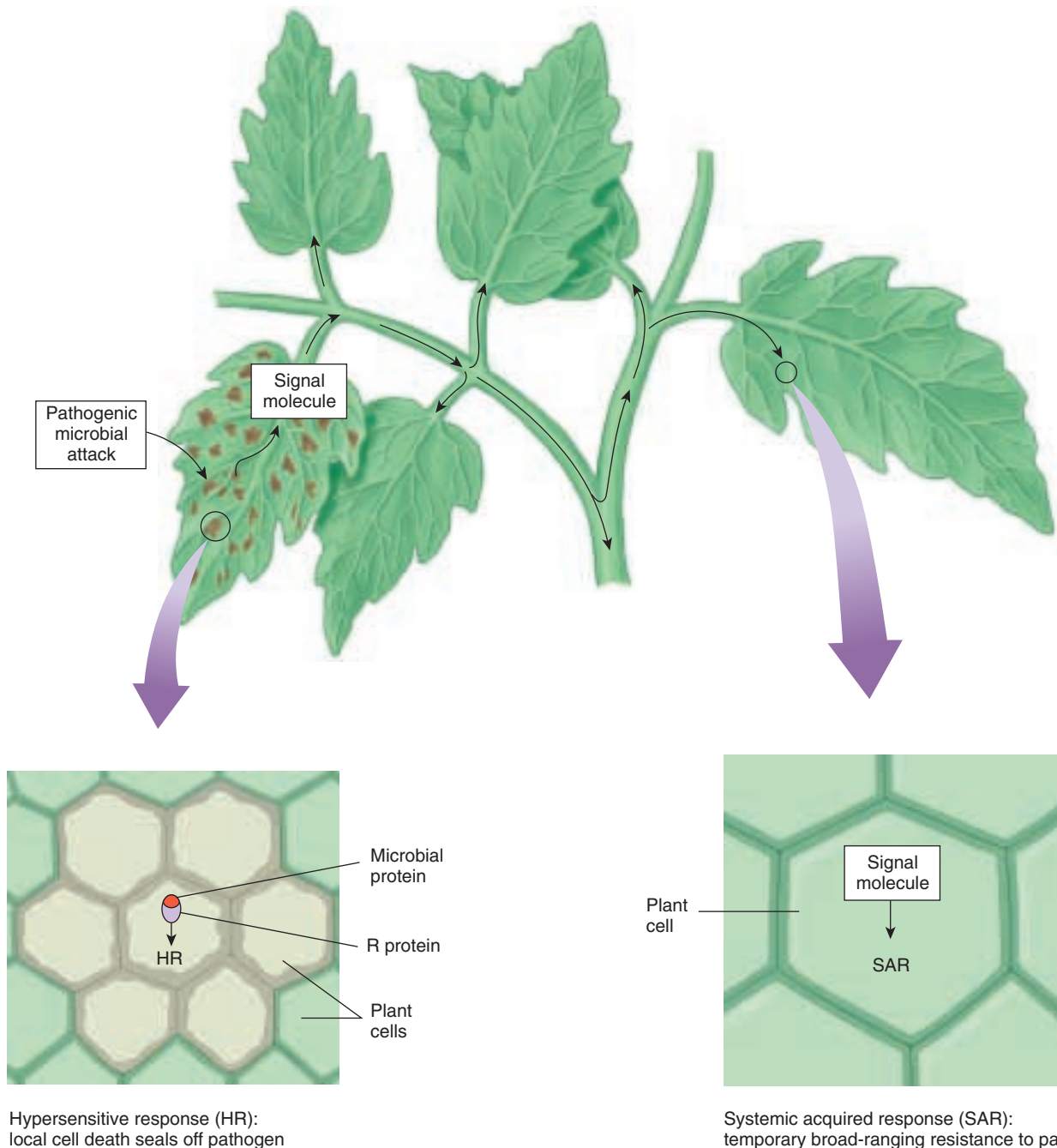


FIGURE 41.36

Plant defense responses. In the gene-for-gene response, a cascade of events is triggered leading to local cell death (HR response) and longer-term resistance in the rest of the plant (SAR).



Summary

Questions

Media Resources

41.1 Plant growth is often guided by environmental cues.

- Tropisms in plants are growth responses to external stimuli, such as light, gravity, or contact.
- Dormancy is a plant adaptation that carries a plant through unfavorable seasons or periods of drought.

1. In general, which part of a plant is positively phototropic? What is the adaptive significance of this reaction?



• Photoperiod

41.2 The hormones that guide growth are keyed to the environment.

- Auxin migrates away from light and promotes the elongation of plant cells on the dark side, causing stems to bend in the direction of light.
- Cytokinins are necessary for mitosis and cell division in plants. They promote growth of lateral buds and inhibit formation of lateral roots.
- Gibberellins, along with auxin, play a major role in stem elongation in most plants. They also tend to hasten the germination of seeds and to break dormancy in buds.

2. How does auxin affect the plasticity of the plant cell walls?

3. Where are most cytokinins produced? From what biomolecule do cytokinins appear to be derived?

4. What plant hormones could be lacking in genetically dwarfed plants?



• Hormones



• Student Research: Plant Growth

41.3 The environment influences flowering.

- The transition of a shoot meristem from vegetative to adult development is called phase change. During phase change, plants gain competence to produce a floral signal(s) and/or perceive a signal.
- The light-dependent pathway uses information from light receptor molecules integrated with a biological clock to determine if the length of night is sufficient for flowering.
- The autonomous path functions independently of environmental cues. Internal floral inhibitor(s) from roots and leaves and floral promoter(s) from leaves move through the plant.

5. A plant has undergone phase change. Although it is an adult, it does not flower. How might you get this plant to flower?

6. You have recently moved from Canada to Mexico and brought some seeds from your favorite plants. They germinate and produce beautiful leaves, but never flower. What went wrong?



• Student Research: Selection in Flowering Plants

41.4 Many short-term responses to the environment do not require growth.

- Changes in turgor pressure reflect responses to environmental signals that can protect plants from predation.
- Other reversible movements in plants are caused by changes in turgor pressure that are regulated by internal circadian rhythms.
- Plants have the ability to recognize and respond to invaders through cellular level recognition and response.

7. How are motor cells involved in the function of the pulvinus? What happens in the motor cells of the sensitive plant (*Mimosa pudica*) when its leaves are touched?

8. In what ways can a plant protect itself from pathogenic microbes? From animals?

42

Plant Reproduction

Concept Outline

42.1 Angiosperms have been incredibly successful, in part, because of their reproductive strategies.

Rise of the Flowering Plants. Animal and wind dispersal of pollen increases genetic variability in a species. Seed and fruit dispersal mechanisms allow offspring to colonize distant regions. Other features such as shortened life cycles may also have been responsible for the rapid diversification of the flowering plants.

Evolution of the Flower. A complete flower has four whorls, containing protective sepals, attractive petals, male stamens, and female ovules.

42.2 Flowering plants use animals or wind to transfer pollen between flowers.

Formation of Angiosperm Gametes. The male gametophytes are the pollen grains, and the female gametophyte is the embryo sac.

Pollination. Evolutionary modifications of flowers have enhanced effective pollination.

Self-Pollination. Self-pollination is favored in stable environments, but outcrossing enhances genetic variability.

Fertilization. Angiosperms use two sperm cells, one to fertilize the egg, the other to produce a nutrient tissue called endosperm.

42.3 Many plants can clone themselves by asexual reproduction.

Asexual Reproduction. Some plants do without sexual reproduction, instead cloning new individuals from parts of themselves.

42.4 How long do plants and plant organs live?

The Life Span of Plants. Clonal plants can live indefinitely through their propagules. Parts of plants senesce and die. Some plants reproduce sexually only once and die.



FIGURE 42.1

Reproductive success in flowering plants. Unique reproductive systems and strategies have coevolved between plants and animals, accounting for almost 250,000 flowering plants inhabiting all but the harshest environments on earth.

The remarkable evolutionary success of flowering plants can be linked to their reproductive strategies (figure 42.1). The evolution and development of flowers has been discussed in chapters 37 and 41. Here we explore reproductive strategies in the angiosperms and how their unique features, flowers and fruits, have contributed to their success. This is, in part, a story of coevolution between plants and animals that ensures greater genetic diversity by dispersing plant gametes widely. However, in a stable environment, there are advantages to maintaining the status quo genetically. Asexual reproduction is a strategy to clonally propagate individuals. An unusual twist to sexual reproduction in some flowering plants is that senescence and death of the parent plant follows.

42.1 Angiosperms have been incredibly successful, in part, because of their reproductive strategies.

Rise of the Flowering Plants

Most of the plants we see daily are angiosperms. The 250,000 species of flowering plants range in size from almost microscopic herbs to giant *Eucalyptus* trees, and their

form varies from cacti, grasses, and daisies to aquatic pondweeds. Most shrubs and trees (other than conifers and *Ginkgo*) are also in this phylum. This chapter focuses on reproduction in angiosperms (figure 42.2) because of their tremendous success and many uses by humans. Virtually all

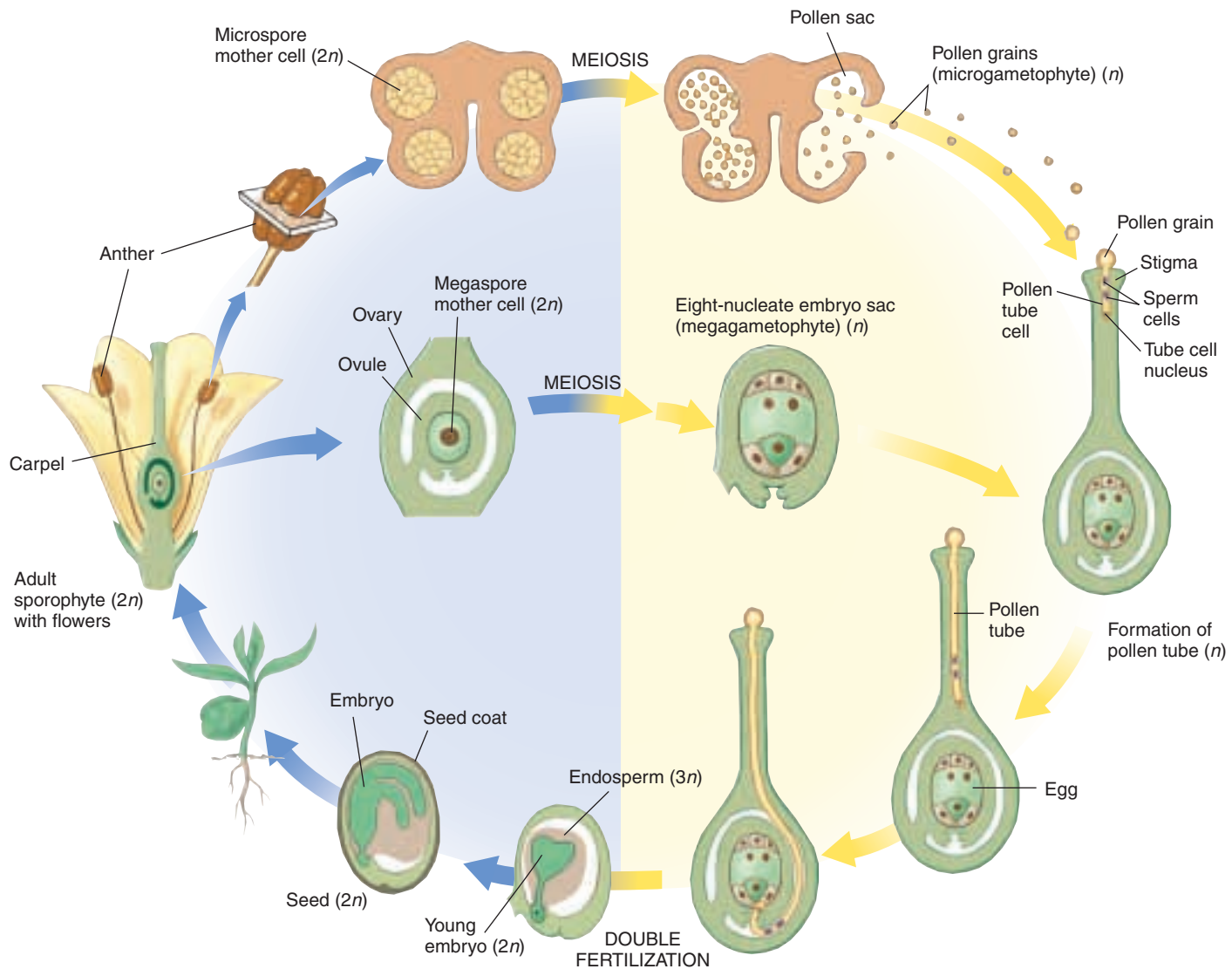


FIGURE 42.2

Angiosperm life cycle. Eggs form within the embryo sac inside the ovules, which, in turn, are enclosed in the carpels. The pollen grains, meanwhile, are formed within the sporangia of the anthers and are shed. Fertilization is a double process. A sperm and an egg come together, producing a zygote; at the same time, another sperm fuses with the polar nuclei to produce the endosperm. The endosperm is the tissue, unique to angiosperms, that nourishes the embryo and young plant.

of our food is derived, directly or indirectly, from flowering plants; in fact, more than 90% of the calories we consume come from just over 100 species. Angiosperms are also sources of medicine, clothing, and building materials. While the other plant phyla also provide resources, they are outnumbered seven to one by the angiosperms. For example, there are only about 750 extant gymnosperm species!

Why Are the Angiosperms Successful?

When flowering plants originated, Africa and South America were still connected to each other, as well as to Antarctica and India, and, via Antarctica, to Australia and New Zealand (figure 42.3). These landmasses formed the great continent known as Gondwanaland. In the north, Eurasia and North America were united, forming another supercontinent called Laurasia. The huge landmass formed by the union of South America and Africa spanned the equator and probably had a climate characterized by extreme temperatures and aridity in its interior. Similar climates occur in the interiors of major continents at present. Much of the early evolution of angiosperms may have taken place in patches of drier and less favorable habitat found in the interior of Gondwanaland. Many features of flowering plants seem to correlate with successful growth under arid and semiarid conditions.

The transfer of pollen between flowers of separate plants, sometimes over long distances, ensures *outcrossing* (cross-pollination between individuals of the same species) and may have been important in the early success of angiosperms. The various means of effective fruit dispersal that evolved in the group were also significant in the success of angiosperms (see chapter 40). The rapid life cycle of some of the angiosperms (*Arabidopsis* can go from seed to adult flowering plant in 24 days) was another factor. Asexual reproduction has given many invasive species a competitive edge. Xylem vessels and other anatomical and morphological features of the angiosperms correlate with their biological success. As early angiosperms evolved, all of these advantageous features became further elaborated and developed, and the pace of their diversification accelerated.

The Rise to Dominance

Angiosperms began to dominate temperate and tropical terrestrial communities about 80 to 90 million years ago, during the second half of the Cretaceous Period. We can document the relative abundance of different groups of plants by studying fossils that occur at the same time and place. In rocks more than 80 million years old, the fossil remains of plant phyla other than angiosperms, including lycopods, horsetails, ferns, and gymnosperms, are

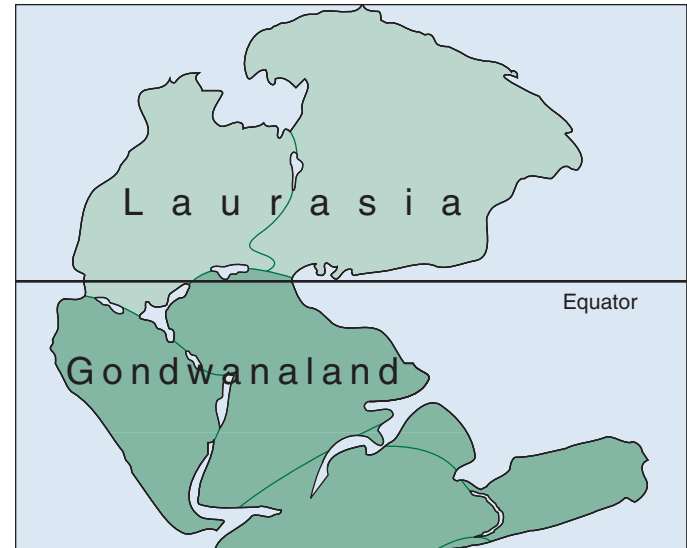


FIGURE 42.3

The alignment of the continents when the angiosperms first appeared in the fossil record about 130 million years ago. Africa, Madagascar, South America, India, Australia, and Antarctica were all connected and part of the huge continent of Gondwanaland, which eventually separated into the discrete landmasses we have today.

most common. Angiosperms arose in temperate and tropical terrestrial communities in a relatively short time.

At about the time that angiosperms became abundant in the fossil record, pollen, leaves, flowers, and fruits of some families that still survive began to appear. For example, representatives of the magnolia, beech, and legume families, which were in existence before the end of the Cretaceous Period (65 million years ago), are alive and flourishing today.

A number of insect orders that are particularly associated with flowers, such as Lepidoptera (butterflies and moths) and Diptera (flies), appeared or became more abundant during the rise of angiosperms. Plants and insects have clearly played a major role in each other's patterns of evolution, and their interactions continue to be of fundamental importance. Additional animals, including birds and mammals, now assist in pollination and seed dispersal.

By 80 to 90 million years ago, angiosperms were dominant in terrestrial habitats throughout the world.

Evolution of the Flower

Pollination in angiosperms does not involve direct contact between the pollen grain and the ovule. Pollen matures within the anthers and is transported, often by insects, birds, or other animals, to the stigma of another flower. When pollen reaches the stigma, it germinates, and a pollen tube grows down, carrying the sperm nuclei to the embryo sac. After double fertilization takes place, development of the embryo and endosperm begins. The seed matures within the ripening fruit; the germination of the seed initiates another life cycle.

Successful pollination in many angiosperms depends on the regular attraction of **pollinators** such as insects, birds, and other animals, so that pollen is transferred between plants of the same species. When animals disperse pollen, they perform the same functions for flowering plants that they do for themselves when they actively search out mates. The relationship between plant and pollinator can be quite intricate. Mutations in either partner can block reproduction. If a plant flowers at the “wrong” time, the pollinator may not be available. If the morphology of the flower or pollinator is altered, there may be physical barriers to pollination. Clearly floral morphology has coevolved with pollinators and the result is much more complex and diverse than the initiation of four distinct whorls of organs described in chapter 40.

Characteristics of Floral Evolution

The evolution of the angiosperms is a focus of chapter 37. Here we need to keep in mind that the diversity of angiosperms is partly due to the evolution of a great variety of floral phenotypes that may enhance the effectiveness of pollination. All floral organs are thought to have evolved from leaves. In early angiosperms, these organs maintain the spiral phyllotaxy often found in leaves. The trend has been toward four distinct whorls. A *complete flower* has four whorls of parts (calyx, corolla, androecium, and gynoecium), while an *incomplete flower* lacks one or more of the whorls (figure 42.4).

In both complete and incomplete flowers, the **calyx** usually constitutes the outermost whorl; it consists of flattened appendages, called **sepals**, which protect the flower in the bud. The petals collectively make up the **corolla** and may be fused. Petals function to attract pollinators. While these two outer whorls of floral organs are sterile, they can enhance reproductive success.

Androecium (from the Greek *andros*, “man,” + *oikos*, “house”) is a collective term for all the **stamens** (male structures) of a flower. Stamens are specialized structures that bear the angiosperm microsporangia. There are similar structures bearing the microsporangia in the pollen cones of gymnosperms. Most living angiosperms have stamens whose **filaments** (“stalks”) are slender and often threadlike, and whose four microsporangia are evident at

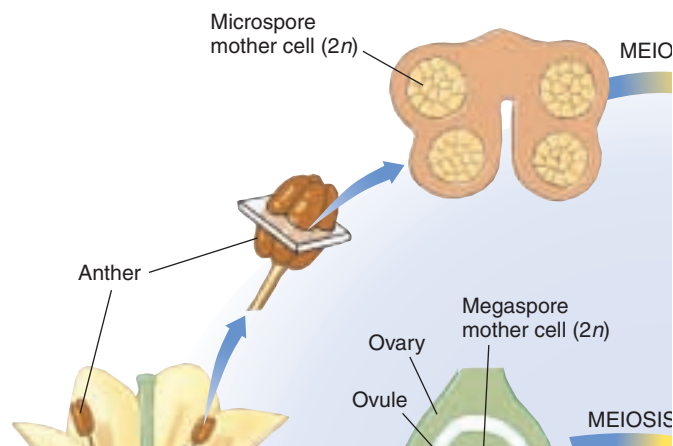


FIGURE 42.4
Structure of an angiosperm flower.

the apex in a swollen portion, the **anther**. Some of the more primitive angiosperms have stamens that are flattened and leaflike, with the sporangia producing from the upper or lower surface.

The **gynoecium** (from the Greek *gyne*, “woman,” + *oikos*, “house”) is a collective term for all the female parts of a flower. In most flowers, the gynoecium, which is unique to angiosperms, consists of a single **carpel** or two or more fused carpels. Single or fused carpels are often referred to as the simple or compound pistils, respectively. Most flowers with which we are familiar—for example, those of tomatoes and oranges—have a single compound pistil. In other mostly primitive flowers—for example, buttercups and stonecups—there may be several to many separate pistils, each formed from a single carpel. **Ovules** (which develop into seeds) are produced in the pistil’s swollen lower portion, the **ovary**, which usually narrows at the top into a slender, necklike **style** with a pollen-receptive **stigma** at its apex. Sometimes the stigma is divided, with the number of stigma branches indicating how many carpels are in the particular pistil. Carpels are essentially inrolled floral leaves with ovules along the margins. It is possible that the first carpels were leaf blades that folded longitudinally; the margins, which had hairs, did not actually fuse until the fruit developed, but the hairs interlocked and were receptive to pollen. In the course of evolution, there is evidence the hairs became localized into a stigma, a style was formed, and the fusing of the carpel margins ultimately resulted in a pistil. In many modern flowering plants, the carpels have become highly modified and are not visually distinguishable from one another unless the pistil is cut open.

Trends of Floral Specialization

Two major evolutionary trends led to the wide diversity of modern flowering plants: (1) separate floral parts have

grouped together, or fused, and (2) floral parts have been lost or reduced (figure 42.5). In the more advanced angiosperms, the number of parts in each whorl has often been reduced from many to few. The spiral patterns of attachment of all floral parts in primitive angiosperms have, in the course of evolution, given way to a single whorl at each level. The central axis of many flowers has shortened, and the whorls are close to one another. In some evolutionary lines, the members of one or more whorls have fused with one another, sometimes joining into a tube. In other kinds of flowering plants, different whorls may be fused together. Whole whorls may even be lost from the flower, which may lack sepals, petals, stamens, carpels, or various combinations of these structures. Modifications often relate to pollination mechanisms and, in some cases like the grasses, wind has replaced animals for pollen dispersal.

While much floral diversity is the result of natural selection related to pollination, it is important to recognize the impact breeding (artificial selection) has had on flower morphology. Humans have selected for practical or aesthetic traits that may have little adaptive value to species in the wild. Maize (corn), for example, has been selected to satisfy the human palate. Human intervention ensures the reproductive success of each generation; while in a natural setting modern corn would not have the same protection from herbivores as its ancestors, and the fruit dispersal mechanism would be quite different (see figure 21.13). Floral shops sell heavily bred species with modified petals, often due to polyploidy, that enhance their economic value, but not their ability to attract pollinators. In making inferences about symbioses between flowers and pollinators, be sure to look at native plants that have not been genetically altered by human intervention.

Trends in Floral Symmetry

Other trends in floral evolution have affected the symmetry of the flower (figure 42.6). Primitive flowers such as those of buttercups are *radially symmetrical*; that is, one could draw a line anywhere through the center and have two roughly equal halves. Flowers of many advanced groups are *bilaterally symmetrical*; that is, they are divisible into two equal parts along only a single plane. Examples of such flowers are snapdragons, mints, and orchids. Such bilaterally symmetrical flowers are also common among violets and peas. In these groups, they are often associated with advanced and highly precise pollination systems. Bilateral symmetry has arisen independently many times. In snapdragons, the *cyclodia* gene regulates floral symmetry, and in its absence flowers are more radial (figure 42.7). Here the evolutionary introduction of a single gene is sufficient to cause a dramatic change in morphology. Whether the same gene or functionally similar genes arose in parallel in other species is an open question.



FIGURE 42.5
Trends in floral specialization. Wild geranium, *Geranium maculatum*. The petals are reduced to five each, the stamens to ten.



FIGURE 42.6
Bilateral symmetry in an orchid. While primitive flowers are usually radially symmetrical, flowers of many advanced groups, such as the orchid family (Orchidaceae), are bilaterally symmetrical.



FIGURE 42.7
Genetic regulation of asymmetry in flowers. (left) Snapdragon flowers normally have bilateral symmetry. (right) The *cyclodia* gene regulates floral symmetry, and *cyclodia* mutant snapdragons have radially symmetrical flowers.

The first angiosperms likely had numerous free, spirally arranged flower parts. Modification of floral parts appears to be closely tied to pollination mechanisms. More recently, horticulturists have bred plants for aesthetic reasons resulting in an even greater diversity of flowers.

42.2 Flowering plants use animals or wind to transfer pollen between flowers.

Formation of Angiosperm Gametes

Reproductive success depends on uniting the gametes (egg and sperm) found in the embryo sacs and pollen grains of flowers. As mentioned previously, plant sexual life cycles are characterized by an alternation of generations, in which a diploid sporophyte generation gives rise to a haploid gametophyte generation. In angiosperms, the gametophyte generation is very small and is completely enclosed within the tissues of the parent sporophyte. The male gametophytes, or microgametophytes, are **pollen grains**. The female gametophyte, or megagametophyte, is the **embryo sac**. Pollen grains and the embryo sac both are produced in separate, specialized structures of the angiosperm flower.

Like animals, angiosperms have separate structures for producing male and female gametes (figure 42.8), but the reproductive organs of angiosperms are different from those of animals in two ways. First, in angiosperms, both male and female structures usually occur together in the same individual flower (with exceptions noted in chapter 38). Second, angiosperm reproductive structures are not permanent parts of the adult individual. Angiosperm flowers and reproductive organs develop seasonally, at times of the year most favorable for pollination. In some cases, reproductive structures are produced only once and the parent plant dies. It is significant that the germ line for angiosperms is not set aside early in development, but forms quite late, as detailed in chapter 40.

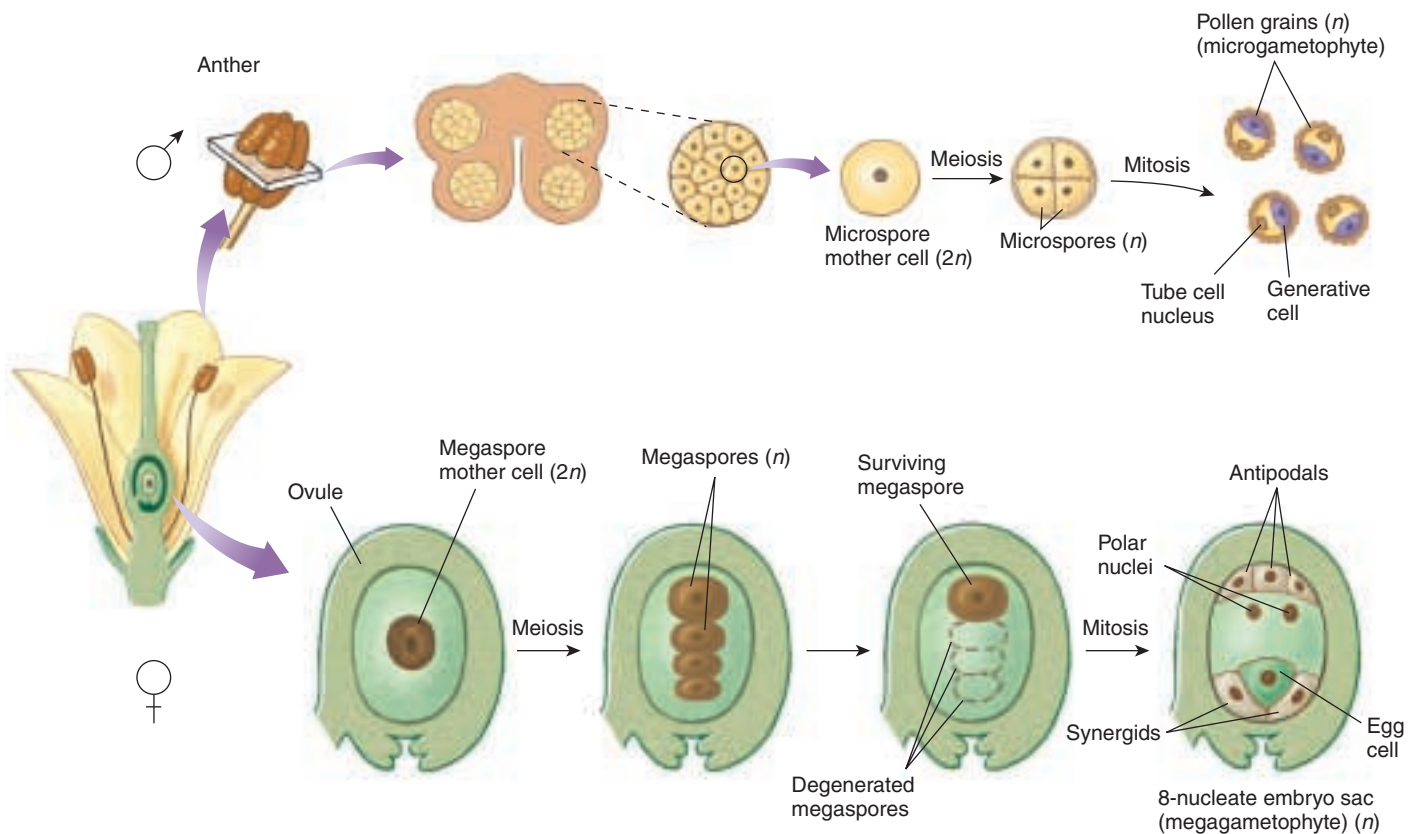
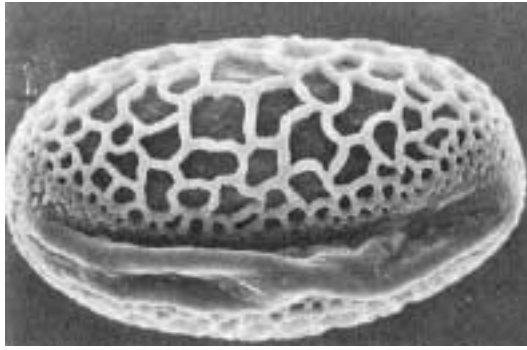


FIGURE 42.8

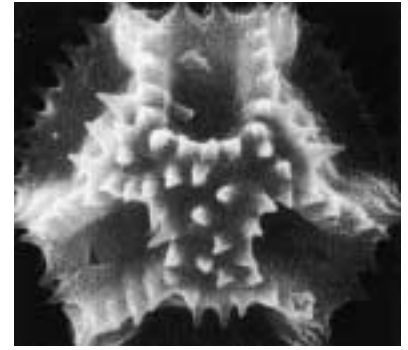
Formation of pollen grains and the embryo sac. Diploid ($2n$) microspore mother cells are housed in the anther and divide by meiosis to form four haploid (n) microspores. Each microspore develops by mitosis into a pollen grain. The generative cell within the pollen grain will later divide to form two sperm cells. Within the ovule, one diploid megaspore mother cell divides by meiosis to produce four haploid megaspores. Usually only one of the megaspores will survive, and the other three will degenerate. The surviving megaspore divides by mitosis to produce an embryo sac with eight nuclei.

FIGURE 42.9

Pollen grains. (a) In the Easter lily, *Lilium candidum*, the pollen tube emerges from the pollen grain through the groove or furrow that occurs on one side of the grain. (b) In a plant of the sunflower family, *Hyoseris longiloba*, three pores are hidden among the ornamentation of the pollen grain. The pollen tube may grow out through any one of them.



(a)



(b)

Pollen Formation

Pollen grains form in the two pollen sacs located in the anther. Each pollen sac contains specialized chambers in which the *microspore mother cells* are enclosed and protected. The microspore mother cells undergo meiosis to form four haploid microspores. Subsequently, mitotic divisions form four pollen grains. Inside each pollen grain is a generative cell; this cell will later divide to produce two sperm cells.

Pollen grain shapes are specialized for specific flower species. As discussed in more detail later in the chapter, fertilization requires that the pollen grain grow a tube that penetrates the style until it encounters the ovary. Most pollen grains have a furrow from which this pollen tube emerges; some grains have three furrows (figure 42.9).

Embryo Sac Formation

Eggs develop in the ovules of the angiosperm flower. Within each ovule is a megaspore mother cell. Each megaspore mother cell undergoes meiosis to produce four haploid megaspores. In most plants, only one of these megaspores, however, survives; the rest are absorbed by the ovule. The lone remaining megaspore undergoes repeated mitotic divisions to produce eight haploid nuclei that are enclosed within a seven-celled embryo sac. Within the embryo sac, the eight nuclei are arranged in precise positions. One nucleus is located near the opening of the embryo sac in the egg cell. Two are located in a single cell in the middle of the embryo sac and are called polar nuclei; two nuclei are contained in cells called synergids that flank the egg cell; and the other three nuclei reside in cells called the antipodals, located at the end of the sac, opposite the egg cell (figure 42.10). The first step in uniting the sperm in the pollen grain with the egg and polar nuclei is to get pollen germinating on the stigma of the carpel and growing toward the embryo sac.

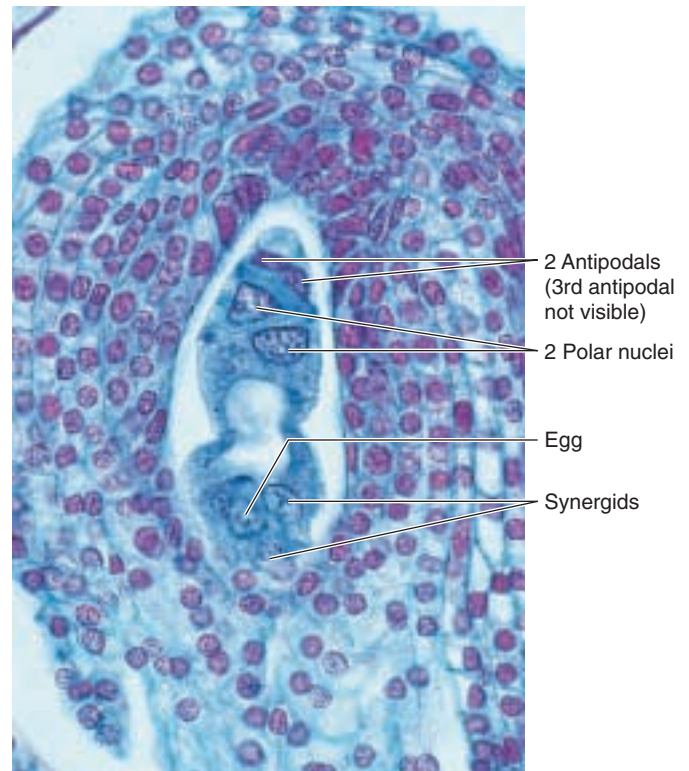


FIGURE 42.10

A mature embryo sac of a lily. The eight haploid nuclei produced by mitotic divisions of the haploid megaspore are labeled.

In angiosperms, both male and female structures often occur together in the same individual flower. These reproductive structures are not a permanent part of the adult individual and the germ line is not set aside early in development.

Pollination

Pollination is the process by which pollen is placed on the stigma. Pollen may be carried to the flower by wind or by animals, or it may originate within the individual flower itself. When pollen from a flower's anther pollinates the same flower's stigma, the process is called *self-pollination*.

Pollination in Early Seed Plants

Early seed plants were pollinated passively, by the action of the wind. As in present-day conifers, great quantities of pollen were shed and blown about, occasionally reaching the vicinity of the ovules of the same species. Individual plants of any given species must grow relatively close to one another for such a system to operate efficiently. Otherwise, the chance that any pollen will arrive at the appropriate destination is very small. The vast majority of windblown pollen travels less than 100 meters. This short distance is significant compared with the long distances pollen is routinely carried by certain insects, birds, and other animals.

Pollination by Animals

The spreading of pollen from plant to plant by pollinators visiting flowers of specific angiosperm species has played an important role in the evolutionary success of the group. It now seems clear that the earliest angiosperms, and perhaps their ancestors also, were insect-pollinated, and the coevolution of insects and plants has been important for both groups for over 100 million years. Such interactions have also been important in bringing about increased floral specialization. As flowers become increasingly specialized, so do their relationships with particular groups of insects and other animals.

Bees. Among insect-pollinated angiosperms, the most numerous groups are those pollinated by bees (figure 42.11). Like most insects, bees initially locate sources of food by odor, then orient themselves on the flower or group of flowers by its shape, color, and texture. Flowers that bees characteristically visit are often blue or yellow. Many have stripes or lines of dots that indicate the location of the nectaries, which often occur within the throats of specialized flowers. Some bees collect nectar, which is used as a source of food for adult bees and occasionally for larvae. Most of the approximately 20,000 species of bees visit flowers to obtain pollen. Pollen is used to provide food in cells where bee larvae complete their development.

Only a few hundred species of bees are social or semi-social in their nesting habits. These bees live in colonies, as do the familiar honeybee, *Apis mellifera*, and the bumble-



FIGURE 42.11
Pollination by a bumblebee. As this bumblebee, *Bombus*, squeezes into the bilaterally symmetrical, advanced flower of a member of the mint family, the stigma contacts its back and picks up any pollen that the bee may have acquired during a visit to a previous flower.

bee, *Bombus*. Such bees produce several generations a year and must shift their attention to different kinds of flowers as the season progresses. To maintain large colonies, they also must use more than one kind of flower as a food source at any given time.

Except for these social and semi-social bees and about 1000 species that are parasitic in the nests of other bees, the great majority of bees—at least 18,000 species—are solitary. Solitary bees in temperate regions characteristically have only a single generation in the course of a year. Often they are active as adults for as little as a few weeks a year.

Solitary bees often use the flowers of a given group of plants almost exclusively as sources of their larval food. The highly constant relationships of such bees with those flowers may lead to modifications, over time, in both the flowers and the bees. For example, the time of day when the flowers open may correlate with the time when the bees appear; the mouthparts of the bees may become elongated in relation to tubular flowers; or the bees' pollen-collecting apparatuses may be adapted to the pollen of the plants that they normally visit. When such relationships are established, they provide both an efficient mechanism of pollination for the flowers and a constant source of food for the bees that “specialize” on them.

Insects Other Than Bees. Among flower-visiting insects other than bees, a few groups are especially prominent. Flowers such as phlox, which are visited regularly by butterflies, often have flat “landing platforms” on which butterflies perch. They also tend to have long, slender floral tubes filled with nectar that is accessible to the long, coiled proboscis characteristic of Lepidoptera, the order of insects that includes butterflies and moths. Flowers like jimsonweed, evening primrose, and others visited regularly by moths are often white, yellow, or some other pale color; they also tend to be heavily scented, thus serving to make the flowers easy to locate at night.

Birds. Several interesting groups of plants are regularly visited and pollinated by birds, especially the hummingbirds of North and South America and the sunbirds of Africa (figure 42.12). Such plants must produce large amounts of nectar because if the birds do not find enough food to maintain themselves, they will not continue to visit flowers of that plant. Flowers producing large amounts of nectar have no advantage in being visited by insects because an insect could obtain its energy requirements at a single flower and would not cross-pollinate the flower. How are these different selective forces balanced in flowers that are “specialized” for hummingbirds and sunbirds?

Ultraviolet light is highly visible to insects. Carotenoids, yellow or orange pigments frequently found in plants, are responsible for the colors of many flowers, such as sunflowers and mustard. Carotenoids reflect both in the yellow range and in the ultraviolet range, the mixture resulting in a distinctive color called “bee’s purple.” Such yellow flowers may also be marked in distinctive ways normally invisible to us, but highly visible to bees and other insects (figure 42.13). These markings can be in the form of a bull’s-eye or a landing strip.

Red does not stand out as a distinct color to most insects, but it is a very conspicuous color to birds. To most insects, the red upper leaves of poinsettias look just like the other leaves of the plant. Consequently, even though the flowers produce abundant supplies of nectar and attract hummingbirds, insects tend to bypass them. Thus, the red color both signals to birds the presence of abundant nectar and makes that nectar as inconspicuous as possible to insects. Red is also seen again in fruits that are dispersed by birds.

Other Animals. Other animals including bats and small rodents may aid in pollination. The signals here also are species specific. These animals also assist in dispersing the seeds and fruits that result from pollination. Monkeys are attracted to orange and yellow and will be effective in dispersing those fruits.

Wind-Pollinated Angiosperms

Many angiosperms, representing a number of different groups, are wind-pollinated—a characteristic of early seed plants. Among them are such familiar plants as oaks, birches, cottonwoods, grasses, sedges, and nettles. The flowers of these plants are small, greenish, and odorless; their corollas are reduced or absent (see figures 42.14 and 42.15). Such flowers often are grouped together in fairly large numbers and may hang down in tassels that wave about in the wind and shed pollen freely. Many wind-pollinated plants have stamen- and carpel-containing flowers separated among individuals or on a single individual. If the pollen-producing and ovule-bearing flowers are separated, it is certain that pollen released to the wind will reach a flower other than the one that sheds it, a strategy



FIGURE 42.12

Hummingbirds and flowers. A long-tailed hermit hummingbird extracts nectar from the flowers of *Heliconia imbricata* in the forests of Costa Rica. Note the pollen on the bird’s beak. Hummingbirds of this group obtain nectar primarily from long, curved flowers that more or less match the length and shape of their beaks.

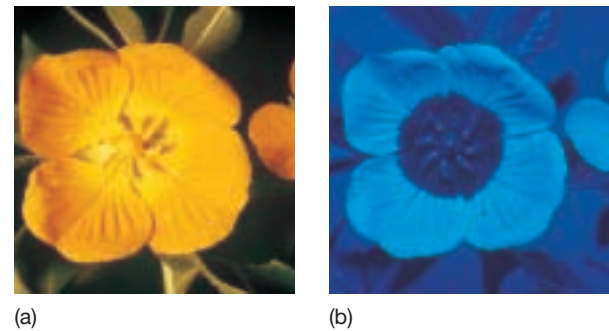


FIGURE 42.13

How a bee sees a flower. (a) The yellow flower of *Ludwigia peruviana* (Onagraceae) photographed in normal light and (b) with a filter that selectively transmits ultraviolet light. The outer sections of the petals reflect both yellow and ultraviolet, a mixture of colors called “bee’s purple”; the inner portions of the petals reflect yellow only and therefore appear dark in the photograph that emphasizes ultraviolet reflection. To a bee, this flower appears as if it has a conspicuous central bull’s-eye.

that greatly promotes outcrossing. Some wind-pollinated plants, especially trees and shrubs, flower in the spring, before the development of their leaves can interfere with the wind-borne pollen. Wind-pollinated species do not depend on the presence of a pollinator for species survival.

Bees are the most frequent and characteristic pollinators of flowers. Insects often are attracted by the odors of flowers. Bird-pollinated flowers are characteristically odorless and red, with the nectar not readily accessed by insects.

Self-Pollination

All of the modes of pollination that we have considered thus far tend to lead to outcrossing, which is as highly advantageous for plants as it is for eukaryotic organisms generally. Nevertheless, self-pollination also occurs among angiosperms, particularly in temperate regions. Most of the self-pollinating plants have small, relatively inconspicuous flowers that shed pollen directly onto the stigma, sometimes even before the bud opens. You might logically ask why there are many self-pollinated plant species if outcrossing is just as important genetically for plants as it is for animals. There are two basic reasons for the frequent occurrence of self-pollinated angiosperms:

1. Self-pollination obviously is ecologically advantageous under certain circumstances because self-pollinators do not need to be visited by animals to produce seed. As a result, self-pollinated plants expend less energy in the production of pollinator attractants and can grow in areas where the kinds of insects or other animals that might visit them are absent or very scarce—as in the Arctic or at high elevations.
2. In genetic terms, self-pollination produces progenies that are more uniform than those that result from outcrossing. Remember that because meiosis is involved, there is still recombination and the offspring will not be identical to the parent. However, such progenies may contain high proportions of individuals well-adapted to particular habitats. Self-pollination in normally outcrossing species tends to produce large numbers of ill-adapted individuals because it brings together deleterious recessive genes; but some of these combinations may be highly advantageous in particular habitats. In such habitats, it may be advantageous for the plant to continue self-pollinating indefinitely. This is the main reason many self-pollinating plant species are weeds—not only have humans made weed habitats uniform, but they have also spread the weeds all over the world.

Factors That Promote Outcrossing

Outcrossing, as we have stressed, is of critical importance for the adaptation and evolution of all eukaryotic organisms. Often flowers contain both stamens and pistils, which increase the likelihood of self-pollination. One strategy to promote outcrossing is to separate stamens and pistils.

In various species of flowering plants—for example, willows and some mulberries—staminate and pistillate flowers may occur on separate plants. Such plants, which produce only ovules or only pollen, are called **dioecious**, from the Greek words for “two houses.” Obviously, they cannot self-pollinate and must rely exclusively on outcrossing. In other kinds of plants, such as oaks, birches, corn (maize), and pumpkins, separate male and female flowers may both be produced on the same plant. Such plants are called **monoecious**, meaning “one house” (figure 42.14). In monoecious plants, the separation of pistillate and staminate flowers, which may mature at different times, greatly enhances the probability of outcrossing.



FIGURE 42.14
Staminate and pistillate flowers of a birch, *Betula*. Birches are monoecious; their staminate flowers hang down in long, yellowish tassels, while their pistillate flowers mature into clusters of small, brownish, conelike structures.



FIGURE 42.15
Wind-pollinated flowers. The large yellow anthers, dangling on very slender filaments, are hanging out, about to shed their pollen to the wind; later, these flowers will become pistillate, with long, feathery stigmas—well suited for trapping windblown pollen—sticking far out of them. Many grasses, like this one, are therefore dichogamous.

Even if, as usually is the case, functional stamens and pistils are both present in each flower of a particular plant species, these organs may reach maturity at different times. Plants in which this occurs are called **dichogamous**. If the stamens mature first, shedding their pollen before the stigmas are receptive, the flower is effectively staminate at that time. Once the stamens have finished shedding pollen, the stigma or stigmas may then become receptive, and the flower may become essentially pistillate (figures 42.15 and 42.16). This has the same effect as if the flower completely lacked either functional stamens or functional pistils; its outcrossing rate is thereby significantly increased.

Many flowers are constructed such that the stamens and stigmas do not come in contact with each other. With such an arrangement, there is a natural tendency for the pollen to be transferred to the stigma of another flower rather than to the stigma of its own flower, thereby promoting outcrossing.

Even when a flower's stamens and stigma mature at the same time, genetic **self-incompatibility**, which is widespread in flowering plants, increases outcrossing. Self-incompatibility results when the pollen and stigma recognize each other as being genetically related and fertilization is blocked (figure 42.17). Self-incompatibility is controlled by the *S* (self-incompatibility) locus. There are many alleles at the *S* locus that regulate recognition responses between the pollen and stigma. There are two types of self-incompatibility. Gametophytic self-incompatibility depends on the haploid *S* locus of the pollen and the diploid *S* locus of the stigma. If either of the *S* alleles in the stigma match the pollen *S* allele, pollen tube growth stops before it reaches the embryo sac. Petunias have gametophytic self-incompatibility. In the case of sporophytic self-incompatibility, such as in broccoli, both *S* alleles of the pollen parent are important; if the alleles in the stigma match with either of the pollen parent *S* alleles, the haploid pollen will not germinate.

Much is being learned about the cellular basis of this recognition and the signal transduction pathways that block the successful growth of the pollen tube. These pollen recognition mechanisms may have had their origins in a common ancestor of the gymnosperms. Fossils with pollen tubes from the Carboniferous are consistent



(a)



(b)

FIGURE 42.16

Dichogamy, as illustrated by the flowers of fireweed, *Epilobium angustifolium*. More than 200 years ago (in the 1790s) fireweed, which is outcrossing, was one of the first plant species to have its process of pollination described. First, the anthers shed pollen, and then the style elongates above the stamens while the four lobes of the stigma curl back and become receptive. Consequently, the flowers are functionally staminate at first, becoming pistillate about two days later. The flowers open progressively up the stem, so that the lowest are visited first. Working up the stem, the bees encounter pollen-shedding, staminate-phase flowers and become covered with pollen, which they then carry to the lower, functionally pistillate flowers of another plant. Shown here are flowers in (a) the staminate phase and (b) the pistillate phase.

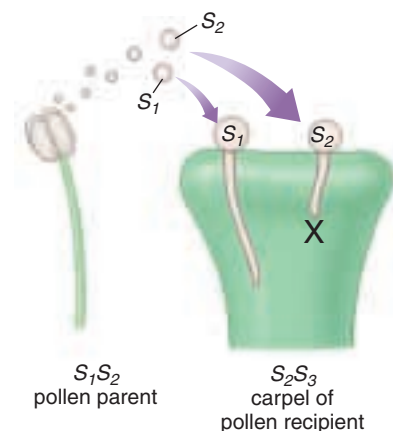
with the hypothesis that they had highly evolved pollen-recognition systems. These may have been systems that recognized foreign pollen that predated self-recognition systems.

Self-pollinated angiosperms are frequent where there is a strong selective pressure to produce large numbers of genetically uniform individuals adapted to specific, relatively uniform habitats. Outcrossing in plants may be promoted through dioecism, monoecism, self-incompatibility, or the physical separation or different maturation times of the stamens and pistils. Outcrossing promotes genetic diversity.

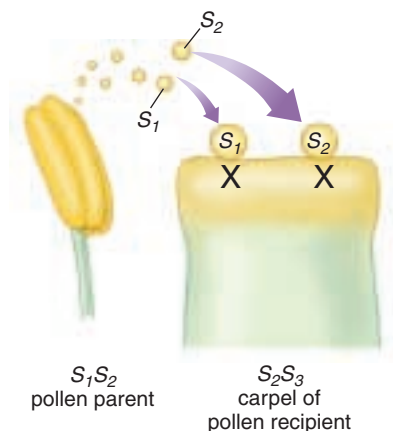
FIGURE 42.17

Self-pollination can be genetically controlled so self-pollen is not recognized.

(a) Gametophytic self-incompatibility is determined by the haploid pollen genotype. (b) Sporophytic self-incompatibility recognizes the genotype of the diploid pollen parent, not just the haploid pollen genotype. In both cases, the recognition is based on the *S* locus, which has many different alleles. The subscript numbers indicate the *S* allele genotype. In gametophytic self-incompatibility, the block comes after pollen tube germination. In sporophytic self-incompatibility, the pollen tube fails to germinate.



(a) Gametophytic self-incompatibility



(b) Sporophytic self-incompatibility

Fertilization

Fertilization in angiosperms is a complex, somewhat unusual process in which two sperm cells are utilized in a unique process called **double fertilization**. Double fertilization results in two key developments: (1) the fertilization of the egg, and (2) the formation of a nutrient substance called endosperm that nourishes the embryo. Once a pollen grain has been spread by wind, by animals, or through self-pollination, it adheres to the sticky, sugary substance that covers the stigma and begins to grow a **pollen tube** that pierces the style (figure 42.18). The pollen tube, nourished by the sugary substance, grows until it reaches the ovule in the ovary. Meanwhile, the generative cell within the pollen grain tube cell divides to form two sperm cells.

The pollen tube eventually reaches the embryo sac in the ovule. At the entry to the embryo sac, the tip of the pollen tube bursts and releases the two sperm cells. Simul-

taneously, the two nuclei that flank the egg cell disintegrate, and one of the sperm cells fertilizes the egg cell, forming a zygote. The other sperm cell fuses with the two polar nuclei located at the center of the embryo sac, forming the triploid ($3n$) primary endosperm nucleus. The primary endosperm nucleus eventually develops into the endosperm.

Once fertilization is complete, the embryo develops by dividing numerous times. Meanwhile, protective tissues enclose the embryo, resulting in the formation of the seed. The seed, in turn, is enclosed in another structure called the fruit. These typical angiosperm structures evolved in response to the need for seeds to be dispersed over long distances to ensure genetic variability.

In double fertilization, angiosperms utilize two sperm cells. One fertilizes the egg, while the other helps form a substance called endosperm that nourishes the embryo.

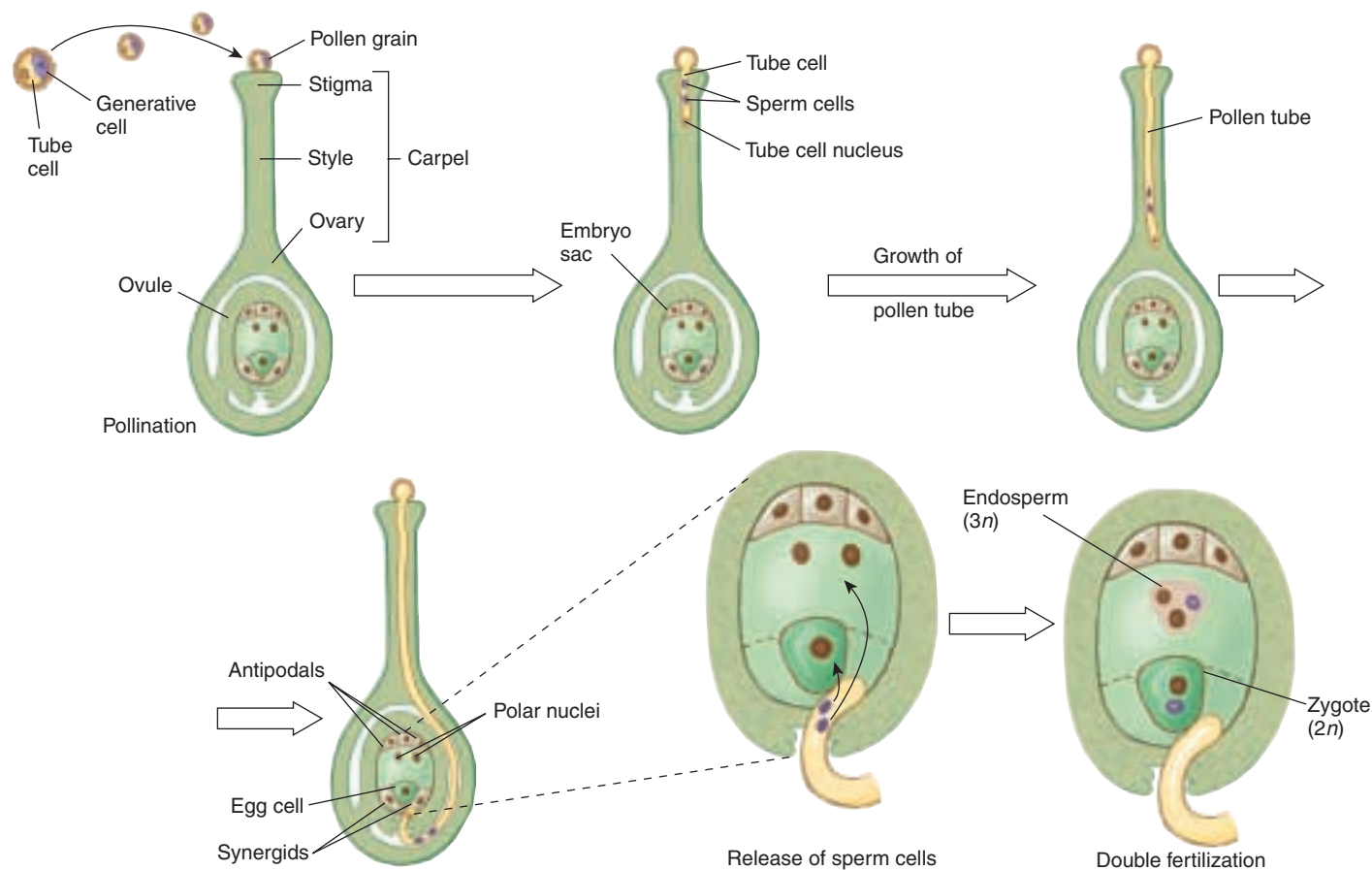


FIGURE 42.18

The formation of the pollen tube and double fertilization. When pollen lands on the stigma of a flower, the pollen tube cell grows toward the embryo sac, forming a pollen tube. While the pollen tube is growing, the generative cell divides to form two sperm cells. When the pollen tube reaches the embryo sac, it bursts through one of the synergids and releases the sperm cells. In a process called double fertilization, one sperm cell nucleus fuses with the egg cell to form the diploid ($2n$) zygote, and the other sperm cell nucleus fuses with the two polar nuclei to form the triploid ($3n$) endosperm nucleus.

42.3 Many plants can clone themselves by asexual reproduction.

Asexual Reproduction

While self-pollination reduces genetic variability, asexual reproduction results in genetically identical individuals because only mitotic cell divisions occur. In the absence of meiosis, individuals that are highly adapted to a relatively unchanging environment persist for the same reasons that self-pollination is favored. Should conditions change dramatically, there will be less variation in the population for natural selection to act upon and the species may be less likely to survive. Asexual reproduction is also used in agriculture and horticulture to propagate a particularly desirable plant whose traits would be altered by sexual reproduction, even self-pollination. Most roses and potatoes for example, are vegetatively propagated.

Vegetative Reproduction

In a very common form of asexual reproduction called vegetative reproduction, new plant individuals are simply cloned from parts of adults (figure 42.19). The forms of vegetative reproduction in plants are many and varied.

Stolons. Some plants reproduce by means of runners, or stolons—long, slender stems that grow along the surface of the soil. In the cultivated strawberry, for example, leaves, flowers, and roots are produced at every other node on the runner. Just beyond each second node, the tip of the runner turns up and becomes thickened. This thickened portion first produces adventitious roots and then a new shoot that continues the runner.

Rhizomes. Underground stems, or rhizomes, are also important reproductive structures, particularly in grasses and sedges. Rhizomes invade areas near the parent plant, and each node can give rise to a new flowering shoot. The noxious character of many weeds results from this type of growth pattern, and many garden plants, such as irises, are propagated almost entirely from rhizomes. Corms, bulbs, and tubers are rhizomes specialized for storage and reproduction. White potatoes are propagated artificially from tuber segments, each with one or more “eyes.” The eyes, or “seed pieces,” of potato give rise to the new plant.

Suckers. The roots of some plants—for example, cherry, apple, raspberry, and blackberry—produce “suckers,” or sprouts, which give rise to new plants. Commercial varieties of banana do not produce seeds and are propagated by suckers that develop from buds on underground stems. When the root of a dandelion is broken, as it may be if one attempts to pull it from the ground, each root fragment may give rise to a new plant.

Adventitious Leaves. In a few species, even the leaves are reproductive. One example is the house plant *Kalanchoë daigremontiana*.



FIGURE 42.19.

Vegetative reproduction. Small plants arise from notches along the leaves of the house plant *Kalanchoë daigremontiana*.

daigremontiana, familiar to many people as the “maternity plant,” or “mother of thousands.” The common names of this plant are based on the fact that numerous plantlets arise from meristematic tissue located in notches along the leaves. The maternity plant is ordinarily propagated by means of these small plants, which, when they mature, drop to the soil and take root.

Apomixis

In certain plants, including some citruses, certain grasses (such as Kentucky bluegrass), and dandelions, the embryos in the seeds may be produced asexually from the parent plant. This kind of asexual reproduction is known as *apomixis*. The seeds produced in this way give rise to individuals that are genetically identical to their parents. Thus, although these plants reproduce asexually by cloning diploid cells in the ovule, they also gain the advantage of seed dispersal, an adaptation usually associated with sexual reproduction. As you will see in chapter 43, embryos can also form via mitosis when plant tissues are cultured. In general, vegetative reproduction, apomixis, and other forms of asexual reproduction promote the exact reproduction of individuals that are particularly well suited to a certain environment or habitat. Asexual reproduction among plants is far more common in harsh or marginal environments, where there is little margin for variation. There is a greater proportion of asexual plants in the arctic, for example, than in temperate regions.

Plants that reproduce asexually clone new individuals from portions of the root, stem, leaves, or ovules of adult individuals. The asexually produced progeny are genetically identical to the parent individual.

42.4 How long do plants and plant organs live?

The Life Span of Plants

Plant Life Spans Vary Greatly

Once established, plants live for highly variable periods of time, depending on the species. Life span may or may not correlate with reproductive strategy. Woody plants, which have extensive secondary growth, nearly always live longer than herbaceous plants, which have limited or no secondary growth. Bristlecone pine, for example, can live upward of 4000 years. Some herbaceous plants send new stems above the ground every year, producing them from woody underground structures. Others germinate and grow, flowering just once before they die. Shorter-lived plants rarely become very woody because there is not enough time for the accumulation of secondary tissues. Depending on the length of their life cycles, herbaceous plants may be annual, biennial, or perennial, while woody plants are generally perennial (figure 42.20). Determining life span is even more complicated for clonally reproducing organisms. Aspen trees form huge clones from asexual reproduction of their roots. Collectively, an aspen clone may form the largest “organism” on earth. Other asexually reproducing plants may cover less territory but live for thousands of years. Creosote bushes in the Mojave Desert have been identified that are up to 12,000 years old!

Annual Plants

Annual plants grow, flower, and form fruits and seeds within one growing season; they then die when the process is complete. Many crop plants are annuals, including corn, wheat, and soybeans. Annuals generally grow rapidly under favorable conditions and in proportion to the availability of water or nutrients. The lateral meristems of some annuals, like sunflowers or giant ragweed, do produce poorly developed secondary tissues, but most are entirely herbaceous. Annuals typically die after flowering once, the developing flowers or embryos using hormonal signaling to reallocate nutrients so the parent plant literally starves to death. This can be demonstrated by comparing a population of bean plants where the beans are continually picked with a population where the beans are left on the plant. The frequently picked population will continue to grow and yield beans much longer than the untouched population. The process that leads to the death of a plant is called *senescence*.



FIGURE 42.20

Annual and perennial plants. Plants live for very different lengths of time. (a) Desert annuals complete their entire life span in a few weeks. (b) Some trees, such as the giant redwood (*Sequoiadendron giganteum*), which occurs in scattered groves along the western slopes of the Sierra Nevada in California, live 2000 years or more.

Biennial Plants

Biennial plants, which are much less common than annuals, have life cycles that take two years to complete. During the first year, biennials store photosynthate in underground storage organs. During the second year of growth, flowering stems are produced using energy stored in the underground parts of the plant. Certain crop plants, including carrots, cabbage, and beets, are biennials, but these plants generally are harvested for food during their first season, before they flower. They are grown for their leaves or roots, not for their fruits or seeds. Wild biennials include evening primroses, Queen Anne’s lace, and mullein. Many plants that are considered biennials actually do not flower until they are three or more years of age, but all biennial plants flower only once before they die.

Perennial Plants

Perennial plants continue to grow year after year and may be herbaceous, as are many woodland, wetland, and prairie wildflowers, or woody, as are trees and shrubs. The majority of vascular plant species are perennials. Herbaceous perennials rarely experience any secondary growth in their stems; the stems die each year after a period of relatively rapid growth and food accumulation. Food is often stored in the plants’ roots or underground stems, which can become quite large in comparison to their less substantial aboveground counterparts.

Trees and shrubs generally flower repeatedly, but there are exceptions. Bamboo lives for many seasons as a vegetative plant, but senesces and dies after flowering. The same is true for at least one tropical tree which achieves great heights before flowering and senescing. Considering the tremendous amount of energy that goes into the growth of a tree, this particular reproductive strategy is quite curious.

Trees and shrubs are either deciduous, with all the leaves falling at one particular time of year and the plants remaining bare for a period, or evergreen, with the leaves dropping throughout the year and the plants never appearing completely bare. In northern temperate regions, conifers are the most familiar evergreens; but in tropical and subtropical regions, most angiosperms are evergreen, except where there is a severe seasonal drought. In these areas, many angiosperms are deciduous, losing their leaves during the drought and thus conserving water.

Organ Abscission

Senescence is an important developmental process that leads to the death of an organ, shoot, or the whole plant. Annual and biennial plants undergo whole plant senescence, but individual organs on any plant can also senesce and be shed. The process by which leaves or petals are shed is called **abscission**.

One advantage to organ senescence is that nutrient sinks can be dispensed with. For example, shaded leaves that are no longer photosynthetically productive can be shed. Petals, which are modified leaves, may senesce once pollination occurs. Orchid flowers remain fresh for long periods of time, even in a florist shop. However, once pollination occurs, a hormonal change is triggered that leads to petal senescence. This makes sense in terms of allocation of energy resources, as the petals are no longer necessary to attract a pollinator. On a larger scale, deciduous plants in temperate areas produce new leaves in the spring and then lose them in the fall. In the tropics, however, the production and subsequent loss of leaves in some species is correlated with wet and dry seasons. Evergreen plants, such as most conifers, usually have a complete change of leaves every two to seven years, periodically losing some but not all of their leaves.

Abscission involves changes that take place in an *abscission zone* at the base of the petiole (figure 42.21). Young leaves produce hormones (especially cytokinins) that inhibit the development of specialized layers of cells in the abscission zone. Hormonal changes take place as the leaf ages, however, and two layers of cells become differentiated. (Despite the name, abscisic acid is not involved in this process.) A *protective layer*, which may be several cells wide, develops on the stem side of the petiole base. These cells become impregnated with *suberin*, which, as you will recall, is a fatty substance that is impervious to moisture. A *separation layer* develops on the side of the

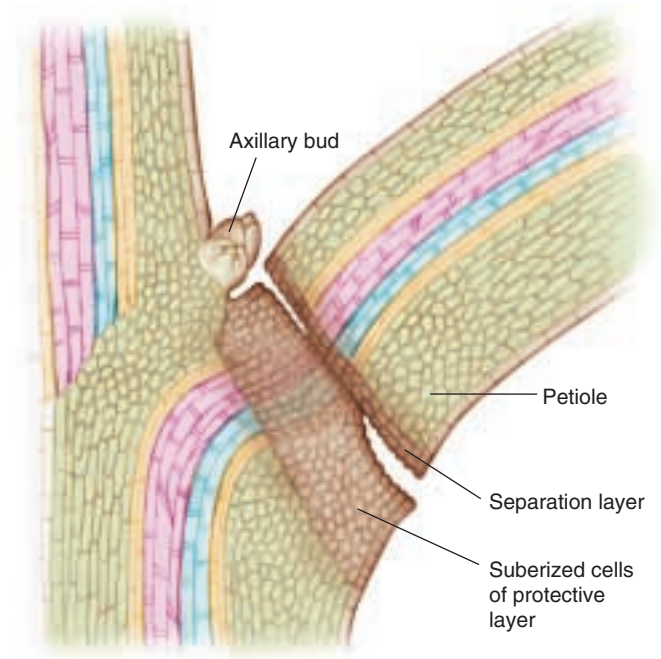


FIGURE 42.21

Leaf abscission. The abscission zone of a leaf. Hormonal changes in this zone cause abscission. Two layers of cells in the abscission zone differentiate into a protective layer and a separation layer. As pectins in the separation layer break down, wind and rain can easily separate the leaf from the stem.

leaf blade; the cells of the separation layer sometimes divide, swell, and become gelatinous. When temperatures drop, the duration and intensity of light diminishes as the days grow shorter, or other environmental changes occur, enzymes break down the pectins in the middle lamellae of the separation cells. Wind and rain can then easily separate the leaf from the stem. Left behind is a sealed leaf scar that is protected from bacteria and other disease organisms.

As the abscission zone develops, the green chlorophyll pigments present in the leaf break down, revealing the yellows and oranges of other pigments, such as carotenoids, that previously had been masked by the intense green colors. At the same time, water-soluble red or blue pigments called *anthocyanins* and *betacyanins* may also accumulate in the vacuoles of the leaf cells—all contributing to an array of fall colors in leaves (see figure 41.7a).

Annual plants complete their whole growth cycle within a single year. Biennial plants flower only once, normally after two seasons of growth. Perennials flower repeatedly and live for many years. Abscission occurs when a plant sheds its organs.



Summary

Questions

Media Resources

42.1 Angiosperms have been incredibly successful, in part, because of their reproductive strategies.

- Angiosperms have been successful because they can be relatively drought-resistant, and smaller herbaceous angiosperms have relatively short life cycles. Most important, however, are their flowers and fruits. Flowers make possible the precise transfer of pollen and, therefore, outcrossing, even when the stationary individual plants are widely separated. Fruits, with their complex adaptations, facilitate the wide dispersal of angiosperms.
 - Modification of floral parts, especially petals, has been key in facilitating pollination. Bilateral symmetry has evolved independently, multiple times.
1. What characteristics of early angiosperms are thought to contribute to their success.
 2. What flower whorl is collectively made up of petals? With which other flower parts are the petals of most flowers homologous?
 3. What is an androecium? Of which flower parts is it composed?

42.2 Flowering plants use animals or wind to transfer pollen between flowers.

- Bees are the most frequent and constant pollinators of flowers. Insects often are attracted by the odors of flowers rather than color. Birds are attracted to red flowers, but not odors.
 - Self-pollination reduces genetic variability among offspring. Outcrossing increases genetic diversity.
 - Outcrossing in different angiosperms is promoted by the separation of stamens and carpels into different flowers, or even into different individuals.
4. What does it mean if a plant is dichogamous? Of what advantage is it to the plant?
 5. Is it more likely that a flower visited by a social or a solitary bee will become highly specialized toward that bee? Why?

42.3 Many plants can clone themselves by asexual reproduction.

- In asexual reproduction, plants clone new individuals from portions of adult roots, stems, leaves, or ovules.
 - The progeny produced by asexual reproduction are all genetically identical to the parent individual, even when they are produced in the ovules (apomixis).
6. Why would a plant capable of sexual reproduction reproduce asexually?
 7. You have just cloned a gene responsible for apomixis. Several corn breeders are very interested in your gene. Why?



- Asexual Reproduction

42.4 How long do plants and plant organs live?

- Plants can live for a single season or thousands of years.
 - For annual and biennial plants, reproduction triggers senescence and death.
 - Asexually reproducing plants can form clones that cover huge areas and/or live for many thousands of years.
 - Plant organs and shoots can senesce and die while the whole plant thrives. Organ senescence is an efficient way to maximize the use of energy resources.
8. Some plants flower once and die; others flower multiple times, reaching great heights and diameters. What are the relative advantages of the two strategies?
 9. How and why does leaf abscission occur?



- Gamete formation
- Fertilization

43

Plant Genomics

Concept Outline

43.1 Genomic organization is much more varied in plants than in animals.

Overview of Plant Genomics. As agrarian societies formed, people began to select for desirable traits. Until relatively recently, plant biologists focused their research efforts on variation in chromosomes, but work is now shifting increasingly to the molecular level.

Organization of Plant Genomes. Plant genomes are more complex than those of other eukaryotic organisms due to the presence of multiple chromosome copies and extensive amounts of DNA with repetitive sequences.

Comparative Genome Mapping and Model Systems. RFLP and AFLP techniques are useful for mapping traits in plant genomes. Despite the technical success in sequencing the *Arabidopsis* genome and other genomes, we still don't know what most of these genes do and how the proteins they encode function in physiology and development.

43.2 Advances in plant tissue culture are revolutionizing agriculture.

Overview of Plant Tissue Culture. Because plants are totipotent, bits of tissue can be used to regenerate whole plants.

Types of Plant Tissue Cultures. Plant cells, tissues, and organs can be grown in an artificial culture medium, and some cells can be directed to generate whole plants.

Applications of Plant Tissue Culture. Plant tissue cultures can be used for the production of plant products, propagation of horticultural plants, and crop improvement.

43.3 Plant biotechnology now affects every aspect of agriculture.

World Population in Relation to Advances Made in Crop Production. It is uncertain whether advances made in crop production by improved farming practices and crop breeding can provide for an increasing world population.

Plant Biotechnology for Agricultural Improvement. Plants can be genetically engineered to have altered levels of oils and amino acids and to provide vaccines.

Methods of Plant Transformation. The genetic engineering of plants is based upon introduction of foreign DNA into plant cells.

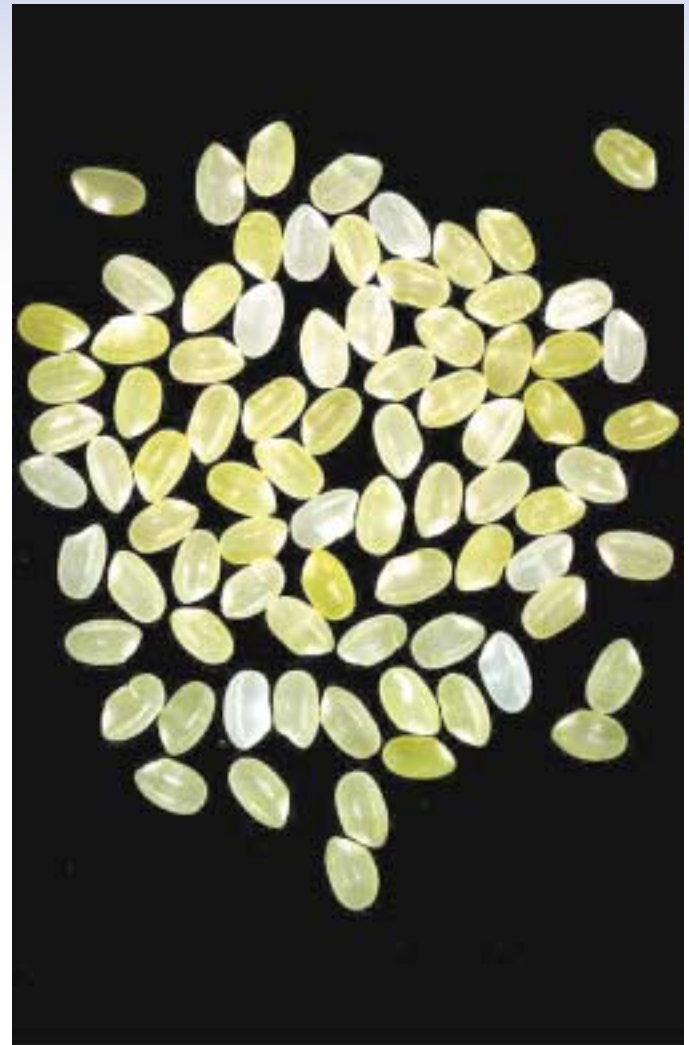


FIGURE 43.1

Golden rice. Rice is the dietary staple of almost half the world's population, but it lacks vitamin A. Vitamin A deficiency leads to vision and immunity problems. Genetically engineered rice that produces vitamin A has now been developed. The rice is golden because a biosynthetic pathway has been genetically modified to produce gold-colored beta-carotene, a precursor to vitamin A. Here, while rice is mixed in with golden rice. The intensity of golden color indicates the amount of pro-vitamin A present.

By selective breeding favoring desired traits, people have been genetically modifying plants since agrarian societies began. All of our key modern crops are the result of this long effort. Today, we have even more powerful tools, recombinant DNA technologies that are the subject of this chapter. This chapter looks ahead to the impact of these new technologies on the future of plants and our study of plant biology (figure 43.1). Both the *Arabidopsis* and rice genomes are essentially sequenced. Not only can we expect to learn much about the molecular basis of plant physiology and development from these rich databases; we will surely gain a far deeper understanding of plant evolution.

43.1 Genomic organization is much more varied in plants than in animals.

Overview of Plant Genomics

Early Approaches

While the term genetic engineering is commonly used to describe plants and animals modified using recombinant DNA technology, people have actually been genetic engineers for thousands of years. As agrarian societies formed, changes in the gene pool within crop species began. For example, seed dispersal was selected against in maize and wheat. Without the ability to disperse seed, these domesticated plants are completely dependent on humans for seed dispersal. Rice was converted from a perennial plant to an annual plant without the seed dormancy mechanisms present in wild rice. Parts of the plant that were of most dietary value to humans and domesticated animals have been selected for increased size. These include seeds, fruits, and storage organs like roots in the case of carrots. All of these changes were accomplished without knowledge of particular genes, by selecting and propagating individuals with the desired traits.

Breeding Strategies to Enhance Yield

At the beginning of the twentieth century, a growing understanding of genetics increased the rate of crop improvement. Among the most dramatic agricultural developments was the introduction of hybrid corn. As corn breeding progressed, highly inbred lines began to have decreased yield as deleterious recessive genes became homozygous. George Harrison Shull found that crossing two different inbred lines gave rise to offspring with “hybrid vigor.” The yield increased fourfold! Hybrid corn now grows in almost all fields in the United States. Hybrid rice developed by the International Rice Research Institute in the Philippines has increased yield 20%.

Breeders have now turned to specific genes to optimize food quality (see figure 43.1). Only a small percentage of the genes and their function have been identified, but we start this century with technologically powerful new ways to understand genomes.

Studying Plant Genomes

Plant genomes are more complex than other eukaryotic genomes, and analysis reveals many evolutionary flips and turns of the DNA sequences over time. Plants show widely different chromosome numbers and varied ploidy levels (figure 43.2). Overall, the size of plant genomes (both number of chromosomes and total nucleotide base-pairs) exhibits the greatest variation of any kingdom in the biological world. For example, tulips contain over 170 times as much DNA as the small weed *Arabidopsis thaliana* (table 43.1). The DNA of plants, like animals, can also contain

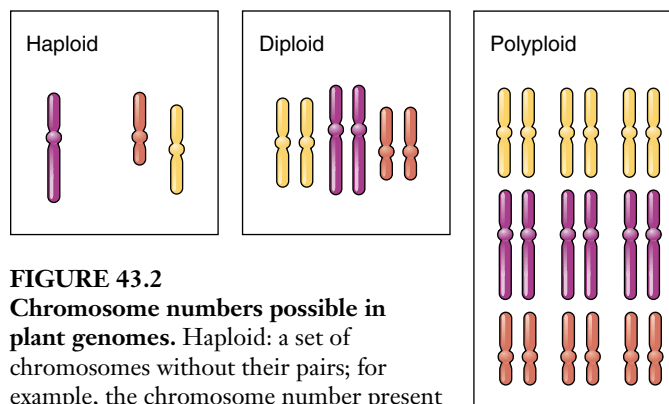


FIGURE 43.2
Chromosome numbers possible in plant genomes. Haploid: a set of chromosomes without their pairs; for example, the chromosome number present in a gamete. Diploid: a single set of chromosome pairs. Polyploid: multiple sets of chromosome pairs; for example, bananas have a triple set of chromosomes and are therefore polyploid.

Table 43.1 Genome Size of Plants

Scientific Name	Common Name	Genome Size (Millions of Base-Pairs)
<i>Arabidopsis thaliana</i>	Arabidopsis	145
<i>Prunus persica</i>	Peach	262
<i>Ricinus communis</i>	Castor bean	323
<i>Citrus sinensis</i>	Orange	367
<i>Oryza sativa</i> spp. <i>javanica</i>	Rice	424
<i>Petunia parodii</i>	Petunia	1,221
<i>Pisum sativum</i>	Garden pea	3,947
<i>Avena sativa</i>	Oats	11,315
<i>Tulipa</i> spp.	Garden tulip	24,704

Source: From *Plant Biochemistry and Molecular Biology*, by P. J. Lea and R. C. Leegods, eds. Copyright © 1993 John Wiley & Sons, Limited. Reproduced with permission.

regions of sequence repeats, sequence inversions, or transposable element insertions, which further modify their genetic content. Traditionally, variation in chromosome inversions and ploidy has been used to build up a picture of how plant species have evolved (figure 43.3). Increasingly, researchers are turning to studying the organization of plant DNA sequences to obtain important information about the evolutionary history of a plant species.

People have been genetically engineering plants for centuries by selecting for desired traits. Traditionally, biologists have examined variation among plants at the chromosome level; today, researchers are focusing more of their efforts at the DNA sequence level.

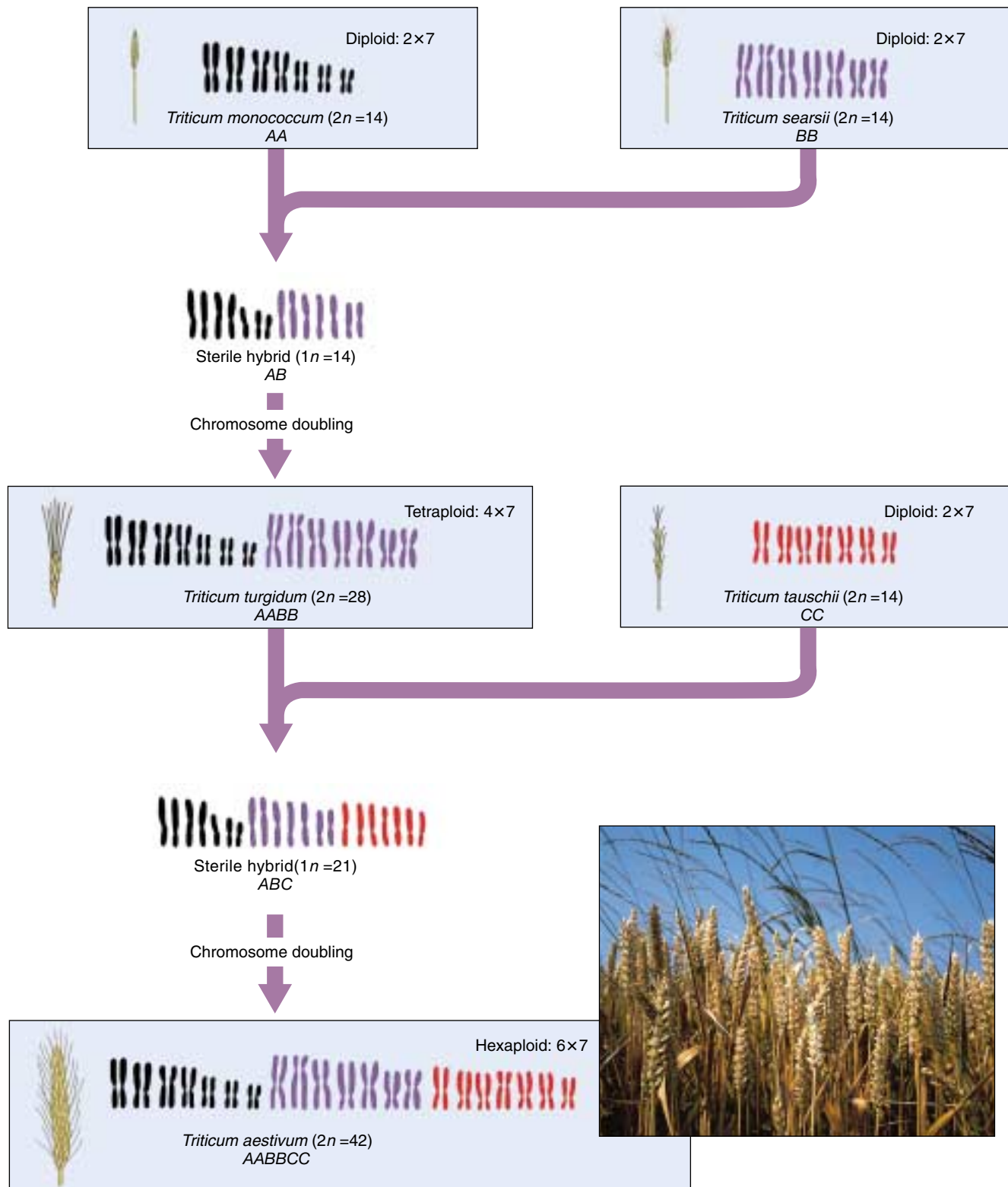


FIGURE 43.3

Evolutionary history of wheat. Domestic wheat arose in southwestern Asia in the hilly country of what is now Iraq. In this region, there is a rich assembly of grasses of the genus *Triticum*. Domestic wheat (*T. aestivum*) is a polyploid species of *Triticum* that arose through two so-called “allopolyploid” events. (1) Two different diploid species, *AA* and *BB*, hybridized to form an *AB* polyploid; the species were so different that *A* and *B* chromosomes could not pair in meiosis, so the *AB* polyploid was sterile. However, in some plants the chromosome number spontaneously doubled due to a failure of chromosomes to separate in meiosis, producing a fertile tetraploid species *AABB*. This wheat is used in the production of pasta. (2) In a similar fashion, the tetraploid species *AABB* hybridized with another diploid species *CC* to produce the hexaploid *T. aestivum*, *AABBCC*. This bread wheat is commonly used throughout the world.

Organization of Plant Genomes

Low-, Medium-, and High-Copy-Number DNA

Most seed plants contain quantities of DNA that greatly exceed their needs for coding and regulatory function. Hence, for plants, a very small percentage of the genome may actually encode genes involved in the production of protein. This portion of the genome which encodes most of the transcribed genes is often referred to as “low-copy-number DNA,” because the DNA sequences comprising these genes are present in single or small numbers of copies. How do plants function with so much extra DNA inserted into the genome? It appears that most of these sequence alterations occur in noncoding regions.

“Medium-copy-number DNA” is composed largely of DNA sequences that encode ribosomal RNA (rRNA), a key element of the cellular machinery that translates transcribed messenger RNA (mRNA) into protein. In plant genomes, rRNA genes may be repeated several hundred to several thousand times. This is in contrast to animal cells, where only 100 to 200 rRNA genes are normally present. The extent of variability in plant genomes with respect to the number of rRNA genes and mutations in them has provided a useful tool for analyzing the evolutionary patterns of plant species.

Plant cells may also contain excess DNA in their genomes in the form of highly repetitive sequences, or “high-copy-number DNA.” At present, the function of this high-copy-number DNA in plant genomes is unknown. Roughly half the maize genome is composed of such retroviral-like DNA. RNA retroviruses like HIV use their host genomes to make DNA copies that then insert into the host genome. Clearly, the effects of some retroviruses can be lethal. How maize came to tolerate such a large amount of this foreign DNA is an evolutionary mystery.

Sequence Replication and Inversion

High-copy-number DNA sequences in the plant genome may be short, such as the nucleotide sequence “GAA,” or much longer, involving up to several hundred nucleotides. Moreover, the number of copies of an individual high-copy repetitive DNA sequence can total from 10,000 to 100,000. There are several possibilities for how high-copy

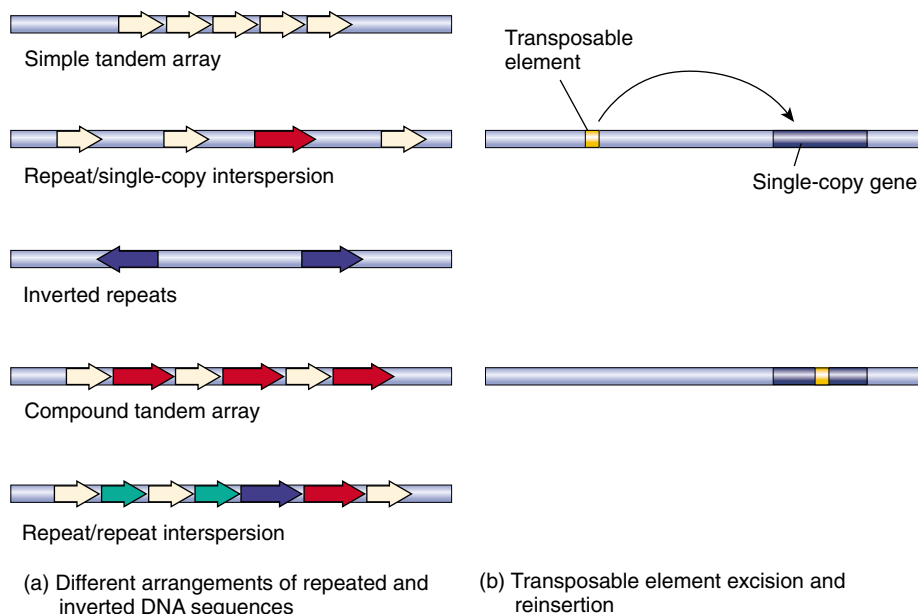


FIGURE 43.4

Organization of repeated DNA sequences and the mechanism of transposable elements in altering gene function. (a) Repeated DNA sequences can occur in plant genomes in several different arrangements. The arrows represent repeated DNA sequences. Arrows of the same size and color represent DNA sequences which are identical to each other. The direction of the arrowhead indicates the orientation of the DNA sequence. (b) Transposable elements can be a source of repetitive DNA that alters gene function. Following excision from its original location, a transposable element may reinsert in the single-copy DNA sequence comprising a gene and alter the gene’s function.

repetitive DNA sequences may be organized within a plant genome (figure 43.4a). Several copies of a single repetitive DNA sequence may be present together in the same orientation, in a pattern called “simple tandem array.” Alternatively, repetitive DNA sequences can be dispersed among single-copy DNA in the same orientation (“repeat/single-copy interspersion”) or the opposite orientation (“inverted repeats”). In addition, groups of repetitive DNA sequences can also occur together in plant genomes in a variety of possible arrangements, such as a “compound tandem array” or a “repeat/repeat interspersion.” The presence of repetitive DNA can vastly increase the size of a plant genome, making it difficult to find and characterize individual single-copy genes. Characterizing single-copy genes can thus become a sort of “needle-in-the-haystack” hunt.

A variety of mechanisms can account for the presence of highly repetitive DNA sequences in plant genomes. Repetitive sequences can be generated by DNA sequence amplification in which multiple rounds of DNA replication occur for specific chromosomal regions. Unequal crossing over of the chromosomes during meiosis or mitosis (translocation) or the action of transposable elements (see next section) can also generate repetitive sequences.

Transposable Elements

Transposable elements, described in chapter 18, are special sequences of DNA with the ability to move from place to place in the genome. They can excise from one site at unpredictable times and reinsert in another site. For this reason, transposable elements have been called “jumping genes.” Transposable elements often insert into coding regions or regulatory regions of a gene and so affect expression of that gene, resulting in a mutation that may or may not be detectable (figure 43.4*b*). Barbara McClintock won the Nobel Prize in 1983 for her work describing transposable elements in corn (see figure 18.23).

Due to their capacity to replicate independently and to move through the genome, transposable elements can also be involved in generating repetitive DNA sequences. This is believed to be the case with the extensive retroviral-like insertions in maize. Retention of the repetitive DNA sequence at a particular site in the genome would involve in each instance a mutation in the transposable element itself which removes its capacity to transpose.

Chloroplast Genome and Its Evolution

The chloroplast is a plant organelle that functions in photosynthesis, and it can independently replicate in the plant cell. Plant chloroplasts have their own specific DNA, which is separate from that present in the nucleus. This DNA is maternally inherited and encodes unique chloroplast proteins. Many of the proteins encoded by chloroplast DNA are involved in photosynthesis. Chloroplasts are thought to have originated from a photosynthetic prokaryote that became part of a plant cell by endosymbiosis. In support of this concept, research has shown that chloroplast DNA has many prokaryote-like features. Chloroplast DNA is present as circular loops of double-stranded DNA similar to prokaryotic chromosomal DNA. Moreover, chloroplast DNA contains genes for ribosomes that are very similar to those present in prokaryotes.

The DNA in chloroplasts of all land plants has about the same number of genes (~100), and they are present in about the same order (figure 43.5). In contrast to the evolution of the DNA in the plant cell nucleus, chloroplast DNA has evolved at a more conservative pace, and therefore shows a more interpretable evolutionary pattern when scientists study DNA sequence similarities. Chloroplast DNA is also not subject to modification caused by transposable elements and mutations due to recombination. Over time, there appears to have been some genetic exchange between the nuclear and chloroplast genomes. For example, the key enzyme in the Calvin cycle of photosynthesis (RUBISCO) consists of a large and small subunit. The small subunit is encoded in the nuclear genome. The protein it encodes has a targeting sequence that allows it to enter the chloroplast and combine with large subunits. The evolutionary history of the localization of these genes is a puzzle.

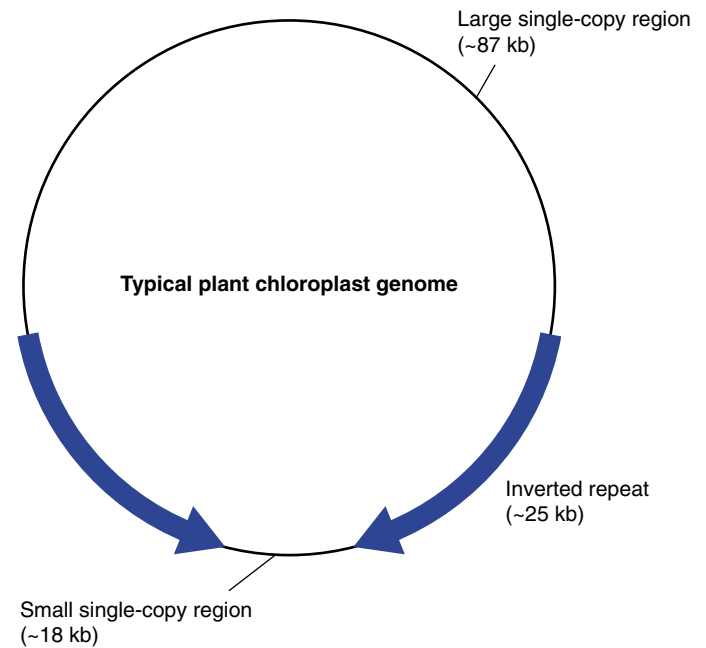


FIGURE 43.5

Chloroplast genome. A schematic drawing of a typical plant chloroplast genome indicates two regions containing single-copy genes, one containing about 87,000 nucleotides (87 kb) and another about 18 kb, and two symmetrical inverted repeats, each containing about 25 kb. Chloroplast DNA does not show recombination events that are common in the nuclear genome. It is thus a good subject for DNA phylogenetic analysis.

A characteristic feature of the chloroplast genome is the presence of two identical inverted repeats in the DNA sequence. Other DNA sequence inversions or deletions occur rarely, but when they do occur, they provide a character or a tool to analyze evolutionary relationships between plants. For instance, a large inversion in chloroplast DNA is found in the Asteraceae, or sunflower family, and not in other plant families. While previous work on the evolutionary relationships between plants has emphasized the comparative analysis of plant anatomy or morphology, there is increasing use of plant molecular data such as chloroplast DNA sequences. When considered together, morphological and molecular information can provide a clearer understanding of the evolutionary processes that govern biological diversity.

Plant nuclear genomes may contain large amounts of DNA in comparison to other eukaryotic organisms, but only a small amount of this DNA represents functional genes. Excess DNA in plant genomes can result from increased chromosome copy number (polyploidy), and DNA sequence repeats. Chloroplast genomes evolve more slowly than nuclear genomes and can provide important evolutionary information.

Comparative Genome Mapping and Model Systems

Knowledge of plant genomes has been growing with the advent of new techniques to study DNA sequences, such as gene mapping and chromosome synteny. An increased understanding of plant genomes can lead to better manipulation of genetic traits such as crop yield, disease resistance, growth abilities, nutritive qualities, or drought tolerance. Multiple genes could encode each of these traits. By genome mapping model plants, plant biologists can lay a foundation for future plant breeding and for an understanding of plant evolution at the genetic level. One such model system, rice, has been chosen because it has a high level of synteny with other grains. In a genomic sense, “rice is wheat.” This provides a strong argument for rice as a model system. The other model system that has been selected in plants is *Arabidopsis*. This small weed that is a member of the mustard family has an unusually small genome with only 20% repetitive DNA (see table 43.1) which has made it possible to sequence the entire genome. Getting down to the level of individual base pairs is a step-wise process, as described below.

RFLP and AFLP as Tools to Map Genomes and Detect Polymorphisms

The classical approach to locating genes in linear order on chromosomes involves making crosses between plants with known genes identified by mutations. The frequency of recombination is used to calculate distance (see chapter 13). The result is a genetic or linkage map. This approach is limited to genes with alleles that can be phenotypically identified. Much more of the genome can be mapped using RFLPs (restriction fragment length polymorphisms) which need not have a macroscopic phenotype. This approach, described in detail in chapter 19 (see figures 19.2, 19.4, 19.9, and 19.10), involves analysis of the RFLP map, or the pattern of DNA fragments, produced when DNA is treated with restriction enzymes that cleave at specific sites. RFLP mapping can identify important regions of the genome at a glance, while sequence data require sophisticated computer-based searching and matching systems. A comparison of the RFLP maps of parents and progeny can give an indication of the heritability of gene traits and of heritable loci that are characteristic of traits. If the trait and the RFLP co-segregate, you have a direct link between the trait and the DNA sequence. Moreover, after full genomes are sequenced at the nucleotide level, the genetic identification of RFLP markers in regions of interest will be facilitated. Remember, RFLPs are chunks of DNA that may contain a part of one or more genes. Currently, the most dense RFLP map is in rice where 2000 DNA sequences have been mapped onto 12 chromosomes.

Another tool that utilizes sequence variability is AFLPs, or amplified fragment length polymorphisms.

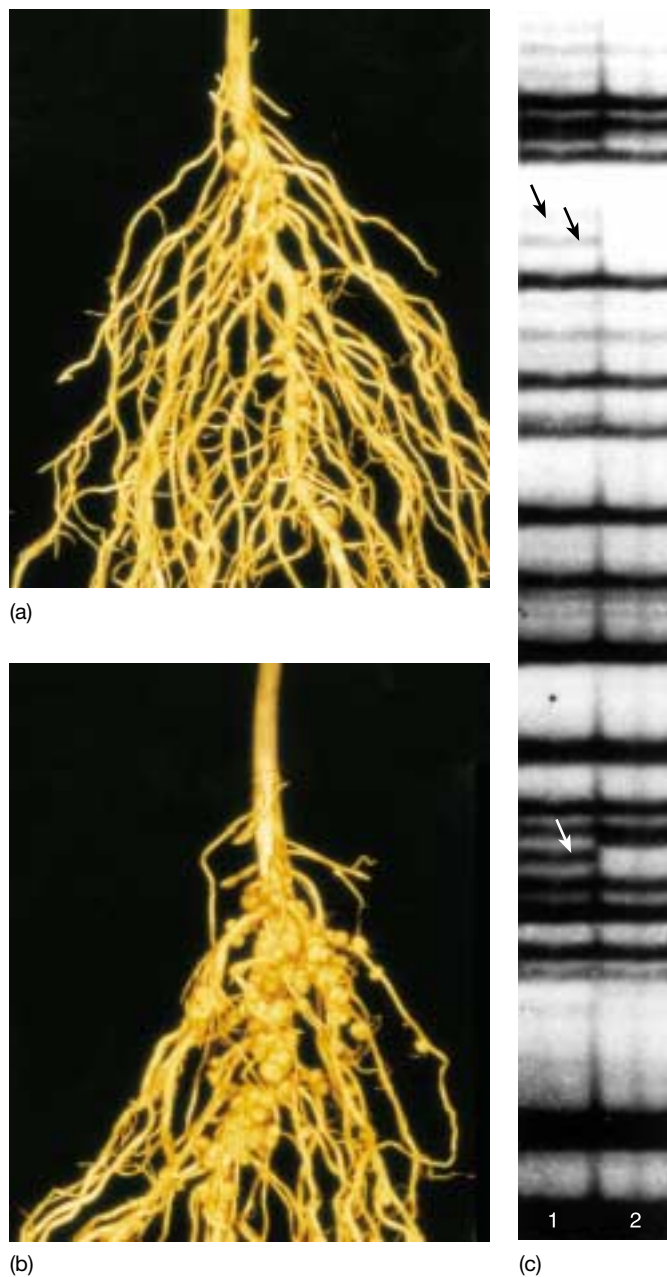


FIGURE 43.6
AFLP fingerprint pattern from normal and “hypernodulating” soybeans. It is still not known what determines the nodule number in (a) a normal soybean root versus (b) a “hypernodulating” mutant. The slight genetic differences between these plants can be evaluated by AFLP (c). The banding pattern changes indicate what genetic markers are linked to the “hypernodulation” mutation. Lane 1: normal soybean DNA; Lane 2: “hypernodulating” soybean DNA.

Hybridizing DNA primers with genomic DNA fragments that have been cut with restriction enzymes, usually *EcoRI* and *MseI*, and then subsequently amplified using the polymerase chain reaction (PCR) generates AFLP maps. The resulting PCR products, which represent each piece of DNA cut by a restriction enzyme, are separated by size

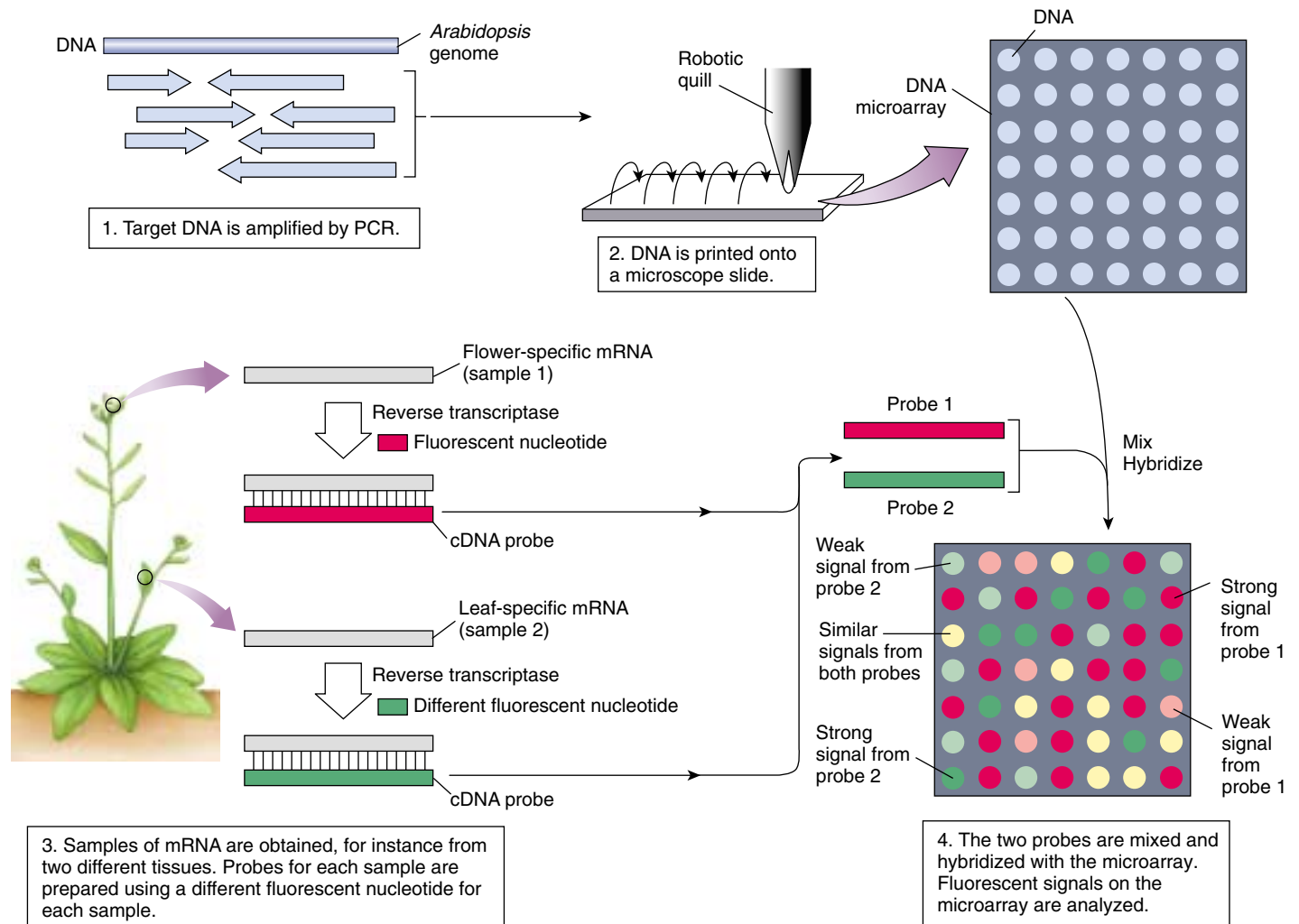


FIGURE 43.7

Microarrays. Microarrays are created by robotically placing DNA onto a microscope slide. The microarray can then be probed with RNA from tissues of interest to identify expressed DNA. The microarray with hybridized probes is analyzed and often displayed as a false-color image. If a gene is frequently expressed in one of the samples, the fluorescent signal will be strong (red or green) where the gene is located on the microarray. If a gene is rarely expressed in one of the samples, the signal will be weak (pink or light green). A yellow color indicates genes that are expressed at similar levels in each sample.

via gel electrophoresis. The band sizes on an AFLP gel tend to show more polymorphisms than those found with RFLP mapping because the entire genome is visible on the gel (figure 43.6). Both RFLPs and AFLPs (among many other tools for genome analysis) can provide markers of traits which are inherited from parents to progeny through crosses.

DNA Microarrays

How can DNA sequences be made available to researchers, other than as databases of electronic information? DNA microarrays are a way to link sequences with the study of gene

function and make DNA sequences available to many. Also called biochips or “genes on chips,” these convenient assays for the presence of a particular version of a gene were discussed in chapter 19. To prepare a particular DNA microarray, fragments of DNA are deposited on a microscope slide by a robot at indexed locations. Up to 10,000 spots can be displayed over an area of only 3.24 cm² (figure 43.7). The primary applications of microarrays are to determine which genes are expressed developmentally in certain tissues or in response to environmental factors. RNA from these tissues can be isolated and used as a probe for these microarrays. Only those sequences that are expressed in the tissues will be present to hybridize to the spot on the microarray.

Plant Genome Projects

The potential of having complete genomic sequences of plants is tremendous and about to be realized now that the *Arabidopsis* Genome Project is essentially complete. This project represents a new paradigm in the way biology is done. The international effort brought together research teams with the expertise and tenacity to apply new sequencing technology to an entire genome, rather than single genes. Powerful databases are being constructed to make this information accessible to all. The completely sequenced *Arabidopsis* genome will have far-reaching uses in agricultural breeding and evolutionary analysis. This information can be expected to help plant breeders in the future because the localization of genes in one plant species can help indicate where that gene might also be located in another species (figure 43.8). In plant genomes, local gene order seems to be more conserved than the nucleotide sequences of homologous genes. Thus, the complete genomic sequence of *Arabidopsis thaliana* will facilitate gene cloning from many plant species, using information on relative genomic location as well as similarity of sequences.

Sequencing the rice genome provides a model for a small monocot genome. Rice was selected, in part, because its genome is 6, 10, and 40 times smaller than maize, barley, and wheat. These grains represent a major food source for humans. By understanding the rice genome at the level of its DNA sequence, it should be much easier to identify and isolate genes from grains with larger genomes. Even though these plants diverged more than 50 million years ago, the chromosomes of rice, corn, barley, wheat, and other grass crops show extensive conserved arrangements of segments (synteny) (figure 43.9). DNA sequence analysis of cereal grains will be important for identifying genes associated with disease resistance, crop yield, nutritional quality, and growth capacity. It will also be possible to construct an approximate map of the ancestral cereal genome.

Functional Genomics and Proteomics

Sequencing the *Arabidopsis* and rice genome represent major technological accomplishments. A new field of bioinformatics takes advantage of high-end computer technology to analyze the growing gene databases, look for relationships among genomes, and hypothesize functions of genes based on sequence. Genomics (the study of genomes) is now shifting gears and moving back to

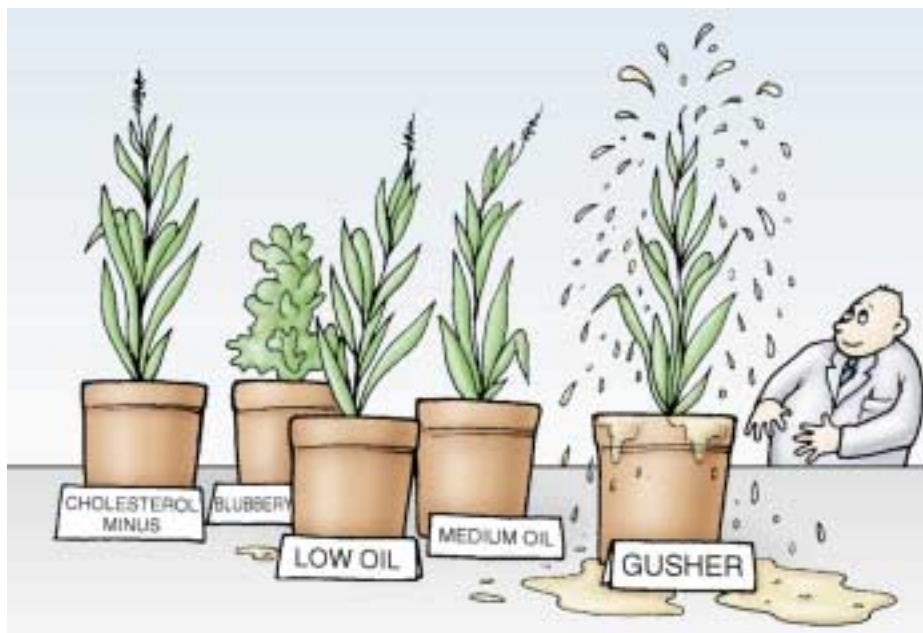


FIGURE 43.8
Future directions in the genetic engineering of vegetable oils?

hypothesis-driven science. Again, an international community of researchers has come together with a plan to assign function to all of the 20,000 to 25,000 *Arabidopsis* genes by 2010 (Project 2010). In many ways, the goal is to ultimately answer the questions we have raised in chapters 37 through 42. One of the first steps is to determine when and where these genes are expressed. Each step beyond that will require additional enabling technology. Research will move from genomics to proteomics (the study of all proteins in an organism). Proteins are much more difficult to study because of posttranslational modification and formation of complexes of proteins. This information will be essential in understanding cell biology, physiology, development, and evolution. For example, how are similar genes used in different plants to create biochemically and morphologically distinct organisms? So, in many ways, we continue to ask the same questions that even Mendel asked, but at a much different level of organization.

Restriction fragment length polymorphisms (RFLPs) and amplified fragment length polymorphisms (AFLPs) represent important tools for mapping genetic traits in plant genomes. Due to its short life cycle, small size, and small genome, the mustard relative *Arabidopsis thaliana* is being used as a model plant for genetic studies. The genome of rice is also essentially sequenced and will be a valuable model for other monocot cereal grains such as wheat, barley, oats, and corn. Assigning function to these genes is the next challenge.

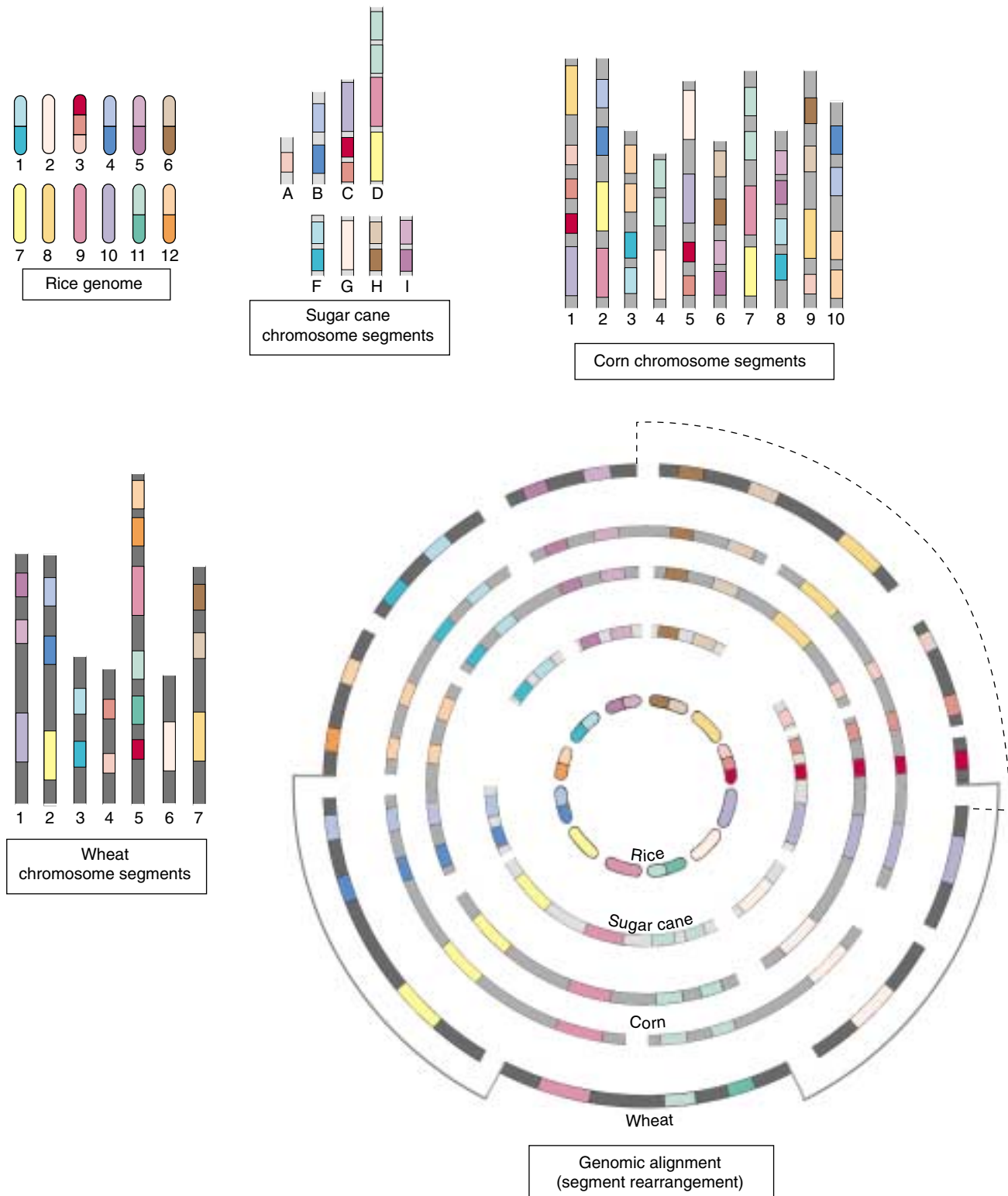


FIGURE 43.9

Grain genomes are rearrangements of similar chromosome segments. Shades of the same color represent pieces of DNA that are conserved among the different species but have been rearranged. By splitting the individual chromosomes of major grass species into segments, and rearranging the segments, researchers have found that the genome components of rice, sugar cane, corn, and wheat are highly conserved. This implies that the order of the segments in the ancestral grass genome has been rearranged by recombination as the grasses have evolved. Data: G. Moore, K. M. Devos, Z. Wang, and M. D. Gale: "Grasses, line up and form a circle," *Current Biology* 1995, vol. 5, pp. 737-739.

43.2 Advances in plant tissue culture are revolutionizing agriculture.

Overview of Plant Tissue Culture

One of the major hopes for the plant genome projects is using newly identified genes for biotechnology, and advances in tissue culture are facilitating this. Having an agriculturally valuable gene in hand is just the beginning. With methods discussed in chapter 19 and below, desirable genes can be introduced into plants, yielding **transgenic** cells and tissues. Whole plants can then be regenerated using tissue culture. While animals can now be cloned, the process is much simpler in plants. Many somatic (not germ-line) plant cells are totipotent, which means they can express portions of their previously unexpressed genes and develop into whole plants under the right conditions.

The successful culture of plant cells, tissues, or organs requires utilizing the proper plant starting material, appropriate nutrient medium, and timing of hormonal treatments to maximize growth potential and drive differentiation (figure 43.10). Most plant tissue cultures are initiated from *explants*, or small sections of tissue removed from an intact plant under sterile conditions. After being placed on a sterile growth medium containing nutrients, vitamins, and combinations of plant growth regulators, cells present in the explant will begin to divide and proliferate. Under appropriate culture conditions, plant cells can multiply and form organs (roots, shoots, embryos, leaf primordia, and so on) and can even regenerate a whole plant. The regeneration of a whole plant from tissue-cultured plant cells represents an important step in the production of genetically engineered plants. Using plant tissue cultures, genetic manipulation can be conducted at the level of single cells in culture, and whole plants can then be produced bearing the introduced genetic trait.

Plant cells growing in culture can also be used for the mass production of genetically identical plants (clones) with valuable inheritable traits. For example, this approach of clonal propagation using plant tissue culture is commonly used in the commercial production of many ornamental plants such as chrysanthemums and ferns. As we will describe next, different types of cultures can be generated based on the initial type of plant tissue used for the explant and on the composition of the growth medium.

It is often possible to regenerate an entire plant from one or a few cells. Depending on the plant tissue used and the growth medium selected, different types of cultures may be produced.



(a)



(b)



(c)

FIGURE 43.10

Culture of orchid plants. While the natural maturation of a single orchid plant may take up to seven years, commercial growers can produce thousands of cultured orchids in a relatively short time. (a) The apical meristem is removed from an orchid plant. (b) The meristematic tissue is grown in flasks containing hormone-enhanced media, and roots and shoots begin to form. (c) The plantlets are then separated and grown to maturity.

Types of Plant Tissue Cultures

Depending on the type of plant tissue used as the explant and the composition of the growth medium, a variety of different types of plant tissue cultures can be generated. These different types of plant tissue cultures have applications both in basic plant research and commercial plant production.

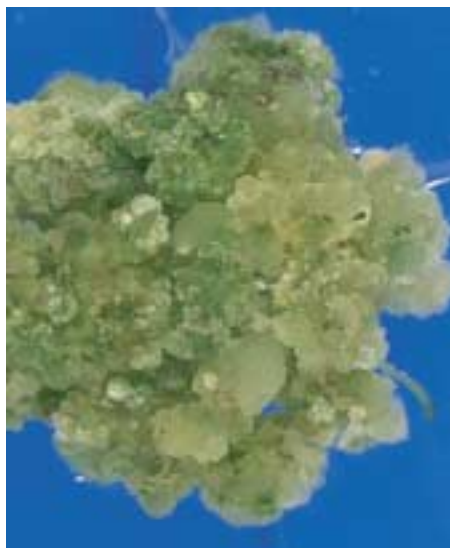
Callus Culture

Callus culture refers to the growth of unorganized masses of plant cells in culture. To generate a callus culture, an explant, usually containing a region of meristematic cells, is incubated on a growth medium containing certain plant growth regulators such as auxin and cytokinin (figure

43.11). The cells grow from the explant and divide to form an undifferentiated mass of cells called a callus. This unorganized mass of growing cells is analogous to a plant tumor. Cells can proliferate indefinitely if they are periodically transferred to fresh growth media. However, if the callus cells are transferred to a growth medium containing a different combination of plant growth regulators, the cells can be directed to differentiate into roots and/or shoots. This process of converting unorganized growth into the production of shoots and roots is called **organogenesis**, and it represents one means by which a whole plant can be regenerated from tissue culture cells. When a plantlet produced by organogenesis is large enough, it can be transferred to a large container with nutrients or soil and grown to maturity.



(a)



(b)



(c)



(d)

FIGURE 43.11

Callus culture. (a) An explant is incubated on growth media. (b) The cells grow and divide and form a callus. (c) The callus cells, grown on media containing new plant growth regulators, differentiate into plant parts. (d) After the plantlet is large enough, it is grown to maturity in soil.

Cell Suspension Culture

Plant cell suspension culture involves the growth of single or small groups of plant cells in a liquid growth medium. Cell suspension cultures are usually initiated by the transfer of plant callus cells into a liquid medium containing a combination of plant growth regulators and chemicals that promote the disaggregation of the cells into single cells or small clumps of cells (figure 43.12). Continued cell growth requires that the liquid cultures be shaken at low speed to promote aeration and chemical exchange with the medium. Suspension cell cultures are often used in research applications where access to single cells is important. The suspension bath can provide an efficient means for selecting out cells with desirable traits such as herbicide tolerance or salt tolerance because the bath is in uniform contact with all the cells at once. This differs from callus culture, where only those cells in contact with the solid medium can be selected by chemical additions to the medium. Suspension cultures can also provide a convenient means for producing and collecting the plant chemicals cells secrete. These can include important plant metabolites, such as food products, oils, and medicinal chemicals. In addition, plant suspension cell cultures can often be used to produce whole plants via a process known as **somatic cell embryogenesis** (figure 43.13). For some plants, this provides a more convenient means of regenerating a whole plant after genetic engineering takes place at the single-cell level. In somatic cell embryogenesis, plant



FIGURE 43.12

Cell suspension culture. Plant cells can be grown as individual cells or small groups of cells in a liquid culture medium. Liquid suspension culture of plant cells ensures that most cells are in contact with the growth medium.

suspension culture cells are transferred to a medium containing a combination of growth regulators that drive differentiation and organization of the cells to form individual embryos. Under a dissection microscope, these embryos can be isolated and transferred to a new growth medium, where they grow into individual plants.



(a)



(b)



(c)



(d)



(e)

FIGURE 43.13

Somatic cell embryogenesis. A large number of plants can be cloned from a single soybean seed via somatic cell embryogenesis. (a) Immature soybean seeds placed on culture medium. (b) Embryos appear on the seeds after two weeks in culture. (c) Four embryos at different stages of development (globular, heart, torpedo, and plantlet). (d) Seedlings with shoots and roots. (e) Mature soybean plants.

Protoplast Isolation and Culture

Protoplasts are plant cells that have had their thick cell walls removed by an enzymatic process, leaving behind a plant cell enclosed only by the plasma membrane. Plant protoplasts have been extremely useful in research on the plant plasma membrane, a structure normally inaccessible due to its close association with the cell wall. Within hours of their isolation, plant protoplasts usually begin to resynthesize cell walls, so this process has also been useful in studies on cell wall production in plants. Plant protoplasts are also more easily transformed with foreign DNA using approaches such as electroporation (see the subsequent section). In addition, protoplasts isolated from different

plants can be forced to fuse together to form a hybrid. If they are regenerated into whole plants, these hybrids formed from protoplast fusion can represent genetic combinations that would never occur in nature. Hence, protoplast fusion can provide an additional means of genetic engineering, allowing beneficial traits from one plant to be incorporated into another plant despite broad differences between the species. When either single or fused protoplasts are transferred to a culture growth medium, cell wall regeneration takes place, followed by cell division to form a callus (figure 43.14). Once a callus is formed, whole plants can be produced either by organogenesis or by somatic cell embryogenesis in culture.

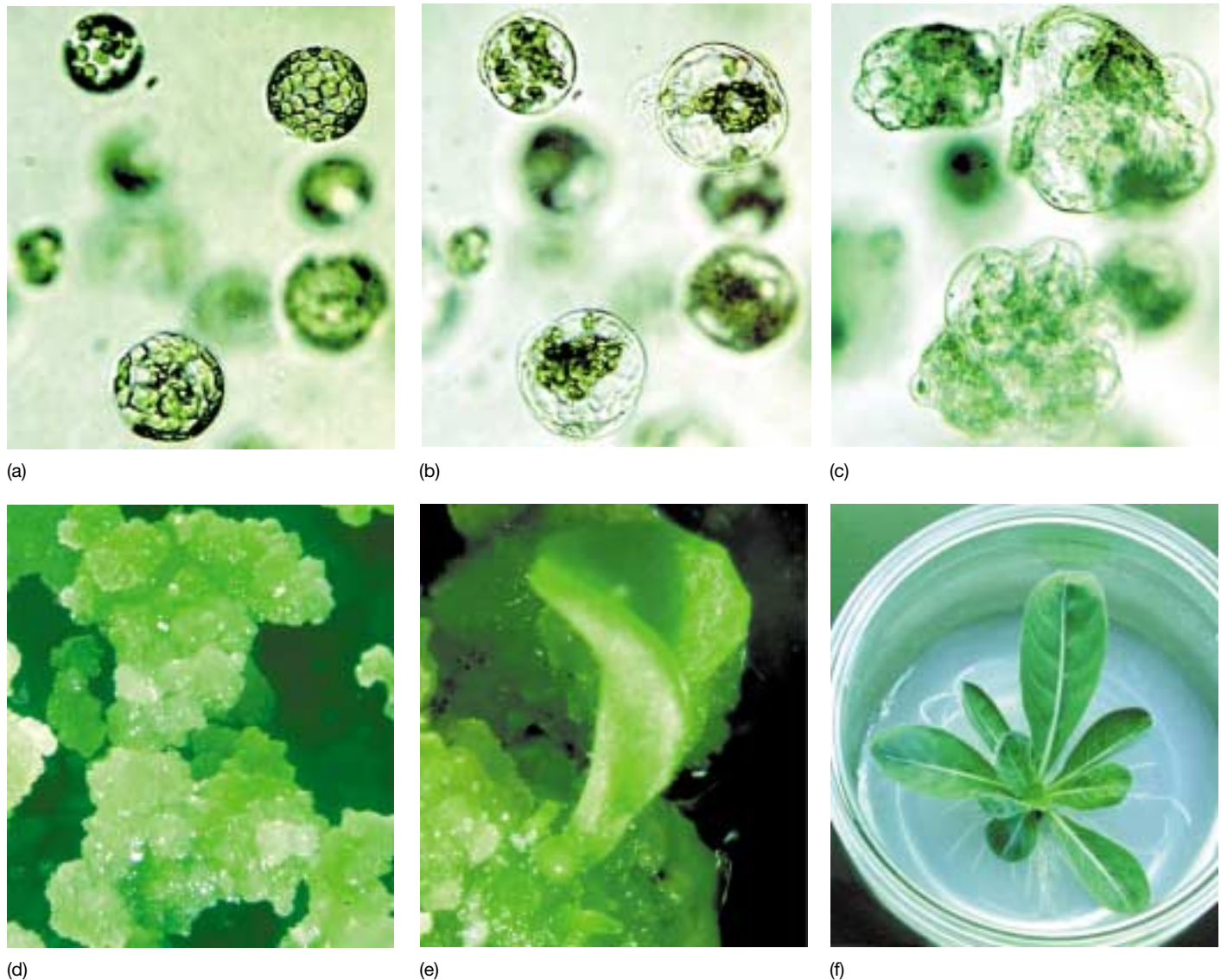


FIGURE 43.14

Protoplast regeneration. Different stages in the recovery of intact plants from single plant protoplasts of evening primrose. (a) Individual plant protoplasts. (b) Regeneration of the cell wall and the beginning of cell division. (c, d) Aggregates of plant cells resulting from cell division which can form a callus. (e) Production of somatic cell embryos from the callus. (f) Recovery of a plantlet from the somatic cell embryo through the process described in figure 43.13.

Anther/Pollen Culture

In flowers, the anthers are the anatomical structures that contain the pollen. In normal flower development, the anthers mature and open to allow pollen dispersal. In anther culture, anthers are excised from the flowers of a plant and then transferred to an appropriate growth medium. After a short period of time, pollen cells can be manipulated to form individual plantlets, which can be grown in culture and used to produce mature plants. The development of these plantlets usually proceeds through the formation of embryos (figure 43.15). Plants produced by anther/pollen culture can be haploid because they were originally derived from pollen cells that have undergone meiosis. However, these plants may be sterile and thus not useful for breeding or genetic manipulation. On the other hand, plants derived from anther/pollen culture can be treated at an early stage with chemical agents such as colchicine, which allows chromosome duplication. Chromosome duplication results in the conversion of sterile haploid plants into fertile diploid organisms. Under these conditions, plants can be produced that are homozygous for every single trait, even those which tend to be recessive traits. On a cautionary note, not every cell exposed to colchicine becomes diploid. Some have unusual ploidy levels, and they can be screened for chromosome number. The homozygous plants are useful tools, allowing breeders to introduce a normally recessive trait.

Plant Organ Culture

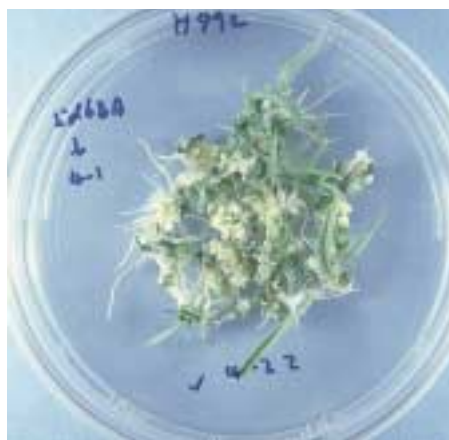
Plant organs can also be grown under culture conditions, and this has provided a useful tool in the study of plant organ development. For example, pollinated flowers of a plant such as a tomato can be excised and transferred to a culture flask containing an appropriate medium. Over time, the ovular portion of the plant will develop into a tomato fruit that will eventually turn red and ripen. Sections of plant roots can also be excised and transferred to a liquid growth medium. In this medium, the roots can proliferate extensively, forming both primary and secondary root branches (figure 43.16).



(a)



(b)



(c)

FIGURE 43.15

Anther culture. Callus formation from maize pollen. Anthers containing pollen can be regenerated on tissue culture medium. The pollen in the anthers contain a haploid set of chromosomes, which can be doubled to form a homozygous diploid cell. Regenerated homozygous diploid plants are important for plant breeding purposes. (a) Maize anthers in culture medium. (b) Callus formation from pollen. (c) Callus and shoot formation.



FIGURE 43.16

Plant organ culture. Plant roots growing in a liquid culture medium. From small excised sections of plant roots, the roots will grow and proliferate with extensive lateral root formation (branching).

Many plant cells are totipotent; a whole plant may be regenerated from a single plant cell. Depending upon the explant type, culture medium, and combinations of plant growth regulators, it is possible to grow plant cells, tissues, or organs in sterile cultures.

Applications of Plant Tissue Culture

In addition to the applications already described, plant tissue cultures have a variety of uses both in agriculture and in industry.

Suspension Cultures as Biological Factories

An important industrial application of plant tissue culture involves the use of plant cells as biological factories. Large-scale suspension cultures can be grown to produce antimicrobial compounds, antitumor alkaloids, vitamins, insecticides, and food flavors. Plant roots can also be grown in liquid culture, creating a mesh of roots that can produce a number of useful plant compounds.

Horticultural Uses

Plants with valuable traits can be mass propagated through tissue-culture cloning. In this application of plant tissue culture, hundreds or even thousands of genetically identical plants can be produced by vegetative asexual propagation from one plant source. This has been extensively used in the flower industry where genetically identical plants can be produced from a superior parent plant. Propagation of plant tissue in the sterile environment of the growth medium can also help in the production disease-free plants, such as those cultured from the meristematic (apical dome) tissue untouched by viruses or other diseases because it is new growth. This approach has been particularly useful in the culture of disease-free orchids and raspberries.

Somaclonal Variation

Plant tissue culture also has a problematic side effect that can be used as an asset under certain conditions. During periods of extended growth of plant cells in callus or suspension cell culture, various parts of the plant genome may become more or less “active” due to a release of control over gene expression. Transposable elements may also become more active, and chromosomal rearrangements may occur. Sometimes you end up with unusual numbers of chromosomes. This altered control provides a new source of genetic diversity that can result in novel traits which

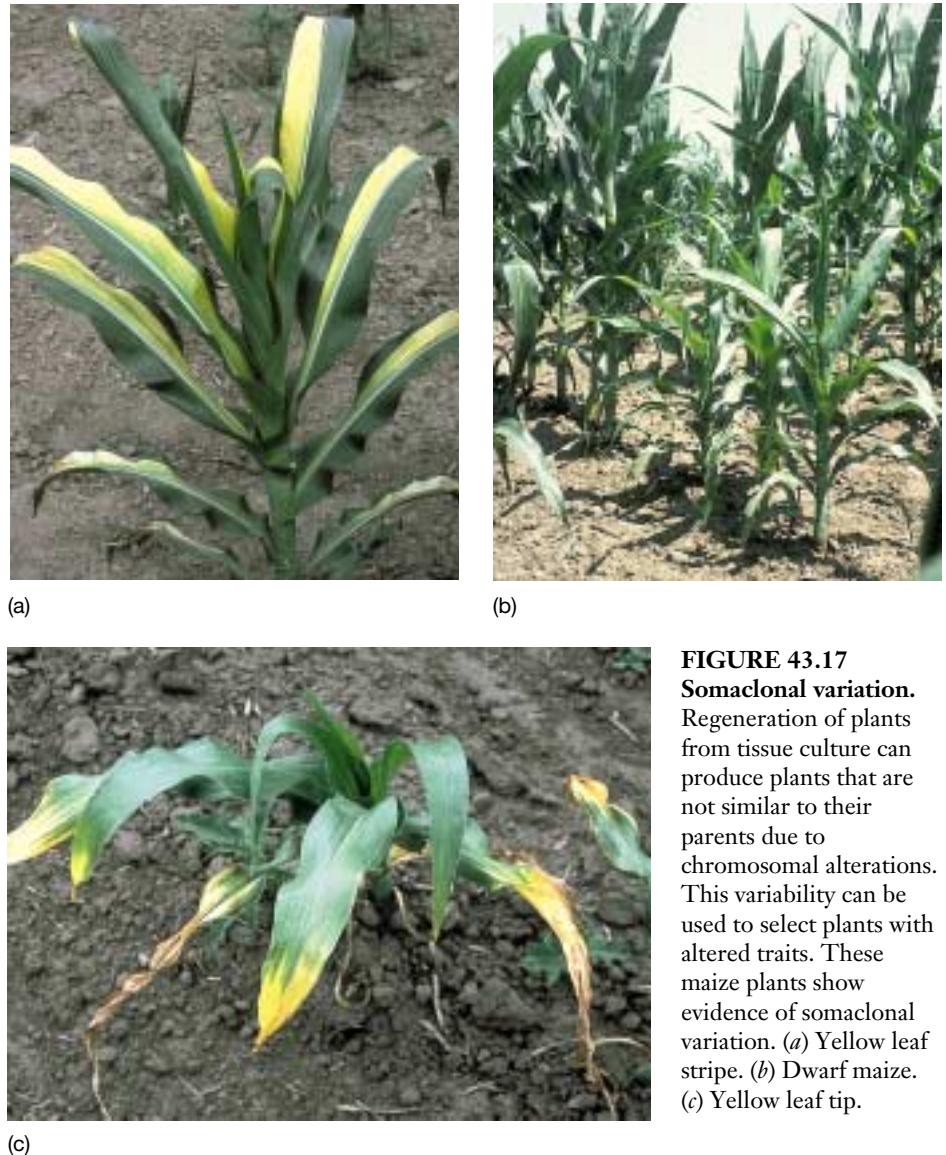


FIGURE 43.17
Somaclonal variation. Regeneration of plants from tissue culture can produce plants that are not similar to their parents due to chromosomal alterations. This variability can be used to select plants with altered traits. These maize plants show evidence of somaclonal variation. (a) Yellow leaf stripe. (b) Dwarf maize. (c) Yellow leaf tip.

were not even present in the original plant material used as the explant to start the cultures (figure 43.17). This increased genetic diversity following extended time in tissue culture is called **somaclonal variation**. It can be problematic if the desired goal is the propagation or production of identical plant clones. However, somaclonal variation, induced by intentionally growing plant cells in tissue over a longer time period, can be very useful to generate novel plants with traits not currently present in a given gene pool. These traits can be identified either at the tissue culture stage (for example, disease resistance or heat tolerance) or following the regeneration of whole plants by either organogenesis or embryogenesis (plant size, photosynthetic rates, and so forth).

Plant cell, tissue, and organ cultures have important applications in agriculture and industry.

43.3 Plant biotechnology now affects every aspect of agriculture.

Plant biotechnology provides an efficient means to produce an array of novel products and tools for use by our global society. Agricultural biotechnology has the potential to increase farming revenue, lower the cost of raw materials, and improve environmental quality. Plant genetic engineering is becoming a key tool for improving crop production.

World Population in Relation to Advances Made in Crop Production

Due in a large part to scientific advances in crop breeding and farming techniques, world food production has doubled since 1960. Moreover, productivity of agricultural land and water usage has tripled over this time period. While major genetic improvements have been made in crops through crop breeding, this can be a slow process. Furthermore, most crops grown in the United States produce less than 50% of their genetic potential. These shortfalls in yield are due in large part to the inability of crops to tolerate or adapt to environmental stresses (salt, water, and temperature), pests, and disease (figure 43.18).

The world now farms an area the size of South America, but without the scientific advances of the past 30 years, farmland equaling the entire western hemisphere would be required to feed the world. Nevertheless, the world population is expected to double to 12 billion by the first half of this century, and it is not clear whether current levels of food production can keep pace with this rate of population growth. Many believe the exploitation of conventional crop breeding programs may have reached their limit. The question is how best to feed billions of additional people without destroying much of the planet in the process. In this respect, the disappearance of tropical rain forests, wetlands, and other vital habitats will accelerate unless agriculture becomes more productive and less taxing to the environment. Advances in our understanding of plant reproduction from the molecular to the ecosystems levels are providing tools to further protect natural environments by preventing the spread of modified genes to wild populations.

Although improved farming practices and crop breeding have increased crop yields, it is uncertain whether these approaches can keep pace with the food demands of an ever-increasing world population.

FIGURE 43.18
Corn crop
productivity well
below its genetic
potential due to
drought stress. Corn
production can be
limited by water
deficiencies due to
drought during the
growing season in dry
climates.



Plant Biotechnology for Agricultural Improvement

It seems certain that plant genetic engineering will play a major role in resolving the problem of feeding an increasing world population. The nutritional quality of crop plants is being improved by increasing the levels of nutrients they contain, such as beta-carotene and vitamins A, C, and E, which may protect people from health problems such as cancer and heart disease. Biotechnology is now being employed to improve the quality of seed grains, increase protein levels in forage crops, and transform plants to improve their resistance to disease, insects, herbicides, and viruses. Other stresses on plants, such as heat or salt, can be improved by engineering higher tolerance levels.

Compared with approaches that rely on plant breeding, genetic engineering can compress the time frame required for the development of improved crop varieties. Moreover, in genetic engineering, genetic barriers, such as pollen compatibility with the pistil, no longer limit the introduction of advantageous traits from diverse species into crop plants. Once a useful trait has been identified at the level of individual genes and their DNA sequences, the incorporation of this trait into a crop plant requires only the introduction of the DNA bearing these genes into the crop plant genome. The process of incorporating foreign DNA into an existing plant genome is called **plant transformation**. At present, there are several approaches for plant transformation; the use of *Agrobacterium tumefaciens* in this process was described in chapter 19. This approach works best if the plant being transformed is a dicot. However, many food crops, such as the cereal grains (rice, wheat, corn, barley, oats, and so on) are monocots. Two additional plant transformation methods that can be used with both dicots and monocots are discussed in the next section.

Useful Traits That Can Be Introduced into Plants

Although plant transformation represents a relatively new technology, extensive efforts are underway to utilize this approach to develop plants and food products with beneficial characteristics. We discussed a variety of biotechnological applications for crop improvement in chapter 19. Further applications of this approach involve modifications of nutritional quality of foods, phytoremediation, production of plastics, and using plants as “edible vaccines.”

Improved Nutritional Quality of Food Crops. Approximately 75% of the world’s production of oils and fats come from plant sources. For medical and dietary reasons, there is a trend away from the use of animal fats and toward the use of high-quality vegetable oils. Genetic engineering has allowed researchers to modify seed oil biochemistry to produce “designer oils” for edible and nonedible products. One technique modifies canola oil to replace cocoa butter

as a source of saturated fatty acids; others modify the enzyme ACP desaturase for the creation of monounsaturated fatty acids in transgenic plants. High-lauric acid canola has been planted in several countries and used in both foods and soaps.

Attempts are also underway to modify the amino acid contents of various plant seeds to present a more complete nutritional diet to the consumer. A high-lysine corn seed is being developed; this would cut down on the need for lysine supplements that are currently added to livestock feed. Biotechnology has the potential to make plant foods healthier and more nutritious for human consumption. Fruits and vegetables, such as tomatoes, may be engineered to contain increased levels of vitamins A and C and beta-carotene, which, when included in the human diet, may help protect against chronic diseases.

Phytoremediation. Cleaning up environmental toxins to reclaim polluted land is an ongoing challenge. Genetically modified plants offer an enticing solution. Work is progressing on plants that accumulate heavy metals at high concentrations. These plants can then be harvested. Because most of their biomass is water, the dried plants allow for the collection of the metals in a small area. Organic compounds that pose hazards to human health have the potential to be taken up by plants and broken down into harmless components. Modified biochemical pathways are being used to break down toxic substances. Modified poplars, for example, have been engineered to break down TNT.

Plants Bearing Vaccines for Human Diseases. Another very interesting application of plant genetic engineering includes the introduction of “vaccine genes” into edible plants. Here, genes encoding the antigen (for example, a viral coat protein) for a particular human pathogen is introduced into the genome of an edible plant such as a banana, tomato, or apple via plant transformation. This antigen protein would then be present in the cells of the edible plant, and a human individual that consumed the plant would develop antibodies against the pathogenic organism. Currently, researchers are trying to develop such edible vaccines for a coat protein of hepatitis B, an enterotoxin B of *E. coli*, and a viral capsid protein of the Norwalk virus. The measles gene has been introduced into tobacco as a model system and is now being introduced into lettuce and rice. This is a terrific advance for tropical areas where it is difficult to keep the traditional vaccine cold (remember that proteins degrade rapidly as the temperature increases).

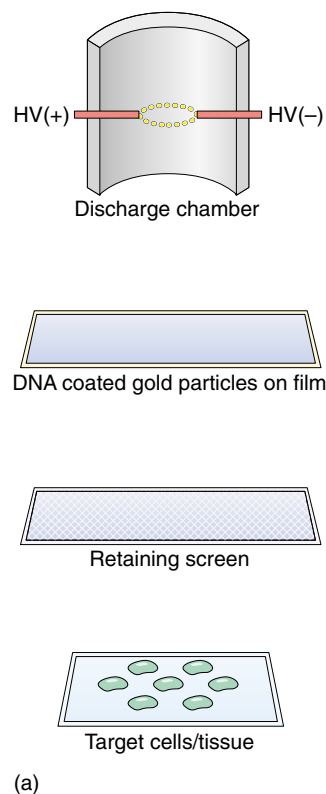
Genetic engineering of crop plants has allowed researchers to alter the oil content, amino acid composition, and vitamin content of food crops.

Genetic engineering may also allow the production of food crops bearing “edible vaccines.”

Methods of Plant Transformation

Plant Transformation Using the Particle Gun

Using a “gun” to blast plant cells does not seem like a suitable method for introducing foreign DNA into a plant genome. However, it works, and many whole plants have been regenerated after foreign DNA is shot through the cell wall and then integrated into the plant genome. The *particle gun* utilizes microscopic gold particles coated with the foreign DNA, shooting these particles into plant cells at high velocity. Acceleration of the particles to a sufficient velocity to pass through the plant cell wall can be achieved by a burst of high-pressure helium gas or an electrical discharge (figure 43.19*a*). Only a few cells actually receive the foreign DNA and survive this treatment. These cells are identified with the help of a **selectable marker** also present on the foreign DNA. The selectable marker allows only those cells receiving the foreign DNA to survive on a particular growth medium (figure 43.20). The selectable markers include genes for resistance to a herbicide or antibiotic. Plant cells which survive growth in the selection medium are then tested for the presence of the foreign gene(s) of interest.



Plant Transformation Using Electroporation

Foreign DNA can also be “shocked” into cells that lack a cell wall, such as the plant protoplasts described earlier. A pulse of high-voltage electricity in a solution containing plant protoplasts and DNA briefly opens up small pores in the protoplasts’ plasma membranes, allowing the foreign DNA to enter the cell (figure 43.19*b*). Ideally, the DNA incorporates into one of the plant’s chromosomes. Following electroporation, the protoplasts are transferred to a growth medium for cell wall regeneration, cell division, and, eventually, the regeneration of whole plants. As with the use of the particle gun, a selectable marker is typically present in the foreign DNA, and protoplasts containing foreign DNA are selected based upon their ability to survive and proliferate in a growth medium containing the selection treatment (antibiotic or herbicide). Once regenerated from electroporated protoplasts, whole plants can then be evaluated for the presence of the beneficial trait.

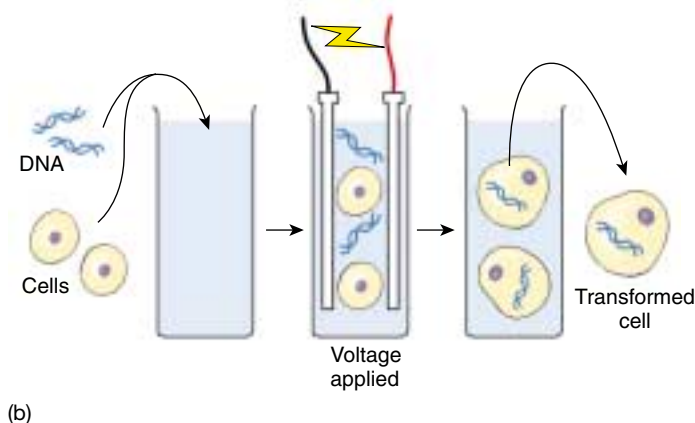
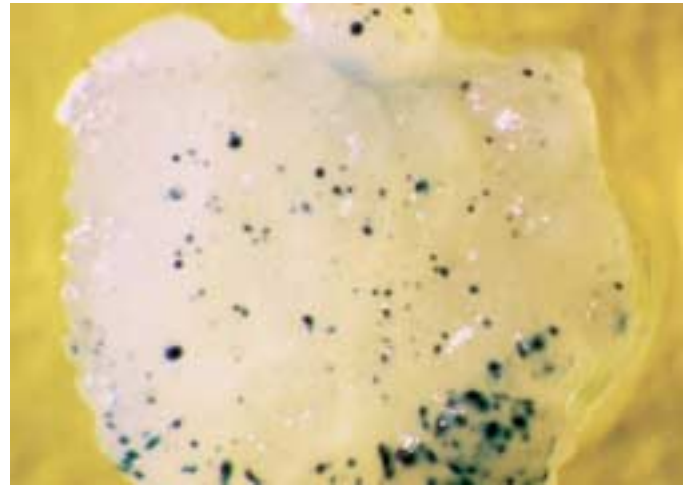


FIGURE 43.19
Methods for plant transformation. (a) The particle gun is one method for introducing foreign DNA into plant cells. Here an electrical discharge propels DNA-coated gold particles into plant cells or tissue. A retaining screen reduces cellular damage associated with bombardment by only allowing the DNA-coated particles to pass and retaining fragments of the mounting film. (b) Foreign DNA can also be introduced into plant protoplasts by electroporation. A brief pulse of electricity generates pores in the plasma membrane, allowing DNA to enter the cells.

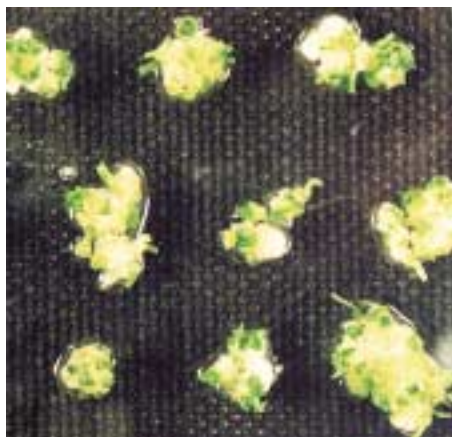
Plant biotechnology may play an important role in the further improvement of crop plants. The particle gun and electroporation are useful methods for introducing foreign DNA into plants.



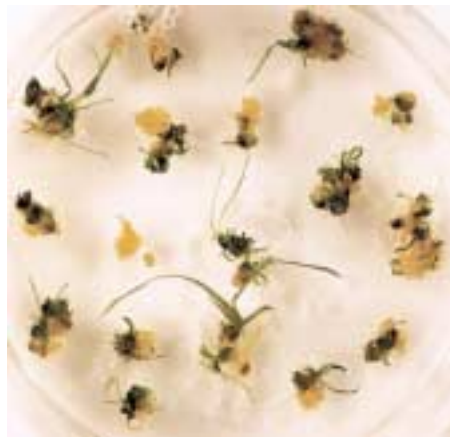
(a)



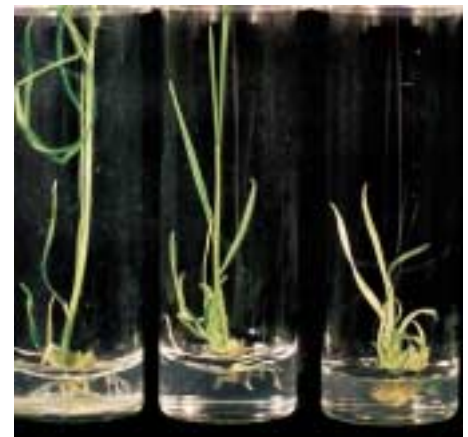
(b)



(c)



(d)



(e)



(f)

FIGURE 43.20

Regeneration after transformation with the use of a selectable marker. Stages in the recovery of a plant containing foreign DNA introduced by the “particle gun” method for plant cell transformation. A selectable marker, in this case a gene for resistance to herbicide, aids in the identification and recovery of plants containing the DNA insert. (a) Embryonic callus just prior to particle gun bombardment. (b) Following bombardment, callus cells containing the foreign DNA are indicated by color from the *gus* gene used as a tag or label on the foreign DNA. (c) Shoot formation in the transformed plants growing on a selective medium. Here, the gene for herbicide resistance in the transformed plants allows growth on the selective medium containing the herbicide. Nontransformed plants do not contain the herbicide resistance gene and do not grow well. (d) Production of plantlets from transformed plants growing on the selective medium. (e) Comparison of growth on the selection medium for transformed plants bearing the herbicide resistance gene (*left*) and a nontransformed plant (*right*). (f) Mature transgenic plants resulting from this process.



Summary

Questions

Media Resources

43.1 Genomic organization is much more varied in plants than in animals.

- Plant genomes are very large in comparison to other eukaryotes, mainly due to a high amount of repetitive DNA.
- Plant genomes can be compared with one another by mapping the locations of certain genes or gene traits in various plants. RFLPs and AFLPs can be used to map plant DNA.
- *Arabidopsis thaliana* has a small genome, for a plant. This complete genome is essentially sequenced, so all genes and their positions are known.
- The molecular maps of the genomes of rice and other grains demonstrate remarkable similarity.
- Functional genomics and proteomics will allow us to understand and utilize the information in fully sequenced plant genomes.

1. Describe mechanisms for the generation of highly repetitive DNA in plants.
2. What characteristics of *Arabidopsis thaliana* make it useful as a model system in genetic studies and for the sequencing of its entire genome? Why is rice useful as a model system for the analysis of the genome of a monocot plant?
3. Why will microarrays be useful in functional genomics?
4. What type of questions can be asked now that the *Arabidopsis* and rice genomes are essentially sequenced?



- Scientists on Science: Plant Biotechnology

43.2 Advances in plant tissue culture are revolutionizing agriculture.

- With the addition of appropriate combinations of plant growth regulators (auxin, cytokinin), plant cells in culture can be directed to form organs, embryos, or whole plants.
- Anther cultures can produce haploid plants or plants that are homozygous for all traits.
- Plant tissue culture has a number of practical applications, including the industrial production of plant chemicals, clonal propagation of horticultural plants, and the generation of disease-free plants.
- Growth of plant cells in tissue culture over extended time results in an increase in genetic variation called somaclonal variation. This variation can extend beyond the traits present in the gene pool and can generate novel genetic variations in breeding studies.

5. Describe how whole plants can be regenerated from tissue-cultured plant cells using either organogenesis or somatic cell embryogenesis. Which approach requires the use of suspension cell cultures?
6. How are plant protoplasts generated, and what is protoplast fusion? How can plant protoplasts be used to generate hybrid plants that would not occur in nature?

43.3 Plant biotechnology now affects every aspect of agriculture.

- Genetic engineering and biotechnology can be utilized to improve the quality of food crops, increase disease resistance, and improve the tolerance of crops to environmental stress.
- A key aspect of plant genetic engineering is the introduction of foreign DNA into plant cells. This can be achieved using a particle gun or electroporation.

7. Describe how the particle gun and electroporation can be used to introduce foreign DNA into plant cells. Which approach requires the use of plant protoplasts? Why?
8. How can a plant be “engineered” to produce an edible vaccine?



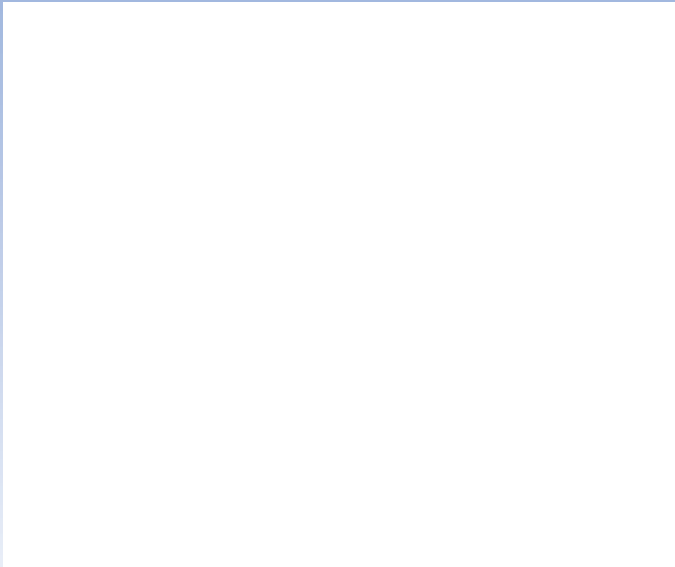
- Student Research: Plant Crop Protection

Animal Diversity

Part Opener Title

Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.
Text to come. Text to come. Text to come. Text to come.

Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to



Part opener figure 1 title. Legend to come.

come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come. Text to come. Text to come. Text to come. Text to
come.

44

The Noncoelomate Animals

Concept Outline

44.1 Animals are multicellular heterotrophs without cell walls.

Some General Features of Animals. Animals lack cell walls and move more rapidly and in more complex ways than other organisms. The animal kingdom is divided into animals without symmetry and tissues, and animals with symmetry and tissues.

Five Key Transitions in Body Plan. Over the course of animal evolution, the animal body plan has undergone many changes, five of them significant.

44.2 The simplest animals are not bilaterally symmetrical.

Parazoa. Sponges are the most primitive animals, without either tissues or, for the most part, symmetry.

Eumetazoa: The Radiata. Cnidarians and ctenophorans have distinct tissues and radial symmetry.

44.3 Acoelomates are solid worms that lack a body cavity.

Eumetazoa: The Bilaterian Acoelomates. Flatworms are the simplest bilaterally symmetrical animals; they lack a body cavity, but possess true organs.

44.4 Pseudocoelomates have a simple body cavity.

The Pseudocoelomates. Nematodes and rotifers possess a simple body cavity.

44.5 The coming revolution in animal taxonomy will likely alter traditional phylogenies.

Reevaluating How the Animal Body Plan Evolved. The use of molecular data will most likely generate changes to traditional animal phylogenies.

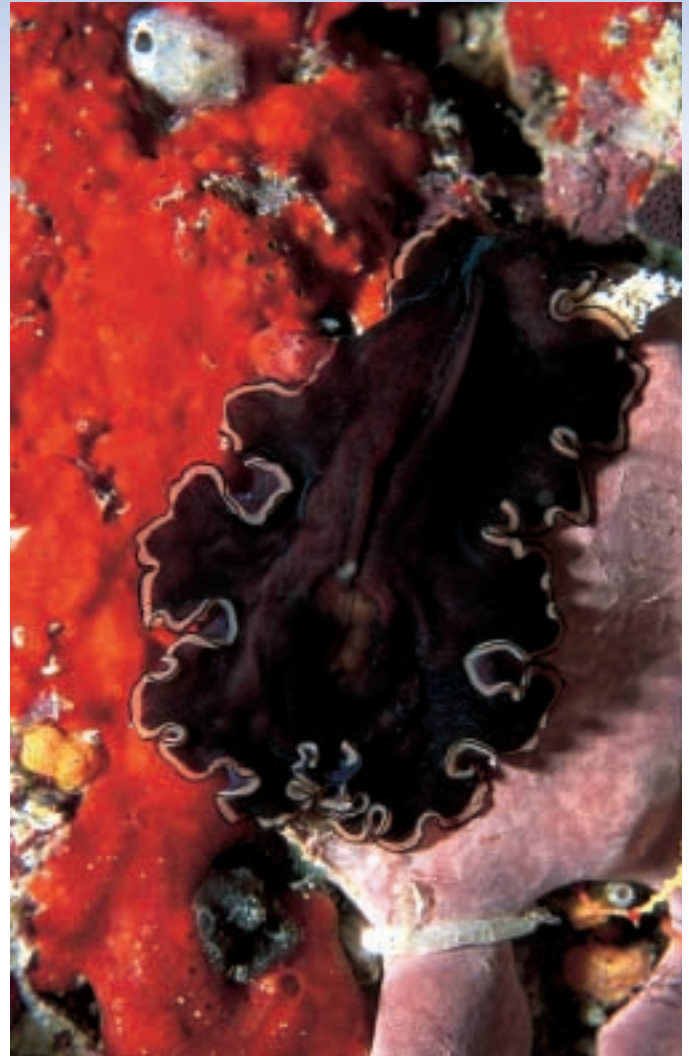


FIGURE 44.1

A noncoelomate: a marine flatworm. Some of the earliest invertebrates to evolve, marine flatworms possess internal organs but lack a true cavity called a coelom.

We will now explore the great diversity of animals, the result of a long evolutionary history. Animals, constituting millions of species, are among the most abundant living things. Found in every conceivable habitat, they bewilder us with their diversity. We will start with the simplest members of the animal kingdom—sponges, jellyfish, and simple worms. These animals lack a body cavity called a coelom, and are thus called noncoelomates (figure 44.1). The major organization of the animal body first evolved in these animals, a basic body plan upon which all the rest of animal evolution has depended. In chapters 45 through 48, we will consider the more complex animals. Despite their great diversity, you will see that all animals have much in common.

44.1 Animals are multicellular heterotrophs without cell walls.

Some General Features of Animals

Animals are the eaters or consumers of the earth. They are heterotrophs and depend directly or indirectly on plants, photosynthetic protists (algae), or autotrophic bacteria for nourishment. Animals are able to move from place to place in search of food. In most, ingestion of food is followed by digestion in an internal cavity.

Multicellular Heterotrophs. All animals are multicellular heterotrophs. The unicellular heterotrophic organisms called Protozoa, which were at one time regarded as simple animals, are now considered to be members of the kingdom Protista, the large and diverse group we discussed in chapter 35.

Diverse in Form. Almost all animals (99%) are **invertebrates**, lacking a backbone. Of the estimated 10 million living animal species, only 42,500 have a backbone and are referred to as **vertebrates**. Animals are very diverse in form, ranging in size from ones too small to see with the naked eye to enormous whales and giant squids. The animal kingdom includes about 35 phyla, most of which occur in the sea. Far fewer phyla occur in fresh water and fewer still occur on land. Members of three phyla, Arthropoda (spiders and insects), Mollusca (snails), and Chordata (vertebrates), dominate animal life on land.

No Cell Walls. Animal cells are distinct among multicellular organisms because they lack rigid cell walls and are usually quite flexible. The cells of all animals but sponges are organized into structural and functional units called **tissues**, collections of cells that have joined together and are specialized to perform a specific function; muscles and nerves are tissues types, for example.

Active Movement. The ability of animals to move more rapidly and in more complex ways than members of other kingdoms is perhaps their most striking characteristic and one that is directly related to the flexibility of their cells and the evolution of nerve and muscle tissues. A remarkable form of movement unique to animals is flying, an ability that is well developed among both insects and vertebrates. Among vertebrates, birds, bats, and pterosaurs (now-extinct flying reptiles) were or are all strong fliers. The only terrestrial vertebrate group never to have had flying representatives is amphibians.

Sexual Reproduction. Most animals reproduce sexually. Animal eggs, which are nonmotile, are much larger than the small, usually flagellated sperm. In animals, cells formed in meiosis function directly as gametes. The haploid cells do not divide by mitosis first, as they do in plants and fungi, but rather fuse directly with each other to form

the zygote. Consequently, with a few exceptions, there is no counterpart among animals to the alternation of haploid (gametophyte) and diploid (sporophyte) generations characteristic of plants (see chapter 32).

Embryonic Development. Most animals have a similar pattern of embryonic development. The zygote first undergoes a series of mitotic divisions, called *cleavage*, and becomes a solid ball of cells, the **morula**, then a hollow ball of cells, the **blastula**. In most animals, the blastula folds inward at one point to form a hollow sac with an opening at one end called the **blastopore**. An embryo at this stage is called a **gastrula**. The subsequent growth and movement of the cells of the gastrula produce the digestive system, also called the gut or intestine. The details of embryonic development differ widely from one phylum of animals to another and often provide important clues to the evolutionary relationships among them.

The Classification of Animals

Two subkingdoms are generally recognized within the kingdom Animalia: **Parazoa**—animals that for the most part lack a definite symmetry and possess neither tissues nor organs, mostly comprised of the sponges, phylum Porifera; and **Eumetazoa**—animals that have a definite shape and symmetry and, in most cases, tissues organized into organs and organ systems. Although very different in structure, both types evolved from a common ancestral form (figure 44.2) and possess the most fundamental animal traits.

All eumetazoans form distinct embryonic layers during development that differentiate into the tissues of the adult animal. Eumetazoans of the subgroup Radiata (having radial symmetry) have two layers, an outer **ectoderm** and an inner **endoderm**, and thus are called **diploblastic**. All other eumetazoans, the Bilateria (having bilateral symmetry), are **triploblastic** and produce a third layer, the **mesoderm**, between the ectoderm and endoderm. No such layers are present in sponges.

The major phyla of animals are listed in table 44.1. The simplest invertebrates make up about 14 phyla. In this chapter, we will discuss 8 of these 14 phyla and focus in detail on 4 major phyla: phylum Porifera (sponges), which lacks any tissue organization; phylum Cnidaria (radially symmetrical jellyfish, hydroids, sea anemones, and corals); phylum Platyhelminthes (bilaterally symmetrical flatworms); and phylum Nematoda (nematodes), a phylum that includes both free-living and parasitic roundworms.

Animals are complex multicellular organisms typically characterized by high mobility and heterotrophy. Most animals also possess internal tissues and organs and reproduce sexually.

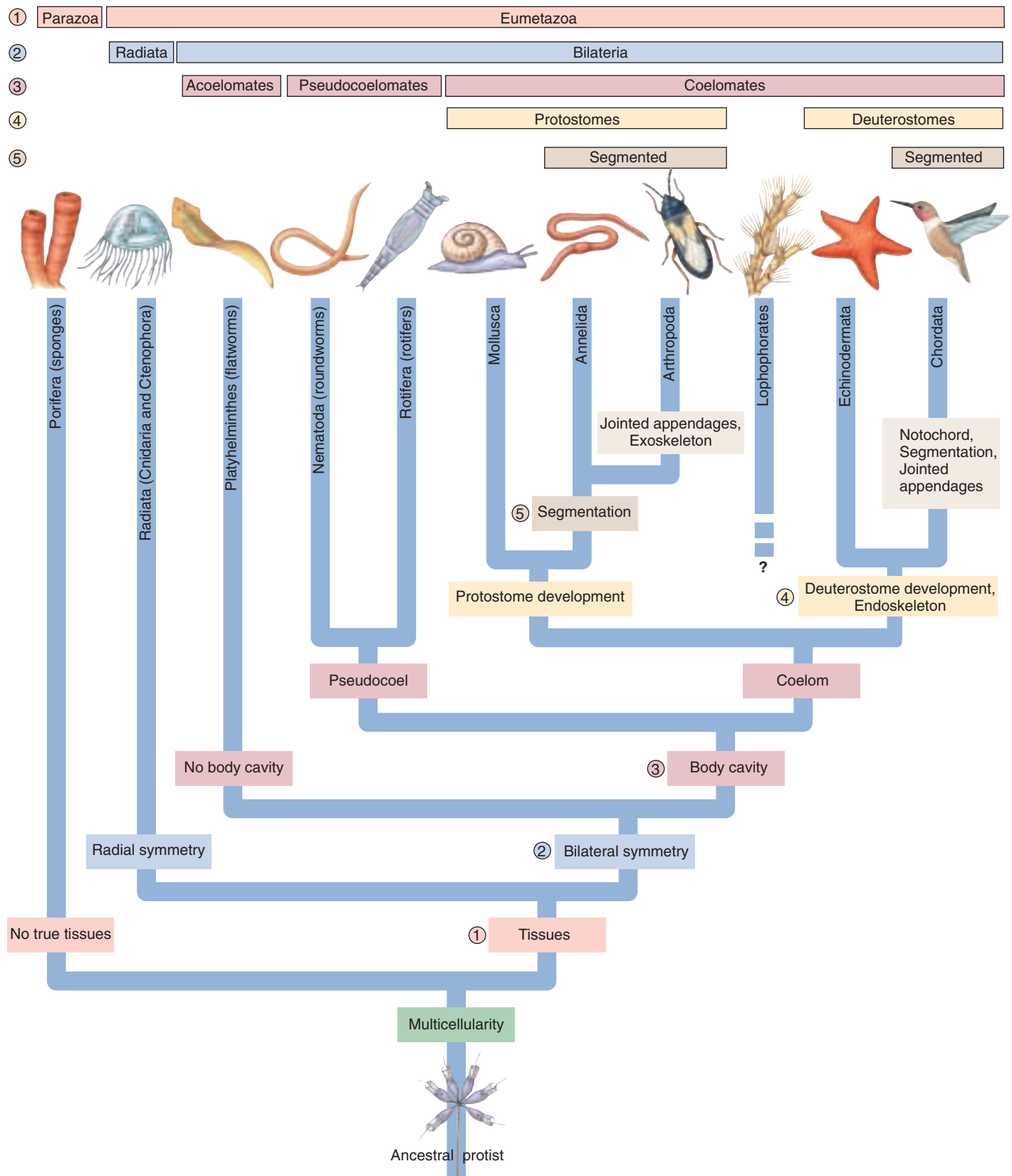


FIGURE 44.2

A possible phylogeny of the major groups of the kingdom Animalia. Transitions in the animal body plan are identified along the branches; the five key advances are the evolution of tissues, bilateral symmetry, a body cavity, protostome and deuterostome development, and segmentation.

Table 44.1 The Major Animal Phyla

Phylum	Typical Examples		Key Characteristics	Approximate Number of Named Species
Arthropoda (arthropods)	Beetles, other insects, crabs, spiders		Most successful of all animal phyla; chitinous exoskeleton covering segmented bodies with paired, jointed appendages; many insect groups have wings	1,000,000
Mollusca (mollusks)	Snails, oysters, octopuses, nudibranchs		Soft-bodied coelomates whose bodies are divided into three parts: head-foot, visceral mass, and mantle; many have shells; almost all possess a unique rasping tongue, called a radula; 35,000 species are terrestrial	110,000
Chordata (chordates)	Mammals, fish, reptiles, birds, amphibians		Segmented coelomates with a notochord; possess a dorsal nerve cord, pharyngeal slits, and a tail at some stage of life; in vertebrates, the notochord is replaced during development by the spinal column; 20,000 species are terrestrial	42,500
Platyhelminthes (flatworms)	<i>Planaria</i> , tapeworms, liver flukes		Solid, unsegmented, bilaterally symmetrical worms; no body cavity; digestive cavity, if present, has only one opening	20,000
Nematoda (roundworms)	<i>Ascaris</i> , pinworms, hookworms, <i>Filaria</i>		Pseudocoelomate, unsegmented, bilaterally symmetrical worms; tubular digestive tract passing from mouth to anus; tiny; without cilia; live in great numbers in soil and aquatic sediments; some are important animal parasites	12,000+
Annelida (segmented worms)	Earthworms, polychaetes, beach tube worms, leeches		Coelomate, serially segmented, bilaterally symmetrical worms; complete digestive tract; most have bristles called setae on each segment that anchor them during crawling	12,000

Table 44.1 The Major Animal Phyla (continued)

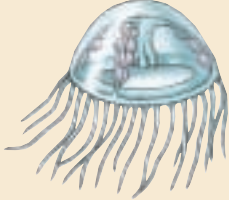
Phylum	Typical Examples		Key Characteristics	Approximate Number of Named Species
Cnidaria (cnidarians)	Jellyfish, hydra, corals, sea anemones		Soft, gelatinous, radially symmetrical bodies whose digestive cavity has a single opening; possess tentacles armed with stinging cells called cnidocytes that shoot sharp harpoons called nematocysts; almost entirely marine	10,000
Echinodermata (echinoderms)	Sea stars, sea urchins, sand dollars, sea cucumbers		Deuterostomes with radially symmetrical adult bodies; endoskeleton of calcium plates; five-part body plan and unique water vascular system with tube feet; able to regenerate lost body parts; marine	6,000
Porifera (sponges)	Barrel sponges, boring sponges, basket sponges, vase sponges		Asymmetrical bodies without distinct tissues or organs; saclike body consists of two layers breached by many pores; internal cavity lined with food-filtering cells called choanocytes; most marine (150 species live in fresh water)	5,150
Bryozoa (moss animals)	<i>Bowerbankia</i> , <i>Plumatella</i> , sea mats, sea moss		Microscopic, aquatic deuterostomes that form branching colonies, possess circular or U-shaped row of ciliated tentacles for feeding called a lophophore that usually protrudes through pores in a hard exoskeleton; also called Ectoprocta because the anus or proct is external to the lophophore; marine or freshwater	4,000
Rotifera (wheel animals)	Rotifers		Small, aquatic pseudocoelomates with a crown of cilia around the mouth resembling a wheel; almost all live in fresh water	2,000

Table 44.1 The Major Animal Phyla (continued)

Phylum	Typical Examples		Key Characteristics	Approximate Number of Named Species
Five phyla of minor worms	Velvet worms, acorn worms, arrow worms, giant tube worms		Chaetognatha (arrow worms): coelomate deuterostomes; bilaterally symmetrical; large eyes (some) and powerful jaws	980
			Hemichordata (acorn worms): marine worms with dorsal <i>and</i> ventral nerve cords	
			Onychophora (velvet worms): protostomes with a chitinous exoskeleton; evolutionary relics	
			Pogonophora (tube worms): sessile deep-sea worms with long tentacles; live within chitinous tubes attached to the ocean floor	
			Nemertea (ribbon worms): acoelomate, bilaterally symmetrical marine worms with long extendable proboscis	
Brachiopoda (lamp shells)	<i>Lingula</i>		Like bryozoans, possess a lophophore, but within two clamlike shells; more than 30,000 species known as fossils	250
Ctenophora (sea walnuts)	Comb jellies, sea walnuts		Gelatinous, almost transparent, often bioluminescent marine animals; eight bands of cilia; largest animals that use cilia for locomotion; complete digestive tract with anal pore	100
Phoronida (phoronids)	<i>Phoronis</i>		Lophophorate tube worms; often live in dense populations; unique U-shaped gut, instead of the straight digestive tube of other tube worms	12
Loricifera (loriciferans)	<i>Nanaloricus mysticus</i>		Tiny, bilaterally symmetrical, marine pseudocoelomates that live in spaces between grains of sand; mouthparts include a unique flexible tube; a recently discovered animal phylum (1983)	6

Five Key Transitions in Body Plan

1. Evolution of Tissues

The simplest animals, the Parazoa, lack both defined tissues and organs. Characterized by the sponges, these animals exist as aggregates of cells with minimal intercellular coordination. All other animals, the Eumetazoa, have distinct tissues with highly specialized cells. The evolution of tissues is the first key transition in the animal body plan.

2. Evolution of Bilateral Symmetry

Sponges also lack any definite symmetry, growing asymmetrically as irregular masses. Virtually all other animals have a definite shape and symmetry that can be defined along an imaginary axis drawn through the animal's body. Animals with symmetry belong to either the Radiata, animals with radial symmetry, or the Bilateria, animals with bilateral symmetry.

Radial Symmetry. Symmetrical bodies first evolved in marine animals belonging to two phyla: Cnidaria (jellyfish, sea anemones, and corals) and Ctenophora (comb jellies). The bodies of members of these two phyla, the Radiata, exhibit **radial symmetry**, a body design in which the parts of the body are arranged around a central axis in such a way that any plane passing through the central axis divides the organism into halves that are approximate mirror images (figure 44.3a).

Bilateral Symmetry. The bodies of all other animals, the Bilateria, are marked by a fundamental **bilateral symmetry**, a body design in which the body has a right and a left half that are mirror images of each other (figure 44.3b). A bilaterally symmetrical body plan has a top and a bottom, better known respectively as the *dorsal* and *ventral* portions of the body. It also has a front, or *anterior* end, and a back, or *posterior* end. In some higher animals like echinoderms (starfish), the adults are radially symmetrical, but even in them the larvae are bilaterally symmetrical.

Bilateral symmetry constitutes the second major evolutionary advance in the animal body plan. This unique form of organization allows parts of the body to evolve in different ways, permitting different organs to be located in different parts of the body. Also, bilaterally symmetrical animals move from place to place more efficiently than radially symmetrical ones, which, in general, lead a sessile or passively floating existence. Due to their increased mobility, bilaterally symmetrical animals are efficient in seeking food, locating mates, and avoiding predators.

During the early evolution of bilaterally symmetrical animals, structures that were important to the organism in monitoring its environment, and thereby capturing prey or avoiding enemies, came to be grouped at the anterior end. Other functions tended to be located farther back in the body. The number and complexity of sense organs are

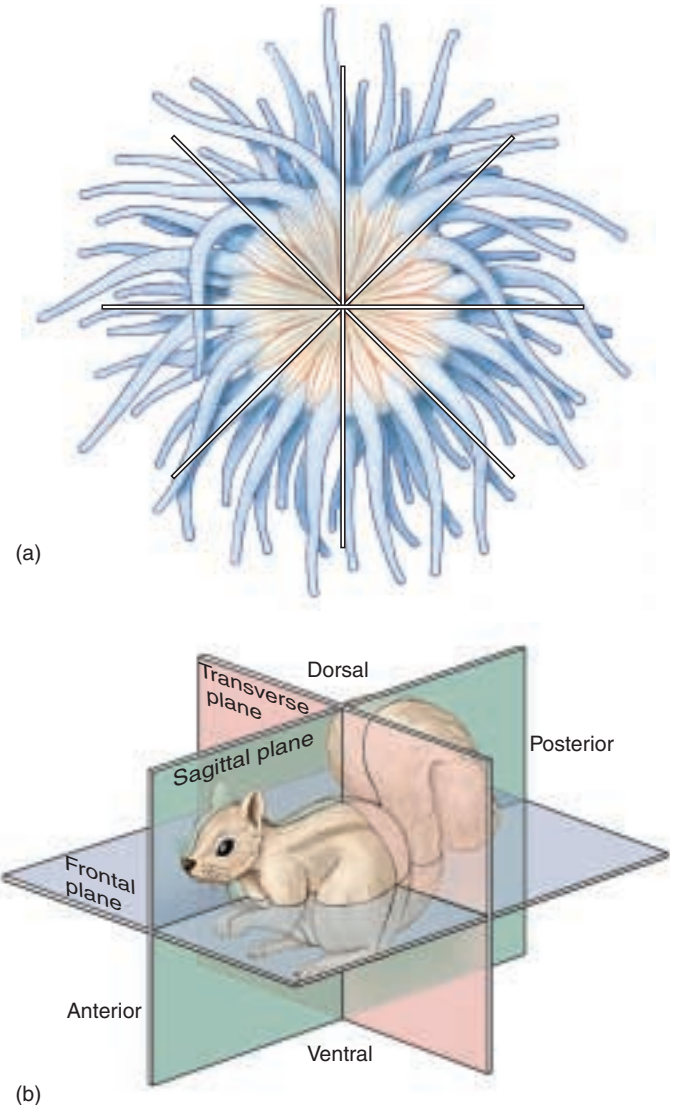


FIGURE 44.3

A comparison of radial and bilateral symmetry. (a) Radially symmetrical animals, such as this sea anemone, can be bisected into equal halves in any two-dimensional plane. (b) Bilaterally symmetrical animals, such as this squirrel, can only be bisected into equal halves in one plane (the sagittal plane).

much greater in bilaterally symmetrical animals than they are in radially symmetrical ones.

Much of the nervous system in bilaterally symmetrical animals is in the form of major longitudinal nerve cords. In a very early evolutionary advance, nerve cells became grouped around the anterior end of the body. These nerve cells probably first functioned mainly to transmit impulses from the anterior sense organs to the rest of the nervous system. This trend ultimately led to the evolution of a definite head and brain area, a process called **cephalization**, as well as to the increasing dominance and specialization of these organs in the more advanced animal phyla.

3. Evolution of a Body Cavity

A third key transition in the evolution of the animal body plan was the evolution of the body cavity. The evolution of efficient organ systems within the animal body was not possible until a body cavity evolved for supporting organs, distributing materials, and fostering complex developmental interactions.

The presence of a body cavity allows the digestive tract to be larger and longer. This longer passage allows for storage of undigested food, longer exposure to enzymes for more complete digestion, and even storage and final processing of food remnants. Such an arrangement allows an animal to eat a great deal when it is safe to do so and then to hide during the digestive process, thus limiting the animal's exposure to predators. The tube within the body cavity architecture is also more flexible, thus allowing the animal greater freedom to move.

An internal body cavity also provides space within which the gonads (ovaries and testes) can expand, allowing the accumulation of large numbers of eggs and sperm. Such storage capacity allows the diverse modifications of breeding strategy that characterize the more advanced phyla of animals. Furthermore, large numbers of gametes can be stored and released when the conditions are as favorable as possible for the survival of the young animals.

Kinds of Body Cavities. Three basic kinds of body plans evolved in the Bilateria. **Acoelomates** have no body cavity. **Pseudocoelomates** have a body cavity called the **pseudocoel** located between the mesoderm and endoderm. A third way of organizing the body is one in which the fluid-filled body cavity develops not between endoderm and mesoderm, but rather entirely within the mesoderm. Such a body cavity is called a **coelom**, and animals that possess such a cavity are called **coelomates**. In coelomates, the gut is suspended, along with other organ systems of the animal, within the coelom; the coelom, in turn, is surrounded by a layer of epithelial cells entirely derived from the mesoderm. The portion of the epithelium that lines the outer wall of the coelom is called the **parietal peritoneum**, and the portion that covers the internal organs suspended within the cavity is called the **visceral peritoneum** (figure 44.4).

The development of the coelom poses a problem—circulation—solved in pseudocoelomates by churning the fluid within the body cavity. In coelomates, the gut is again surrounded by tissue that presents a barrier to diffusion, just as it was in solid worms. This problem is solved among coelomates by the development of a **circulatory system**, a network of vessels that carries fluids to parts of the body. The circulating fluid, or blood, carries

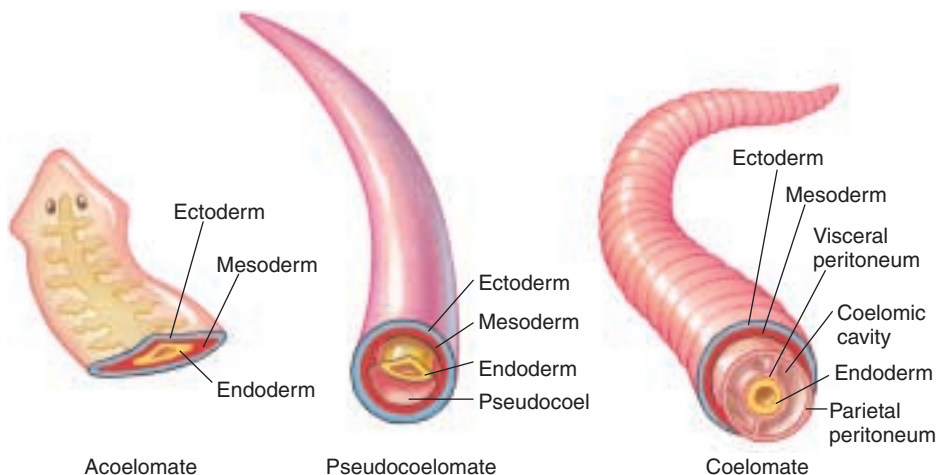


FIGURE 44.4
Three body plans for bilaterally symmetrical animals.

nutrients and oxygen to the tissues and removes wastes and carbon dioxide. Blood is usually pushed through the circulatory system by contraction of one or more muscular hearts. In an **open circulatory system**, the blood passes from vessels into sinuses, mixes with body fluid, and then reenters the vessels later in another location. In a **closed circulatory system**, the blood is physically separated from other body fluids and can be separately controlled. Also, blood moves through a closed circulatory system faster and more efficiently than it does through an open system.

The evolutionary relationship among coelomates, pseudocoelomates, and acoelomates is not clear. Acoelomates, for example, could have given rise to coelomates, but scientists also cannot rule out the possibility that acoelomates were derived from coelomates. The different phyla of pseudocoelomates form two groups that do not appear to be closely related.

Advantages of a Coelom. What is the functional difference between a pseudocoel and a coelom? The answer has to do with the nature of animal embryonic development. In animals, development of specialized tissues involves a process called **primary induction** in which one of the three primary tissues (endoderm, mesoderm, and ectoderm) interacts with another. The interaction requires physical contact. A major advantage of the coelomate body plan is that it allows contact between mesoderm and endoderm, so that primary induction can occur during development. For example, contact between mesoderm and endoderm permits localized portions of the digestive tract to develop into complex, highly specialized regions like the stomach. In pseudocoelomates, mesoderm and endoderm are separated by the body cavity, limiting developmental interactions between these tissues that ultimately limits tissue specialization and development.

4. The Evolution of Protostome and Deuterostome Development

Two outwardly dissimilar large phyla, Echinodermata (starfish) and Chordata (vertebrates), together with two smaller phyla, have a series of key embryological features different from those shared by the other animal phyla. Because it is extremely unlikely that these features evolved more than once, it is believed that these four phyla share a common ancestry. They are the members of a group called the **deuterostomes**. Members of the other coelomate animal phyla are called **protostomes**. Deuterostomes evolved from protostomes more than 630 million years ago.

Deuterostomes, like protostomes, are coelomates. They differ fundamentally from protostomes, however, in the way in which the embryo grows. Early in embryonic growth, when the embryo is a hollow ball of cells, a portion invaginates inward to form an opening called the blastopore. The blastopore of a protostome becomes the animal's mouth, and the anus develops at the other end. In a deuterostome, by contrast, the blastopore becomes the animal's anus, and the mouth develops at the other end (figure 44.5).

Deuterostomes differ in many other aspects of embryo growth, including the plane in which the cells divide. Perhaps most importantly, the cells that make up an embryonic protostome each contain a different portion of the regulatory signals present in the egg, so no one cell of the embryo (or adult) can develop into a complete organism. In marked contrast, any of the cells of a deuterostome can develop into a complete organism.

5. The Evolution of Segmentation

The fifth key transition in the animal body plan involved the subdivision of the body into **segments**. Just as it is efficient for workers to construct a tunnel from a series of

identical prefabricated parts, so segmented animals are “assembled” from a succession of identical segments. During the animal's early development, these segments become most obvious in the mesoderm but later are reflected in the ectoderm and endoderm as well. Two advantages result from early embryonic segmentation:

1. In annelids and other highly segmented animals, each segment may go on to develop a more or less complete set of adult organ systems. Damage to any one segment need not be fatal to the individual because the other segments duplicate that segment's functions.
2. Locomotion is far more effective when individual segments can move independently because the animal as a whole has more flexibility of movement. Because the separations isolate each segment into an individual skeletal unit, each is able to contract or expand autonomously in response to changes in hydrostatic pressure. Therefore, a long body can move in ways that are often quite complex.

Segmentation, also referred to as *metamerism*, underlies the organization of all advanced animal body plans. In some adult arthropods, the segments are fused, but segmentation is usually apparent in their embryological development. In vertebrates, the backbone and muscular areas are segmented, although segmentation is often disguised in the adult form. True segmentation is found in only three phyla: the annelids, the arthropods, and the chordates, although this trend is evident in many phyla.

Five key transitions in body design are responsible for most of the differences we see among the major animal phyla: the evolution of (1) tissues, (2) bilateral symmetry, (3) a body cavity, (4) protostome and deuterostome development, and (5) segmentation.

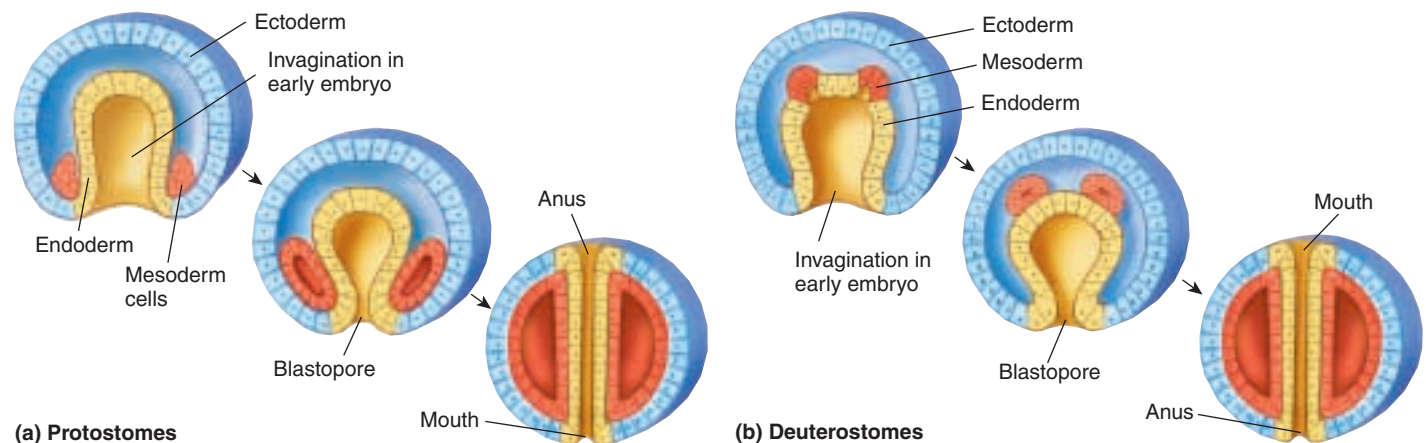


FIGURE 44.5

The fate of the blastopore. (a) In protostomes, the blastopore becomes the animal's mouth. (b) In deuterostomes, the blastopore becomes the animal's anus.

44.2 The simplest animals are not bilaterally symmetrical.

Parazoa

The sponges are Parazoans, animals that lack tissues and organs and a definite symmetry. However, sponges, like all animals, have true, complex *multicellularity*, unlike their protistan ancestors. The body of a sponge contains several distinctly different types of cells whose activities are loosely coordinated with one another. As we will see, the coordination between cell types in the eumetazoans increases and becomes quite complex.

The Sponges

There are perhaps 5000 species of marine sponges, phylum Porifera, and about 150 species that live in fresh water. In the sea, sponges are abundant at all depths. Although some sponges are tiny, no more than a few millimeters across, some, like the loggerhead sponges, may reach 2 meters or more in diameter. A few small ones are radially symmetrical, but most members of this phylum completely lack symmetry. Many sponges are colonial. Some have a low and encrusting form, while others may be erect and lobed, sometimes in complex patterns. Although larval sponges are free-swimming, adults are **sessile**, or anchored in place to submerged objects.

Sponges, like all animals, are composed of multiple cell types (see figure 44.7), but there is relatively little coordination among sponge cells. A sponge seems to be little more than a mass of cells embedded in a gelatinous matrix, but these cells recognize one another with a high degree of fidelity and are specialized for different functions of the body.

The basic structure of a sponge can best be understood by examining the form of a young individual. A small, anatomically simple sponge first attaches to a substrate and then grows into a vase-like shape. The walls of the “vase” have three functional layers. First, facing into the internal cavity are specialized flagellated cells called **choanocytes**, or collar cells. These cells line either the entire body cavity or, in many large and more complex

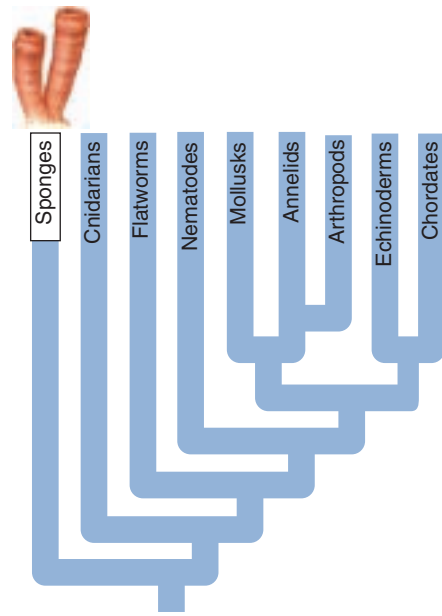


FIGURE 44.6
Aplysina longissima. This beautiful, bright blue and yellow elongated sponge is found on deep regions of coral reefs. The oscula are ringed with yellow.

sponges, specialized chambers. Second, the bodies of sponges are bounded by an outer epithelial layer consisting of flattened cells somewhat like those that make up the epithelia, or outer layers, of other animal phyla. Some portions of this layer contract when touched or exposed to appropriate chemical stimuli, and this contraction may cause some of the pores to close. Third, between these two layers, sponges consist mainly of a gelatinous, protein-rich matrix called the **mesohyl**, within which various types of amoeboid cells occur. In addition, many kinds of sponges have minute needles of calcium carbonate or silica known as **spicules**, or fibers of a tough protein called **spongin**, or both, within this matrix. Spicules and spongin strengthen the bodies of the sponges in which they occur. A spongin skeleton is the model for the bathtub sponge, once the skeleton of a real animal, but now largely known from its cellulose and plastic mimics.

Sponges feed in a unique way. The beating of flagella that line the inside of the sponge draws water in through numerous small pores; the name of the phylum, Porifera, refers to this system of pores. Plankton and other small organisms are filtered from the water, which flows through passageways and eventually is forced out through an **osculum**, a specialized, larger pore (figure 44.6).

Choanocytes. Each choanocyte closely resembles a protist with a single flagellum (figure 44.7), a similarity that reflects its evolutionary derivation. The beating of the flagella of

many choanocytes that line the body cavity draws water in through the pores and through the sponge, thus bringing in food and oxygen and expelling wastes. Each choanocyte flagellum beats independently, and the pressure they create collectively in the cavity forces water out of the osculum. In some sponges, the inner wall of the body cavity is highly convoluted, increasing the surface area and, therefore, the number of flagella that can drive the water. In such a sponge, 1 cubic centimeter of sponge can propel more than 20 liters of water a day.

PHYLUM PORIFERA: Multicellularity

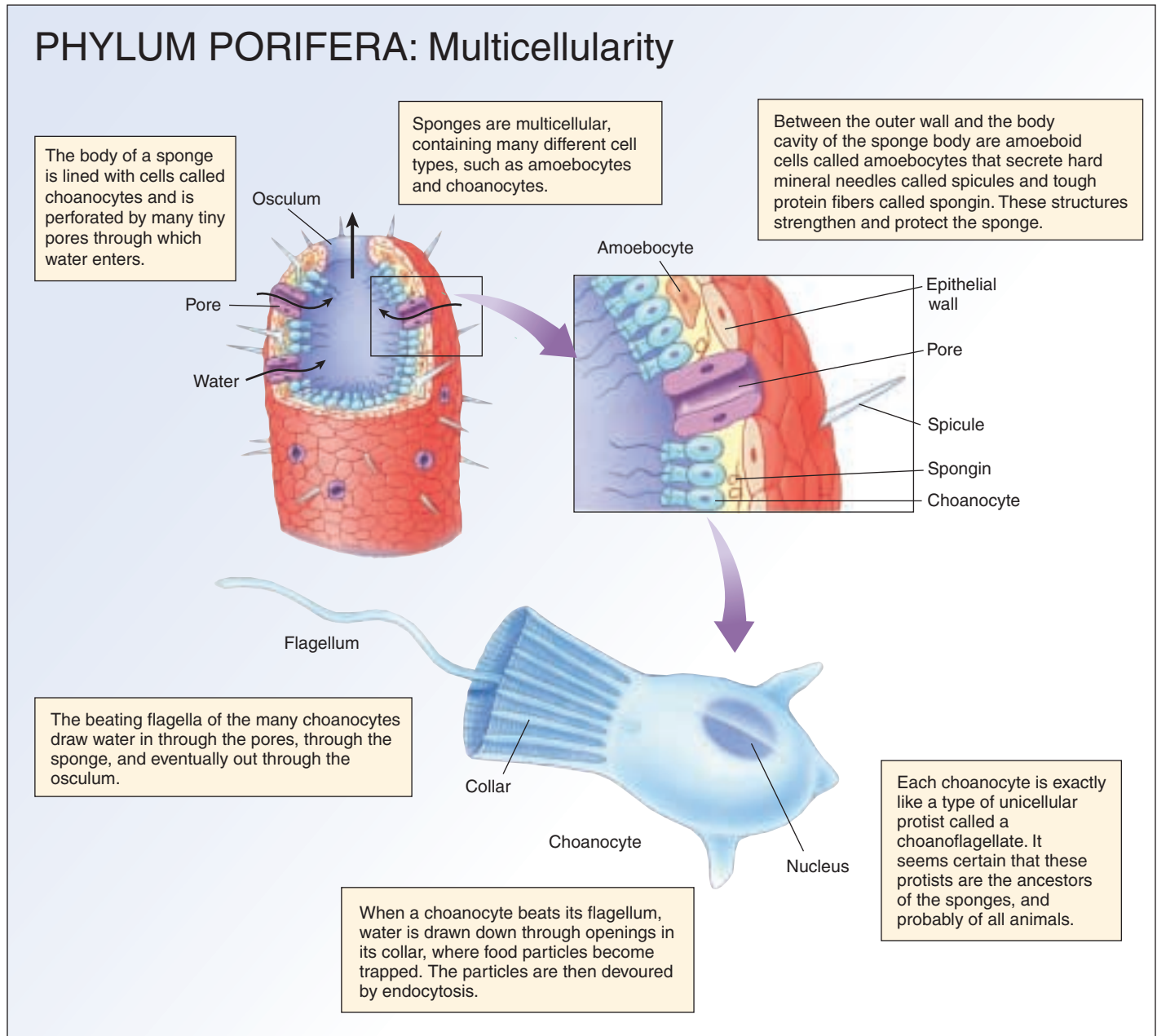


FIGURE 44.7

The body of a sponge is multicellular. The first evolutionary advance seen in animals is complex multicellularity, in which individuals are composed of many highly specialized kinds of cells.

Reproduction in Sponges. Some sponges will re-form themselves once they have passed through a silk mesh. Thus, as you might suspect, sponges frequently reproduce by simply breaking into fragments. If a sponge breaks up, the resulting fragments usually are able to re-constitute whole new individuals. Sexual reproduction is also exhibited by sponges, with some mature individuals producing eggs and sperm. Larval sponges may undergo their initial stages of development within the parent. They have numerous external, flagellated cells and are free-swimming. After a short planktonic stage, they settle

down on a suitable substrate, where they begin their transformation into adults.

Sponges probably represent the most primitive animals, possessing multicellularity but neither tissue-level development nor body symmetry. Their cellular organization hints at the evolutionary ties between the unicellular protists and the multicellular animals. Sponges are unique in the animal kingdom in possessing choanocytes, special flagellated cells whose beating drives water through the body cavity.

Eumetazoa: The Radiata

The subkingdom Eumetazoa contains animals that evolved the first key transition in the animal body plan: distinct *tissues*. Two distinct cell layers form in the embryos of these animals: an outer ectoderm and an inner endoderm. These embryonic tissues give rise to the basic body plan, differentiating into the many tissues of the adult body. Typically, the outer covering of the body (called the epidermis) and the nervous system develop from the ectoderm, and the layer of digestive tissue (called the **gastrodermis**) develops from the endoderm. A layer of gelatinous material, called the **mesoglea**, lies between the epidermis and gastrodermis and contains the muscles in most eumetazoans.

Eumetazoans also evolved true body symmetry and are divided into two major groups. The Radiata includes two phyla of radially symmetrical organisms, Cnidaria (pronounced ni-DAH-ree-ah), the cnidarians—hydroids, jellyfish, sea anemones, and corals—and Ctenophora (pronounced tea-NO-for-ah), the comb jellies, or ctenophores. All other eumetazoans are in the Bilateria and exhibit a fundamental bilateral symmetry.

The Cnidarians

Cnidarians are nearly all marine, although a few live in fresh water. These fascinating and simply constructed animals are basically gelatinous in composition. They differ markedly from the sponges in organization; their bodies are made up of distinct tissues, although they have not evolved true organs. These animals are carnivores. For the most part, they do not actively move from place to place, but rather capture their prey (which includes fishes, crustaceans, and many other kinds of animals) with the tentacles that ring their mouths.

Cnidarians may have two basic body forms, polyps and medusae (figure 44.8). Polyps are cylindrical and are usually found attached to a firm substrate. They may be solitary or colonial. In a polyp, the mouth faces away from the substrate on which the animal is growing, and, therefore, often faces upward. Many polyps build up a chitinous or



FIGURE 44.8
Two body forms of cnidarians, the medusa and the polyp. These two phases alternate in the life cycles of many cnidarians, but a number—including the corals and sea anemones, for example—exist only as polyps. Both forms have two fundamental layers of cells, separated by a jellylike layer called the mesoglea.

calcareous (made up of calcium carbonate) external or internal skeleton, or both. Only a few polyps are free-floating. In contrast, most medusae are free-floating and are often umbrella-shaped. Their mouths usually point downward, and the tentacles hang down around them. Medusae, particularly those of the class Scyphozoa, are commonly known as jellyfish because their mesoglea is thick and jellylike.

Many cnidarians occur only as polyps, while others exist only as medusae; still others alternate between these two phases during their life cycles. Both phases consist of diploid individuals. Polyps may reproduce asexually by budding; if they do, they may produce either new polyps or medusae. Medusae reproduce sexually. In most cnidarians, fertilized eggs give rise to free-swimming, multicellular, ciliated larvae known as **planulae**. Planulae are common in the plankton at times and may be dispersed widely in the currents.

A major evolutionary innovation in cnidarians, compared with sponges, is the internal extracellular digestion of food (figure 44.9). Digestion takes place within a gut cavity, rather than only within individual cells. Digestive enzymes are released from cells lining the walls of the cavity and partially break down food. Cells lining the gut subsequently engulf food fragments by phagocytosis.

The extracellular fragmentation that precedes phagocytosis and intracellular digestion allows cnidarians to digest animals larger than individual cells, an important improvement over the strictly intracellular digestion of sponges.

Nets of nerve cells coordinate contraction of cnidarian muscles, apparently with little central control. Cnidarians have no blood vessels, respiratory system, or excretory organs.

On their tentacles and sometimes on their body surface, cnidarians bear specialized cells called **cnidocytes**. The name of the phylum Cnidaria refers to these cells, which are highly distinctive and occur in no other group of organisms. Within each cnidocyte is a **nematocyst**, a small but powerful “harpoon.” Each nematocyst features a coiled, threadlike tube. Lining the inner wall of the tube is a series of barbed spines.

PHYLUM CNIDARIA: Tissues and radial symmetry

Hydra and other jellyfish are radially symmetrical, with parts arranged around a central axis like petals of a daisy.

The cells of cnidarians are organized into tissues. A major innovation of hydra and jellyfish is extracellular digestion of food—that is, digestion within a gut cavity.

Tentacles and body have stinging cells (cnidocytes) that contain small but very powerful harpoons called nematocysts.

The harpoon is propelled by osmotic pressure and is one of the fastest and most powerful processes in nature.

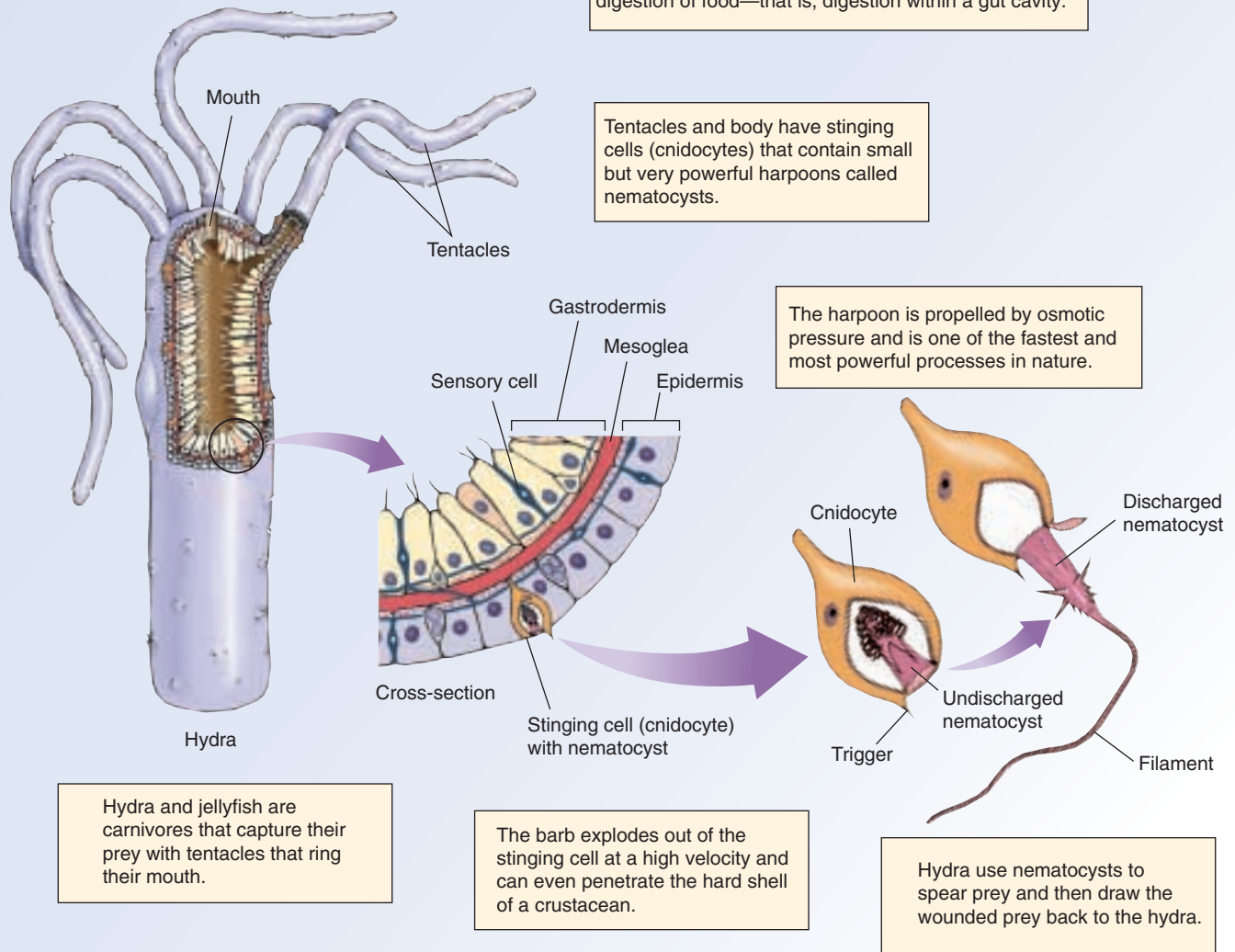


FIGURE 44.9

Eumetazoans all have tissues and symmetry. The cells of a cnidarian like this *Hydra* are organized into specialized tissues. The interior gut cavity is specialized for extracellular digestion—that is, digestion within a gut cavity rather than within individual cells. Cnidarians are also radially symmetrical.

Cnidarians use the threadlike tube to spear their prey and then draw the harpooned prey back with the tentacle containing the cnidocyte. Nematocysts may also serve a defensive purpose. To propel the harpoon, the cnidocyte uses water pressure. Before firing, the cnidocyte builds up a very high internal osmotic pressure. This is done by using active transport to build a high concentration of ions inside, while keeping its wall impermeable to water. Within the undischarged nematocyst, osmotic pressure reaches about 140 atmospheres.

When a flagellum-like trigger on the cnidocyte is stimulated to discharge, its walls become permeable to water, which rushes inside and violently pushes out the barbed filament. Nematocyst discharge is one of the fastest cellular processes in nature. The nematocyst is pushed outward so explosively that the barb can penetrate even the hard shell of a crab. A toxic protein often produced a stinging sensation, causing some cnidarians to be called “stinging nettles.”

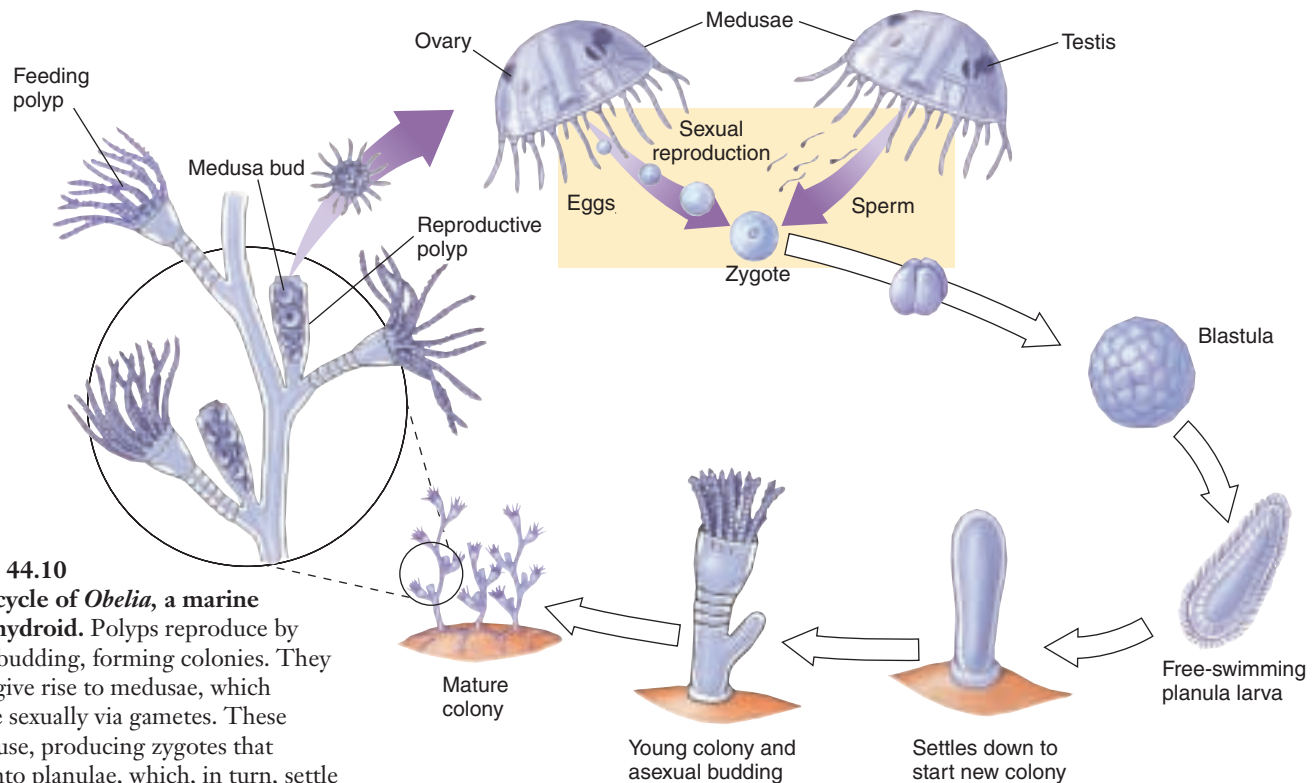


FIGURE 44.10
The life cycle of *Obelia*, a marine colonial hydroid. Polyps reproduce by asexually budding, forming colonies. They may also give rise to medusae, which reproduce sexually via gametes. These gametes fuse, producing zygotes that develop into planulae, which, in turn, settle down to produce polyps.

Classes of Cnidarians

There are four classes of cnidarians: Hydrozoa (hydroids), Scyphozoa (jellyfish), Cubozoa (box jellyfish), and Anthozoa (anemones and corals).

Class Hydrozoa: The Hydroids. Most of the approximately 2700 species of hydroids (class Hydrozoa) have both polyp and medusa stages in their life cycle (figure 44.10). Most of these animals are marine and colonial, such as *Obelia* and the already mentioned, very unusual Portuguese man-of-war. Some of the marine hydroids are bioluminescent.

A well-known hydroid is the abundant freshwater genus *Hydra*, which is exceptional in that it has no medusa stage and exists as a solitary polyp. Each polyp sits on a basal disk, which it can use to glide around, aided by mucous secretions. It can also move by somersaulting, bending over and attaching itself to the substrate by its tentacles, and then looping over to a new location. If the polyp detaches itself from the substrate, it can float to the surface.

Class Scyphozoa: The Jellyfish. The approximately 200 species of jellyfish (class Scyphozoa) are transparent or translucent marine organisms, some of a striking orange, blue, or pink color (figure 44.11). These animals spend most of their time floating near the surface of the sea. In all of them, the medusa stage is dominant—much larger and more complex than the polyps. The medusae are bell-



FIGURE 44.11
Class Scyphozoa. Jellyfish, *Aurelia aurita*.

shaped, with hanging tentacles around their margins. The polyp stage is small, inconspicuous, and simple in structure.

The outer layer, or epithelium, of a jellyfish contains a number of specialized epitheliomuscular cells, each of which can contract individually. Together, the cells form a muscular ring around the margin of the bell that pulses rhythmically and propels the animal through the water. Jellyfish have separate male and female individuals. After fertilization, planulae form, which then attach and develop into polyps. The polyps can reproduce asexually as well as budding off medusae. In some jellyfish that live in the open ocean, the polyp stage is suppressed, and planulae develop directly into medusae.

Class Cubozoa: The Box Jellyfish.

Until recently the cubozoa were considered an order of Scyphozoa. As their name implies, they are box-shaped medusa (the polyp stage is inconspicuous and in many cases not known). Most are only a few cm in height, although some are 25 cm tall. A tentacle or group of tentacles is found at each corner of the box (figure 44.12). Box jellies are strong swimmers and voracious predators of fish. Stings of some species can be fatal to humans.

Class Anthozoa: The Sea Anemones and Corals. By far the largest class of cnidarians is Anthozoa, the “flower animals” (from the Greek *anthos*, meaning “flower”). The approximately 6200 species of this group are solitary or colonial marine animals. They include stonelike corals, soft-bodied sea anemones, and other groups known by such fanciful names as sea pens, sea pansies, sea fans, and sea whips (figure 44.13). All of these names reflect a plantlike body topped by a tuft or crown of hollow tentacles. Like other cnidarians, anthozoans use these tentacles in feeding. Nearly all members of this class that live in shallow waters harbor symbiotic algae, which supplement the nutrition of their hosts through photosynthesis. Fertilized eggs of anthozoans usually develop into planulae that settle and develop into polyps; no medusae are formed.

Sea anemones are a large group of soft-bodied anthozoans that live in coastal waters all over the world and are especially abundant in the tropics. When touched, most sea anemones retract their tentacles into their bodies and fold up. Sea anemones are highly muscular and relatively complex organisms, with greatly divided internal cavities. These animals range from a few millimeters to about 10 centimeters in diameter and are perhaps twice that high.

Corals are another major group of anthozoans. Many of them secrete tough outer skeletons, or exoskeletons, of calcium carbonate and are thus stony in texture. Others, includ-



FIGURE 44.12
Class Cubozoa. Box jelly, *Chironex fleckeri*.



FIGURE 44.13
Class Anthozoa. The sessile soft-bodied sea anemone.



ing the gorgonians, or soft corals, do not secrete exoskeletons. Some of the hard corals help form coral reefs, which are shallow-water limestone ridges that occur in warm seas. Although the waters where coral reefs develop are often nutrient-poor, the coral animals are able to grow actively because of the abundant algae found within them.

The Ctenophorans (Comb Jellies)

The members of this small phylum range from spherical to ribbonlike and are known as comb jellies or sea walnuts. Traditionally, the roughly 90 marine species of ctenophores (phylum Ctenophora) were considered closely related to the cnidarians. However, ctenophores are structurally more complex than cnidarians. They have anal pores, so that water and other substances pass completely through the animal. Comb jellies, abundant in the open ocean, are transparent and usually only a few centimeters long. The members of one group have two long, retractable tentacles that they use to capture their prey.

Ctenophores propel themselves through water with eight comblike plates of fused cilia that beat in a coordinated fashion (figure 44.14). They are the largest animals that use cilia for locomotion. Many ctenophores are bioluminescent, giving off bright flashes of light particularly evident in the open ocean at night.

Cnidarians and ctenophores have tissues and radial symmetry.
Cnidarians have a specialized kind of cell called a cnidocyte.
Ctenophores propel themselves through the water by means of eight comblike plates of fused cilia.

FIGURE 44.14
A comb jelly (phylum Ctenophora). Note the comblike plates along the ridges of the base.

44.3 Acoelomates are solid worms that lack a body cavity.

Eumetazoa: The Bilateral Acoelomates

The Bilateria are characterized by the second key transition in the animal body plan, *bilateral symmetry*, which allowed animals to achieve high levels of specialization within parts of their bodies. The simplest bilaterians are the acoelomates; they lack any internal cavity other than the digestive tract. As discussed earlier, all bilaterians have three embryonic layers during development: ectoderm, endoderm, and mesoderm. We will focus our discussion of the acoelomates on the largest phylum of the group, the flatworms.

Phylum Platyhelminthes: The Flatworms

Phylum Platyhelminthes consists of some 20,000 species. These ribbon-shaped, soft-bodied animals are flattened dorsoventrally, from top to bottom. Flatworms are among the simplest of bilaterally symmetrical animals, but they do have a definite head at the anterior end and they do possess organs. Their bodies are solid: the only internal space consists of the digestive cavity (figure 44.15).

Flatworms range in size from a millimeter or less to many meters long, as in some tapeworms. Most species of flatworms are parasitic, occurring within the bodies of

many other kinds of animals (figure 44.16). Other flatworms are free-living, occurring in a wide variety of marine and freshwater habitats, as well as moist places on land. Free-living flatworms are carnivores and scavengers; they eat various small animals and bits of organic debris. They move from place to place by means of ciliated epithelial cells, which are particularly concentrated on their ventral surfaces.

Those flatworms that have a digestive cavity have an incomplete gut, one with only one opening. As a result, they cannot feed, digest, and eliminate undigested particles of food simultaneously, and thus, flatworms cannot feed continuously, as more advanced animals can. Muscular contractions in the upper end of the gut

cause a strong sucking force allowing flatworms to ingest their food and tear it into small bits. The gut is branched and extends throughout the body, functioning in both digestion and transport of food. Cells that line the gut engulf most of the food particles by phagocytosis and digest them; but, as in the cnidarians, some of these particles are partly digested extracellularly. Tapeworms, which are parasitic flatworms, lack digestive systems. They absorb their food directly through their body walls.

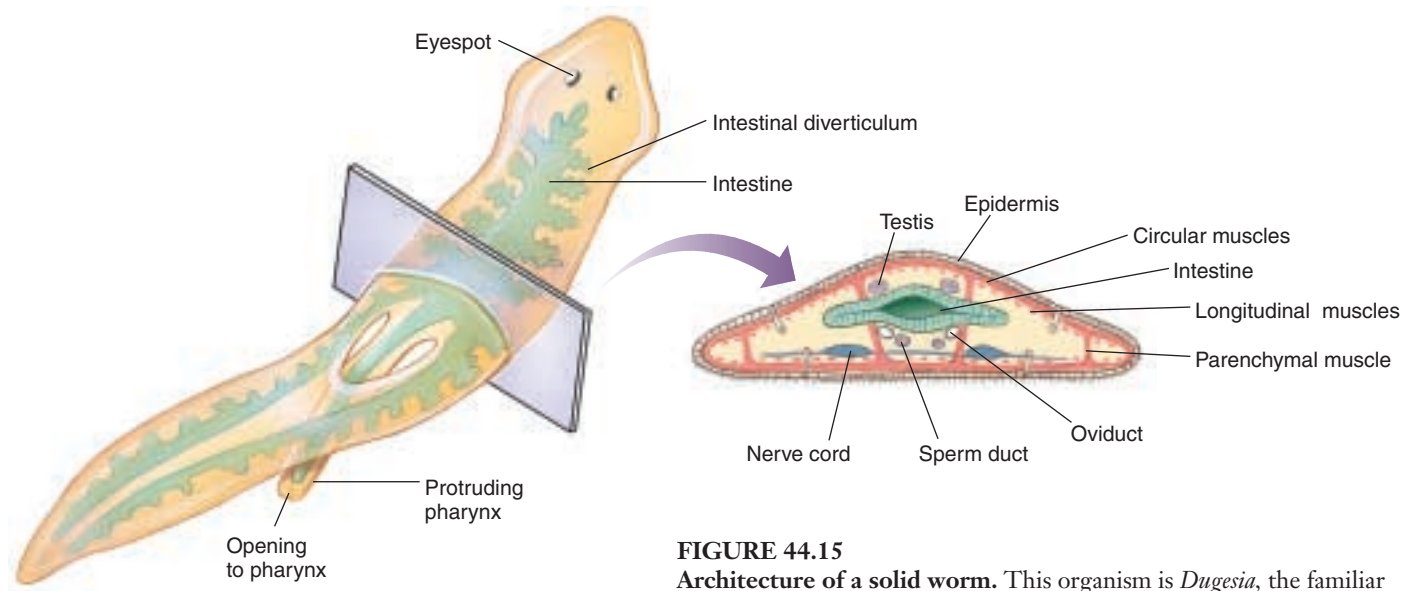
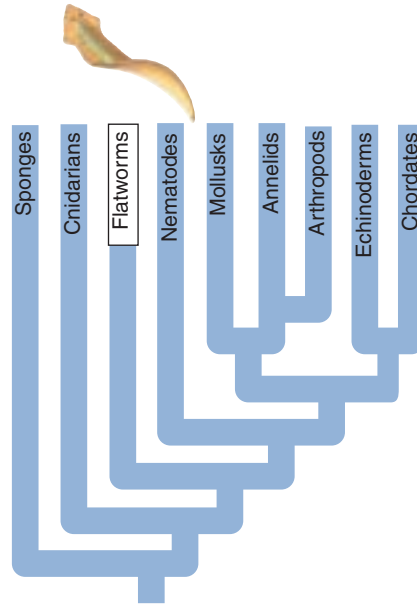


FIGURE 44.15

Architecture of a solid worm. This organism is *Dugesia*, the familiar freshwater “planaria” of many biology laboratories.

PHYLUM PLATYHELMINTHES: Bilateral symmetry

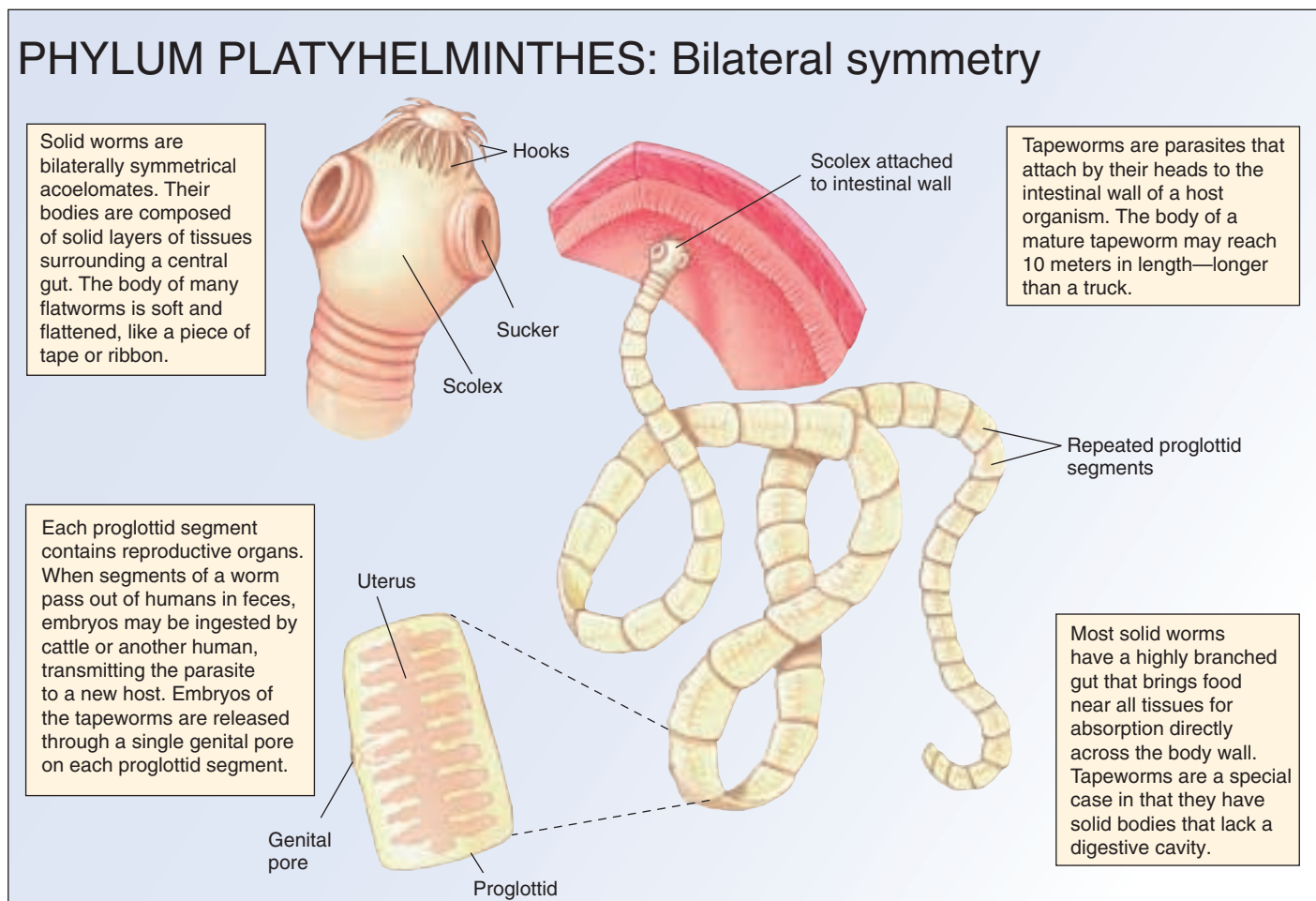


FIGURE 44.16

The evolution of bilateral symmetry. Acoelomate solid worms like this beef tapeworm, *Taenia saginata*, are bilaterally symmetrical. In addition, all bilaterians have three embryonic layers and exhibit cephalization.

Unlike cnidarians, flatworms have an excretory system, which consists of a network of fine tubules (little tubes) that runs throughout the body. Cilia line the hollow centers of bulblike **flame cells** located on the side branches of the tubules (see figure 58.9). Cilia in the flame cells move water and excretory substances into the tubules and then to exit pores located between the epidermal cells. Flame cells were named because of the flickering movements of the tuft of cilia within them. They primarily regulate the water balance of the organism. The excretory function of flame cells appears to be a secondary one. A large proportion of the metabolic wastes excreted by flatworms diffuses directly into the gut and is eliminated through the mouth.

Like sponges, cnidarians, and ctenophorans, flatworms lack circulatory systems for the transport of oxygen and food molecules. Consequently, all flatworm cells must be within diffusion distance of oxygen and food. Flatworms have thin bodies and highly branched digestive cavities that make such a relationship possible.

The nervous system of flatworms is very simple. Like cnidarians, some primitive flatworms have only a nerve

net. However, most members of this phylum have longitudinal nerve cords that constitute a simple central nervous system.

Free-living members of this phylum have eyespots on their heads. These are inverted, pigmented cups containing light-sensitive cells connected to the nervous system. These eyespots enable the worms to distinguish light from dark; worms move away from strong light.

The reproductive systems of flatworms are complex. Most flatworms are **hermaphroditic**, with each individual containing both male and female sexual structures. In many of them, fertilization is internal. When they mate, each partner deposits sperm in the copulatory sac of the other. The sperm travel along special tubes to reach the eggs. In most free-living flatworms, fertilized eggs are laid in cocoons strung in ribbons and hatch into miniature adults. In some parasitic flatworms, there is a complex succession of distinct larval forms. Flatworms are also capable of asexual regeneration. In some genera, when a single individual is divided into two or more parts, each part can regenerate an entirely new flatworm.

Class Turbellaria: Turbellarians.

Only one of the three classes of flatworms, the turbellarians (class Turbellaria) are free-living. One of the most familiar is the freshwater genus *Dugesia*, the common planaria used in biology laboratory exercises. Other members of this class are widespread and often abundant in lakes, ponds, and the sea. Some also occur in moist places on land.

Class Trematoda: The Flukes.

Two classes of parasitic flatworms live within the bodies of other animals: flukes (class Trematoda) and tapeworms (class Cestoda). Both groups of worms have epithelial layers resistant to the digestive enzymes and immune defenses produced by their hosts—an important feature in their parasitic way of life. However, they lack certain features of the free-living flatworms, such as cilia in the adult stage, eyespots, and other sensory organs that lack adaptive significance for an organism that lives within the body of another animal.

Flukes take in food through their mouth, just like their free-living relatives. There are more than 10,000 named species, ranging in length from less than 1 millimeter to more than 8 centimeters. Flukes attach themselves within the bodies of their hosts by means of suckers, anchors, or hooks. Some have a life cycle that involves only one host, usually a fish. Most have life cycles involving two or more hosts. Their larvae almost always occur in snails, and there may be other intermediate hosts. The final host of these flukes is almost always a vertebrate.

To human beings, one of the most important flatworms is the human liver fluke, *Clonorchis sinensis*. It lives in the bile passages of the liver of humans, cats, dogs, and pigs. It is especially common in Asia. The worms are 1 to 2 centimeters long and have a complex life cycle. Although they are hermaphroditic, cross-fertilization usually occurs between different individuals. Eggs, each containing a complete, ciliated first-stage larva, or **miracidium**, are passed in the feces (figure 44.17). If they reach water, they may be ingested by a snail. Within the snail an egg transforms into a *sporocyst*—a baglike structure with embryonic germ cells. Within the sporocysts are produced **rediae**, which are elongated, nonciliated larvae. These larvae continue growing within the snail, giving rise to several individuals of the tadpole-like next larval stage, **cercariae**.

Cercariae escape into the water, where they swim about freely. If they encounter a fish of the family Cyprinidae—

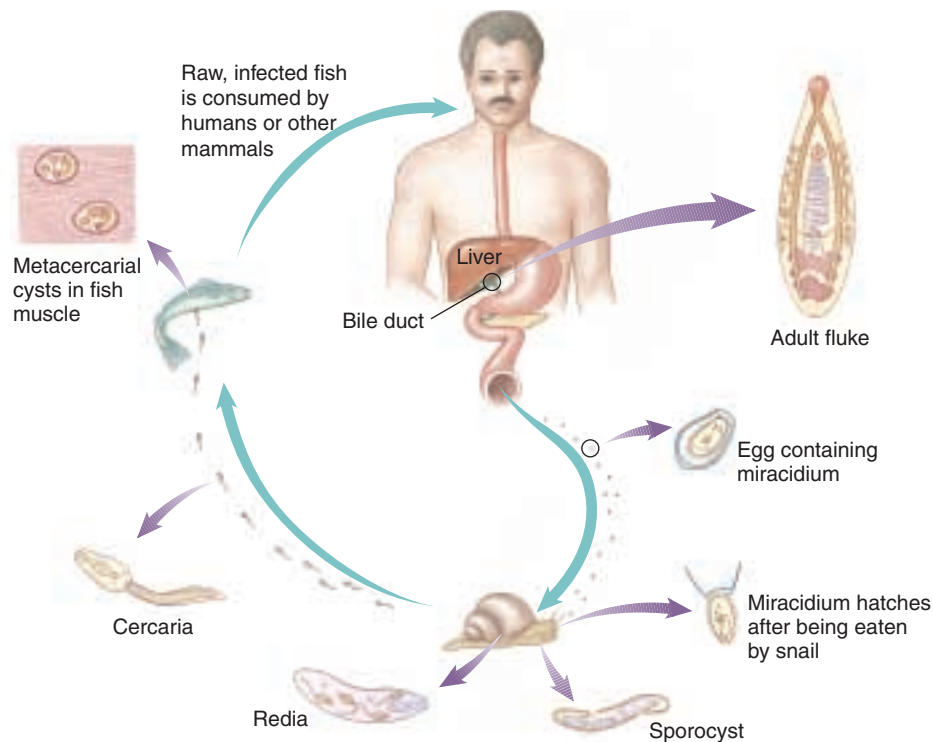


FIGURE 44.17
Life cycle of the human liver fluke, *Clonorchis sinensis*.

the family that includes carp and goldfish—they bore into the muscles or under the scales, lose their tails, and transform into **metacercariae** within cysts in the muscle tissue. If a human being or other mammal eats raw infected fish, the cysts dissolve in the intestine, and the young flukes migrate to the bile duct, where they mature. An individual fluke may live for 15 to 30 years in the liver. In humans, a heavy infestation of liver flukes may cause cirrhosis of the liver and death.

Other very important flukes are the blood flukes of genus *Schistosoma*. They afflict about 1 in 20 of the world's population, more than 200 million people throughout tropical Asia, Africa, Latin America, and the Middle East. Three species of *Schistosoma* cause the disease called schistosomiasis, or bilharzia. Some 800,000 people die each year from this disease.

Recently, there has been a great deal of effort to control schistosomiasis. The worms protect themselves in part from the body's immune system by coating themselves with a variety of the host's own antigens that effectively render the worm immunologically invisible (see chapter 57). Despite this difficulty, the search is on for a vaccine that would cause the host to develop antibodies to one of the antigens of the young worms before they protect themselves with host antigens. This vaccine would prevent humans from infection. The disease can be cured with drugs after infection.

Class Cestoda: The Tapeworms. Class Cestoda is the third class of flatworms; like flukes, they live as parasites within the bodies of other animals. In contrast to flukes, tapeworms simply hang on to the inner walls of their hosts by means of specialized terminal attachment organs and absorb food through their skins. Tapeworms lack digestive cavities as well as digestive enzymes. They are extremely specialized in relation to their parasitic way of life. Most species of tapeworms occur in the intestines of vertebrates, about a dozen of them regularly in humans.

The long, flat bodies of tapeworms are divided into three zones: the **scolex**, or attachment organ; the unsegmented **neck**; and a series of repetitive segments, the **proglottids** (see figure 44.16). The scolex usually bears several suckers and may also have hooks. Each proglottid is a complete hermaphroditic unit, containing both male and female reproductive organs. Proglottids are formed continuously in an actively growing zone at the base of the neck, with maturing ones moving farther back as new ones are formed in front of them. Ultimately the proglottids near the end of the body form mature eggs. As these eggs are fertilized, the zygotes in the very last segments begin to differentiate, and these segments fill with embryos, break off, and leave their host with the host's feces. Embryos, each surrounded by a shell, emerge from the proglottid through a pore or the ruptured body wall. They are deposited on leaves, in water, or in other places where they may be picked up by another animal.

The beef tapeworm *Taenia saginata* occurs as a juvenile in the intermuscular tissue of cattle but as an adult in the intestines of human beings. A mature adult beef tapeworm may reach a length of 10 meters or more. These worms attach themselves to the intestinal wall of their host by a scolex with four suckers. The segments that are shed from the end of the worm pass from the human in the feces and

may crawl onto vegetation. The segments ultimately rupture and scatter the embryos. Embryos may remain viable for up to five months. If they are ingested by cattle, they burrow through the wall of the intestine and ultimately reach muscle tissues through the blood or lymph vessels. About 1% of the cattle in the United States are infected, and some 20% of the beef consumed is not federally inspected. When infected beef is eaten rare, infection of humans by these tapeworms is likely. As a result, the beef tapeworm is a frequent parasite of humans.

Phylum Nemertea: The Ribbon Worms

The phylogenetic relationship of phylum Nemertea (figure 44.18) to other free-living flatworms is unclear. Nemertean worms are often called ribbon worms or proboscis worms. These aquatic worms have the body plan of a flatworm, but also possess a fluid-filled sac that may be a primitive coelom. This sac serves as a hydraulic power source for their proboscis, a long muscular tube that can be thrust out quickly from a sheath to capture prey. Shaped like a thread or a ribbon, ribbon worms are mostly marine and consist of about 900 species. Ribbon worms are large, often 10 to 20 centimeters and sometimes many meters in length. They are the simplest animals that possess a **complete digestive system**, one that has two separate openings, a mouth and an anus. Ribbon worms also exhibit a circulatory system in which blood flows in vessels. Many important evolutionary trends that become fully developed in more advanced animals make their first appearance in the Nemertea.

The acelomates, typified by flatworms, are the most primitive bilaterally symmetrical animals and the simplest animals in which organs occur.



FIGURE 44.18
A ribbon worm, *Lineus* (phylum Nemertea). This is the simplest animal with a complete digestive system.

44.4 Pseudocoelomates have a simple body cavity.

The Pseudocoelomates

All bilaterians except solid worms possess an internal *body cavity*, the third key transition in the animal body plan. Seven phyla are characterized by their possession of a pseudocoel (see figure 44.4). Their evolutionary relationships remain unclear, with the possibility that the pseudocoelomate condition arose independently many times. The pseudocoel serves as a hydrostatic skeleton—one that gains its rigidity from being filled with fluid under pressure. The animals' muscles can work against this “skeleton,” thus making the movement of pseudocoelomates far more efficient than that of the acoelomates.

Pseudocoelomates lack a defined circulatory system; this role is performed by the fluids that move within the pseudocoel. Most pseudocoelomates have a complete, one-way digestive tract that acts like an assembly line. Food is first broken down, then absorbed, and then treated and stored.

Phylum Nematoda: The Roundworms

Nematodes, eelworms, and other roundworms constitute a large phylum, Nematoda, with some 12,000 recognized species. Scientists estimate that the actual number might approach 100 times that many. Members of this phylum are found everywhere. Nematodes are abundant and diverse in marine and freshwater habitats, and many members of this phylum are parasites of animals (figure 44.19) and plants. Many nematodes are microscopic and live in soil. It has been estimated that a spadeful of fertile soil may contain, on the average, a million nematodes.

Nematodes are bilaterally symmetrical, unsegmented worms. They are covered by a flexible, thick cuticle, which is molted as they grow. Their muscles constitute a layer beneath the epidermis and extend along the length of the worm, rather than encircling its body. These longitudinal muscles pull both against the cuticle and the pseudocoel, which forms a hydrostatic skeleton. When

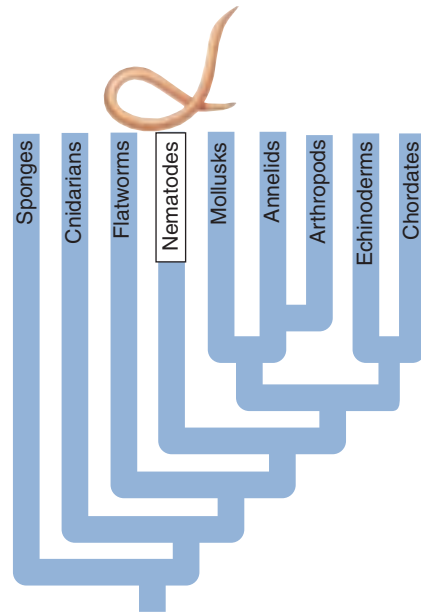


FIGURE 44.19
Trichinella nematode encysted in pork.
The serious disease trichinosis can result from eating undercooked pork or bear meat containing such cysts.

nematodes move, their bodies whip about from side to side.

Near the mouth of a nematode, at its anterior end, are usually 16 raised, hairlike, sensory organs. The mouth is often equipped with piercing organs called **stylets**. Food passes through the mouth as a result of the sucking action of a muscular chamber called the **pharynx**. After passing through a short corridor into the pharynx, food continues through the other portions of the digestive tract, where it is broken down and then digested. Some of the water with which the food has been mixed is reabsorbed near the end of the digestive tract, and material that has not been digested is eliminated through the anus (figure 44.20).

Nematodes completely lack flagella or cilia, even on sperm cells. Reproduction in nematodes is sexual, with sexes usually separate. Their development is simple, and the adults consist of very few cells. For this reason, nematodes have become extremely important subjects for genetic and developmental studies (see chapter 17). The 1-millimeter-long *Caenorhabditis elegans* matures in only three days, its body is transparent, and it has only 959 cells. It is the only animal whose complete developmental cellular anatomy is known.

About 50 species of nematodes, including several that are rather common in the United States, regularly parasitize human beings. The most serious common nematode-caused disease in temperate regions is trichinosis, caused by worms of the genus *Trichinella*. These worms live in the

small intestine of pigs, where fertilized female worms burrow into the intestinal wall. Once it has penetrated these tissues, each female produces about 1500 live young. The young enter the lymph channels and travel to muscle tissue throughout the body, where they mature and form highly resistant, calcified cysts. Infection in human beings or other animals arises from eating undercooked or raw pork in which the cysts of *Trichinella* are present. If the worms are abundant, a fatal infection can result, but such infections are rare; only about 20 deaths in the United States have been attributed to trichinosis during the past decade.

PHYLUM NEMATODA: Body cavity

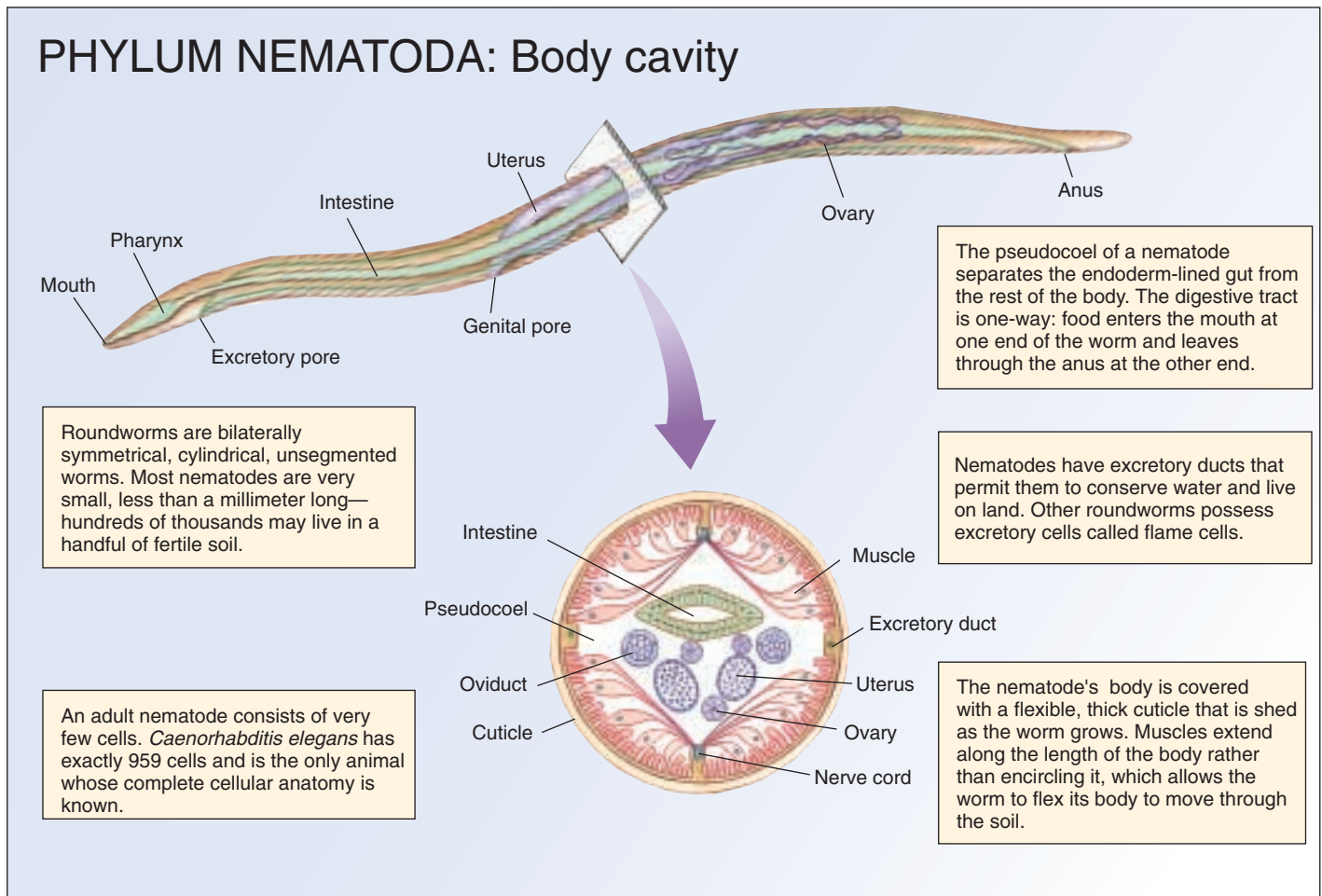


FIGURE 44.20

The evolution of a simple body cavity. The major innovation in body design in roundworms (phylum Nematoda) is a body cavity between the gut and the body wall. This cavity is the pseudocoel. It allows chemicals to circulate throughout the body and prevents organs from being deformed by muscle movements.

Phylum Rotifera: Rotifers

Phylum Rotifera includes common, small, bilaterally symmetrical, basically aquatic animals that have a crown of cilia at their heads. Rotifers are pseudocoelomates but are very unlike nematodes. They have several features that suggest their ancestors may have resembled flatworms. There are about 2000 species of this phylum. While a few rotifers live in soil or in the capillary water in cushions of mosses, most occur in fresh water, and they are common everywhere. Very few rotifers are marine. Most rotifers are between 50 and 500 micrometers in length, smaller than many protists.

Rotifers have a well-developed food-processing apparatus. A conspicuous organ on the tip of the head called the corona gathers food. It is composed of a circle of cilia which sweeps their food into their mouths, as well as being used for locomotion. Rotifers are often called “wheel animals” because the cilia, when they are beating together, resemble the movement of spokes radiating from a wheel.

A Relatively New Phylum: Cycliophora

In December 1995, two Danish biologists reported the discovery of a strange new kind of creature, smaller than a period on a printed page. The tiny organism had a striking circular mouth surrounded by a ring of fine, hairlike cilia and has so unusual a life cycle that they assigned it to an entirely new phylum, Cycliophora (Greek for “carrying a small wheel”). There are only about 35 known animal phyla, so finding a new one is extremely rare! When the lobster to which it is attached starts to molt, the tiny symbiont begins a bizarre form of sexual reproduction. Dwarf males emerge, with nothing but brains and reproductive organs. Each dwarf male seeks out another female symbiont on the molting lobster and fertilizes its eggs, generating free-swimming individuals that can seek out another lobster and renew the life cycle.

The pseudocoelomates, including nematodes and rotifers, all have fluid-filled pseudocoels.

44.5 The coming revolution in animal taxonomy will likely alter traditional phylogenies.

Reevaluating How the Animal Body Plan Evolved

The great diversity seen in the body plan of animals is difficult to fit into any one taxonomic scheme. Biologists have traditionally inferred the general relationships among the 35 animal phyla by examining what seemed to be fundamental characters—segmentation, possession of a coelom, and so on. The general idea has been that such characters are most likely to be conserved during a group's evolution. Animal phyla that share a fundamental character are more likely to be closely related to each other than to other phyla that do not exhibit the character. The phylogeny presented in figure 44.2 is a good example of the sort of taxonomy this approach has generated.

However, not every animal can be easily accommodated by this approach. Take, for example, the myzostomids (figure 44.21), an enigmatic and anatomically bizarre group of marine animals that are parasites or symbionts of echinoderms. Myzostomid fossils are found associated with echinoderms since the Ordovician, so the myzostomid-echinoderm relationship is a very ancient one. Their long history of obligate association has led to the loss or simplification of many myzostomid body elements, leaving them, for example, with no body cavity (they are acoelomates) and only incomplete segmentation.

This character loss has led to considerable disagreement among taxonomists. However, while taxonomists have disagreed about the details, all have generally allied myzostomids in some fashion with the annelids, sometimes within the polychaetes, sometimes as a separate phylum closely allied to the annelids.

Recently, this view has been challenged. New taxonomical comparisons using molecular data have come to very different conclusions. Researchers examined two components of the protein synthesis machinery, the small ribosomal subunit rRNA gene, and an elongation factor gene (called 1 alpha). The phylogeny they obtain does not place the myzostomids in with the annelids. Indeed, they find that the myzostomids have no close links to the annelids at all. Instead, surprisingly, they are more closely allied with the flatworms!

This result hints strongly that the key morphological characters that biologists have traditionally used to construct animal phylogenies—segmentation, coeloms, jointed appendages, and the like—are not the conservative characters we had supposed. Among the myzostomids these features appear to have been gained and lost again during the course of their evolution. If this unconservative evolutionary pattern should prove general, our view of the evolution of the animal body plan, and how the various animal phyla relate to one another, will soon be in need of major revision.



FIGURE 44.21

A taxonomic puzzle. *Myzostoma martenseni* has no body cavity and incomplete segmentation. Animals such as this present a classification challenge, causing taxonomists to reconsider traditional animal phylogenies based on fundamental characters.

Molecular Phylogenies

The last decade has seen a wealth of new molecular sequence data on the various animal groups. The animal phylogenies that these data suggest are often significantly at odds with the traditional phylogeny used in this text and presented in figure 44.2. One such phylogeny, developed from ribosomal RNA studies, is presented in figure 44.22. It is only a rough outline; in the future more data should allow us to resolve relationships within groupings. Still, it is clear that major groups are related in very different ways in the molecular phylogeny than in the more traditional one.

At present, molecular phylogenetic analysis of the animal kingdom is in its infancy. Molecular phylogenies developed from different molecules often tend to suggest different evolutionary relationships. However, the childhood of this approach is likely to be short. Over the next few years, a mountain of additional molecular data can be anticipated. As more data are brought to bear, we can hope that the confusion will lessen, and that a consensus phylogeny will emerge. When and if this happens, it is likely to be very different from the traditional view.

The use of molecular data to construct phylogenies is likely to significantly alter our understanding of relationships among the animal phyla.

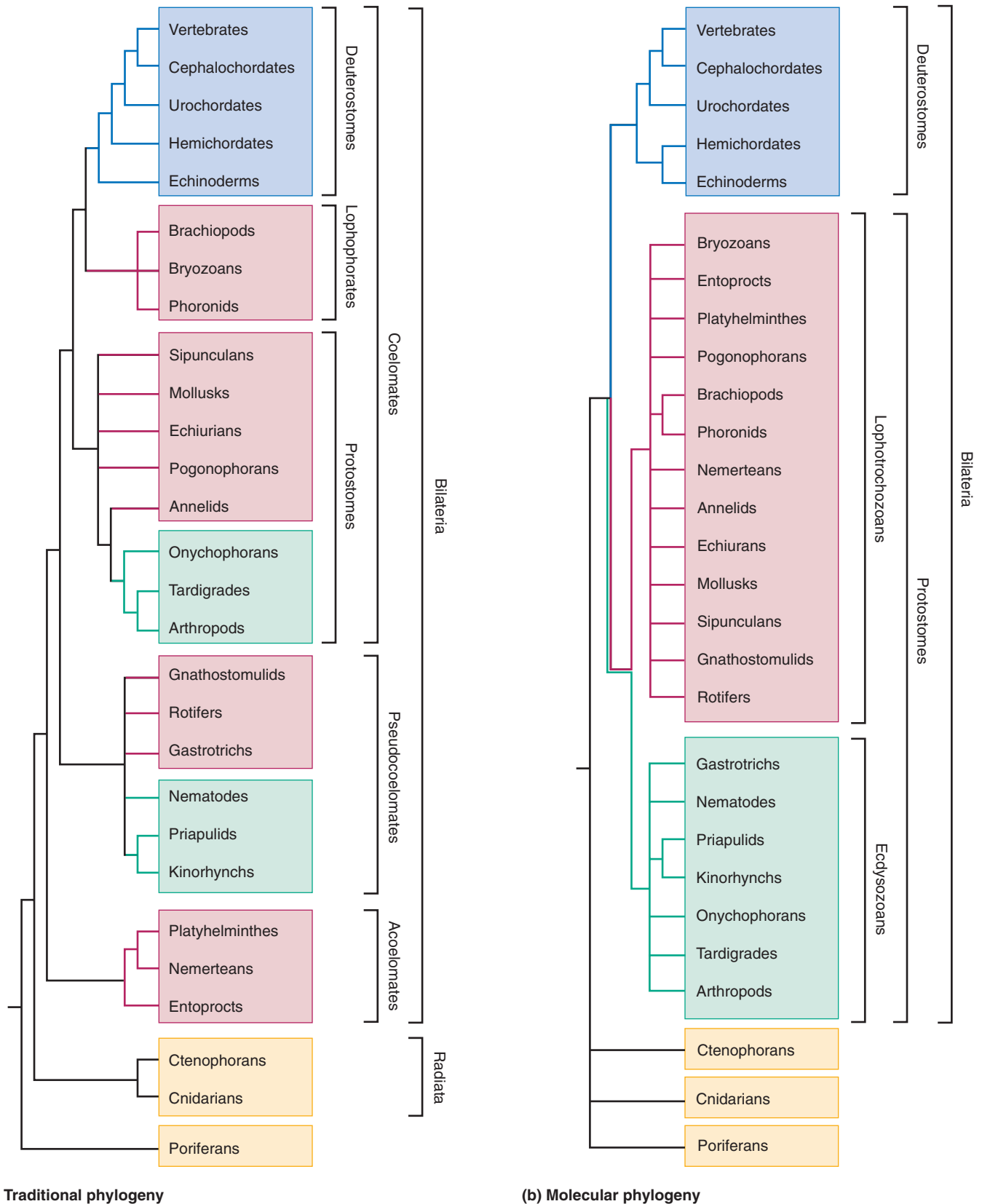


FIGURE 44.22

Traditional versus molecular animal phylogenies. (a) Traditional phylogenies are based on fundamental morphological characters. (After L. H. Hyman, *The Invertebrates*, 1940.) (b) More recent phylogenies are often based on molecular analyses, this one on comparisons of rRNA sequence differences among the animal phyla. (After Adoutte, et al., *Proc. Nat. Acad. Sci* 97: p. 4454, 2000.)



Summary

Questions

Media Resources

44.1 Animals are multicellular heterotrophs without cell walls.

- Animals are heterotrophic, multicellular, and usually have the ability to move. Almost all animals reproduce sexually. Animal cells lack rigid cell walls and digest their food internally.
- The kingdom Animalia is divided into two subkingdoms: Parazoa, which includes only the asymmetrical phylum Porifera, and Eumetazoa, characterized by body symmetry.

1. What are the characteristics that distinguish animals from other living organisms?
2. What are the two subkingdoms of animals? How do they differ in terms of symmetry and body organization?



- Activity: Invertebrates
- Characteristics of Invertebrates
- Body Organization



- Symmetry in Nature
- Posterior to Anterior
- Sagittal Plane
- Frontal to Coronal Plane
- Transverse/Cross-sectional Planes

44.2 The simplest animals are not bilaterally symmetrical.

- The sponges (phylum Porifera) are characterized by specialized, flagellated cells called choanocytes. They do not possess tissues or organs, and most species lack symmetry in their body organization.
- Cnidarians (phylum Cnidaria) are predominantly marine animals with unique stinging cells called cnidocytes, each of which contains a specialized harpoonlike apparatus, or nematocyst.

3. From what kind of ancestor did sponges probably evolve?
4. What are the specialized cells used by a sponge to capture food?
5. What are the two ways sponges reproduce? What do larval sponges look like?
6. What is a planula?



- Sponges
- Radical Phyla

44.3 Acoelomates are solid worms that lack a body cavity.

- Acoelomates lack an internal cavity, except for the digestive system, and are the simplest animals that have organs.
- The most prominent phylum of acoelomates, Platyhelminthes, includes the free-living flatworms and the parasitic flukes and tapeworms.
- Ribbon worms (phylum Nemertea) are similar to free-living flatworms, but have a complete digestive system and a circulatory system in which the blood flows in vessels.

7. What body plan do members of the phylum Platyhelminthes possess? Are these animals parasitic or free-living? How do they move from place to place?
8. How are tapeworms different from flukes? How do tapeworms reproduce?



- Bilateral Acoelomates



- Student Research: Parasitic Flatworms

44.4 Pseudocoelomates have a simple body cavity.

- Pseudocoelomates, exemplified by the nematodes (phylum Nematoda), have a body cavity that develops between the mesoderm and the endoderm.
- Rotifers (phylum Rotifera), or wheel animals, are very small freshwater pseudocoelomates.

9. Why are nematodes structurally unique in the animal world?
10. How do rotifers capture food?



- Pseudocoelomates

44.5 The coming revolution in animal taxonomy will likely alter traditional phylogenies.

- Molecular data are suggesting animal phylogenies that are in considerable disagreement with traditional phylogenies.

11. With what group are myzostomids most closely allied?



- Student Research: Molecular Phylogeny of Gastropods

45

Mollusks and Annelids

Concept Outline

45.1 Mollusks were among the first coelomates.

Coelomates. Next to the arthropods, mollusks comprise the second most diverse phylum and include snails, clams, and octopuses. There are more terrestrial mollusk species than terrestrial vertebrates!

Body Plan of the Mollusks. The mollusk body plan is characterized by three distinct sections, a unique rasping tongue, and distinctive free-swimming larvae also found in annelid worms.

The Classes of Mollusks. The three major classes of mollusks are the gastropods (snails and slugs), the bivalves (oysters and clams), and the cephalopods (octopuses and squids). While they seem very different at first glance, on closer inspection, they all have the same basic mollusk body plan.

45.2 Annelids were the first segmented animals.

Segmented Animals. Annelids are segmented coelomate worms, most of which live in the sea. The annelid body is composed of numerous similar segments.

Classes of Annelids. The three major classes of annelids are the polychaetes (marine worms), the oligochaetes (earthworms and related freshwater worms), and the hirudines (leeches).

45.3 Lophophorates appear to be a transitional group.

Lophophorates. The three phyla of lophophorates share a unique ciliated feeding structure, but differ in many other ways.



FIGURE 45.1

An annelid, the Christmas-tree worm, *Spirobranchus giganteus*. Mollusks and annelids inhabit both terrestrial and aquatic habitats. They are large and successful groups, with some of their most spectacular members represented in marine environments.

Although acoelomates and pseudocoelomates have proven very successful, a third way of organizing the animal body has also evolved, one that occurs in the bulk of the animal kingdom. We will begin our discussion of the coelomate animals with mollusks, which include such animals as clams, snails, slugs, and octopuses. Annelids (figure 45.1), such as earthworms, leeches, and seaworms, are also coelomates, but in addition, were the earliest group of animals to evolve segmented bodies. The lophophorates, a group of marine animals united by a distinctive feeding structure called the lophophore, have features intermediate between those of protostomes and deuterostomes and will also be discussed in this chapter. The remaining groups of coelomate animals will be discussed in chapters 46, 47, and 48.

45.1 Mollusks were among the first coelomates.

Coelomates

The evolution of the *coelom* was a significant advance in the structure of the animal body. Coelomates have a new body design that repositions the fluid and allows the development of complex tissues and organs. This new body plan also made it possible for animals to evolve a wide variety of different body architectures and to grow to much larger sizes than acoelomate animals. Among the earliest groups of coelomates were the mollusks and the annelids.

Mollusks

Mollusks (phylum Mollusca) are an extremely diverse animal phylum, second only to the arthropods, with over 110,000 described species. Mollusks include snails, slugs, clams, scallops, oysters, cuttlefish, octopuses, and many other familiar animals (figure 45.2). The durable shells of some mollusks are often beautiful and elegant; they have long been favorite objects for professional scientists and amateurs alike to collect, preserve, and study. Chitons and nudibranchs are less familiar marine mollusks. Mollusks are characterized by a coelom, and while there is extraordinary diversity in this phylum, many of the basic components of the mollusk body plan can be seen in figure 45.3.

Mollusks evolved in the oceans, and most groups have remained there. Marine mollusks are widespread and often abundant. Some groups of mollusks have invaded freshwater and terrestrial habitats, including the snails and slugs that live in your garden. Terrestrial mollusks are often abundant in places that are at least seasonally moist. Some of these places, such as the crevices of desert rocks, may appear very dry, but even these habitats have at least a temporary supply of water at certain times. There are so many terrestrial mollusks that only the arthropods have more species adapted to a terrestrial way of life. The 35,000 species of terrestrial mollusks far outnumber the roughly 20,000 species of terrestrial vertebrates.

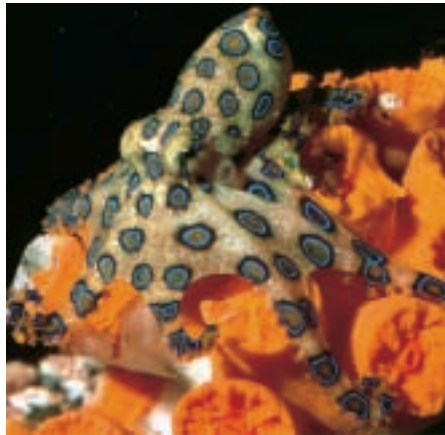
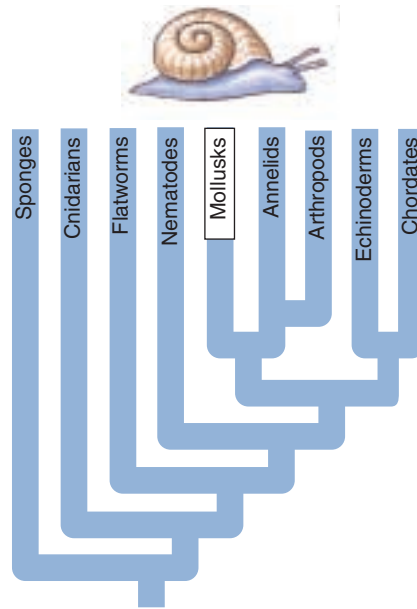


FIGURE 45.2

A mollusk. The blue-ringed octopus is one of the few mollusks dangerous to humans. Strikingly beautiful, it is equipped with a sharp beak and poison glands—divers give it a wide berth!

As a group, mollusks are an important source of food for humans. Oysters, clams, scallops, mussels, octopuses, and squids are among the culinary delicacies that belong to this large phylum. Mollusks are also of economic significance to us in many other ways. For example, pearls are produced in oysters, and the material called mother-of-pearl, often used in jewelry and other decorative objects, is produced in the shells of a number of different mollusks, but most notably in the snail called abalone. Mollusks are not wholly beneficial to humans, however. Bivalve mollusks called shipworms burrow through wood submerged in the sea, damaging boats, docks, and pilings. The zebra mussel has recently invaded North American ecosystems from Europe via the ballast water of cargo ships from Europe, wreaking havoc in many aquatic ecosystems. Slugs and terrestrial snails often cause extensive damage to garden flowers, vegetables, and crops. Other mollusks serve as hosts to the intermediate stages of many serious parasites, including several nematodes and flatworms, which we discussed in chapter 44.

Mollusks range in size from almost microscopic to huge, although most measure a few centimeters in their largest dimension. Some, however, are minute, while others reach formidable sizes. The giant squid, which is occasionally cast ashore but has rarely been observed in its natural environment, may grow up to 21 meters long! Weighing up to 250 kilograms, the giant squid is the largest invertebrate

and, along with the giant clam (figure 45.4), the heaviest. Millions of giant squid probably inhabit the deep regions of the ocean, even though they are seldom caught. Another large mollusk is the bivalve *Tridacna maxima*, the giant clam, which may be as long as 1.5 meters and may weigh as much as 270 kilograms.

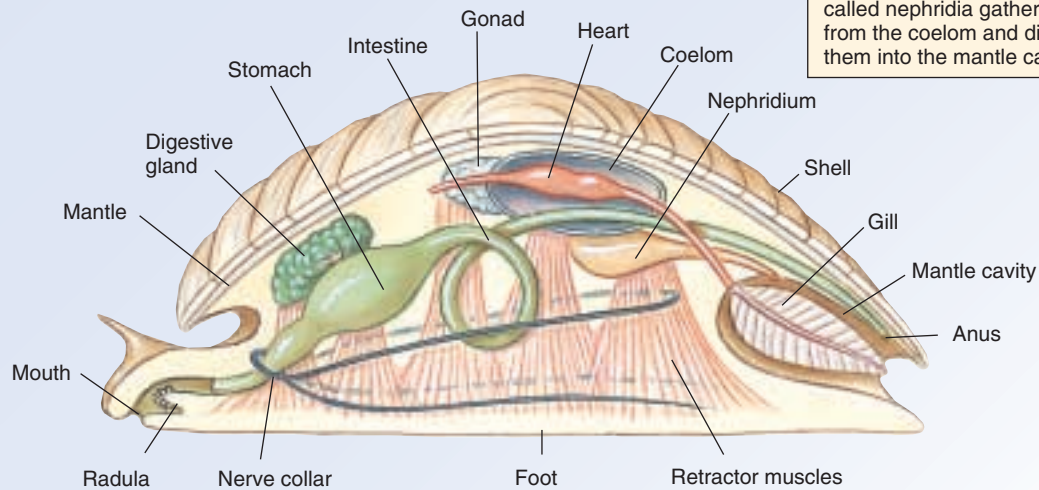
Mollusks are the second-largest phylum of animals in terms of named species; mollusks exhibit a variety of body forms and live in many different environments.

PHYLUM MOLLUSCA: Coelom

The mantle is a heavy fold of tissue wrapped around the mollusk body like a cape. The cavity between the mantle and the body contains gills, which capture oxygen from the water passing through the mantle cavity. In some mollusks, like snails, the mantle secretes a hard outer shell.

Snails have a three-chambered heart and an open circulation system. The coelom is confined to a small cavity around the heart.

Mollusks were among the first animals to develop an efficient excretory system. Tubular structures called nephridia gather wastes from the coelom and discharge them into the mantle cavity.



Many mollusks are carnivores. They locate prey using chemosensory structures. Within the mouth of a snail are horny jaws and a unique rasping tongue called a radula.

Snails creep along the ground on a muscular foot. Squid can shoot through the water by squeezing water out of the mantle cavity, in a kind of jet propulsion.

FIGURE 45.3

Evolution of the coelom. A generalized mollusk body plan is shown above. The body cavity of a mollusk is a coelom, which is completely enclosed within the mesoderm. This allows physical contact between the mesoderm and the endoderm, permitting interactions that lead to development of highly specialized organs such as a stomach.

FIGURE 45.4

Giant clam. Second only to the arthropods in number of described species, members of the phylum Mollusca occupy almost every habitat on earth. This giant clam, *Tridacna maxima*, has a green color that is caused by the presence of symbiotic dinoflagellates (zooxanthellae). Through photosynthesis, the dinoflagellates probably contribute most of the food supply of the clam, although it remains a filter feeder like most bivalves. Some individual giant clams may be nearly 1.5 meters long and weigh up to 270 kilograms.



Body Plan of the Mollusks

In their basic body plan (figure 45.5), mollusks have distinct bilateral symmetry. Their digestive, excretory, and reproductive organs are concentrated in a **visceral mass**, and a muscular **foot** is their primary mechanism of locomotion. They may also have a differentiated **head** at the anterior end of the body. Folds (often two) arise from the dorsal body wall and enclose a cavity between themselves and the visceral mass; these folds constitute the **mantle**. In some mollusks the mantle cavity acts as a lung; in others it contains gills. **Gills** are specialized portions of the mantle that usually consist of a system of filamentous projections rich in blood vessels. These projections greatly increase the surface area available for gas exchange and, therefore, the animal's overall respiratory potential. Mollusk gills are very efficient, and many gilled mollusks extract 50% or more of the dissolved oxygen from the water that passes through the mantle cavity. Finally, in most members of this phylum, the outer surface of the mantle also secretes a protective shell.

A mollusk shell consists of a horny outer layer, rich in protein, which protects the two underlying calcium-rich layers from erosion. The middle layer consists of densely packed crystals of calcium carbonate. The inner layer is pearly and increases in thickness throughout the animal's life. When it reaches a sufficient thickness, this layer is used as mother-of-pearl. Pearls themselves are formed when a foreign object, like a grain of sand, becomes lodged between the mantle and the inner shell layer of **bivalve mollusks** (two-shelled), including clams and oysters. The mantle coats the foreign object with layer upon layer of shell material to reduce irritation caused by the object. The shell of mollusks serves primarily for protection. Many species can withdraw for protection into their shell if they have one.

In aquatic mollusks, a continuous stream of water passes into and out of the mantle cavity, drawn by the cilia on the gills. This water brings in oxygen and, in the case of the bivalves, also brings in food; it also carries out waste materials. When the gametes are being produced, they are frequently carried out in the same stream.

The foot of a mollusk is muscular and may be adapted for locomotion, attachment, food capture (in squids and octopuses), or various combinations of these functions. Some mollusks secrete mucus, forming a path that they glide along on their foot. In cephalopods—squids and octopuses—the foot is divided into arms, also called tentacles. In some *pelagic* forms, mollusks that are perpetually free-swimming, the foot is modified into wing-like projections or thin fins.

One of the most characteristic features of all the mollusks except the bivalves is the **radula**, a rasping, tongue-like organ used for feeding. The radula consists primarily of dozens to thousands of microscopic, chitinous teeth arranged in rows (figure 45.6). Gastropods (snails and their relatives) use their radula to scrape algae and other food materials off their substrates and then to convey this food to the digestive tract. Other gastropods are active predators, some using a modified radula to drill through the shells of prey and extract the food. The small holes often seen in oyster shells are produced by gastropods that have bored holes to kill the oyster and extract its body for food.

The circulatory system of all mollusks except cephalopods consists of a heart and an open system in which blood circulates freely. The mollusk heart usually has three chambers, two that collect aerated blood from the gills, while the third pumps it to the other body tissues. In mollusks, the coelom takes the form of a small cavity around the heart.

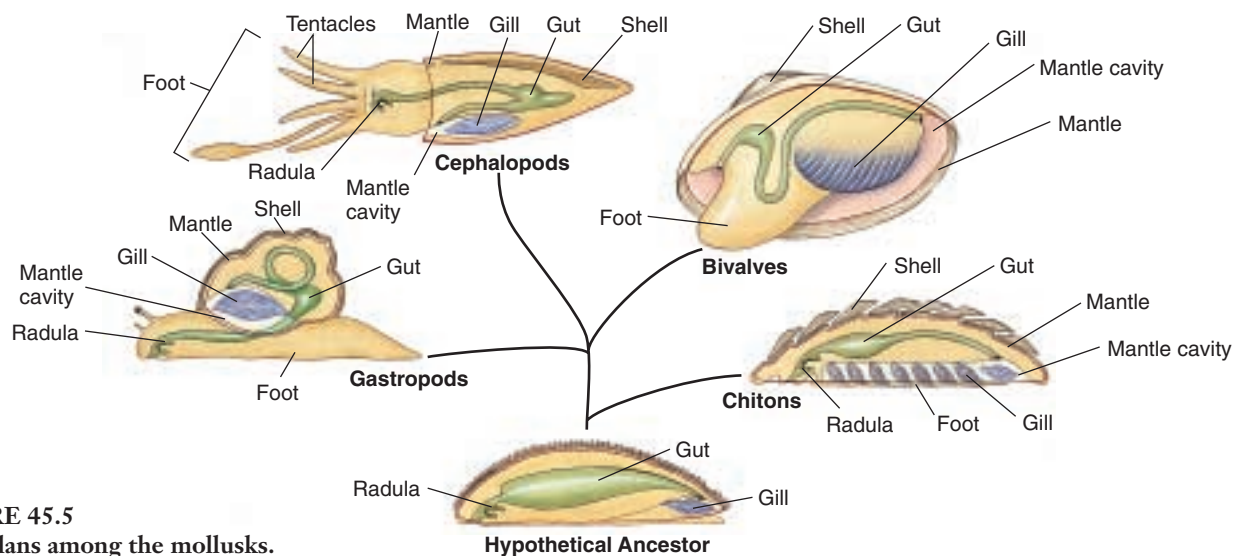


FIGURE 45.5
Body plans among the mollusks.

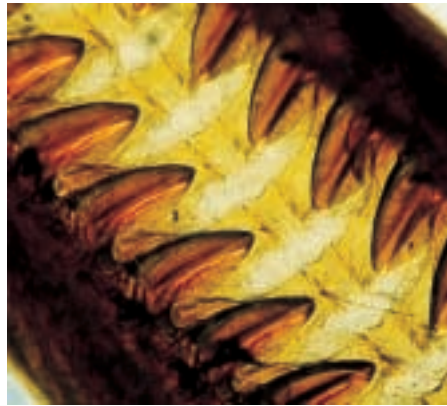
Nitrogenous wastes are removed from the mollusk by one or two tubular structures called **nephridia**. A typical nephridium has an open funnel, the **nephrostome**, which is lined with cilia. A coiled tubule runs from the nephrostome into a bladder, which in turn connects to an excretory pore. Wastes are gathered by the nephridia from the coelom and discharged into the mantle cavity. The wastes are then expelled from the mantle cavity by the continuous pumping of the gills. Sugars, salts, water, and other materials are reabsorbed by the walls of the nephridia and returned to the animal's body as needed to achieve an appropriate osmotic balance.

In animals with a closed circulatory system, such as annelids, cephalopod mollusks, and vertebrates, the coiled tubule of a nephridium is surrounded by a network of capillaries. Wastes are extracted from the circulatory system through these capillaries and are transferred into the nephridium, then subsequently discharged. Salts, water, and other associated materials may also be reabsorbed from the tubule of the nephridium back into the capillaries. For this reason, the excretory systems of these coelomates are much more efficient than the flame cells of the acoelomates, which pick up substances only from the body fluids. Mollusks were one of the earliest evolutionary lines to develop an efficient excretory system. Other than chordates, coelomates with closed circulation have similar excretory systems.

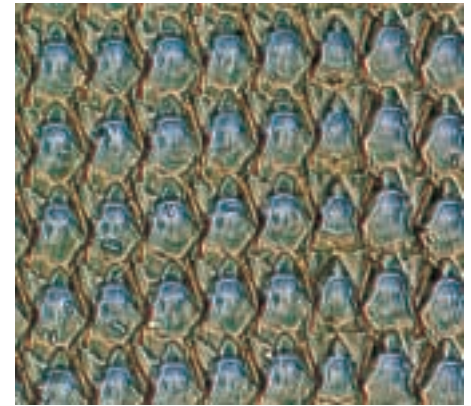
Reproduction in Mollusks

Most mollusks have distinct male and female individuals, although a few bivalves and many gastropods are hermaphroditic. Even in hermaphroditic mollusks, cross-fertilization is most common. Remarkably, some sea slugs and oysters are able to change from one sex to the other several times during a single season.

Most aquatic mollusks engage in external fertilization. The males and females release their gametes into the water, where they mix and fertilization occurs. Gastropods more often have internal fertilization, however, with the male inserting sperm directly into the female's body. Internal fertilization is one of the key adaptations that allowed gastropods to colonize the land.



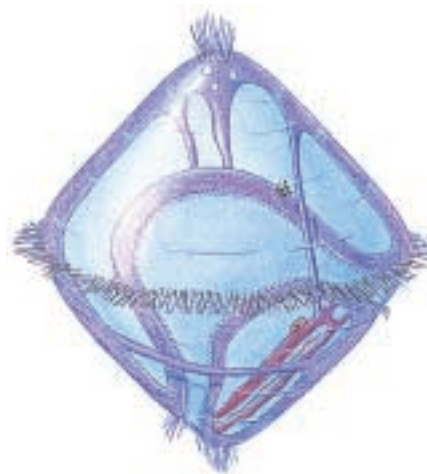
(a)



(b)

FIGURE 45.6

Structure of the radula in a snail. (a) The radula consists of chitin and is covered with rows of teeth. (b) Enlargement of the rasping teeth on a radula.



(a)



(b)

FIGURE 45.7

Stages in the molluscan life cycle. (a) The trochophore larva of a mollusk. Similar larvae, as you will see, are characteristic of some annelid worms as well as a few other phyla. (b) Veliger stage of a mollusk.

Many marine mollusks have free-swimming larvae called **trochophores** (figure 45.7a), which closely resemble the larval stage of many marine annelids. Trochophores swim by means of a row of cilia that encircles the middle of their body. In most marine snails and in bivalves, a second free-swimming stage, the veliger, follows the trochophore stage. This **veliger** stage, has the beginnings of a foot, shell, and mantle (figure 45.7b). Trochophores and veligers drift widely in the ocean currents, dispersing mollusks to new areas.

Mollusks were among the earliest animals to evolve an efficient excretory system. The mantle of mollusks not only secretes their protective shell, but also forms a cavity that is essential to respiration.

The Classes of Mollusks

There are seven classes of mollusks. We will examine four classes of mollusks as representatives of the phylum: (1) Polyplacophora—chitons; (2) Gastropoda—snails, slugs, limpets, and their relatives; (3) Bivalvia—clams, oysters, scallops, and their relatives; and (4) Cephalopoda—squids, octopuses, cuttlefishes, and nautilus. By studying living mollusks and the fossil record, some scientists have deduced that the ancestral mollusk was probably a dorsoventrally flattened, unsegmented, wormlike animal that glided on its ventral surface. This animal may also have had a chitinous cuticle and overlapping calcareous scales. Other scientists believe that mollusks arose from segmented ancestors and became unsegmented secondarily.

Class Polyplacophora: The Chitons

Chitons are marine mollusks that have oval bodies with eight overlapping calcareous plates. Underneath the plates, the body is not segmented. Chitons creep along using a broad, flat foot surrounded by a groove or mantle cavity in which the gills are arranged. Most chitons are grazing herbivores that live in shallow marine habitats, but some live at depths of more than 7000 meters.

Class Gastropoda: The Snails and Slugs

The class Gastropoda contains about 40,000 described species of snails, slugs, and similar animals. This class is primarily a marine group, but it also contains many freshwater and terrestrial mollusks (figure 45.8). Most gastropods have a shell, but some, like slugs and nudibranchs, have lost their shells through the course of evolution. Gastropods generally creep along on a foot, which may be modified for swimming.

The heads of most gastropods have a pair of tentacles with eyes at the ends. These tentacles have been lost in some of the more advanced forms of the class. Within the mouth cavity of many members of this class are horny jaws and a radula.

During embryological development, gastropods undergo **torsion**. Torsion is the process by which the mantle cavity and anus are moved from a posterior location to the front of the body, where the mouth is located. Torsion is brought about by a disproportionate growth of the lateral muscles; that is, one side of the larva grows much more rapidly than the other. A 120-degree rotation of the visceral mass brings the mantle cavity above the head and twists many internal structures. In some groups of gastropods, varying degrees of detorsion have taken place. The **coiling**, or spiral winding, of the shell is a separate process. This process has led to the loss of the right gill and right nephridium in most gastropods. Thus, the visceral mass of gastropods has become bilaterally asymmetrical during the course of evolution.



FIGURE 45.8
A gastropod mollusk. The terrestrial snail, *Allogona townsendiana*.

Gastropods display extremely varied feeding habits. Some are predatory, others scrape algae off rocks (or aquarium glass), and others are scavengers. Many are herbivores, and some terrestrial ones are serious garden and agricultural pests. The radula of oyster drills is used to bore holes in the shells of other mollusks, through which the contents of the prey can be removed. In cone shells, the radula has been modified into a kind of poisonous harpoon, which is shot with great speed into the prey.

Sea slugs, or nudibranchs, are active predators; a few species of nudibranchs have the extraordinary ability to extract the nematocysts from the cnidarian polyps they eat and transfer them through their digestive tract to the surface of their gills intact and use them for their own protection. Nudibranchs are interesting in that they get their name from their gills, which instead of being enclosed within the mantle cavity are exposed along the dorsal surface (*nudi*, “naked”; *branch*, “gill”).

In terrestrial gastropods, the empty mantle cavity, which was occupied by gills in their aquatic ancestors, is extremely rich in blood vessels and serves as a lung, in effect. This structure evolved in animals living in environments with plentiful oxygen; it absorbs oxygen from the air much more effectively than a gill could, but is not as effective under water.

Class Bivalvia: The Bivalves

Members of the class Bivalvia include the clams, scallops, mussels, and oysters. Bivalves have two lateral (left and right) shells (valves) hinged together dorsally (figure 45.9). A ligament hinges the shells together and causes them to gape open. Pulling against this ligament are one or two large adductor muscles that can draw the shells together.

The mantle secretes the shells and ligament and envelops the internal organs within the pair of shells. The mantle is frequently drawn out to form two siphons, one for an incoming and one for an outgoing stream of water. The siphons often function as snorkels to allow bivalves to filter water through their body while remaining almost completely buried in sediments. A complex folded gill lies on each side of the visceral mass. These gills consist of pairs of filaments that contain many blood vessels. Rhythmic beating of cilia on the gills creates a pattern of water circulation. Most bivalves are sessile filter-feeders. They extract small organisms from the water that passes through their mantle cavity.

Bivalves do not have distinct heads or radulas, differing from gastropods in this respect (see figure 45.5). However, most have a wedge-shaped foot that may be adapted, in different species, for creeping, burrowing, cleansing the animal, or anchoring it in its burrow. Some species of clams can dig into sand or mud very rapidly by means of muscular contractions of their foot.

Bivalves disperse from place to place largely as larvae. While most adults are adapted to a burrowing way of life, some genera of scallops can move swiftly through the water by using their large adductor muscles to clap their shells together. These muscles are what we usually eat as “scallops.” The edge of a scallop’s body is lined with tentacle-like projections tipped with complex eyes.

There are about 10,000 species of bivalves. Most species are marine, although many also live in fresh water. Over 500 species of pearly freshwater mussels, or naiads, occur in the rivers and lakes of North America.

Class Cephalopoda: The Octopuses, Squids, and Nautilus

The more than 600 species of the class Cephalopoda—octopuses, squids, and nautilus—are the most intelligent of the invertebrates. They are active marine predators that swim, often swiftly, and compete successfully with fish. The foot has evolved into a series of tentacles equipped with suction cups, adhesive structures, or hooks that seize prey efficiently. Squids have 10 tentacles (figure 45.10); octopuses, as indicated by their name, have eight; and the nautilus, about 80 to 90. Once the tentacles have snared the prey, it is bitten with strong, beaklike paired jaws and pulled into the mouth by the tonguelike action of the radula.

Cephalopods have highly developed nervous systems, and their brains are unique among mollusks. Their eyes are very elaborate, and have a structure much like that of vertebrate eyes, although they evolved separately (see chapter 55). Many cephalopods exhibit complex patterns of behavior and a high level of intelligence; octopuses can be easily trained to distinguish among classes of objects. Most members of this class have closed circulatory systems and are the only mollusks that do.



FIGURE 45.9
A bivalve. The file shell, *Lima scabra*, opened, showing tentacles.



FIGURE 45.10
A cephalopod. Squids are active predators, competing effectively with fish for prey.

Although they evolved from shelled ancestors, living cephalopods, except for the few species of nautilus, lack an external shell. Like other mollusks, cephalopods take water into the mantle cavity and expel it through a siphon. Cephalopods have modified this system into a means of jet propulsion. When threatened, they eject water violently and shoot themselves through the water.

Most octopus and squid are capable of changing color to suit their background or display messages to one another. They accomplish this feat through the use of their *chromatophores*, pouches of pigments embedded in the epithelium.

Gastropods typically live in a hard shell. Bivalves have hinged shells but do not have a distinct head area. Cephalopods possess well-developed brains and are the most intelligent invertebrates.

45.2 Annelids were the first segmented animals.

Segmented Animals

A key transition in the animal body plan was *segmentation*, the building of a body from a series of similar segments. The first segmented animals to evolve were most likely **annelid worms**, phylum Annelida (figure 45.11). One advantage of having a body built from repeated units (segments) is that the development and function of these units can be more precisely controlled, at the level of individual segments or groups of segments. For example, different segments may possess different combinations of organs or perform different functions relating to reproduction, feeding, locomotion, respiration, or excretion.

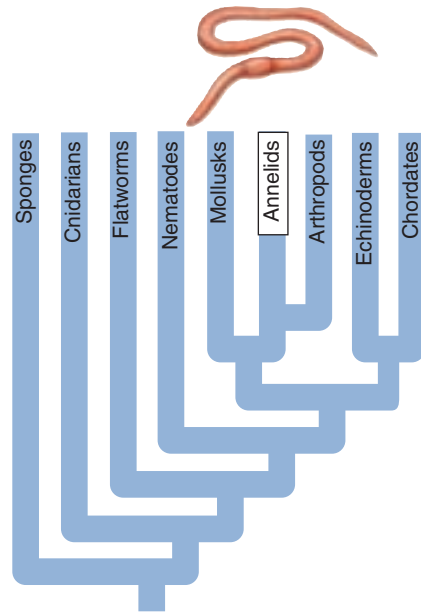


FIGURE 45.11

A polychaete annelid. *Nereis virens* is a wide-ranging, predatory, marine polychaete worm equipped with feathery parapodia for movement and respiration, as well as jaws for hunting. You may have purchased *Nereis* as fishing bait!

Annelids

Two-thirds of all annelids live in the sea (about 8000 species), and most of the rest, some 3100 species, are earthworms. Annelids are characterized by three principal features:

- 1. Repeated segments.** The body of an annelid worm is composed of a series of ring-like segments running the length of the body, looking like a stack of donuts or roll of coins (figure 45.12). Internally, the segments are divided from one another by partitions called **septa**, just as bulkheads separate the segments of a submarine. In each of the cylindrical segments, the excretory and locomotor organs are repeated. The fluid within the coelom of each segment creates a hydrostatic (liquid-supported) skeleton that gives the segment rigidity, like an inflated balloon. Muscles within each segment push against the fluid in the coelom. Because each segment is separate, each can expand or contract independently. This lets the worm move in complex ways.
- 2. Specialized segments.** The anterior (front) segments of annelids have become modified to contain specialized sensory organs. Some are sensitive to light, and elaborate eyes with lenses and retinas have evolved in some annelids. A well-developed cerebral ganglion, or brain, is contained in one anterior segment.
- 3. Connections.** Although partitions separate the segments, materials and information do pass between segments. Annelids have a closed circulatory system that carries blood from one segment to another. A

ventral nerve cord connects the nerve centers or ganglia in each segment with one another and the brain. These neural connections are critical features that allow the worm to function and behave as a unified and coordinated organism.

Body Plan of the Annelids

The basic annelid body plan is a tube within a tube, with the internal digestive tract—a tube running from mouth to anus—suspended within the coelom. The tube that makes up the digestive tract has several portions—the pharynx, esophagus, crop, gizzard, and intestine—that are specialized for different functions.

Annelids make use of their hydrostatic skeleton for locomotion. To move, annelids contract circular muscles running around each segment. Doing so squeezes the segment, causing the coelomic fluid to squirt outwards, like a tube of toothpaste. Because the fluid is trapped in the segment by the septa, instead of escaping like toothpaste, the fluid causes the segment to elongate and get much thinner. By then contracting longitudinal muscles that run along the length of the worm, the segment is returned to its original shape. In most annelid groups, each segment typically possesses **setae**, bristles of chitin that help anchor the worms during locomotion. By extending the setae in some segments so that they anchor in the substrate and retracting them in other segments, the worm can squirt its body, section by section, in either direction.

PHYLUM ANNELIDA: Segmentation

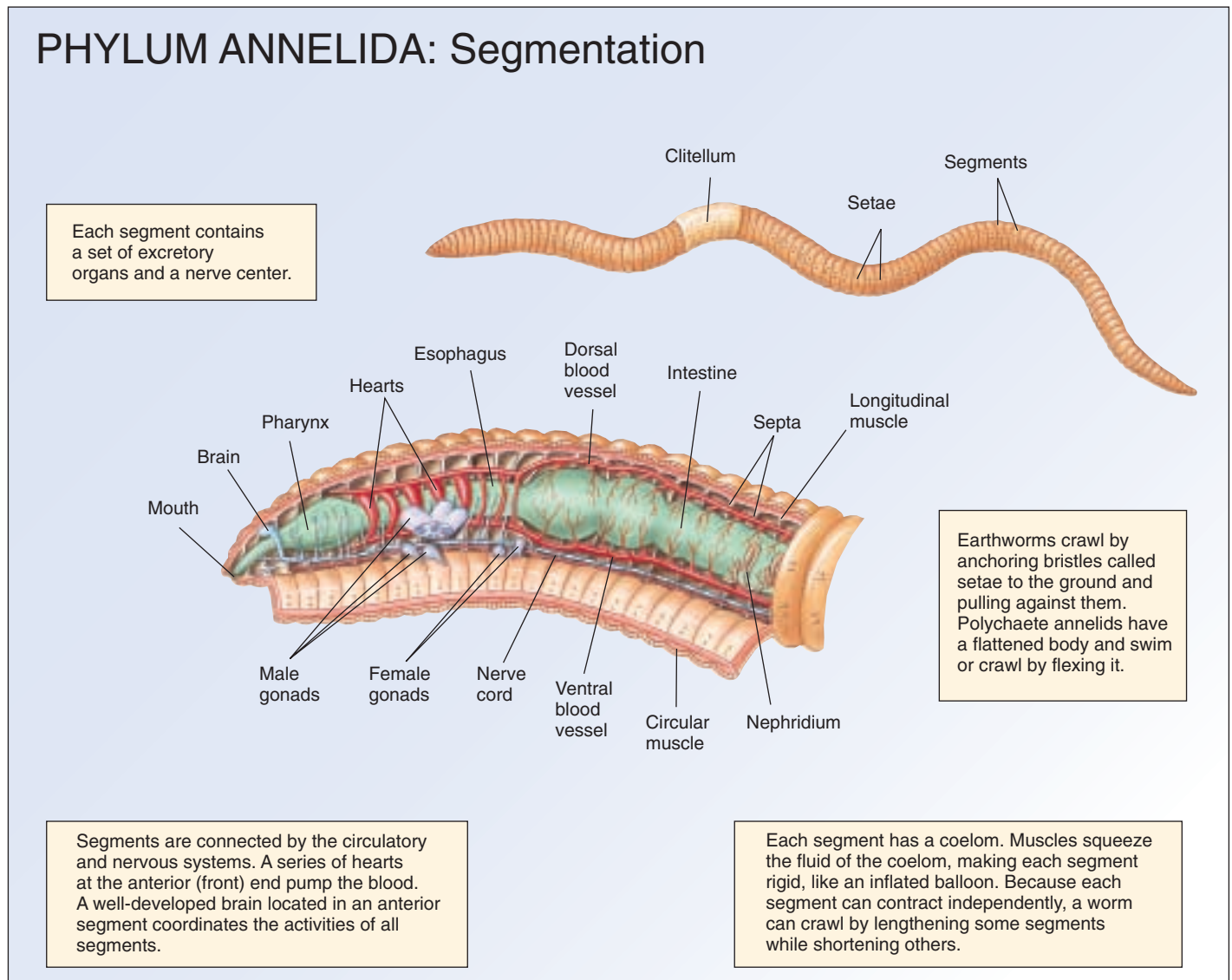


FIGURE 45.12

The evolution of segmentation. Marine polychaetes and earthworms (phylum Annelida) were most likely the first organisms to evolve a body plan based on partly repeated body segments. Segments are separated internally from each other by septa.

Unlike the arthropods and most mollusks, most annelids have a closed circulatory system. Annelids exchange oxygen and carbon dioxide with the environment through their body surfaces; most lack gills or lungs. However, much of their oxygen supply reaches the different parts of their bodies through their blood vessels. Some of these vessels at the anterior end of the worm body are enlarged and heavily muscular, serving as hearts that pump the blood. Earthworms have five pulsating blood vessels on each side that serve as hearts, helping to pump blood from the main dorsal vessel, which is their major pumping structure, to the main ventral vessel.

The excretory system of annelids consists of ciliated, funnel-shaped nephridia generally similar to those of mollusks. These nephridia—each segment has a pair—collect waste products and transport them out of the body through the coelom by way of specialized excretory tubes.

Annelids are a diverse group of coelomate animals characterized by serial segmentation. Each segment in the annelid body has its own circulatory, excretory, neural elements, and setae.

Classes of Annelids

The roughly 12,000 described species of annelids occur in many different habitats. They range in length from as little as 0.5 millimeter to the more than 3-meter length of some polychaetes and giant Australian earthworms. There are three classes of annelids: (1) Polychaeta, which are free-living, almost entirely marine bristleworms, comprising some 8000 species; (2) Oligochaeta, terrestrial earthworms and related marine and freshwater worms, with some 3100 species; and (3) Hirudinea, leeches, mainly freshwater predators or bloodsuckers, with about 500 species. The annelids are believed to have evolved in the sea, with polychaetes being the most primitive class. Oligochaetes seem to have evolved from polychaetes, perhaps by way of brackish water to estuaries and then to streams. Leeches share with oligochaetes an organ called a **clitellum**, which secretes a cocoon specialized to receive the eggs. It is generally agreed that leeches evolved from oligochaetes, specializing in their bloodsucking lifestyle as external parasites.

Class Polychaeta: The Polychaetes

Polychaetes (class Polychaeta) include clamworms, plume worms, scaleworms, lugworms, twin-fan worms, sea mice, peacock worms, and many others. These worms are often surprisingly beautiful, with unusual forms and sometimes iridescent colors (figure 45.13; see also figure 45.1). Polychaetes are often a crucial part of marine food chains, as they are extremely abundant in certain habitats.

Some polychaetes live in tubes or permanent burrows of hardened mud, sand, mucuslike secretions, or calcium carbonate. These sedentary polychaetes are primarily filter feeders, projecting a set of feathery tentacles from the tubes in which they live that sweep the water for food. Other polychaetes are active swimmers, crawlers, or burrowers. Many are active predators.

Polychaetes have a well-developed head with specialized sense organs; they differ from other annelids in this respect. Their bodies are often highly organized into distinct regions formed by groups of segments related in function and structure. Their sense organs include eyes, which range from simple eyespots to quite large and conspicuous stalked eyes.

Another distinctive characteristic of polychaetes is the paired, fleshy, paddlelike flaps, called **parapodia**, on most of their segments. These parapodia, which bear bristlelike setae, are used in swimming, burrowing, or crawling. They also play an important role in gas exchange because they greatly increase the surface area of the body. Some polychaetes that live in burrows or tubes may have parapodia featuring hooks to help anchor the worm. Slow crawling is carried out by means of the parapodia. Rapid crawling and swimming is by undulating the body. In addition, the polychaete epidermis often includes ciliated cells which aid in respiration and food procurement.



FIGURE 45.13

A polychaete. The shiny bristleworm, *Oenone fulgida*.

The sexes of polychaetes are usually separate, and fertilization is often external, occurring in the water and away from both parents. Unlike other annelids, polychaetes usually lack permanent **gonads**, the sex organs that produce gametes. They produce their gametes directly from germ cells in the lining of the coelom or in their septa. Fertilization results in the production of ciliated, mobile *trochophore larvae* similar to the larvae of mollusks. The trochophores develop for long periods in the plankton before beginning to add segments and thus changing to a juvenile form that more closely resembles the adult form.

Class Oligochaeta: The Earthworms

The body of an earthworm (class Oligochaeta) consists of 100 to 175 similar segments, with a mouth on the first and an anus on the last. Earthworms seem to eat their way through the soil because they suck in organic and other material by expanding their strong pharynx. Everything that they ingest passes through their long, straight digestive tracts. One region of this tract, the gizzard, grinds up the organic material with the help of soil particles.

The material that passes through an earthworm is deposited outside of its burrow in the form of castings that

consist of irregular mounds at the opening of a burrow. In this way, earthworms aerate and enrich the soil. A worm can eat its own weight in soil every day.

In view of the underground lifestyle that earthworms have evolved, it is not surprising that they have no eyes. However, earthworms do have numerous light-, chemo-, and touch-sensitive cells, mostly concentrated in segments near each end of the body—those regions most likely to encounter light or other stimuli. Earthworms have fewer setae than polychaetes and no parapodia or head region.

Earthworms are hermaphroditic, another way in which they differ from most polychaetes. When they mate (figure 45.14), their anterior ends point in opposite directions, and their ventral surfaces touch. The *clitellum* is a thickened band on an earthworm's body; the mucus it secretes holds the worms together during copulation. Sperm cells are released from pores in specialized segments of one partner into the sperm receptacles of the other, the process going in both directions simultaneously.

Two or three days after the worms separate, the clitellum of each worm secretes a mucous cocoon, surrounded by a protective layer of chitin. As this sheath passes over the female pores of the body—a process that takes place as the worm moves—it receives eggs. As it subsequently passes along the body, it incorporates the sperm that were deposited during copulation. Fertilization of the eggs takes place within the cocoon. When the cocoon finally passes over the end of the worm, its ends pinch together. Within the cocoon, the fertilized eggs develop directly into young worms similar to adults.

Class Hirudinea: The Leeches

Leeches (class Hirudinea) occur mostly in fresh water, although a few are marine and some tropical leeches occupy terrestrial habitats. Most leeches are 2 to 6 centimeters long, but one tropical species reaches up to 30 centimeters. Leeches are usually flattened dorsoventrally, like flatworms. They are hermaphroditic, and develop a clitellum during the breeding season; cross-fertilization is obligatory as they are unable to self-fertilize.

A leech's coelom is reduced and continuous throughout the body, not divided into individual segments as in the polychaetes and oligochaetes. Leeches have evolved suckers at one or both ends of the body. Those that have suckers at both ends move by attaching first one and then the other end to the substrate, looping along. Many species are also capable of swimming. Except for one species, leeches have no setae.

Some leeches have evolved the ability to suck blood from animals. Many freshwater leeches live as external parasites. They remain on their hosts for long periods and suck their blood from time to time.

The best-known leech is the medicinal leech, *Hirudo medicinalis* (figure 45.15). Individuals of *Hirudo* are 10 to 12 centimeters long and have bladelike, chitinous jaws that



FIGURE 45.14

Earthworms mating. The anterior ends are pointing in opposite directions.



FIGURE 45.15

Hirudo medicinalis, the medicinal leech, is seen here feeding on a human arm. Leeches use chitinous, bladelike jaws to make an incision to access blood and secrete an anticoagulant to keep the blood from clotting. Both the anticoagulant and the leech itself have made important contributions to modern medicine.

rasp through the skin of the victim. The leech secretes an anticoagulant into the wound to prevent the blood from clotting as it flows out, and its powerful sucking muscles pump the blood out quickly once the hole has been opened. Leeches were used in medicine for hundreds of years to suck blood out of patients whose diseases were mistakenly believed to be caused by an excess of blood. Today, European pharmaceutical companies still raise and sell leeches, but they are used to remove excess blood after certain surgeries. Following the surgery, blood may accumulate because veins may function improperly and fail to circulate the blood. The accumulating blood “turns off” the arterial supply of fresh blood, and the tissue often dies. When leeches remove the excess blood, new capillaries form in about a week, and the tissues remain healthy.

Segmented annelids evolved in the sea. Earthworms are their descendents, as are parasitic leeches.

45.3 Lophophorates appear to be a transitional group.

Lophophorates

Three phyla of marine animals—Phoronida, Ectoprocta, and Brachiopoda—are characterized by a **lophophore**, a circular or U-shaped ridge around the mouth bearing one or two rows of ciliated, hollow tentacles. Because of this unusual feature, they are thought to be related to one another. The lophophore presumably arose in a common ancestor. The coelomic cavity of lophophorates extends into the lophophore and its tentacles. The lophophore functions as a surface for gas exchange and as a food-collection organ. Lophophorates use the cilia of their lophophore to capture the organic detritus and plankton on which they feed. Lophophorates are attached to their substrate or move slowly.

Lophophorates share some features with mollusks, annelids and arthropods (all protostomes) and share others with deuterostomes. Cleavage in lophophorates is mostly radial, as in deuterostomes. The formation of the coelom varies; some lophophorates resemble protostomes in this respect, others deuterostomes. In the Phoronida, the mouth forms from the blastopore, while in the other two phyla, it forms from the end of the embryo opposite the blastopore. Molecular evidence shows that the ribosomes of all lophophorates are decidedly protostome-like, lending strength to placing them within the protostome phyla. Despite the differences among the three phyla, the unique structure of the lophophore seems to indicate that the members share a common ancestor. Their relationships continue to present a fascinating puzzle.

Phylum Phoronida: The Phoronids

Phoronids (phylum Phoronida) superficially resemble common polychaete tube worms seen on dock pilings but have many important differences. Each phoronid secretes a chitinous tube and lives out its life within it (figure 45.16). They also extend tentacles to feed and quickly withdraw them when disturbed, but the resemblance to the tube worm ends there. Instead of a straight tube-within-a-tube body plan, phoronids have a U-shaped gut. Only about 10 phoronid species are known, ranging in length from a few millimeters to 30 centimeters. Some species lie buried in sand, others are attached to rocks either singly or in groups. Phoronids develop as protostomes, with radial cleavage and the anus developing secondarily.

Phylum Ectoprocta: The Bryozoans

Ectoprocts (phylum Ectoprocta) look like tiny, short versions of phoronids (figure 45.17). They are small—usually

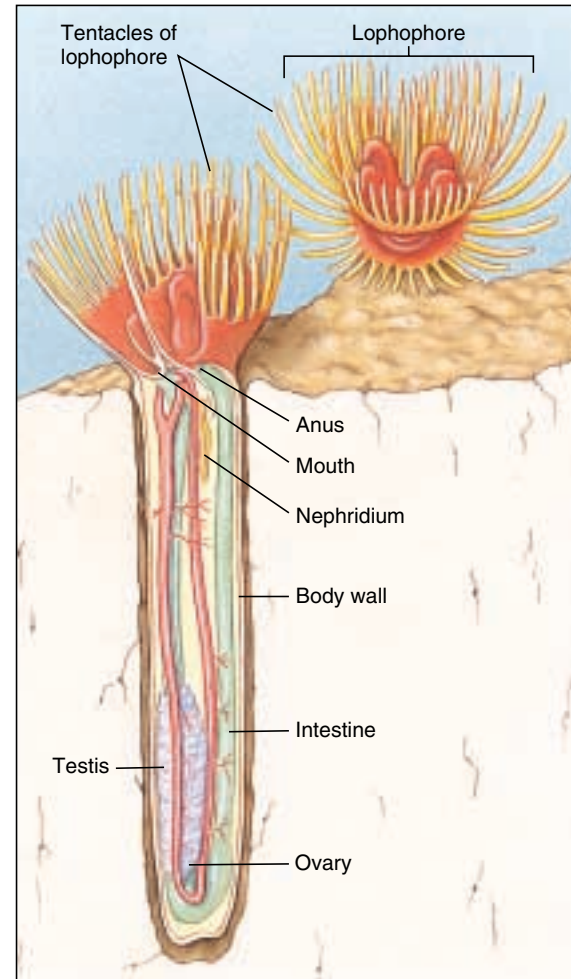


FIGURE 45.16

Phoronids (phylum Phoronida). A phoronid, such as *Phoronis*, lives in a chitinous tube that the animal secretes to form the outer wall of its body. The lophophore consists of two parallel, horseshoe-shaped ridges of tentacles and can be withdrawn into the tube when the animal is disturbed.

less than 0.5 millimeter long—and live in colonies that look like patches of moss on the surfaces of rocks, seaweed, or other submerged objects (in fact, their common name bryozoans translates from Greek as “moss-animals”). The name Ectoprocta refers to the location of the anus (proct), which is external to the lophophore. The 4000 species include both marine and freshwater forms—the only nonmarine lophophorates. Individual ectoprocts secrete a tiny chitinous chamber called a **zoecium** that attaches to rocks and other members of the colony. Individuals communicate chemically through pores between chambers. Ectoprocts develop as deuterostomes, with the mouth developing secondarily; cleavage is radial.

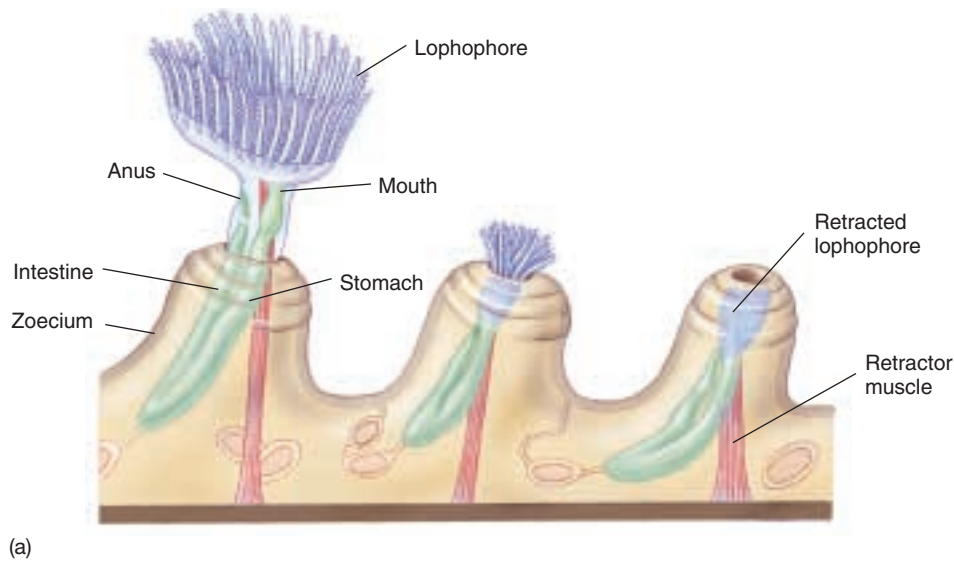


FIGURE 45.17

Ectoprocts (phylum Ectoprocta). (a) A small portion of a colony of the freshwater ectoproct, *Plumatella* (phylum Ectoprocta), which grows on the underside of rocks. The individual at the left has a fully extended lophophore, the structure characteristic of the three lophophorate phyla. The tiny individuals of *Plumatella* disappear into their shells when disturbed. (b) *Plumatella repens*, a freshwater bryozoan.

Phylum Brachiopoda: The Brachiopods

Brachiopods, or lamp shells, superficially resemble clams, with two calcified shells (figure 45.18). Many species attach to rocks or sand by a stalk that protrudes through an opening in one shell. The lophophore lies within the shell and functions when the brachiopod's shells are opened slightly. Although a little more than 300 species of brachiopods (phylum Brachiopoda) exist today, more than 30,000 species of this phylum are known as fossils. Because brachiopods were common in the earth's oceans for millions of years and because their shells fossilize readily, they are often used as index fossils to define a particular time period or sediment type. Brachiopods develop as deuterostomes and show radial cleavage.

The three phyla of lophophorates probably share a common ancestor, and they show a mixture of protostome and deuterostome characteristics.

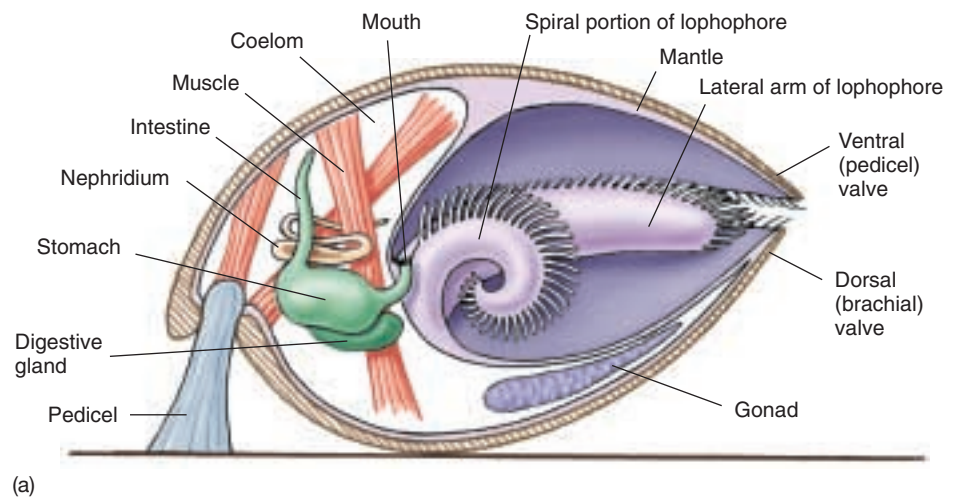


FIGURE 45.18

Brachiopods (phylum Brachiopoda). (a) The lophophore lies within two calcified shells, or valves. (b) The brachiopod, *Terebratulina septentrionalis*, is shown here slightly opened so that the lophophore is visible.



Summary

Questions

Media Resources

45.1 Mollusks were among the first coelomates.

- Mollusks contain a true body cavity, or coelom, within the embryonic mesoderm and were among the first coelomate animals.
- The mollusks constitute the second largest phylum of animals in terms of named species. Their body plan consists of distinct parts: a head, a visceral mass, and a foot.
- Of the seven classes of mollusks, the gastropods (snails and slugs), bivalves (clams and scallops), and cephalopods (octopuses, squids, and nautilus), are best known.
- Gastropods typically live in a hard shell. During development, one side of the embryo grows more rapidly than the other, producing a characteristic twisting of the visceral mass.
- Members of the class Bivalvia have two shells hinged together dorsally and a wedge-shaped foot. They lack distinct heads and radulas. Most bivalves are filter-feeders.
- Octopuses and other cephalopods are efficient and often large predators. They possess well-developed brains and are the most intelligent invertebrates.

1. What is the basic body plan of a mollusk? Where is the mantle located? Why is it important in the mollusks? What occurs in the mantle cavity of aquatic mollusks?
2. What is a radula? Do all classes of mollusks possess this structure? How is it used in different types of mollusks?
3. How does the mollusk excretory structure work? Why is it better than the flame cells of acoelomates?
4. What is a trochophore? What is a veliger?
5. Do bivalves generally disperse as larvae or adults? Explain.



- Mollusks

45.2 Annelids were the first segmented animals.

- Segmentation is a characteristic seen only in coelomate animals at the annelid evolutionary level and above. Segmentation, or the repetition of body regions, greatly facilitates the development of specialized regions of the body.
- Annelids are worms with bodies composed of numerous similar segments, each with its own circulatory, excretory, and neural elements, and array of setae. There are three classes of annelids, the largely marine Polychaeta, the largely terrestrial Oligochaeta, and the largely freshwater Hirudinea.

6. What evolutionary advantages does segmentation confer upon an organism?
7. What are annelid setae? What function do they serve? What are parapodia? What class of annelids possess them?
8. How do earthworms obtain their nutrients? What sensory structures do earthworms possess? How do these animals reproduce?



- Annelids



- Student Research: Growth in Earthworms

45.3 Lophophorates appear to be a transitional group.

- The lophophorates consist of three phyla of marine animals—Phoronida, Ectoprocta, and Brachiopoda—characterized by a circular or U-shaped ridge, the lophophore, around the mouth.
- Some lophophorates have characteristics like protostomes, others like deuterostomes. All are characterized by a lophophore and are thought to share a common ancestor.

9. What prominent feature characterizes the lophophorate animals? What are the functions of this feature?

46

Arthropods

Concept Outline

46.1 The evolution of jointed appendages has made arthropods very successful.

Jointed Appendages and an Exoskeleton. Arthropods probably evolved from annelids, and with their jointed appendages and an exoskeleton, have successfully invaded practically every habitat on earth.

Classification of Arthropods. Arthropods have been traditionally divided into three groups based on morphological characters. However, recent research suggests a restructuring of arthropod classification is needed.

General Characteristics of Arthropods. Arthropods have segmented bodies, a chitinous exoskeleton, and often have compound eyes. They have open circulatory systems. In some groups, a series of tubes carry oxygen to the organs, and unique tubules eliminate waste.

46.2 The chelicerates all have fangs or pincers.

Class Arachnida: The Arachnids. Spiders and scorpions are predators, while most mites are herbivores.

Class Merostomata: Horseshoe Crabs. Among the most ancient of living animals, horseshoe crabs are thought to have evolved from trilobites.

Class Pycnogonida: The Sea Spiders. The spiders that are common in marine habitats differ greatly from terrestrial spiders.

46.3 Crustaceans have branched appendages.

Crustaceans. Crustaceans are unique among living arthropods because virtually all of their appendages are branched.

46.4 Insects are the most diverse of all animal groups.

Classes Chilopoda and Diplopoda: The Centipedes and Millipedes. Centipedes and millipedes are highly segmented, with legs on each segment.

Class Insecta: The Insects. Insects are the largest group of organisms on earth. They are the only invertebrate animals that have wings and can fly.

Insect Life Histories. Insects undergo simple or complete metamorphosis.



FIGURE 46.1

An arthropod. One of the major arthropod groups is represented here by *Polistes*, the common paper wasp (class Insecta).

The evolution of segmentation among annelids marked the first major innovation in body structure among coelomates. An even more profound innovation was the development of jointed appendages in arthropods, a phylum that almost certainly evolved from an annelid ancestor. Arthropod bodies are segmented like those of annelids, but the individual segments often exist only during early development and fuse into functional groups as adults. In arthropods like the wasp above (figure 46.1), jointed appendages include legs, antennae, and a complex array of mouthparts. The functional flexibility provided by such a broad array of appendages has made arthropods the most successful of animal groups.

46.1 The evolution of jointed appendages has made arthropods very successful.

Jointed Appendages and an Exoskeleton

With the evolution of the first annelids, many of the major innovations of animal structure had already appeared: the division of tissues into three primary types (endoderm, mesoderm, and ectoderm), bilateral symmetry, a coelom, and segmentation. With arthropods, two more innovations arose—the development of *jointed appendages* and an *exoskeleton*. Jointed appendages and an exoskeleton have allowed arthropods (phylum Arthropoda) to become the most diverse phylum.

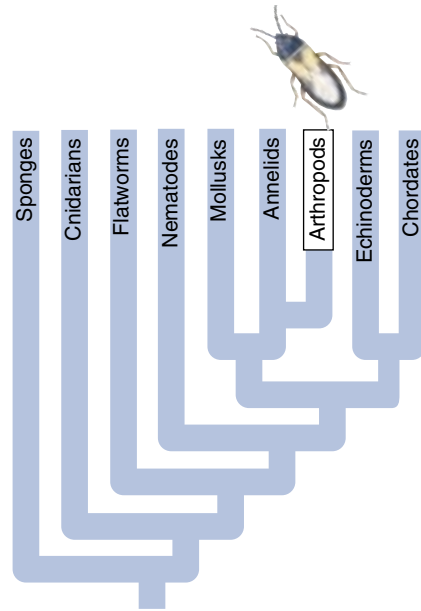
Jointed Appendages

The name “arthropod” comes from two Greek words, *arthros*, “jointed,” and *podes*, “feet.” All arthropods have jointed appendages. The numbers of these appendages are reduced in the more advanced members of the phylum. Individual appendages may be modified into antennae, mouthparts of various kinds, or legs. Some appendages, such as the wings of certain insects, are not homologous to the other appendages; insect wings evolved separately.

To gain some idea of the importance of jointed appendages, imagine yourself without them—no hips, knees, ankles, shoulders, elbows, wrists, or knuckles. Without jointed appendages, you could not walk or grasp any object. Arthropods use jointed appendages such as legs for walking, antennae to sense their environment, and mouthparts for feeding.

Exoskeleton

The arthropod body plan has a second major innovation: a rigid external skeleton, or **exoskeleton**, made of chitin and protein. In any animal, the skeleton functions to provide places for muscle attachment. In arthropods, the muscles attach to the interior surface of their hard exoskeleton, which also protects the animal from predators and impedes water loss. Chitin is chemically similar to cellulose, the dominant structural



component of plants, and shares similar properties of toughness and flexibility. Together, the chitin and protein provide an external covering that is both very strong and capable of flexing in response to the contraction of muscles attached to it. In most crustaceans, the exoskeleton is made even tougher, although less flexible, with deposits of calcium salts. However, there is a limitation. The exoskeleton must be much thicker to bear the pull of the muscles in large insects than in small ones. That is why you don't see beetles as big as birds, or crabs the size of a cow—the exoskeleton would be so thick the animal couldn't move its great weight. Because this size limitation is inherent in the body design of arthropods, there are no large arthropods—few are larger than your thumb.

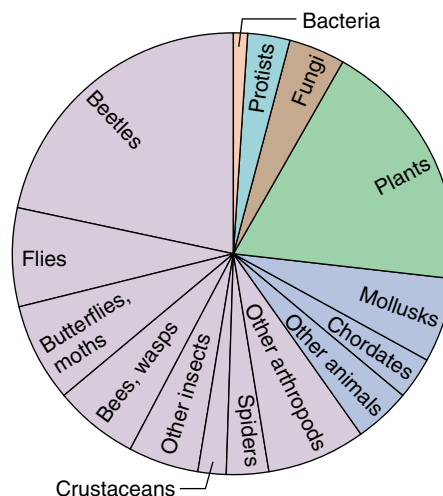


FIGURE 46.2

Arthropods are a successful group.

About two-thirds of all named species are arthropods. About 80% of all arthropods are insects, and about half of the named species of insects are beetles.

The Arthropods

Arthropods, especially the largest class—insects—are by far the most successful of all animals. Well over 1,000,000 species—about two-thirds of all the named species on earth—are members of this phylum (figure 46.2). One scientist recently estimated, based on the number and diversity of insects in tropical forests, that there might be as many as 30 million species in this one class alone. About 200 million insects are alive at any one time for each human! Insects and other arthropods (figure 46.3) abound in every habitat on the planet, but they especially dominate the land, along with flowering plants and vertebrates.

The majority of arthropod species consist of small animals, mostly about a millimeter in length. Members of

the phylum range in adult size from about 80 micrometers long (some parasitic mites) to 3.6 meters across (a gigantic crab found in the sea off Japan).

Arthropods, especially insects, are of enormous economic importance and affect all aspects of human life. They compete with humans for food of every kind, play a key role in the pollination of certain crops, and cause bil-

PHYLUM ARTHROPODA: Jointed appendages and exoskeleton

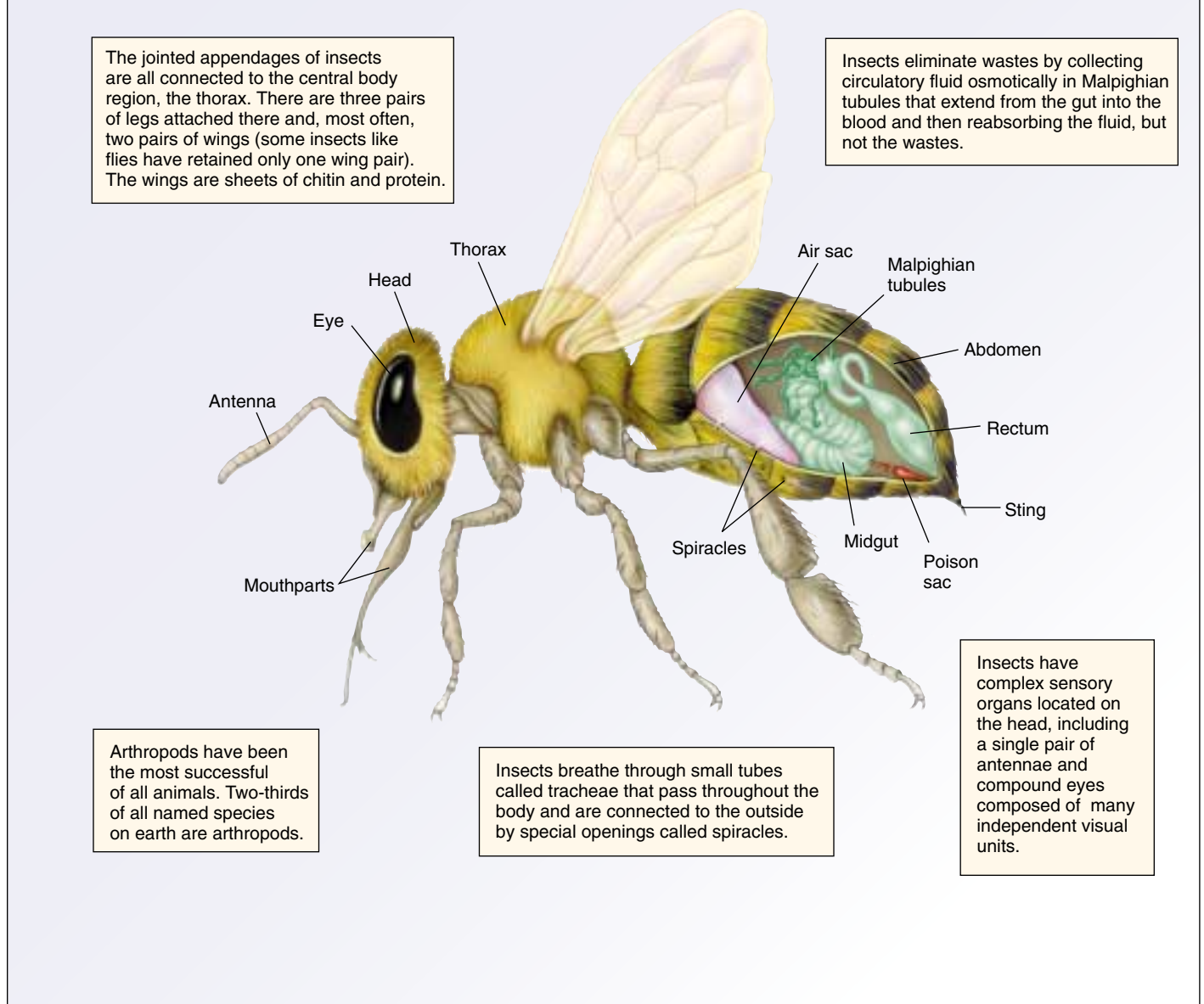


FIGURE 46.3

The evolution of jointed appendages and an exoskeleton. Insects and other arthropods (phylum Arthropoda) have a coelom, segmented bodies, and jointed appendages. The three body regions of an insect (head, thorax, and abdomen) are each actually composed of a number of segments that fuse during development. All arthropods have a strong exoskeleton made of chitin. One class, the insects, has evolved wings that permit them to fly rapidly through the air.

lions of dollars of damage to crops, before and after harvest. They are by far the most important herbivores in all terrestrial ecosystems and are a valuable food source as well. Virtually every kind of plant is eaten by one or more species of insect. Diseases spread by insects cause enormous financial damage each year and strike every

kind of domesticated animal and plant, as well as human beings.

Arthropods are segmented protostomes with jointed appendages. Arthropods are the most successful of all animal groups.

Classification of the Arthropods

Arthropods are among the oldest of animals, first appearing in the Precambrian over 600 million years ago. Ranging in size from enormous to microscopic, all arthropods share a common heritage of segmented bodies and jointed appendages, a powerful combination for generating novel evolutionary forms. Arthropods are the most diverse of all the animal phyla, with more species than all other animal phyla combined, most of them insects.

Origin of the Arthropods

Taxonomists have long held that there is a close relationship between the annelids and the arthropods, the two great segmented phyla. Velvet worms (phylum Onychophora), known from the Burgess Shale (where upside-down fossils were called *Hallucigenia*) and many other early Cambrian deposits, have many features in common with both annelids and arthropods. Some recent molecular studies have supported the close relationship between annelids and arthropods, others have not.

Traditional Classification

Members of the phylum Arthropoda have been traditionally divided into three subphyla, based largely on morphological characters.

1. Trilobites (the extinct trilobites). Trilobites, common in the seas 250 million years ago, were the first animals whose eyes were capable of a high degree of resolution.
2. Chelicerates (spiders, horseshoe crabs, sea spiders). These arthropods lack jaws. The foremost appendages of their bodies are mouthparts called **chelicerae** (figure 46.4a) that function in feeding, usually pincers or fangs.
3. Mandibulates (crustaceans, insects, centipedes, millipedes). These arthropods have biting jaws, called **mandibles** (figure 46.4b). In mandibulates, the most anterior appendages are one or more pairs of sensory antennae, and the next appendages are the mandibles. Among the mandibulates, insects have traditionally been set apart from the crustaceans, grouped instead with the myriapods (centipedes and millipedes) in a taxon called Tracheata. This phylogeny, still widely employed, dates back to benchmark work by the great comparative biologist Robert Snodgrass in the 1930s. He pointed out that insects, centipedes, and millipedes are united by several seemingly powerful attributes:

A tracheal respiratory system. Trachea are small, branched air ducts that transmit oxygen from openings in the exoskeleton to every cell of the body.

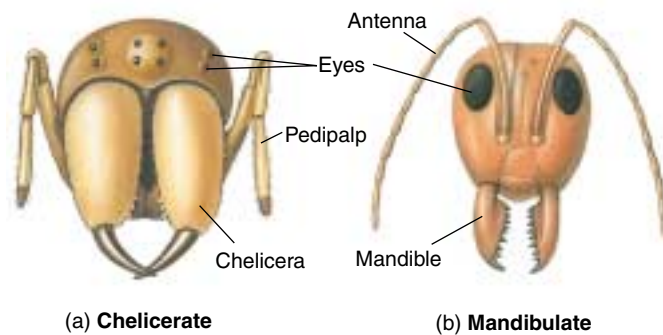


FIGURE 46.4

Chelicerates and mandibulates. In the chelicerates, such as a spider (a), the chelicerae are the foremost appendages of the body. In contrast, the foremost appendages in the mandibulates, such as an ant (b), are the antennae, followed by the mandibles.

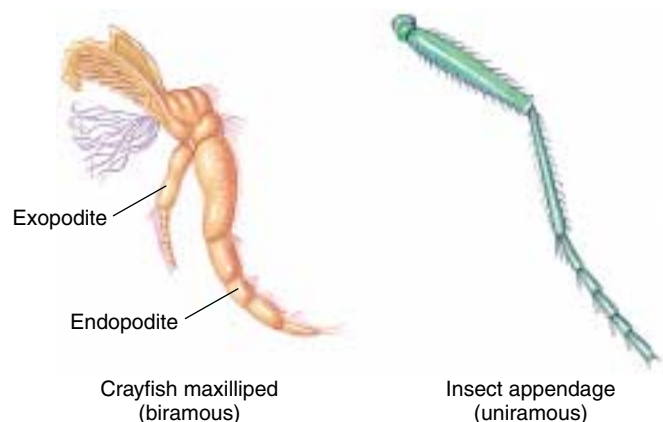


FIGURE 46.5

Mandibulate appendages. A biramous leg in a crustacean (crayfish) and a uniramous leg in an insect.

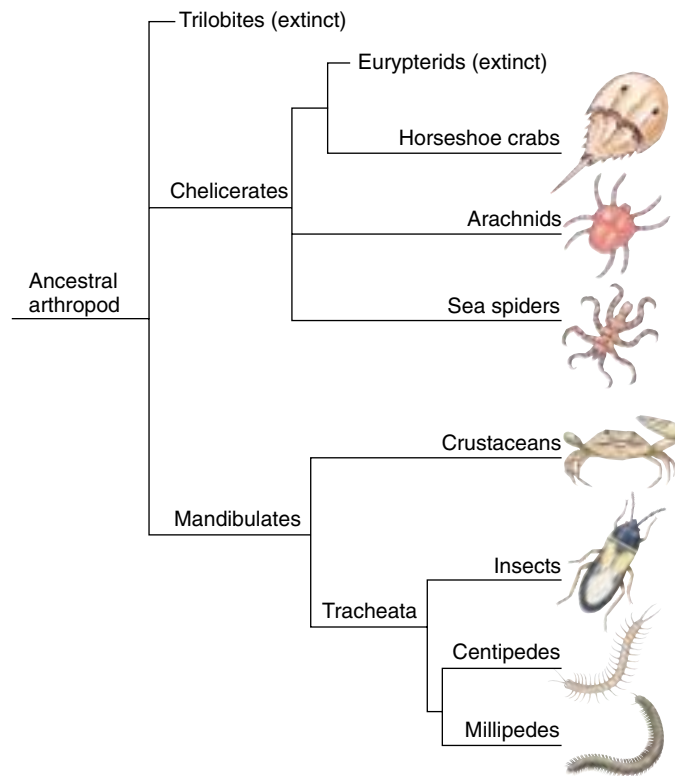
Use of Malpighian tubules for excretion. Malpighian tubules are slender projections from the digestive tract which collect and filter body fluids, emptying wastes into the hindgut.

Uniramous (single-branched) legs. All crustacean appendages are basically biramous, or “two-branched” (figure 46.5), although some of these appendages have become single-branched by reduction in the course of their evolution. Insects, by contrast, have uniramous, or single-branched, mandibles and other appendages.

Doubts about the Traditional Approach

Recent research is casting doubt on the wisdom of this taxonomic decision. The problem is that the key morphological traits used to define the Tracheata are not as powerful taxonomically as had been assumed. Taxonomists have traditionally assumed a character like branching ap-

Traditional Phylogeny



Revised Phylogeny

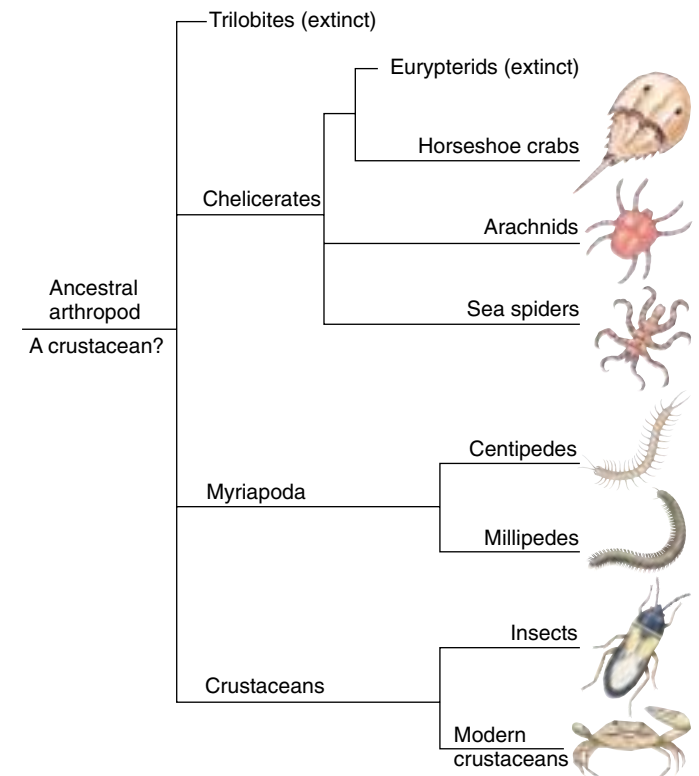


FIGURE 46.6

A proposed revision of arthropod phylogeny. Accumulating evidence supports the hypothesis that insects and modern crustaceans are sister groups, having evolved from the same ancient crustacean ancestor in the Precambrian. This implies that insects may be viewed as “flying crustaceans,” and that the traditional Tracheata taxon, which places centipedes, millipedes, and insects together, is in fact a polyphyletic group.

pendages to be a fundamental one, conserved over the course of evolution, and thus suitable for making taxonomic distinctions.

However, modern molecular biology now tells us that this is not a valid assumption. The branching of arthropod legs, for example, turns out to be controlled by a single gene. The pattern of appendages among arthropods is orchestrated by a family of genes called homeotic (*Hox*) genes, described in detail in chapter 17. A single one of these *Hox* genes, called *Distal-less*, has recently been shown to initiate development of unbranched limbs in insects and branched limbs in crustaceans. The same *Distal-less* gene is found in many animal phyla, including vertebrates.

A Revolutionary New Phylogeny

In recent years a mass of accumulating morphological and molecular data has led many taxonomists to suggest new arthropod taxonomies. The most revolutionary of these, championed by Richard Brusca of Columbia University, considers crustaceans to be the basic arthropod group, and insects a close sister group (figure 46.6).

Morphological Evidence. The most recent morphological study of arthropod phylogeny, reported in 1998, was based on 100 conserved anatomical features of the central nervous system. It concluded insects were more closely related to crustaceans than to any other arthropod group. They share a unique pattern of segmental neurons, and many other features.

Molecular Evidence. Molecular phylogenies based on 18S rRNA sequences, the 18S rDNA gene, the elongation factor EF-1a, and the RNA polymerase II gene, all place insects as a sister group to crustaceans, not myriapods, and arising from within the crustaceans. In conflict with 150 years of morphology-based thinking, these conclusions are certain to engender lively discussion.

Arthropods have traditionally been classified into arachnids and other chelicerates that lack jaws and have fang mouthparts, and mandibulates (crustaceans and tracheates) with biting jaws. A revised arthropod taxonomy considers Tracheata to be the products convergent evolution, with insects and crustaceans sister groups.

General Characteristics of Arthropods

Arthropod bodies are segmented like annelids, a phylum to which at least some arthropods are clearly related. Members of some classes of arthropods have many body segments. In others, the segments have become fused together into functional groups, or **tagmata** (singular, **tagma**), such as the head and thorax of an insect (figure 46.7). This fusing process, known as **tagmatization**, is of central importance in the evolution of arthropods. In most arthropods, the original segments can be distinguished during larval development. All arthropods have a distinct head, sometimes fused with the thorax to form a tagma called the **cephalothorax**.

Exoskeleton

The bodies of all arthropods are covered by an exoskeleton, or cuticle, that contains chitin. This tough outer covering, against which the muscles work, is secreted by the epidermis and fused with it. The exoskeleton remains fairly flexible at specific points, allowing the exoskeleton to bend and appendages to move. The exoskeleton protects arthropods from water loss and helps to protect them from predators, parasites, and injury.

Molting. Arthropods periodically undergo **ecdysis**, or molting, the shedding of the outer cuticular layer. When they outgrow their exoskeleton, they form a new one underneath. This process is controlled by hormones. When the new exoskeleton is complete, it becomes separated from the old one by fluid. This fluid dissolves the chitin and protein and, if it is present, calcium carbonate, from the old exoskeleton. The fluid increases in volume until, finally, the original exoskeleton cracks open, usually along the

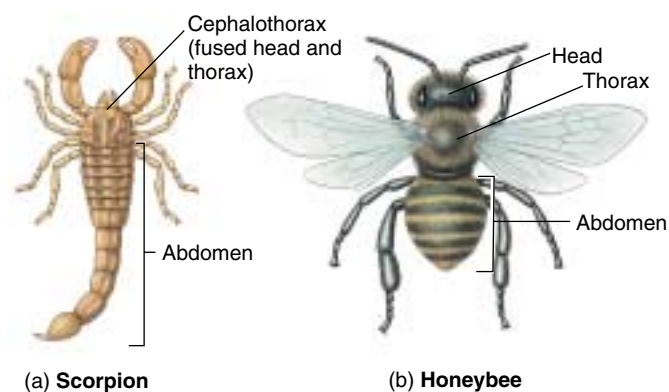


FIGURE 46.7
Arthropod evolution from many to few body segments. The (a) scorpion and the (b) honeybee are arthropods with different numbers of body segments.

back, and is shed. The arthropod emerges, clothed in a new, pale, and still somewhat soft exoskeleton. The arthropod then “puffs itself up,” ultimately expanding to full size. The blood circulation to all parts of the body aids them in this expansion, and many insects and spiders take in air to assist them. The expanded exoskeleton subsequently hardens. While the exoskeleton is soft, the animal is especially vulnerable. At this stage, arthropods often hide under stones, leaves, or branches.

Compound Eye

Another important structure in many arthropods is the **compound eye** (figure 46.8a). Compound eyes are composed of many independent visual units, often thousands of them, called **ommatidia**. Each ommatidium is covered with a lens and linked to a complex of eight reticular cells

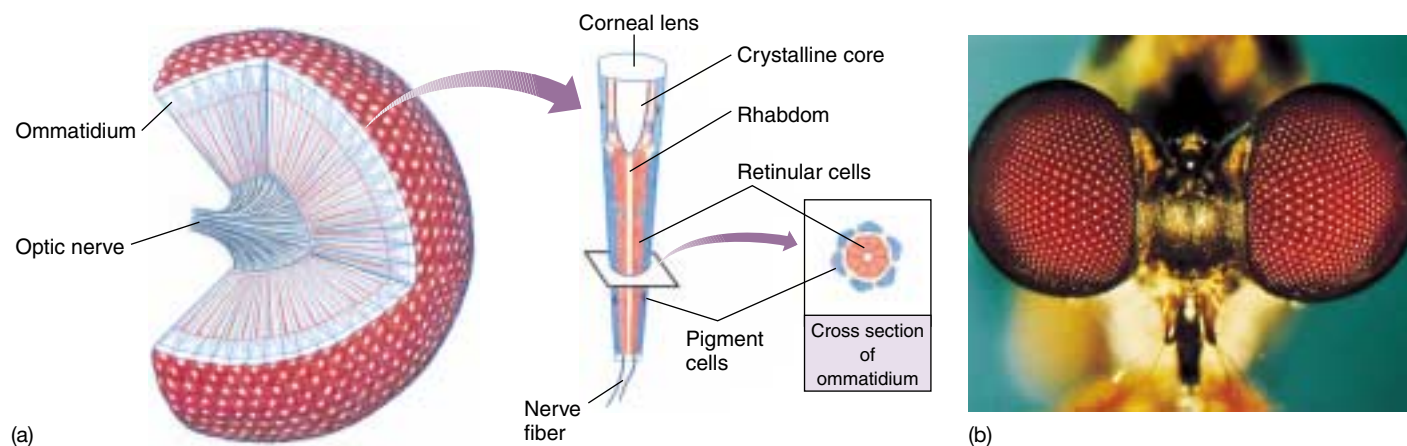


FIGURE 46.8
The compound eye. (a) The compound eyes found in insects are complex structures. (b) Three ocelli are visible between the compound eyes of the robberfly (order Diptera).

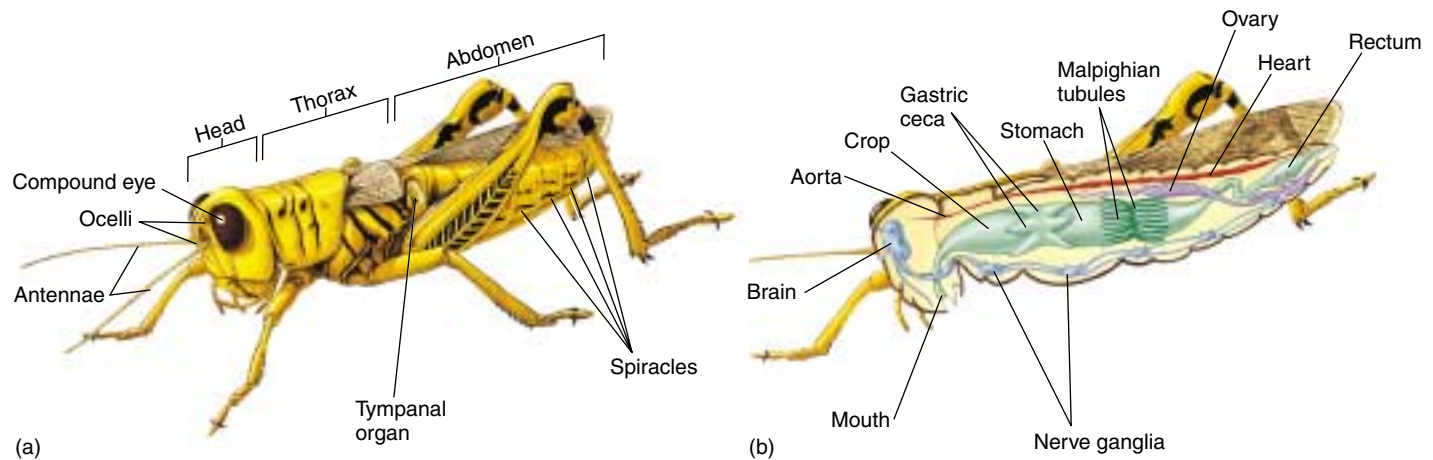


FIGURE 46.9

A grasshopper (order Orthoptera). This grasshopper illustrates the major structural features of the insects, the most numerous group of arthropods. (a) External anatomy. (b) Internal anatomy.

and a light-sensitive central core, or **rhabdom**. Compound eyes among insects are of two main types: **apposition eyes** and **superposition eyes**. Apposition eyes are found in bees and butterflies and other insects that are active during the day. Each ommatidium acts in isolation, surrounded by a curtain of pigment cells that blocks the passage of light from one to another. Superposition eyes, such as those found in moths and other insects that are active at night, are designed to maximize the amount of light that enters each ommatidium. At night, the pigment in the pigment cells is concentrated at the top of the cells so that the low levels of light can be received by many different ommatidia. During daylight, the pigment in the pigment cells is evenly dispersed throughout the cells, allowing the eye to function much like an apposition eye. The pigment in the pigment cells gives the arthropod eye its color, but it is not the critical pigment needed for vision. The visual pigment is located in an area called the rhabdom found in the center of the ommatidium. The individual images from each ommatidium are combined in the arthropod's brain to form its image of the external world.

Simple eyes, or ocelli, with single lenses are found in the other arthropod groups and sometimes occur together with compound eyes, as is often the case in insects (figure 46.8b). Ocelli function in distinguishing light from darkness. The ocelli of some flying insects, namely locusts and dragonflies, function as horizon detectors and help the insect visually stabilize its course in flight.

Circulatory System

In the course of arthropod evolution, the coelom has become greatly reduced, consisting only of cavities that house the reproductive organs and some glands. Arthropods completely lack cilia, both on the external surfaces of

the body and on the internal organs. Like annelids, arthropods have a tubular gut that extends from the mouth to the anus. In the next paragraphs we will discuss the circulatory, respiratory, excretory, and nervous systems of the arthropods (figure 46.9).

The circulatory system of arthropods is open; their blood flows through cavities between the internal organs and not through closed vessels. The principal component of an insect's circulatory system is a longitudinal vessel called the heart. This vessel runs near the dorsal surface of the thorax and abdomen. When it contracts, blood flows into the head region of the insect.

When an insect's heart relaxes, blood returns to it through a series of valves. These valves are located in the posterior region of the heart and allow the blood to flow inward only. Thus, blood from the head and other anterior portions of the insect gradually flows through the spaces between the tissues toward the posterior end and then back through the one-way valves into the heart. Blood flows most rapidly when the insect is running, flying, or otherwise active. At such times, the blood efficiently delivers nutrients to the tissues and removes wastes from them.

Nervous System

The central feature of the arthropod nervous system is a double chain of segmented ganglia running along the animal's ventral surface. At the anterior end of the animal are three fused pairs of dorsal ganglia, which constitute the brain. However, much of the control of an arthropod's activities is relegated to ventral ganglia. Therefore, the animal can carry out many functions, including eating, movement, and copulation, even if the brain has been removed. The brain of arthropods seems to be a control point, or inhibitor, for various actions, rather than a stimulator, as it is in vertebrates.

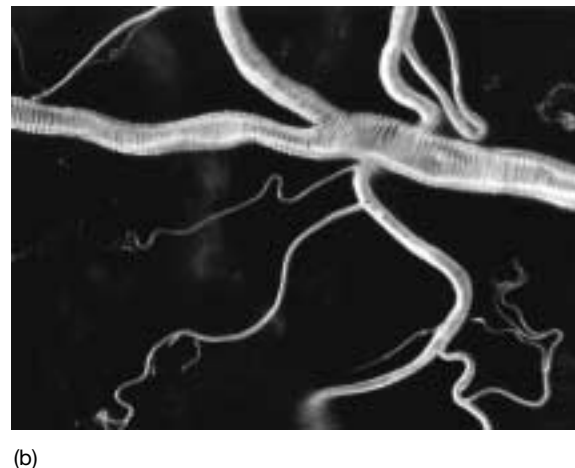
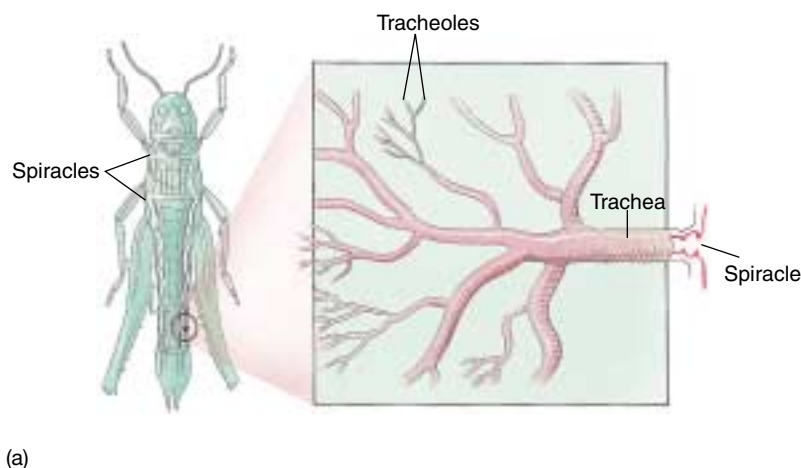


FIGURE 46.10

Tracheae and tracheoles. Tracheae and tracheoles are connected to the exterior by specialized openings called spiracles and carry oxygen to all parts of a terrestrial insect's body. (a) The tracheal system of a grasshopper. (b) A portion of the tracheal system of a cockroach.

Respiratory System

Insects and other members of subphylum Uniramia, which are fundamentally terrestrial, depend on their respiratory rather than their circulatory system to carry oxygen to their tissues. In vertebrates, blood moves within a closed circulatory system to all parts of the body, carrying the oxygen with it. This is a much more efficient arrangement than exists in arthropods, in which all parts of the body need to be near a respiratory passage to obtain oxygen. As a result, the size of the arthropod body is much more limited than that of the vertebrates. Along with the brittleness of their chitin exoskeletons, this feature of arthropod design places severe limitations on size.

Unlike most animals, the arthropods have no single major respiratory organ. The respiratory system of most terrestrial arthropods consists of small, branched, cuticle-lined air ducts called **tracheae** (figure 46.10). These tracheae, which ultimately branch into very small **tracheoles**, are a series of tubes that transmit oxygen throughout the body. Tracheoles are in direct contact with individual cells, and oxygen diffuses directly across the cell membranes. Air passes into the tracheae by way of specialized openings in the exoskeleton called **spiracles**, which, in most insects, can be opened and closed by valves. The ability to prevent water loss by closing the spiracles was a key adaptation that facilitated the invasion of the land by arthropods. In many insects, especially larger ones, muscle contraction helps to increase the flow of gases in and out of the tracheae. In other terrestrial arthropods, the flow of gases is essentially a passive process.

Many spiders and some other chelicerates have a unique respiratory system that involves **book lungs**, a series of leaflike plates within a chamber. Air is drawn in and expelled out of this chamber by muscular contraction. Book lungs may exist alongside tracheae, or they may function

instead of tracheae. One small class of marine chelicerates, the horseshoe crabs, have book gills, which are analogous to book lungs but function in water. Tracheae, book lungs, and book gills are all structures found only in arthropods and in the phylum Onychophora, which have tracheae. Crustaceans lack such structures and have gills.

Excretory System

Though there are various kinds of excretory systems in different groups of arthropods, we will focus here on the unique excretory system consisting of **Malpighian tubules** that evolved in terrestrial uniramians. Malpighian tubules are slender projections from the digestive tract that are attached at the junction of the midgut and hindgut (see figure 46.3). Fluid passes through the walls of the Malpighian tubules to and from the blood in which the tubules are bathed. As this fluid passes through the tubules toward the hindgut, nitrogenous wastes are precipitated as concentrated uric acid or guanine. These substances are then emptied into the hindgut and eliminated. Most of the water and salts in the fluid are reabsorbed by the hindgut and rectum and returned to the arthropod's body. Malpighian tubules are an efficient mechanism for water conservation and were another key adaptation facilitating invasion of the land by arthropods.

All arthropods have a rigid chitin and protein exoskeleton that provides places for muscle attachment, protects the animal from predators and injury, and, most important, impedes water loss. Many arthropods have compound eyes. Arthropods have an open circulatory system. Many arthropods eliminate metabolic wastes by a unique system of Malpighian tubules. Most terrestrial insects have a network of tubes called tracheae that transmit oxygen from the outside to the organs.

46.2 The chelicerates all have fangs or pincers.

Class Arachnida: The Arachnids

Chelicerates (subphylum Chelicerata) are a distinct evolutionary line of arthropods in which the most anterior appendages have been modified into chelicerae, which often function as fangs or pincers. By far the largest of the three classes of chelicerates is the largely terrestrial Arachnida, with some 57,000 named species; it includes spiders, ticks, mites, scorpions, and daddy longlegs. Arachnids have a pair of chelicerae, a pair of pedipalps, and four pairs of walking legs. The chelicerae are the foremost appendages; they consist of a stout basal portion and a movable fang often connected to a poison gland.

The next pair of appendages, **pedipalps**, resemble legs but have one less segment and are not used for locomotion. In male spiders, they are specialized copulatory organs. In scorpions, the pedipalps are large pincers.

Most arachnids are carnivorous. The main exception is mites, which are largely herbivorous. Most arachnids can ingest only preliquified food, which they often digest externally by secreting enzymes into their prey. They can then suck up the digested material with their muscular, pumping pharynx. Arachnids are primarily, but not exclusively, terrestrial. Some 4000 known species of mites and one species of spider live in fresh water, and a few mites live in the sea. Arachnids breathe by means of tracheae, book lungs, or both.

Order Opiliones: The Daddy Longlegs

A familiar group of arachnids consists of the daddy longlegs, or harvestmen (order Opiliones). Members of this order are easily recognized by their oval, compact bodies and extremely long, slender legs (figure 46.11). They respire by means of a primary pair of tracheae and are unusual among the arachnids in that they engage in direct copulation. The males have a penis, and the females an **ovipositor**, or egg-laying organ which deposits their eggs in cracks and crevices. Most daddy longlegs are predators of insects and other arachnids, but some live on plant juices and many scavenge dead animal matter. The order includes about 5000 species.

Order Scorpiones: The Scorpions

Scorpions (order Scorpiones) are arachnids whose pedipalps are modified into pincers. Scorpions use these pincers to handle and tear apart their food (figure 46.12). The venomous stings of scorpions are used mainly to stun their prey and less commonly in self-defense. The stinging apparatus is located in the terminal segment of the abdomen. A scorpion holds its abdomen folded forward over its body



FIGURE 46.11
A harvestman, or daddy longlegs.



FIGURE 46.12
The scorpion, *Uroctonus mordax*. This photograph shows the characteristic pincers and segmented abdomen, ending in a stinging apparatus, raised over the animal's back. The white mass is comprised of the scorpion's young.

when it is moving about. The elongated, jointed abdomens of scorpions are distinctive; in most chelicerates, the abdominal segments are more or less fused together and appear as a single unit.

Scorpions are probably the most ancient group of terrestrial arthropods; they are known from the Silurian Period, some 425 million years ago. Adults of this order of arachnids range in size from 1 to 18 centimeters. There are some 1200 species of scorpions, all terrestrial, which occur throughout the world. They are most common in tropical, subtropical, and desert regions. The young are born alive, with 1 to 95 in a given brood.

Order Araneae: The Spiders

There are about 35,000 named species of spiders (order Araneae). These animals play a major role in virtually all terrestrial ecosystems. They are particularly important as predators of insects and other small animals. Spiders hunt their prey or catch it in silk webs of remarkable diversity. The silk is formed from a fluid protein that is forced out of spinnerets on the posterior portion of the spider's abdomen. The webs and habits of spiders are often distinctive. Some spiders can spin gossamer floats that allow them to drift away in the breeze to a new site.

Many kinds of spiders, like the familiar wolf spiders and tarantulas, do not spin webs but instead hunt their prey actively. Others, called trap-door spiders, construct silk-lined burrows with lids, seizing their prey as it passes by. One species of spider, *Argyroneta aquatica*, lives in fresh water, spending most of its time below the surface. Its body is surrounded by a bubble of air, while its legs, which are used both for underwater walking and for swimming, are not. Several other kinds of spiders walk about freely on the surface of water.

Spiders have poison glands leading through their chelicerae, which are pointed and used to bite and paralyze prey. Some members of this order, such as the black widow and brown recluse (figure 46.13), have bites that are poisonous to humans and other large mammals.

Order Acari: Mites and Ticks

The order Acari, the mites and ticks, is the largest in terms of number of species and the most diverse of the arachnids. Although only about 30,000 species of mites and ticks have been named, scientists that study the group estimate that there may be a million or more members of this order in existence.

Most mites are small, less than 1 millimeter long, but adults of different species range from 100 nanometers to 2 centimeters. In most mites, the cephalothorax and abdomen are fused into an unsegmented ovoid body. Respiration occurs either by means of tracheae or directly through the exoskeleton. Many mites pass through several distinct stages during their life cycle. In most, an inactive eight-legged prelarva gives rise to an active six-legged larva, which in turn produces a succession of three eight-legged stages and, finally, the adult males and females.

Mites and ticks are diverse in structure and habitat. They are found in virtually every terrestrial, freshwater, and marine habitat known and feed on fungi, plants, and animals. They act as predators and as internal and external parasites of both invertebrates and vertebrates.



(a)



(b)

FIGURE 46.13

Two common poisonous spiders. (a) The black widow spider, *Latrodectus mactans*. (b) The brown recluse spider, *Loxosceles reclusa*. Both species are common throughout temperate and subtropical North America, but bites are rare in humans.



FIGURE 46.14

Ticks (order Acari). Ticks (the large one is engorged) on the hide of a tapir in Peru. Many ticks spread diseases in humans and other vertebrates.

Many mites produce irritating bites and diseases in humans. Mites live in the hair follicles and wax glands of your forehead and nose, but usually cause no symptoms.

Ticks are blood-feeding **ectoparasites**, parasites that occur on the surface of their host (figure 46.14). They are larger than most other mites and cause discomfort by sucking the blood of humans and other animals. Ticks can carry many diseases, including some caused by viruses, bacteria, and protozoa. The spotted fevers (Rocky Mountain spotted fever is a familiar example) are caused by bacteria carried by ticks. Lyme disease is apparently caused by spirochaetes transmitted by ticks. Red-water fever, or Texas fever, is an important tick-borne protozoan disease of cattle, horses, sheep, and dogs.

Scorpions, spiders, and mites are all arachnids, the largest class of chelicerates.



FIGURE 46.15
Limulus. Horseshoe crabs, emerging from the sea to mate at the edge of Delaware Bay, New Jersey, in early May.

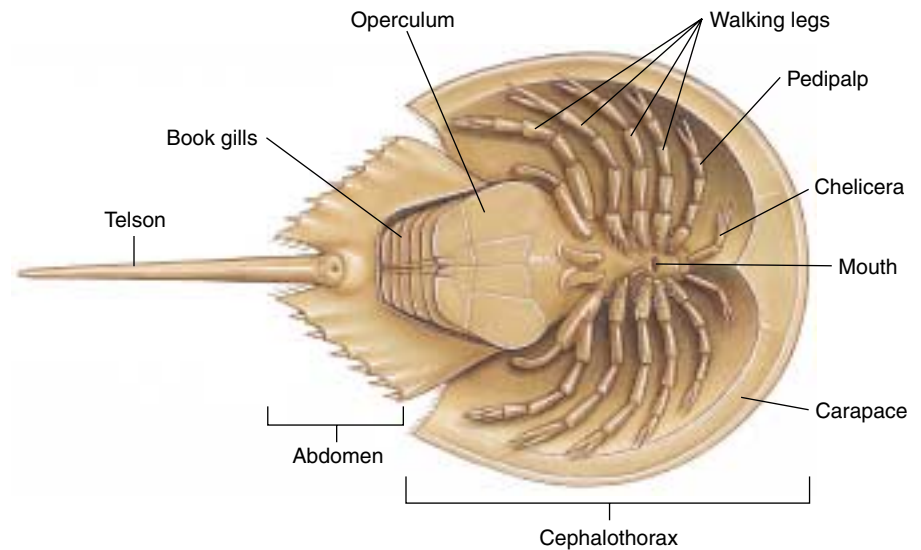


FIGURE 46.16
Diagram of a horseshoe crab, *Limulus*, from below. This diagram illustrates the principal features of this archaic animal.

Class Merostomata: Horseshoe Crabs

A second class of chelicerates is the horseshoe crabs (class Merostomata). There are three genera of horseshoe crabs. One, *Limulus* (figure 46.15), is common along the East Coast of North America. The other two genera live in the Asian tropics. Horseshoe crabs are an ancient group, with fossils virtually identical to *Limulus* dating back 220 million years to the Triassic Period. Other members of the class, the now-extinct eurypterans, are known from 400 million years ago. Horseshoe crabs may have been derived from trilobites, a relationship suggested by the appearance of their larvae. Individuals of *Limulus* grow up to 60 centimeters long. They mature in 9 to 12 years and have a life span of 14 to 19 years. *Limulus* individuals live in deep water, but they migrate to shallow coastal waters every spring, emerging from the sea to mate on moonlit nights when the tide is high.

Horseshoe crabs feed at night, primarily on mollusks and annelids. They swim on their backs by moving their abdominal plates. They can also walk on their four pairs of legs, protected along with chelicerae and pedipalps by their shell (figure 46.16).

Horseshoe crabs are a very ancient group.

Class Pycnogonida: The Sea Spiders

The third class of chelicerates is the sea spiders (class Pycnogonida). Sea spiders are common in coastal waters, with more than 1000 species in the class. These animals are not often observed because many are small, only about 1 to 3 centimeters long, and rather inconspicuous. They are found in oceans throughout the world but are most abundant in the far north and far south. Adult sea spiders are mostly external parasites or predators of other animals like sea anemones (figure 46.17).

Sea spiders have a sucking proboscis in a mouth located at its end. Their abdomen is much reduced, and their body appears to consist almost entirely of the cephalothorax, with no well-defined head. Sea spiders usually have four, or less commonly five or six, pairs of legs. Male sea spiders carry the eggs on their legs until they hatch, thus providing a measure of parental care. Sea spiders completely lack excretory and respiratory systems. They appear to carry out these functions by direct diffusion, with waste products flowing outward through the cells and oxygen flowing inward through them. Sea spiders are not closely related to either of the other two classes of chelicerates.



FIGURE 46.17
A marine pycnogonid. The sea spider, *Pycnogonum littorale* (yellow animal), crawling over a sea anemone.

Sea spiders are very common in marine habitats. They are not closely related to terrestrial spiders.

46.3 Crustaceans have branched appendages.

Crustaceans

The crustaceans (subphylum Crustacea) are a large group of primarily aquatic organisms, consisting of some 35,000 species of crabs, shrimps, lobsters, crayfish, barnacles, water fleas, pillbugs, and related groups (table 46.1). Most crustaceans have two pairs of antennae, three types of chewing appendages, and various numbers of pairs of legs. All crustacean appendages, with the possible exception of the first pair of antennae, are basically biramous. In some crustaceans, appendages appear to have only a single branch; in those cases, one of the branches has been lost during the course of evolutionary specialization. The **nauplius** larva stage through which all crustaceans pass (figure 46.18) provides evidence that all members of this diverse group are descended from a common ancestor. The nauplius hatches with three pairs of appendages and metamorphoses through several stages before reaching maturity. In many groups, this nauplius stage is passed in the egg, and development of the hatchling to the adult form is direct.

Crustaceans differ from insects but resemble centipedes and millipedes in that they have appendages on their abdomen as well as on their thorax. They are the only arthropods with two pairs of antennae. Their mandibles likely originated from a pair of limbs that took on a chewing function during the course of evolution, a process that apparently occurred independently in the common ancestor of the terrestrial mandibulates. Many crustaceans have compound eyes. In addition, they have delicate tactile hairs that project from the cuticle all over the body. Larger crustaceans have feathery gills near the bases of their legs. In smaller members of this class, gas exchange takes place directly through the thinner areas of the cuticle or the entire body. Most crustaceans have separate sexes. Many different kinds of specialized copulation occur among the crustaceans, and the members of some orders carry their eggs with them, either singly or in egg pouches, until they hatch.

Decapod Crustaceans

Large, primarily marine crustaceans such as shrimps, lobsters, and crabs, along with their freshwater relatives, the crayfish, are collectively called *decapod crustaceans* (figure 46.19). The term *decapod* means “ten-footed.” In these animals, the exoskeleton is usually reinforced with calcium carbonate. Most of their body segments are fused into a cephalothorax covered by a dorsal shield, or **carapace**, which arises from the head. The crushing pincers common in many decapod crustaceans are used in obtaining food, for example, by crushing mollusk shells.

Table 46.1 Traditional Classification of the Phylum Arthropoda

Subphylum	Characteristics	Members
Chelicerata	Mouthparts are chelicerae	The chelicerates: spiders, horseshoe crabs, mites
Crustacea	Mouthparts are mandibles; biramous appendages	The crustaceans: lobsters, crabs, shrimp, isopods, barnacles
Uniramia	Mouthparts are mandibles; uniramous appendages	Chilopods (centipedes), diplopods (millipedes), and insects

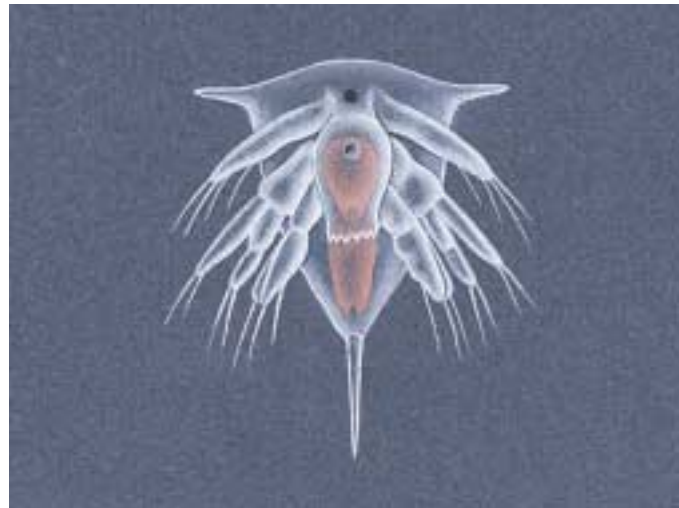


FIGURE 46.18
Although crustaceans are diverse, they have fundamentally similar larvae. The nauplius larva of a crustacean is an important unifying feature found in all members of this group.

In lobsters and crayfish, appendages called **swimmerets** occur in lines along the ventral surface of the abdomen and are used in reproduction and swimming. In addition, flattened appendages known as **uropods** form a kind of compound “paddle” at the end of the abdomen. These animals may also have a **telson**, or tail spine. By snapping its abdomen, the animal propels itself through the water rapidly and forcefully. Crabs differ from lobsters and crayfish in proportion; their carapace is much larger and broader and the abdomen is tucked under it.

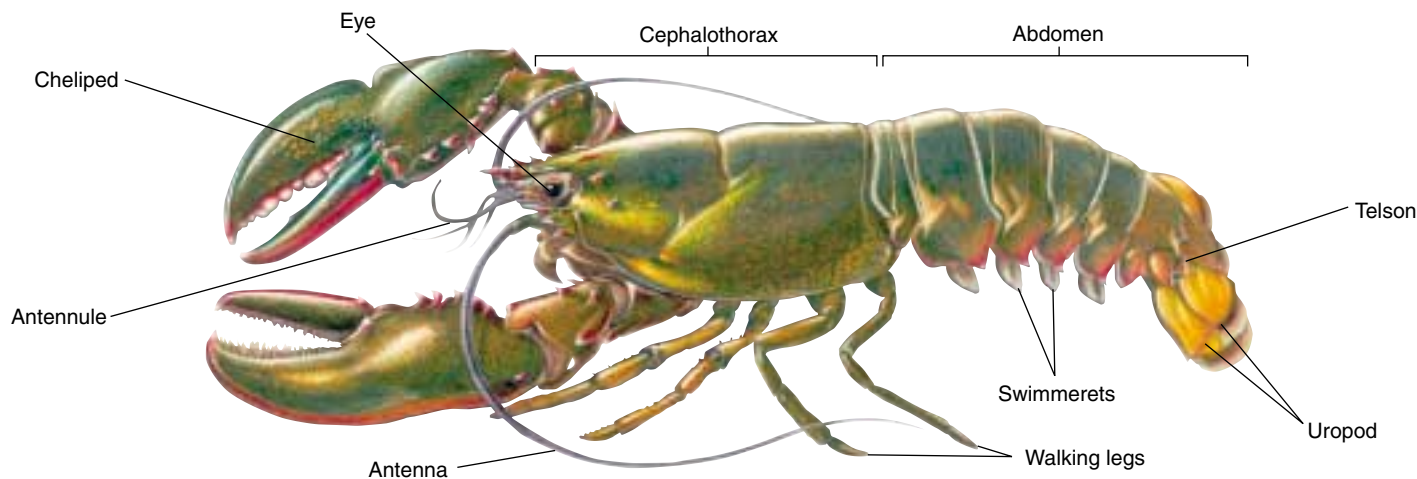


FIGURE 46.19
Decapod crustacean. A lobster, *Homarus americanus*. The principal features are labeled.

Terrestrial and Freshwater Crustaceans

Although most crustaceans are marine, many occur in fresh water and a few have become terrestrial. These include pillbugs and sowbugs, the terrestrial members of a large order of crustaceans known as the isopods (order Isopoda). About half of the estimated 4500 species of this order are terrestrial and live primarily in places that are moist, at least seasonally. Sand fleas or beach fleas (order Amphipoda) are other familiar crustaceans, many of which are semiterrestrial (intertidal) species.

Along with the larvae of larger species, minute crustaceans are abundant in the plankton. Especially significant are the tiny copepods (order Copepoda; figure 46.20), which are among the most abundant multicellular organisms on earth.

Sessile Crustaceans

Barnacles (order Cirripedia; figure 46.21) are a group of crustaceans that are sessile as adults. Barnacles have free-swimming larvae, which ultimately attach their heads to a piling, rock, or other submerged object and then stir food into their mouth with their feathery legs. Calcareous plates protect the barnacle's body, and these plates are usually attached directly and solidly to the substrate. Although most crustaceans have separate sexes, barnacles are hermaphroditic, but they generally cross-fertilize.

Crustaceans include marine, freshwater, and terrestrial forms. All possess a nauplius larval stage and branched appendages.

FIGURE 46.21
Gooseneck barnacles, *Lepas anatifera*, feeding.
These are stalked barnacles; many others lack a stalk.



FIGURE 46.20
Freshwater crustacean. A copepod with attached eggs, a member of an abundant group of marine and freshwater crustaceans (order Copepoda), most of which are a few millimeters long. Copepods are important components of the plankton.



46.4 Insects are the most diverse of all animal groups.

Millipedes, centipedes, and insects, three distinct classes, are uniramian mandibulates. They respire by means of tracheae and excrete their waste products through Malpighian tubules. These groups were certainly derived from annelids, probably ones similar to the oligochaetes, which they resemble in their embryology.

Classes Chilopoda and Diplopoda: The Centipedes and Millipedes

The centipedes (class Chilopoda) and millipedes (class Diplopoda) both have bodies that consist of a head region followed by numerous segments, all more or less similar and nearly all bearing paired appendages. Although the name *centipede* would imply an animal with a hundred legs and the name *millipede* one with a thousand, adult centipedes usually have 30 or more legs, adult millipedes 60 or more. Centipedes have one pair of legs on each body segment (figure 46.22), millipedes two (figure 46.23). Each segment of a millipede is a tagma that originated during the group's evolution when two ancestral segments fused. This explains why millipedes have twice as many legs per segment as centipedes.

In both centipedes and millipedes, fertilization is internal and takes place by direct transfer of sperm. The sexes are separate, and all species lay eggs. Young millipedes usually hatch with three pairs of legs; they experience a number of growth stages, adding segments and legs as they mature, but do not change in general appearance.

Centipedes, of which some 2500 species are known, are all carnivorous and feed mainly on insects. The appendages of the first trunk segment are modified into a pair of poison fangs. The poison is often quite toxic to human beings, and many centipede bites are extremely painful, sometimes even dangerous.

In contrast, most millipedes are herbivores, feeding mainly on decaying vegetation. A few millipedes are carnivorous. Many millipedes can roll their bodies into a flat coil or sphere because the dorsal area of each of their body segments is much longer than the ventral one. More than 10,000 species of millipedes have been named, but this is estimated to be no more than one-sixth of the actual number of species that exists. In each segment of their body, most millipedes have a pair of complex glands that produces a bad-smelling fluid. This fluid is ex-



FIGURE 46.22

A centipede. Centipedes, like this member of the genus *Scolopendra*, are active predators.

uded for defensive purposes through openings along the sides of the body. The chemistry of the secretions of different millipedes has become a subject of considerable interest because of the diversity of the compounds involved and their effectiveness in protecting millipedes from attack. Some produce cyanide gas from segments near their head end. Millipedes live primarily in damp, protected places, such as under leaf litter, in rotting logs, under bark or stones, or in the soil.

Centipedes are segmented hunters with one pair of legs on each segment. Millipedes are segmented herbivores with two pairs of legs on each segment.



FIGURE 46.23

A millipede. Millipedes, such as this *Sigmoria* individual, are herbivores.

Class Insecta: The Insects

The insects, class Insecta, are by far the largest group of organisms on earth, whether measured in terms of numbers of species or numbers of individuals. Insects live in every conceivable habitat on land and in fresh water, and a few have even invaded the sea. More than half of all the named animal species are insects, and the actual proportion is doubtless much higher because millions of additional forms await detection, classification, and naming. Approximately 90,000 described species occur in the United States and Canada, and the actual number of species in this area probably approaches 125,000. A hectare of lowland tropical forest is estimated to be inhabited by as many as 41,000 species of insects, and many suburban gardens may have 1500 or more species. It has been estimated that approximately a billion billion (10^{18}) individual insects are alive at any one time. A glimpse at the enormous diversity of insects is presented in figure 46.24 and later in table 46.2.



(a)



(b)



(c)



(d)



(e)



(f)

FIGURE 46.24

Insect diversity. (a) Luna moth, *Actias luna*. Luna moths and their relatives are among the most spectacular insects (order Lepidoptera). (b) Soldier fly, *Ptericus trivittatus* (order Diptera). (c) Boll weevil, *Anthonomus grandis*. Weevils are one of the largest groups of beetles (order Coleoptera). (d) A thorn-shaped leafhopper, *Umberica crassicornis* (order Hemiptera). (e) Copulating grasshoppers (order Orthoptera). (f) Termite, *Macrotermes bellicosus* (order Isoptera). The large, sausage-shaped individual is a queen, specialized for laying eggs; most of the smaller individuals around the queen are nonreproductive workers, but the larger individual at the lower left is a reproductive male.

External Features

Insects are primarily a terrestrial group, and most, if not all, of the aquatic insects probably had terrestrial ancestors. Most insects are relatively small, ranging in size from 0.1 millimeter to about 30 centimeters in length or wingspan. Insects have three body sections, the head, thorax, and abdomen; three pairs of legs, all attached to the thorax; and one pair of antennae. In addition, they may have one or two pairs of wings. Insect mouthparts all have the same basic structure but are modified in different groups in relation to their feeding habits (figure 46.25). Most insects have compound eyes, and many have ocelli as well.

The insect thorax consists of three segments, each with a pair of legs. Legs are completely absent in the larvae of certain groups, for example, in most members of the order Diptera (flies) (figure 46.26). If two pairs of wings are present, they attach to the middle and posterior segments of the thorax. If only one pair of wings is present, it usually attaches to the middle segment. The thorax is almost entirely filled with muscles that operate the legs and wings.

The wings of insects arise as saclike outgrowths of the body wall. In adult insects, the wings are solid except for the veins. Insect wings are not homologous to the other appendages. Basically, insects have two pairs of wings, but in some groups, like flies, the second set has been reduced to a pair of balancing knobs called halteres during the course of evolution. Most insects can fold their wings over their abdomen when they are at rest; but a few, such as the dragonflies and damselflies (order Odonata), keep their wings erect or outstretched at all times.

Insect forewings may be tough and hard, as in beetles. If they are, they form a cover for the hindwings and usually open during flight. The tough forewings also serve a protective function in the order Orthoptera, which includes grasshoppers and crickets. The wings of insects are made of sheets of chitin and protein; their strengthening veins are tubules of chitin and protein. Moths and butterflies have wings that are covered with detachable scales that provide most of their bright colors (figure 46.27). In some wingless insects, such as the springtails or silverfish, wings never evolved. Other wingless groups, such as fleas and lice, are derived from ancestral groups of insects that had wings.

Internal Organization

The internal features of insects resemble those of the other arthropods in many ways. The digestive tract is a tube, usually somewhat coiled. It is often about the same length as the body. However, in the order Hemiptera, which consists of the leafhoppers, cicadas, and related groups, and in many flies (order Diptera), the digestive tube may be greatly coiled and several times longer than the body. Such long digestive tracts are generally found in

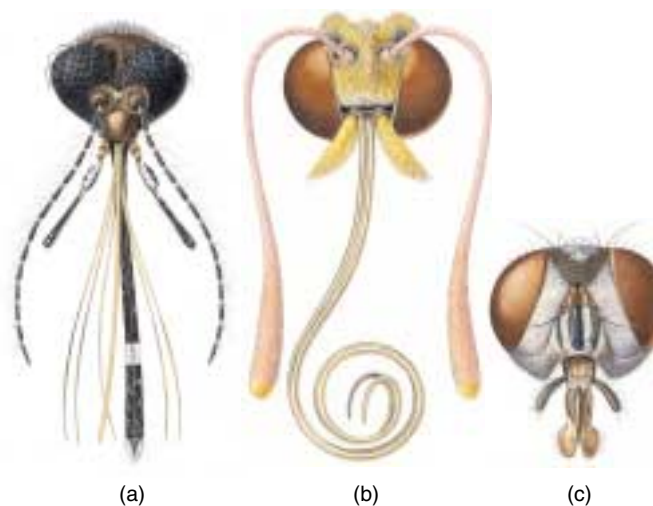


FIGURE 46.25

Modified mouthparts in three kinds of insects. Mouthparts are modified for (a) piercing in the mosquito, *Culex*, (b) sucking nectar from flowers in the alfalfa butterfly, *Colias*, and (c) sopping up liquids in the housefly, *Musca domestica*.



FIGURE 46.26

Larvae of a mosquito, *Culex pipiens*. The aquatic larvae of mosquitoes are quite active. They breathe through tubes from the surface of the water, as shown here. Covering the water with a thin film of oil causes them to drown.

insects that have sucking mouthparts and feed on juices rather than on protein-rich solid foods because they offer a greater opportunity to absorb fluids and their dissolved nutrients. The digestive enzymes of the insect also are more dilute and thus less effective in a highly liquid medium than in a more solid one. Longer digestive tracts give these enzymes more time to work while food is passing through.

The anterior and posterior regions of an insect's digestive tract are lined with cuticle. Digestion takes place primarily in the stomach, or midgut; and excretion takes place through Malpighian tubules. Digestive enzymes are mainly secreted from the cells that line the midgut, although some are contributed by the salivary glands near the mouth.

The tracheae of insects extend throughout the body and permeate its different tissues. In many winged insects, the tracheae are dilated in various parts of the body, forming air sacs. These air sacs are surrounded by muscles and form a kind of bellows system to force air deep into the tracheal system. The spiracles, a maximum of 10 on each side of the insect, are paired and located on or between the segments along the sides of the thorax and abdomen. In most insects, the spiracles can be opened by muscular action. Closing the spiracles at times may be important in retarding water loss. In some parasitic and aquatic groups of insects, the spiracles are permanently closed. In these groups, the tracheae run just below the surface of the insect, and gas exchange takes place by diffusion.

The **fat body** is a group of cells located in the insect body cavity. This structure may be quite large in relation to the size of the insect, and it serves as a food-storage reservoir, also having some of the functions of a vertebrate liver. It is often more prominent in immature insects than in adults, and it may be completely depleted when metamorphosis is finished. Insects that do not feed as adults retain their fat bodies and live on the food stored in them throughout their adult lives (which may be very short).

Sense Receptors

In addition to their eyes, insects have several characteristic kinds of sense receptors. These include **sensory hairs**, which are usually widely distributed over their bodies. The



FIGURE 46.27
Scales on the wing of *Parnassius imperator*, a butterfly from China. Scales of this sort account for most of the colored patterns on the wings of butterflies and moths.

sensory hairs are linked to nerve cells and are sensitive to mechanical and chemical stimulation. They are particularly abundant on the antennae and legs—the parts of the insect most likely to come into contact with other objects.

Sound, which is of vital importance to insects, is detected by tympanal organs in groups such as grasshoppers and crickets, cicadas, and some moths. These organs are paired structures composed of a thin membrane, the **tympanum**, associated with the tracheal air sacs. In many other groups of insects, sound waves are detected by sensory hairs. Male mosquitoes use thousands of sensory hairs on their antennae to detect the sounds made by the vibrating wings of female mosquitoes.

Sound detection in insects is important not only for protection but also for communication. Many insects communicate by making sounds, most of which are quite soft, very high-pitched, or both, and thus inaudible to humans. Only a few groups of insects, especially grasshoppers, crickets, and cicadas, make sounds that people can hear. Male crickets and longhorned grasshoppers produce sounds by rubbing their two front wings together. Shorthorned grasshoppers do so by rubbing their hind legs over specialized areas on their wings. Male cicadas vibrate the membranes of air sacs located on the lower side of the most anterior abdominal segment.

In addition to using sound, nearly all insects communicate by means of chemicals or mixtures of chemicals known as **pheromones**. These compounds, extremely diverse in their chemical structure, are sent forth into the environment, where they are active in very small amounts and convey a variety of messages to other individuals. These messages not only convey the attraction and recognition of members of the same species for mating, but they also mark trails for members of the same species, as in the ants.

All insects possess three body segments (tagmata): the head, the thorax, and the abdomen. The three pairs of legs are attached to the thorax. Most insects have compound eyes, and many have one or two pairs of wings. Insects possess sophisticated means of sensing their environment, including sensory hairs, tympanal organs, and chemoreceptors.

Insect Life Histories

Most young insects hatch from fertilized eggs laid outside their mother's body. The zygote develops within the egg into a young insect, which escapes by chewing through or bursting the shell. Some immature insects have specialized projections on the head that assist in this process. In a few insects, eggs hatch within the mother's body.

During the course of their development, young insects undergo ecdysis a number of times before they become adults. Most insects molt four to eight times during the course of their development, but some may molt as many as 30 times. The stages between the molts are called **instars**. When an insect first emerges following ecdysis, it is pale, soft, and especially susceptible to predators. Its exoskeleton generally hardens in an hour or two. It must grow to its new size, usually by taking in air or water, during this brief period.

There are two principal kinds of metamorphosis in insects: **simple metamorphosis** and **complete metamorphosis** (figure 46.28). In insects with simple metamorphosis, immature stages are often called **nymphs**. Nymphs are usually quite similar to adults, differing mainly in their smaller size, less well-developed wings, and sometimes color. In insect orders with simple metamorphosis, such as mayflies and dragonflies, nymphs are aquatic and extract oxygen from the water through gills. The adult stages are terrestrial and look very different from the nymphs. In other groups, such as grasshoppers and their relatives, nymphs and adults live in the same habitat. Such insects usually change gradually during their life cycles with respect to wing development, body proportions, the appearance of ocelli, and other features.

In complete metamorphosis, the wings develop internally during the juvenile stages and appear externally only during the resting stage that immediately precedes the final molt (figure 46.28*b*). During this stage, the insect is called a **pupa** or **chrysalis**, depending on the group to which it belongs. A pupa or chrysalis does not normally move around much, although mosquito pupae do move around freely. A large amount of internal body reorganization takes place while the insect is a pupa or chrysalis.

More than 90% of the insects, including the members of all of the largest and most successful orders, display complete metamorphosis. The juvenile stages and adults often live in distinct habitats, have different habits, and are usually extremely different in form. In these insects, development is indirect. The immature stages, called **larvae**, are often wormlike, differing greatly in appearance from the adults. Larvae do not have compound eyes. They may be legless or have legs or leg-like appendages on the abdomen. They usually have chewing mouthparts, even in those orders in which the adults have sucking mouthparts; chewing mouthparts are the primitive condition in these groups. When larvae and adults play different ecological roles, they

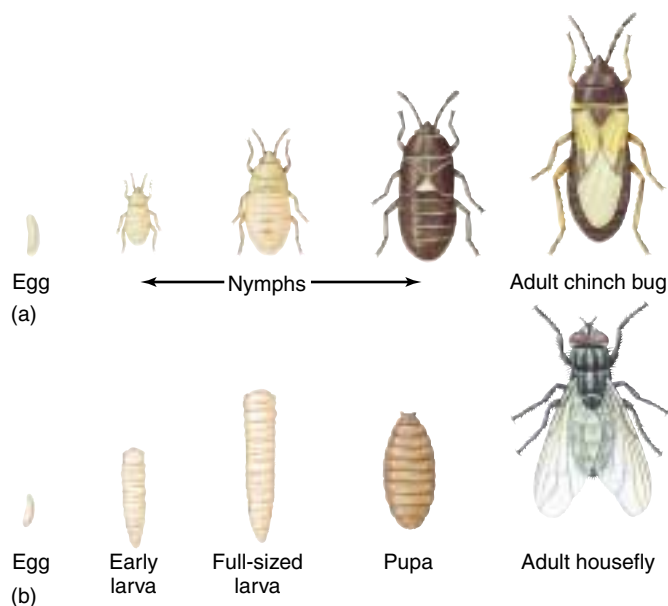


FIGURE 46.28

Metamorphosis. (a) Simple metamorphosis in a chinch bug (order Hemiptera), and (b) complete metamorphosis in a housefly, *Musca domestica* (order Diptera).

do not compete directly for the same resources, an advantage to the species.

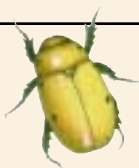
Pupae do not feed and are usually relatively inactive. As pupae, insects are extremely vulnerable to predators and parasites, but they are often covered by a cocoon or some other protective structure. Groups of insects with complete metamorphosis include moths and butterflies; beetles; bees, wasps, and ants; flies; and fleas.

Some species of insects exhibit no dramatic change in form from immature stages to adult. This type of development is called **ametabolus** (meaning without change) and is seen in the most primitive orders of insects such as the silverfish and springtails.

Hormones control both ecdysis and metamorphosis. **Molting hormone**, or **ecdysone**, is released from a gland in the thorax when that gland has been stimulated by brain hormone, which in turn is produced by neurosecretory cells and released into the blood. The effects of the molting induced by ecdysone are determined by juvenile hormone, which is present during the immature stages but declines in quantity as the insect passes through successive molts. When the level of juvenile hormone is relatively high, the molt produces another larva; when it is lower, it produces the pupa and then the final development of the adult.

Insects undergo either simple or complete metamorphosis.

Table 46.2 Major Orders of Insects

Order	Typical Examples		Key Characteristics	Approximate Number of Named Species
Coleoptera	Beetles		The most diverse animal order; two pairs of wings; front pair of wings is a hard cover that partially protects the transparent rear pair of flying wings; heavily armored exoskeleton; biting and chewing mouthparts; complete metamorphosis	350,000
Diptera	Flies		Some that bite people and other mammals are considered pests; front flying wings are transparent; hind wings are reduced to knobby balancing organs; sucking, piercing, and lapping mouthparts; complete metamorphosis	120,000
Lepidoptera	Butterflies, moths		Often collected for their beauty; two pairs of broad, scaly, flying wings, often brightly colored; hairy body; tubelike, sucking mouthparts; complete metamorphosis	120,000
Hymenoptera	Bees, wasps, ants		Often social, known to many by their sting; two pairs of transparent flying wings; mobile head and well-developed eyes; often possess stingers; chewing and sucking mouthparts; complete metamorphosis	100,000
Hemiptera	True bugs, bedbugs, leafhoppers		Often live on blood; two pairs of wings, or wingless; piercing, sucking mouthparts; simple metamorphosis	60,000
Orthoptera	Grasshoppers, crickets, roaches		Known for their jumping; two pairs of wings or wingless; among the largest insects; biting and chewing mouthparts in adults; simple metamorphosis	20,000
Odonata	Dragonflies		Among the most primitive of the insect order; two pairs of transparent flying wings; large, long, and slender body; chewing mouthparts; simple metamorphosis	5,000
Isoptera	Termites		One of the few types of animals able to eat wood; two pairs of wings, but some stages are wingless; social insects; there are several body types with division of labor; chewing mouthparts; simple metamorphosis	2,000
Siphonaptera	Fleas		Small, known for their irritating bites; wingless; small flattened body with jumping legs; piercing and sucking mouthparts; complete metamorphosis	1,200



Summary

Questions

Media Resources

46.1 The evolution of jointed appendages has made arthropods very successful.

- Jointed appendages and an exoskeleton greatly expanded locomotive and manipulative capabilities for the arthropod phyla, the most successful of all animals in terms of numbers of individuals, species, and ecological diversification.
- Traditionally, arthropods have been grouped into three subphyla based on morphological characters but new research is calling this classification of the arthropods into question.
- Like annelids, arthropods have segmented bodies, but some of their segments have become fused into tagmata during the course of evolution. All possess a rigid external skeleton, or exoskeleton.

1. What are the advantages of an exoskeleton? What occurs during ecdysis? What controls this process?
2. What type of circulatory system do arthropods have? Describe the direction of blood flow. What helps to maintain this one-way flow?
3. What are Malpighian tubules? How do they work? What other system are they connected to? How does this system process wastes? How does it regulate water loss?



- Arthropods



- Enhancement
Chapter: Arthropod Taxonomy, Sections 1 and 2

46.2 The chelicerates all have fangs or pincers.

- Chelicerates consist of three classes: Arachnida (spiders, ticks, mites, and scorpions); Merostomata (horseshoe crabs); and Pycnogonida (sea spiders).
- Spiders, the best known arachnids, have a pair of chelicerae, a pair of pedipalps, and four pairs of walking legs. Spiders secrete digestive enzymes into their prey, then suck the contents out.

4. Into what two groups are arthropods traditionally divided? Describe each group in terms of its mouthparts and appendages, and give several examples of each.

46.3 Crustaceans have branched appendages.

- Crustaceans comprise some 35,000 species of crabs, shrimps, lobsters, barnacles, sowbugs, beach fleas, and many other groups. Their appendages are basically biramous, and their embryology is distinctive.

5. On which parts of the body do crustaceans possess legs?
6. How do biramous and uniramous appendages differ?



- Enhancement
Chapter: Arthropod Taxonomy, Section 3

46.4 Insects are the most diverse of all animal groups.

- Centipedes and millipedes are segmented uniramia. Centipedes are hunters with one pair of legs per segment, and millipedes are herbivores with two pairs of legs per segment.
- Insects have three body segments, three pairs of legs, and often one or two pairs of wings. Many have complex eyes and other specialized sensory structures.
- Insects exhibit either simple metamorphosis, moving through a succession of forms relatively similar to the adult, or complete metamorphosis, in which an often wormlike larva becomes a usually sedentary pupa, and then an adult.

7. How are millipedes and centipedes similar to each other? How do they differ?
8. What type of digestive system do most insects possess? What digestive adaptations occur in insects that feed on juices low in protein? Why?
9. What is an instar as it relates to insect metamorphosis? What are the two different kinds of metamorphosis in insects? How do they differ?



- Enhancement
Chapter: Arthropod Taxonomy, Section 4
- Student Research:
Insect Behavior

47

Echinoderms

Concept Outline

47.1 The embryos of deuterostomes develop quite differently from those of protostomes.

Protostomes and Deuterostomes. Deuterostomes—the echinoderms, chordates, and a few other groups—share a mode of development that is quite different from other animals.

47.2 Echinoderms are deuterostomes with an endoskeleton.

Deuterostomes. Echinoderms are bilaterally symmetrical as larvae but metamorphose to radially symmetrical adults.

Echinoderm Body Plan. Echinoderms have an endoskeleton and a unique water-vascular system seen in no other phylum.

47.3 The six classes of echinoderms are all radially symmetrical as adults.

Class Crinoidea: The Sea Lilies and Feather Stars.

Crinoids are the only echinoderms that are attached to the sea bottom for much of their lives.

Class Asteroidea: The Sea Stars. Sea stars, also called starfish, are five-armed mobile predators.

Class Ophiuroidea: The Brittle Stars. Brittle stars are quite different from the sea stars for whom they are sometimes mistaken.

Class Echinoidea: The Sea Urchins and Sand Dollars. Sea urchins and sand dollars have five-part radial symmetry but lack arms.

Classes Holothuroidea and Concentricycloidea: Sea Cucumbers and Sea Daisies. Sea cucumbers are soft-bodied echinoderms without arms. The most recently discovered class of echinoderms, sea daisies are tiny, primitive echinoderms that live at great depths.

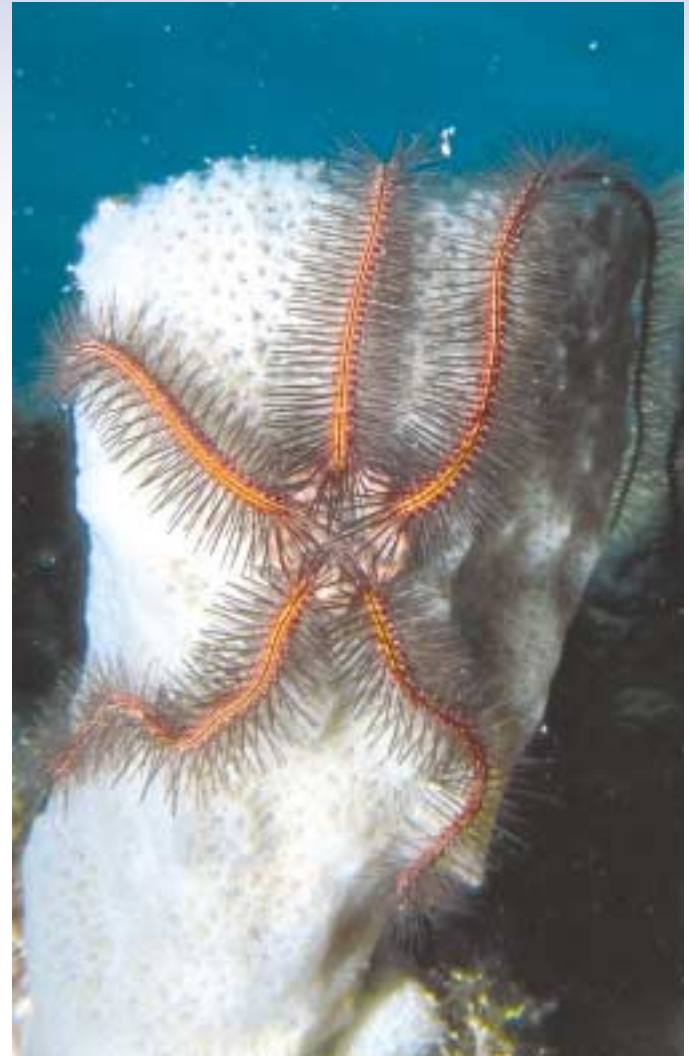


FIGURE 47.1

An echinoderm. Brittle star, *Ophiothrix*, a member of the largest group of echinoderms.

Echinoderms, which include the familiar starfish, have been described as a “noble group especially designed to puzzle the zoologist.” They are bilaterally symmetrical as larvae, but undergo a bizarre metamorphosis to a radially symmetrical adult (figure 47.1). A compartment of the coelom is transformed into a unique water-vascular system that uses hydraulic power to operate a multitude of tiny tube feet that are used in locomotion and food capture. Some echinoderms have an endoskeleton of dermal plates beneath the skin, fused together like body armor. Many have miniature jawlike pincers scattered over their body surface, often on stalks and sometimes bearing poison glands. This collection of characteristics is unique in the animal kingdom.

47.1 The embryos of deuterostomes develop quite differently from those of protostomes.

Protostomes and Deuterostomes

The coelomates we have met so far—the mollusks, annelids, and arthropods—exhibit essentially the same kind of embryological development, starting as a hollow ball of cells, a blastula, which indents to form a two-layer-thick ball with a blastopore opening to the outside. Also in this group, the mouth (stoma) develops from or near the blastopore (figure 47.2). This same pattern of development, in a general sense, is seen in all noncoelomate animals. An animal whose mouth develops in this way is called a **protostome** (from the Greek words *protos*, “first,” and *stoma*, “mouth”). If such an animal has a distinct anus or anal pore, it develops later in another region of the embryo. The fact that this kind of developmental pattern is so widespread in diverse phyla suggests that it is the original pattern for animals as a whole and that it was characteristic of the common ancestor of all eumetazoan animals.

A second distinct pattern of embryological development occurs in the echinoderms, the chordates, and a few other smaller related phyla. The consistency of this pattern of development, and its distinctiveness from that of the protostomes suggests that it evolved once, in a common ancestor

to all of the phyla that exhibit it. In **deuterostome** (Greek, *deuteros*, “second,” and *stoma*, “mouth”) development, the blastopore gives rise to the organism’s anus, and the mouth develops from a second pore that arises in the blastula later in development.

Deuterostomes represent a revolution in embryonic development. In addition to the pattern of blastopore formation, deuterostomes differ from protostomes in a number of other fundamental embryological features:

1. The progressive division of cells during embryonic growth is called *cleavage*. The cleavage pattern relative to the embryo’s polar axis determines how the cells will array. In nearly all protostomes, each new cell buds off at an angle oblique to the polar axis. As a result, a new cell nestles into the space between the older ones in a closely packed array. This pattern is called **spiral cleavage** because a line drawn through a sequence of dividing cells spirals outward from the polar axis (figure 47.2).

In deuterostomes, the cells divide parallel to and at right angles to the polar axis. As a result, the pairs of cells from each division are positioned directly above

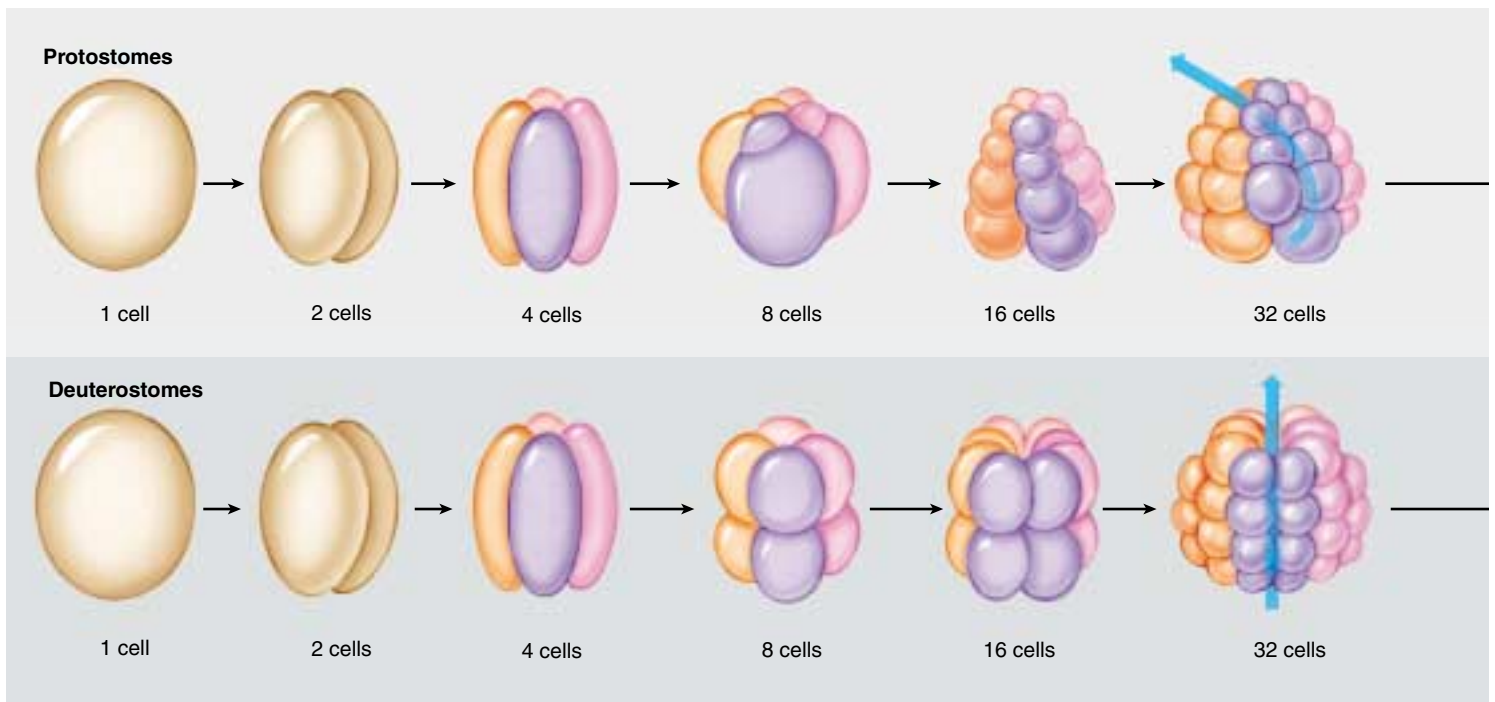


FIGURE 47.2

Embryonic development in protostomes and deuterostomes. Cleavage of the egg produces a hollow ball of cells called the blastula. Invagination of the blastula produces the blastopore and archenteron. In protostomes, embryonic cells cleave in a spiral pattern and become tightly packed. The blastopore becomes the animal’s mouth, and the coelom originates from a mesodermal split.

and below one another; this process gives rise to a loosely packed array of cells. This pattern is called **radial cleavage** because a line drawn through a sequence of dividing cells describes a radius outward from the polar axis.

2. Protostomes exhibit **determinate** development. In this type of development, each embryonic cell has a predetermined fate in terms of what kind of tissue it will form in the adult. Before cleavage begins, the chemicals that act as developmental signals are localized in different parts of the egg. Consequently, the cell divisions that occur after fertilization separate different signals into different daughter cells. This process specifies the fate of even the very earliest embryonic cells. Deuterostomes, on the other hand, display **indeterminate** development. The first few cell divisions of the egg produce identical daughter cells. Any one of these cells, if separated from the others, can develop into a complete organism. This is possible because the chemicals that signal the embryonic cells to develop differently are not localized until later in the animal's development.
3. In all coelomates, the coelom originates from mesoderm. In protostomes, this occurs simply and directly: the cells simply move away from one another as the coelomic cavity expands within the mesoderm. How-

ever, in deuterostomes, whole groups of cells usually move around to form new tissue associations. The coelom is normally produced by an evagination of the **archenteron**—the central tube within the gastrula, also called the primitive gut. This tube, lined with endoderm, opens to the outside via the blastopore and eventually becomes the gut cavity.

The first abundant and well-preserved animal fossils are nearly 600 million years old; they occur in the Ediacara series of Australia and similar formations elsewhere. Among these fossils, many represent groups of animals that no longer exist. In addition, these ancient rocks bear evidence of the coelomates, the most advanced evolutionary line of animals, and it is remarkable that their two major subdivisions were differentiated so early. In the coelomates, it seems likely that all deuterostomes share a common protostome ancestor—a theory that is supported by evidence from comparison of rRNA and other molecular studies. The event, however, occurred very long ago and presumably did not involve groups of organisms that closely resemble any that are living now.

In deuterostomes, the egg cleaves radially, and the blastopore becomes the anus. In protostomes, the egg cleaves spirally, and the blastopore becomes the mouth.

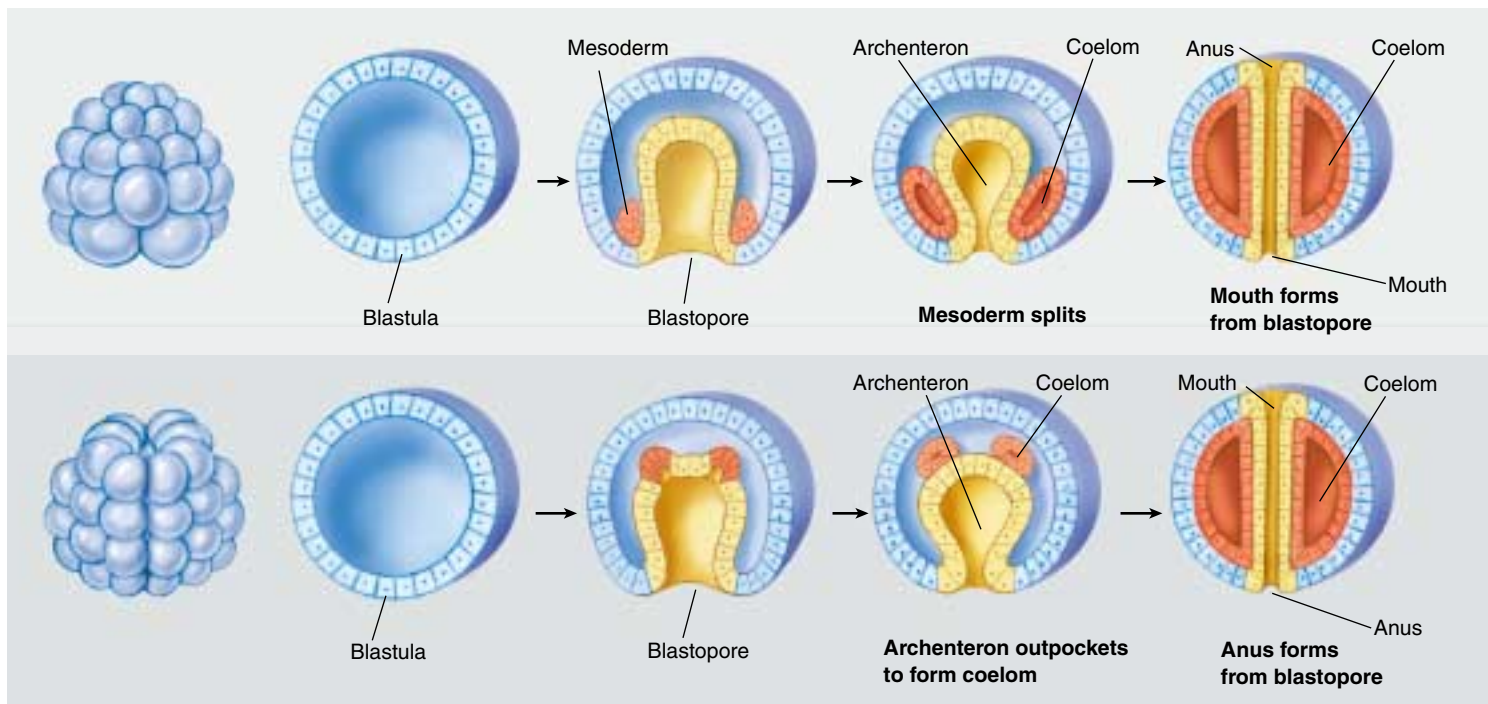


FIGURE 47.2 (continued)

In deuterostomes, embryonic cells cleave radially and form a loosely packed array. The blastopore becomes the animal's anus, and the mouth develops at the other end. The coelom originates from an evagination, or outpouching, of the archenteron in deuterostomes.

47.2 Echinoderms are deuterostomes with an endoskeleton.

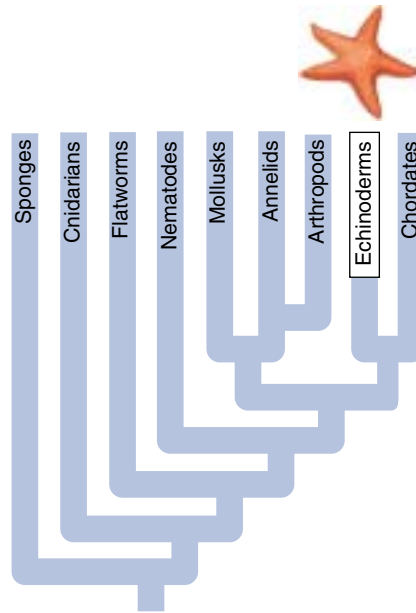
Deuterostomes

Mollusks, annelids, and arthropods are protostomes. However, the echinoderms are characterized by *deuterostome development*, a key evolutionary advance. The *endoskeleton* makes its first appearance in the echinoderms also.

The Echinoderms

Deuterostomate marine animals called **echinoderms** appeared nearly 600 million years ago (figure 47.3). Echinoderms (phylum Echinodermata) are an ancient group of marine animals consisting of about 6000 living species and are also well represented in the fossil record. The term *echinoderm* means “spiny skin” and refers to an **endoskeleton** composed of hard calcium-rich plates just beneath the delicate skin (figure 47.4). When they first form, the plates are enclosed in living tissue and so are truly an endoskeleton, although in adults they frequently fuse, forming a hard shell. Another innovation in echinoderms is the development of a hydraulic system to aid in movement or feeding. Called a **water-vascular system**, this fluid-filled system is composed of a central ring canal from which five radial canals extend out into the body and arms.

Many of the most familiar animals seen along the seashore, sea stars (starfish), brittle stars, sea urchins, sand dollars, and sea cucumbers, are echinoderms. All are radially symmetrical as adults. While some other kinds of animals are radially symmetrical, none have the complex organ systems of adult echinoderms. Echinoderms are well



represented not only in the shallow waters of the sea but also in its abyssal depths. In the oceanic trenches, which are the deepest regions of the oceans, sea cucumbers account for more than 90% of the biomass! All of them are bottom-dwellers except for a few swimming sea cucumbers. The adults range from a few millimeters to more than a meter in diameter (for one species of sea star) or in length (for a species of sea cucumber).

There is an excellent fossil record of the echinoderms, extending back into the Cambrian. However, despite this wealth of information, the origin of echinoderms remains unclear. They are thought to have evolved from bilaterally symmetrical ancestors because echinoderm larvae are bilateral. The radial symmetry that is

the hallmark of echinoderms develops later, in the adult body. Many biologists believe that early echinoderms were sessile and evolved radially as an adaptation to the sessile existence. Bilaterality is of adaptive value to an animal that travels through its environment, while radially is of value to an animal whose environment meets it on all sides. Echinoderms attached to the sea bottom by a central stalk were once common, but only about 80 such species survive today.

Echinoderms are a unique, exclusively marine group of organisms in which deuterostome development and an endoskeleton are seen for the first time.



(a)



(b)



(c)

FIGURE 47.3

Diversity in echinoderms. (a) Sea star, *Oreaster occidentalis* (class Asterozoidea), in the Gulf of California, Mexico. (b) Warty sea cucumber, *Parastichopus parvimensis* (class Holothurozoidea), Philippines. (c) Sea urchin (class Echinozoidea).

PHYLUM ECHINODERMATA: Deuterostome development and endoskeleton

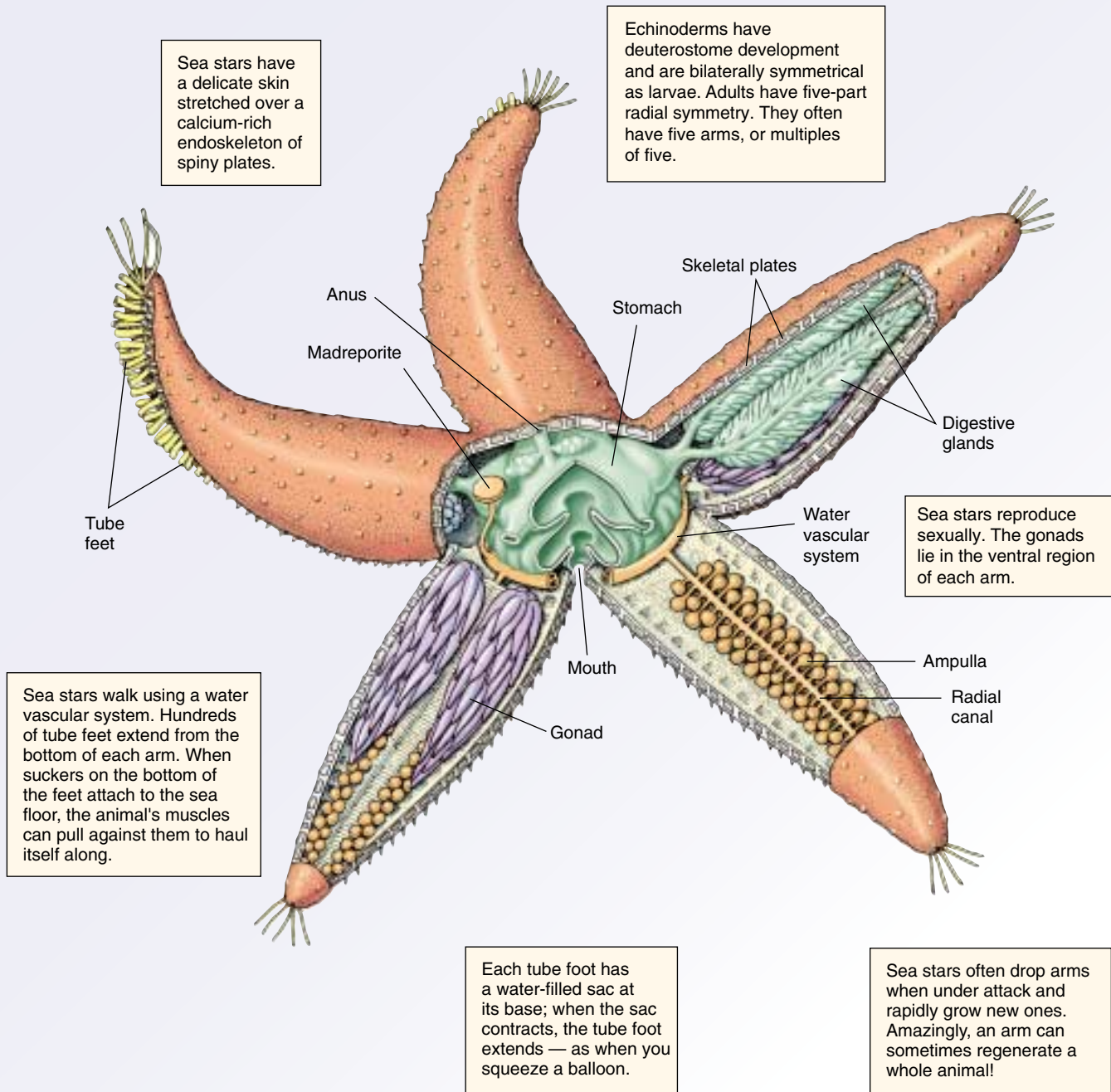


FIGURE 47.4

The evolution of deuterostome development and an endoskeleton. Echinoderms, such as sea stars (phylum Echinodermata), are coelomates with a deuterostome pattern of development. A delicate skin stretches over an endoskeleton made of calcium-rich plates, often fused into a continuous, tough, spiny layer.

Echinoderm Body Plan

The body plan of echinoderms undergoes a fundamental shift during development. All echinoderms have **secondary radial symmetry**, that is, they are bilaterally symmetrical during larval development but become radially symmetrical as adults. Because of their radially symmetrical bodies, the usual terms used to describe an animal's body are not applicable: dorsal, ventral, anterior, and posterior have no meaning without a head or tail. Instead, the body structure of echinoderms is discussed in reference to their mouths which are located on the oral surface. Most echinoderms crawl along on their oral surfaces, although in sea cucumbers and some other echinoderms, the animal's axis lies horizontally and they crawl with the oral surface in front.

Echinoderms have a five-part body plan corresponding to the arms of a sea star or the design on the "shell" of a sand dollar. These animals have no head or brain. Their nervous systems consist of a central **nerve ring** from which branches arise. The animals are capable of complex response patterns, but there is no centralization of function.

Endoskeleton

Echinoderms have a delicate epidermis, containing thousands of neurosensory cells, stretched over an endoskeleton composed of either movable or fixed calcium-rich (calcite) plates called ossicles. The animals grow more or less contin-

uously, but their growth slows down with age. When the plates first form, they are enclosed in living tissue. In some echinoderms, such as asteroids and holothuroids, the ossicles are widely scattered and the body wall is flexible. In others, especially the echinoids, the ossicles become fused and form a rigid shell. In many cases, these plates bear spines. In nearly all species of echinoderms, the entire skeleton, even the long spines of sea urchins, is covered by a layer of skin. Another important feature of this phylum is the presence of mutable collagenous tissue which can range from being tough and rubbery to weak and fluid. This amazing tissue accounts for many of the special attributes of echinoderms, such as the ability to rapidly autotomize body parts. The plates in certain portions of the body of some echinoderms are perforated by pores. Through these pores extend **tube feet**, part of the water-vascular system that is a unique feature of this phylum.

The Water-Vascular System

The water-vascular system of an echinoderm radiates from a **ring canal** that encircles the animal's esophagus. Five **radial canals**, their positions determined early in the development of the embryo, extend into each of the five parts of the body and determine its basic symmetry (figure 47.5). Water enters the water-vascular system through a **madreporite**, a sieve-like plate on the animal's surface, and flows to the ring canal through a tube, or stone canal, so named because of

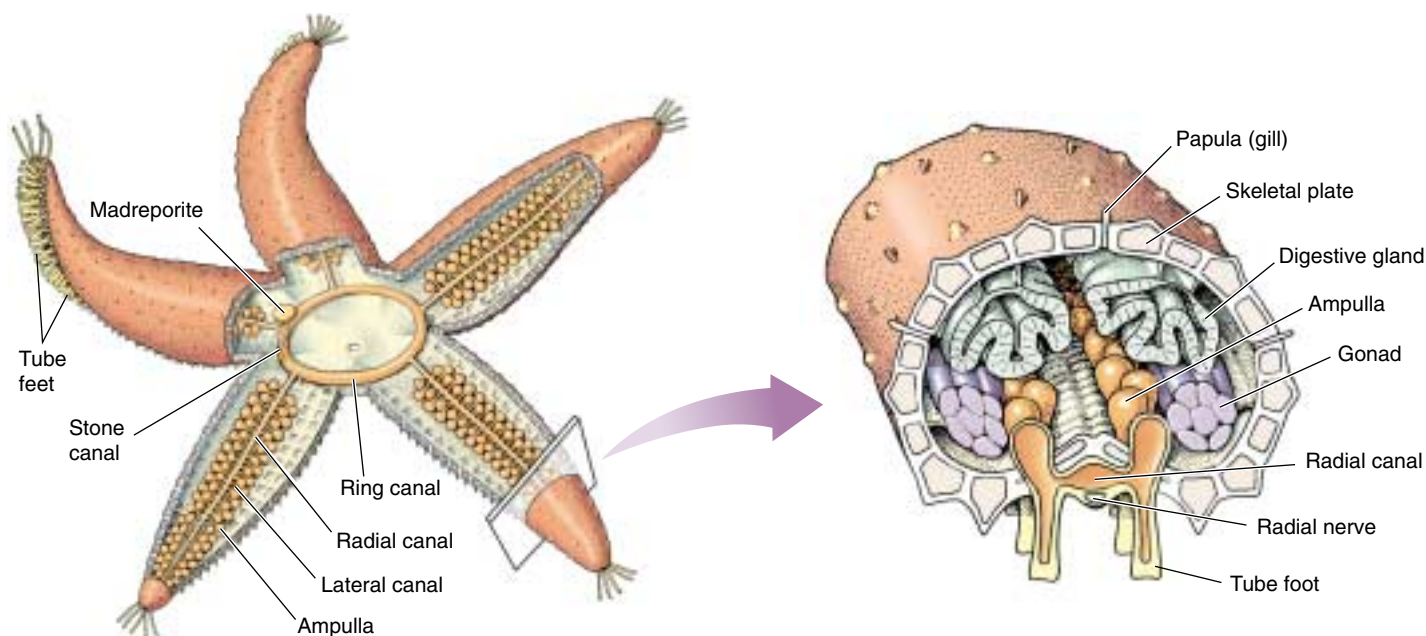


FIGURE 47.5

The water-vascular system of an echinoderm. Radial canals allow water to flow into the tube feet. As the ampulla in each tube foot contracts, the tube extends and can attach to the substrate. Subsequently, muscles in the tube feet contract bending the tube foot and pulling the animal forward.



FIGURE 47.6
Tube feet. The nonsuckered tube feet of the sea star, *Ludia magnifica*, are extended.

the surrounding rings of calcium carbonate. The five radial canals in turn extend out through short side branches into the hollow tube feet (figure 47.6). In some echinoderms, each tube foot has a sucker at its end; in others, suckers are absent. At the base of each tube foot is a muscular sac, the **ampulla**, which contains fluid. When the ampulla contracts, the fluid is prevented from entering the radial canal by a one-way valve and is forced into the tube foot, thus extending it. When extended, the foot can attach itself to the substrate. Longitudinal muscles in the tube foot wall then contract causing the tube feet to bend. Relaxation of the muscles in the ampulla allows the fluid to flow back into the ampulla which moves the foot. By the concerted action of a very large number of small individually weak tube feet, the animal can move across the sea floor.

Sea cucumbers (see figure 47.3*b*) usually have five rows of tube feet on the body surface that are used in locomotion. They also have modified tube feet around their mouth cavity that are used in feeding. In sea lilies, the tube feet arise from the branches of the arms, which extend from the margins of an upward-directed cup. With these tube feet, the animals take food from the surrounding water. In brittle stars (see figure 47.1), the tube feet are pointed and specialized for feeding.

Body Cavity

In echinoderms, the coelom, which is proportionately large, connects with a complicated system of tubes and helps provide circulation and respiration. In many echinoderms, respiration and waste removal occurs across the skin through small, fingerlike extensions of the coelom called papulae (see figure 47.5). They are covered with a thin layer of skin and protrude through the body wall to function as gills.

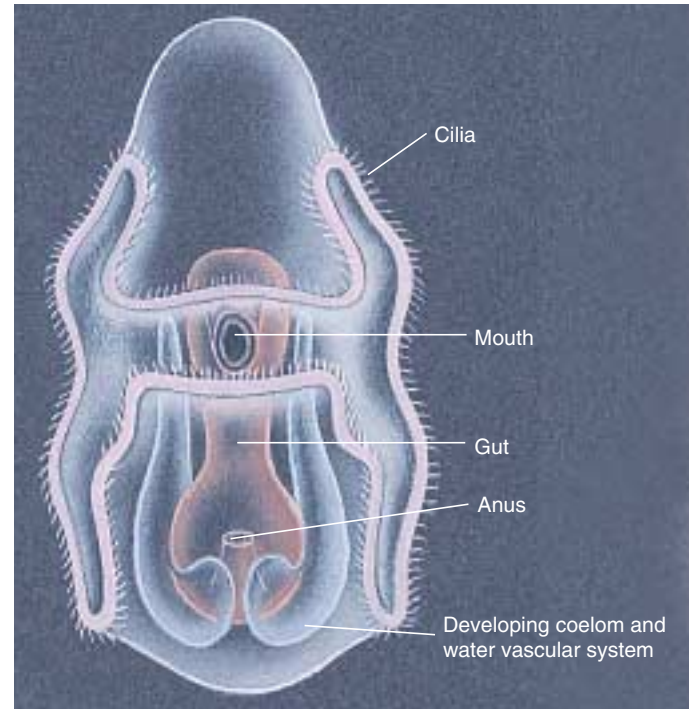


FIGURE 47.7
The free-swimming larva of an echinoderm. The bands of cilia by which the larva moves are prominent in this drawing. Such bilaterally symmetrical larvae suggest that the ancestors of the echinoderms were not radially symmetrical, like the living members of the phylum.

Reproduction

Many echinoderms are able to regenerate lost parts, and some, especially sea stars and brittle stars, drop various parts when under attack. In a few echinoderms, asexual reproduction takes place by splitting, and the broken parts of sea stars can sometimes regenerate whole animals. Some of the smaller brittle stars, especially tropical species, regularly reproduce by breaking into two equal parts; each half then regenerates a whole animal.

Despite the ability of many echinoderms to break into parts and regenerate new animals from them, most reproduction in the phylum is sexual and external. The sexes in most echinoderms are separate, although there are few external differences. Fertilized eggs of echinoderms usually develop into free-swimming, bilaterally symmetrical larvae (figure 47.7), which differ from the trochophore larvae of mollusks and annelids. These larvae form a part of the plankton until they metamorphose through a series of stages into the more sedentary adults.

Echinoderms are characterized by a secondary radial symmetry and a five-part body plan. They have characteristic calcium-rich plates called ossicles and a unique water-vascular system that includes hollow tube feet.

47.3 The six classes of echinoderms are all radially symmetrical as adults.

There are more than 20 extinct classes of echinoderms and an additional 6 with living members: (1) Crinoidea, sea lilies and feather stars; (2) Asteroidea, sea stars, or starfish; (3) Ophiuroidea, brittle stars; (4) Echinoidea, sea urchins and sand dollars; (5) Holothuroidea, sea cucumbers, and (6) Concentricycloidea, sea daisies. Sea daisies were recently discovered living on submerged wood in the deep sea.

Class Crinoidea: The Sea Lilies and Feather Stars

Sea lilies and feather stars, or crinoids (class Crinoidea) differ from all other living echinoderms in that the mouth and anus are located on their upper surface in an open disc. The two structures are connected by a simple gut. These animals have simple excretory and reproductive systems and an extensive water-vascular system. The arms, which are the food-gathering structures of crinoids, are located around the margins of the disc. Different species of crinoids may have from 5 to more than 200 arms extending upward from their bodies, with smaller structures called pinnules branching from the arms. In all crinoids, the number of arms is initially small. Species with more than 10 arms add additional arms progressively during growth. Crinoids are filter feeders, capturing the microscopic organisms on which they feed by means of the mucus that coats their tube feet, which are abundant on the animals' pinnules.

Scientists that study echinoderms believe that the common ancestors of this phylum were sessile, sedentary, radially symmetrical animals that resembled crinoids. Crinoids were abundant in ancient seas, and were present when the Burgess Shale was deposited about 515 million years ago. More than 6000 fossil species of this class are known, in comparison with the approximately 600 living species.



FIGURE 47.8
Sea lilies, *Cenocrinus asterius*. Two specimens showing a typical parabola of arms forming a “feeding net.” The water current is flowing from right to left, carrying small organisms to the stalked crinoid’s arms. Prey, when captured, are passed down the arms to the central mouth. This photograph was taken at a depth of about 400 meters in the Bahamas from the Johnson-Sea-Link Submersible of the Harbor Branch Foundation, Inc.



FIGURE 47.9
Feather star. This feather star is on the Great Barrier Reef in Australia.

Sea Lilies

There are two basic crinoid body plans. In sea lilies, the flower-shaped body is attached to its substrate by a stalk that is from 15 to 30 cm long, although in some species the stalk may be as much as a meter long (figure 47.8). Some fossil species had stalks up to 20 meters long. If they are detached from the substrate, some sea lilies can move slowly by means of their feather-like arms. All of the approximately 80 living species of sea lilies are found below a depth of 100 meters in the ocean. Sea lilies are the only living echinoderms that are fully sessile.

Feather Stars

In the second group of crinoids, the 520 or so species of feather stars, the disc detaches from the stalk at an early stage of development (figure 47.9). Adult feather stars have long, many-branched arms and usually anchor themselves to their substrate by claw-like structures. However, some feather stars are able to swim for short distances, and many of them can move along the substrate. Feather stars range into shallower water than do sea lilies, and only a few species of either group are found at depths greater than 500 meters. Along with sea cucumbers, crinoids are the most abundant and conspicuous large invertebrates in the warm waters and among the coral reefs of the western Pacific Ocean. They have separate sexes, with the sex organs simple masses of cells in special cavities of the arms and pinnules. Fertilization is usually external, with the male and female gametes shed into the water, but brooding—in which the female shelters the young—occurs occasionally.

Crinoids, the sea lilies and feather stars, were once far more numerous. Crinoids are the only echinoderms attached for much of their lives to the sea bottom.

Class Asteroidea: The Sea Stars

Sea stars, or starfish (class Asteroidea; figure 47.10), are perhaps the most familiar echinoderms. Among the most important predators in many marine ecosystems, they range in size from a centimeter to a meter across. They are abundant in the intertidal zone, but they also occur at depths as great as 10,000 meters. Around 1500 species of sea stars occur throughout the world.

The body of a sea star is composed of a central disc that merges gradually with the arms. Although most sea stars have five arms, the basic symmetry of the phylum, members of some families have many more, typically in multiples of five. The body is somewhat flattened, flexible, and covered with a pigmented epidermis.

Endoskeleton

Beneath the epidermis is an endoskeleton of small calcium-rich plates called ossicles, bound together with connective tissue. From these ossicles project spines that make up the spiny upper surface. Around the base of the spines are minute, pincerlike *pedicellariae*, bearing tiny jaws manipulated by muscles. These keep the body surface free of debris and may aid in food capture.

The Water-Vascular System

A deep groove runs along the oral (bottom) surface of each arm from the central mouth out to the tip of the arm. This groove is bordered by rows of tube feet, which the animal uses to move about. Within each arm, there is a radial canal that connects the tube feet to a ring canal in the central body. This system of piping is used by sea stars to power a unique hydraulic system. Contraction of small chambers called ampullae attached to the tube feet forces water into the podium of the feet, extending them. Conversely, contraction of muscles in the tube foot retracts the podium, forcing fluid back into the ampulla. Small muscles at the end of each tube foot can raise the center of the disklike end, creating suction when the foot is pressed against a substrate. Hundreds of tube feet, moving in unison, pull the arm along the surface.

Feeding

The mouth of a sea star is located in the center of its oral surface. Some sea stars have an extraordinary way of feeding on bivalve mollusks. They can open a small gape between the shells of bivalves by exerting a muscular pull on the shells (figure 47.11). Eventually, muscular fatigue in the bivalve results in a very narrow gape, sufficient enough for the sea star to insert its stomach gape through its mouth into the bivalve. Within the mollusk, the sea star secretes its digestive enzymes and digests the soft tissues of its prey, retracting its stomach when the process is complete.



FIGURE 47.10

Class Asteroidea. This class includes the familiar starfish, or sea stars.



FIGURE 47.11

A sea star attacking a clam. The tube feet, each of which ends in a suction cup, are located along grooves on the underside of the arms.

Reproduction

Most sea stars have separate sexes, with a pair of gonads lying in the ventral region inside each arm. Eggs and sperm are shed into the water so that fertilization is external. In some species, fertilized eggs are brooded in special cavities or simply under the animal. They mature into larvae that swim by means of conspicuous bands of cilia.

Sea stars, also called starfish, are five-armed, mobile predators.

Class Ophiuroidea: The Brittle Stars

Brittle stars (class Ophiuroidea; figure 47.12) are the largest class of echinoderms in numbers of species (about 2000) and they are probably the most abundant also. Secretive, they avoid light and are more active at night.

Brittle stars have slender, branched arms. The most mobile of echinoderms, brittle stars move by pulling themselves along, “rowing” over the substrate by moving their arms, often in pairs or groups, from side to side. Some brittle stars use their arms to swim, a very unusual habit among echinoderms.

Brittle stars feed by capturing suspended microplankton and organic detritus with their tube feet, climbing over objects on the ocean floor. In addition, the tube feet are important sensory organs and assist in directing food into the mouth once the animal has captured it. As implied by their common name, the arms of brittle stars detach easily, a characteristic that helps to protect the brittle stars from their predators.

Like sea stars, brittle stars have five arms. More closely related to the sea stars than to the other classes of the phylum, on closer inspection they are surprisingly different. They have no pedicellariae, as sea stars have, and the groove running down the length of each arm is closed over and covered with ossicles. Their tube feet lack ampullae, have no suckers, and are used for feeding, not locomotion.

Brittle stars usually have separate sexes, with the male and female gametes in most species being released into the water and fusing there. Development takes place in the plankton and the larvae swim and feed using elaborate bands of cilia. Some species brood their young in special cavities and fully developed juvenile brittle stars emerge at the end of development.

Brittle stars, very secretive, pull themselves along with their arms.



FIGURE 47.12
Class Ophiuroidea. Brittle stars crawl actively across their marine substrates.



(a)



(b)

FIGURE 47.13
Class Echinoidea. (a) Sand dollar, *Echinarachnius parma*. (b) Giant red sea urchin, *Strongylocentrotus franciscanus*.

Class Echinoidea: The Sea Urchins and Sand Dollars

The members of the class Echinoidea, sand dollars and sea urchins, lack distinct arms but have the same five-part body plan as all other echinoderms (figure 47.13). Five rows of tube feet protrude through the plates of the calcareous skeleton, and there are also openings for the mouth and anus. These different openings can be seen in the globular skeletons of sea urchins and in the flat skeletons of sand dollars. Both types of endoskeleton, often common along the seashore, consist of fused calcareous plates. About 950 living species constitute the class Echinoidea.

Echinoids walk by means of their tube feet or their movable spines, which are hinged to the skeleton by a joint that makes free rotation possible. Sea urchins and sand dollars move along the sea bottom, feeding on algae and small fragments of organic material. They scrape these off the substrate with the large, triangular teeth that ring their mouths. The gonads of sea urchins are considered a great delicacy by people in different parts of the world. Because of their calcareous plates, sea urchins and sand dollars are well preserved in the fossil record, with more than 5000 additional species described.

As with most other echinoderms, the sexes of sea urchins and sand dollars are separate. The eggs and sperm are shed separately into the water, where they fuse. Some brood their young, and others have free-swimming larvae, with bands of cilia extending onto their long, graceful arms.

Sand dollars and sea urchins lack arms but have a five-part symmetry.

Classes Holothuroidea and Concentricycloidea: Sea Cucumbers and Sea Daisies

Sea Cucumbers

Sea cucumbers (class Holothuroidea) are shaped somewhat like their plant namesakes. They differ from the preceding classes in that they are soft, sluglike organisms, often with a tough, leathery outside skin (figure 47.14). The class consists of about 1500 species found worldwide. Except for a few forms that swim, sea cucumbers lie on their sides at the bottom of the ocean. Their mouth is located at one end and is surrounded by eight to 30 modified tube feet called tentacles; the anus is at the other end. The tentacles around the mouth may secrete mucus, used to capture the small planktonic organisms on which the animals feed. Each tentacle is periodically wiped off within the esophagus and then brought out again, covered with a new supply of mucus.

Sea cucumbers are soft because their calcareous skeletons are reduced to widely separated microscopic plates. These animals have extensive internal branching systems, called respiratory trees, which arise from the **cloaca**, or anal cavity. Water is pulled into and expelled from the respiratory tree by contractions of the cloaca; gas exchange takes place as this process occurs. The sexes of most cucumbers are separate, but some of them are hermaphroditic.

Most kinds of sea cucumbers have tube feet on the body in addition to tentacles. These additional tube feet, which might be restricted to five radial grooves or scattered over the surface of the body, may enable the animals to move about slowly. On the other hand, sea cucumbers may simply wriggle along whether or not they have additional tube feet. Most sea cucumbers are quite sluggish, but some, especially among the deep-sea forms, swim actively. Sea cucumbers, when irritated, sometimes eject a portion of their intestines by a strong muscular contraction that may send the intestinal fragments through the anus or even rupture the body wall.

Sea Daisies

The most recently described class of echinoderms (1986), sea daisies are strange little disc-shaped animals, less than 1 cm in diameter, discovered in waters over 1000 m deep off New Zealand (figure 47.15). Only two species are known so far. They have five-part radial symmetry, but no arms. Their tube feet are located around the periphery of the disc, rather than along radial lines, as in other echinoderms. One species has a shallow, saclike stomach but no intestine or anus; the other species has no digestive tract at all—the surface of its mouth is covered by a membrane through which it apparently absorbs nutrients.



FIGURE 47.14
Class Holothuroidea. Sea cucumber.



FIGURE 47.15
Class Concentricycloidea. Sea daisy.

Sea cucumbers are soft-bodied, sluglike animals without arms. The newly discovered sea daisies are the most mysterious echinoderms. Tiny and simple in form, they live at great depths, absorbing food from their surroundings.



Summary

Questions

Media Resources

47.1 The embryos of deuterostomes develop quite differently from those of protostomes.

- The two major evolutionary lines of coelomate animals—the protostomes and the deuterostomes—are both represented among the oldest known fossils of multicellular animals, dating back some 650 million years.
 - In the protostomes, the mouth develops from or near the blastopore, and the early divisions of the embryo are spiral. At early stages of development, the fate of the individual cells is already determined, and they cannot develop individually into a whole animal.
 - In the deuterostomes, the anus develops from or near the blastopore, and the mouth forms subsequently on another part of the gastrula. The early divisions of the embryo are radial. At early stages of development, each cell of the embryo can differentiate into a whole animal.
1. What patterns of embryonic development related to cleavage and the blastopore occur in protostome coelomates? What patterns occur in deuterostome coelomates?
 2. Which major coelomate phyla are protostomes and which are deuterostomes? How does the early developmental fate of cells differ between the two groups? How is the development of the coelom from mesodermal tissue different between them?

47.2 Echinoderms are deuterostomes with an endoskeleton.

- Echinoderms are exclusively marine deuterostomes that are radially symmetrical as adults.
 - The epidermis of an echinoderm stretches over an endoskeleton made of separate or fused calcium-rich plates.
 - Echinoderms use a unique water-vascular system that includes tube feet for locomotion and feeding.
3. What type of symmetry and body plan do adult echinoderms exhibit?
 4. What is the composition and location of the echinoderm skeleton?
 5. How do echinoderms respire? How developed is their digestive system?



• Echinoderms

47.3 The six classes of echinoderms are all radially symmetrical as adults.

- Crinoids are sessile for some or all of their lives and have a mouth and anus located on the upper surface of the animal.
 - Sea stars are active predators that move about on their tube feet.
 - Brittle stars use their tube feet for feeding and move about using two arms at a time.
 - The endoskeletons of sea urchins and sand dollars consist of fused calcareous plates that have been well preserved in the fossil record.
 - The endoskeletons of sea cucumbers are drastically reduced and separated, making them soft-bodied.
 - Sea daisies are a newly described class of echinoderms with disc-shaped bodies.
6. In what two ways do members of the phylum Echinodermata reproduce? What type of larva do they possess?
 7. How do sea cucumbers superficially differ from other echinoderms? How are some of their tube feet specially modified? What is the extent of their skeleton? What is the function of their unique respiratory tree? How is their reproduction different from that of other echinoderms?

48

Vertebrates

Concept Outline

48.1 Attaching muscles to an internal framework greatly improves movement.

The Chordates. Chordates have an internal flexible rod, the first stage in the evolution of a truly internal skeleton.

48.2 Nonvertebrate chordates have a notochord but no backbone.

The Nonvertebrate Chordates. Lancelets are thought to resemble the ancestors of vertebrates.

48.3 The vertebrates have an interior framework of bone.

Characteristics of Vertebrate. Vertebrates have a true, usually bony endoskeleton, with a backbone encasing the spinal column, and a skull-encased brain.

48.4 The evolution of vertebrates involves invasions of sea, land, and air.

Fishes. Over half of all vertebrate species are fishes, which include the group from which all other vertebrates evolved.

History of the Fishes. Swim bladders have made bony fishes a particularly successful group.

Amphibians. The key innovation that made life on land possible for vertebrates was the pulmonary vein.

History of the Amphibians. In the past, amphibians were far more diverse, and included many large, armored terrestrial forms.

Reptiles. Reptiles were the first vertebrates to completely master the challenge of living on dry land.

The Rise and Fall of Dominant Reptile Groups. Now-extinct forms of reptiles dominated life on land for 250 million years. Four orders survive today.

Birds. Birds are much like reptiles, but with feathers.

History of the Birds. Birds are thought to have evolved from dinosaurs with adaptations of feathers and flight.

Mammals. Mammals are the only vertebrates that possess hair and milk glands.

History of the Mammals. Mammals evolved at the same time as dinosaurs, but only became common when dinosaurs disappeared.



FIGURE 48.1

A typical vertebrate. Today mammals, like this snow leopard, *Panthera uncia*, dominate vertebrate life on land, but for over 200 million years in the past they were a minor group in a world dominated by reptiles.

Members of the phylum Chordata (figure 48.1) exhibit great improvements in the endoskeleton over what is seen in echinoderms. As we saw in the previous chapter, the endoskeleton of echinoderms is functionally similar to the exoskeleton of arthropods; it is a hard shell that encases the body, with muscles attached to its inner surface. Chordates employ a very different kind of endoskeleton, one that is truly internal. Members of the phylum Chordata are characterized by a flexible rod that develops along the back of the embryo. Muscles attached to this rod allowed early chordates to swing their backs from side to side, swimming through the water. This key evolutionary advance, attaching muscles to an internal element, started chordates along an evolutionary path that led to the vertebrates—and, for the first time, to truly large animals.

48.1 Attaching muscles to an internal framework greatly improves movement.

The Chordates

Chordates (phylum Chordata) are deuterostome coelomates whose nearest relations in the animal kingdom are the echinoderms, the only other deuterostomes. However, unlike echinoderms, chordates are characterized by a *notochord*, *jointed appendages*, and *segmentation*. There are some 43,000 species of chordates, a phylum that includes birds, reptiles, amphibians, fishes, and mammals.

Four features characterize the chordates and have played an important role in the evolution of the phylum (figure 48.2):

1. A single, hollow **nerve cord** runs just beneath the dorsal surface of the animal. In vertebrates, the dorsal nerve cord differentiates into the brain and spinal cord.
2. A flexible rod, the **notochord**, forms on the dorsal side of the primitive gut in the early embryo and is present at some developmental stage in all chordates. The notochord is located just below the nerve cord. The notochord may persist throughout the life cycle of some chordates or be displaced during embryological development as in most vertebrates by the vertebral column that forms around the nerve cord.
3. **Pharyngeal slits** connect the **pharynx**, a muscular tube that links the mouth cavity and the esophagus, with the outside. In terrestrial vertebrates, the slits do not actually connect to the outside and are better termed pharyngeal pouches. Pharyngeal pouches are present in the embryos of all vertebrates. They become slits, open to the outside in animals with gills, but disappear in those lacking gills. The presence of these structures in all vertebrate embryos provides evidence to their aquatic ancestry.
4. Chordates have a **postanal tail** that extends beyond the anus, at least during their embryonic development. Nearly all other animals have a terminal anus.

All chordates have all four of these characteristics at some time in their lives. For example, humans have pharyngeal slits, a dorsal nerve cord, and a notochord as embryos. As adults, the nerve cord remains while the notochord is replaced by the vertebral column and all but one pair of pharyngeal slits are lost. This remaining pair forms the Eustachian tubes that connect the throat to the middle ear.

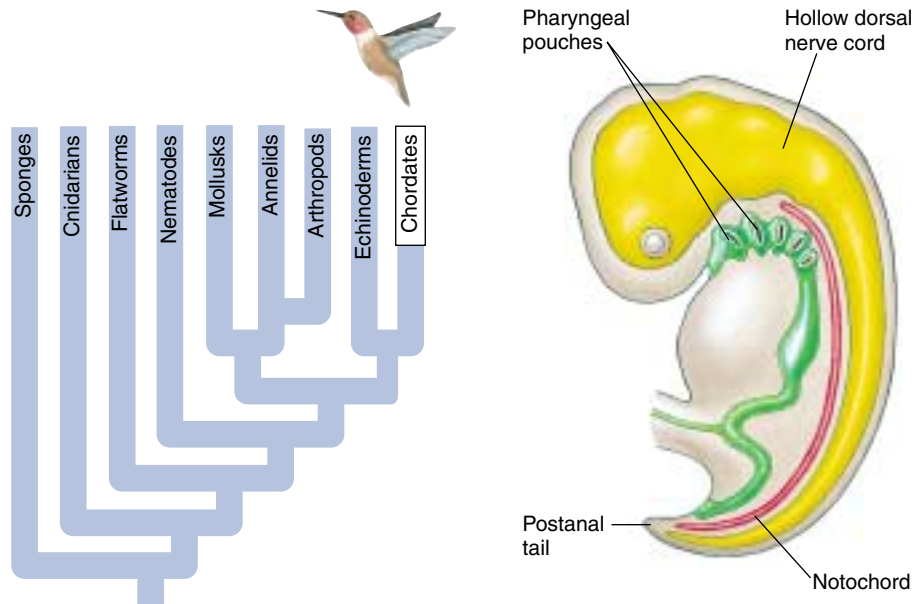


FIGURE 48.2
Some of the principal features of the chordates, as shown in a generalized embryo.



FIGURE 48.3
A mouse embryo. At 11.5 days of development, the mesoderm is already divided into segments called somites (stained dark in this photo), reflecting the fundamentally segmented nature of all chordates.

PHYLUM CHORDATA: Notochord

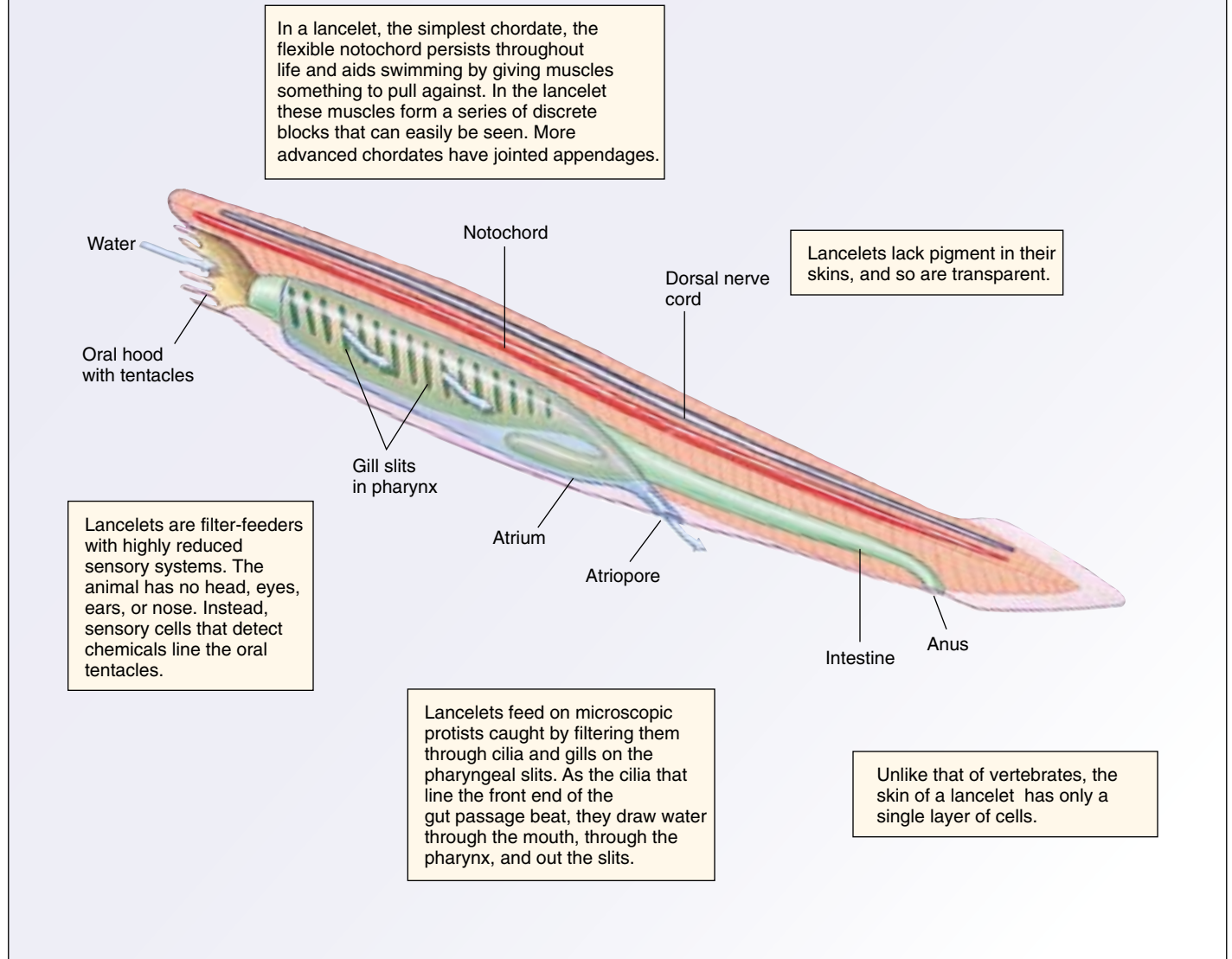


FIGURE 48.4

Evolution of a notochord. Vertebrates, tunicates, and lancelets are chordates (phylum Chordata), coelomate animals with a flexible rod, the notochord, that provides resistance to muscle contraction and permits rapid lateral body movements. Chordates also possess pharyngeal slits (reflecting their aquatic ancestry and present habitat in some) and a hollow dorsal nerve cord. In vertebrates, the notochord is replaced during embryonic development by the vertebral column.

A number of other characteristics also distinguish the chordates fundamentally from other animals. Chordates' muscles are arranged in segmented blocks that affect the basic organization of the chordate body and can often be clearly seen in embryos of this phylum (figure 48.3). Most chordates have an internal skeleton against which the muscles work. Either this internal skeleton or the notochord (figure 48.4) makes possible the extraordinary

powers of locomotion that characterize the members of this group.

Chordates are characterized by a hollow dorsal nerve cord, a notochord, pharyngeal gill slits, and a postanal tail at some point in their development. The flexible notochord anchors internal muscles and allows rapid, versatile movement.

48.2 Nonvertebrate chordates have a notochord but no backbone.

The Nonvertebrate Chordates

Tunicates

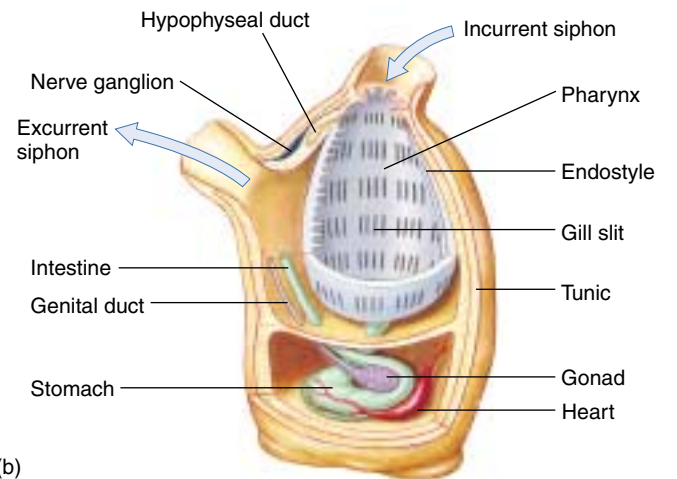
The tunicates (subphylum Urochordata) are a group of about 1250 species of marine animals. Most of them are sessile as adults (figure 48.5*a,b*), with only the larvae having a notochord and nerve cord. As adults, they exhibit neither a major body cavity nor visible signs of segmentation. Most species occur in shallow waters, but some are found at great depths. In some tunicates, adults are colonial, living in masses on the ocean floor. The pharynx is lined with numerous cilia, and the animals obtain their food by ciliary action. The cilia beat, drawing a stream of water into the pharynx, where microscopic food particles are trapped in a mucous sheet secreted from a structure called an *endostyle*.

The tadpolelike larvae of tunicates plainly exhibit all of the basic characteristics of chordates and mark the tunicates as having the most primitive combination of features found in any chordate (figure 48.5*c*). The larvae do not feed and have a poorly developed gut. They remain free-swimming for only a few days before settling to the bottom and attaching themselves to a suitable substrate by means of a sucker.

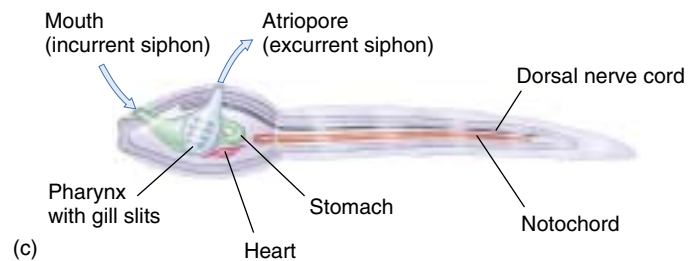
Tunicates change so much as they mature and adjust developmentally to a sessile, filter-feeding existence that it would be difficult to discern their evolutionary relationships by examining an adult. Many adult tunicates secrete a **tunic**, a tough sac composed mainly of cellulose. The tunic surrounds the animal and gives the subphylum its name. Cellulose is a substance frequently found in the cell walls of plants and algae but is rarely found in ani-



(a)



(b)



(c)

FIGURE 48.5

Tunicates (phylum Chordata, subphylum Urochordata).

(a) The sea peach, *Halocynthia aurantiu*. (b) Diagram of the structure of an adult tunicate. (c) Diagram of the structure of a larval tunicate, showing the characteristic tadpolelike form. Larval tunicates resemble the postulated common ancestor of the chordates.

mals. In colonial tunicates, there may be a common sac and a common opening to the outside. There is a group of Urochordates, the Larvacea, which retains the tail and notochord into adulthood. One theory of vertebrate origins involves a larval form, perhaps that of a tunicate, which acquires the ability to reproduce.

Lancelets

Lancelets are scaleless, fishlike marine chordates a few centimeters long that occur widely in shallow water throughout the oceans of the world. Lancelets (subphylum Cephalochordata) were given their English name because they resemble a lancet—a small, two-edged surgical knife. There are about 23 species of this subphylum. Most of them belong to the genus *Branchiostoma*, formerly called *Amphioxus*, a name still used widely. In lancelets, the notochord runs the entire length of the dorsal nerve cord and persists throughout the animal's life.

Lancelets spend most of their time partly buried in sandy or muddy substrates, with only their anterior ends protruding (figure 48.6). They can swim, although they rarely do so. Their muscles can easily be seen as a series of discrete blocks. Lancelets have many more pharyngeal gill slits than fishes, which they resemble in overall shape. They lack pigment in their skin, which has only a single layer of cells, unlike the multilayered skin of vertebrates. The lancelet body is pointed at both ends. There is no distinguishable head or sensory structures other than pigmented light receptors.

Lancelets feed on microscopic plankton, using a current created by beating cilia that lines the oral hood, pharynx, and gill slits (figure 48.7). The gill slits provide an exit for the water and are an adaptation for filter feeding. The oral hood projects beyond the mouth and bears sensory tentacles, which also ring the mouth. Males and females are separate, but no obvious external differences exist between them.

Biologists are not sure whether lancelets are primitive or are actually degenerate fishes whose structural features have been reduced and simplified during the course of evolution. The fact that lancelets feed by means of cilia and have a single-layered skin, coupled with distinctive features of their excretory systems, suggest that this is an ancient



FIGURE 48.6

Lancelets. Two lancelets, *Branchiostoma lanceolatum* (phylum Chordata, subphylum Cephalochordata), partly buried in shell gravel, with their anterior ends protruding. The muscle segments are clearly visible; the square objects along the side of the body are gonads, indicating that these are male lancelets.

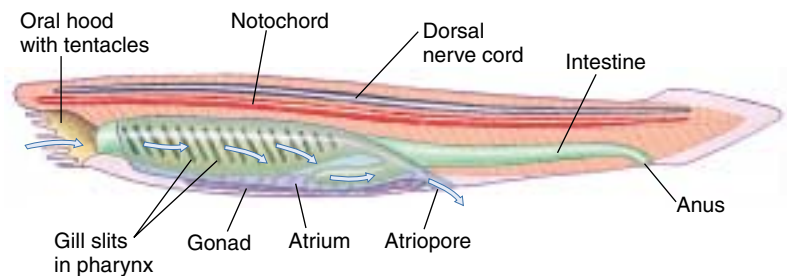


FIGURE 48.7

The structure of a lancelet. This diagram shows the path through which the lancelet's cilia pull water.

group of chordates. The recent discovery of fossil forms similar to living lancelets in rocks 550 million years old—well before the appearance of any fishes—also argues for the antiquity of this group. Recent studies by molecular systematists further support the hypothesis that lancelets are vertebrates' closest ancestors.

Nonvertebrate chordates, including tunicates and lancelets, have notochords but not vertebrae. They are the closest relatives of vertebrates.

48.3 The vertebrates have an interior framework of bone.

Characteristics of Vertebrates

Vertebrates (subphylum Vertebrata) are chordates with a spinal column. The name *vertebrate* comes from the individual bony segments called vertebrae that make up the spine. Vertebrates differ from the tunicates and lancelets in two important respects:

1. **Vertebral column.** In vertebrates, the notochord is replaced during the course of embryonic development by a bony vertebral column. The column is a series of bones that encloses and protects the dorsal nerve cord like a sleeve (figure 48.8).
2. **Head.** In all vertebrates but the earliest fishes, there is a distinct and well-differentiated head, with a skull and brain. For this reason, the vertebrates are sometimes called the **craniate chordates** (Greek *kranion*, “skull”).

In addition to these two key characteristics, vertebrates differ from other chordates in other important respects:

1. **Neural crest.** A unique group of embryonic cells called the neural crest contributes to the development of many vertebrate structures. These cells develop on the crest of the neural tube as it forms by an invagination and pinching together of the neural plate (see chapter 60 for a detailed account). Neural crest cells then migrate to various locations in the developing embryo, where they participate in the development of a variety of structures.
2. **Internal organs.** Among the internal organs of vertebrates, livers, kidneys, and endocrine glands are characteristic of the group. The ductless endocrine glands secrete hormones that help regulate many of the body's functions. All vertebrates have a heart and a closed circulatory system. In both their circulatory and their excretory functions, vertebrates differ markedly from other animals.
3. **Endoskeleton.** The endoskeleton of most vertebrates is made of cartilage or bone. Cartilage and bone are specialized tissue containing fibers of the protein collagen compacted together. Bone also contains crystals of a calcium phosphate salt. Bone forms in two stages. First, collagen is laid down in a matrix of fibers along stress lines to provide flexibility, and then calcium minerals infiltrate the fibers, providing rigidity. The great advantage of bone over chitin as a structural material is that bone is strong without being brittle. The vertebrate endoskeleton makes possible the great size and extraordinary powers of movement that characterize this group.

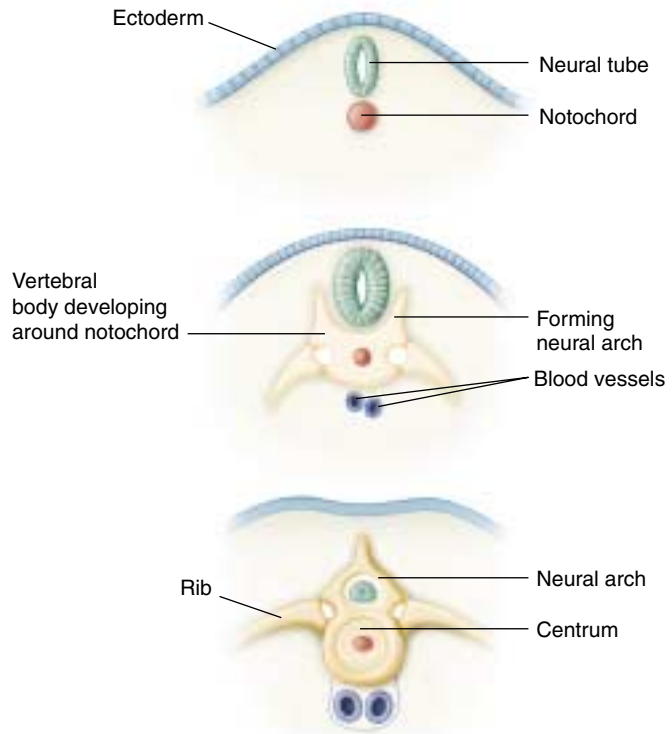


FIGURE 48.8

Embryonic development of a vertebra. During the course of evolution of animal development, the flexible notochord is surrounded and eventually replaced by a cartilaginous or bony covering, the centrum. The neural tube is protected by an arch above the centrum, and the vertebra may also have a hemal arch, which protects major blood vessels below the centrum. The vertebral column functions as a strong, flexible rod that the muscles pull against when the animal swims or moves.

Overview of the Evolution of Vertebrates

The first vertebrates evolved in the oceans about 470 million years ago. They were jawless fishes with a single caudal fin. Many of them looked like a flat hot dog, with a hole at one end and a fin at the other. The appearance of a hinged jaw was a major advancement, opening up new food options, and jawed fishes became the dominant creatures in the sea. Their descendants, the amphibians, invaded the land. Salamander-like amphibians and other, much larger now-extinct amphibians were the first vertebrates to live successfully on land. Amphibians, in turn, gave rise to the first reptiles about 300 million years ago. Within 50 million years, reptiles, better suited than amphibians to living out of water, replaced them as the dominant land vertebrates.

With the success of reptiles, vertebrates truly came to dominate the surface of the earth. Many kinds of reptiles evolved, ranging in size from smaller than a chicken to big-

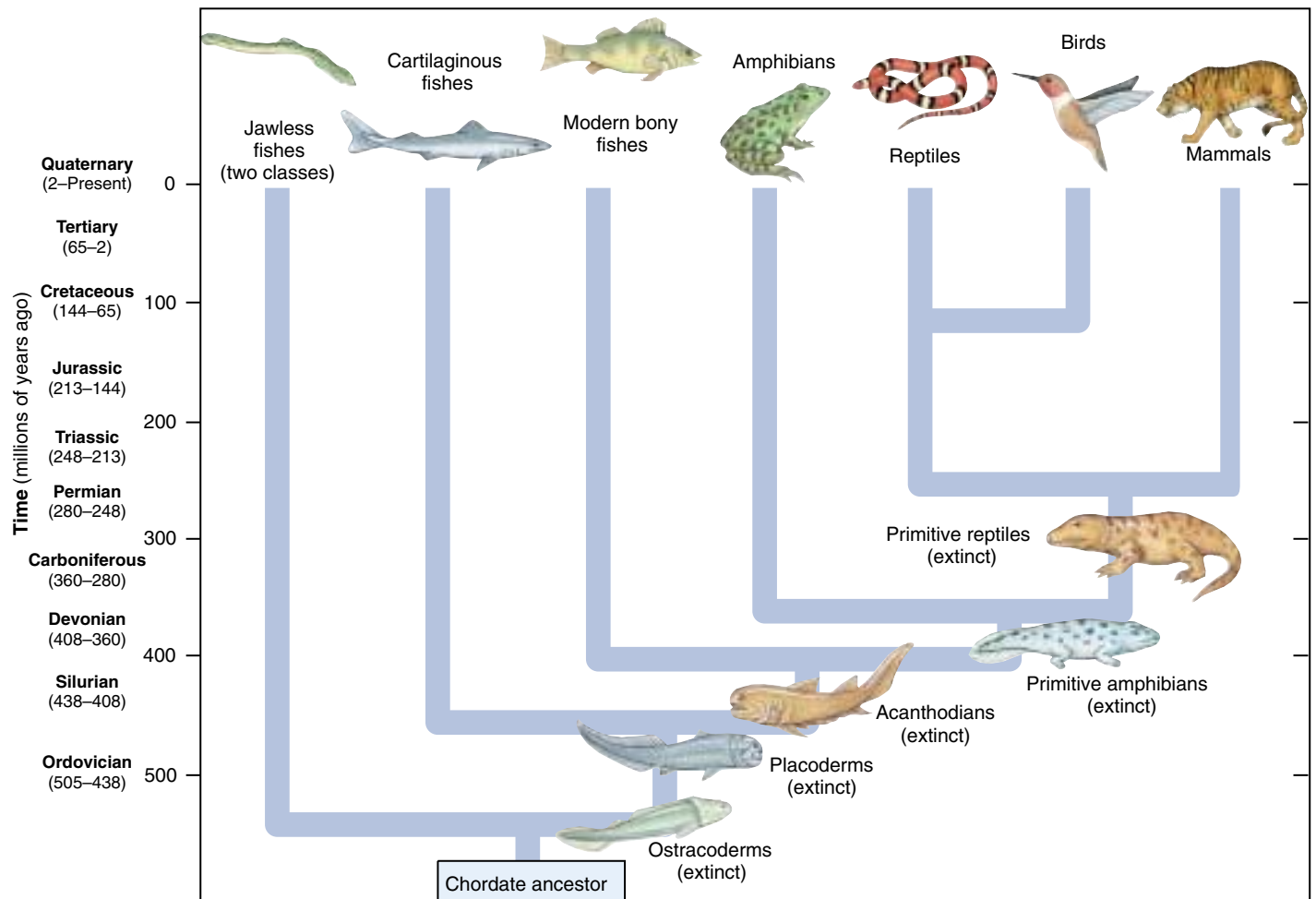


FIGURE 48.9

Vertebrate family tree. Two classes of vertebrates comprise the Agnatha, or jawless fishes. Primitive amphibians arose from fish. Primitive reptiles arose from amphibians and gave rise to mammals and to dinosaurs, which survive today as birds.

ger than a truck. Some flew, and others swam. Among them evolved reptiles that gave rise to the two remaining great lines of terrestrial vertebrates, birds (descendants of the dinosaurs) and mammals. Dinosaurs and mammals appear at about the same time in the fossil record, 220 million years ago. For over 150 million years, dinosaurs dominated the face of the earth. Over all these centuries (think of it—over *a million centuries!*) the largest mammal was no bigger than a cat. Then, about 65 million years ago, the dinosaurs abruptly disappeared, for reasons that are still hotly debated. In their absence, mammals and birds quickly took their place, becoming in turn abundant and diverse.

The history of vertebrates has been a series of evolutionary advances that have allowed vertebrates to first invade the sea and then the land. In this chapter, we will examine the key evolutionary advances that permitted vertebrates to invade the land successfully. As you will see, this invasion was a staggering evolutionary achievement, involving fundamental changes in many body systems.

Vertebrates are a diverse group, containing members adapted to life in aquatic habitats, on land, and in the air. There are eight principal classes of living vertebrates (figure 48.9). Four of the classes are fishes that live in the water, and four are land-dwelling **tetrapods**, animals with four limbs. (The name *tetrapod* comes from two Greek words meaning “four-footed.”) The extant classes of fishes are the superclass Agnatha (the jawless fishes), which includes the class Myxini, the hagfish, and the class Cephalaspidomorphi, the lampreys; Chondrichthyes, the cartilaginous fishes, sharks, skates, and rays; and Osteichthyes, the bony fishes that are dominant today. The four classes of tetrapods are Amphibia, the amphibians; Reptilia, the reptiles; Aves, the birds; and Mammalia, the mammals.

Vertebrates, the principal chordate group, are characterized by a vertebral column and a distinct head.

48.4 The evolution of vertebrates involves invasions of sea, land, and air.

Fishes

Over half of all vertebrates are fishes. The most diverse and successful vertebrate group (figure 48.10), they provided the evolutionary base for invasion of land by amphibians. In many ways, amphibians, the first terrestrial vertebrates, can be viewed as transitional—fish out of water. In fact, fishes and amphibians share many similar features, among the host of obvious differences. First, let us look at the fishes (table 48.1).

The story of vertebrate evolution started in the ancient seas of the Cambrian Period (570 to 505 million years ago), when the first backboned animals appeared (figure 48.11). Wriggling through the water, jawless and toothless, these first fishes sucked up small food particles from the ocean floor like miniature vacuum cleaners. Most were less than a foot long, respired with gills, and had no paired fins—just a primitive tail to push them through the water. For 50 million years, during the Ordovician Period (505 to 438 million years ago), these simple fishes were the only vertebrates. By the end of this period, fish had developed primitive fins to help them swim and massive shields of



FIGURE 48.10 Fish are diverse and include more species than all other kinds of vertebrates combined.

bone for protection. Jawed fishes first appeared during the Silurian Period (438 to 408 million years ago) and along with them came a new mode of feeding. Later, both the cartilaginous and bony fishes appeared.

Table 48.1 Major Classes of Fishes				
Class	Typical Examples		Key Characteristics	Approximate Number of Living Species
Placodermi	Armored fishes		Jawed fishes with heavily armored heads; often quite large	Extinct
Acanthodii	Spiny fishes		Fishes with jaws; all now extinct; paired fins supported by sharp spines	Extinct
Osteichthyes	Ray-finned fishes		Most diverse group of vertebrates; swim bladders and bony skeletons; paired fins supported by bony rays	20,000
	Lobe-finned fishes		Largely extinct group of bony fishes; ancestral to amphibians; paired lobed fins	7
Chondrichthyes	Sharks, skates, rays		Streamlined hunters; cartilaginous skeletons; no swim bladders; internal fertilization	850
Myxini	Hagfishes		Jawless fishes with no paired appendages; scavengers; mostly blind, but a well-developed sense of smell	43
Cephalaspidomorphi	Lampreys		Largely extinct group of jawless fishes with no paired appendages; parasitic and nonparasitic types; all breed in fresh water	17

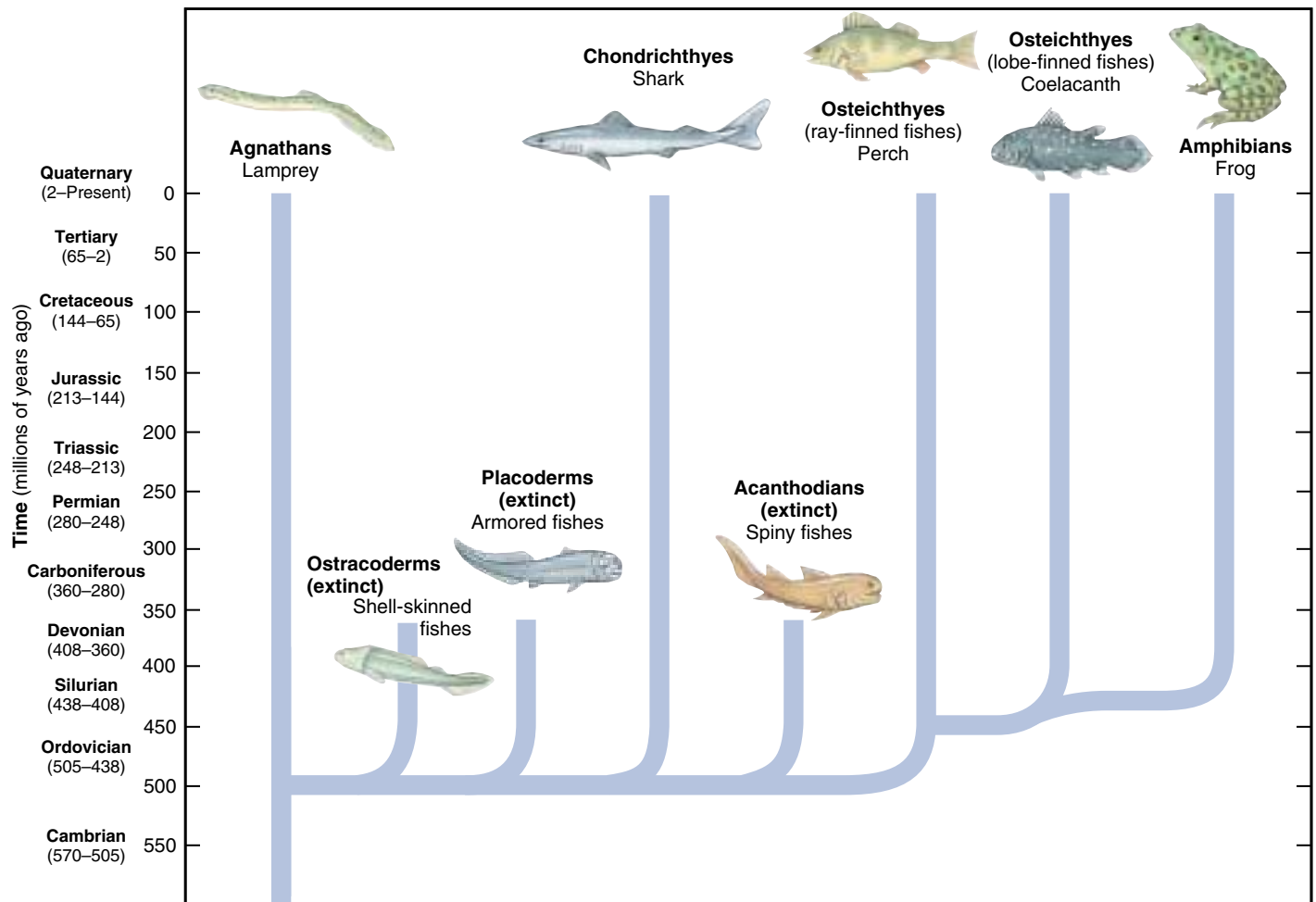


FIGURE 48.11

Evolution of the fishes. The evolutionary relationships among the different groups of fishes as well as between fishes and amphibians is shown. The spiny and armored fishes that dominated the early seas are now extinct.

Characteristics of Fishes

From whale sharks that are 18 meters long to tiny cichlids no larger than your fingernail, fishes vary considerably in size, shape, color, and appearance. Some live in freezing Arctic seas, others in warm freshwater lakes, and still others spend a lot of time out of water entirely. However varied, all fishes have important characteristics in common:

1. **Gills.** Fishes are water-dwelling creatures and must extract oxygen dissolved in the water around them. They do this by directing a flow of water through their mouths and across their gills. The gills are composed of fine filaments of tissue that are rich in blood vessels. They are located at the back of the pharynx and are supported by arches of cartilage. Blood moves through the gills in the opposite direction to the flow of water in order to maximize the efficiency of oxygen absorption.

2. **Vertebral column.** All fishes have an internal skeleton with a spine surrounding the dorsal nerve cord, although it may not necessarily be made of bone. The brain is fully encased within a protective box, the skull or cranium, made of bone or cartilage.
3. **Single-loop blood circulation.** Blood is pumped from the heart to the gills. From the gills, the oxygenated blood passes to the rest of the body, then returns to the heart. The heart is a muscular tube-pump made of four chambers that contract in sequence.
4. **Nutritional deficiencies.** Fishes are unable to synthesize the aromatic amino acids and must consume them in their diet. This inability has been inherited by all their vertebrate descendants.

Fishes were the first vertebrates to make their appearance, and today they are still the largest vertebrate class. They are the vertebrate group from which all other vertebrates evolved.

History of the Fishes

The First Fishes

The first fishes were members of the five Ostracoderm orders (the word means “shell-skinned”). Only their head-shields were made of bone; their elaborate internal skeletons were constructed of cartilage. Many ostracoderms were bottom dwellers, with a jawless mouth underneath a flat head, and eyes on the upper surface. Ostracoderms thrived in the Ordovician Period and in the period which followed, the Silurian Period (438 to 408 million years ago), only to become almost completely extinct at the close of the following Devonian Period (408 to 360 million years ago). One group, the jawless Agnatha, survive today as hagfish and parasitic lampreys (figure 48.12).

A fundamentally important evolutionary advance occurred in the late Silurian Period, 410 million years ago—the development of jaws. Jaws evolved from the most anterior of a series of arch-supports made of cartilage that were used to reinforce the tissue between gill slits, holding the slits open (figure 48.13). This transformation was not as radical as it might at first appear. Each gill arch was formed by a series of several cartilages (later to become bones) arranged somewhat in the shape of a V turned on its side, with the point directed outward. Imagine the fusion of the front pair of arches at top and bottom, with hinges at the points, and you have the primitive vertebrate jaw. The top half of the jaw is not attached to the skull directly except at the rear. Teeth developed on the jaws from modified scales on the skin that lined the mouth.

Armored fishes called placoderms and spiny fishes called acanthodians both had jaws. Spiny fishes were very common during the early Devonian, largely replacing ostracoderms, but became extinct themselves at the close of the Permian. Like ostracoderms, they had internal skeletons made of cartilage, but their scales contained small plates of bone, foreshadowing the much larger role bone would play in the future of vertebrates. Spiny fishes were predators and far better swimmers than ostracoderms, with as many as seven fins to aid them swimming. All of these fins were reinforced with strong spines, giving these fishes their name. No spiny fishes survive today.

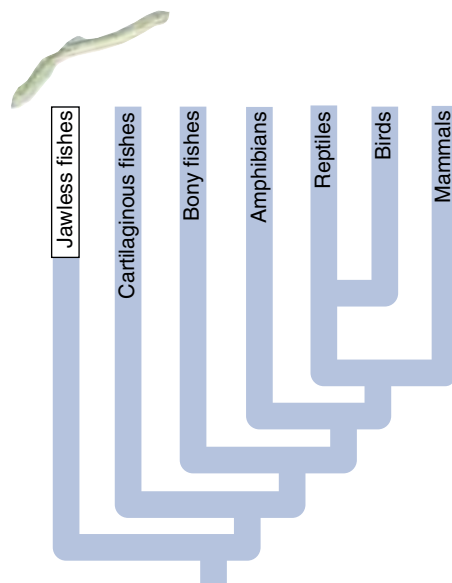


FIGURE 48.12
Specialized mouth of a lamprey.

Lampreys use their suckerlike mouths to attach themselves to the fishes on which they prey. When they have done so, they bore a hole in the fish with their teeth and feed on its blood.

By the mid-Devonian, the heavily armored placoderms became common. A very diverse and successful group, seven orders of placoderms dominated the seas of the late Devonian, only to become extinct at the end of that period. The front of the placoderm body was more heavily armored than the rear. The placoderm jaw was much improved from the primitive jaw of spiny fishes, with the upper jaw fused to the skull and the skull hinged on the shoulder. Many of the placoderms grew to enormous sizes, some over 30 feet long, with two-foot skulls that had an enormous bite.

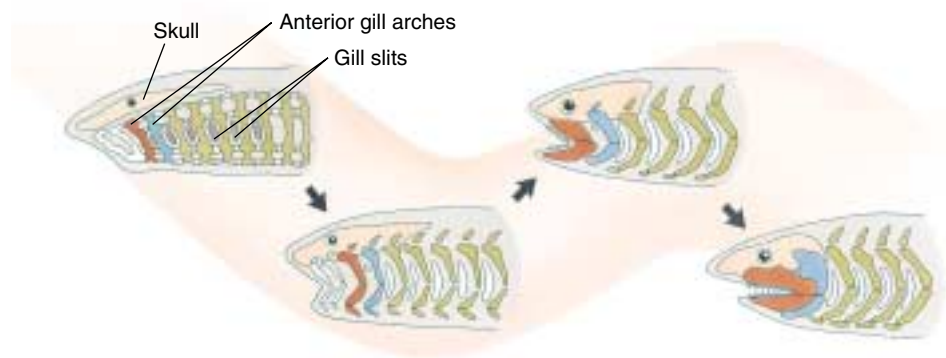


FIGURE 48.13
Evolution of the jaw. Jaws evolved from the anterior gill arches of ancient, jawless fishes.

The Rise of Active Swimmers

At the end of the Devonian, essentially all of these pioneer vertebrates disappeared, replaced by sharks and bony fishes. Sharks and bony fishes first evolved in the early Devonian, 400 million years ago. In these fishes, the jaw was improved even further, with the first gill arch behind the jaws being transformed into a supporting strut or prop, joining the rear of the lower jaw to the rear of the skull. This allowed the mouth to open very wide, into almost a full circle. In a great white shark, this wide-open mouth can be a very efficient weapon.

The major factor responsible for the replacement of primitive fishes by sharks and bony fishes was that they had a superior design for swimming. The typical shark and bony fish is streamlined. The head of the fish acts as a wedge to cleave through the water, and the body tapers back to the tail, allowing the fish to slip through the water with a minimum amount of turbulence.

In addition, sharks and bony fishes have an array of mobile fins that greatly aid swimming. First, there is a propulsion fin: a large and efficient tail (caudal) fin that helps drive the fish through the water when it is swept side-to-side, pushing against the water and thrusting the fish forward. Second, there are stabilizing fins: one (or sometimes two) dorsal fins on the back that act as a stabilizer to prevent rolling as the fish swims through the water, while another ventral fin acts as a keel to prevent side-slip. Third, there are the paired fins at shoulder and hip (“A fin at each corner”), consisting of a front (pectoral) pair and a rear (pelvic) pair. These fins act like the elevator flaps of an airplane to assist the fish in going up or down through the water, as rudders to help it turn sharply left or right, and as brakes to help it stop quickly.

Sharks Become Top Predators

In the period following the Devonian, the Carboniferous Period (360 to 280 million years ago), sharks became the dominant predator in the sea. Sharks (class Chondrichthyes) have a skeleton made of cartilage, like primitive fishes, but it is “calcified,” strengthened by granules of calcium carbonate deposited in the outer layers of cartilage. The result is a very light and strong skeleton. Streamlined, with paired fins and a light, flexible skeleton, sharks are superior swimmers (figure 48.14). Their pectoral fins are particularly large, jutting out stiffly like airplane wings—and that is how they function, adding lift to compensate for the downward thrust of the tail fin. Very aggressive predators, some sharks reached enormous size.

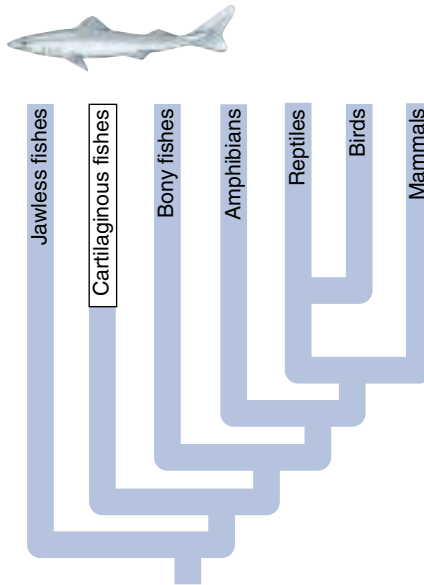


FIGURE 48.14

Chondrichthyes. Members of the class Chondrichthyes, such as this bull shark, are mainly predators or scavengers and spend most of their time in graceful motion. As they move, they create a flow of water past their gills, extracting oxygen from the water.

Sharks were among the first vertebrates to develop teeth. These teeth evolved from rough scales on the skin and are not set into the jaw, as yours are, but rather sit atop it. The teeth are not firmly anchored and are easily lost. In a shark’s mouth, the teeth are arrayed in up to 20 rows, the teeth in front doing the biting and cutting, while behind them other teeth grow and await their turn. When a tooth breaks or is worn down, a replacement from the next row moves forward. One shark may eventually use more than 20,000 teeth. This programmed loss of teeth offers a great advantage: the teeth in use are always new and sharp. The skin is covered with tiny teeth-like scales, giving it a rough “sandpaper” texture. Like the teeth, these scales are constantly replaced throughout the shark’s life.

Reproduction among the Chondrichthyes is the most advanced of any fishes. Shark eggs are fertilized internally. During mating, the male grasps the female with modified fins called claspers. Sperm run from the male into the female through grooves in the claspers. Although a few species lay fertilized eggs, the eggs of most species develop within the female’s body, and the pups are born alive.

Many of the early evolutionary lines of sharks died out during the great extinction at the end of the Permian Period (280 to 248 million years ago). The survivors thrived and underwent a burst of diversification during the Mesozoic era, when most of the modern groups of sharks appeared. Skates and rays (flattened sharks that are bottom-dwellers) evolved at this time, some 200 million years after the sharks first appeared. Sharks competed successfully with the marine reptiles of that time and are still the dominant predators of the sea. Today there are 275 species of sharks, more kinds than existed in the Carboniferous.

Bony Fishes Dominate the Water

Bony fishes (members of the class Osteichthyes, figure 48.15) evolved at the same time as sharks, some 400 million years ago, but took quite a different evolutionary road. Instead of gaining speed through lightness, as sharks did, bony fishes adopted a heavy internal skeleton made completely of bone. Such an internal skeleton is very strong, providing a base against which very strong muscles could pull. The process of *ossification* (the evolutionary replacement of cartilage by bone) happened suddenly in evolutionary terms, completing a process started by sharks, who lay down a thin film of bone over their cartilage. Not only is the internal skeleton ossified, but also the external skeleton, the outer covering of plates and scales. Many scientists believe bony fishes evolved from spiny sharks, which also had bony plates set in their skin. Bony fishes are the most successful of all fishes, indeed of all vertebrates. There are several dozen orders containing more than 20,000 living species.

Unlike sharks, bony fishes evolved in fresh water. The most ancient fossils of bony fishes are found in freshwater lake beds from the middle Devonian. These first bony fishes were small and possessed paired air sacs connected to the back of the throat. These sacs could be inflated with air to buoy the fish up or deflated to sink it down in the water.

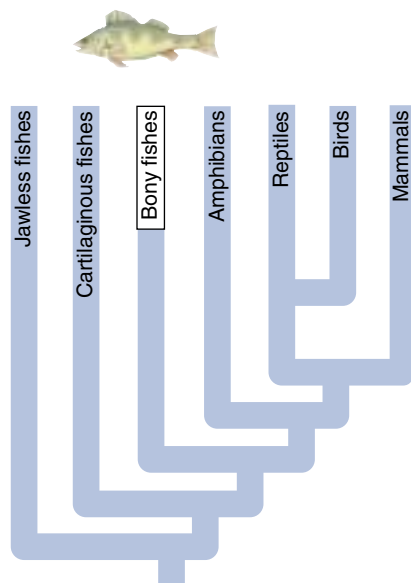
Most bony fishes have highly mobile fins, very thin scales, and completely symmetrical tails (which keep the fish on a straight course as it swims through the water). This is a very successful design for a fish. Two great groups arose from these pioneers: the lobe-finned fishes, ancestors of the first tetrapods, and the ray-finned fishes, which include the vast majority of today's fishes.

The characteristic feature of all ray-finned fishes is an internal skeleton of parallel bony rays that support and stiffen each fin. There are no muscles within the fins; they are moved by muscles within the body. In ray-finned fishes, the primitive air sacs are transformed into an air pouch, which provides a remarkable degree of control over buoyancy.

Important Adaptations of Bony Fishes

The remarkable success of the bony fishes has resulted from a series of significant adaptations that have enabled them to dominate life in the water. These include the swim bladder, lateral line system, and gill cover.

Swim Bladder. Although bones are heavier than cartilaginous skeletons, bony fishes are still buoyant because



they possess a swim bladder, a gas-filled sac that allows them to regulate their buoyant density and so remain suspended at any depth in the water effortlessly (figure 48.16). Sharks, by contrast, must move through the water or sink, as their bodies are denser than water. In primitive bony fishes, the swim bladder is a ventral outpocketing of the pharynx behind the throat, and these species fill the swim bladder by simply gulping air at the surface of the water. In most of today's bony fishes, the swim bladder is an independent organ that is filled and drained of gases, mostly nitrogen and oxygen, internally. How do bony fishes manage this remarkable trick? It turns out that the gases are released from their blood. Gas exchange occurs across the wall of the swim bladder and the blood

vessels located near the swim bladder. A variety of physiological factors controls the exchange of gases between the blood stream and the swim bladder.

Lateral Line System. Although precursors are found in sharks, bony fishes possess a fully developed lateral line system. The lateral line system consists of a series of sensory organs that project into a canal beneath the surface of the skin. The canal runs the length of the fish's body and is open to the exterior through a series of sunken pits. Move-



FIGURE 48.15

Bony fishes. The bony fishes (class Osteichthyes) are extremely diverse. This Korean angelfish in Fiji is one of the many striking fishes that live around coral reefs in tropical seas.

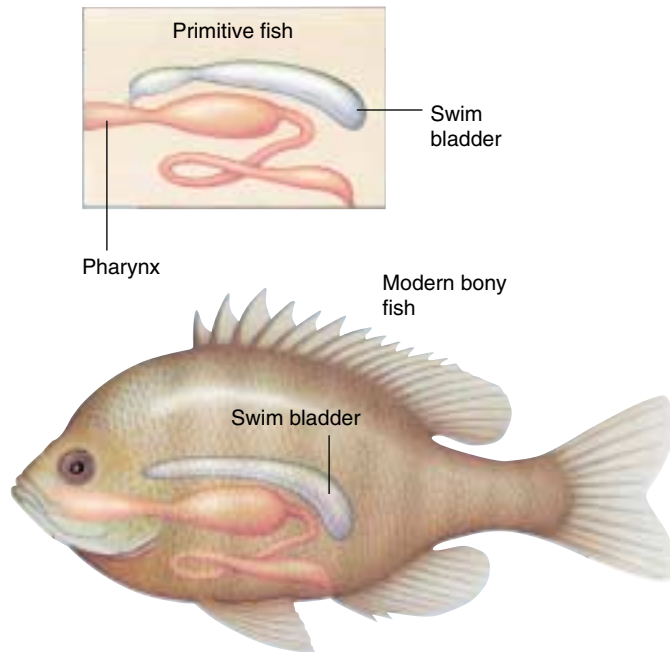


FIGURE 48.16

Diagram of a swim bladder. The bony fishes use this structure, which evolved as a ventral outpocketing of the pharynx, to control their buoyancy in water.

ment of water past the fish forces water through the canal. The sensory organs consist of clusters of cells with hairlike projections called cilia, embedded in a gelatinous cap. The hairs are deflected by the slightest movement of water over them. The pits are oriented so that some are stimulated no matter what direction the water moves (see chapter 55). Nerve impulses from these sensory organs permit the fish to assess its rate of movement through water, sensing the movement as pressure waves against its lateral line. This is how a trout orients itself with its head upstream.

The lateral line system also enables a fish to detect motionless objects at a distance by the movement of water reflected off the object. In a very real sense, this is the fish equivalent of hearing. The basic mechanism of cilia deflection by pressure waves is very similar to what happens in human ears (see chapter 55).

Gill Cover. Most bony fishes have a hard plate called the operculum that covers the gills on each side of the head. Flexing the operculum permits bony fishes to pump water over their gills. The gills are suspended in the pharyngeal slits that form a passageway between the pharynx and the outside of the fish's body. When the operculum is closed, it seals off the exit. When the mouth is open, closing the operculum increases the volume of the mouth cavity, so that water is drawn into the mouth. When the mouth is closed, opening the operculum decreases the volume of the mouth



FIGURE 48.17

The living coelacanth, *Latimeria chalumnae*. Discovered in the western Indian Ocean in 1938, this coelacanth represents a group of fishes that had been thought to be extinct for about 70 million years. Scientists who studied living individuals in their natural habitat at depths of 100 to 200 meters observed them drifting in the current and hunting other fishes at night. Some individuals are nearly 3 meters long; they have a slender, fat-filled swim bladder. *Latimeria* is a strange animal, and its discovery was a complete surprise.

cavity, forcing water past the gills to the outside. Using this very efficient bellows, bony fishes can pass water over the gills while stationary in the water. That is what a goldfish is doing when it seems to be gulping in a fish tank.

The Path to Land

Lobe-finned fishes (figure 48.17) evolved 390 million years ago, shortly after the first bony fishes appeared. Only seven species survive today, a single species of coelacanth and six species of lungfish. Lobe-finned fishes have paired fins that consist of a long fleshy muscular lobe (hence their name), supported by a central core of bones that form fully articulated joints with one another. There are bony rays only at the tips of each lobed fin. Muscles within each lobe can move the fin rays independently of one another, a feat no ray-finned fish could match. Although rare today, lobe-finned fishes played an important part in the evolutionary story of vertebrates. Amphibians almost certainly evolved from the lobe-finned fishes.

Fishes are characterized by gills and a simple, single-loop circulatory system. Cartilaginous fishes, such as sharks, are fast swimmers, while the very successful bony fishes have unique characteristics such as swim bladders and lateral line systems.

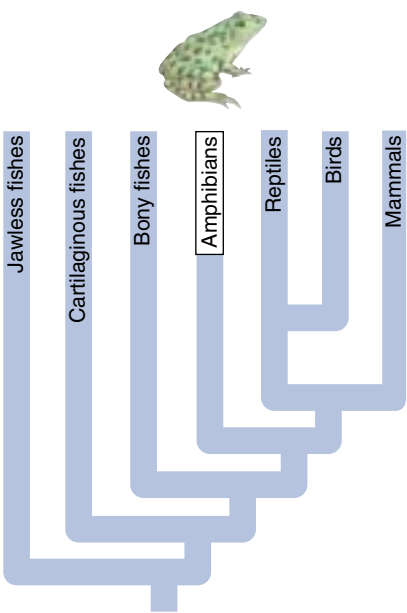
Amphibians

Frogs, salamanders, and caecilians, the damp-skinned vertebrates, are direct descendants of fishes. They are the sole survivors of a very successful group, the amphibians, the first vertebrates to walk on land. Most present-day amphibians are small and live largely unnoticed by humans. Amphibians are among the most numerous of terrestrial animals; there are more species of amphibians than of mammals. Throughout the world amphibians play key roles in terrestrial food chains.

Characteristics of Living Amphibians

Biologists have classified living species of amphibians into three orders (table 48.2): 3680 species of frogs and toads in 22 families make up the order Anura (“without a tail”); 369 species of salamanders and newts in 9 families make up the order Urodela or Caudata (“visible tail”); and 168 species (6 families) of wormlike, nearly blind organisms called caecilians that live in the tropics make up the order Apoda or Gymnophiona (“without legs”). They have key characteristics in common:

- 1. **Legs.** Frogs and salamanders have four legs and can move about on land quite well. Legs were one of the key adaptations to life on land. Caecilians have lost their legs during the course of adapting to a burrowing existence.
- 2. **Cutaneous respiration.** Frogs, salamanders, and caecilians all supplement the use of lungs by respiring directly across their skin, which is kept moist and provides an extensive surface area. This mode of respiration is only efficient for a high surface-to-volume ratio in an animal.



3. **Lungs.** Most amphibians possess a pair of lungs, although the internal surfaces are poorly developed, with much less surface area than reptilian or mammalian lungs. Amphibians still breathe by lowering the floor of the mouth to suck air in, then raising it back to force the air down into the lungs.

4. **Pulmonary veins.** After blood is pumped through the lungs, two large veins called pulmonary veins return the aerated blood to the heart for repumping. This allows the aerated blood to be pumped to the tissues at a much higher pressure than when it leaves the lungs.

5. **Partially divided heart.** The initial chamber of the fish heart is absent in amphibians, and the second and last chambers are separated by a dividing wall that helps prevent aerated blood from the lungs from mixing with non-

aerated blood being returned to the heart from the rest of the body. This separates the blood circulation into two separate paths, pulmonary and systemic. The separation is imperfect; the third chamber has no dividing wall.

Several other specialized characteristics are shared by all present-day amphibians. In all three orders, there is a zone of weakness between the base and the crown of the teeth. They also have a peculiar type of sensory rod cell in the retina of the eye called a “green rod.” The exact function of this rod is unknown.

Amphibians, with legs and more efficient blood circulation than fishes, were the first vertebrates to walk on land.

Table 48.2 Orders of Amphibians				
Order	Typical Examples		Key Characteristics	Approximate Number of Living Species
Anura	Frogs, toads		Compact tailless body; large head fused to the trunk; rear limbs specialized for jumping	3680
Caudata	Salamanders, newts		Slender body; long tail and limbs set out at right angles to the body	369
Apoda (Gymnophiona)	Caecilians		Tropical group with a snakelike body; no limbs; little or no tail	168

History of the Amphibians

The word *amphibia* (a Greek word meaning “both lives”) nicely describes the essential quality of modern day amphibians, referring to their ability to live in two worlds: the aquatic world of their fish ancestors and in the terrestrial world that they first invaded. In this section, we will review the checkered history of this group, almost all of whose members have been extinct for the last 200 million years. Then, in the following section, we will examine in more detail what the few kinds of surviving amphibians are like.

Origin of Amphibians

Paleontologists (scientists who study fossils) agree that amphibians must have evolved from the lobe-finned fishes, although for some years there has been considerable disagreement about whether the direct ancestors were coelacanths, lungfish, or the extinct rhipidistian fishes. Good arguments can be made for each. Many details of amphibian internal anatomy resemble those of the coelacanth. Lungfish and rhipidistians have openings in the tops of their mouths similar to the internal nostrils of amphibians. In addition, lungfish have paired lungs, like those of

amphibians. Recent DNA analysis indicates lungfish are in fact far more closely related to amphibians than are coelacanths. Most paleontologists consider that amphibians evolved from rhipidistian fishes, rather than lungfish, because the pattern of bones in the early amphibian skull and limbs bears a remarkable resemblance to the rhipidistians. They also share a particular tooth structure.

Their successful invasion of land by vertebrates involved a number of major adaptations:

1. Legs were necessary to support the body's weight as well as to allow movement from place to place (figure 48.18).
2. Lungs were necessary to extract oxygen from air. Even though there is far more oxygen available to gills in air than in water, the delicate structure of fish gills requires the buoyancy of water to support them and they will not function in air.
3. The heart had to be redesigned to make full use of new respiratory systems and to deliver the greater amounts of oxygen required by walking muscles.
4. Reproduction had to be carried out in water until methods evolved to prevent eggs from drying out.
5. Most importantly, a system had to be developed to prevent the body itself from drying out.

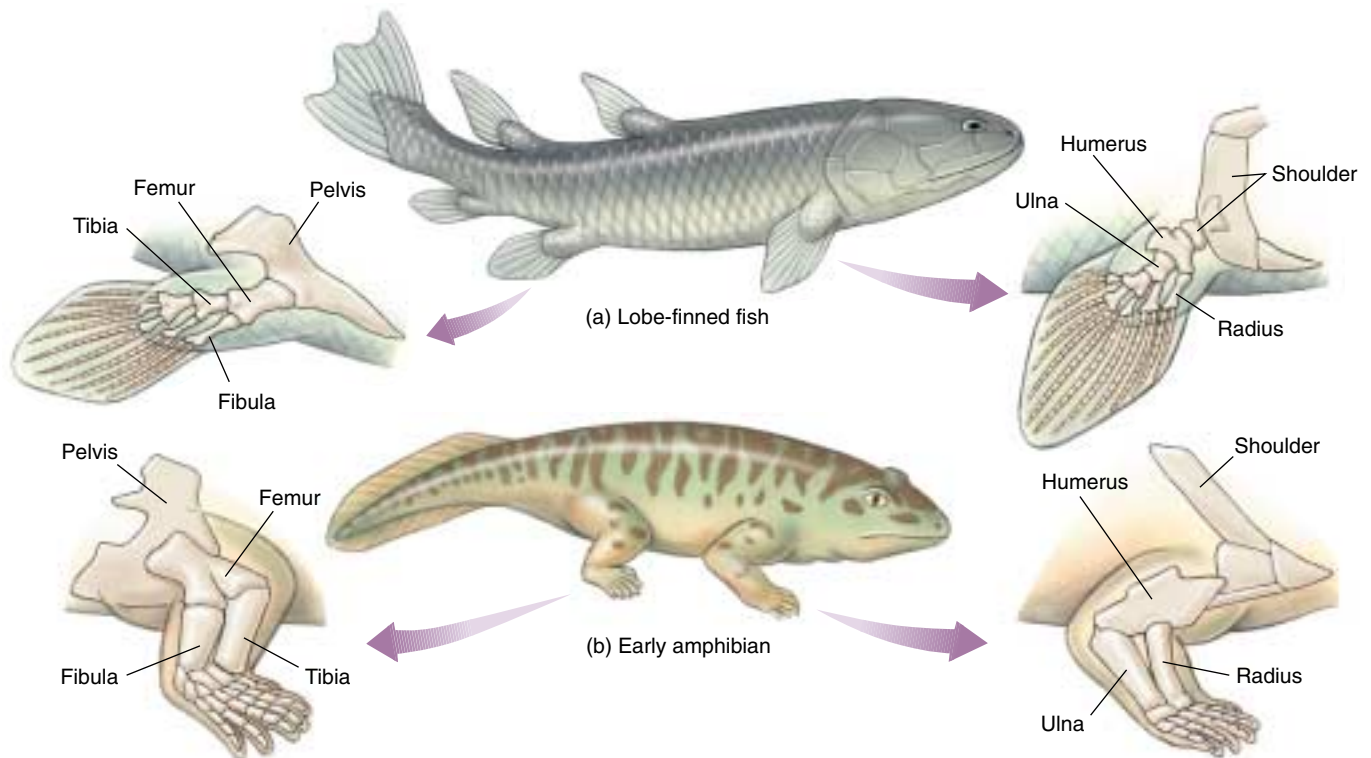


FIGURE 48.18

A comparison between the limbs of a lobe-finned fish and those of a primitive amphibian. (a) A lobe-finned fish. Some of these animals could probably move onto land. (b) A primitive amphibian. As illustrated by their skeletal structure, the legs of such an animal could clearly function on land much better than the fins of the lobe-finned fish.

The First Amphibian

Amphibians solved these problems only partially, but their solutions worked well enough that amphibians have survived for 350 million years. Evolution does not insist on perfect solutions, only workable ones.

Ichthyostega, the earliest amphibian fossil (figure 48.19) was found in a 370-million-year-old rock in Greenland. At that time, Greenland was part of the North American continent and lay near the equator. For the next 100 million years, all amphibian fossils are found in North America. Only when Asia and the southern continents all merged with North America to form the supercontinent Pangaea did amphibians spread throughout the world.

Ichthyostega was a strongly built animal, with four sturdy legs well supported by hip and shoulder bones. The shoulder bones no longer attached to the skull as in fish. The hipbones were braced against the backbone unlike in fish, so the limbs could support the animal's weight. To strengthen the backbone further, long, broad ribs that overlap each other formed a solid cage for the lungs and heart. The rib cage was so solid that it probably couldn't expand and contract for breathing. Instead, *Ichthyostega* obtained oxygen somewhat as a fish does, by lowering the floor of the mouth to draw air in, then raising it to push air down the windpipe into the lungs.

The Rise and Fall of Amphibians

Amphibians first became common during the Carboniferous Period (360 to 280 million years ago). Fourteen families of amphibians are known from the early Carboniferous, nearly all aquatic or semiaquatic, like *Ichthyostega*. By the late Carboniferous, much of North America was covered by low-lying tropical swamplands, and 34 families of amphibians thrived in this wet terrestrial environment, sharing it with pelycosaurs and other early reptiles. In the early Permian Period that followed (280 to 248 million years ago), a remarkable change occurred among amphibians—they began to leave the marshes for dry uplands. Many of these terrestrial amphibians had bony plates and armor covering their bodies and grew to be very large, some as big as a pony (figure 48.20). Both their large size and the complete covering of their bodies indicate that these amphibians did not use the skin respiratory system of present-day amphibians, but rather had an impermeable leathery skin to prevent water loss. By the mid-Permian, there were 40 families of amphibians. Only 25% of them were still semiaquatic like *Ichthyostega*; 60% of the amphibians were fully terrestrial, 15% were semiterrestrial.

This was the peak of amphibian success. By the end of the Permian, a reptile called a therapsid had become common, ousting the amphibians from their newly acquired niche on land. Following the mass extinction event at the end of the Permian, therapsids were the dominant land vertebrate and most amphibians were aquatic. This trend continued in the following Triassic Period (248 to 213 million



FIGURE 48.19

Amphibians were the first vertebrates to walk on land.

Reconstruction of *Ichthyostega*, one of the first amphibians with efficient limbs for crawling on land, an improved olfactory sense associated with a lengthened snout, and a relatively advanced ear structure for picking up airborne sounds. Despite these features, *Ichthyostega*, which lived about 350 million years ago, was still quite fishlike in overall appearance and represents a very early amphibian.



FIGURE 48.20

A terrestrial amphibian of the Permian. *Cacops*, a large, extinct amphibian, had extensive body armor.

years ago), which saw the virtual extinction of amphibians from land. By the end of the Triassic, there were only 15 families of amphibians (including the first frog), and almost without exception they were aquatic. Some of these grew to great size; one was 3 meters long. Only two groups of amphibians are known from the following Jurassic Period (213 to 144 million years ago), the anurans (frogs and toads) and the urodeles (salamanders and newts). The Age of Amphibians was over.

Amphibians Today

All of today's amphibians descended from the two families of amphibians that survived the Age of the Dinosaurs. During the Tertiary Period (65 to 2 million years ago), these moist-skinned amphibians underwent a highly successful invasion of wet habitats all over the world, and today there are over 4200 species of amphibians in 37 different families.

Anura. Frogs and toads, amphibians without tails, live in a variety of environments from deserts and mountains to ponds and puddles (figure 48.21*a*). Frogs have smooth, moist skin, a broad body, and long hind legs that make them excellent jumpers. Most frogs live in or near water, although some tropical species live in trees. Unlike frogs, toads have a dry, bumpy skin, short legs, and are well adapted to dry environments. All adult anurans are carnivores, eating a wide variety of invertebrates.

Most frogs and toads return to water to reproduce, laying their eggs directly in water. Their eggs lack water-tight external membranes and would dry out quickly out of the water. Eggs are fertilized externally and hatch into swimming larval forms called tadpoles. Tadpoles live in the water, where they generally feed on minute algae. After considerable growth, the body of the tadpole gradually changes into that of an adult frog. This process of abrupt change in body form is called **metamorphosis**.

Urodela (Caudata). Salamanders have elongated bodies, long tails, and smooth moist skin (figure 48.21*b*). They typically range in length from a few inches to a foot, although giant Asiatic salamanders of the genus *Andrias* are as much as 1.5 meters long and weigh up to 33 kilograms. Most salamanders live in moist places, such as under stones or logs, or among the leaves of tropical plants. Some salamanders live entirely in water.

Salamanders lay their eggs in water or in moist places. Fertilization is usually external, although a few species practice a type of internal fertilization in which the female picks up sperm packets deposited by the male. Unlike anurans, the young that hatch from salamander eggs do not undergo profound metamorphosis, but are born looking like small adults and are carnivorous.

Apoda (Gymnophiona). Caecilians, members of the order Apoda (Gymnophiona), are a highly specialized group of tropical burrowing amphibians (figure 48.21*c*). These legless, wormlike creatures average about 30 centimeters long, but can be up to 1.3 meters long. They have very small eyes and are often blind. They resemble worms but have jaws with teeth. They eat worms and other soil invertebrates. The caecilian male deposits sperm directly into the female, and the female usually bears live young. Mud eels, small amphibians with tiny forelimbs and no hind limbs that live in the eastern United States, are not apodans, but highly specialized urodelians.



(a)



(b)



(c)

FIGURE 48.21

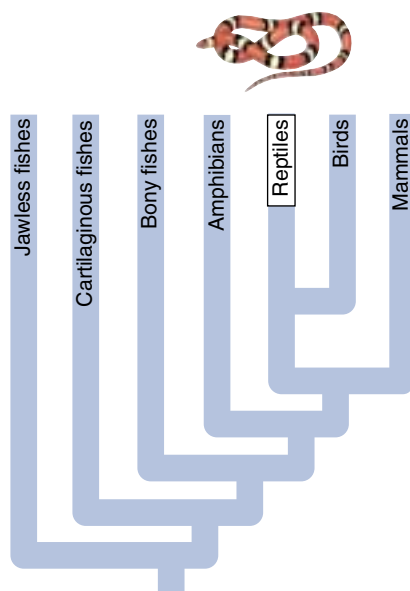
Class Amphibia. (a) Red-eyed tree frog, *Agalychnis callidryas* (order Anura). (b) An adult barred tiger salamander, *Ambystoma tigrinum* (order Caudata). (c) A XXXXXXXX caecilian, XXXXXXXX xxxxxxxxxxxx (order Gymnophiona).

Amphibians ventured onto land some 370 million years ago. They are characterized by moist skin, legs (secondarily lost in some species), lungs (usually), and a more complex and divided circulatory system. They are still tied to water for reproduction.

Reptiles

If one thinks of amphibians as a first draft of a manuscript about survival on land, then reptiles are the finished book. For each of the five key challenges of living on land, reptiles improved on the innovations first seen in amphibians. Legs were arranged to support the body's weight more effectively, allowing reptile bodies to be bigger and to *run*. Lungs and heart were altered to make them more efficient. The skin was covered with dry plates or scales to minimize water loss, and eggs were encased in watertight covers (figure 48.22). Reptiles were the first truly *terrestrial* vertebrates.

Over 7000 species of reptiles (class Reptilia) now live on earth (table 48.3). They are a highly successful group in today's world, more common than mammals. There are three reptile species for every two mammal species. While it is traditional to think of reptiles as more primitive than mammals, the great majority of reptiles that live today evolved from lines that appeared after therapsids did (the line that leads directly to mammals).



yolk sac provides food from the yolk for the embryo via blood vessels connecting to the embryo's gut. The **allantois** surrounds a cavity into which waste products from the embryo are excreted. All modern reptiles (as well as birds and mammals) show exactly this same pattern of membranes within the egg. These three classes are called amniotes.

2. Dry skin. Living amphibians have a moist skin and must remain in moist places to avoid drying out. Reptiles have dry, watertight skin. A layer of scales or armor covers their bodies, preventing water loss. These scales develop as surface cells fill with keratin, the same protein that forms claws, fingernails, hair, and bird feathers.

3. Thoracic breathing. Amphibians breathe by squeezing their throat to pump air into their lungs; this limits their breathing capacity to the volume of their mouth. Reptiles developed pulmonary breathing, expanding and contracting the rib cage to suck air into the lungs and then force it out. The capacity of this system is limited only by the volume of the lungs.

Key Characteristics of Reptiles

All living reptiles share certain fundamental characteristics, features they retain from the time when they replaced amphibians as the dominant terrestrial vertebrates. Among the most important are:

1. Amniotic egg. Amphibians never succeeded in becoming fully terrestrial because amphibian eggs must be laid in water to avoid drying out. Most reptiles lay watertight eggs that contain a food source (the yolk) and a series of four membranes—the yolk sac, the amnion, the allantois, and the chorion (figure 48.22). Each membrane plays a role in making the egg an independent life-support system. The outermost membrane of the egg is the **chorion**, which lies just beneath the porous shell. It allows respiratory gases to pass through, but retains water within the egg. Within, the **amnion** encases the developing embryo within a fluid-filled cavity. The

Reptiles were the first vertebrates to completely master the challenge of living on dry land.

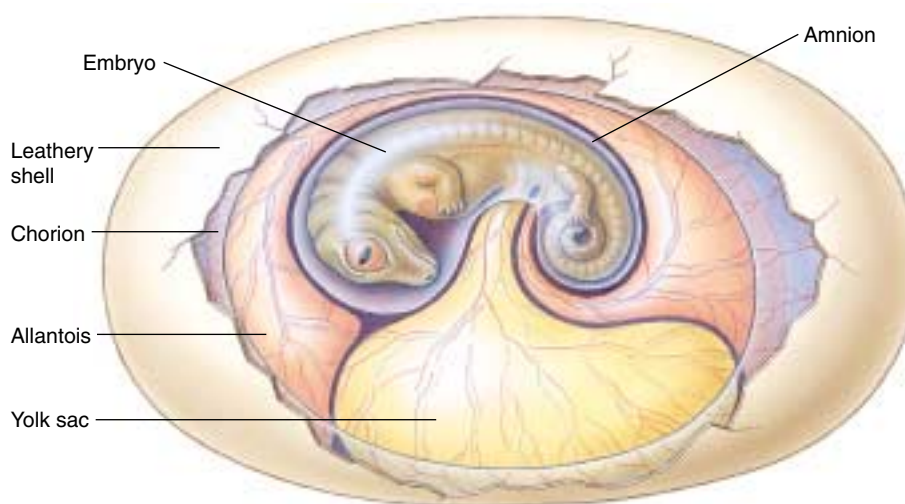


FIGURE 48.22

The watertight egg. The amniotic egg is perhaps the most important feature that allows reptiles to live in a wide variety of terrestrial habitats.

Table 48.3 Major Orders of Reptiles

Order	Typical Examples		Key Characteristics	Approximate Number of Living Species
Ornithischia	Stegosaur		Dinosaurs with two pelvic bones facing backward, like a bird's pelvis; herbivores, with turtlelike upper beak; legs under body	Extinct
Saurischia	Tyrannosaur		Dinosaurs with one pelvic bone facing forward, the other back, like a lizard's pelvis; both plant- and flesh-eaters; legs under body	Extinct
Pterosauria	Pterosaur		Flying reptiles; wings were made of skin stretched between fourth fingers and body; wingspans of early forms typically 60 centimeters, later forms nearly 8 meters	Extinct
Plesiosauroidea	Plesiosaur		Barrel-shaped marine reptiles with sharp teeth and large, paddle-shaped fins; some had snakelike necks twice as long as their bodies	Extinct
Ichthyosauria	Ichthyosaur		Streamlined marine reptiles with many body similarities to sharks and modern fishes	Extinct
Squamata, suborder Sauria	Lizards		Lizards; limbs set at right angles to body; anus is in transverse (sideways) slit; most are terrestrial	3800
Squamata, suborder Serpentes	Snakes		Snakes; no legs; move by slithering; scaly skin is shed periodically; most are terrestrial	3000
Chelonia	Turtles, tortoises, sea turtles		Ancient armored reptiles with shell of bony plates to which vertebrae and ribs are fused; sharp, horny beak without teeth	250
Crocodylia	Crocodiles, alligators, gavials, caimans		Advanced reptiles with four-chambered heart and socketed teeth; anus is a longitudinal (lengthwise) slit; closest living relatives to birds	25
Rhynchocephalia	Tuataras		Sole survivors of a once successful group that largely disappeared before dinosaurs; fused, wedgelike, socketless teeth; primitive third eye under skin of forehead	2

The Rise and Fall of Dominant Reptile Groups

During the 250 million years that reptiles were the dominant large terrestrial vertebrates, four major forms of reptiles took turns as the dominant type: pelycosaurs, therapsids, thecodonts, and dinosaurs.

Pelycosaurs: Becoming a Better Predator

Early reptiles like *pelycosaurs* were better adapted to life on dry land than amphibians because they evolved watertight eggs. They had powerful jaws because of an innovation in skull design and muscle arrangement. Pelycosaurs were **synapsids**, meaning that their skulls had a pair of temporal holes behind the openings for the eyes. An important feature of reptile classification is the presence and number of openings behind the eyes (see figure 48.27). Their jaw muscles were anchored to these holes, which allowed them to bite more powerfully. An individual pelycosaur weighed about 200 kilograms. With long, sharp, “steak knife” teeth, pelycosaurs were the first land vertebrates to kill beasts their own size (figure 48.23). Dominant for 50 million years, pelycosaurs once made up 70% of all land vertebrates. They died out about 250 million years ago, replaced by their direct descendants—the therapsids.

Therapsids: Speeding Up Metabolism

Therapsids (figure 48.24) ate ten times more frequently than their pelycosaur ancestors (figure 48.24). There is evidence that they may have been endotherms, able to regulate their own body temperature. The extra food consumption would have been necessary to produce body heat. This would have permitted therapsids to be far more active than other vertebrates of that time, when winters were cold and long. For 20 million years, therapsids (also called “mammallike reptiles”) were the dominant land vertebrate, until largely replaced 230 million years ago by a cold-blooded, or ectothermic, reptile line—the thecodonts. Therapsids became extinct 170 million years ago, but not before giving rise to their descendants—the mammals.

Thecodonts: Wasting Less Energy

Thecodonts were **diapsids**, their skulls having two pairs of temporal holes, and like amphibians and early reptiles, they were ectotherms (figure 48.25). Thecodonts largely replaced therapsids when the world’s climate warmed 230 million years ago. In the warm climate, the therapsid’s endothermy no longer offered a competitive advantage, and ectothermic thecodonts needed only a tenth as much food. Thecodonts were the first land vertebrates to be bipedal—to stand and walk on two feet. They were dominant through the Triassic and survived for 15 million years, until replaced by their direct descendants—the dinosaurs.



FIGURE 48.23

A pelycosaur. *Dimetrodon*, a carnivorous pelycosaur, had a dorsal sail that is thought to have been used to dissipate body heat or gain it by basking.



FIGURE 48.24

A therapsid. This small weaslelike cynodont therapsid, *Megazostrodon*, may have had fur. From the late Triassic, it is so similar to modern mammals that some paleontologists consider it the first mammal.



FIGURE 48.25

A thecodont. *Euparkeria*, a thecodont, had rows of bony plates along the sides of the backbone, as seen in modern crocodiles and alligators.

Dinosaurs: Learning to Run Upright

Dinosaurs evolved from thecodonts about 220 million years ago. Unlike the thecodonts, their legs were positioned directly underneath their bodies, a significant improvement in body design (figure 48.26). This design placed the weight of the body directly over the legs, which allowed dinosaurs to run with great speed and agility. A dinosaur fossil can be distinguished from a thecodont fossil by the presence of a hole in the side of the hip socket. Because the dinosaur leg is positioned underneath the socket, the force is directed upward, not inward, so there was no need for bone on the side of the socket. Dinosaurs went on to become the most successful of all land vertebrates, dominating for 150 million years. All dinosaurs became extinct rather abruptly 65 million years ago, apparently as a result of an asteroid's impact.

Figures 48.27 and 48.28 summarize the evolutionary relationships among the extinct and living reptiles.

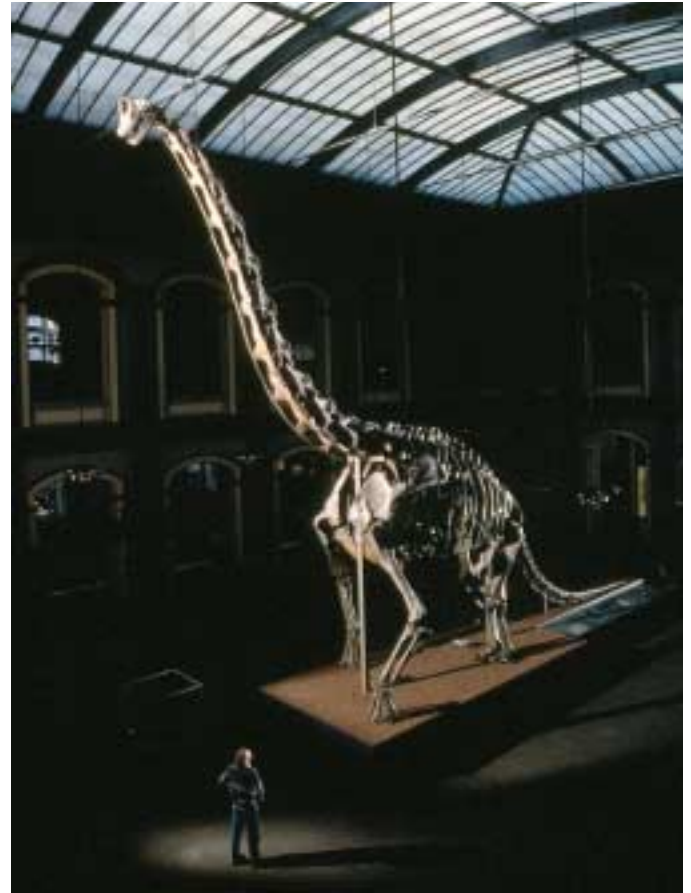
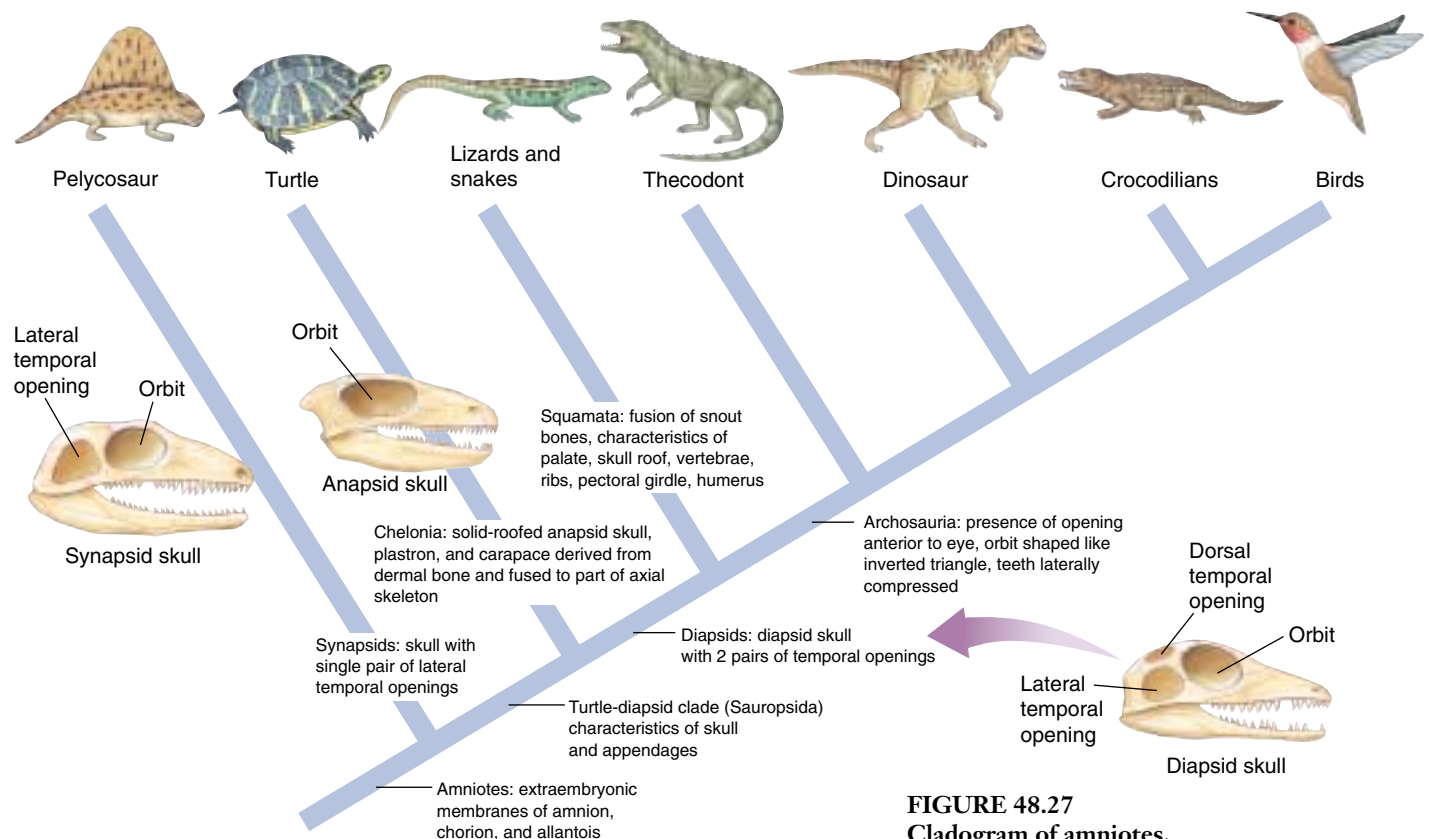


FIGURE 48.26

The largest mounted dinosaur in the world. This 145-million-year-old *Brachiosaurus*, a plant-eating sauropod over 80 feet long, lived in East Africa.



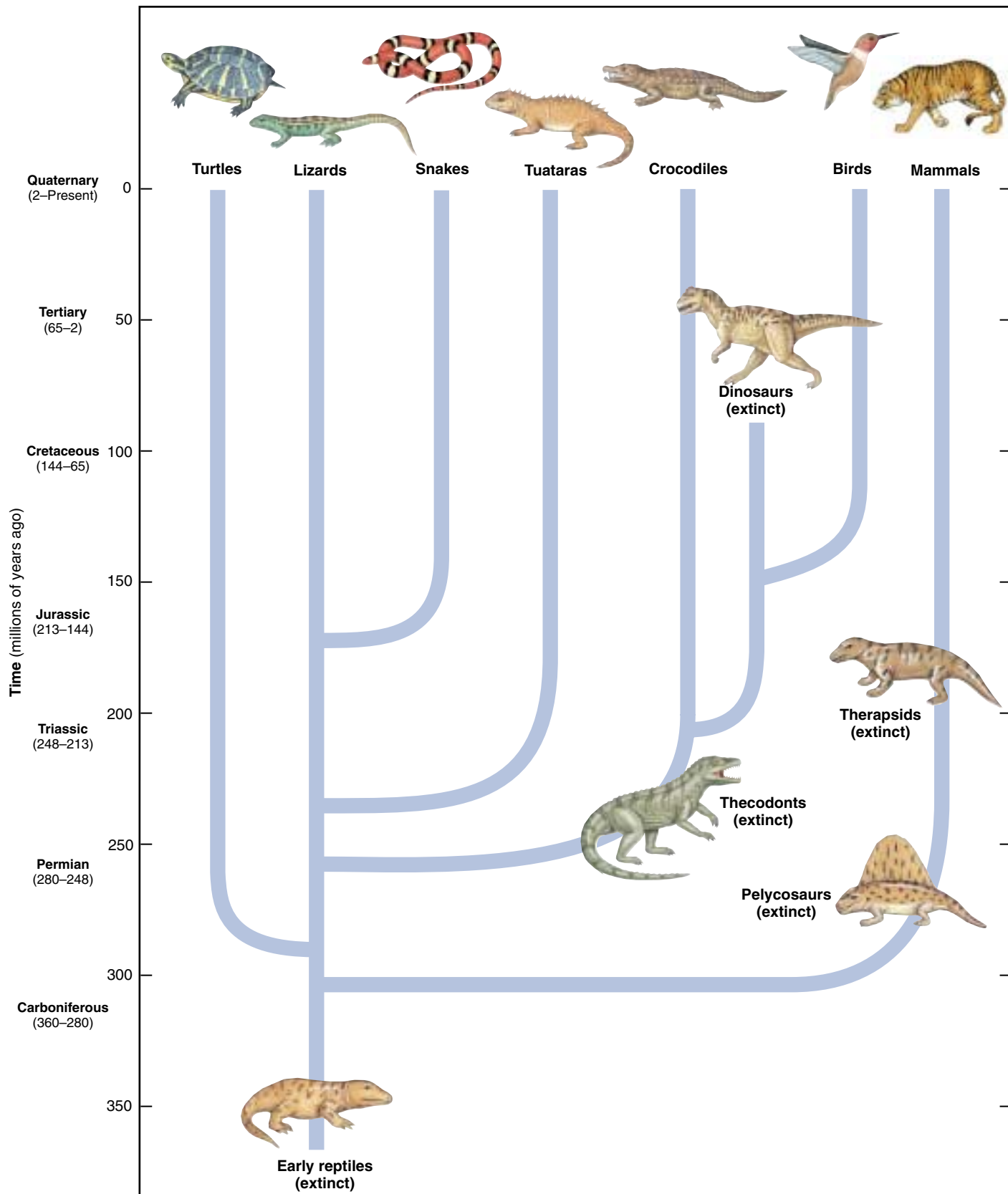
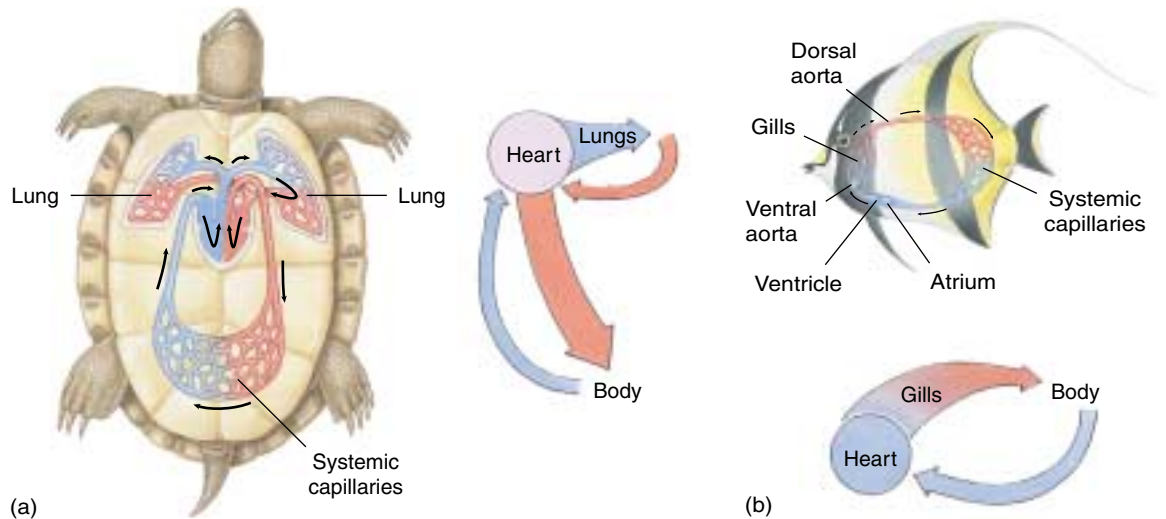


FIGURE 48.28
Evolutionary relationships among the reptiles. There are four orders of living reptiles: turtles, lizards and snakes, tuataras, and crocodiles. This phylogenetic tree shows how these four orders are related to one another and to dinosaurs, birds, and mammals.

FIGURE 48.29

A comparison of reptile and fish circulation.

(a) In reptiles such as this turtle, blood is repumped after leaving the lungs, and circulation to the rest of the body remains vigorous. (b) The blood in fishes flows from the gills directly to the rest of the body, resulting in slower circulation.



Today's Reptiles

Most of the major reptile orders are now extinct. Of the 16 orders of reptiles that have existed, only 4 survive.

Turtles. The most ancient surviving lineage of reptiles is that of turtles. Turtles have anapsid skulls much like those of the first reptiles. Turtles have changed little in the past 200 million years.

Lizards and snakes. Most reptiles living today belong to the second lineage to evolve, the lizards and snakes. Lizards and snakes are descended from an ancient lineage of lizardlike reptiles that branched off the main line of reptile evolution in the late Permian, 250 million years ago, before the thecodonts appeared (figure 48.28). Throughout the Mesozoic era, during the dominance of the dinosaurs, these reptiles survived as minor elements of the landscape, much as mammals did. Like mammals, lizards and snakes became diverse and common only after the dinosaurs disappeared.

Tuataras. The third lineage of surviving reptiles to evolve were the Rhynchocephalons, small diapsid reptiles that appeared shortly before the dinosaurs. They lived throughout the time of the dinosaurs and were common in the Jurassic. They began to decline in the Cretaceous, apparently unable to compete with lizards, and were already rare by the time dinosaurs disappeared. Today only two species of the order Rhynchocephalia survive, both tuataras living on small islands near New Zealand.

Crocodiles. The fourth lineage of living reptile, crocodiles, appeared on the evolutionary scene much later than other living reptiles. Crocodiles are descended from the same line of thecodonts that gave rise to the dinosaurs and resemble dinosaurs in many ways. They have changed very little in over 200 million years. Crocodiles, pterosaurs, thecodonts, and dinosaurs together make up a group called archosaurs ("ruling reptiles").

Other Important Characteristics

As you might imagine from the structure of the amniotic egg, reptiles and other amniotes do not practice external fertilization as most amphibians do. There would be no way for a sperm to penetrate the membrane barriers protecting the egg. Instead, the male places sperm inside the female, where they fertilize the egg before the membranes are formed. This is called internal fertilization.

The circulatory system of reptiles is improved over that of fish and amphibians, providing oxygen to the body more efficiently (figure 48.29). The improvement is achieved by extending the septum within the heart from the atrium partway across the ventricle. This septum creates a partial wall that tends to lessen mixing of oxygen-poor blood with oxygen-rich blood within the ventricle. In crocodiles, the septum completely divides the ventricle, creating a four-chambered heart, just as it does in birds and mammals (and probably in dinosaurs).

All living reptiles are **ectothermic**, obtaining their heat from external sources. In contrast, **endothermic** animals are able to generate their heat internally. In addition, **homeothermic** animals have a constant body temperature, and **poikilothermic** animals have a body temperature that fluctuates with ambient temperature. Thus, a deep-sea fish may be an ectothermic homeotherm because its heat comes from an external source, but its body temperature is constant. Reptiles are largely ectothermic poikilotherms; their body temperature is largely determined by their surroundings. Reptiles also regulate their temperature through behavior. They may bask in the sun to warm up or seek shade to prevent overheating. The thecodont ancestors of crocodiles were ectothermic, as crocodiles are today. The later dinosaurs from which birds evolved were endothermic. Crocodiles and birds differ in this one important respect. Ectothermy is a principal reason why crocodiles have been grouped among the reptiles.

Kinds of Living Reptiles

The four surviving orders of reptiles contain about 7000 species. Reptiles occur worldwide except in the coldest regions, where it is impossible for ectotherms to survive. Reptiles are among the most numerous and diverse of terrestrial vertebrates. The four living orders of the class Reptilia are Chelonia, Rhynchocephalia, Squamata, and Crocodylia.

Order Chelonia: Turtles and Tortoises. The order Chelonia consists of about 250 species of turtles (most of which are aquatic; figure 48.30) and tortoises (which are terrestrial). They differ from all other reptiles because their bodies are encased within a protective shell. Many of them can pull their head and legs into the shell as well, for total protection from predators. Turtles and tortoises lack teeth but have sharp beaks.

Today's turtles and tortoises have changed very little since the first turtles appeared 200 million years ago. Turtles are **anapsid**—they lack the temporal openings in the skull characteristic of other living reptiles, which are diapsid. This evolutionary stability of turtles may reflect the continuous benefit of their basic design—a body covered with a shell. In some species, the shell is made of hard plates; in other species, it is a covering of tough, leathery skin. In either case, the shell consists of two basic parts. The carapace is the dorsal covering, while the plastron is the ventral portion. In a fundamental commitment to this shell architecture, the vertebrae and ribs of most turtle and tortoise species are fused to the inside of the carapace. All of the support for muscle attachment comes from the shell.

While most tortoises have a domed-shaped shell into which they can retract their head and limbs, water-dwelling turtles have a streamlined, disc-shaped shell that permits rapid turning in water. Freshwater turtles have webbed toes, and in marine turtles, the forelimbs have evolved into flippers. Although marine turtles spend their lives at sea, they must return to land to lay their eggs. Many species migrate long distances to do this. Atlantic green turtles migrate from their feeding grounds off the coast of Brazil to Ascension Island in the middle of the South Atlantic—a distance of more than 2000 kilometers—to lay their eggs on the same beaches where they hatched.

Order Rhynchocephalia: Tuatara. The order Rhynchocephalia contains only two species today, the tuataras, large, lizardlike animals about half a meter long. The only place in the world where these endangered species are found is on a cluster of small islands off the coast of New Zealand. The native Maoris of New Zealand named the tuatara for the conspicuous spiny crest running down its back.

An unusual feature of the tuatara (and some lizards) is the inconspicuous “third eye” on the top of its head, called a parietal eye. Concealed under a thin layer of scales, the eye has a lens and retina and is connected by nerves to the



FIGURE 48.30
Red-bellied turtles, *Pseudemys rubriventris*. This turtle is common in the northeastern United States.

brain. Why have an eye, if it is covered up? The parietal eye may function to alert the tuatara when it has been exposed to too much sun, protecting it against overheating. Unlike most reptiles, tuataras are most active at low temperatures. They burrow during the day and feed at night on insects, worms, and other small animals.

Order Squamata: Lizards and Snakes. The order Squamata (figure 48.31) consists of three suborders: Sauria, some 3800 species of lizards, Amphisbaenia, about 135 species of worm lizards, and Serpentes, about 3000 species of snakes. The distinguishing characteristics of this order are the presence of paired copulatory organs in the male and a lower jaw that is not joined directly to the skull. A movable hinge with five joints (your jaw has only one) allows great flexibility in the movements of the jaw. In addition, the loss of the lower arch of bone below the lower opening in the skull of lizards makes room for large muscles to operate their jaws. Most lizards and snakes are carnivores, preying on insects and small animals, and these improvements in jaw design have made a major contribution to their evolutionary success.

The chief difference between lizards and snakes is that most lizards have limbs and snakes do not. Snakes also lack movable eyelids and external ears. Lizards are a more ancient group than modern snakes, which evolved only 20 million years ago. Common lizards include iguanas, chameleons, geckos, and anoles. Most are small, measuring less than a foot in length. The largest lizards belong to the monitor family. The largest of all monitors is the Komodo dragon of Indonesia, which reaches 3 meters in length and weighs up to 100 kilograms. Snakes also vary in size from only a few inches long to those that reach nearly 10 meters in length.

Lizards and snakes rely on agility and speed to catch prey and elude predators. Only two species of lizard are venomous, the Gila monster of the southwestern United States and the beaded lizard of western Mexico. Similarly, most species of snakes are nonvenomous. Of the 13 families of snakes, only 4 are venomous: the elapids (cobras, kraits,



(a)



(b)

FIGURE 48.31

Representatives from the order Squamata. (a) An Australian skink, *Sphenomorphus*. Some burrowing lizards lack legs, and the snakes evolved from one line of legless lizards. (b) A smooth green snake, *Liophorophis vernalis*.

and coral snakes); the sea snakes; the vipers (adders, bush-masters, rattlesnakes, water moccasins, and copperheads); and some colubrids (African boomslang and twig snake).

Many lizards, including skinks and geckos, have the ability to lose their tails and then regenerate a new one. This apparently allows these lizards to escape from predators.

Order Crocodilia: Crocodiles and Alligators. The order Crocodilia is composed of 25 species of large, primarily aquatic, primitive-looking reptiles (figure 48.32). In addition to crocodiles and alligators, the order includes two less familiar animals: the caimans and gavials. Crocodilians have remained relatively unchanged since they first evolved.

Crocodiles are largely nocturnal animals that live in or near water in tropical or subtropical regions of Africa, Asia, and South America. The American crocodile is found in southern Florida and Cuba to Columbia and Ecuador. Nile crocodiles and estuarine crocodiles can grow to enormous size and are responsible for many human fatalities each



FIGURE 48.32

River crocodile, *Crocodilus acutus*. Most crocodiles resemble birds and mammals in having four-chambered hearts; all other living reptiles have three-chambered hearts. Crocodiles, like birds, are more closely related to dinosaurs than to any of the other living reptiles.

year. There are only two species of alligators: one living in the southern United States and the other a rare endangered species living in China. Caimans, which resemble alligators, are native to Central America. Gavials are a group of fish-eating crocodilians with long, slender snouts that live only in India and Burma.

All crocodilians are carnivores. They generally hunt by stealth, waiting in ambush for prey, then attacking ferociously. Their bodies are well adapted for this form of hunting: their eyes are on top of their heads and their nostrils on top of their snouts, so they can see and breathe while lying quietly submerged in water. They have enormous mouths, studded with sharp teeth, and very strong necks. A valve in the back of the mouth prevents water from entering the air passage when a crocodilian feeds underwater.

Crocodiles resemble birds far more than they do other living reptiles. Alone among living reptiles, crocodiles care for their young (a trait they share with at least some dinosaurs) and have a four-chambered heart, as birds do. There are also many other points of anatomy in which crocodiles differ from all living reptiles and resemble birds. Why are crocodiles more similar to birds than to other living reptiles? Most biologists now believe that birds are in fact the direct descendants of dinosaurs. Both crocodiles and birds are more closely related to dinosaurs, and each other, than they are related to lizards and snakes.

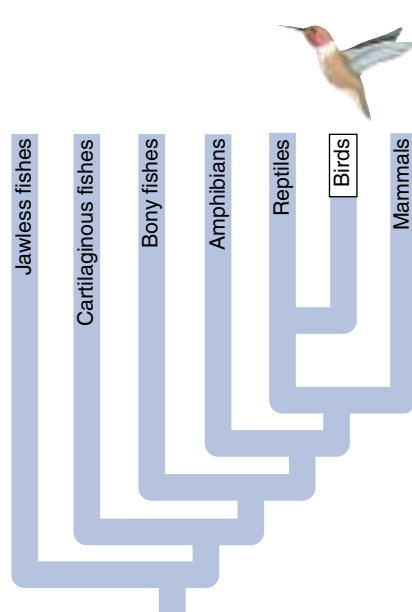
Many major reptile groups that dominated life on land for 250 million years are now extinct. The four living orders of reptiles include the turtles, lizards and snakes, tuataras, and crocodiles.

Birds

Only four groups of animals have evolved the ability to fly—insects, pterosaurs, birds, and bats. Pterosaurs, flying reptiles, evolved from gliding reptiles and flew for 130 million years before becoming extinct with the dinosaurs. There are startling similarities in how these very different animals meet the challenges of flight. Like water running downhill through similar gullies, evolution tends to seek out similar adaptations. There are major differences as well. The success of birds lies in the development of a structure unique in the animal world—the feather. Developed from reptilian scales, feathers are the ideal adaptation for flight—lightweight airfoils that are easily replaced if damaged (unlike the vulnerable skin wings of pterosaurs and bats). Today, birds (class Aves) are the most successful and diverse of all terrestrial vertebrates, with 28 orders containing a total of 166 families and about 8800 species (table 48.4).

Key Characteristics of Birds

Modern birds lack teeth and have only vestigial tails, but they still retain many reptilian characteristics. For instance, birds lay amniotic eggs, although the shells of bird eggs are hard rather than leathery. Also, reptilian scales are present on the feet and lower legs of birds. What makes birds unique? What distinguishes them from living reptiles?



1. Feathers. Feathers are modified reptilian scales that serve two functions: providing lift for flight and conserving heat. The structure of feathers combines maximum flexibility and strength with minimum weight (figure 48.33). Feathers develop from tiny pits in the skin called follicles. In a typical flight feather, a shaft emerges from the follicle, and pairs of vanes develop from its opposite sides. At maturity, each vane has many branches called barbs. The barbs, in turn, have many projections called barbules that are equipped with microscopic hooks. These hooks link the barbs to one another, giving the feather a continuous surface and a sturdy but flexible shape. Like scales, feathers can be replaced. Feathers are unique

to birds among living animals. Recent fossil finds suggest that some dinosaurs may have had feathers.

2. Flight skeleton. The bones of birds are thin and hollow. Many of the bones are fused, making the bird skeleton more rigid than a reptilian skeleton. The fused sections of backbone and of the shoulder and hip girdles form a sturdy frame that anchors muscles during flight. The power for active flight comes from large breast muscles that can make up 30% of a bird's total body weight. They stretch down from the wing and attach to the breastbone, which is greatly enlarged and bears a prominent keel for muscle attachment. They also attach to the fused collarbones that form the so-called “wishbone.” No other living vertebrates have a fused collarbone or a keeled breastbone.

Birds are the most diverse of all terrestrial vertebrates. They are closely related to reptiles, but unlike reptiles or any other animals, birds have feathers.

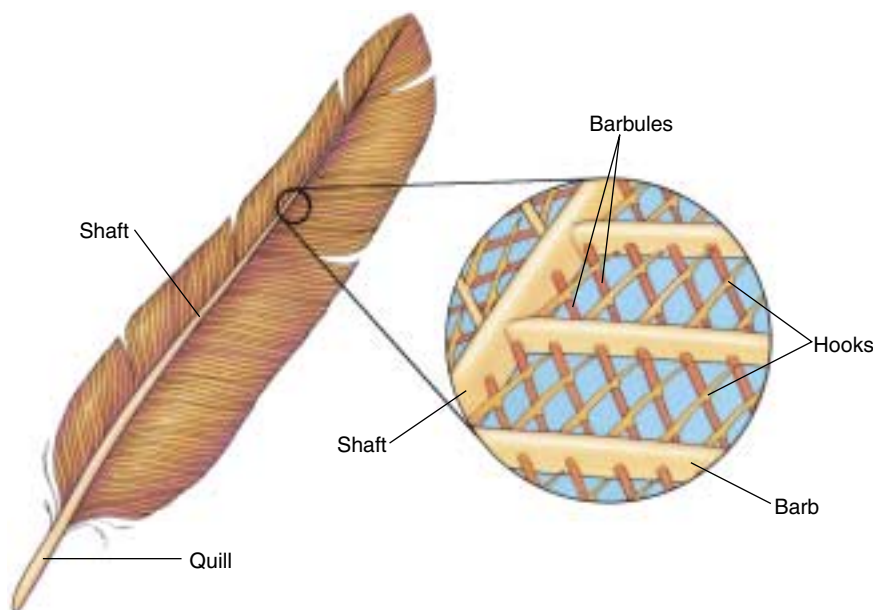


FIGURE 48.33

A feather. This enlargement shows how the vanes, secondary branches and barbs, are linked together by microscopic barbules.

Table 48.4 Major Orders of Birds

Order	Typical Examples		Key Characteristics	Approximate Number of Living Species
Passeriformes	Crows, mockingbirds, robins, sparrows, starlings, warblers		Songbirds Well-developed vocal organs; perching feet; dependent young	5276 (largest of all bird orders; contains over 60% of all species)
Apodiformes	Hummingbirds, swifts		<i>Fast fliers</i> Short legs; small bodies; rapid wing beat	428
Piciformes	Honeyguides, toucans, woodpeckers		<i>Woodpeckers or toucans</i> Grasping feet; chisel-like, sharp bills can break down wood	383
Psittaciformes	Cockatoos, parrots		<i>Parrots</i> Large, powerful bills for crushing seeds; well-developed vocal organs	340
Charadriiformes	Auks, gulls, plovers, sandpipers, terns		<i>Shorebirds</i> Long, stiltlike legs; slender probing bills	331
Columbiformes	Doves, pigeons		<i>Pigeons</i> Perching feet; rounded, stout bodies	303
Falconiformes	Eagles, falcons, hawks, vultures		<i>Birds of prey</i> Carnivorous; keen vision; sharp, pointed beaks for tearing flesh; active during the day	288
Galliformes	Chickens, grouse, pheasants, quail		<i>Gamebirds</i> Often limited flying ability; rounded bodies	268
Gruiformes	Bitterns, coots, cranes, rails		<i>Marsh birds</i> Long, stiltlike legs; diverse body shapes; marsh-dwellers	209
Anseriformes	Ducks, geese, swans		<i>Waterfowl</i> Webbed toes; broad bill with filtering ridges	150
Strigiformes	Barn owls, screech owls		<i>Owls</i> Nocturnal birds of prey; strong beaks; powerful feet	146
Ciconiiformes	Herons, ibises, storks		<i>Waders</i> Long-legged; large bodies	114
Procellariiformes	Albatrosses, petrels		<i>Seabirds</i> Tube-shaped bills; capable of flying for long periods of time	104
Sphenisciformes	Emperor penguins, crested penguins		<i>Penguins</i> Marine; modified wings for swimming; flightless; found only in southern hemisphere; thick coats of insulating feathers	18
Dinornithiformes	Kiwis		<i>Kiwis</i> Flightless; small; primitive; confined to New Zealand	2
Struthioniformes	Ostriches		<i>Ostriches</i> Powerful running legs; flightless; only two toes; very large	1

History of the Birds

A 150-million-year-old fossil of the first known bird, *Archaeopteryx* (figure 48.34)—pronounced “archie-op-ter-ichs”—was found in 1862 in a limestone quarry in Bavaria, the impression of its feathers stamped clearly into the rocks.

Birds Are Descended from Dinosaurs

The skeleton of *Archaeopteryx* shares many features with small theropod dinosaurs. About the size of a crow, its skull has teeth, and very few of its bones are fused to one another—dinosaurian features, not avian. Its bones are solid, not hollow like a bird’s. Also, it has a long reptilian tail, and no enlarged breastbone such as modern birds use to anchor flight muscles. Finally, it has the forelimbs of a dinosaur. Because of its many dinosaur features, several *Archaeopteryx* fossils were originally classified as the coelurosaur *Compsognathus*, a small theropod dinosaur of similar size—until feathers were discovered on the fossils. What makes *Archaeopteryx* distinctly avian is the presence of feathers on its wings and tail. It also has other birdlike features, notably the presence of a wishbone. Dinosaurs lack a wishbone, although thecodonts had them.

The remarkable similarity of *Archaeopteryx* to *Compsognathus* has led almost all paleontologists to conclude that *Archaeopteryx* is the direct descendant of dinosaurs—indeed, that today’s birds are “feathered dinosaurs.” Some even speak flippantly of “carving the dinosaur” at Thanksgiving dinner. The recent discovery of feathered dinosaurs in China lends strong support to this inference. The dinosaur *Caudipteryx*, for example, is clearly intermediate between *Archaeopteryx* and dinosaurs, having large feathers on its tail and arms but also many features of velociraptor di-



FIGURE 48.34

Archaeopteryx. An artist’s reconstruction of *Archaeopteryx*, an early bird about the size of a crow. Closely related to its ancestors among the bipedal dinosaurs, *Archaeopteryx* lived in the forests of central Europe 150 million years ago. The true feather colors of *Archaeopteryx* are not known.

nosaurus (figure 48.35). Because the arms of *Caudipteryx* were too short to use as wings, feathers probably didn’t evolve for flight. Instead, they probably served as insulation, much as fur does for animals. Flight is something that certain kinds of dinosaurs achieved as they evolved longer arms. We call these dinosaurs birds.

Despite their close affinity to dinosaurs, biologists continue to classify birds as Aves, a separate class, because of the key evolutionary novelties of birds: feathers, hollow bones, and physiological mechanisms such as superefficient lungs that permit sustained, powered flight. It is because of their unique adaptations and great diversity that

FIGURE 48.35

The evolutionary path to the birds. Almost all paleontologists now accept the theory that birds are the direct descendents of theropod dinosaurs.

birds are assigned to a separate class. This practical judgment should not conceal the basic agreement among almost all biologists that birds are the direct descendants of theropod dinosaurs, as closely related to coelurosaurs as are other theropods (see figure 48.35).

By the early Cretaceous, only a few million years after *Archaeopteryx*, a diverse array of birds had evolved, with many of the features of modern birds. Fossils in Mongolia, Spain, and China discovered within the last few years reveal a diverse collection of toothed birds with the hollow bones and breastbones necessary for sustained flight. Other fossils reveal highly specialized, flightless diving birds. The diverse birds of the Cretaceous shared the skies with pterosaurs for 70 million years.

Because the impression of feathers is rarely fossilized and modern birds have hollow, delicate bones, the fossil record of birds is incomplete. Relationships among the 166 families of modern birds are mostly inferred from studies of the degree of DNA similarity among living birds. These studies suggest that the most ancient living birds are the flightless birds, like the ostrich. Ducks, geese, and other waterfowl evolved next, in the early Cretaceous, followed by a diverse group of woodpeckers, parrots, swifts, and owls. The largest of the bird orders, Passeriformes, or songbirds (60% of all species of birds today), evolved in the mid-Cretaceous. The more specialized orders of birds, such as shorebirds, birds of prey, flamingos, and penguins, did not appear until the late Cretaceous. All but a few of the modern orders of toothless birds are thought to have arisen before the disappearance of the pterosaurs and dinosaurs at the end of the Cretaceous 65 million years ago.

Birds Today

You can tell a great deal about the habits and food of a bird by examining its beak and feet. For instance, carnivorous birds such as owls have curved talons for seizing prey and sharp beaks for tearing apart their meal. The beaks of ducks are flat for shoveling through mud, while the beaks of finches are short, thick seed-crushers. There are 28 orders of birds, the largest consisting of over 5000 species (figure 48.36).

Many adaptations enabled birds to cope with the heavy energy demands of flight:

1. **Efficient respiration.** Flight muscles consume an enormous amount of oxygen during active flight. The reptilian lung has a limited internal surface



FIGURE 48.36
Class Aves. This Western tanager, *Piranga ludoviciana*, is a member of the largest order of birds, the Passeriformes, with over 5000 species.

area, not nearly enough to absorb all the oxygen needed. Mammalian lungs have a greater surface area, but as we will see in chapter 53, bird lungs satisfy this challenge with a radical redesign. When a bird inhales, the air goes past the lungs to a series of air sacs located near and within the hollow bones of the back; from there the air travels to the lungs and then to a set of anterior air sacs before being exhaled. Because air always passes through the lungs in the same direction, and blood flows past the lung at right angles to the airflow, gas exchange is highly efficient.

2. **Efficient circulation.** The revved-up metabolism needed to power active flight also requires very efficient blood circulation, so

that the oxygen captured by the lungs can be delivered to the flight muscles quickly. In the heart of most living reptiles, oxygen-rich blood coming from the lungs mixes with oxygen-poor blood returning from the body because the wall dividing the ventricle into two chambers is not complete. In birds, the wall dividing the ventricle is complete, and the two blood circulations do not mix, so flight muscles receive fully oxygenated blood.

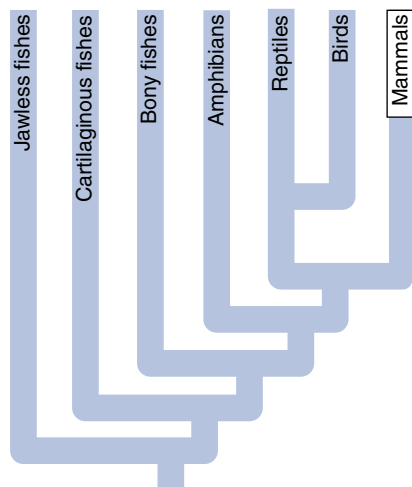
In comparison with reptiles and most other vertebrates, birds have a rapid heartbeat. A hummingbird's heart beats about 600 times a minute. An active chickadee's heart beats 1000 times a minute. In contrast, the heart of the large, flightless ostrich averages 70 beats per minute—the same rate as the human heart.

3. **Endothermy.** Birds, like mammals, are endothermic. Many paleontologists believe the dinosaurs that birds evolved from were endothermic as well. Birds maintain body temperatures significantly higher than most mammals, ranging from 40° to 42°C (your body temperature is 37°C). Feathers provide excellent insulation, helping to conserve body heat. The high temperatures maintained by endothermy permit metabolism in the bird's flight muscles to proceed at a rapid pace, to provide the ATP necessary to drive rapid muscle contraction.

The class Aves probably debuted 150 million years ago with *Archaeopteryx*. Modern birds are characterized by feathers, scales, a thin, hollow skeleton, auxiliary air sacs, and a four-chambered heart. Birds lay amniotic eggs and are endothermic.

Mammals

There are about 4100 living species of mammals (class Mammalia), the smallest number of species in any of the five classes of vertebrates. Most large, land-dwelling vertebrates are mammals (figure 48.37), and they tend to dominate terrestrial communities, as did the dinosaurs that they replaced. When you look out over an African plain, you see the big mammals, the lions, zebras, gazelles, and antelope. Your eye does not as readily pick out the many birds, lizards, and frogs that live in the grassland community with them. But the typical mammal is not all that large. Of the 4100 species of mammals, 3200 are rodents, bats, shrews, or moles (table 48.5).



are very sensitive to touch. Mammals that are active at night or live underground often rely on their whiskers to locate prey or to avoid colliding with objects. Hair can also serve as a defense weapon. Porcupines and hedgehogs protect themselves with long, sharp, stiff hairs called quills.

2. **Mammary glands.** All female mammals possess mammary glands that secrete milk. Newborn mammals, born without teeth, suckle this milk. Even baby whales are nursed by their mother's milk. Milk is a fluid rich in fat, sugar, and protein. A liter of human milk contains 11 grams of protein, 49 grams of fat, 70 grams of carbohydrate (chiefly the sugar lactose), and 2 grams of minerals critical to early growth, such as calcium. About 95% of the volume is water, critical to avoid dehydration. Milk is a very high calorie food (human milk has 750 kcal per liter), important because of the high energy needs of a rapidly growing newborn mammal. About 50% of the energy in the milk comes from fat.

Key Mammalian Characteristics

Mammals are distinguished from all other classes of vertebrates by two fundamental characteristics that are unique to mammals:

1. **Hair.** All mammals have hair. Even apparently naked whales and dolphins grow sensitive bristles on their snouts. Evolution of fur and the ability to regulate body temperature enabled mammals to invade colder climates that ectothermic reptiles could not inhabit, and the insulation fur provided may have ensured the survival of mammals when the dinosaurs perished.

Unlike feathers, which evolved from modified reptilian scales, mammalian hair is a completely different form of skin structure. An individual mammalian hair is a long, protein-rich filament that extends like a stiff thread from a bulblike foundation beneath the skin known as a hair follicle. The filament is composed mainly of dead cells filled with the fibrous protein keratin.

One of the most important functions of hair is insulation against heat loss. Mammals are endothermic animals, and typically maintain body temperatures higher than the temperature of their surroundings. The dense undercoat of many mammals reduces the amount of body heat that escapes.

Another function of hair is camouflage. The coloration and pattern of a mammal's coat usually matches its background. A little brown mouse is practically invisible against the brown leaf litter of a forest floor, while the orange and black stripes of a Bengal tiger disappear against the orange-brown color of the tall grass in which it hunts. Hairs also function as sensory structures. The whiskers of cats and dogs are stiff hairs that

Mammals first appeared 220 million years ago, evolving to their present position of dominance in modern terrestrial ecosystems. Mammals are the only vertebrates that possess hair and milk glands.



FIGURE 48.37

Mammals. African elephants, *Loxodonta africana*, at a water hole (order Proboscidea).

Table 48.5 Major Orders of Mammals

Order	Typical Examples		Key Characteristics	Approximate Number of Living Species
Rodentia	Beavers, mice, porcupines, rats		<i>Small plant-eaters</i> Chisel-like incisor teeth	1814
Chiroptera	Bats		<i>Flying mammals</i> Primarily fruit- or insect-eaters; elongated fingers; thin wing membrane; nocturnal; navigate by sonar	986
Insectivora	Moles, shrews		<i>Small, burrowing mammals</i> Insect-eaters; most primitive placental mammals; spend most of their time underground	390
Marsupialia	Kangaroos, koalas		<i>Pouched mammals</i> Young develop in abdominal pouch	280
Carnivora	Bears, cats, raccoons, weasels, dogs		<i>Carnivorous predators</i> Teeth adapted for shearing flesh; no native families in Australia	240
Primates	Apes, humans, lemurs, monkeys		<i>Tree-dwellers</i> Large brain size; binocular vision; opposable thumb; end product of a line that branched off early from other mammals	233
Artiodactyla	Cattle, deer, giraffes, pigs		<i>Hoofed mammals</i> With two or four toes; mostly herbivores	211
Cetacea	Dolphins, porpoises, whales		<i>Fully marine mammals</i> Streamlined bodies; front limbs modified into flippers; no hind limbs; blowholes on top of head; no hair except on muzzle	79
Lagomorpha	Rabbits, hares, pikas		<i>Rodentlike jumpers</i> Four upper incisors (rather than the two seen in rodents); hind legs often longer than forelegs; an adaptation for jumping	69
Pinnipedia	Sea lions, seals, walruses		<i>Marine carnivores</i> Feed mainly on fish; limbs modified for swimming	34
Edentata	Anteaters, armadillos, sloths		<i>Toothless insect-eaters</i> Many are toothless, but some have degenerate, peglike teeth	30
Perissodactyla	Horses, rhinoceroses, zebras		<i>Hoofed mammals with one or three toes</i> Herbivorous teeth adapted for chewing	17
Proboscidea	Elephants		<i>Long-trunked herbivores</i> Two upper incisors elongated as tusks; largest living land animal	2

History of the Mammals

Mammals have been around since the time of the dinosaurs, although they were never common until the dinosaurs disappeared. We have learned a lot about the evolutionary history of mammals from their fossils.

Origin of Mammals

The first mammals arose from therapsids in the mid-Triassic about 220 million years ago, just as the first dinosaurs evolved from thecodonts. Tiny, shrewlike creatures that lived in trees eating insects, mammals were only a minor element in a land that quickly came to be dominated by dinosaurs. Fossils reveal that these early mammals had large eye sockets, evidence that they may have been active at night. Early mammals had a single lower jawbone. Therapsid fossils show a change from the reptile lower jaw with several bones to a jaw closer to the mammalian-type jaw. Two of the bones forming the therapsid jaw joint retreated into the middle ear of mammals, linking with a bone already there producing a three-bone structure that amplifies sound better than the reptilian ear.

Early Divergence in Mammals

For 155 million years, while the dinosaurs flourished, mammals were a minor group of small insectivores and herbivores. Only five orders of mammals arose in that time, and their fossils are scarce, indicating that mammals were not abundant. However, the two groups to which present-day mammals belong did appear. The most primitive mammals, direct descendents of therapsids, were members of the subclass Prototheria. Most prototherians were small and resembled modern shrews. All prototherians laid eggs, as did their therapsid ancestors. The only prototherians surviving today are the monotremes—the duckbill platypus and the echidnas, or spiny anteaters. The other major mammalian group is the subclass Theria. All of the mammals you are familiar with, including humans, are therians. Therians are viviparous (that is, their young are born alive). The two major living therian groups are marsupials, or pouched mammals, and placental mammals. Kangaroos, opossums, and koalas are marsupials. Dogs, cats, humans, horses, and most other mammals are placentals.

Table 48.6 Some Groups of Extinct Mammals

Group	Description	
Cave bears	Numerous in the ice ages; this enormous vegetarian bear slept through the winter in large groups.	
Irish elk	Neither Irish nor an elk (it is a kind of deer), <i>Megaloceros</i> was the largest deer that ever lived, with horns spanning 12 feet. Seen in French cave paintings, they became extinct about 2500 years ago.	
Mammoths	Although only two species of elephants survive today, the elephant family was far more diverse during the late Tertiary. Many were cold-adapted mammoths with fur.	
Giant ground sloths	<i>Megatherium</i> was a giant 20-foot ground sloth that weighed three tons and was as large as a modern elephant.	
Sabertooth cats	The jaws of these large, lionlike cats opened an incredible 120 degrees to allow the animal to drive its huge upper pair of saber teeth into prey.	

The Age of Mammals

At the end of the Cretaceous Period 65 million years ago, the dinosaurs and numerous other land and marine animals became extinct, but mammals survived, possibly because of the insulation their fur provided. In the Tertiary Period (lasting from 65 million years to 2 million years ago), mammals rapidly diversified, taking over many of the ecological roles once dominated by dinosaurs (table 48.6). Mammals reached their maximum diversity late in the Tertiary Period, about 15 million years ago. At that time, tropical conditions existed over much of the world. During the last 15 million years, world climates have deteriorated, and the area covered by tropical habitats has decreased, causing a decline in the total number of mammalian species. There are now 19 orders of mammals.

Characteristics of Modern Mammals

Endothermy. Mammals are endothermic, a crucial adaptation that has allowed mammals to be active at any time of the day or night and to colonize severe environments, from deserts to ice fields. Many characteristics, such as hair that provides insulation, played important roles in making endothermy possible. Also, the more efficient blood circulation provided by the four-chambered heart and the more efficient respiration provided by the *diaphragm* (a special sheet of muscles below the rib cage that aids breathing) make possible the higher metabolic rate upon which endothermy depends.

Placenta. In most mammal species, females carry their young in a uterus during development, nourishing them through a placenta, and give birth to live young. The placenta is a specialized organ within the uterus of the pregnant mother that brings the bloodstream of the fetus into close contact with the bloodstream of the mother (figure 48.38). Food, water, and oxygen can pass across from mother to child, and wastes can pass over to the mother's blood and be carried away.

Teeth. Reptiles have homodont dentition: their teeth are all the same. However, mammals have heterodont dentition, with different types of teeth that are highly specialized to match particular eating habits (figure 48.39). It is usually possible to determine a mammal's diet simply by examining its teeth. Compare the skull of a dog (a carnivore) and a deer (an herbivore). The dog's long canine teeth are well suited for biting and holding prey, and some of its premolar and molar teeth are triangular and sharp for ripping off chunks of flesh. In contrast, canine teeth are absent in deer; instead the deer clips off mouthfuls of plants with flat, chisel-like incisors on its lower jaw. The deer's molars are large and covered with ridges to effectively grind and break up tough plant tissues. Rodents, such as beavers, are gnawers and have long incisors for chewing through branches or stems. These incisors are ever-growing; that is, the ends wear down, but new incisor growth maintains the length.

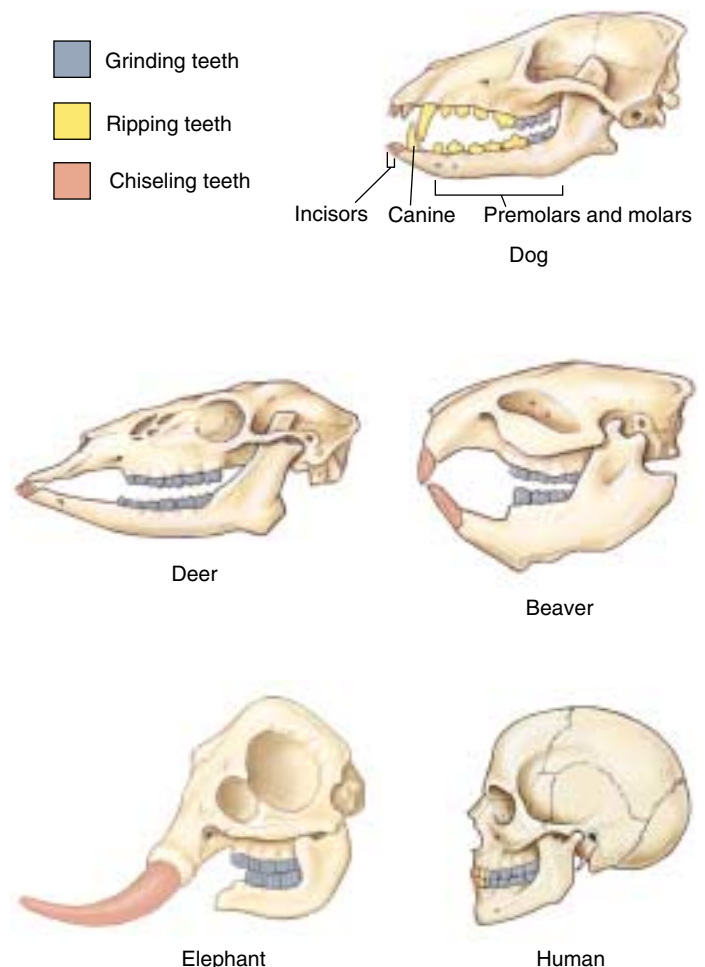
FIGURE 48.39

Mammals have different types of specialized teeth. While reptiles have all the same kind of teeth, mammals have different types of teeth specialized for different feeding habits. Carnivores such as dogs, have *canine* teeth that are able to rip food; some of the *premolars* and *molars* in dogs are also ripping teeth. Herbivores, such as deer, have *incisors* to chisel off vegetation and molars designed to grind up the plant material. In the beaver, the chiseling incisors dominate. In the elephant, the incisors have become specialized weapons, and molars grind up vegetation. Humans are omnivores; we have ripping, chiseling, and grinding teeth.



FIGURE 48.38

The placenta. The placenta is characteristic of the largest group of mammals, the placental mammals. It evolved from membranes in the amniotic egg. The umbilical cord evolved from the allantois. The chorion, or outermost part of the amniotic egg, forms most of the placenta itself. The placenta serves as the provisional lungs, intestine, and kidneys of the embryo, without ever mixing maternal and fetal blood.



Digesting Plants. Most mammals are herbivores, eating mostly or only plants. Cellulose, the major component of plant cell walls, forms the bulk of a plant's body and is a major source of food for mammalian herbivores. The cellulose molecule has the structure of a pearl necklace, with each pearl a glucose sugar molecule. Mammals do not have enzymes that can break the links between the pearls to release the glucose elements for use as food. Herbivorous mammals rely on a mutualistic partnership with bacteria that have the necessary cellulose-splitting enzymes to digest cellulose into sugar for them.

Mammals such as cows, buffalo, antelopes, goats, deer, and giraffes have huge, four-chambered stomachs that function as storage and fermentation vats. The first chamber is the largest and holds a dense population of cellulose-digesting bacteria. Chewed plant material passes into this chamber, where the bacteria attack the cellulose. The material is then digested further in the rest of the stomach.

Rodents, horses, rabbits, and elephants are herbivores that employ mutualistic bacteria to digest cellulose in a different way. They have relatively small stomachs, and instead digest plant material in their large intestine, like a termite. The bacteria that actually carry out the digestion of the cellulose live in a pouch called the cecum that branches from the end of the small intestine.

Even with these complex adaptations for digesting cellulose, a mouthful of plant is less nutritious than a mouthful of flesh. Herbivores must consume large amounts of plant material to gain sufficient nutrition. An elephant eats 135 to 150 kg (300 to 400 pounds) each day.

Horns and Hooves. Keratin, the protein of hair, is also the structural building material in claws, fingernails, and hooves. Hooves are specialized keratin pads on the toes of horses, cows, sheep, antelopes, and other running mammals. The pads are hard and horny, protecting the toe and cushioning it from impact.

The horns of cattle and sheep are composed of a core of bone surrounded by a sheath of keratin. The bony core is attached to the skull, and the horn is not shed. The horn that you see is the outer sheath, made of hairlike fibers of keratin compacted into a very hard structure. Deer antlers are made not of keratin but of bone. Male deer grow and shed a set of antlers each year. While growing during the summer, antlers are covered by a thin layer of skin known as velvet. A third type of horn, the rhinoceros horn, is composed only of keratinized fibers with no bony core.



FIGURE 48.40
Greater horseshoe bat, *Rhinolophus ferrumequinum*. The bat is the only mammal capable of true flight.

Flying Mammals. Bats are the only mammals capable of powered flight (figure 48.40). Like the wings of birds, bat wings are modified forelimbs. The bat wing is a leathery membrane of skin and muscle stretched over the bones of four fingers. The edges of the membrane attach to the side of the body and to the hind leg. When resting, most bats prefer to hang upside down by their toe claws. Bats are the second largest order of mammals, after rodents. They have been a particularly successful group because many species have been able to utilize a food resource that most birds do not have access to—night-flying insects.

How do bats navigate in the dark? Late in the eighteenth century, the Italian biologist Lazzaro Spallanzani showed that a blinded bat could fly without crashing into things and still capture insects. Clearly another sense other than vision was being used by bats to navigate in the dark. When Spallanzani plugged the ears of a bat, it was unable to navigate and collided with objects. Spallanzani concluded that bats “hear” their way through the night world.

We now know that bats have evolved a sonar system that functions much like the sonar devices used by ships and submarines to locate underwater objects. As a bat flies, it emits a very rapid series of extremely high-pitched “clicking” sounds well above our range of human hearing. The high-frequency pulses are emitted either through the mouth or, in some cases, through the nose. The soundwaves bounce off obstacles or flying insects, and the bat hears the echo. Through sophisticated processing of this echo within its brain, a bat can determine not only the direction of an object but also the distance to the object.

The Orders of Mammals

There are 19 orders of mammals. Seventeen of them (containing 94% of the species) are placental. The other two are the primitive monotremes and the marsupials.

Monotremes: Egg-laying Mammals. The duck-billed platypus and two species of echidna, or spiny anteater, are the only living monotremes (figure 48.41*a*). Among living mammals, only monotremes lay shelled eggs. The structure of their shoulder and pelvis is more similar to that of the early reptiles than to any other living mammal. Also like reptiles, monotremes have a cloaca, a single opening through which feces, urine, and reproductive products leave the body. Monotremes are more closely related to early mammals than are any other living mammal.

In addition to many reptilian features, monotremes have both defining mammalian features: fur and functioning mammary glands. Young monotremes drink their mother's milk after they hatch from eggs. Females lack well-developed nipples so the babies cannot suckle. Instead, the milk oozes onto the mother's fur, and the babies lap it off with their tongues.

The platypus, found only in Australia, lives much of its life in the water and is a good swimmer. It uses its bill much as a duck does, rooting in the mud for worms and other soft-bodied animals. Echidnas of Australia and New Guinea have very strong, sharp claws, which they use for burrowing and digging. The echidna probes with its long, beaklike snout for insects, especially ants and termites.

Marsupials: Pouched Mammals. The major difference between marsupials (figure 48.41*b*) and other mammals is their pattern of embryonic development. In marsupials, a fertilized egg is surrounded by chorion and amniotic membranes, but no shell forms around the egg as it does in monotremes. During most of its early development, the marsupial embryo is nourished by an abundant yolk within the egg. Shortly before birth, a short-lived placenta forms from the chorion membrane. Soon after, sometimes within eight days of fertilization, the embryonic marsupial is born. It emerges tiny and hairless, and crawls into the marsupial pouch, where it latches onto a nipple and continues its development.

Marsupials evolved shortly before placental mammals, about 100 million years ago. Today, most species of marsupials live in Australia and South America, areas that have been historically isolated. Marsupials in Australia and New Guinea have diversified to fill ecological positions occupied by placental mammals elsewhere in the world. For example, kangaroos are the Australian grazers, playing the role antelope, horses, and buffalo perform elsewhere. The placental mammals in Australia and New Guinea today arrived relatively recently and include some introduced by humans. The only marsupial found in North America is the Virginia opossum.

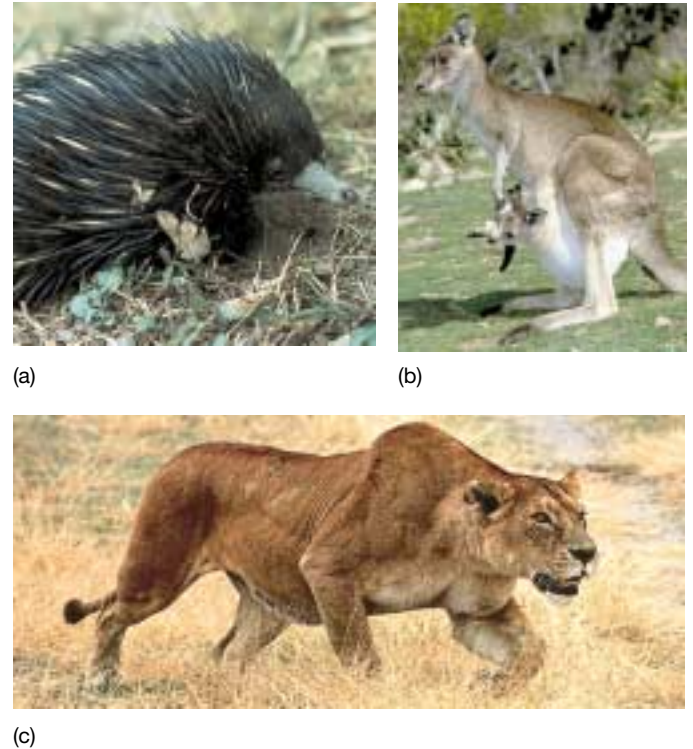


FIGURE 48.41
Three types of mammals. (a) This echidna, *Tachyglossus aculeatus*, is a monotreme. (b) Marsupials include kangaroos, like this adult with young in its pouch. (c) This female African lion, *Panthera leo* (order Carnivora), is a placental mammal.

Placental Mammals. Mammals that produce a true placenta that nourishes the embryo throughout its entire development are called placental mammals (figure 48.41*c*). Most species of mammals living today, including humans, are in this group. Of the 19 orders of living mammals, 17 are placental mammals. They are a very diverse group, ranging in size from 1.5 g pygmy shrews to 100,000 kg whales.

Early in the course of embryonic development, the placenta forms. Both fetal and maternal blood vessels are abundant in the placenta, and substances can be exchanged efficiently between the bloodstreams of mother and offspring. The fetal placenta is formed from the membranes of the chorion and allantois. The maternal side of the placenta is part of the wall of the uterus, the organ in which the young develop. In placental mammals, unlike marsupials, the young undergo a considerable period of development before they are born.

Mammals were not a major group until the dinosaurs disappeared. Mammal specializations include the placenta, a tooth design suited to diet, and specialized sensory systems.



Chapter 48

Summary

Questions

Media Resources

48.1 Attaching muscles to an internal framework greatly improves movement.

- The chordates are characterized by a dorsal nerve cord and by the presence, at least early in development, of a notochord, pharyngeal slits, and a postanal tail. In vertebrates, a bony endoskeleton provides attachment sites for skeletal muscle.

1. What are the four primary characteristics of the chordates?



- Chordates

48.2 Nonvertebrate chordates have a notochord but no backbone.

- Tunicates and the lancelets seem to represent ancient evolutionary Chordate offshoots.

2. What are the three subphyla of the chordates? Give an example of each.

48.3 The vertebrates have an interior framework of bone.

- Vertebrates differ from other chordates in that they possess a vertebral column, a distinct and well-differentiated head, and a bony skeleton.

3. What is the relationship between the notochord and the vertebral column in vertebrates?



- Introduction to Vertebrates



- Enhancement Chapter: Dinosaurs, Sections 6 and 7

48.4 The evolution of vertebrates involves successful invasions of sea, land, and air.

- Members of the group Agnatha differ from other vertebrates because they lack jaws.
- Jawed fishes constitute more than half of the estimated 42,500 species of vertebrates and are dominant in fresh and salt water everywhere.
- The first land vertebrates were the amphibians. Amphibians are dependent on water and lay their eggs in moist places.
- Reptiles were the first vertebrates fully adapted to terrestrial habitats. Scales and amniotic eggs represented significant adaptations to the dry conditions on land.
- Birds and mammals were derived from reptiles and are now among the dominant groups of animals on land. The members of these two classes have independently become endothermic, capable of regulating their own body temperatures; all other living animals are ectothermic, their temperatures set by external conditions.
- The living mammals are divided into three major groups: (1) the monotremes, or egg-laying mammals, consisting only of the echidnas and the duck-billed platypus; (2) the marsupials, in which the young are born at a very early stage of development and complete their development in a pouch; and (3) the placental mammals, which lack pouches and suckle their young.

4. What is one advantage of possessing jaws? From what existing structures did jaws evolve?



- Activity: Lamprey
- Activity: Fin Fish
- Fish
- Amphibians
- Reptiles
- Birds
- Mammals

5. What is the primary disadvantage of a bony skeleton compared to one made of cartilage?



- Enhancement Chapter: Dinosaurs, Sections 5
- Book Review: *The Pope's Rhinoceros* by Norfolk

6. What is the lateral line system in fishes? How does it function?

7. The successful invasion of land by amphibians involved five major innovations. What were they, and why was each important?

8. How does the embryo obtain nutrients and excrete wastes while contained within the egg?



9. From what reptilian structure are feathers derived?

10. How do amphibian, reptile, and mammal legs differ?

11. Exactly how would you distinguish a cat from a dog? (be specific)

- Student Research: Phylogeny of Hyliid Frogs
- Student Research: Metamorphosis in Flatfish
- Evolution of Fish



Animal Form and Function

Why Some Lizards Take a Deep Breath

Sometimes, what is intended as a straightforward observational study about an animal turns out instead to uncover an odd fact, something that doesn't at first seem to make sense. Teasing your understanding with the unexpected, this kind of tantalizing finding can be fun and illuminating to investigate. Just such an unexpected puzzle comes to light when you look very carefully at how lizards run.

A lizard runs a bit like a football fullback, swinging his shoulder forward to take a step as the opposite foot pushes off the ground. This produces a lateral undulating gait, the body flexing from side to side with each step. This sort of gait uses the body to aid the legs in power running. By contracting the chest (intercostal) muscles on the side of the body opposite the swinging shoulder, the lizard literally thrusts itself forward with each flex of its body.

The odd fact, the thing that at first doesn't seem to make sense, is that running lizards should be using these same intercostal chest muscles for something else.

At rest, lizards breathe by expanding their chest, much as you do. The greater volume of the expanded thorax lowers the interior air pressure, causing fresh air to be pushed into the lungs from outside. You expand your chest by contracting a diaphragm at the bottom of the chest. Lizards do not have a diaphragm. Instead, they expand their chest by contracting the intercostal chest muscles on both sides of the chest simultaneously. This contraction rotates the ribs, causing the chest to expand.

Do you see the problem? A running lizard cannot contract its chest muscles on both sides simultaneously for effective breathing at the same time that it is contracting the same chest muscles alternatively for running. This apparent conflict has led to a controversial hypothesis about how running lizards breathe. Called the axial constraint hypothesis, it states that lizards are subject to a speed-dependent axial constraint that prevents effective lung ventilation while they are running.

This constraint, if true, would be rather puzzling from an evolutionary perspective, because it suggests that when a lizard needs more oxygen because it is running, it breathes less effectively.

Dr. Elizabeth Brainerd of the University of Massachusetts, Amherst, is one of a growing cadre of young re-



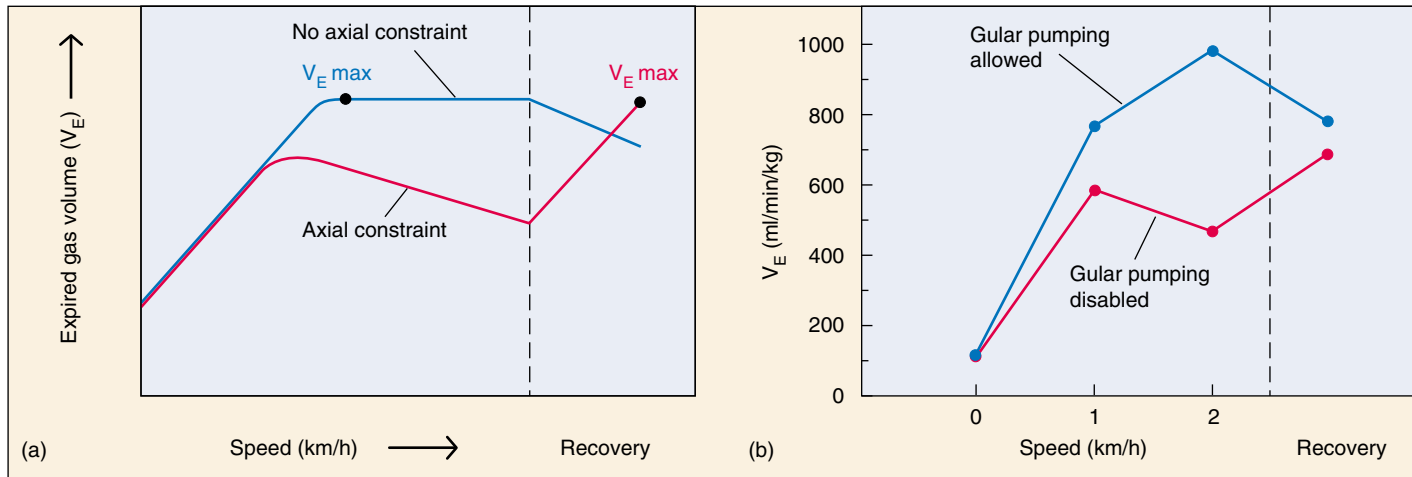
Some species of lizard breathe better than others. The savannah monitor lizard *Varanus exanthematicus* breathes more efficiently than some of its relatives by pumping air into its lungs from the gular folds over its throat.

searchers around the country that study the biology of lizards. She set out to investigate this puzzle several years ago, first by examining oxygen consumption.

Looking at oxygen consumption seemed a very straightforward approach. If the axial constraint hypothesis is correct, then running lizards should exhibit a lower oxygen consumption because of lowered breathing efficiency. This is just what her research team found with green iguanas (*Iguana iguana*). Studying fast-running iguanas on treadmills, oxygen consumption went down as running proceeded, as the axial constraint hypothesis predicted.

Unexpectedly, however, another large lizard gave a completely different result. The savannah monitor lizard (*Varanus exanthematicus*) exhibited *elevated* oxygen consumption with increasing speeds of locomotion! This result suggests that something else is going on in monitor lizards. Somehow, they seem to have found a way to beat the axial constraint.

How do they do it? Taking a more detailed look at running monitor lizards, Dr. Brainerd's research team ran a series of experiments to sort this out. First, they used videoradiography to directly observe lung ventilation in monitor lizards while the lizards were running on a treadmill. The X-ray negative video images revealed the monitor's trick: the breathing cycle began with an inhalation that did not completely fill the lungs, just as the axial constraint hypothesis predicts. But then something else kicks in. The gular cavity located in the throat area also fills with air, and as inhalation proceeds the gular cavity compresses, forcing this air into the lungs. Like an afterburner on a jet, this added air increases the efficiency of breathing, making up for the lost contribution of the intercostal chest muscles.



Effects of gular pumping in lizards. (a) THEORY: The axial constraint hypothesis predicts that, above a threshold speed, ventilation, measured by expired gas volume (V_E), will decrease with increasing speed, and only reach a maximum during the recovery period after locomotion ceases. Without axial constraint, ventilation should reach its maximum during locomotion. (b) EXPERIMENT: Monitor lizards typically show no axial constraint while running. Axial constraint is evident, however, if gular pumping of air is disabled. So, it seems that some species of monitor lizards are able to use gular pumping to overcome the axial constraint on ventilation.

The Experiment

Brainerd set out to test this gular pumping hypothesis. Gular pumping occurs after the initial inhalation because the lizard closes its mouth, sealing shut internal nares (nostril-like structures). Air is thus trapped in the gular cavity. By contracting muscles that compress the gular cavity, this air is forced into the lungs. This process can be disrupted by propping the mouth open so that, when the gular cavity is compressed, its air escapes back out of the mouth. The lizards were trained to run on a treadmill. A plastic mask was placed over the animal's mouth and nostrils and air was drawn through the mask. The mask permitted the measurement of oxygen and CO_2 levels as a means of monitoring gas consumption. The expired gas volume (V_E) was measured in the last minutes of locomotion and the first minute of recovery at each speed. The speeds ranged from 0 km/hr to 2 km/hr. The maximum running speed of these lizards on a treadmill is 6.6 km/hr.

To disable gular pumping, the animal's mouth was propped open with a retainer made of plastic tubing. In parallel experiments that allow gular pumping, the same animals wore the masks, but no retainer was used to disrupt the oral seal necessary for gular pumping.

The Results

Parallel experiments were conducted on monitor lizards with and without gular pumping:

1. *Gular pumping allowed.* When the gular pumping mechanism was not obstructed, the V_E increased to a maximum at a speed of 2 km/hr and decreased during the recovery period (see blue line in graph *b* above). This result is predicted under conditions where there is no axial constraint on the animal (see graph *a* above).

2. *Gular pumping disabled.* When the gular pumping mechanism is obstructed, V_E increased above the resting

value up to a speed of 1 km/hr. The value began to decrease between 1 and 2 km/hr indicating that there was constraint on ventilation. During the recovery period, V_E increased as predicted by the axial constraint hypothesis, because there was no longer constraint on the intercostal muscles. V_E increased to pay back an oxygen debt that occurred during the period of time when anaerobic metabolism took over.

Comparing the V_E measurements under control and experimental conditions, the researchers concluded that monitor lizards are indeed subject to speed-dependent axial constraint, just as theory had predicted, but can circumvent this constraint when running by using an accessory gular pump to enhance ventilation. When the gular pump was experimentally disrupted, the speed-dependent axial constraint condition became apparent.

Although the researchers have not conducted a more complete comparative analysis using the methods shown here, they have found correlations between gular pumping and increased locomotor activity. Six highly active species exhibited gular pumping while six less active species did not exhibit gular pumping in lung ventilation. It is interesting to speculate that gular pumping evolved in lizards as a means of enhancing breathing to allow greater locomotor endurance. The gular pumping seen in lizards is similar to the breathing mechanism found in amphibians and air-breathing fish. In these animals, the air first enters a cavity in the mouth called the buccal cavity. The mouth and nares close and the buccal cavity collapses, forcing air into the lungs. The similarities in these two mechanisms suggest that one might have arisen from the other.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab13.mhtml

49

Organization of the Animal Body

Concept Outline

49.1 The bodies of vertebrates are organized into functional systems.

Organization of the Body. Cells are organized into tissues, and tissues are organized into organs. Several organs can cooperate to form organ systems.

49.2 Epithelial tissue forms membranes and glands.

Characteristics of Epithelial Tissue. Epithelial membranes cover all body surfaces, and thus can serve for protection or for transport of materials. Glands are also epithelial tissue. Epithelial membranes may be composed of one layer or many.

49.3 Connective tissues contain abundant extracellular material.

Connective Tissue Proper. Connective tissues have abundant extracellular material. In connective tissue proper, this material consists of protein fibers within an amorphous ground substance.

Special Connective Tissues. These tissues include cartilage, bone, and blood, each with their own unique form of extracellular material.

49.4 Muscle tissue provides for movement, and nerve tissue provides for control.

Muscle Tissue. Muscle tissue contains the filaments actin and myosin, which enable the muscles to contract. There are three types of muscle: smooth, cardiac, and skeletal.

Nerve Tissue. Nerve cells, or neurons, have specialized regions that produce and conduct electrical impulses. Neuroglia cells support neurons but do not conduct electrical impulses.



FIGURE 49.1

Bone. Like most of the tissues in the vertebrate body, bone is a dynamic structure, constantly renewing itself.

When most people think of animals, they think of their pet dogs and cats and the animals that they've seen in a zoo, on a farm, in an aquarium, or out in the wild. When they think about the diversity of animals, they may think of the differences between the predatory lions and tigers and the herbivorous deer and antelope, between a ferocious-looking shark and a playful dolphin. Despite the differences among these animals, they are all vertebrates. All vertebrates share the same basic body plan, with the same sorts of organs operating in much the same way. In this chapter, we will begin a detailed consideration of the biology of vertebrates and of the fascinating structure and function of their bodies (figure 49.1).

49.1 The bodies of vertebrates are organized into functional systems.

Organization of the Body

The bodies of all mammals have the same general architecture (figure 49.2), and are very similar to the general body plan of other vertebrate groups. This body plan is basically a tube suspended within a tube. Starting from the inside, it is composed of the digestive tract, a long tube that travels from one end of the body to the other (mouth to anus). This tube is suspended within an internal body cavity, the *coelom*. In fishes, amphibians, and most reptiles, the coelom is subdivided into two cavities, one housing the heart and the other the liver, stomach, and intestines. In mammals and some reptiles, a sheet of muscle, the *diaphragm*, separates the **peritoneal cavity**, which contains the stomach, intestines, and liver, from the **thoracic cavity**; the thoracic cavity is further subdivided into the *pericardial cavity*, which contains the heart, and *pleural cavities*, which contain the lungs. All vertebrate bodies are supported by an internal **skeleton** made of jointed bones or cartilage blocks that grow as the body grows. A bony *skull* surrounds the brain, and a column of bones, the *vertebrae*, surrounds the dorsal nerve cord, or *spinal cord*.

There are four levels of organization in the vertebrate body: (1) cells, (2) tissues, (3) organs, and (4) organ systems. Like those of all animals, the bodies of vertebrates are composed of different cell types. In adult vertebrates, there are between 50 and several hundred different kinds of cells.

Tissues

Groups of cells similar in structure and function are organized into **tissues**. Early in development, the cells of the growing embryo differentiate (specialize) into three fundamental embryonic tissues, called *germ layers*. From innermost to outermost layers, these are the **endoderm**, **mesoderm**, and **ectoderm**. These germ layers, in turn, differentiate into the scores of different cell types and tissues that are characteristic of the vertebrate body. In adult vertebrates, there are four principal kinds of tissues, or *primary tissues*: epithelial, connective, muscle, and nerve (figure 49.3), each discussed in separate sections of this chapter.

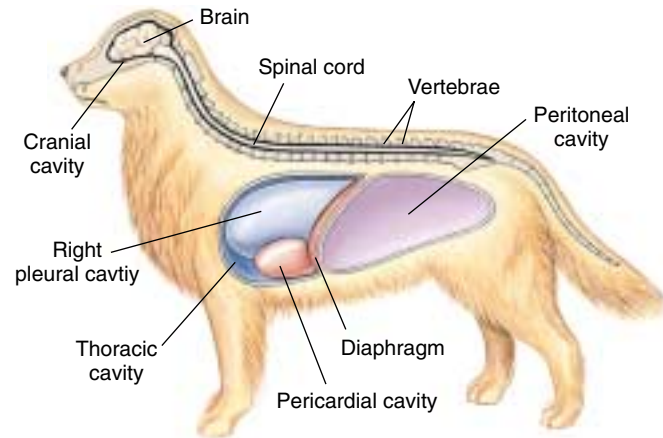


FIGURE 49.2

Architecture of the vertebrate body. All vertebrates have a dorsal central nervous system. In mammals and some reptiles, a muscular diaphragm divides the coelom into the thoracic cavity and the peritoneal cavity.

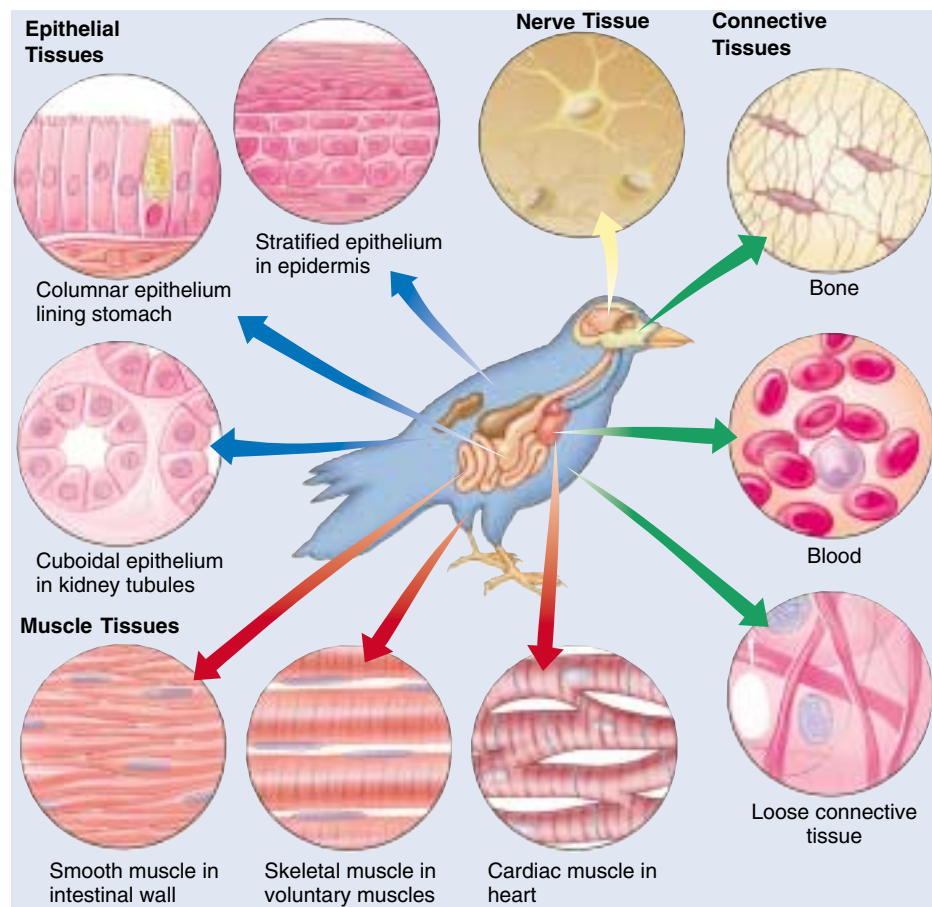


FIGURE 49.3

Vertebrate tissue types. Epithelial tissues are indicated by blue arrows, connective tissues by green arrows, muscle tissues by red arrows, and nerve tissue by a yellow arrow.

Organs and Organ Systems

Organs are body structures composed of several different tissues that form a structural and functional unit (figure 49.4). One example is the heart, which contains cardiac muscle, connective tissue, and epithelial tissue and is laced with nerve tissue that helps regulate the heartbeat. An **organ system** is a group of organs that function together to carry out the major activities of the body. For example, the digestive system is composed of the digestive tract, liver, gallbladder, and pancreas. These organs cooperate in the digestion of food and the absorption of digestion products into the body. The vertebrate body contains 11 principal organ systems (table 49.1 and figure 49.5).

The bodies of humans and other mammals contain a cavity divided by the diaphragm into thoracic and abdominal cavities. The body’s cells are organized into tissues, which are, in turn, organized into organs and organ systems.

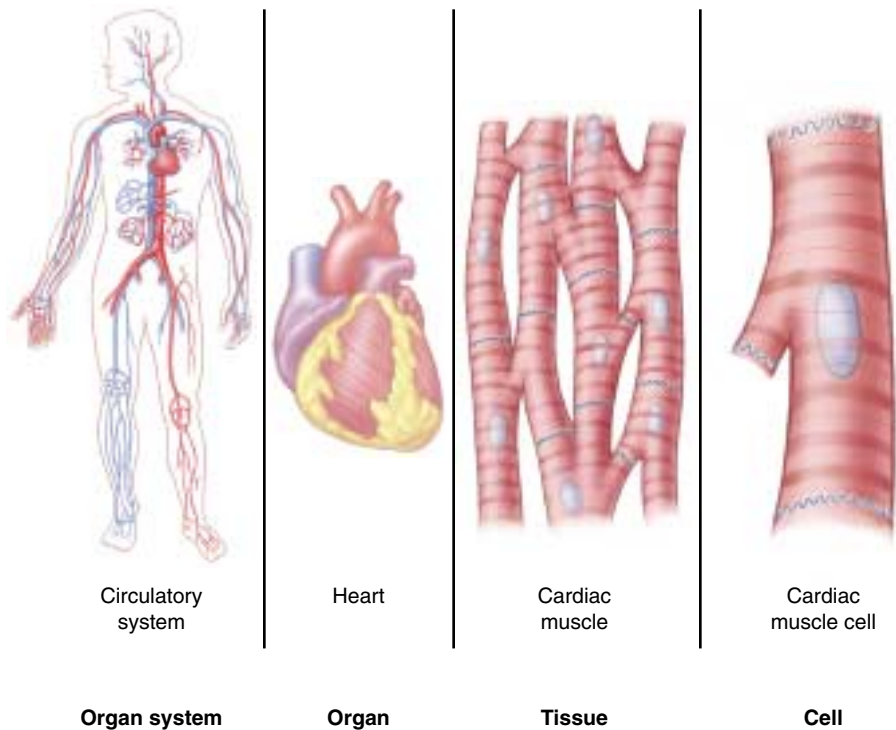


FIGURE 49.4
Levels of organization within the body. Similar cell types operate together and form tissues. Tissues functioning together form organs. Several organs working together to carry out a function for the body are called an organ system. The circulatory system is an example of an organ system.

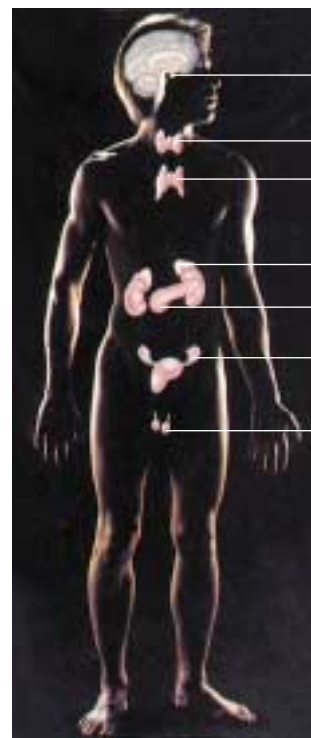
Table 49.1 The Major Vertebrate Organ Systems			
System	Functions	Components	Detailed Treatment
Circulatory	Transports cells, respiratory gases, and chemical compounds throughout the body	Heart, blood vessels, lymph, and lymph structures	Chapter 52
Digestive	Captures soluble nutrients from ingested food	Mouth, esophagus, stomach, intestines, liver, and pancreas	Chapter 51
Endocrine	Coordinates and integrates the activities of the body	Pituitary, adrenal, thyroid, and other ductless glands	Chapter 56
Integumentary	Covers and protects the body	Skin, hair, nails, scales, feathers, and sweat glands	Chapter 57
Lymphatic/Immune	Vessels transport extracellular fluid and fat to circulatory system; lymph nodes and lymphatic organs provide defenses to microbial infection and cancer	Lymphatic vessels, lymph nodes, thymus, tonsils, spleen	Chapter 57
Muscular	Produces body movement	Skeletal muscle, cardiac muscle, and smooth muscle	Chapter 50
Nervous	Receives stimuli, integrates information, and directs the body	Nerves, sense organs, brain, and spinal cord	Chapters 54, 55
Reproductive	Carries out reproduction	Testes, ovaries, and associated reproductive structures	Chapter 59
Respiratory	Captures oxygen and exchanges gases	Lungs, trachea, gills, and other air passageways	Chapter 53
Skeletal	Protects the body and provides support for locomotion and movement	Bones, cartilage, and ligaments	Chapter 50
Urinary	Removes metabolic wastes from the bloodstream	Kidney, bladder, and associated ducts	Chapter 58



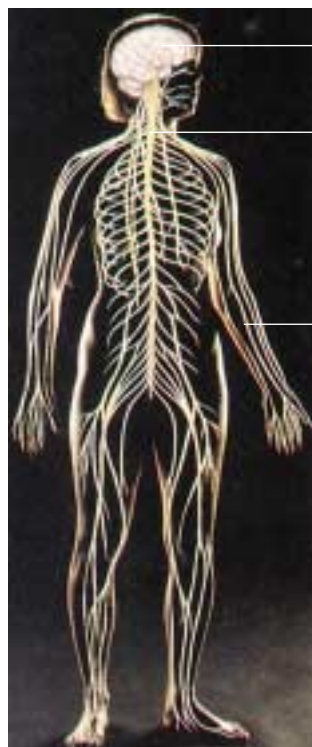
Skeletal system



Circulatory system



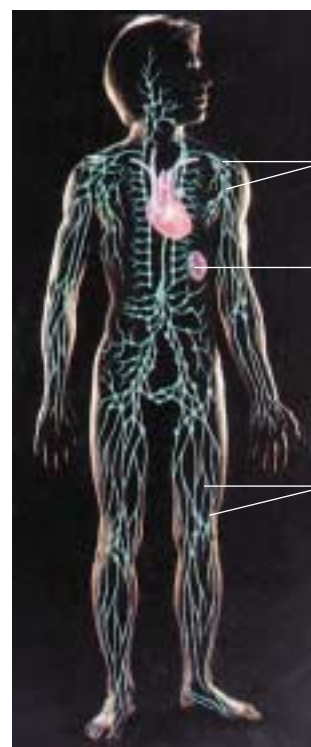
Endocrine system



Nervous system



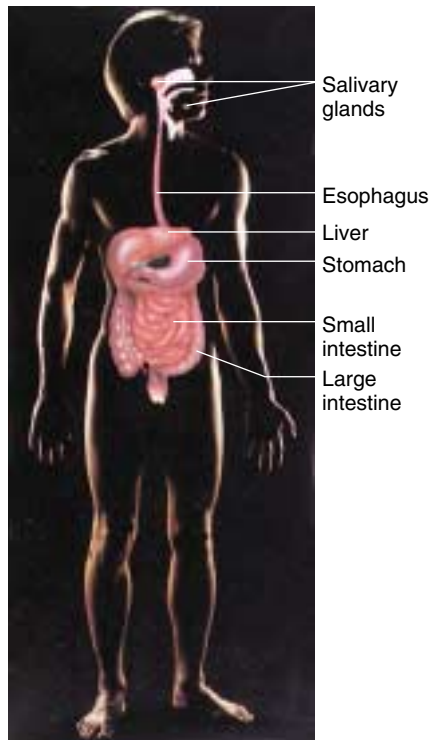
Respiratory system



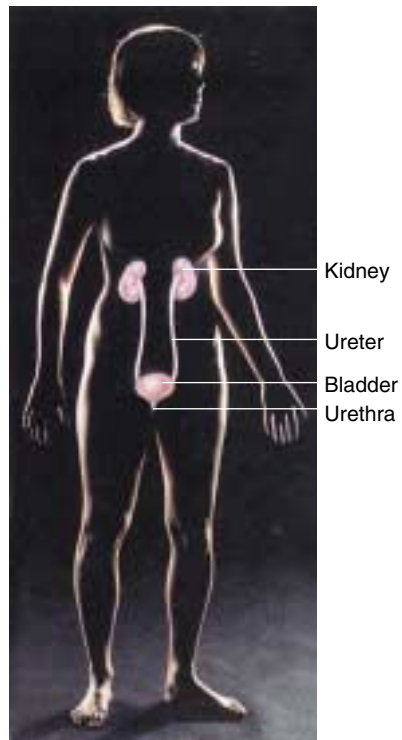
Lymphatic/Immune system

FIGURE 49.5

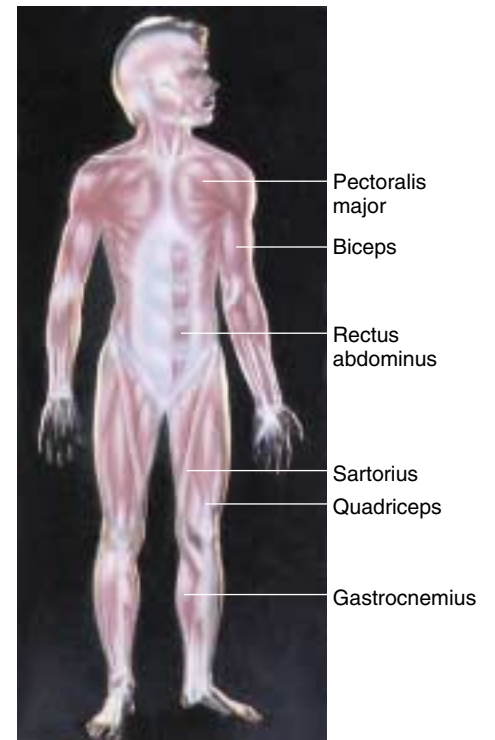
Vertebrate organ systems. The 11 principal organ systems of the human body are shown, including both male and female reproductive systems.



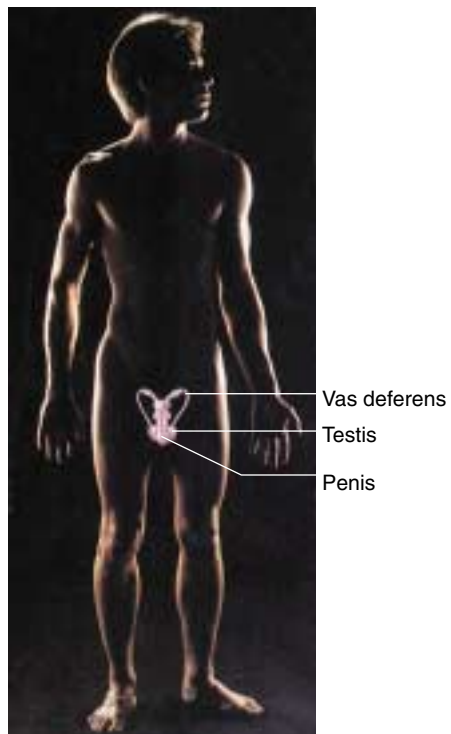
Digestive system



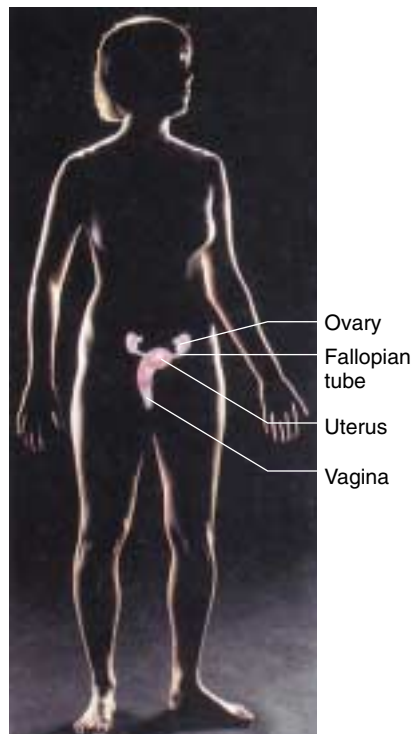
Urinary system



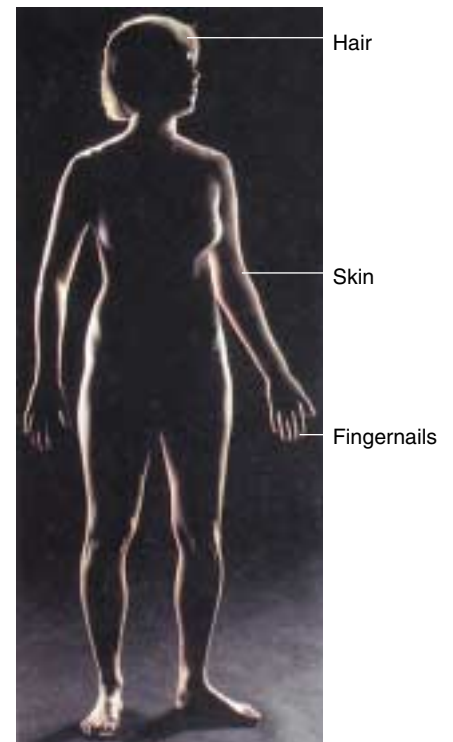
Muscular system



Reproductive system
(male)



Reproductive system
(female)



Integumentary system

FIGURE 49.5 (continued)